

INTERNATIONAL STUDENT EDITION

FUNDAMENTALS OF PROBABILITY

FIFTH EDITION

SAEED GHAHRAMANI



CRC Press
Taylor & Francis Group

A CHAPMAN & HALL BOOK

Fundamentals of Probability

Praise for the fourth edition:

“This book is an excellent primer on probability The flow of the text aids its readability, and the book is indeed a treasure trove of set and solved problems.

--**Dalia Chakrabarty**, *Brunel University*, UK

“This textbook provides a thorough and rigorous treatment of fundamental probability, including both discrete and continuous cases. The book’s ample collection of exercises gives instructors and students a great deal of practice and tools to sharpen their understanding.”

--**Joshua Stangle**, *University of Wisconsin – Superior*, USA

This one- or two-term calculus-based basic probability text is written for majors in mathematics, physical sciences, engineering, statistics, actuarial science, business and finance, operations research, and computer science. It presents probability in a natural way: through interesting and instructive examples and exercises that motivate the theory, definitions, theorems, and methodology. This book is mathematically rigorous and, at the same time, closely matches the historical development of probability. Whenever appropriate, historical remarks are included, and the 2096 examples and exercises have been carefully designed to arouse curiosity and hence encourage students to delve into the theory with enthusiasm.

New to the Fifth Edition:

In this edition, a significant change has been made in the order of material presentation.

The topics such as the joint probability mass function, joint probability density functions, independence of random variables, sums of random variables, the central limit theorem, and certain other materials have been covered earlier in the book, enabling students to grasp these crucial concepts from the start. These changes have considerable merit, particularly the idea of covering the celebrated central limit theorem immediately after discussing the normal distribution. Additionally, discussions on sigma fields are provided and an in-depth section on characteristic functions is added. The central limit theorem has been proven using both moment-generating functions and characteristic functions.

In the present edition, numerous new figures are included that were drawn for the first time, specifically to aid in students’ understanding of the material. These fresh illustrations, along with all the previous ones in the book, have been meticulously crafted by the technical support team at CRC.

Instructors who prefer the content arrangement used in previous editions can still teach the material in the same order as those editions. Moreover, the homepage of this book contains a whole chapter with comprehensive coverage on Stochastic Processes as well as additional contents for Chapters 1 to 10, such as extra examples, supplementary topics, and practical applications to facilitate in-depth exploration. Furthermore, it offers thorough solutions for all self-tests and self-quiz problems, empowering students to assess their progress and grasp of this demanding subject.

In this new edition, at the end of select chapters, sections are included dedicated to exploring approximate solutions for complex probabilistic problems using simulation techniques. These simulations are conducted using the R software, a powerful tool well-suited for probabilistic simulations due to its extensive collection of built-in functions and numerous specialized libraries designed for various simulation purposes. In the homepage of the book, a chapter, titled “Algorithm-Driven Simulations,” is presented in which we delve deeply into the concept of simulation using algorithms exclusively, without being tied to any specific programming language.

Fundamentals of Probability

Fifth Edition

Saeed Ghahramani



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To Lili, Adam, and Andrew



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Instructor & Student Resources

The *Instructor and Student Resources* website, accessible to both students and instructors at

<https://www.routledge.com/9781032366081>,

features comprehensive materials including an in-depth chapter on stochastic processes. It also offers supplementary material for each chapter, encompassing vital topics, a wealth of instructive and challenging examples, and complete solutions for all self-quiz and self-test problems. For ease of reference, the website's table of contents is provided below.

Contents of the Instructor and Student Resources

► Chapter 11 Stochastic Processes

- 11.1 Introduction
- 11.2 More on Poisson Processes
 - What Is a Queuing System?*
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- Chapter 11 Summary
- Review Problems
- Self-Test on Chapter 11

► Chapter 12 Algorithm-Driven Simulation

- 12.1 Introduction
- 12.2 Simulation of Combinatorial Problems
- 12.3 Simulation of Conditional Probabilities
- 12.4 Simulation of Random Variables
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Axioms of Probability

- 1A Applications of Probability to Genetics
- 1B Additional Examples
- 1C Solutions to Self-Quiz Problems
- 1D Solutions to Self-Test Problems

► **Supplementary Resources for Chapter 2**
Combinatorial Methods

- 2A Additional Examples
- 2B Solutions to Self-Quiz Problems
- 2C Solutions to Self-Test Problems

► **Supplementary Resources for Chapter 3**
Conditional Probability and Independence

- 3A More on Applications of Probability to Genetics
Hardy-Weinberg Law
Sex-Linked Genes
- 3B Additional Examples
- 3C Solutions to Self-Quiz Problems
- 3D Solutions to Self-Test Problems

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Distribution Functions and Discrete Random Variables

- 4A Pólya's Urn Model
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► **Supplementary Resources for Chapter 8**
**Understanding Relationships: Covariance, Correlations,
and Conditional Distributions**

- 8A Transformations of $n > 2$ Random Variables
- 8B Additional Examples
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► **Supplementary Resources for Chapter 9**
More Bivariate and Multivariate Topics

- 9A More on Memoryless Property of Exponential Random Variables
- 9B Conditioning on More than One Random Variable
- 9C Additional Examples
- 9D Solutions to Self-Quiz Problems
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► **Supplementary Resources for Chapter 10**
Inequalities and Limit Theorems

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- 10B Additional Examples
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► **Supplementary Resources for Chapter 11**
Stochastic Processes

- 11A Additional Examples
- 11B Solutions to Self-Quiz Problems
- 11C Solutions to Self-Test Problems

► **Answers to Odd-Numbered Exercises included on the**
Instructor and Student Resources Website



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Preface

This one- or two-term basic probability text is written for majors in mathematics, physical sciences, engineering, statistics, actuarial science, business and finance, operations research, and computer science. It can also be used by students who have completed a basic calculus course. Our aim is to present probability in a natural way: through interesting and instructive examples and exercises that motivate the theory, definitions, theorems, and methodology. Examples and exercises have been carefully crafted to arouse curiosity and hence encourage the students to engage deeply and eagerly with the subject matter.

Authors are usually faced with two opposing impulses. One is a tendency to put too much into the book, because *everything* is important and *everything* has to be said the author's way! On the other hand, authors must also keep in mind a clear definition of the focus, the level, and the audience for the book, thereby choosing carefully what should be “in” and what “out.” Hopefully, this book is a resolution of the tension generated by these opposing forces.

New to the Fifth Edition:

For the present edition, I have included numerous new figures that were drawn for the first time, specifically to aid in students' understanding of the material. These fresh illustrations, along with all the previous ones in the book, have been meticulously crafted by the technical support team at CRC.

In this edition, a significant change has been made in the order of material presentation. The fourth edition of this book underwent thorough review by 25 distinguished professors from around the world, who provided valuable input for the fifth edition. Many of them recommended that topics such as the joint probability mass function, joint probability density functions, independence of random variables, sums of random variables, the central limit theorem, and certain other materials should be covered earlier in the book, enabling students to grasp these crucial concepts from the start. Those suggestions had considerable merit, particularly the idea of covering the celebrated central limit theorem immediately after discussing the normal distribution. I have taken all of these recommendations into account, and they have been fully incorporated.

Additionally, there were suggestions to include discussions on sigma fields and to provide an in-depth section on characteristic functions, which I have duly incorporated. Instructors have the flexibility to skip these sections if time constraints arise.

In [Chapter 10](#), the central limit theorem has been proven using both moment-generating functions and characteristic functions. Instructors who prefer to teach only one of these topics can make use of the corresponding proof that aligns with their teaching approach.

Instructors who prefer the content arrangement used in previous editions can still teach the material in the same order as those editions. I have provided guidance on how to do this in the “Sample Syllabi” section of this preface.

In this fifth edition, as in the fourth edition, you will find self-quizzes at the end of each section and self-tests at the end of each chapter, allowing students to quiz and test themselves.

Additionally, this edition comes with access to a comprehensive *Instructor and Student Resources* website, specifically designed to enhance students' knowledge and proficiency in probability. I have personally authored all the material on this website, which includes two supplementary chapters, Chapters 11 and 12, that expand upon the content of the textbook. Furthermore, it enhances the content for [Chapters 1 to 10](#) with sections specifically titled *Supplementary Resources for Chapter 1*, *Supplementary Resources for Chapter 2*, and so on, for each respective chapter. These sections provide extra examples, supplementary topics, and practical applications to facilitate in-depth exploration. In addition, the website offers comprehensive solutions for all self-test and self-quiz problems, empowering students to assess their progress and absorb the intricacies of this challenging subject.

The website for Instructor and Student Resources is accessible to both instructors and students who are using the fifth edition of the book. You can visit it at this URL:

<https://www.routledge.com/9781032366081>.

Moreover, I have compiled an “Instructor’s Solutions Manual” that offers comprehensive solutions to nearly all of the 1650+ exercises found both in the book and on the Instructor and Student Resources website. It is important to note that this manual is exclusively available to instructors and is password protected. To access it, simply follow the URL provided above.

At the end of select chapters in this book, you will find sections dedicated to exploring approximate solutions for intricate probabilistic problems through the use of simulation techniques. These simulations are conducted using the R software, a powerful tool well-suited for probabilistic simulations due to its extensive collection of built-in functions and numerous specialized libraries designed for various simulation purposes.

In Chapter 12, titled “Algorithm-Driven Simulations,” which is available on the Instructor and Student Resources website for this book, we delve deeply into the concept of simulation using algorithms exclusively, without being tied to any specific programming language.

Since the release of the fourth edition of this book, I have not only benefited from 25 comprehensive expert reviews but also received substantial feedback from educators and students both domestically and internationally. The valuable comments, discussions, recommendations, and reviews have played a crucial role in enhancing the book. All identified errors have been rectified, and the text has been refined for accuracy. Furthermore, additional explanations and clarifications have been included in specific sections, and more insightful and improved solutions have been provided for numerous problems and exercises.

To differentiate the content within the book from that on the Instructor and Student Resources website, the book uses numbers to denote sections, theorems, examples, figures, etc., within each chapter. This convention is maintained for Chapters 11 and 12, which extend the textbook’s coverage on the website. Conversely, for the additional materials on the website beyond these chapters, sections are designated with capital letters, while examples, theorems, figures, etc., are marked with lowercase letters. For instance, within the textbook and on the website for Chapters 11 and 12, [Section 5.3](#), Example 11.2, and Figure 8.5 correspond to the third section of [Chapter 5](#), the second example in Chapter 11, and the fifth figure in [Chapter 8](#), respectively. In contrast, for the supplementary content on the website, Section 5B, Example 7a, and Figure 8d denote the second section in the Supplementary Resources for [Chapter 5](#), the first example in the Supplementary Resources for [Chapter 7](#), and the fourth figure in the Supplementary Resources for [Chapter 8](#), respectively.

Core Features Retained:

- Instructors can fully appreciate the flexibility offered by this text and its Instructor and Student Resources website. They have the freedom to select their preferred examples and exercises from a wide-ranging collection of over 2150, and if needed, they can omit specific sections or theorems to tailor their teaching to the appropriate level.
- I have endeavored to maintain an approach that combines mathematical rigor with a close alignment to the historical development of probability. Whenever relevant, historical remarks have been included, alongside discussions of various probability problems that have been published in journals like *Mathematics Magazine* and *American Mathematical Monthly*. These problems are not only intriguing but also serve as valuable teaching materials for classroom discussions.
- The book starts by exploring probability and its definition, instead of commencing with combinatorics. I hold the view that combinatorics should be introduced after students grasp the foundational concepts of probability. This method naturally brings to light the necessity for counting techniques, making the link between the two disciplines evident from the start. Furthermore, this approach makes combinatorics more engaging and enjoyable.
- Exercises for most sections are categorized into two groups: A and B. Category A comprises routine exercises, while category B includes challenging ones. However, it's worth noting that not all exercises in category B are uniformly challenging; some of them are included because students may find them somewhat difficult.
- The book and the Instructor and Student Resources present practical, meaningful, and relevant applications of probability and stochastic processes in finance, economics, and actuarial sciences."
- The historical roots and applications of many of the theorems and definitions are presented in detail, accompanied by suitable examples or counterexamples.
- Throughout the book, there is a rich array of insurance-related probability examples and exercises. These, combined with hundreds of other general indeterministic problems that are also central to actuarial studies, make the book an ideal resource for actuarial students.
- Borel's normal number theorem is discussed in [Chapter 10](#), and a version of a famous set that is not an event is presented in [Chapter 1](#).
- Whenever a new concept is introduced, its relationship to preceding concepts and theorems is explained.
- Although the usual analytic proofs are given, simple probabilistic arguments are presented to promote deeper understanding of the subject.
- Students beginning their study of probability have a tendency to think that sample spaces always have a finite number of sample points. To minimize this proclivity, the concept of *random selection of a point from an interval* is introduced in [Chapter 1](#) and applied where appropriate throughout the book. Moreover, since the basis of simulating indeterministic problems is selection of random points from $(0, 1)$, in order to understand simulations, students should possess a solid comprehension of this concept.

- Often, when we think of a collection of events, we have a tendency to think about them in either temporal or logical sequence. So, if, for example, a sequence of events $A_1, A_2, \dots, A_n$ occur in time or in some logical order, we can usually immediately write down the probabilities $P(A_1), P(A_2 \mid A_1), \dots, P(A_n \mid A_1 A_2 \cdots A_{n-1})$ without much computation. However, we may be interested in probabilities of the intersection of events, or probabilities of events unconditional on the rest, or probabilities of earlier events, given later events. These three questions motivated the need for the law of multiplication, the law of total probability, and Bayes' theorem. I have given the law of multiplication a section of its own so that each of these fundamental uses of conditional probability would have its full share of attention and coverage.
- Significant emphasis is placed on the Poisson distribution, highlighting its frequent applicability for two key reasons. Firstly, it serves as an approximation to the binomial distribution. Secondly, it stands as the mathematical model for a wide array of phenomena. The thorough coverage of the Poisson process and its practical applications is accessible to both junior- and senior-level students.
- Students often have difficulties understanding functions or quantities such as the density function of a continuous random variable and the formula for mathematical expectation. For example, they may wonder why $\int x f(x) dx$ is the appropriate definition for $E(X)$ and why correction for continuity is necessary. I have explained the reason behind such definitions, theorems, and concepts, and have demonstrated why they are the natural extensions of discrete cases.
- The initial six chapters feature numerous examples and exercises related to the selection of random points from intervals. As a result, when we delve into the topic of uniform random variables in [Chapter 7](#), we can easily derive the probability distribution and, through differentiation, the probability density function of X , which represents a random point from an interval (a, b) . This approach serves to seamlessly introduce the concept of a uniform random variable and lays the foundation for defining its probability density function.
- The normal density, often regarded as one of the most crucial density functions, finds a natural motivation through De Moivre's theorem. In [Section 7.2](#), I have introduced the standard normal density, an elementary version of the central limit theorem, and the normal density, presenting them in the historical context in which they were developed. The general version of the central limit theorem is subsequently covered in [Section 7.3](#), following discussions of normal random variables. This pedagogical approach has proven effective, as it renders the normal density a natural concept, rather than a bizarre function appearing seemingly out of the blue.
- Exponential random variables inherently arise as “times between consecutive events in Poisson processes.” The time of occurrence of the n th event of a Poisson process has a gamma distribution. For these reasons I have motivated exponential and gamma distributions by Poisson processes. In this way we can obtain many examples of exponential and gamma random variables from the abundant examples of Poisson processes already known. Another advantage is that it helps us visualize memoryless random variables by looking at the interevent times of Poisson processes.

- Joint distributions and conditioning can often pose challenges for students. However, a comprehensive explanation along with numerous practical applications of these concepts and techniques has made these materials more accessible and understandable for students.
- A section dedicated to pattern appearance can be found in Section 4B of the Supplementary Resources for [Chapter 4](#). While the method discussed in this section is intuitive and probabilistic, it can greatly assist students in comprehending seemingly paradoxical results, such as the following: On average, it requires nearly twice as many flips of a fair coin to achieve a sequence of five successive heads as it does to obtain a tail followed by four heads.
- If a fair coin is tossed a very large number of times, the general perception is that heads occurs as often as tails. In Section 10A of the Supplementary Resources for [Chapter 10](#), I have explained what is meant by “heads occurs as often as tails.”
- To study the risk or rate of “failure,” per unit of time of “lifetimes” that have already survived a certain length of time, I have included a section, Survival Analysis and Hazard Functions, to [Chapter 7](#).
- Ever since 1853, when Gregor Johann Mendel (1822–1884) began his breeding experiments with the garden pea *Pisum sativum*, probability has played an important role in the understanding of the principles of heredity. In the Instructor and Student Resources website, I have included sections and examples on genetics to demonstrate the extent of that role.
- Chapter 11, titled Stochastic Processes, which is accessible on the Instructor and Student Resources website for this book, delves deeper into Poisson processes and introduces the fundamentals of Markov chains, continuous-time Markov chains, and a comprehensive section on Brownian motion. Consequently, the current edition, in conjunction with its Instructor and Student Resources website, provides sufficient material for a second course in probability. The level of difficulty in the chapter on stochastic processes aligns with the rest of the book, aiming to make challenging concepts more accessible to undergraduate and early graduate students. The only prerequisite assumed is calculus. Throughout the chapter, we explore various examples and discuss significant results from areas such as queuing theory, random walks, branching processes, superposition of Poisson processes, and compound Poisson processes. Additionally, I elucidate the meaning of the celebrated theorem, PASTA (Poisson Arrivals See Time Average). In essence, the chapter on stochastic processes lays the foundation upon which students can build their further studies and work in both pure and applied probability.
- For random sums of random variables, I have discussed Wald’s equation in [Chapter 9](#) and its analogous case for variance. Certain applications of Wald’s equation have been discussed in the exercises, as well as in Chapter 11, Stochastic Processes.
- In this book and its Instructor and Student Resources website, in mathematical discussions of a topic, there is no distinction between normal and boldface text.
- The answers to the odd-numbered exercises of the book are included at the end of the book. Likewise, the answers to the odd-numbered exercises featured on the Instructor and Student Resources website are provided at the website’s conclusion.

Sample Syllabi

For a one-term course on probability, instructors have the flexibility to omit certain sections without difficulty. The book has been designed to accommodate students with varying levels of ability, making it suitable for a wide range of probability courses, whether they are focused on applied or pure aspects of the subject. A typical one-semester probability course may cover the following sections:

- Sections 1.1–1.7
- Sections 2.2–2.4
- Sections 3.1–3.5
- Sections 4.1–4.8
- Sections 5.1–5.5
- Sections 6.1–6.4
- Sections 7.1–7.5
- Sections 8.2–8.5
- Section 9.1
- Sections 10.1 and/or 10.3
- Section 10.2
- Sections 10.4–10.6

An alternative syllabus follows the approach of previous editions and may include:

- Sections 1.1–1.7
- Sections 2.2–2.4
- Sections 3.1–3.5
- Sections 4.1–4.6
- Sections 6.1–6.3
- Sections 5.1–5.3
- Sections 7.1–7.2
- Sections 7.3–7.5
- Sections 4.7–4.8

- Sections 5.4–5.5
- Section 6.4
- Section 7.3
- Sections 8.2–8.5
- Section 9.1
- Sections 10.1 and/or 10.3
- Section 10.2
- Sections 10.4–10.6

For a discrete probability course, the typical curriculum includes:

- Sections 1.1–1.5
- Chapters 2, 3, 4, and 6
- Conditional Distributions: Discrete Case from Chapter 8
- Selected discrete topics from Chapters 9 and 10

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Chapter 1

Axioms of Probability

1.1 INTRODUCTION

In search of natural laws that govern a phenomenon, science often faces “events” that may or may not occur. The event of *disintegration of a given atom of radium* is one such example because, in any given time interval, such an atom may or may not disintegrate. The event of *finding no defect during inspection of a microwave oven* is another example, since an inspector may or may not find defects in the microwave oven. The event that an *orbital satellite in space is at a certain position* is a third example. In any experiment, an event that may or may not occur is called **random**. If the occurrence of an event is inevitable, it is called **certain**, and if it can never occur, it is called **impossible**. For example, the event that an object travels faster than light is impossible, and the event that in a thunderstorm flashes of lightning precede any thunder echoes is certain.

Knowing that an event is random determines only that the existing conditions under which the experiment is being performed do not guarantee its occurrence. Therefore, the knowledge obtained from randomness itself is hardly decisive. It is highly desirable to determine quantitatively the exact value, or an estimate, of the chance of the occurrence of a random event. The theory of probability has emerged from attempts to deal with this problem. In many different fields of science and technology, it has been observed that, under a long series of experiments, the proportion of the time that an event occurs may appear to approach a constant. It is these constants that probability theory (and statistics) aims at predicting and describing as quantitative measures of the chance of occurrence of events. For example, if a fair coin is tossed repeatedly, the proportion of the heads approaches $1/2$. Hence probability theory postulates that the number $1/2$ be assigned to the event of *getting heads in a toss of a fair coin*.

Historically, from the dawn of civilization, humans have been interested in games of chance and gambling. However, the advent of probability as a mathematical discipline is relatively recent. Ancient Egyptians, about 3500 B.C., were using astragali, a four-sided die-shaped bone found in the heels of some animals, to play a game now called *hounds and jackals*. The ordinary six-sided die was made about 1600 B.C. and since then has been used in all kinds of games. The ordinary deck of playing cards, probably the most popular tool in games and gambling, is much more recent than dice. Although it is not known where and when dice originated, there are reasons to believe that they were invented in China sometime between the seventh and tenth centuries. Clearly, through gambling and games of chance people have gained intuitive ideas about the frequency of occurrence of certain events and, hence, about probabilities. But surprisingly, studies of the chances of events were not begun until the fifteenth century. The Italian scholars

Luca Paccioli (1445–1514), Niccolò Tartaglia (1499–1557), Girolamo Cardano (1501–1576), and especially Galileo Galilei (1564–1642) were among the first prominent mathematicians who calculated probabilities concerning many different games of chance. They also tried to construct a mathematical foundation for probability. Cardano even published a handbook on gambling, with sections discussing methods of cheating. Nevertheless, real progress started in France in 1654, when Blaise Pascal (1623–1662) and Pierre de Fermat (1601–1665) exchanged several letters in which they discussed general methods for the calculation of probabilities. In 1655, the Dutch scholar Christian Huygens (1629–1695) joined them. In 1657 Huygens published the first book on probability, *De Ratiocinates in Aleae Ludo* (*On Calculations in Games of Chance*). This book marked the birth of probability. Scholars who read it realized that they had encountered an important theory. Discussions of solved and unsolved problems and these new ideas generated readers interested in this challenging new field.

After the work of Pascal, Fermat, and Huygens, the book written by James Bernoulli (1654–1705) and published in 1713 and that by Abraham de Moivre (1667–1754) in 1730 were major breakthroughs. In the eighteenth century, studies by Pierre-Simon Laplace (1749–1827), Siméon Denis Poisson (1781–1840), and Karl Friedrich Gauss (1777–1855) expanded the growth of probability and its applications very rapidly and in many different directions. In the nineteenth century, prominent Russian mathematicians Pafnuty Chebyshev (1821–1894), Andrei Markov (1856–1922), and Aleksandr Lyapunov (1857–1918) advanced the works of Laplace, De Moivre, and Bernoulli considerably. By the early twentieth century, probability was already a developed theory, but its foundation was not firm. A major goal was to put it on firm mathematical grounds. Until then, among other interpretations perhaps the **relative frequency interpretation** of probability was the most satisfactory. According to this interpretation, to define p , the probability of the occurrence of an event A of an experiment, we study a series of sequential or simultaneous performances of the experiment and observe that the proportion of times that A occurs approaches a constant. Then we count $n(A)$, the number of times that A occurs during n performances of the experiment, and we define $p = \lim_{n \rightarrow \infty} n(A)/n$.

As an example, consider a list of movie genres including action, adventure, comedy, drama, horror, romance, and so on. Let A represent the event where a randomly chosen individual from a specific region prefers action movies. To estimate the probability of A , we assess the relative frequency of individuals who favor action movies. As n , the number of participants in the survey, increases, we observe whether this frequency approaches a consistent value. Imagine that a study produces a relative frequency for event A plotted against n , as depicted in [Figure 1.1](#). This graph reveals that as n grows, the relative frequency for A , or the preference for action movies, converges to 0.63. This suggests that roughly 63% of the population in the examined region prefer action movies, and thus $P(A) \approx 0.63$.

The interpretation of probability based on relative frequency, along with the definition $p = \lim_{n \rightarrow \infty} n(A)/n$ presents serious mathematical challenges and cannot serve as the foundation for a rigorous theory of probability. Some challenges posed by this definition include:

1. In practice, $\lim_{n \rightarrow \infty} n(A)/n$ cannot be computed since it is impossible to repeat an experiment infinitely many times. Moreover, if for a large n , $n(A)/n$ is taken as an approximation for the probability of A , there is no way to analyze the error.
2. There is no reason to believe that the limit of $n(A)/n$, as $n \rightarrow \infty$, exists. Also, if the existence of this limit is accepted as an axiom, many dilemmas arise that cannot be solved. For example, there is no reason to believe that, in a different series of experiments and

for the same event A , this ratio approaches the same limit. Hence the uniqueness of the probability of event A is not guaranteed.

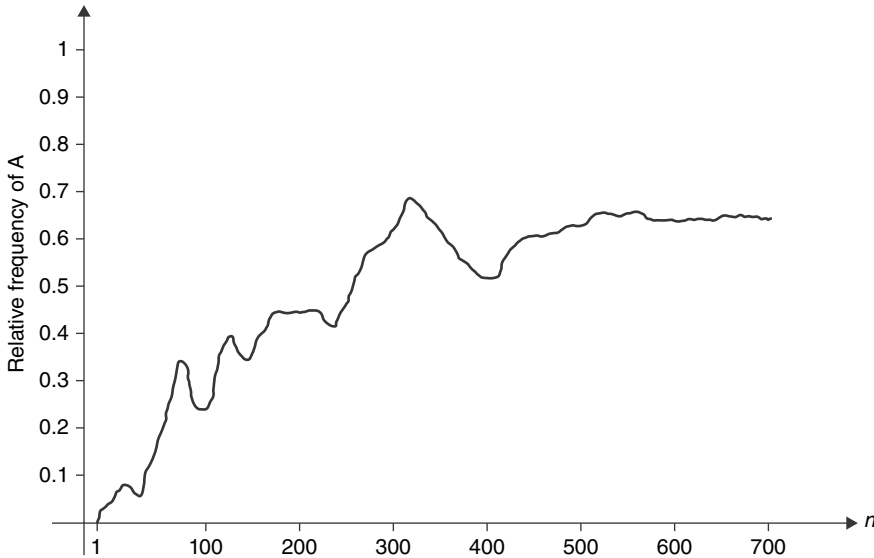


Figure 1.1 Relative frequency of individuals who favor action movies.

3. By this definition, probabilities that are based on our personal belief and knowledge are not justifiable. Thus statements such as the following would be meaningless.

- The probability that the price of oil will be raised in the next 6 months is 60%.
- The probability that the 50,000th decimal figure of the number π is seven exceeds 10%.
- The probability that it will snow next Christmas is 30%.
- The probability that Mozart was poisoned by Salieri is 18%.

In 1900, at the International Congress of Mathematicians in Paris, David Hilbert (1862–1943) proposed 23 problems whose solutions were, in his opinion, crucial to the advancement of mathematics. One of these problems was the axiomatic treatment of the theory of probability. In his lecture, Hilbert quoted Weierstrass, who had said, “The final object, always to be kept in mind, is to arrive at a correct understanding of the foundations of the science.” Hilbert added that a thorough understanding of special theories of a science is necessary for successful treatment of its foundation. Probability had reached that point and was studied enough to warrant the creation of a firm mathematical foundation. Some work toward this goal had been done by Émile Borel (1871–1956), Serge Bernstein (1880–1968), and Richard von Mises (1883–1953), but it was not until 1933 that Andrei Kolmogorov (1903–1987), a prominent Russian mathematician, successfully axiomatized the theory of probability. In Kolmogorov’s work, which is now universally accepted, three self-evident and indisputable properties of probability (discussed later) are taken as **axioms**, and the entire theory of probability is developed and

rigorously based on these axioms. In particular, the existence of a constant p , as the limit of the proportion of the number of times that the event A occurs when the number of experiments increases to ∞ , in some sense, is shown. Subjective probabilities based on our personal knowledge, feelings, and beliefs may also be modeled and studied by this axiomatic approach.

In this book we study the mathematics of probability based on the axiomatic approach. Since in this approach the concepts of **sample space** and **event** play a central role, we now delve into these concepts in detail.

1.2 SAMPLE SPACE AND EVENTS

If the outcome of an experiment is not certain but all of its possible outcomes are predictable in advance, then the set of all these possible outcomes is called the **sample space** of the experiment and is usually denoted by S . Therefore, the sample space of an experiment consists of all possible outcomes of the experiment. These outcomes are sometimes called **sample points**, or simply **points**, of the sample space. In the language of probability, certain subsets of S are referred to as **events**. So events are sets of points in the sample space that represent potential outcomes or occurrences. In other words, events encompass the various possibilities that can unfold. Some examples follow.

Example 1.1 For the experiment of *tossing a coin once*, the sample space S consists of two points (outcomes), “heads” (H) and “tails” (T). Thus $S = \{H, T\}$. ♦

Example 1.2 Suppose that an experiment consists of two steps. First a coin is flipped. If the outcome is tails, a die is tossed. If the outcome is heads, the coin is flipped again. The sample space of this experiment is $S = \{T1, T2, T3, T4, T5, T6, HT, HH\}$. For this experiment, the event of *heads in the first flip of the coin* is $E = \{HT, HH\}$, and the event of *an odd outcome* when the die is tossed is $F = \{T1, T3, T5\}$. ♦

Example 1.3 Consider measuring the lifetime of a light bulb. Since any nonnegative real number can be considered as the lifetime of the light bulb (in hours), the sample space is $S = \{x: x \geq 0\}$. In this experiment, $E = \{x: x \geq 100\}$ is the event that *the light bulb lasts at least 100 hours*, $F = \{x: x \leq 1000\}$ is the event that *it lasts at most 1000 hours*, and $G = \{505.5\}$ is the event that *it lasts exactly 505.5 hours*. ♦

Example 1.4 Suppose that a study is being done on all families with one, two, or three children. Let the outcomes of the study be the genders of the children in descending order of their ages. Then

$$S = \{b, g, bg, gb, bb, gg, bbb, bgb, bbg, bgg, ggg, gbg, ggb, gbb\}.$$

Here the outcome b means that the child is a boy, and g means that it is a girl. The events $F = \{b, bg, bb, bbb, bgb, bbg, bgg\}$ and $G = \{gg, bgg, gbg, ggb\}$ represent families where the eldest child is a boy and families with exactly two girls, respectively. ♦

Example 1.5 A bus with a capacity of 34 passengers stops at a station some time between 11:00 A.M. and 11:40 A.M. every day. The sample space of the experiment, consisting of counting the number of passengers on the bus and measuring the arrival time of the bus, is

$$S = \left\{ (i, t) : 0 \leq i \leq 34, \quad 11 \leq t \leq 11\frac{2}{3} \right\}, \quad (1.1)$$

where i represents the number of passengers and t the arrival time of the bus in hours and fractions of hours. The subset of S defined by $F = \{(27, t) : 11\frac{1}{3} < t < 11\frac{2}{3}\}$ is the event that the bus arrives between 11:20 A.M. and 11:40 A.M. with 27 passengers. ♦

Remark 1.1 Different manifestations of outcomes of an experiment might lead to different representations for the sample space of the same experiment. For instance, in Example 1.5, the outcome that the *bus arrives at t with i passengers* is represented by (i, t) , where t is expressed in hours and fractions of hours. By this representation, (1.1) is the sample space of the experiment. Now if the same outcome is denoted by (i, t) , where t is the number of minutes after 11 A.M. that the bus arrives, then the sample space takes the form

$$S_1 = \{(i, t) : 0 \leq i \leq 34, 0 \leq t \leq 40\}.$$

To the outcome that the *bus arrives at 11:20 A.M. with 31 passengers*, in S the corresponding point is $(31, 11\frac{1}{3})$, while in S_1 it is $(31, 20)$. ♦

Example 1.6 (Round-Off Error) Suppose that each time Jay charges an item to his credit card, he will round the amount to the nearest dollar in his records. Therefore, the round-off error, which is the true value charged minus the amount recorded, is random, with the sample space

$$S = \{0, 0.01, 0.02, \dots, 0.49, -0.50, -0.49, \dots, -0.01\},$$

where we have assumed that for any integer dollar amount a , Jay rounds $a.50$ to $a + 1$. The event of rounding off at most 3 cents in a random charge is given by

$$\{0, 0.01, 0.02, 0.03, -0.01, -0.02, -0.03\}. \quad \blacklozenge$$

If the outcome of an experiment belongs to an event E , we say that the event E has **occurred**. For example, if we draw two cards from an ordinary deck of 52 cards and observe that one is a spade and the other a heart, all of the events $\{sh\}$, $\{sh, dd\}$, $\{cc, dh, sh\}$, $\{hc, sh, ss, hh\}$, and $\{cc, hh, sh, dd\}$ have occurred because sh , the outcome of the experiment, belongs to all of them. However, none of the events $\{dh, sc\}$, $\{dd\}$, $\{ss, hh, cc\}$, and $\{hd, hc, dc, sc, sd\}$ has occurred because sh does not belong to any of them.

In the study of probability theory the relations between different events of an experiment play a central role. In the remainder of this section we study these relations. In all of the following definitions the events belong to a fixed sample space S .

Subset An event E is said to be a **subset** of the event F if, whenever E occurs, F also occurs. This means that all of the sample points of E are contained in F . Hence considering E and F solely as two sets, E is a subset of F in the usual set-theoretic sense: that is, $E \subseteq F$.

Equality Events E and F are said to be **equal** if the occurrence of E implies the occurrence of F , and vice versa; that is, if $E \subseteq F$ and $F \subseteq E$, hence $E = F$.

Intersection An event is called the **intersection** of two events E and F if it occurs only whenever E and F occur simultaneously. In the language of sets this event is denoted by EF or $E \cap F$ because it is the set containing exactly the common points of E and F .

- Union** An event is called the **union** of two events E and F if it occurs whenever at least one of them occurs. This event is $E \cup F$ since all of its points are in E or F or both.
- Complement** An event is called the **complement** of the event E if it only occurs whenever E does not occur. The complement of E is denoted by E^c .
- Difference** An event is called the **difference** of two events E and F if it occurs whenever E occurs but F does not. The difference of the events E and F is denoted by $E - F$. It is clear that $E^c = S - E$ and $E - F = EF^c$.
- Certainty** An event is called **certain** if its occurrence is inevitable. Thus the sample space is a certain event.
- Impossibility** An event is called **impossible** if there is certainty in its nonoccurrence. Therefore, the empty set $\emptyset$, which is S^c , is an impossible event.
- Mutually Exclusiveness** If the joint occurrence of two events E and F is impossible, we say that E and F are **mutually exclusive**. Thus E and F are mutually exclusive if the occurrence of E precludes the occurrence of F , and vice versa. Since the event representing the joint occurrence of E and F is EF , their intersection, E and F , are mutually exclusive if $EF = \emptyset$. A set of events $\{E_1, E_2, \dots\}$ is called **mutually exclusive** if the joint occurrence of any two of them is impossible, that is, if $\forall i \neq j, E_i E_j = \emptyset$ (see Figure 1.2). Thus $\{E_1, E_2, \dots\}$ is mutually exclusive if and only if every pair of them is mutually exclusive.

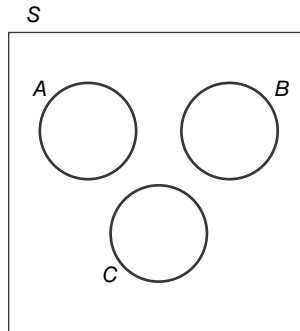


Figure 1.2 Venn diagram for mutually exclusive events A , B , and C .

The events $\bigcup_{i=1}^n E_i$, $\bigcap_{i=1}^n E_i$, $\bigcup_{i=1}^{\infty} E_i$, and $\bigcap_{i=1}^{\infty} E_i$ are defined in a way similar to $E_1 \cup E_2$ and $E_1 \cap E_2$. For example, if $\{E_1, E_2, \dots, E_n\}$ is a set of events, by $\bigcup_{i=1}^n E_i$ we mean the event in which at least one of the events E_i , $1 \leq i \leq n$, occurs. By $\bigcap_{i=1}^n E_i$ we mean an event that occurs only when all of the events E_i , $1 \leq i \leq n$, occur.

Example 1.7 At a busy international airport, arriving planes land on a first-come, first-served basis. Let

E = there are at least five planes waiting to land,

F = there are at most three planes waiting to land,

H = there are exactly two planes waiting to land.

Then

1. E^c is the event that at most four planes are waiting to land.
2. F^c is the event that at least four planes are waiting to land.
3. E is a subset of F^c ; that is, if E occurs, then F^c occurs. Therefore, $EF^c = E$.
4. H is a subset of F ; that is, if H occurs, then F occurs. Therefore, $FH = H$.
5. E and F are mutually exclusive; that is, $EF = \emptyset$. E and H are also mutually exclusive since $EH = \emptyset$.
6. FH^c is the event that the number of planes waiting to land is zero, one, or three. ♦

Sometimes Venn diagrams are used to represent the relations among events of a sample space. The sample space S of the experiment is usually shown as a large rectangle and, inside S , circles or other geometrical objects are drawn to indicate the events of interest. Figure 1.3 presents Venn diagrams for EF , $E \cup F$, E^c , and $(E^cG) \cup F$. The shaded regions are the indicated events.

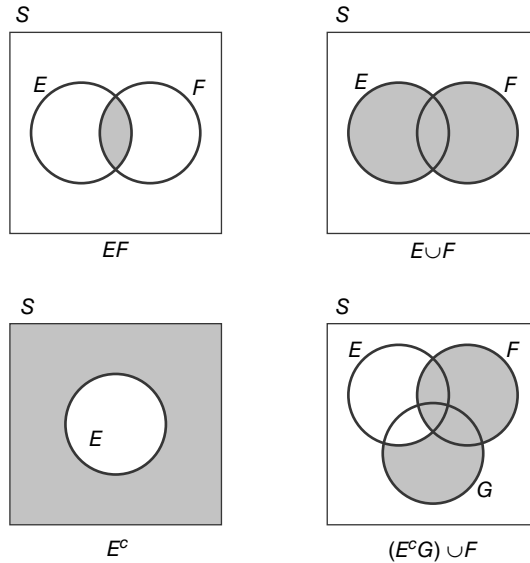


Figure 1.3 Venn diagrams of the events specified.

Example 1.8 Figure 1.4 shows an electric circuit in which, at any random time, each of the switches located at 1, 2, 3, 4, and 5 is either closed or open. For $1 \leq i \leq 5$, let E_i be the event that, at a random time, the switch at location i is closed. In terms of E_i 's, describe the event that a signal fed to the input is transmitted to the output.

Solution: The event that a signal fed to the input is transmitted to the output is

$$F = E_1 E_5 \cup E_1 E_3 E_4 \cup E_2 E_4 \cup E_2 E_3 E_5. \quad \blacklozenge$$

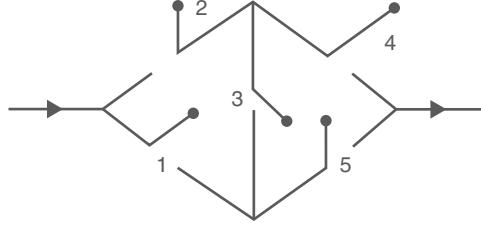


Figure 1.4 Electric circuit of Example 1.8.

Unions, intersections, and complementations satisfy many useful relations between events. A few of these relations are as follows:

$$(E^c)^c = E, \quad E \cup E^c = S, \quad ES = E, \quad \text{and} \quad EE^c = \emptyset.$$

Commutative laws:

$$E \cup F = F \cup E, \quad EF = FE.$$

Associative laws:

$$E \cup (F \cup G) = (E \cup F) \cup G, \quad E(FG) = (EF)G.$$

Distributive laws:

$$(EF) \cup H = (E \cup H)(F \cup H), \quad (E \cup F)H = (EH) \cup (FH).$$

De Morgan's first law:

$$(E \cup F)^c = E^c F^c, \quad \left(\bigcup_{i=1}^n E_i \right)^c = \bigcap_{i=1}^n E_i^c, \quad \left(\bigcup_{i=1}^{\infty} E_i \right)^c = \bigcap_{i=1}^{\infty} E_i^c.$$

De Morgan's second law:

$$(EF)^c = E^c \cup F^c, \quad \left(\bigcap_{i=1}^n E_i \right)^c = \bigcup_{i=1}^n E_i^c, \quad \left(\bigcap_{i=1}^{\infty} E_i \right)^c = \bigcup_{i=1}^{\infty} E_i^c.$$

Another useful relation between E and F , two arbitrary events of a sample space S , is

$$E = EF \cup EF^c.$$

This equality readily follows from $E = ES$ and distributivity:

$$E = ES = E(F \cup F^c) = EF \cup EF^c.$$

These and similar identities are usually proved by the **elementwise method**. The idea is to show that the events on both sides of the equation are formed of the same sample points. To use this method, we prove set inclusion in both directions. That is, sample points belonging to the event on the left also belong to the event on the right, and vice versa. An example follows.

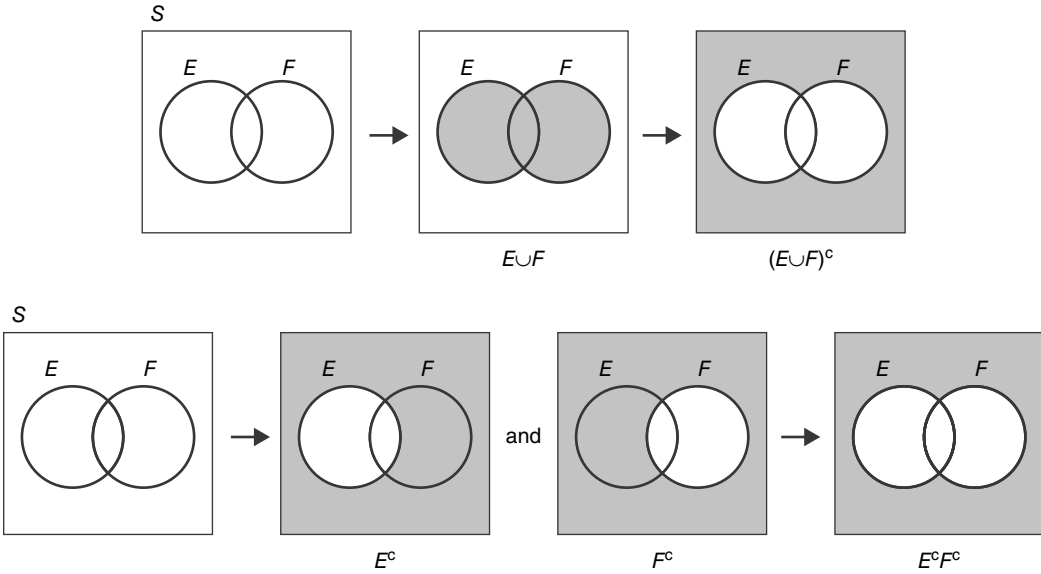


Figure 1.5 Venn diagram illustrating the equality $(E \cup F)^c = E^c F^c$.

Example 1.9 Prove De Morgan's first law: For E and F , two events of a sample space S , $(E \cup F)^c = E^c F^c$. (Figure 1.5 provides an intuitive rationale for this law's validity.)

Proof: First we show that $(E \cup F)^c \subseteq E^c F^c$, then we prove the reverse inclusion $E^c F^c \subseteq (E \cup F)^c$. To show that $(E \cup F)^c \subseteq E^c F^c$, let x be an outcome that belongs to $(E \cup F)^c$. Then x does not belong to $E \cup F$, meaning that x is neither in E nor in F . So x belongs to both E^c and F^c and hence to $E^c F^c$. To prove the reverse inclusion, let $x \in E^c F^c$. Then $x \in E^c$ and $x \in F^c$, implying that $x \notin E$ and $x \notin F$. Therefore, $x \notin E \cup F$ and thus $x \in (E \cup F)^c$. ♦

Note that Venn diagrams are an excellent way to give intuitive justification for the validity of relations or to create counterexamples and show the invalidity of relations. However, they are not appropriate to prove relations. This is because of the large number of cases that must be considered (particularly if more than two events are involved). For example, suppose that by means of Venn diagrams, we want to prove the identity $(EF)^c = E^c \cup F^c$. First, we must draw appropriate representations for all possible ways that E and F can be related: cases such as $EF = \emptyset$, $EF \neq \emptyset$, $E = F$, $E = \emptyset$, $F = S$, and so on. Then in each particular case, we should find the regions that represent $(EF)^c$ and $E^c \cup F^c$ and observe that they are the same. Even if these two sets have different representations in only one case, the identity would be false.

Fields and σ -Fields[†]

As stated at the beginning of this section, “certain subsets of a sample space are refereed as events.” Even though all subsets of finite or countably infinite sample spaces are events, not all

[†] Skipping this subsection does not affect the overall continuity. If you choose to skip it, it is advisable to skip all exercises or discussions related to Fields and σ -Fields throughout the book as well.

subsets of an uncountable sample space are events. In Example 1.29, we present an experiment and provide a subset of its sample space which is not an event.

Our objective is to assign probabilities exclusively to sets that qualify as events. It is neither possible nor our intention to assign a probability to a set that does not meet the criteria of being an event. Moreover, as set operations generate new events from existing ones, it is imperative that we can assign probabilities to these derived events resulting from operations such as combinations and complementations. Throughout this process, it is crucial to ensure that non-event sets do not arise as a consequence of manipulating set operations.

Let S denote the sample space of an experiment. The collection of all subsets of S is denoted by $\mathcal{P}(S)$ and referred to as the **power set** of S . Now, consider $\mathcal{F}$, which is a collection of elements from $\mathcal{P}(S)$, encompassing both events and non-events. This collection $\mathcal{F}$ is defined as a **field** or an **algebra** if

- (i) If $S \in \mathcal{F}$;
- (ii) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- (iii) If $A, B \in \mathcal{F}$, then $A \cup B \in \mathcal{F}$.

By property (ii), the empty set, $= S^c$ also belongs to $\mathcal{F}$.

A straightforward induction shows that

$$\text{if } A_1, A_2, \dots, A_n \in \mathcal{F}, \quad \text{then} \quad A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i \in \mathcal{F}.$$

For $n = 2$, by property (i), if $A_1, A_2 \in \mathcal{F}$, then $A_1 \cup A_2 \in \mathcal{F}$. Now, assume that the relation is true for $n-1$ events; that is $E = A_1 \cup A_2 \cup \dots \cup A_{n-1} \in \mathcal{F}$. We will show that it is also true for n events. This follows from property (i) since $E \cup A_n \in \mathcal{F}$ and $E \cup A_n = A_1 \cup A_2 \cup \dots \cup A_n$. Furthermore, by property (ii) above, for each A_i , we have that $A_i^c \in \mathcal{F}$, hence $\bigcup_{i=1}^n A_i^c \in \mathcal{F}$.

As a result, $\left(\bigcup_{i=1}^n A_i^c\right)^c \in \mathcal{F}$, and by De Morgan's first law,

$$\bigcap_{i=1}^n A_i = \bigcap_{i=1}^n (A_i^c)^c = \left(\bigcup_{i=1}^n A_i^c\right)^c \in \mathcal{F}.$$

Thus,

$$\text{if } A_1, A_2, \dots, A_n \in \mathcal{F}, \quad \text{then} \quad A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i \in \mathcal{F}.$$

Therefore, fields are closed under complementation, finite unions, and finite intersections.

If we require that $\mathcal{F}$, a collection of elements of $\mathcal{P}(S)$, be also closed under countable unions, then $\mathcal{F}$ is called a **σ -field** (sigma field) or a **σ -algebra** (sigma algebra). Thus $\mathcal{F}$ is a σ -field if

- (a) $S \in \mathcal{F}$
- (b) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- (c) If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$;

Similarly to fields, in the case of σ -fields, $\emptyset = S^c \in \mathcal{F}$. Moreover, a proof similar to the one for a finite number of events demonstrates that

$$\text{if } A_1, A_2, \dots \in \mathcal{F}, \quad \text{then} \quad \bigcap_{i=1}^{\infty} A_i \in \mathcal{F}.$$

Therefore, $\mathcal{F}$ is closed under countable intersections as well.

Let S be the sample space of an experiment, and $\mathcal{F}$ be the subcollection of $\mathcal{P}(S)$, the power set of S , consisting of subsets that represent potential outcomes or occurrences, i.e., events. It is evident that $\mathcal{F}$ satisfies properties (a), (b), and (c) making it a σ -field. Our objective is to assign probabilities to the elements within this σ -field, aiming to establish a consistent and coherent framework for measuring and evaluating probabilities of different events. In this context, events can be considered as subsets of S for which a probability can be calculated.

Example 1.10 Let $\mathcal{F}$ be a σ -field on a set S . Show that if A and B belong to $\mathcal{F}$, then $A - B$ and hence $A \Delta B = (A - B) \cup (B - A)$ also belong to $\mathcal{F}$.

Solution: Note that $A - B = A \cap B^c$. Since A and $B \in \mathcal{F}$, $A \cap B^c$ is also in $\mathcal{F}$. ♦

Example 1.11 For a set S , $\mathcal{F} = \{S, \emptyset\}$ is a σ -field since it satisfies properties (a), (b), and (c). It is called the *trivial σ -field* and is the smallest σ -field associated with S . ♦

Example 1.12 For a set S , the power set $\mathcal{P}(S)$ satisfies properties (a), (b), and (c). Therefore, it is a sigma field. ♦

Example 1.13 Let A be a subset of S . Then $\mathcal{F} = \{S, A, A^c, \emptyset\}$ possesses properties (a), (b), and (c). So it is a σ -field. ♦

Example 1.14 For a set S and σ -fields $\mathcal{F}_1, \mathcal{F}_2, \dots$ defined on S , show that $\bigcap_{i=1}^{\infty} \mathcal{F}_i$ and $\bigcup_{i=1}^{\infty} \mathcal{F}_i$ are also σ -fields on S .

Proof: We will prove that $\bigcap_{i=1}^{\infty} \mathcal{F}_i$ is a σ -field. The proof that $\bigcup_{i=1}^{\infty} \mathcal{F}_i$ is also a σ -field is similar. Since for all $i \geq 1$, $S \in \mathcal{F}_i$, $S \in \bigcap_{i=1}^{\infty} \mathcal{F}_i$ and property (a) is satisfied. To show property (b), let $A \in \bigcap_{i=1}^{\infty} \mathcal{F}_i$, then $A \in \mathcal{F}_i$, for all i , and hence $A^c \in \mathcal{F}_i$, for all i , so $A^c \in \bigcap_{i=1}^{\infty} \mathcal{F}_i$. As for property (c), if $A_1, A_2, \dots \in \bigcap_{i=1}^{\infty} \mathcal{F}_i$, then, for all $i \geq 1$, $A_1, A_2, \dots \in \mathcal{F}_i$. Thus, for all $i \geq 1$, $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}_i$ implying that $\bigcup_{i=1}^{\infty} A_i \in \bigcap_{i=1}^{\infty} \mathcal{F}_i$. Thus, the intersection of the σ -fields $\mathcal{F}_1, \mathcal{F}_2, \dots$ is a σ -field. ♦

Example 1.15 Let $\mathcal{B}$ be a collection of subsets of S . If $\mathcal{B}$ is not a σ -field, then the smallest σ -field containing all elements of $\mathcal{B}$ is called the σ -field *generated* by $\mathcal{B}$ and is denoted by $\sigma(\mathcal{B})$. It is the intersection of all σ -fields on S that contain all elements of $\mathcal{B}$. For example, let $S = \{1, 2, 3, 4\}$ and $\mathcal{B} = \{\{1, 2\}, \{2, 3\}, \{4\}, \emptyset\}$. Clearly, $\mathcal{B}$ is not a σ -field. For instance, it does not include S , and $\{1, 2\} \cap \{2, 3\} = \{2\} \notin \mathcal{B}$. To find the σ -field generated by $\mathcal{B}$, we first calculate $\mathcal{B}_1$, the set of all unions, all intersections, and all complements of the elements of $\mathcal{B}$. Then calculate $\mathcal{B}_2$, the set of all unions, all intersections, and all complements of the elements of $\mathcal{B}_1$. We repeat this process until the first i for which there are no new sets

to include in $\mathcal{B}_i$. The σ -field generated by $\mathcal{B}$ is $\mathcal{B}_i$; that is, $\sigma(\mathcal{B}) = \mathcal{B}_i$. For the collection $\mathcal{B} = \{\{1, 2\}, \{2, 3\}, \{4\}, \emptyset\}$,

$$\mathcal{B}_1 = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2\}, \{2, 3, 4\}, \{2, 3\}, \{4\}, \{2\}, \emptyset, \{3, 4\}, \{1, 4\}, \{1, 2, 3, 4\}\},$$

$$\mathcal{B}_2 = \mathcal{B}_1 \cup \{\{2, 4\}, \{1, 3, 4\}, \{3\}, \{1\}\},$$

$$\mathcal{B}_3 = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \{\{1, 3\}\} = \mathcal{P}(S).$$

This shows that $\sigma(\mathcal{B}) = \mathcal{B}_3 = \mathcal{P}(S)$. Therefore, the smallest σ -field containing all elements of $\mathcal{B} = \{\{1, 2\}, \{2, 3\}, \{4\}, \emptyset\}$ is $\mathcal{P}(S)$, the power set of S . Since the largest σ -field containing $\mathcal{B}$ is also the power set of S , $\mathcal{P}(S)$ is the only σ -field containing all elements of $\mathcal{B}$. ♦

EXERCISES

A

1. From the letters of the word MISSISSIPPI, a letter is chosen at random. What is a sample space for this experiment? What is the event that the outcome is a vowel?
2. Last month, an insurance company sold 57 life insurance policies. Define a sample space for the number of claims that the company will receive from the beneficiaries of this group within the next 15 years. What is the event that the company receives at least 3 but no more than 8 claims?
3. In the experiment of flipping a coin three times, what does the event

$$E = \{\text{THH, HTH, HHT, HHH}\}$$

represent?

4. In the experiment of tossing two dice, what do the following events represent?
 $E = \{(1, 3), (2, 6), (3, 1), (6, 2)\}$ and $F = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$.
5. A deck of six cards consists of three black cards numbered 1, 2, 3, and three red cards numbered 1, 2, 3. First, Vann draws a card at random and without replacement. Then Paul draws a card at random and without replacement from the remaining cards. Let A be the event that Paul's card has a larger number than Vann's card. Let B be the event that Vann's card has a larger number than Paul's card.
 - (a) Are A and B mutually exclusive?
 - (b) Are A and B complements of one another?
6. A box contains three red and five blue balls. Define a sample space for the experiment of recording the colors of three balls that are drawn from the box, one by one, with replacement.

7. Define a sample space for the experiment of choosing a number from the interval $(0, 20)$. Describe the event that such a number is an integer.
8. Define a sample space for the experiment of putting three different books on a shelf in random order. If two of these three books are a two-volume dictionary, describe the event that these volumes stand in increasing order side-by-side (i.e., volume I precedes volume II).
9. Two dice are rolled. Let E be the event that the sum of the outcomes is odd and F be the event of at least one 1. Interpret the events EF , $E^c F$, and $E^c F^c$.
10. Define a sample space for the experiment of drawing two coins from a purse that contains two quarters, three nickels, one dime, and four pennies. For the same experiment describe the following events:
 - (a) drawing 26 cents;
 - (b) drawing more than 9 but less than 25 cents;
 - (c) drawing 29 cents.
11. A telephone call from a certain person is received some time between 7:00 A.M. and 9:10 A.M. every day. Define a sample space for this phenomenon, and describe the event that the call arrives within 15 minutes of the hour.
12. A device that has two components fails if at least one of its components breaks down. The device is observed at a random time. Let A_i , $1 \leq i \leq 2$, denote the outcome that the i th component is operative at the random time. In terms of A_i 's, (a) define a sample space for the status of the system at the random time; (b) describe the event that the device is not operative at that random time.
13. When flipping a coin more than once, what experiment has a sample space defined by the following?

$$S = \{TT, HTT, THTT, HHTT, HHHTT, HTHTT, THHTT, \dots\}.$$

14. Let E , F , and G be three events; explain the meaning of the relations $E \cup F \cup G = G$ and $EF G = G$.
15. A limousine that carries passengers from an airport to three different hotels just left the airport with two passengers. Describe the sample space of the stops and the event that both of the passengers get off at the same hotel.
16. A psychologist, who is interested in human complexion, in a study of hues and shades, shows her subjects three pieces of wood colored almond, lemon, and flax, respectively. She then asks them to identify their favorite colors. Define a sample space for the answer given by a random subject.
17. For a saw blade manufacturer's products, the global demand, per month, for band saws is between 30 and 36 thousands; for reciprocating saws, it is between 28 and 33 thousands; and for hole saws, it is between 300 and 600 thousands. Define a sample space for the demands for these three types of saws by this manufacturer in a random month. Describe the event that, in such a month, the demand for band saws is at least 33 thousands and

for hole saws, it is less than 435 thousands. In this exercise, by “between two numbers a and b ,” we mean inclusive of both a and b .

18. An insurance company sells a joint life insurance policy to Alexia and her husband, Roy. Define a sample space for the death or survival of this couple in 5 years. What is the event that at that time only one of them lives?
19. Find the simplest possible expression for the following events.
 - (a) $(E \cup F)(F \cup G)$.
 - (b) $(E \cup F)(E^c \cup F)(E \cup F^c)$.
20. At a certain university, every year eight to 12 professors are granted University Merit Awards. This year among the nominated faculty are Drs. Jones, Smith, and Brown. Let A , B , and C denote the events, respectively, that these professors will be given awards. In terms of A , B , and C , find an expression for the event that the award goes to (a) only Dr. Jones; (b) at least one of the three; (c) none of the three; (d) exactly two of them; (e) exactly one of them; (f) Drs. Jones or Smith but not both.
21. A device that has n , $n > 1$, components fails to operate if at least one of its components breaks down. The device is observed at a random time. Let A_i , $1 < i \leq n$, denote the event that the i th component is operative at the random time. In terms of A_i 's, describe the event that the device is operative at that random time.
22. Prove that the event B is impossible if and only if for every event A ,

$$A = (B \cap A^c) \cup (B^c \cap A).$$

23. Let E , F , and G be three events. Determine which of the following statements are correct and which are incorrect. Justify your answers.
 - (a) $(E - EF) \cup F = E \cup F$.
 - (b) $F^c G \cup E^c G = G(F \cup E)^c$.
 - (c) $(E \cup F)^c G = E^c F^c G$.
 - (d) $EF \cup EG \cup FG \subset E \cup F \cup G$.
24. Travis picks up darts to shoot toward an 18"-diameter dartboard aiming at the bullseye on the board. Describe a sample space for the point at which a dart hits the board.
25. For the experiment of flipping a coin until a heads occurs, (a) describe the sample space; (b) describe the event that it takes an odd number of flips until a heads occurs.
26. In an experiment, cards are drawn, one by one, at random and successively from an ordinary deck of 52 cards. Let A_n be the event that no face card or ace appears on the first $n - 1$ drawings, and the n th draw is an ace. In terms of A_n 's, find an expression for the event that an ace appears before a face card, (a) if the cards are drawn with replacement; (b) if they are drawn without replacement.
27. A point is chosen at random from the interval $(-1, 1)$. Let E_1 be the event that it falls in the interval $(-1/2, 1/2)$, E_2 be the event that it falls in the interval $(-1/4, 1/4)$ and, in

general, for $1 \leq i < \infty$, E_i be the event that the point is in the interval $(-1/2^i, 1/2^i)$. Find $\bigcup_{i=1}^{\infty} E_i$ and $\bigcap_{i=1}^{\infty} E_i$.

Exercises for Fields and σ -Fields

28. Let $S = \{1, 2, 3\}$, $\mathcal{B}_1 = \{\{1\}\}$, and $\mathcal{B}_2 = \{\{1, 2\}\}$. Show that $\sigma(\mathcal{B}_1) \cup \sigma(\mathcal{B}_2)$ is not a σ -field. That is, the union of two σ -fields is not necessarily a sigma field.
29. Let $S = \mathbb{R} = \{x: -\infty < x < \infty\}$ represent the real line, and let $\mathcal{F}$ be the collection of subsets A of $\mathbb{R}$ such that either A is countable or its complement A^c is countable. Show that $\mathcal{F}$ is a σ -Field.
30. Let $S = [0, 1] = \{x \in \mathbb{R}: 0 \leq x \leq 1\}$. Define

$$\mathcal{B} = \{[a, b): 0 \leq a < b \leq 1 \text{ and } [a, b) \subset [0, 1]\}$$

and let $\mathcal{F} = \sigma(\mathcal{B})$ be the σ -field generated by $\mathcal{B}$. $\mathcal{F}$ is referred to as the **Borel σ -Field** on S . Consider the set $\mathbb{Q}$, which consists of all rational numbers between 0 and 1. Then $\mathbb{Q}^c$ denotes the set of all irrational numbers between 0 and 1. Is $\mathbb{Q}$ an element of $\mathcal{F}$? What about $\mathbb{Q}^c$?

31. Let $S = \mathbb{R} = \{x: -\infty < x < \infty\}$, $\mathcal{B} = \{(a, b): -\infty < a < b < \infty\}$, and $\mathcal{C} = \{(a, b]: -\infty < a < b < \infty\}$. Show that $\sigma(\mathcal{B}) = \sigma(\mathcal{C})$. That is, the σ -fields generated by $\mathcal{B}$ and $\mathcal{C}$ are equal.
32. Let $S = (0, 1] = \{x: 0 < x \leq 1\}$, and let $\mathcal{F}$ be the set comprising the empty set and all finite unions of disjoint intervals in the form of $(a, b]$. That is,

$$\mathcal{F} = \emptyset \cup \left\{ \bigcup_{i=1}^n (a_i, b_i], 0 \leq a_i < b_i \leq 1, n = 1, 2, \dots, \text{where for } i \neq j, (a_i, b_i] \cap (a_j, b_j] = \emptyset \right\}.$$

Show that $\mathcal{F}$ is a field but not a σ -field.

B

33. Prove De Morgan's second law, $(AB)^c = A^c \cup B^c$, (a) by elementwise proof; (b) by applying De Morgan's first law to A^c and B^c .
34. Let A and B be two events. Prove the following relations by the elementwise method.
- (a) $(A - AB) \cup B = A \cup B$.
- (b) $(A \cup B) - AB = AB^c \cup A^c B$.
35. Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of events. Prove that for every event B ,
- (a) $B \left(\bigcup_{i=1}^{\infty} A_i \right) = \bigcup_{i=1}^{\infty} BA_i$.
- (b) $B \cup \left(\bigcap_{i=1}^{\infty} A_i \right) = \bigcap_{i=1}^{\infty} (B \cup A_i)$.
36. Define a sample space for the experiment of putting in a random order seven different books on a shelf. If three of these seven books are a three-volume dictionary, describe the event that these volumes stand in increasing order side by side (i.e., volume I precedes volume II and volume II precedes volume III).

37. In a mathematics department of 31 voting faculty members, there are three candidates running for the chair position. The voting procedure adopted by the department is **approval voting**, in which the eligible voters can vote for as many candidates as they wish. The candidate with the maximum votes will win. Let A_{ij} , $1 \leq i \leq 31$, $1 \leq j \leq 3$, be the event that, in the next election, the i th faculty member votes for the j th candidate. In terms of A_{ij} 's, describe the sample space for the next election of this department.
38. Let $\{A_1, A_2, A_3, \dots\}$ be a sequence of events. Find an expression for the event that infinitely many of the A_i 's occur.
39. Let $\{A_1, A_2, A_3, \dots\}$ be a sequence of events of a sample space S . Find a sequence $\{B_1, B_2, B_3, \dots\}$ of mutually exclusive events such that for all $n \geq 1$, $\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n B_i$.

Self-Quiz on Section 1.2

Time allotted: 20 Minutes

Each problem is worth 2.5 points.

- Jody, Ann, Bill, and Karl line up in a random order to get their photo taken. Describe the event that, on the line, males and females alternate.
- In a large hospital, there are 100 patients scheduled to have heart bypass surgery. Let E_i , $1 \leq i \leq 100$, be the event that the i th patient lives through the postoperative period of the surgery. In terms of E_i 's, (a) describe the event that all patients survive the critical postoperative period; (b) describe the event that exactly one patient dies during that period.
- Find the simplest possible expression for the event $(E \cup F)(F \cup G)(EG \cup F^c)$.
- Consider the system shown by the diagram of Figure 1.6, consisting of 7 components denoted by 1, 2, $\dots$, 7. Suppose that each component is either functioning or not functioning, with no other capabilities. Suppose that the system itself also has two performance capabilities, functioning and not functioning. For $1 \leq i \leq 7$, let E_i be the event that component i is functioning. In terms of E_i 's, describe the event that a signal fed to the input is transmitted to the output.

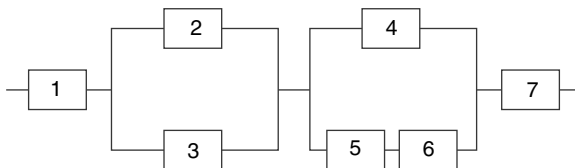


Figure 1.6 A diagram for the system of Self-Quiz Problem 4.

1.3 AXIOMS OF PROBABILITY

In mathematics, the goals of researchers are to obtain new results and prove their correctness, create simple proofs for already established results, discover or create connections between different fields of mathematics, construct and solve mathematical models for real-world problems, and so on. To discover new results, mathematicians use trial and error, instinct and inspired guessing, inductive analysis, studies of special cases, and other methods. But when a new result is discovered, its validity remains subject to skepticism until it is rigorously proven. Sometimes attempts to prove a result fail and contradictory examples are found. Such examples that invalidate a result are called **counterexamples**. No mathematical proposition is settled unless it is either proven or refuted by a counterexample. If a result is false, a counterexample exists to refute it. Similarly, if a result is valid, a proof must be found for its validity, although in some cases it might take years, decades, or even centuries to find it.

Proofs in probability theory (and virtually any other theory) are done in the framework of the **axiomatic method**. By this method, if we want to convince any rational person, say Sonya, that a statement L_1 is correct, we will show her how L_1 can be deduced logically from another statement L_2 that might be acceptable to her. However, if Sonya does not accept L_2 , we should demonstrate how L_2 can be deduced logically from a simpler statement L_3 . If she disputes L_3 , we must continue this process until, somewhere along the way we reach a statement that, without further justification, is acceptable to her. This statement will then become the basis of our argument. Its existence is necessary since otherwise the process continues ad infinitum without any conclusions. Therefore, in the axiomatic method, first we adopt certain simple, indisputable, and consistent statements without justifications. These are **axioms** or **postulates**. Then we agree on how and when one statement is a logical consequence of another one and, finally, using the terms that are already clearly understood, axioms and definitions, we obtain new results. New results found in this manner are called **theorems**. Theorems are statements that can be proved. Upon establishment, they are used for discovery of new theorems, and the process continues and a theory evolves.

In this book, our approach is based on the axiomatic method. There are three axioms upon which probability theory is based and, except for them, everything else needs to be proved. We will now explain these axioms.

Definition 1.1 (Probability Axioms) *Let S be the sample space of a random phenomenon. Suppose that to each event A of S , a number denoted by $P(A)$ is associated with A . If P satisfies the following axioms, then it is called a **probability** and the number $P(A)$ is said to be the **probability of A** .*

Axiom 1 $P(A) \geq 0$.

Axiom 2 $P(S) = 1$.

Axiom 3 *If $\{A_1, A_2, A_3, \dots\}$ is a sequence of mutually exclusive events (i.e., the joint occurrence of every pair of them is impossible: $A_i A_j = \emptyset$ when $i \neq j$), then*

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Note that the axioms of probability are a set of rules that must be satisfied before S and P can be considered a probability model.

Remark 1.2 Let $\mathcal{F}$ be the σ -field consisting of all events in the sample space S . According to this definition, a probability P on $(S, \mathcal{F})$ is a function $P: \mathcal{F} \rightarrow [0, 1]$ that satisfies the three axioms stated above. P is commonly referred to as a **probability function** or a **probability measure**. The triple $(S, \mathcal{F}, P)$, where S is the sample space, $\mathcal{F}$ the σ -field of all events, and P a probability measure, is called a **probability space**.

Axiom 1 states that the probability of the occurrence of an event is always nonnegative. Axiom 2 guarantees that the probability of the occurrence of the event S that is certain is 1. Axiom 3 states that for a sequence of mutually exclusive events the probability of the occurrence of at least one of them is equal to the sum of their probabilities.

Axiom 2 is merely a convenience to make things definite. It would be equally reasonable to have $P(S) = 100$ and interpret probabilities as percentages (which we frequently do).

Let S be the sample space of an experiment. Let A and B be events of S . We say that A and B are **equally likely** if $P(A) = P(B)$. Let ω_1 and ω_2 be sample points of S . We say that ω_1 and ω_2 are **equally likely** if the events $\{\omega_1\}$ and $\{\omega_2\}$ are equally likely, that is, if $P(\{\omega_1\}) = P(\{\omega_2\})$.

We will now prove some immediate implications of the axioms of probability.

Theorem 1.1 *The probability of the empty set $\emptyset$ is 0. That is, $P(\emptyset) = 0$.*

Proof: Let $A_1 = S$ and $A_i = \emptyset$ for $i \geq 2$, then $A_1, A_2, A_3, \dots$ is a sequence of mutually exclusive events. Thus, by Axiom 3,

$$P(S) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = P(S) + \sum_{i=2}^{\infty} P(\emptyset),$$

implying that $\sum_{i=2}^{\infty} P(\emptyset) = 0$. This is possible only if $P(\emptyset) = 0$. ♦

Axiom 3 is stated for a *countably infinite* collection of mutually exclusive events. For this reason, it is also called the *axiom of countable additivity*. We will now show that the same property holds for a *finite* collection of mutually exclusive events as well. That is, P also satisfies *finite additivity*.

Theorem 1.2 *Let $\{A_1, A_2, \dots, A_n\}$ be a mutually exclusive set of events. Then*

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

Proof: For $i > n$, let $A_i = \emptyset$. Then $A_1, A_2, A_3, \dots$ is a sequence of mutually exclusive events. From Axiom 3 and Theorem 1.1 we get

$$\begin{aligned}
 P\left(\bigcup_{i=1}^n A_i\right) &= P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \\
 &= \sum_{i=1}^n P(A_i) + \sum_{i=n+1}^{\infty} P(A_i) = \sum_{i=1}^n P(A_i) + \sum_{i=n+1}^{\infty} P(\emptyset) \\
 &= \sum_{i=1}^n P(A_i). \quad \blacklozenge
 \end{aligned}$$

Letting $n = 2$, Theorem 1.2 implies that if A_1 and A_2 are mutually exclusive, then

$$P(A_1 \cup A_2) = P(A_1) + P(A_2). \quad (1.2)$$

The intuitive meaning of (1.2) is that if an experiment can be repeated indefinitely, then for two mutually exclusive events A_1 and A_2 , the proportion of times that $A_1 \cup A_2$ occurs is equal to the sum of the proportion of times that A_1 occurs and the proportion of times that A_2 occurs. For example, for the experiment of tossing a fair die, $S = \{1, 2, 3, 4, 5, 6\}$ is the sample space. Let A_1 be the event that the outcome is 6, and A_2 be the event that the outcome is odd. Then $A_1 = \{6\}$ and $A_2 = \{1, 3, 5\}$. Since all sample points are equally likely to occur (by the definition of a fair die) and the number of sample points of A_1 is $1/6$ of the number of sample points of S , we expect that $P(A_1) = 1/6$. Similarly, we expect that $P(A_2) = 3/6$. Now $A_1 A_2 = \emptyset$ implies that the number of sample points of $A_1 \cup A_2$ is $(1/6 + 3/6)$ th of the number of sample points of S . Hence we should expect that $P(A_1 \cup A_2) = 1/6 + 3/6$, which is the same as $P(A_1) + P(A_2)$. This and many other examples suggest that (1.2) is a reasonable relation to be taken as Axiom 3. However, if we do this, difficulties arise when a sample space contains infinitely many sample points, that is, when the number of possible outcomes of an experiment is not finite. For example, in successive throws of a die let A_n be the event that the first 6 occurs on the n th throw. Then we would be unable to find the probability of $\bigcup_{n=1}^{\infty} A_n$, which represents the event that eventually a 6 occurs. For this reason, Axiom 3, which is the infinite analog of (1.2), is required. It by no means contradicts our intuitive ideas of probability, and one of its great advantages is that Theorems 1.1 and 1.2 are its immediate consequences.

A significant implication of (1.2) is that for any event A , $P(A) \leq 1$. To see this, note that

$$P(A \cup A^c) = P(A) + P(A^c).$$

Now, by Axiom 2,

$$P(A \cup A^c) = P(S) = 1;$$

therefore, $P(A) + P(A^c) = 1$. This and Axiom 1 imply that $P(A) \leq 1$. Hence

The probability of the occurrence of an event is always some number between 0 and 1. That is,

$$0 \leq P(A) \leq 1.$$

Example 1.16 A coin is called **unbiased** or **fair** if, whenever it is flipped, the probability of obtaining heads equals that of obtaining tails. Suppose that in an experiment an unbiased coin

is flipped. The sample space of such an experiment is $S = \{T, H\}$. Since the events $\{H\}$ and $\{T\}$ are equally likely to occur, $P(\{T\}) = P(\{H\})$, and since they are mutually exclusive,

$$P(\{T, H\}) = P(\{T\}) + P(\{H\}).$$

Hence Axioms 2 and 3 imply that

$$1 = P(S) = P(\{H, T\}) = P(\{H\}) + P(\{T\}) = P(\{H\}) + P(\{H\}) = 2P(\{H\}).$$

This gives that $P(\{H\}) = 1/2$ and $P(\{T\}) = 1/2$. Now suppose that an experiment consists of flipping a biased coin where the outcome of tails is twice as likely as heads, then $P(\{T\}) = 2P(\{H\})$. Hence

$$1 = P(S) = P(\{H, T\}) = P(\{H\}) + P(\{T\}) = P(\{H\}) + 2P(\{H\}) = 3P(\{H\}).$$

This shows that $P(\{H\}) = 1/3$; thus $P(\{T\}) = 2/3$. ♦

Example 1.17 Sharon has baked five loaves of bread that are identical except that one of them is underweight. Sharon's husband chooses one of these loaves at random. Let B_i , $1 \leq i \leq 5$, be the event that he chooses the i th loaf. Since all five loaves are equally likely to be drawn, we have

$$P(\{B_1\}) = P(\{B_2\}) = P(\{B_3\}) = P(\{B_4\}) = P(\{B_5\}).$$

But the events $\{B_1\}$, $\{B_2\}$, $\{B_3\}$, $\{B_4\}$, and $\{B_5\}$ are mutually exclusive, and the sample space is $S = \{B_1, B_2, B_3, B_4, B_5\}$. Therefore, by Axioms 2 and 3,

$$1 = P(S) = P(\{B_1\}) + P(\{B_2\}) + P(\{B_3\}) + P(\{B_4\}) + P(\{B_5\}) = 5 \cdot P(\{B_1\}).$$

This gives $P(\{B_1\}) = 1/5$ and hence $P(\{B_i\}) = 1/5$, $1 \leq i \leq 5$. Therefore, the probability that Sharon's husband chooses the underweight loaf is $1/5$. ♦

From Examples 1.16 and 1.17 it should be clear that if a sample space contains N points that are equally likely to occur, then the probability of each outcome (sample point) is $1/N$. In general, this can be shown as follows. Let $S = \{s_1, s_2, \dots, s_N\}$ be the sample space of an experiment, then, if all of the sample points are equally likely to occur, we have

$$P(\{s_1\}) = P(\{s_2\}) = \dots = P(\{s_N\}).$$

But $P(S) = 1$, and the events $\{s_1\}$, $\{s_2\}$, $\dots$, $\{s_N\}$ are mutually exclusive. Therefore,

$$\begin{aligned} 1 = P(S) &= P(\{s_1, s_2, \dots, s_N\}) \\ &= P(\{s_1\}) + P(\{s_2\}) + \dots + P(\{s_N\}) = NP(\{s_1\}). \end{aligned}$$

This shows that $P(\{s_1\}) = 1/N$. Thus $P(\{s_i\}) = 1/N$ for $1 \leq i \leq N$.

One simple consequence of the axioms of probability is that, if the sample space of an experiment contains N points that are all equally likely to occur, then the probability of the

occurrence of any event A is equal to the number of points of A , say $N(A)$, divided by N . Historically, until the introduction of the axiomatic method by A. N. Kolmogorov in 1933, this fact was taken as the definition of the probability of A . It is now called the **classical definition of probability**. The following theorem, which shows that the classical definition is a simple result of the axiomatic approach, is also an important tool for the computation of probabilities of events for experiments with finite sample spaces.

Theorem 1.3 *Let S be the sample space of an experiment. If S has N points that are all equally likely to occur, then for any event A of S ,*

$$P(A) = \frac{N(A)}{N},$$

where $N(A)$ is the number of points of A .

Proof: Let $S = \{s_1, s_2, \dots, s_N\}$, where each s_i is an outcome (a sample point) of the experiment. Since the outcomes are equiprobable, $P(\{s_i\}) = 1/N$ for all i , $1 \leq i \leq N$. Now let $A = \{s_{i_1}, s_{i_2}, \dots, s_{i_{N(A)}}\}$, where $s_{i_j} \in S$ for all i_j . Since $\{s_{i_1}\}, \{s_{i_2}\}, \dots, \{s_{i_{N(A)}}\}$ are mutually exclusive, Axiom 3 implies that

$$\begin{aligned} P(A) &= P(\{s_{i_1}, s_{i_2}, \dots, s_{i_{N(A)}}\}) \\ &= P(\{s_{i_1}\}) + P(\{s_{i_2}\}) + \dots + P(\{s_{i_{N(A)}}\}) \\ &= \underbrace{\frac{1}{N} + \frac{1}{N} + \dots + \frac{1}{N}}_{N(A) \text{ terms}} = \frac{N(A)}{N}. \quad \blacklozenge \end{aligned}$$

Example 1.18 An elevator with two passengers stops at the second, third, and fourth floors. If it is equally likely that a passenger gets off at any of the three floors, what is the probability that the passengers get off at different floors?

Solution: Let a and b denote the two passengers and a_2b_4 mean that a gets off at the second floor and b gets off at the fourth floor, with similar representations for other cases. Let A be the event that the passengers get off at different floors. Then

$$S = \{a_2b_2, a_2b_3, a_2b_4, a_3b_2, a_3b_3, a_3b_4, a_4b_2, a_4b_3, a_4b_4\}$$

and $A = \{a_2b_3, a_2b_4, a_3b_2, a_3b_4, a_4b_2, a_4b_3\}$. So $N = 9$ and $N(A) = 6$. Therefore, the desired probability is $N(A)/N = 6/9 = 2/3$. $\blacklozenge$

Example 1.19 Let S be the sample space of flipping a fair coin three times and A be the event of at least two heads, then

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

and $A = \{HHH, HHT, HTH, THH\}$. So $N = 8$ and $N(A) = 4$. Therefore, the probability of at least two heads in flipping a fair coin three times is $N(A)/N = 4/8 = 1/2$. Similarly, the chance of getting three heads stands at $1/8$. [Figure 1.7](#), displayed on the following page, illustrates a systematic method for listing all elements of the sample space, ensuring none are omitted. $\blacklozenge$

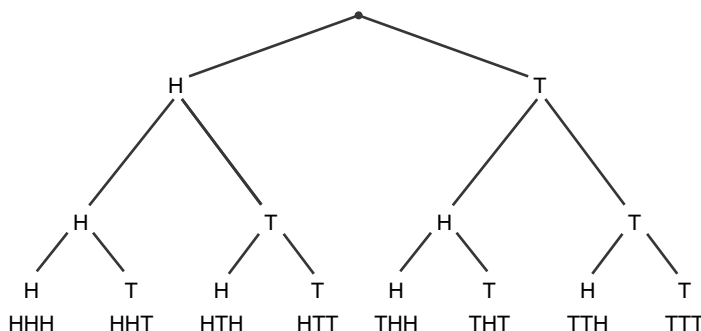


Figure 1.7 Systematic listing of the sample space elements from Example 1.19.

Example 1.20 A number is selected at random from the set of integers $\{1, 2, \dots, 1000\}$. What is the probability that the number is divisible by 3?

Solution: Here the sample space contains 1000 points, so $N = 1000$. Let A be the set of all numbers between 1 and 1000 that are divisible by 3. Then $A = \{3m : 1 \leq m \leq 333\}$. So $N(A) = 333$. Therefore, the probability that a random natural number between 1 and 1000 is divisible by 3 is equal to $333/1000$. ♦

Example 1.21 A number is selected at random from the set $\{1, 2, \dots, N\}$. What is the probability that the number is divisible by k , $1 \leq k \leq N$?

Solution: Here the sample space contains N points. Let A be the event that the outcome is divisible by k . Then $A = \{km : 1 \leq m \leq [N/k]\}$, where $[N/k]$ is the greatest integer less than or equal to N/k (to compute $[N/k]$, just divide N by k and round down). So $N(A) = [N/k]$ and $P(A) = [N/k]/N$. ♦

Example 1.22 (Galton's Paradox) In the scenario where three fair coins are flipped and two of them show the same face, one might argue that the probability of the third coin also showing the same face is $1/2$. If the two coins that landed on the same face are heads, the probability of the third coin being heads is said to be $1/2$. Similarly, if the two coins landed on tails, the probability of the third coin being tails is also said to be $1/2$. However, Francis Galton[†] explained in his 1894 paper that this argument is flawed due to the confusion between a *specific* coin and *any* coin in general. To analyze the experiment of flipping three coins, let's consider the sample space:

$$S = \{\text{HTT}, \text{HHT}, \text{HTH}, \text{HHH}, \text{THH}, \text{TTH}, \text{THT}, \text{TTT}\}.$$

Examining each possible outcome, we find that in all outcomes, at least two coins land on the same face. However, in only two of them (HHH and TTT) do all three coins land on the same face. Therefore, the desired probability is $2/8$, which simplifies to $1/4$ or 0.25 . ♦

[†] Sir Francis Galton (1822–1911), a cousin of Charles Darwin, was a renowned scientist and polymath who made significant contributions to a diverse array of scientific disciplines. Among his notable accomplishments, he pioneered the development of statistical concepts such as regression and correlation. In recognition of his immense achievements, he was knighted in 1909.

Remark 1.3 As explained in Remark 1.1, different manifestations of outcomes of an experiment may lead to different representations for the sample space of the same experiment. Because of this, different sample points of a representation might not have the same probability of occurrence. For example, suppose that a study is being done on families with three children. Let the outcomes of the study be the number of girls and the number of boys in a randomly selected family. Then $S = \{bbb, bgg, bgb, bbg, ggb, gbg, gbb, ggg\}$ and $\Omega = \{bbb, bbg, bgg, ggg\}$ are both reasonable sample spaces for the genders of the children of the family. In S , for example, bgg means that the first child of the family is a boy, the second child is a girl, and the third child is also a girl. In Ω , bgg means that the family has one boy and two girls. Therefore, in S all sample points occur with the same probability, namely, $1/8$. In Ω , however, probabilities associated to the sample points are not equal: $P(\{bbb\}) = P(\{ggg\}) = 1/8$ whereas $P(\{bbg\}) = P(\{bgg\}) = 3/8$. ♦

Finally, we should note that for finite sample spaces, if nonnegative probabilities are assigned to sample points so that they sum to 1, then the probability axioms hold. Let

$$S = \{w_1, w_2, \dots, w_n\}$$

be a sample space. Let $p_1, p_2, \dots, p_n$ be n nonnegative real numbers with $\sum_{i=1}^n p_i = 1$. Let P be defined on subsets of S by $P(\{w_i\}) = p_i$, and

$$P(\{w_{i_1}, w_{i_2}, \dots, w_{i_\ell}\}) = p_{i_1} + p_{i_2} + \dots + p_{i_\ell}.$$

It is straightforward to verify that P satisfies the probability axioms. Hence P defines a probability on the sample space S .

1.4 BASIC THEOREMS

Theorem 1.4 For any event A , $P(A^c) = 1 - P(A)$.

Proof: Since $AA^c = \emptyset$, A and A^c are mutually exclusive. Thus

$$P(A \cup A^c) = P(A) + P(A^c).$$

But $A \cup A^c = S$ and $P(S) = 1$, so

$$1 = P(S) = P(A \cup A^c) = P(A) + P(A^c).$$

Therefore, $P(A^c) = 1 - P(A)$. ♦

This theorem states that the probability of nonoccurrence of the event A is 1 minus the probability of its occurrence. For example, consider $S = \{(i, j): 1 \leq i \leq 6, 1 \leq j \leq 6\}$, the sample space of tossing two fair dice. If A is the event of getting a sum of 4, then $A = \{(1, 3), (2, 2), (3, 1)\}$ and $P(A) = 3/36$. Theorem 1.4 states that the probability of A^c , the event of not getting a sum of 4, which is harder to count, is $1 - 3/36 = 33/36$. As another example, consider the experiment of selecting a random number from the set $\{1, 2, 3, \dots, 1000\}$. By Example 1.20, the probability that the number selected is divisible

by 3 is 333/1000. Thus by Theorem 1.4, the probability that it is not divisible by 3, a quantity harder to find directly, is $1 - 333/1000 = 667/1000$.

Theorem 1.5 *If $A \subseteq B$, then*

$$P(B - A) = P(BA^c) = P(B) - P(A).$$

Proof: $A \subseteq B$ implies that $B = (B - A) \cup A$ (see Figure 1.8). But $(B - A)A = \emptyset$. So the events $B - A$ and A are mutually exclusive, and $P(B) = P((B - A) \cup A) = P(B - A) + P(A)$. This gives $P(B - A) = P(B) - P(A)$. ♦

Corollary *If $A \subseteq B$, then $P(A) \leq P(B)$.*

Proof: By Theorem 1.5, $P(B - A) = P(B) - P(A)$. Since $P(B - A) \geq 0$, we have that $P(B) - P(A) \geq 0$. Hence $P(B) \geq P(A)$. ♦

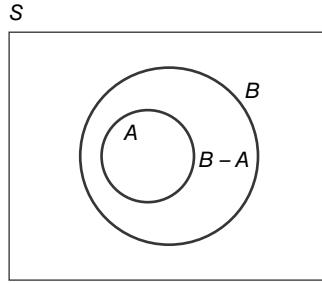


Figure 1.8 $A \subseteq B$ implies that $B = (B - A) \cup A$.

This corollary says that, for instance, it is less likely that a computer has one defect than it has at least one defect. Note that in Theorem 1.5, the condition of $A \subseteq B$ is necessary. The relation $P(B - A) = P(B) - P(A)$ is not true in general. For example, in rolling a fair die, let $B = \{1, 2\}$ and $A = \{3, 4, 5\}$, then $B - A = \{1, 2\}$. Therefore, $P(B - A) = 1/3$, $P(B) = 1/3$, and $P(A) = 1/2$. Hence $P(B - A) \neq P(B) - P(A)$.

Theorem 1.6 $P(A \cup B) = P(A) + P(B) - P(AB)$.

Proof: As depicted in Figure 1.9, $A \cup B = A \cup (B - AB)$ and $A(B - AB) = \emptyset$, so A and $B - AB$ are mutually exclusive events and

$$P(A \cup B) = P(A \cup (B - AB)) = P(A) + P(B - AB). \quad (1.3)$$

Now since $AB \subseteq B$, Theorem 1.5 implies that

$$P(B - AB) = P(B) - P(AB).$$

Therefore, (1.3) gives

$$P(A \cup B) = P(A) + P(B) - P(AB). \quad \blacklozenge$$

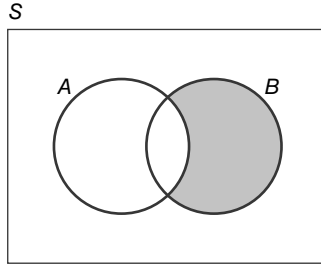


Figure 1.9 $A \cup B = A \cup (B - AB)$.

Example 1.23 Suppose that in a community of 400 adults, 300 bike or swim or do both, 160 swim, and 120 swim and bike. What is the probability that an adult, selected at random from this community, bikes?

Solution: Let A be the event that the person swims and B be the event that he or she bikes, then $P(A \cup B) = 300/400$, $P(A) = 160/400$, and $P(AB) = 120/400$. Hence the relation $P(A \cup B) = P(A) + P(B) - P(AB)$ implies that

$$\begin{aligned} P(B) &= P(A \cup B) + P(AB) - P(A) \\ &= \frac{300}{400} + \frac{120}{400} - \frac{160}{400} = \frac{260}{400} = 0.65. \quad \blacklozenge \end{aligned}$$

Example 1.24 A number is chosen at random from the set of integers $\{1, 2, \dots, 1000\}$. What is the probability that it is divisible by 3 or 5 (i.e., either 3 or 5 or both)?

Solution: The number of integers between 1 and N that are divisible by k is computed by dividing N by k and then rounding down (see Examples 1.21). Therefore, if A is the event that the outcome is divisible by 3 and B is the event that it is divisible by 5, then $P(A) = 333/1000$ and $P(B) = 200/1000$. Now AB is the event that the outcome is divisible by both 3 and 5. Since a number is divisible by 3 and 5 if and only if it is divisible by 15 (3 and 5 are prime numbers), $P(AB) = 66/1000$ (divide 1000 by 15 and round down to get 66). Thus the desired probability is computed as follows:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(AB) \\ &= \frac{333}{1000} + \frac{200}{1000} - \frac{66}{1000} = \frac{467}{1000} = 0.467. \quad \blacklozenge \end{aligned}$$

Theorem 1.6 gives a formula to calculate the probability that at least one of A and B occurs. We may also calculate the probability that at least one of the events $A_1, A_2, A_3, \dots$, and A_n occurs. For three events A_1, A_2 , and A_3 ,

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) &= P(A_1) + P(A_2) + P(A_3) - P(A_1 A_2) \\ &\quad - P(A_1 A_3) - P(A_2 A_3) + P(A_1 A_2 A_3). \end{aligned}$$

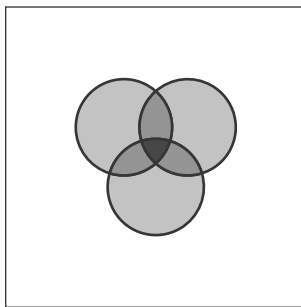


Figure 1.10 Venn diagram illustrating the formula for $P(A_1 \cup A_2 \cup A_3)$.

For four events,

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3 \cup A_4) = & P(A_1) + P(A_2) + P(A_3) + P(A_4) - P(A_1A_2) \\ & - P(A_1A_3) - P(A_1A_4) - P(A_2A_3) - P(A_2A_4) \\ & - P(A_3A_4) + P(A_1A_2A_3) + P(A_1A_2A_4) \\ & + P(A_1A_3A_4) + P(A_2A_3A_4) - P(A_1A_2A_3A_4). \end{aligned}$$

We now explain a procedure, which will enable us to find $P(A_1 \cup A_2 \cup \cdots \cup A_n)$, the probability that at least one of the events $A_1, A_2, \dots, A_n$ occurs.

Inclusion-Exclusion Principle *To calculate $P(A_1 \cup A_2 \cup \cdots \cup A_n)$, first find all of the possible intersections of events from $A_1, A_2, \dots, A_n$ and calculate their probabilities. Then add the probabilities of those intersections that are formed of an odd number of events, and subtract the probabilities of those formed of an even number of events.*

The following formula is an expression for the principle of inclusion-exclusion. It follows by induction. (For an intuitive proof, see Example 2.33.)

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) = & \sum_{i=1}^n P(A_i) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n P(A_iA_j) + \sum_{i=1}^{n-2} \sum_{j=i+1}^{n-1} \sum_{k=j+1}^n P(A_iA_jA_k) \\ & - \cdots + (-1)^{n-1} P(A_1A_2 \cdots A_n). \end{aligned}$$

Example 1.25 Suppose that 25% of the population of a city read newspaper A, 20% read newspaper B, 13% read C, 10% read both A and B, 8% read both A and C, 5% read B and C, and 4% read all three. If a person from this city is selected at random, what is the probability that he or she does not read any of these newspapers?

Solution: Let E , F , and G be the events that the person reads A, B, and C, respectively. The event that the person reads at least one of the newspapers A, B, or C is $E \cup F \cup G$. Therefore, $1 - P(E \cup F \cup G)$ is the probability that he or she reads none of them. Since

$$\begin{aligned} P(E \cup F \cup G) = & P(E) + P(F) + P(G) - P(EF) - P(EG) \\ & - P(FG) + P(EFG) \\ = & 0.25 + 0.20 + 0.13 - 0.10 - 0.08 - 0.05 + 0.04 = 0.39, \end{aligned}$$

the desired probability equals $1 - 0.39 = 0.61$. ♦

Example 1.26 Dr. Grossman, an internist, has 520 patients, of which (i) 230 are hypertensive, (ii) 185 are diabetic, (iii) 35 are hypochondriac and diabetic, (iv) 25 are all three, (v) 150 are none, (vi) 140 are only hypertensive, and finally, (vii) 15 are hypertensive and hypochondriac but not diabetic. Find the probability that Dr. Grossman's next appointment is hypochondriac but neither diabetic nor hypertensive. Assume that appointments are all random. This implies that even hypochondriacs do not make more visits than others.

Solution: Let T , C , and D denote the events that the next appointment of Dr. Grossman is hypertensive, hypochondriac, and diabetic, respectively. The Venn diagram of Figure 1.11 shows that the number of patients with only hypochondria is 30. Therefore, the desired probability is $30/520 \approx 0.06$. ♦

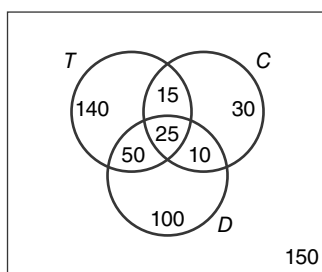


Figure 1.11 Venn diagram of Example 1.26.

Theorem 1.7 $P(A) = P(AB) + P(AB^c)$.

Proof: Clearly, $A = AS = A(B \cup B^c) = AB \cup AB^c$. Since AB and AB^c are mutually exclusive,

$$P(A) = P(AB \cup AB^c) = P(AB) + P(AB^c).$$

See also Figure 1.12. ♦

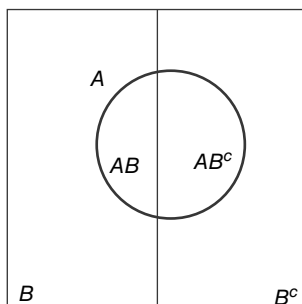


Figure 1.12 Venn diagram illustrating $P(A) = P(AB) + P(AB^c)$.

Example 1.27 In a community, 32% of the population are male smokers; 27% are female smokers. What percentage of the population of this community smoke?

Solution: Let A be the event that a randomly selected person from this community smokes. Let B be the event that the person is male. By Theorem 1.7,

$$P(A) = P(AB) + P(AB^c) = 0.32 + 0.27 = 0.59.$$

Therefore, 59% of the population of this community smoke. ♦

Remark 1.4 In the real world, especially for games of chance, it is common to express probability in terms of **odds**. We say that the *odds in favor of an event A are r to s* if $P(A) = r/(r + s)$. Similarly, the *odds against an event A are r to s* if $P(A) = s/(r + s)$. Therefore, if the odds in favor of an event A are r to s , then the odds against A are s to r . For example, in drawing a card at random from an ordinary deck of 52 cards, the odds against drawing an ace are 48 to 4 or, equivalently, 12 to 1. The odds in favor of an ace are 4 to 48 or, equivalently, 1 to 12. If for an event A , $P(A) = p$, then the odds in favor of A are p to $1 - p$. Therefore, for example, in flipping a fair coin three times, by Example 1.19, the odds in favor of HHH are $1/8$ to $7/8$ or, equivalently, 1 to 7. ♦

EXERCISES

A

1. Gottfried Wilhelm Leibniz (1646–1716), the German mathematician, philosopher, statesman, and one of the supreme intellects of the seventeenth century, believed that in a throw of a pair of fair dice, the probability of obtaining the sum 11 is equal to that of obtaining the sum 12. Do you agree with Leibniz? Explain.
2. Show that if A and B are mutually exclusive, then $P(A) + P(B) \leq 1$.
3. Suppose that 33% of the people have O^+ blood and 7% have O^- . What is the probability that the next president of the United States has type O blood?
4. Sixty-eight minutes through *Morituri*, the 1965 Marlon Brando movie, a group of underground anti-Nazis wanting to escape from a Nazi ship to a nearby island state that the chances of their succeeding are 15-to-1. Eighty-two minutes through the movie, the same group refers to this estimate stating that the odds against successfully escaping to that island are 15-to-1. Are these two statements consistent? Why or why not?
5. The probability that an earthquake will damage a certain structure during a year is 0.015. The probability that a hurricane will damage the same structure during a year is 0.025. If the probability that *both* an earthquake and a hurricane will damage the structure during a year is 0.0073, what is the probability that next year the structure will not be damaged by a hurricane or an earthquake?

6. In a probability test, for two events E and F of a sample space, Tina's calculations resulted in $P(E) = 1/4$, $P(F) = 1/2$, and $P(EF) = 3/8$. Is it possible that Tina made a mistake in her calculations? Why or why not?
7. For events A and B , suppose that the probability that at least one of them occurs is 0.8 and the probability that both of them occur is 0.3. Find the probability that exactly one of them occurs.
8. The admission office of a college admits only applicants whose high school GPA is at least 3.0 or whose SAT score is 1200 or higher. If 38% of the applicants of this college have at least a 3.0 GPA, 30% have a SAT of 1200 or higher, and 15% have both, what percentage of all the applicants are admitted to the college?
9. Jacqueline, Bonnie, and Tina are the only contestants in an athletic race, where it is not possible to tie. The probability that Bonnie wins is $2/3$ that of Jacqueline winning and $4/3$ that of Tina winning. Find the probability of each of these three athletes winning.
10. A motorcycle insurance company has 7000 policyholders of whom 5000 are under 40. If 4100 of the policy holders are males and under 40, 1100 are married and under 40, and 550 are married males who are under 40, find the probability that the next motorcycle policy holder of this company who gets into an accident is a single female who is under 40. Assume that it is equally likely for all policyholders to get into an accident.
11. From an ordinary deck of 52 cards, we draw cards at random and without replacement. What is the probability that at least two cards must be drawn to obtain a face card?
12. Suppose that the probability that a driver is a male, and has at least one motor vehicles accident during a 1-year period, is 0.12. Suppose that the corresponding probability for a female is 0.06. What is the probability of a randomly selected driver having at least one accident during the next 12 months?
13. Suppose that 75% of all investors invest in traditional annuities and 45% of them invest in the stock market. If 85% invest in the stock market and/or traditional annuities, what percentage invest in both?
14. In a horse race, the odds in favor of the first horse winning in an 8-horse race are 2 to 5. The odds against the second horse winning are 7 to 3. What is the probability that one of these horses will win?
15. A professor asks her students to present three methods of generating a random state out of US states. One of the methods a student, Walter, introduces is to draw a congressman from the list of 535 voting members of the House of Representatives and then select the state to which that member belongs. Is the state selected by this method random?
16. Let $S = \{\omega_1, \omega_2, \omega_3\}$ be the sample space of an experiment. If $P(\{\omega_1, \omega_2\}) = 0.5$ and $P(\{\omega_1, \omega_3\}) = 0.7$, find $P(\{\omega_1\})$, $P(\{\omega_2\})$, and $P(\{\omega_3\})$.
17. Suppose that on the mortality rates table provided on the following page, by the probability of the interval $[x, y)$ we mean the probability that, in a certain region, a person dies on or after his or her x th birthday but before his or her y th birthday. What is the probability that a newborn in this region will survive to reach the age of 60? What is the probability that he or she lives to be at least 20 but dies before 50?

Interval	Probability of the interval	Interval	Probability of the interval
[0, 10)	0.0634	[50, 60)	0.1257
[10, 20)	0.0129	[60, 70)	0.2089
[20, 30)	0.0218	[70, 80)	0.2710
[30, 40)	0.0317	[80, 90)	0.1683
[40, 50)	0.0634	[90, 100)	0.0277
		[100, ∞)	0.0052

18. Excerpt from the TV show *The Rockford Files*:

Rockford: There are only two doctors in town. The chances of both autopsies being performed by the same doctor are 50–50.

Reporter: No, that is only for one autopsy. For two autopsies, the chances are 25–75.

Rockford: You’re right.

Was Rockford right to agree with the reporter? Explain why or why not?

19. In a major state university, to study the performance relationship between calculus I and calculus II grades, the mathematics department reviewed the letter grades of all students who had taken both of the calculus courses in the last 20 years. For each pair of letter grades (X, Y) , the department calculated the proportion of all of the students who had earned X in calculus I and Y in calculus II. The results are given in the following table.

Calculus I	Calculus II				
	F	D	C	B	A
F	0.00	0.00	0.00	0.00	0.00
D	0.08	0.05	0.04	0.02	0.01
C	0.02	0.03	0.18	0.08	0.02
B	0.01	0.01	0.09	0.07	0.04
A	0.00	0.01	0.06	0.10	0.08

According to the rules of this university, if a student takes a course several times, only the last grade will be recorded. Based on this table, find the probability that a randomly selected student (a) does better in calculus I than in calculus II; (b) does better in calculus II than in calculus I; (c) earns the same letter grade in both calculus courses.

20. A company has only one position with three highly qualified applicants: John, Barbara, and Marty. However, because the company has only a few women employees, Barbara’s chance to be hired is 20% higher than John’s and 20% higher than Marty’s. Find the probability that Barbara will be hired.
21. In a psychiatric hospital, the number of patients with schizophrenia is three times the number with psychoneurotic reactions, twice the number with alcohol addictions, and 10 times the number with involuntional psychotic reaction. If a patient is selected randomly

from the list of all patients with one of these four diseases, what is the probability that he or she suffers from schizophrenia? Assume that none of these patients has more than one of these four diseases.

22. Let A and B be two events. Prove that

$$P(AB) \geq P(A) + P(B) - 1.$$

23. A card is drawn at random from an ordinary deck of 52 cards. What is the probability that it is (a) a black ace or a red queen; (b) a face or a black card; (c) neither a heart nor a queen?
24. Which of the following statements is true? If a statement is true, prove it. If it is false, give a counterexample.

(a) If $P(A) + P(B) + P(C) = 1$, then the events A , B , and C are mutually exclusive.

(b) If $P(A \cup B \cup C) = 1$, then A , B , and C are mutually exclusive events.

25. Suppose that in the Baltimore metropolitan area 25% of the crimes occur during the day and 80% of the crimes occur in the city. If only 10% of the crimes occur outside the city during the day, what percent occur inside the city during the night? What percent occur outside the city during the night?
26. Suppose that, in a temperate coniferous forest, 40% of randomly selected quarter-acre plots have both cedar and cypress trees, 25% have both cedar and redwood trees, 20% have cypress and redwoods, and 15% have all three. What is the probability that in a randomly selected quarter-acre plot in this forest we can find exactly two of the three types of these trees?
27. Let A , B , and C be three events. Prove that

$$\begin{aligned} P(A \cup B \cup C) \\ = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC). \end{aligned}$$

28. Let A , B , and C be three events. Show that exactly two of these events will occur with probability

$$P(AB) + P(AC) + P(BC) - 3P(ABC).$$

29. Eleven chairs are numbered 1 through 11. Four girls and seven boys sit on these chairs at random. What is the probability that chair 5 is occupied by a boy?
30. A ball is thrown at a square that is divided into n^2 identical squares. The probability that the ball hits the square of the i th column and j th row is p_{ij} , where $\sum_{i=1}^n \sum_{j=1}^n p_{ij} = 1$. In terms of p_{ij} 's, find the probability that the ball hits the j th horizontal strip.
31. Among 33 students in a class, 17 of them earned A 's on the midterm exam, 14 earned A 's on the final exam, and 11 did not earn A 's on either examination. What is the probability that a randomly selected student from this class earned an A on both exams?

32. From a small town 120 persons were selected at random and asked the following question: Which of the three shampoos, A , B , or C , do you use? The following results were obtained: 20 use A and C , 10 use A and B but not C , 15 use all three, 30 use only C , 35 use B but not C , 25 use B and C , and 10 use none of the three. If a person is selected at random from this group, what is the probability that he or she uses (a) only A ; (b) only B ; (c) A and B ? (Draw a Venn diagram.)
33. The coefficients of the quadratic equation $x^2 + bx + c = 0$ are determined by tossing a fair die twice (the first outcome is b , the second one is c). Find the probability that the equation has real roots.
34. Two integers m and n are called *relatively prime* if 1 is their only common positive divisor. Thus 8 and 5 are relatively prime, whereas 8 and 6 are not. A number is selected at random from the set $\{1, 2, 3, \dots, 63\}$. Find the probability that it is relatively prime to 63.
35. A number is selected randomly from the set $\{1, 2, \dots, 1000\}$. What is the probability that (a) it is divisible by 3 but not by 5; (b) it is divisible neither by 3 nor by 5?
36. The secretary of a college has calculated that from the students who took calculus, physics, and chemistry last semester, 78% passed calculus, 80% physics, 84% chemistry, 60% calculus and physics, 65% physics and chemistry, 70% calculus and chemistry, and 55% all three. Show that these numbers are not consistent, and therefore the secretary has made a mistake.

B

37. From an ordinary deck of 52 cards, we draw cards at random and without replacement until only cards of one suit are left. Find the probability that the cards left are all spades.
38. A number is selected at random from the set of natural numbers $\{1, 2, \dots, 1000\}$. What is the probability that it is divisible by 4 but neither by 5 nor by 7?
39. For a Democratic candidate to win an election, she must win districts I, II, and III. Polls have shown that the probability of winning I and III is 0.55, losing II but not I is 0.34, and losing II and III but not I is 0.15. Find the probability that this candidate will win all three districts. (Draw a Venn diagram.)
40. For events E and F , show that

$$P(E \cup F) + P(E \cup F^c) + P(E^c \cup F) + P(E^c \cup F^c) = 3.$$

41. Two numbers are successively selected at random and with replacement from the set $\{1, 2, \dots, 100\}$. What is the probability that the first one is greater than the second?
42. Let $A_1, A_2, A_3, \dots$ be a sequence of events of a sample space. Prove that

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n).$$

This is called **Boole's inequality**.

43. Let $A_1, A_2, A_3, \dots$ be a sequence of events of an experiment. Prove that

$$P\left(\bigcap_{n=1}^{\infty} A_n\right) \geq 1 - \sum_{n=1}^{\infty} P(A_n^c).$$

Hint: Use Boole's inequality, discussed in Exercise 42.

44. In a certain country, the probability is 49/50 that a randomly selected fighter plane returns from a mission without mishap. Mia argues that this means there is one mission with a mishap in every 50 consecutive flights. She concludes that if a fighter pilot returns safely from 49 consecutive missions, he should return home before his fiftieth mission. Is Mia right? Explain why or why not.
45. Let P be a probability defined on a sample space S . For events A of S define $Q(A) = [P(A)]^2$ and $R(A) = P(A)/2$. Is Q a probability on S ? Is R a probability on S ? Why or why not?

In its general case, the following exercise has important applications in coding theory, telecommunications, and computer science.

46. **(The Hat Problem)** A game begins with a team of three players entering a room one at a time. For each player, a fair coin is tossed. If the outcome is heads, a red hat is placed on the player's head, and if it is tails, a blue hat is placed on the player's head. The players are allowed to communicate before the game begins to decide on a strategy. However, no communication is permitted after the game begins. Players cannot see their own hats. But each player can see the other two players' hats. Each player is given the option to guess the color of his or her hat or to pass. The game ends when the three players simultaneously make their choices. The team wins if no player's guess is incorrect and at least one player's guess is correct. Obviously, the team's goal is to develop a strategy that maximizes the probability of winning. A trivial strategy for the team would be for two of its players to pass and the third player to guess red or blue as he or she wishes. This gives the team a 50% chance to win. Can you think of a strategy that improves the chances of the team winning?



Self-Quiz on Section 1.4

Time allotted: 20 Minutes

- Let A and B be mutually exclusive events of an experiment with $P(A) = 1/3$ and $P(B) = 1/4$. What is the probability that neither A occurs nor B ? (3 points)
- Zack has two weeks to read a book assigned by his English teacher and two weeks to write an essay assigned by his philosophy professor. The probability that he completes both assignments on time is 0.6, and the probability that he completes at least one of them on time is 0.95. What is the probability that he ends up meeting the deadline only for one assignment? (3 points)

3. Suppose that two-thirds of Americans traveling to Europe visit at least one of the three countries France, England, and Italy. Furthermore, suppose that one-half of them visit England, one-third visit France, and one-fourth visit Italy. If for each pair of these countries, one-fifth of the travelers visit that pair, what fraction of Americans traveling to Europe visit all of these three countries? (4 points)



1.5 CONTINUITY OF PROBABILITY FUNCTIONS

As with Exercises 29, 30, and 31 in [Section 1.2](#), let the symbol $\mathbb{R}$ represent the set of all real numbers here and throughout the rest of the book. We know from calculus that a function $f: \mathbb{R} \rightarrow \mathbb{R}$ is called continuous at a point $c \in \mathbb{R}$ if $\lim_{x \rightarrow c} f(x) = f(c)$. It is called continuous on $\mathbb{R}$ if it is continuous at all points $c \in \mathbb{R}$. We also know that this definition is equivalent to the sequential criterion $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $\mathbb{R}$ if and only if, for every convergent sequence $\{x_n\}_{n=1}^{\infty}$ in $\mathbb{R}$,

$$\lim_{n \rightarrow \infty} f(x_n) = f\left(\lim_{n \rightarrow \infty} x_n\right). \quad (1.4)$$

This property, in some sense, is shared by the probability function. To explain this, we need to introduce some definitions. But first recall that probability is a set function from $\mathcal{P}(S)$, the set of all possible events of the sample space S , to $[0, 1]$.

A sequence $\{E_n, n \geq 1\}$ of events of a sample space is called **increasing** if

$$E_1 \subseteq E_2 \subseteq E_3 \subseteq \cdots \subseteq E_n \subseteq E_{n+1} \cdots;$$

it is called **decreasing** if

$$E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots \supseteq E_n \supseteq E_{n+1} \supseteq \cdots.$$

For an increasing sequence of events $\{E_n, n \geq 1\}$, by $\lim_{n \rightarrow \infty} E_n$ we mean the event that at least one $E_i, 1 \leq i < \infty$ occurs. Therefore,

$$\lim_{n \rightarrow \infty} E_n = \bigcup_{i=1}^{\infty} E_i.$$

Similarly, for a decreasing sequence of events $\{E_n, n \geq 1\}$, by $\lim_{n \rightarrow \infty} E_n$ we mean the event that every E_i occurs. Thus in this case

$$\lim_{n \rightarrow \infty} E_n = \bigcap_{i=1}^{\infty} E_i.$$

The following theorem expresses the property of probability function that is analogous to (1.4).

Theorem 1.8 (Continuity of Probability Function) For any increasing or decreasing sequence of events, $\{E_n, n \geq 1\}$,

$$\lim_{n \rightarrow \infty} P(E_n) = P\left(\lim_{n \rightarrow \infty} E_n\right).$$

Proof: For the case where $\{E_n, n \geq 1\}$ is increasing, let $F_1 = E_1$, $F_2 = E_2 - E_1$, $F_3 = E_3 - E_2, \dots, F_n = E_n - E_{n-1}, \dots$. Clearly, $\{F_i, i \geq 1\}$ is a mutually exclusive set of events that satisfies the following relations:

$$\bigcup_{i=1}^n F_i = \bigcup_{i=1}^n E_i = E_n, \quad n = 1, 2, 3, \dots,$$

$$\bigcup_{i=1}^{\infty} F_i = \bigcup_{i=1}^{\infty} E_i$$

(see Figure 1.13). Hence

$$\begin{aligned} P(\lim_{n \rightarrow \infty} E_n) &= P\left(\bigcup_{i=1}^{\infty} E_i\right) = P\left(\bigcup_{i=1}^{\infty} F_i\right) = \sum_{i=1}^{\infty} P(F_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(F_i) \\ &= \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n F_i\right) = \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n E_i\right) = \lim_{n \rightarrow \infty} P(E_n), \end{aligned}$$

where the last equality follows since $\{E_n, n \geq 1\}$ is increasing, and hence $\bigcup_{i=1}^n E_i = E_n$. This establishes the theorem for increasing sequences.

If $\{E_n, n \geq 1\}$ is decreasing, then $E_n \supseteq E_{n+1}, \forall n$, implies that $E_n^c \subseteq E_{n+1}^c, \forall n$. Therefore, the sequence $\{E_n^c, n \geq 1\}$ is increasing and

$$\begin{aligned} P(\lim_{n \rightarrow \infty} E_n) &= P\left(\bigcap_{i=1}^{\infty} E_i\right) = 1 - P\left[\left(\bigcap_{i=1}^{\infty} E_i\right)^c\right] = 1 - P\left(\bigcup_{i=1}^{\infty} E_i^c\right) \\ &= 1 - P(\lim_{n \rightarrow \infty} E_n^c) = 1 - \lim_{n \rightarrow \infty} P(E_n^c) = 1 - \lim_{n \rightarrow \infty} [1 - P(E_n)] \\ &= 1 - 1 + \lim_{n \rightarrow \infty} P(E_n) = \lim_{n \rightarrow \infty} P(E_n). \quad \blacklozenge \end{aligned}$$

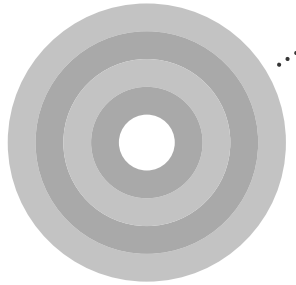


Figure 1.13 The circular disks are the E_i 's and the shaded circular annuli are the F_i 's, except for F_1 , which equals E_1 .

Example 1.28 Suppose that some individuals in a population produce offspring of the same kind. The offspring of the initial population are called second generation, the offspring of the

second generation are called third generation, and so on. Furthermore, suppose that with probability $\exp[-(2n^2 + 7)/(6n^2)]$ the entire population completely dies out by the n th generation before producing any offspring, what is the probability that such a population survives forever?

Solution: Let E_n denote the event of extinction of the entire population by the n th generation, then

$$E_1 \subseteq E_2 \subseteq E_3 \subseteq \cdots \subseteq E_n \subseteq E_{n+1} \subseteq \cdots$$

because if E_n occurs, then E_{n+1} also occurs. Hence, by Theorem 1.8,

$$\begin{aligned} P(\{\text{population survives forever}\}) &= 1 - P(\{\text{population eventually dies out}\}) \\ &= 1 - P\left(\bigcup_{i=1}^{\infty} E_i\right) = 1 - \lim_{n \rightarrow \infty} P(E_n) \\ &= 1 - \lim_{n \rightarrow \infty} \exp\left(-\frac{2n^2 + 7}{6n^2}\right) = 1 - e^{-1/3}. \quad \blacklozenge \end{aligned}$$

1.6 PROBABILITIES 0 AND 1

Events with probabilities 1 and 0 should not be misinterpreted. If E and F are events with probabilities 1 and 0, respectively, it is not correct to say that E is the sample space S and F is the empty set $\emptyset$. In fact, there are experiments in which there exist infinitely many events each with probability 1, and infinitely many events each with probability 0. An example follows.

Suppose that an experiment consists of selecting a random point from the interval $(0, 1)$. Since every point in $(0, 1)$ has a decimal representation such as

$$0.529387043219721 \cdots,$$

the experiment is equivalent to picking an endless decimal from $(0, 1)$ at random (note that if a decimal terminates, all of its digits from some point on are 0). In such an experiment we want to compute the probability of selecting the point $1/3$. In other words, we want to compute the probability of choosing $0.333333 \cdots$ in a random selection of an endless decimal. Let A_n be the event that the selected decimal has 3 as its first n digits, then

$$A_1 \supset A_2 \supset A_3 \supset A_4 \supset \cdots \supset A_n \supset A_{n+1} \supset \cdots,$$

since the occurrence of A_{n+1} guarantees the occurrence of A_n . Now $P(A_1) = 1/10$ because there are 10 choices 0, 1, 2, $\dots$, 9 for the first digit, and we want only one of them, namely 3, to occur. $P(A_2) = 1/100$ since there are 100 choices 00, 01, $\dots$, 09, 10, 11, $\dots$, 19, 20, $\dots$, 99 for the first two digits, and we want only one of them, 33, to occur. $P(A_3) = 1/1000$ because there are 1000 choices 000, 001, $\dots$, 999 for the first three digits, and we want only one of them, 333, to occur. Continuing this argument, we have $P(A_n) = (1/10)^n$. Since $\bigcap_{n=1}^{\infty} A_n = \{1/3\}$, by Theorem 1.8,

$$P\left(\frac{1}{3} \text{ is selected}\right) = P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} \left(\frac{1}{10}\right)^n = 0.$$

Note that there is nothing special about the point $1/3$. For any other point $0.\alpha_1\alpha_2\alpha_3\alpha_4\cdots$ from $(0, 1)$, the same argument could be used to show that the probability of its occurrence is 0 (define A_n to be the event that the first n digits of the selected decimal are $\alpha_1, \alpha_2, \dots, \alpha_n$, respectively, and repeat the same argument). We have shown that in random selection of points from $(0, 1)$, the probability of the occurrence of any particular point is 0. Now for $t \in (0, 1)$, let $B_t = (0, 1) - \{t\}$. Then $P(\{t\}) = 0$ implies that

$$P(B_t) = P(\{t\}^c) = 1 - P(\{t\}) = 1.$$

Therefore, there are infinitely many events, B_t 's, each with probability 1 and none equal to the sample space $(0, 1)$.

1.7 RANDOM SELECTION OF POINTS FROM INTERVALS

In [Section 1.6](#), we showed that the probability of the occurrence of any particular point in a random selection of points from an interval (a, b) is 0. This implies immediately that if $[\alpha, \beta] \subseteq (a, b)$, then the events that the point falls in $[\alpha, \beta]$, (α, β) , $[\alpha, \beta)$, and $(\alpha, \beta]$ are all equiprobable. Now consider the intervals $\left(a, \frac{a+b}{2}\right)$ and $\left(\frac{a+b}{2}, b\right)$; since $\frac{a+b}{2}$ is the midpoint of (a, b) , it is reasonable to assume that

$$p_1 = p_2, \tag{1.5}$$

where p_1 is the probability that the point belongs to $\left(a, \frac{a+b}{2}\right)$ and p_2 is the probability that it belongs to $\left(\frac{a+b}{2}, b\right)$. The events that the random point belongs to $\left(a, \frac{a+b}{2}\right)$ and $\left(\frac{a+b}{2}, b\right)$ are mutually exclusive and

$$\left(a, \frac{a+b}{2}\right) \cup \left[\frac{a+b}{2}, b\right) = (a, b);$$

therefore,

$$p_1 + p_2 = 1.$$

This relation and (1.5) imply that

$$p_1 = p_2 = 1/2.$$

Hence *the probability that a random point selected from (a, b) falls into the interval*

$\left(a, \frac{a+b}{2}\right)$ is $1/2$. The probability that it falls into $\left[\frac{a+b}{2}, b\right)$ is also $1/2$. Note that the

length of each of these intervals is $1/2$ of the length of (a, b) . Now consider the intervals

$\left(a, \frac{2a+b}{3}\right]$, $\left(\frac{2a+b}{3}, \frac{a+2b}{3}\right]$, and $\left(\frac{a+2b}{3}, b\right)$. Since $\frac{2a+b}{3}$ and $\frac{a+2b}{3}$ are the points that divide the interval (a, b) into three subintervals with equal lengths, we can assume that

$$p_1 = p_2 = p_3, \tag{1.6}$$

where p_1 , p_2 , and p_3 are the probabilities that the point falls into $\left(a, \frac{2a+b}{3}\right]$, $\left(\frac{2a+b}{3}, \frac{a+2b}{3}\right]$, and $\left(\frac{a+2b}{3}, b\right)$, respectively. On the other hand, these three intervals are mutually disjoint and

$$\left(a, \frac{2a+b}{3}\right] \cup \left(\frac{2a+b}{3}, \frac{a+2b}{3}\right] \cup \left(\frac{a+2b}{3}, b\right) = (a, b).$$

Hence

$$p_1 + p_2 + p_3 = 1.$$

This relation and (1.6) imply that

$$p_1 = p_2 = p_3 = 1/3.$$

Therefore, the probability that a random point selected from (a, b) falls into the interval

$\left(a, \frac{2a+b}{3}\right]$ is $1/3$. The probability that it falls into $\left(\frac{2a+b}{3}, \frac{a+2b}{3}\right]$ is $1/3$, and the probability that it falls into $\left(\frac{a+2b}{3}, b\right)$ is $1/3$. Note that the length of each of these

intervals is $1/3$ of the length of (a, b) . These and other similar observations indicate that the probability of the event that a random point from (a, b) falls into a subinterval (α, β) is equal to $(\beta - \alpha)/(b - a)$.

Note that in this discussion we have assumed that subintervals of equal lengths are equiprobable. Even though two subintervals of equal lengths may differ by a finite or a countably infinite set (or even a set of measure zero), this assumption is still consistent with our intuitive understanding of choosing random points from intervals. This is because in such an experiment, the probability of the occurrence of a finite or countably infinite set (or a set of measure zero) is 0.

Thus far, we have based our discussion of selecting random points from intervals on our intuitive understanding of this experiment and not on a mathematical definition. Such discussions are often necessary for the creation of appropriate mathematical meanings for unclear concepts. The following definition, which is based on our intuitive analysis, gives an exact mathematical meaning to the experiment of random selection of points from intervals.

Definition 1.2 A point is said to be **randomly selected from an interval** (a, b) if any two subintervals of (a, b) that have the same length are equally likely to include the point. The probability associated with the event that the subinterval (α, β) contains the point is defined to be $(\beta - \alpha)/(b - a)$.

As explained before, choosing a random number from $(0, 1)$ is equivalent to choosing randomly all the decimal digits of the number successively. Since in practice this is impossible, choosing exact random points or numbers from $(0, 1)$ or any other interval is only a theoretical matter. Approximate random numbers, however, can be generated by computers. Most of the computer languages, some scientific computer software, and some calculators are equipped with subroutines that generate approximate random numbers from intervals. However, since it

is difficult to construct good random number generators, there are computer languages, software programs, and calculators that are equipped with poor random number generator algorithms. An excellent reference for construction of good random number generators is *The Art of Computer Programming*, Volume 2, *Seminumerical Algorithms*, third edition, by Donald E. Knuth (Addison Wesley, 1998). Simple mechanical tools can also be used to find such approximations. For example, consider a spinner mounted on a wheel of *unit circumference* (radius $1/2\pi$). Let A be a point on the perimeter of the wheel. Each time that we flick the spinner, it stops, pointing toward some point B on the wheel's circumference. The length of the arc AB (directed, say, counterclockwise) is an approximate random number between 0 and 1 if the spinner does not have any “sticky” spots (see Figure 1.14).

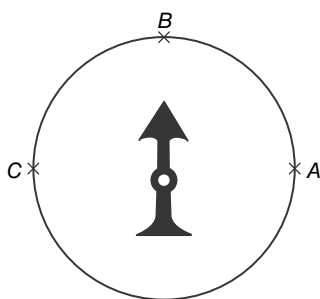


Figure 1.14 Spinner, a model to generate random numbers.

Finally, when selecting a random point from an interval (a, b) , we may think of an extremely large hypothetical box that contains infinitely many indistinguishable balls. Imagine that each ball is marked by a number from (a, b) , each number of (a, b) is marked on exactly one ball, and the balls are completely mixed up, so that in a random selection of balls, any two of them have the same chance of being drawn. With this transcendental model in mind, choosing a random number from (a, b) is then equivalent to drawing a random ball from such a box and looking at its number.

★ Example 1.29 (A Set that Is Not an Event) Let an experiment consist of selecting a point at random from the interval $[-1, 2]$. We will construct a set that is not an event (i.e., it is impossible to associate a probability with the set). We begin by defining an equivalence relation on $[0, 1]$: $x \sim y$ if $x - y$ is a rational number. Let $\mathbf{Q} = \{r_1, r_2, \dots\}$ be the set of rational numbers in $[-1, 1]$. Clearly, x is equivalent to y if $x - y \in \mathbf{Q}$. The fact that this relation is reflexive, symmetric, and transitive is trivial. Therefore, being an equivalence relation, it partitions the interval $[0, 1]$ into disjoint equivalence classes (Λ_α) . These classes are such that if x and $y \in \Lambda_\alpha$ for some α , then $x - y$ is rational. However, if $x \in \Lambda_\alpha$ and $y \in \Lambda_\beta$, and $\alpha \neq \beta$, then $x - y$ is irrational. These observations imply that for each α , the equivalence class Λ_α is countable. Since $\bigcup_\alpha \Lambda_\alpha = [0, 1]$ is uncountable, the number of equivalence classes is uncountable. Let E be a set consisting of *exactly* one point from each equivalence class Λ_α . The existence of such a set is guaranteed by the Axiom of Choice. We will show, by contradiction, that E is not an event. Suppose that E is an event, and let p be the probability associated with E . For each positive integer n , let $E_n = \{r_n + x : x \in E\} \subseteq [-1, 2]$. For

each $r_n \in \mathbf{Q}$, E_n is simply a translation of E . Thus, for all n , the set E_n is also an event, and $P(E_n) = P(E) = p$.

We now make two more observations: (1) For $n \neq m$, $E_n \cap E_m = \emptyset$, (2) $[0, 1] \subset \bigcup_{n=1}^{\infty} E_n$. To prove (1), let $t \in E_n \cap E_m$. We will show that $E_n = E_m$. If $t \in E_n \cap E_m$, then for some $r_n, r_m \in \mathbf{Q}$, and $x, y \in E$, we have that $t = r_n + x = r_m + y$. That is, $x - y = r_m - r_n$ is rational, and $x - y$ belongs to the same equivalence class. Since E has exactly one point from each equivalence class, we must have $x = y$, hence $r_n = r_m$, hence $E_n = E_m$. To prove (2), let $x \in [0, 1]$. Then $x \sim y$ for some $y \in E$. This implies that $x - y$ is a rational number in $\mathbf{Q}$. That is, for some n , $x - y = r_n$, or $x = y + r_n$, or $x \in E_n$. Thus $x \in \bigcup_{n=1}^{\infty} E_n$.

Putting (1) and (2) together, we obtain

$$1/3 = P([0, 1]) \leq P\left(\bigcup_{n=1}^{\infty} E_n\right) \leq 1,$$

or

$$1/3 \leq \sum_{n=1}^{\infty} P(E_n) = \sum_{n=1}^{\infty} p \leq 1.$$

This is a contradiction because $\sum_{n=1}^{\infty} p$ is either 0 or ∞ . Hence E is not an event, and we cannot associate a probability with this set. ♦

EXERCISES

A

1. A bus arrives at a station every day at a random time between 1:00 P.M. and 1:30 P.M. What is the probability that a person arriving at this station at 1:00 P.M. will have to wait at least 10 minutes?
2. Past experience shows that every new book by a certain publisher captures randomly between 4 and 12% of the market. What is the probability that the next book by this publisher captures at most 6.35% of the market?
3. Which of the following statements are true? If a statement is true, prove it. If it is false, give a counterexample.
 - (a) If A is an event with probability 1, then A is the sample space.
 - (b) If B is an event with probability 0, then $B = \emptyset$.
4. Let A and B be two events. Show that if $P(A) = 1$ and $P(B) = 1$, then $P(AB) = 1$.
5. A point is selected at random from the interval $(0, 2000)$. What is the probability that it is an integer?

6. Suppose that a point is randomly selected from the interval $(0, 1)$. Using Definition 1.2, show that all numerals are equally likely to appear as the first digit of the decimal representation of the selected point.
7. For an experiment with sample space $S = (0, 2)$, for $n \geq 1$, let $E_n = \left(0, 1 + \frac{1}{n}\right)$ and $P(E_n) = (3n + 2)/5n$. Find the probability of $E = (0, 1]$. Note that this is *not* the experiment of choosing a point at random from the interval $(0, 2)$ as defined in Section 1.7.
8. A point is chosen at random from the interval $(-1, 1)$. Let E_1 be the event that it falls in the interval $(-1, 1/3]$, E_2 be the event that it falls in the interval $(-1, 1/9]$, E_3 be the event that it falls in the interval $(-1, 1/27]$ and, in general, for $1 \leq i < \infty$, E_i be the event that the point is in the interval $(-1, 1/3^i]$. Find $\bigcup_{i=1}^{\infty} E_i$ and $\bigcap_{i=1}^{\infty} E_i$.
9. For the experiment of choosing a point at random from the interval $(0, 1)$, let $E_n = (1/2 - 1/2n, 1/2 + 1/2n)$, $n \geq 1$.
 - (a) Prove that $\bigcap_{n=1}^{\infty} E_n = \{1/2\}$.
 - (b) Using part (a) and the continuity of probability function, show that the probability of selecting $1/2$ in a random selection of a point from $(0, 1)$ is 0.

B

10. Is it possible to define a probability on a countably infinite sample space so that the outcomes are equally probable?
11. Let $A_1, A_2, \dots, A_n$ be n events. Show that if

$$P(A_1) = P(A_2) = \dots = P(A_n) = 1,$$

then $P(A_1 A_2 \dots A_n) = 1$.

12. A point is selected at random from the interval $(0, 1)$. What is the probability that it is rational? What is the probability that it is irrational?
13. Suppose that a point is randomly selected from the interval $(0, 1)$. Using Definition 1.2, show that all numerals are equally likely to appear as the n th digit of the decimal representation of the selected point.
14. Let $\{A_1, A_2, A_3, \dots\}$ be a sequence of events. Prove that if the series $\sum_{n=1}^{\infty} P(A_n)$ converges, then $P\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n\right) = 0$. This is called the **Borel-Cantelli lemma**. It says that if $\sum_{n=1}^{\infty} P(A_n) < \infty$, the probability that infinitely many of the A_n 's occur is 0.
Hint: Let $B_m = \bigcup_{n=m}^{\infty} A_n$ and apply Theorem 1.8 to $\{B_m, m \geq 1\}$.
15. Show that the result of Exercise 11 is not true for an infinite number of events. That is, show that if $\{E_t : 0 < t < 1\}$ is a collection of events for which $P(E_t) = 1$, it is not necessarily true that $P\left(\bigcap_{t \in (0,1)} E_t\right) = 1$.

16. Let A be the set of rational numbers in $(0, 1)$. Since A is countable, it can be written as a sequence (i.e., $A = \{r_n : n = 1, 2, 3, \dots\}$). Prove that for any $\varepsilon > 0$, A can be covered by a sequence of open balls whose total length is less than ε . That is, $\forall \varepsilon > 0$, there exists a sequence of open intervals (α_n, β_n) such that $r_n \in (\alpha_n, \beta_n)$ and $\sum_{n=1}^{\infty} (\beta_n - \alpha_n) < \varepsilon$. This important result explains why in a random selection of points from $(0, 1)$ the probability of choosing a rational is zero.

Hint: Let $\alpha_n = r_n - \varepsilon/2^{n+2}$, $\beta_n = r_n + \varepsilon/2^{n+2}$.



Self-Quiz on Section 1.7

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. Suppose that for events A and B , $P(AB) = 0$. Does this imply that A and B are mutually exclusive? Why or why not?
2. For the experiment of choosing a point at random from the interval $[0, 1]$, let $E_n = \left[\frac{1}{3} - \frac{1}{n+2}, \frac{1}{3} + \frac{2}{n+2}\right]$, $n \geq 1$. Applying the Continuity of Probability Function to E_n 's, show that $P(1/3 \text{ is selected}) = 0$.



1.8 RANDOMNESS AND UNCERTAINTY: DISTINGUISHING BETWEEN TWO CONCEPTS

The concepts of randomness and uncertainty can sometimes be confused, but they refer to different aspects of probabilistic events. Here we discuss a clarification of their differences. It is important to understand that while “randomness” and “uncertainty” are related concepts, they have distinct meanings. Randomness refers to the lack of pattern or predictability in a sequence of events or outcomes, suggesting that there is no discernible order or purpose behind their occurrence. Randomness can arise from various sources, such as natural processes, chance occurrences, or unpredictable factors. Here are some examples of random phenomena:

- Rolling a fair six-sided die: The outcome of each roll is random and unpredictable.
- Radioactive decay: The decay of a radioactive substance is random, and it is impossible to predict precisely when a particular atom will decay, but the decay rate can be described statistically.
- Lottery drawings: The selection of winning numbers in a lottery is typically done randomly. Each number has an equal chance of being drawn, and the outcome is unpredictable.

- **Brownian motion:** The movement of particles suspended in a fluid exhibits random motion known as Brownian motion. The individual trajectory of each particle is unpredictable due to random collisions with fluid molecules.

On the other hand, uncertainty relates to a lack of knowledge, information, or confidence about the future or a specific event. It reflects a situation where the outcome or consequences are not fully known or predictable. Uncertainty can arise due to incomplete information, limited understanding, ambiguous factors, or the presence of various possible outcomes. Unlike randomness, uncertainty does not necessarily imply that events are governed by a random process. Instead, it highlights the limitations of our knowledge or the complexity of the system under consideration.

Uncertainty can be quantified or managed using probabilistic methods, where different possibilities are assigned probabilities based on available information or assumptions. For example, when predicting the weather, there might be uncertainty regarding the outcome because weather forecasting is a complex process influenced by many variables. The outcome cannot be precisely determined due to limited information, inherent variability in the atmosphere, and the inability to measure all relevant factors accurately. However, the underlying weather system is not random but follows physical laws.

Similarly, predicting the future state of the economy involves considerable uncertainty. Factors such as global events, policy changes, and market fluctuations make it challenging to accurately forecast economic indicators like GDP[†] growth, inflation rates, or stock market performance. While there is uncertainty in these predictions, the economic system operates based on certain principles and is not entirely random.

Determining a diagnosis for a complex medical condition can also involve uncertainty. Doctors often rely on test results, symptoms, and their expertise, but due to the complexity of the human body and the potential for overlapping symptoms, there can be uncertainty in reaching a definitive diagnosis. This uncertainty arises from the limitations of medical knowledge and the uniqueness of individual cases rather than from randomness.

In summary, randomness refers to the lack of predictability or pattern in outcomes, while uncertainty refers to a lack of knowledge or precise understanding of the probabilities associated with different outcomes.

1.9 WHAT IS SIMULATION?

Solving a scientific or an industrial problem usually involves mathematical analysis and/or simulation. To perform a simulation, we repeat an experiment a large number of times to assess the probability of an event or condition occurring. For example, to estimate the probability of at least one 6 occurring within four rolls of a die, we may do a large number of experiments rolling a die four times and calculate the number of times that at least one 6 is obtained. Similarly, to estimate the fraction of time that, in a certain bank all the tellers are busy, we may measure the lengths of such time intervals over a long period X , add them, and then divide by X . Clearly, in simulations, the key to reliable answers is *to perform the experiment a large number of times or over a long period of time, whichever is applicable*. Since manually this is almost impossible,

[†] Gross Domestic Product.

simulations are carried out by computers. Only computers can handle millions of operations in short periods of time.

To simulate a problem that involves random phenomena, generating random numbers from the interval $(0, 1)$ is essential. In almost every simulation of a probabilistic model, we will need to select random points from the interval $(0, 1)$. For example, to simulate the experiment of tossing a fair coin, we draw a random number from $(0, 1)$. If it is in $(0, 1/2)$, we say that the outcome is heads, and if it is in $[1/2, 1)$, we say that it is tails. Similarly, in the simulation of die tossing, the outcomes 1, 2, 3, 4, 5, and 6, respectively, correspond to the events that the random point from $(0, 1)$ is in $(0, 1/6)$, $[1/6, 1/3)$, $[1/3, 1/2)$, $[1/2, 2/3)$, $[2/3, 5/6)$, and $[5/6, 1)$.

As discussed in [Section 1.7](#), choosing a random number from a given interval is, in practice, impossible. In real-world problems, to perform simulation we use **pseudorandom numbers** instead. A pseudorandom number is a number that appears to be random, but it is generated deterministically from a process. This is in contrast to true random numbers, which are generated by genuinely random physical processes, such as radioactive decay or atmospheric noise. To generate n pseudorandom numbers from a uniform distribution on an interval (a, b) , we take an initial value $x_0 \in (a, b)$, called the **seed**, and construct a function ψ so that the sequence $\{x_1, x_2, \dots, x_n\} \subset (a, b)$ obtained recursively from

$$x_{i+1} = \psi(x_i), \quad 0 \leq i \leq n-1, \quad (1.7)$$

satisfies certain statistical tests for randomness. (Choosing the tests and constructing the function ψ are complicated matters beyond the scope of this book.) The function ψ takes a seed and generates a sequence of pseudorandom numbers in the interval (a, b) . Clearly, in any pseudorandom number generating process, the numbers generated are rounded to a certain number of decimal places. Therefore, ψ can only generate a finite number of pseudorandom numbers, which implies that, eventually, some x_j will be generated a second time. From that point on, by (1.7), a pitfall is that the same sequence of numbers that appeared after x_j 's first appearance will reappear. Beyond that point, numbers are not effectively random. One important aspect of the construction of ψ is that the second appearance of any of the x_j 's is postponed as long as possible.

The maximum number of pseudorandom numbers a generator can produce before the sequence begins to repeat is called the period of the random number generator. High-quality pseudorandom number generators have a very long period, making the repetition unnoticeable in practical applications. Common algorithms used for generating pseudorandom numbers include the Linear Congruential Generator, *Mersenne Twister*, and *Xorshift*. The Mersenne Twister, for instance, has a period of $2^{19937} - 1$, which is an extraordinarily long sequence before it repeats. For all practical purposes, this can be considered as a virtually infinite sequence.

It should be noted that passing certain statistical tests for randomness does not mean that the sequence $\{x_1, x_2, \dots, x_n\}$ is a randomly selected sequence from (a, b) in the true mathematical sense discussed in [Section 1.7](#). It is quite surprising that there are deterministic real-valued functions ψ that, for each i , generate an x_{i+1} that is completely determined by x_i , and yet the sequence $\{x_1, x_2, \dots, x_n\}$ passes certain statistical tests for randomness.

For convenience, throughout this chapter, in practical problems, by random number we simply mean pseudorandom number. In general, for good choices of ψ , the generated numbers are usually sufficiently random for practical purposes.

Most of the computer languages and some scientific computer software are equipped with subroutines that generate random numbers from intervals and from sets of integers. For example, in *Mathematica*, from Wolfram Research, Inc. (<http://www.wolfram.com>), the command **Random[]** selects a random number from $(0, 1)$, **Random[Real, { a, b }]** chooses a random number from the interval (a, b) , and **Random[Integer, { $m, m + n$ }]** picks up an integer from $\{m, m + 1, m + 2, \dots, m + n\}$ randomly. However, there are computer languages that are not equipped with a subroutine that generates random numbers, and there are a few that are equipped with poor algorithms. As mentioned in [Section 1.7](#), even though it is difficult to construct good pseudorandom number generators, an excellent reference for construction of such numbers is *The Art of Computer Programming*, Volume 2, 3rd edition, by Donald E. Knuth.

At this point, let us emphasize that the main goal of scientists and engineers is always to solve a problem mathematically. It is a mathematical solution that is accurate, exact, and completely reliable. Simulations cannot take the place of a rigorous mathematical solution. They are widely used (a) to find good estimations for solutions of problems that either cannot be modeled mathematically or whose mathematical models are too difficult to solve; (b) to get a better understanding of the behavior of a complicated phenomenon; and/or (c) to obtain a mathematical solution by acquiring insight into the nature of the problem, its functions, and the magnitude and characteristics of its solution. Intuitively, it is clear why the results that are obtained by simulations are good. Theoretically, most of them can be justified by the strong law of large numbers, which will be discussed in [Section 10.5](#).

It should be noted that *true* random numbers are generated through non-deterministic processes. Unlike pseudorandom number generators, they are not generated by algorithms and do not exhibit periodicity. True random number generation can be achieved through methods such as measuring electronic noise, observing radioactive decay, or leveraging chaotic processes like fluid dynamics.

At the conclusion of certain chapters in this book, there is a section that delves into approximate solutions to challenging probabilistic problems using simulation. For these simulations, we employ the R software. R is a potent tool for indeterministic simulations, owing to its comprehensive suite of built-in functions and the plethora of specialized libraries available for various simulation tasks. In Chapter 12 of this book, titled “Algorithm-Driven Simulations,” available on the Instructor and Student Resources website, we thoroughly explore the concept of simulation using algorithms exclusively, regardless of any specific programming languages.

1.10 A CONCISE STUDY OF R

Introduction

R is a robust, open-source software environment designed for probabilistic simulation, statistical computing, and graphics. It offers a comprehensive range of statistical tools, including linear and nonlinear modeling, classical statistical tests, time-series analysis, classification, clustering, and more. It was co-developed by Robert Gentleman and Ross Ihaka in 1992 at the University of Auckland, New Zealand.

One of R’s primary strengths is its extensibility. Users have the ability to craft their own functions and packages tailored to specific tasks or analyses. Beyond this, thousands of readily

available packages span a wide array of statistical, data manipulation, and visualization techniques. R boasts comprehensive capabilities for graphical displays, catering both to data visualization and the representation of statistical models. It accommodates conditional statements, loops, and intuitive recursive functions. Furthermore, R incorporates robust object-oriented and functional programming features.

It is not only the software but also the dynamic community that elevates R's value. A proactive online community provides support, especially through platforms like Stack Overflow and the R-help mailing list. The Comprehensive R Archive Network (CRAN) serves as the primary repository hosting thousands of R packages available to the public. Events such as the useR! conferences, along with numerous local R user groups, foster networking and knowledge exchange among R aficionados.

Pseudorandom Number Generators in R

One of R's strengths lies in its utilization of the Mersenne Twister as its primary pseudorandom number generator. As highlighted in [Section 1.9](#), the Mersenne Twister boasts a period of $2^{19937} - 1$. This means R can generate a pseudorandom sequence up to this maximum length, which, for all intents and purposes, is tantamount to an infinite sequence. It is nearly inconceivable for any practical application to exhaust this sequence. While the Mersenne Twister is pivotal, R also supports other random number generation algorithms. Users can toggle between different generators with the `RNGkind()` function. Each algorithm possesses its unique period and characteristics. For an in-depth understanding and the latest information on R's pseudorandom number generation methods, one can refer to the R documentation using `?RNGkind` or explore the R website.

The primary random number generator in R is executed using the command `runif()`. By default, this function yields numbers in the interval $[0, 1)$, meaning it produces numbers ranging from 0 (inclusive) up to, but not including, 1. Given the uniform distribution, any number within this range is equally probable. Below are a few illustrative examples

First, generate a single random number between 0 and 1. Next, produce 8 random numbers in the same range (0 to 1). Lastly, create 7 random numbers between 12 and 21, inclusive.

```
> runif(1)
[1] 0.198098
> runif(8)
[1] 0.58487560 0.98493753 0.07469242 0.85701668 0.57945711 0.03643187
[7] 0.05206846 0.65365963
> runif(7, 12, 21)
[1] 13.97913 20.21620 20.47132 20.23147 13.41544 18.78861 14.22891
```

For the command `runif(8)` above, the random numbers generated are displayed on two lines. The notation `[1]` at the beginning of the first line indicates that the first data point begins there, while `[7]` at the beginning of the second line indicates that the seventh data point is the first to appear on that line.

The code `sample(23:35,8)` produces 8 random integers between 23 and 35, inclusive. We will delve into the details of this code later.

```
> sample(23:35, 8)
[1] 27 32 23 33 26 31 25 34
```

Installation

To begin working with R, it's common to install both the core R environment from the CRAN website and, for those seeking a more user-friendly interface, the RStudio Integrated Development Environment (IDE). Both R and RStudio are freely available for Mac OS, Windows, and UNIX systems. Mastering R is a valuable skill. You can download R and access related resources at <https://www.r-project.org>. RStudio can be downloaded from <http://www.rstudio.com>.

Computational Prowess of R

In this book, examples of R input, output, and code are presented in a **typewriter-style** font. The right angle bracket `>` signifies the R input prompt. In R, commands can be separated by typing newline or return. If you type return before completing a command, R will display the `+` prompt instead of the typical `>`, indicating it is awaiting the completion of your command.

While R offers an extensive computing environment, we will focus on introducing only the objects and functions pertinent to the content of this book to ensure clarity and avoid overwhelming readers.

1.10.1 Basic Arithmetic R employs standard symbols for mathematical operations: `+` for addition, `-` for subtraction, `*` for multiplication, `/` for division, and `^` for exponentiation. We can use parentheses `( )` to define the order of operations. The order of basic arithmetic operations in R follows the standard arithmetic precedence: parentheses, exponents (and roots), multiplication and Division (from left to right), addition and subtraction (from left to right).

Certain special constants like `pi`, representing the value of $\pi \approx 3.1415927$, `Inf` representing positive infinity, and `-Inf` representing negative infinity are already defined.

When executing a *single* computation, R displays the result with the `[1]` prefix. Here are some examples:

```
> (2-1/25)^19
[1] 357162.1
> ((1+3/5)/(2-1/3))*5/8
[1] 0.6
```

Note that based on the order of operations, the expression in the parentheses are $1 + (3/5)$ and $2 - (1/3)$, not $4/5$ and $1/3$.

```
> 1/0
[1] Inf
```

```
> -1/0
[1] -Inf
```

1.10.2 Variables In R, a variable is a named location used to store data that can be used and manipulated throughout a program. Variables are objects in R where data, like numbers, texts, strings, vectors, matrices, and even functions can be stored. In R, we typically assign a value to a variable using the `<-` assignment operator (although `=` can also be used). For example, the command `x <- 5` signifies that `x` is a variable holding the value of `5`. R's flexibility with data structures and its object-oriented nature means that variables can hold a wide variety of complex data, from simple scalars to full-fledged data frames or lists. To display the value of a variable `x` on the screen, you can use the command `print(x)` or `show(x)`. While there are scenarios where these commands are necessary, often simply typing `x` will display its value. Examples follow.

```
> x <- 13
> (2+4/x)^x
[1] 52639.62
> print(x)
[1] 13
```

```
> n <- 12
> seq(1,n,n/4)
[1] 1 4 7 10
```

In programming languages, especially in R, we frequently need to update the values of certain variables based on specific algorithmic requirements. For instance, consider the commands `k<- 2`, `k<- k+1`, and `k<- k/10`. Executing them in this sequence, `k` initially takes the value 2, then is updated to $2 + 1 = 3$, and finally becomes $3/10 = 0.3$.

```
> k <- 2
> k <- k + 1
> k <- k/10
> k
[1] 0.3
> y <- (10*k)^k
> show(y)
[1] 1.390389
```

1.10.3 Functions R comes equipped with many built-in functions, including `sin(x)`, `cos(x)`, `tan(x)` (all in radians), `asin(x)`, `acos(x)`, `atan(x)`, `sinh(x)`, `cosh(x)`, `tanh(x)`, `asinh(x)`, `acosh(x)`, `atanh(x)`, `exp(x)`, `abs(x)`, absolute value of x , `sqrt(x)`,

square root of x , **log(x)**, natural logarithm (base e) of x , **log10(x)**, logarithm base 10 of x , **round(x)**, rounds x to the nearest whole number, **floor(x)**, rounds down x to the nearest whole number, **ceiling(x)**, rounds x up to the nearest whole number. Euler's number e , the base of the natural logarithm, is approximately 2.71828182846. It can be obtained in R using the command **exp(1)**.

```
> exp(1)
[1] 2.718282
> sin(pi/4)
[1] 0.7071068
> ceiling(1.4985)
[1] 2
> floor(1.4985)
[1] 1
> round(1.4985)
[1] 1
> tanh(3) + 2*asin(0.5)
[1] 2.042252
> log(1000)
[1] 6.907755
> log10(1000)
[1] 3
```

R computes with high precision, yet it shows only 7 significant digits by default. To display up to k digits, use the command **options(digits=k)**.

```
> options(digits=15)
> exp(1)
[1] 2.71828182845905
> cos(pi/6)
[1] 0.866025403784439
```

Another built-in function in R is **seq**, which generates arithmetic sequences. Here is an illustration of its usage.

```
> seq(from=5,to=20,by=4)
[1] 5 9 13 17
> seq(from=5,to=20)
[1] 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
```

Every function has a predefined argument order. If we supply arguments in this specific order, there is no need to name them.

```

> seq(5,20,4)
[1] 5 9 13 17
> seq(5,20)
[1] 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
> seq(20,1,-4)
[1] 20 16 12 8 4

```

In R, the **function** keyword is used to define custom functions. Here is a breakdown of how it works:

```

> function_name <- function(arguments){
> function body
> return(value_to_return)}

```

The **function_name** is how you will call the function later in your code. Naming follows typical variable naming conventions in R. Inside the parentheses after **function**, you specify any arguments that the function expects. Arguments are inputs to the **function**. The code block within the curly braces `{...}` is the function body. This is where the main logic of the function is written. Functions can optionally return a value using the **return()** function. If **return()** is not specified, the function will return the result of the last expression in the function body by default.

Example: A simple function to add two numbers:

```

> add <- function(a,b){
+ return(a + b)}
> print(add(6,19))
[1] 15

```

Example: A function without explicit return

```

> power <- function(a,b){
+ a^b}
> print(power(3,4))
[1] 81
> print(power(6,19))
[1] 6.093597e+14

```

In R, the **curve()** function is used to plot mathematical functions or expressions over a specified range. It is a useful tool for visualizing functions in a graphical form. Here is how it works:

curve(expr, from=a, to=b, n=2000)

expr is The mathematical expression or function we want to plot. It can be given as a formula or an expression that includes variables, constants, and mathematical operations. **from** is the lower bound of the x -axis range for plotting. If not specified, it will be determined automatically based on the function and the current plotting region. **to** is the upper bound of the x -axis range for plotting. If not specified, it will be determined automatically based on the function and the current plotting region. **n** is the number of points to use when plotting the curve. It controls the smoothness of the curve. More points (higher **n**) result in a smoother curve. When plotting a curve in R, if we do not specify a value for **n**, R will automatically determine an appropriate value for **n** to create a smooth curve. The **abline(h=0)** command adds the horizontal x -axis to the plot, while **abline(v=0)** adds the vertical y -axis. Setting **par(pin=c(6,3))** adjusts the plotting region dimensions using the **pin** parameter to create smaller figures. In this configuration, the plotting region's width is set to 6 inches, and its height is set to 3 inches. These dimensions can be customized according to our preferences. The following code generates a plot of the sine curve, $\sin(x)$, over the range -2π to 2π . The resulting output corresponds to [Figure 1.15](#).

```
> curve(sin(x), -2*pi, 2*pi)
> abline(h=0)
> abline(v=0)
> par(pin=c(6,3))
```

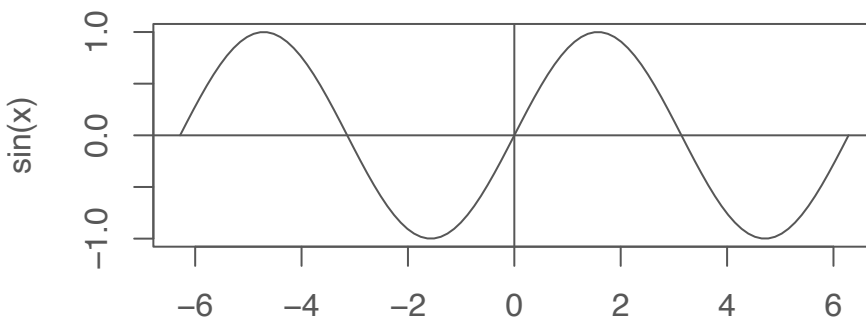


Figure 1.15 Plot of the sine function, $\sin(x)$ using R.

1.10.4 Vectors In R, a vector is one of the primary data structures, and it represents a sequence of elements of the same basic type. When working with R, understanding vectors is fundamental, as they are the foundation for more complex data structures like data frames and lists. To construct a vector in R, we can utilize the **c** function, which is short for “concatenate.” For instance, the expression **v <- c(7,5,11,3,1)** initializes **v** as the vector (7, 5, 11, 3, 1). In R, scalars are treated as vectors with a length of one. Thus, values like 3.18, 42, and 2π are all considered vectors that have a single element.

The command **length(v)** returns the number of entries in vector **v**. Similarly, **max(v)** provides the largest component within the vector, **min(v)** yields the smallest component, and **sum(v)** calculates the sum of all entries in **v**.

In R, the **sort()** function is used to arrange the elements of a vector or a data frame in ascending order by default. When the logical argument **decreasing=TRUE** is specified, it performs sorting in descending order. Here is how it works:

```
> v=c(7,2,14,8,9)
> sort(v)
[1] 2 7 8 9 14
> sort(v,decreasing = TRUE)
[1] 14 9 8 7 2
```

Vectors in R can store diverse data types beyond just numerical values. Consider the non-numeric vector, **names**, as an example. Below are some operations and manipulations we can perform on this vector.

```
> names <- c("Jody", "Ann Marie", "Susan")
> names[2]
[1] "Ann Marie"
> names <- c(names, "Agie")
> length(names)
[1] 4
> show(names)
[1] "Jody" "Ann Marie" "Susan" "Agie"
> toupper(names)
[1] "JODY" "ANN MARIE" "SUSAN" "AGIE"
> sort(names)
[1] "Agie" "Ann Marie" "Jody" "Susan"
```

To quickly generate a vector containing the sequence from 1 to n , simply use **1:n**. More broadly, for any integers m and n , the command **m:n** produces a sequence of integers starting from m and ending at n . If m is less than n , the sequence will be in ascending order; if m is greater than n , it will be in descending order.

Assume **v** is the vector defined as (7, 5, 11, 3, 1). In this context, **v[i]** refers to the i th element of **v**. Using the command **v[c(2,4,5)]**, we can extract a subvector from **v** containing its 2nd, 4th, and 5th elements. Conversely, **v[-c(1,3)]** provides all elements of **v** excluding the 1st and 3rd.

```
> v <- c(7,5,11,3,1)
> v[3]
[1] 11
```

```
> v[c(2,4,5)]
[1] 5 3 1
> v[-c(1,3)]
[1] 5 3 1
```

In R, many operations are applied element-wise to vectors. For instance, if $\mathbf{v}$ is a vector, then $\mathbf{v}^2$ will square each component of $\mathbf{v}$ separately. Likewise, the expression $1/(1 : 7)^3$ produces the vector $(1, 1/2^3, \dots, 1/7^3)$. Let us calculate $\sum_{n=1}^{100} 1/n^2$ using R:

```
> sum(1/(1:100)^2)
[1] 1.634984
```

The series $\sum_{n=1}^{\infty} 1/n^2$ has a well-known value of $\pi^2/6$, which is approximately 1.644934. For $n > 100$, the terms in the series become extremely small, making their contribution to the overall sum negligible. This observation highlights the accuracy of R's computation, given that its result for the sum of the first 100 terms of the series, 1.634984, is expectedly close to 1.644934.

To **add two vectors** of different lengths, say, $v = (a_1, a_2, a_3, a_4, a_5, a_6)$ and $w = (b_1, b_2)$, R *recycles* the shorter vector to match the length of the longer one. This means, R adds them in the following way:

$$v + w = (a_1 + b_1, a_2 + b_2, a_3 + b_1, a_4 + b_2, a_5 + b_1, a_6 + b_2).$$

However, if the length of the longer vector is not a multiple of the shorter one, R will issue a warning.

In R, the **rep** command is used to replicate elements of vectors and lists. Here is a basic breakdown of its usage:

```
> rep(5,times=3)
[1] 5 5 5
> rep(5,3)
[1] 5 5 5
> rep(c(1,3,5), times=2)
[1] 1 3 5 1 3 5
> rep(c(1,3,5), 2)
[1] 1 3 5 1 3 5
> rep(c(1,2,3), each=2)
[1] 1 1 2 2 3 3
> rep(c(1,3,5), times=2, each=3)
[1] 1 1 1 3 3 3 5 5 5 1 1 1 3 3 3 5 5 5
```

In R, the **cumsum** function calculates the cumulative sum of elements in a numeric vector. It returns a vector whose length is the same as the input, with each element representing the sum of the elements up to that position in the original vector.


```

> cumsum(c(2,4,6,8,10))
[1] 2 6 12 20 30
v<- c(1:45)
> cumsum(v)
[1] 1 3 6 10 15 21 28 36 45 55 66 78 91 105 120
[16] 136 153 171 190 210 231 253 276 300 325 351 378 406 435 465
[31] 496 528 561 595 630 666 703 741 780 820 861 903 946 990 1035

```

In this example, in the output for `cumsum(v)`,
[1] indicates that the line starts with the 1st element of the vector.
[16] indicates that the line starts with the 16th element of the vector.
[31] indicates that the line starts with the 31st element of the vector, and so on.

In R, the `all` function tests whether all elements of a logical vector are `TRUE`. It is often used in conjunction with conditions to check if all elements of a vector satisfy a certain condition. The `all` function is especially useful in control structures, data validation, and other situations where you need to verify that every element of a vector meets a particular condition. Here is how it works:

```

all(c(TRUE, TRUE, TRUE))
[1] TRUE
> all(c(FALSE, TRUE, TRUE))
[1] FALSE
> v<- c(4,7,10,13,16)
> all(v>0)
[1] TRUE

```

1.10.5 Sampling In R, the command `sample` is a handy tool for obtaining random samples. Given predefined values for n and k , $k \leq n$, the command `sample(n, k)` produces an ordered random sample of k numbers from the range 1 to n without replacement (i.e. without repetition). To draw a sample with replacement (with repetition), we utilize the command `sample(n, k, replace=TRUE)`.

The command `sample(n,n)` or its simpler variant `sample(n)` in R generates a random permutation of the sequence 1, 2, ..., n .

```

> sample(9)
[1] 7 4 2 6 5 8 3 9 1
> sample(9,9)
[1] 9 5 8 2 6 1 7 3 4
> sample(9,4)

```

```
[1] 3 8 5 2
> sample(9,4,replace=true)
Error: object 'true' not found
> sample(9,4,replace=TRUE)
[1] 1 9 6 4
> sample(9,6,replace=TRUE)
[1] 4 7 2 6 5 8
> sample(9,6,replace=TRUE)
[1] 2 2 3 7 7 3
```

The **sample** command in R can be customized to account for specific probabilities associated with each number. Consider the command:

```
sample(5,3,replace=True,prob=c(0.2,0.25,0.34,0.15,0.06))
```

This draws three numbers from the range 1 to 5, allowing repetition, where the chances of selecting each number are determined by the probabilities 0.2, 0.25, 0.34, 0.15, and 0.06, respectively. If we opt for sampling without replacement, the likelihood of picking a number not selected so far adjusts in proportion to its initial probability.

The **sample** command in R is not limited to numeric vectors; it can also draw from non-numeric ones. For instance, R has a built-in vector, **letters**, representing the 26 lowercase letters of the English alphabet. Therefore, for instance, using **sample(letters,5)**, we can produce a random string of 5 letters by selecting from this alphabet without repetition.

```
> Sample(month.name, 4)
Error in Sample(month.name, 4) : could not find function "Sample"
> sample(month.name, 4)
[1] "January" "October" "March" "November"
> sample(month.name, 5, replace=TRUE)
[1] "August" "September" "October" "May" "June"
> sample(month.name, 5, replace=TRUE)
[1] "November" "April" "September" "September" "August"
> sample(LETTERS, 6)
[1] "R" "I" "U" "X" "P" "W"
> sample(LETTERS, 6, replace=TRUE)
[1] "O" "X" "Z" "V" "P" "C"
```

1.10.6 Control structures in R: the “if,” “for,” and “while” commands To develop programs that execute mathematical algorithms, it’s essential to have the capability to make decisions and iterate over operations. This is accomplished through the use of the

if, **for**, and **while** commands. We will illustrate the functionality of these commands using practical examples.

Although the straightforward command **sum(1:100)** computes the sum $1 + 2 + \cdots + 100$, for educational reasons, we will demonstrate how to calculate it using three alternative methods with the commands **if**, **for**, and **while**.

```
> sum(1:100)
```

```
[1] 5050
```

```
> sum <- 0
```

```
> for (i in 1:100){sum <- sum + i}
```

```
> print(sum)
```

```
[1] 5050
```

```
> sum <- 0
```

```
> i <- 1
```

```
> while (i <= 100){sum <- sum + i  
+ i <- i + 1}
```

```
> print(sum)
```

```
[1] 5050
```

```
> sum <- 0
```

```
> i <- 1
```

```
> repeat{sum <- sum + i  
+ i <- i + 1  
+ if (i > 100){break}}
```

```
> print(sum)
```

```
[1] 5050
```

We will now employ the **if** command to compute $1 + 2 + 4 + 5 + \cdots + 100$. But first, let us familiarize ourselves with the relational operator **==**, which tests for equality. It checks if the values on the left and right of the operator are equal, and returns a logical value **TRUE** or **FALSE**. Note that **==** is used for comparison, where as **=** and **<-** are used for assignment in R. Also, be cautious when comparing floating point numbers using **==** due to precision issues. In such cases, it is often better to use functions that check for equality within a small tolerance, rather than exact equality.

```
> sum <- 0
```

```
> for(i in 1:100){if (i==3){next}
```

```
+ sum <- sum + i}
```

```
> print(sum)
[1] 5047
```

1.10.7 Simulation Producing a very large number of random samples empowers us to simulate probability problems. A handy tool for this task is the **replicate** command. We will employ this command for our initial simulation challenge below. Just like the **if**, **for**, and **while** commands, the most effective way to illustrate the usage of **replicate** is through practical examples.

Example 1.30 Two numbers are selected at random and without replacement from the set $\{1, 2, 3, \dots, n\}$. Write an algorithm in R for approximating the probability that the difference of these numbers is at least k .

Solution: Let **num** represent the total number of trials to be conducted, with the function **t** denoting the execution of the initial trial. We will use the **replicate** command to run the **t** function **num** times.

For $1 \leq i \leq \text{num}$, the i th entry of the vector **results**, **results[i]**, is set to **TRUE**, if on the i th trial, d , the absolute value of the difference between the two numbers meets or exceeds k ; otherwise, it is set to **FALSE**. The vector **results** has a length of **num** and contains entries of either **TRUE** or **FALSE**. Since, in R, **TRUE** is encoded as 1 and **FALSE** as 0, the expression **sum(results)** yields the total count of trials where $d \geq k$. Meanwhile, **mean(results)** calculates the average of all entries in the **results** vector, providing an approximate probability that the difference between the two chosen numbers is at least k . Below is the R code implementation of the described procedure for **num** = 10,000, $n = 500$ and $k = 80$:

```
> t<- function(){
+ s<- sample(500,2)
+ d<- abs(s[1]-s[2])
+ return(d >= 80)}
> num <- 10000
> results<- replicate (num, t())
> p<- mean(results)
> print(p)
[1] 0.7082
```

The precise solution to this problem, derived using analytical methods, is given by

$$\frac{(n-k)(n-k+1)}{n(n-1)}.$$

For $n = 500$ and $k = 80$ the theoretical result is

$$\frac{(420)(421)}{(500)(499)} \approx 0.7087.$$

This comparison reveals that our simulation closely matches the theoretical outcome. ♦

Example 1.31 In a blue state's election, the Democratic candidate received 3586 votes, while the Republican candidate garnered 2958 votes. Craft an R program to simulate and estimate the likelihood that the Democratic candidate maintained a lead throughout the entire vote-counting process.

Solution: Let **demv** and **repv** denote the number of votes received by the Democratic and Republican candidates, respectively. Define **totalv** as the total number of votes cast. To simulate this voting process, we will carry out the following experiment **num** times:

1. Create a vector **v** of length **totalv**. The first **demv** entries of this vector are set to 1 (representing Democratic votes), followed by **repv** entries set to 0 (indicating Republican votes).
2. Generate a vector **randv** by randomizing the order of entries in **v** using **sample(v)**. This shuffled vector represents one possible sequence in which the votes could have been cast.
3. Define a vector **d** where the *i*th entry represents the cumulative votes for the Democratic candidate after the *i*th vote has been counted. This can be computed using **d = cumsum(randv)**.
4. Construct another vector **w**, where the *i*th entry shows the total votes the Republican candidate has received after the *i*th voter has cast his or her vote. This is given by **w = (1:totalv) - d**.
5. Initialize a counter, **aahead**, to track the number of instances where the Democratic candidate was consistently ahead in the vote count. If every entry in **d** is greater than its corresponding entry in **w** (i.e., the Democratic candidate always led), increment **aahead** by 1.

After executing the above steps **num** times, the estimated probability **p** that the Democratic candidate maintained a lead throughout the entire vote-counting process is given by **p=aahead/num**. Below is the R code that implements the outlined simulation with **num** set to 25,000, **demv** as 3586, and **repv** as 2958.

```
> demv <- 3586
> repv <- 2958
> totalv <- demv + repv
> num <- 25000
> aahead <- 0
> for(i in 1:num){
+ v <- c(rep(1,demv), rep(0,repv))
+ randv <- sample(v)
+ d <- cumsum(randv)
+ w <- (1:totalv) - d
```

```

+ if(all(d>w)){aahead <- aahead+1}
+ }
> p <- aahead/num
> print(p)
[1] 0.09664

```

In this simulation, the result closely matched the exact solution derived from the analytical method. ♦

EXERCISES

1. A die is successively rolled until, for the first time, a number appears three consecutive times. Create an R program to run simulations and estimate the probability that it will take at least 50 rolls before this event occurs.
2. In an urn, there are 25 white and 35 red balls. Balls are drawn successively from the urn without replacement. Create an R program that simulates this scenario and computes the probability of a tie occurring when the number of red and white balls drawn becomes equal at some point.
3. The probability of a bank rejecting a loan application is 0.35. Develop an R program to simulate data and perform likelihood estimation to calculate the probability that out of 300 applicants, more than 100 of them will be refused financing.
4. There are five urns, each holding 10 white and 15 red balls. A ball is randomly drawn from the first urn and transferred to the second urn. Subsequently, a ball is randomly drawn from the second urn and placed into the third urn, continuing this process. Utilize R to approximate the probability that the last ball drawn will be red through simulation. *Answer:* 0.6.
5. In a small town with 1000 inhabitants, a person starts a story by telling it to a random individual, who subsequently shares it with another randomly chosen person, and so forth. Develop an R code to conduct likelihood estimation through simulations and estimate the approximate probability that before the story is retold 150 times, it circles back to the initial storyteller.
6. A city has a fleet of taxis numbered from 1 to n . On three separate occasions, a statistician ends up taking taxis numbered 31, 50, and 112. To determine the best estimate for n among the numbers 112 through 120, employ simulation in R.
Hint: Repeatedly conduct the following experiment for each n value within the range of 112 to 120 — select three random numbers from the set $\{1, 2, \dots, n\}$ and assess whether they match 31, 50, and 112.
7. Imagine an airplane passenger with an itinerary that involves changing planes in Istanbul, Turkey. Independently, there is a 4% chance of losing each piece of luggage at Istanbul international Airport. Furthermore, the probability of losing luggage is 5% at

Da Vinci Airport in Rome, 5% at Kennedy Airport in New York, and 4% at O'Hare Airport in Chicago. Dr. Magotra travels from Mumbai to San Francisco with three suitcases, changing planes in Istanbul, Rome, New York, and Chicago. Using R for simulation, determine the approximate probability that one of Dr. Magotra's suitcases does not arrive at his destination with him.



CHAPTER 1 SUMMARY

► If the outcome of an experiment is not certain but all of its possible outcomes are predictable in advance, then the set of all these possible outcomes is called the *sample space* of the experiment and is usually denoted by S . These outcomes are sometimes called *sample points*, or simply *points*, of the sample space. Certain subsets of S are referred to as *events*. So events are sets of points of the sample space. If the outcome of an experiment belongs to an event E , we say that the event E has *occurred*.

► In the study of probability theory the relations between different events of an experiment play a central role. The following are among the most important examples:

De Morgan's first law:

$$(E \cup F)^c = E^c F^c, \quad \left(\bigcup_{i=1}^n E_i \right)^c = \bigcap_{i=1}^n E_i^c, \quad \left(\bigcup_{i=1}^{\infty} E_i \right)^c = \bigcap_{i=1}^{\infty} E_i^c.$$

De Morgan's second law:

$$(EF)^c = E^c \cup F^c, \quad \left(\bigcap_{i=1}^n E_i \right)^c = \bigcup_{i=1}^n E_i^c, \quad \left(\bigcap_{i=1}^{\infty} E_i \right)^c = \bigcup_{i=1}^{\infty} E_i^c.$$

Another useful relation between E and F , two arbitrary events of a sample space S , is

$$E = EF \cup EF^c.$$

► If the joint occurrence of two events E and F is impossible, we say that E and F are *mutually exclusive*. So E and F are mutually exclusive if $EF = \emptyset$. A set of events $\{E_1, E_2, \dots\}$ is called *mutually exclusive* if the joint occurrence of any two of them is impossible, that is, if every pair of them is mutually exclusive.

► **Probability Axioms:** Let S be the sample space of a random phenomenon. Suppose that to each event A of S , a number denoted by $P(A)$ is associated with A . If P satisfies the following axioms, then it is called a *probability* and the number $P(A)$ is said to be the *probability of A* .

Axiom 1 $P(A) \geq 0$.

Axiom 2 $P(S) = 1$.

Axiom 3 If $\{A_1, A_2, A_3, \dots\}$ is a sequence of mutually exclusive events (i.e., the joint occurrence of every pair of them is impossible: $A_i A_j = \emptyset$ when $i \neq j$), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Some immediate implications of the axioms of probability are the following:

- $P(\emptyset) = 0$.
- Let $\{A_1, A_2, \dots, A_n\}$ be a mutually exclusive set of events. Then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

- Let S be the sample space of an experiment. If S has N points that are all equally likely to occur, then for any event A of S , $P(A) = N(A)/N$, where $N(A)$ is the number of points of A .
- For any event A , $P(A^c) = 1 - P(A)$.
- If $A \subseteq B$, then $P(B - A) = P(BA^c) = P(B) - P(A)$.
- If $A \subseteq B$, then $P(A) \leq P(B)$.
- $P(A \cup B) = P(A) + P(B) - P(AB)$.
- $P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 A_2) - P(A_1 A_3) - P(A_2 A_3) + P(A_1 A_2 A_3)$.
- **Inclusion-Exclusion Principle** To calculate $P(A_1 \cup A_2 \cup \dots \cup A_n)$, first find all of the possible intersections of events from $A_1, A_2, \dots, A_n$ and calculate their probabilities. Then add the probabilities of those intersections that are formed of an odd number of events, and subtract the probabilities of those formed of an even number of events. The following formula is an expression for this principle.

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) = & \sum_{i=1}^n P(A_i) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n P(A_i A_j) + \sum_{i=1}^{n-2} \sum_{j=i+1}^{n-1} \sum_{k=j+1}^n P(A_i A_j A_k) \\ & - \dots + (-1)^{n-1} P(A_1 A_2 \dots A_n). \end{aligned}$$

- $P(A) = P(AB) + P(AB^c)$.
- **(Continuity of Probability Function)** For any increasing or decreasing sequence of events, $\{E_n, n \geq 1\}$,

$$\lim_{n \rightarrow \infty} P(E_n) = P\left(\lim_{n \rightarrow \infty} E_n\right).$$

► A point is said to be *randomly selected from an interval* (a, b) if any two subintervals of (a, b) that have the same length are equally likely to include the point. The probability associated with the event that the subinterval (α, β) contains the point is defined to be $(\beta - \alpha)/(b - a)$.

REVIEW PROBLEMS

1. The number of minutes it takes for a certain animal to react to a certain stimulus is a random number between 2 and 4.3. Find the probability that the reaction time of such an animal to this stimulus is no longer than 3.25 minutes.
2. Two dice are rolled. What is the event that the outcomes are consecutive?
3. From a phone book, a phone number is selected at random. (a) What is the event that the last digit is an odd number? (b) What is the event that the last digit is divisible by 3?
4. Let $\mathcal{P}$ be the set of all subsets of $A = \{1, 2\}$. We choose two distinct sets randomly from $\mathcal{P}$. Define a sample space for this experiment, and describe the following events:
 - (a) The intersection of the sets chosen at random is empty.
 - (b) The sets are complements of each other.
 - (c) One of the sets contains more elements than the other.
5. In a tutoring center, there are three computers that can be up or down at any given time. For $1 \leq i \leq 3$, let E_i be the event that computer i is up at a random time. In terms of E_i 's, describe the event that at least two computers are up.
6. Aiden just bought a stock for \$320. Define a sample space for the price of this stock in 2 years. Define the event that he makes money selling this stock at that time.
7. In a certain experiment, whenever the event A occurs, the event B also occurs. Which of the following statements is true and why?
 - (a) If we know that A has not occurred, we can be sure that B has not occurred as well.
 - (b) If we know that B has not occurred, we can be sure that A has not occurred as well.
8. A department store accepts only its own credit card or an American Express card. Customers not carrying one of these two cards must pay with cash. If 47% of the customers of this store carry American Express, 32% carry the store's credit card, and 12% carry both, what percentage of the store customers have no choice but to pay with cash?
9. Kayla has two cars, an Audi and a Jeep. At a given time, whether these cars are operative or totaled, due to accidents, is of concern to her insurance company. Define a sample space for the driving conditions of these cars at a random time. What is the event that at least one of the two cars is operative at that random time? Assume that a car that is not totaled is operative.

10. For a saw blade manufacturer's products, the global demand, per month, for band saws is between 30 and 36 thousands; for reciprocating saws, it is between 28 and 33 thousands; for hole saws, it is between 300 and 600 thousands; and for hacksaws, it is between 500 and 650 thousands. Define a sample space for the demands for these four types of saws by this manufacturer in a random month. Describe the event that, in such a month, the demand for reciprocating saws is no more than 28.5 thousands and for hacksaws, it is at least 570 thousands. In this exercise, by "between two numbers a and b ," we mean inclusive of both a and b .
11. The following relations *are not* always true. In each case give an example to refute them.
- (a) $P(A \cup B) = P(A) + P(B)$.
- (b) $P(AB) = P(A)P(B)$.
12. A coin is tossed until, for the first time, the same result appears twice in succession. Define a sample space for this experiment.
13. In a midwest town, 80% of households have cable TV, 60% have an internet subscription, and 90% have at least one of these. What percentage of the households of this town have both cable TV and an internet subscription?
14. Let A and B be two events of an experiment with $P(A) = 1/3$ and $P(B) = 1/4$. Find the maximum value for $P(A \cup B)$.
15. The number of the patients now in a hospital is 63. Of these 37 are male and 20 are for surgery. If among those who are for surgery 12 are male, how many of the 63 patients are neither male nor for surgery?
16. Let A , B , and C be three events. Prove that

$$P(A \cup B \cup C) \leq P(A) + P(B) + P(C).$$

17. Let A , B , and C be three events. Show that

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

if and only if $P(AB) = P(AC) = P(BC) = 0$.

18. Suppose that 40% of the people in a community drink or serve white wine, 50% drink or serve red wine, and 70% drink or serve red or white wine. What percentage of the people in this community drink or serve both red and white wine?
19. Suppose that, in a temperate coniferous forest, 60% of randomly selected quarter-acre plots have cedar trees, 45% have cypress trees, 30% have redwoods, 40% have both cedar and cypress, 25% have cedar and redwoods, 20% have cypress and redwoods, and 80% of such plots have at least one of these trees. What is the probability that in a randomly selected quarter-acre plot in this forest we can find all three types of these trees?
20. Suppose that $P(E \cup F) = 0.75$ and $P(E \cup F^c) = 0.85$. Find $P(E)$.

21. Anthony, Bob, and Carl, three American race car drivers, will compete in a professional Trans-Am road race. Past records show that the probability that one of these former champions wins is $10/13$. If Anthony is twice as likely to win as Bob, and Bob is three times more likely than Carl to win the race, find the respective probabilities of these three people winning.
22. Five customers enter a wireless corporate store to buy smartphones. If the probability that at least two of them purchase an Android smartphone is 0.6 and the probability that all of them buy non-Android smartphones is 0.17, what is the probability that exactly one Android smartphone is purchased?
23. Answer the following question, asked of Marilyn Vos Savant in the “Ask Marilyn” column of *Parade Magazine*, March 3, 1996.

My dad heard this story on the radio. At Duke University, two students had received A's in chemistry all semester. But on the night before the final exam, they were partying in another state and didn't get back to Duke until it was over. Their excuse to the professor was that they had a flat tire, and they asked if they could take a make-up test. The professor agreed, wrote out a test and sent the two to separate rooms to take it. The first question (on one side of the paper) was worth 5 points, and they answered it easily. Then they flipped the paper over and found the second question, worth 95 points: ‘Which tire was it?’ What was the probability that both students would say the same thing? My dad and I think it's 1 in 16. Is that right?

24. Let A and B be two events. Suppose that $P(A)$, $P(B)$, and $P(AB)$ are given. What is the probability that neither A nor B will occur?
25. Let A and B be two events. The event $(A - B) \cup (B - A)$ is called the **symmetric difference** of A and B and is denoted by $A \Delta B$. Clearly, $A \Delta B$ is the event that exactly one of the two events A and B occurs. Show that

$$P(A \Delta B) = P(A) + P(B) - 2P(AB).$$

26. Let $S = \{\omega_1, \omega_2, \omega_3, \dots\}$ be the sample space of an experiment. Suppose that $P(\{\omega_1\}) = 1/8$ and, for a constant k , $0 < k < 1$, $P(\{\omega_{i+1}\}) = k \cdot P(\{\omega_i\})$ for $i \geq 1$. Find $P(\{\omega_i\})$, for $i > 1$.
27. A bookstore receives six boxes of books per month on six random days of each month. Suppose that two of those boxes are from one publisher, two from another publisher, and the remaining two from a third publisher. Define a sample space for the possible orders in which the boxes are received in a given month by the bookstore. Describe the event that the last two boxes of books received last month are from the same publisher.
28. Mildred, a commuter student studying at Western New England University, reported to his advisor that there are three traffic lights that she needs to pass while driving from home to school. She said that, based on her experience, 10% of the time all of the three traffic lights are green, 35% of the time exactly two of them are green, and 43% of the time exactly one of them is green. Assuming that Mildred's observations are correct,

what is the probability that on a random day she encounters (a) no green traffic lights; (b) at most one non-green (red or yellow) light; (c) at least two non-green lights?

29. Suppose that in a certain town the number of people with blood type O and blood type A are approximately the same. The number of people with blood type B is $1/10$ of those with blood type A and twice the number of those with blood type AB. Find the probability that the next baby born in the town has blood type AB.
30. A number is selected at random from the set of natural numbers $\{1, 2, 3, \dots, 1000\}$. What is the probability that it is not divisible by 4, 7, or 9?
31. A number is selected at random from the set $\{1, 2, 3, \dots, 150\}$. What is the probability that it is relatively prime to 150? See Exercise 34, [Section 1.4](#), for the definition of relatively prime numbers.
32. A point is selected randomly from the interval $(0, 2)$. For $n \geq 2$, let E_n be the event that it is in the interval $(0, \sqrt[n]{3})$. Determine the event $\bigcap_{n=2}^{\infty} E_n$.
33. Suppose that each day the price of a stock moves up $1/8$ of a point, moves down $1/8$ of a point, or remains unchanged. For $i \geq 1$, let U_i and D_i be the events that the price of the stock moves up and down on the i th trading day, respectively. In terms of U_i 's and D_i 's, find an expression for the event that the price of the stock
 - (a) remains unchanged on the i th trading day;
 - (b) moves up every day of the next n trading days;
 - (c) remains unchanged on at least one of the next n trading days;
 - (d) is the same as today after three trading days;
 - (e) does not move down on any of the next n trading days.
34. A bus traveling from Baltimore to New York has breaks down at a random location. What is the probability that the breakdown occurred after passing through Philadelphia? The distances from New York and Philadelphia to Baltimore are, respectively, 199 and 96 miles.
35. The coefficient of the quadratic equation $ax^2 + bx + c = 0$ are determined by tossing a fair die three times (the first outcome is a , the second one b , and the third one c). Find the probability that the equation has no real roots.



Self-Test on Chapter 1

Time allotted: 120 Minutes

Each problem is worth 10 points.

1. To determine who pays for dinner, Crispin, Allison, and Terry each flips an unbiased coin. The one whose flip has a different face up will pay. If all of the flips land on

the same face, they start all over again. (a) What is the probability that Crispin ends up paying for dinner; (b) what is the probability that no more than one round of flips is necessary?

2. Every day, a major ball manufacturer produces at least 800, but no more than 1300 of each of its five products: baseballs, tennis balls, softballs, basketballs, and soccer balls. Define a sample space for the production levels of these five types of balls that this manufacturer manufactures on a random working day. Describe the event that, on this day, the difference between the numbers of baseballs and soccer balls produced will not exceed 100.

Hint: Let x_1, x_2, x_3, x_4 , and x_5 be the production level, in hundreds, for baseballs, tennis balls, softballs, basketballs, and soccer balls, respectively.

3. A device that has three components fails if at least one of its components breaks down. The device is observed at a random time. Let A_i , $1 \leq i \leq 3$, denote the outcome that the i th component is operative at the random time. In terms of A_i 's, (a) define a sample space for the status of the system at the random time; (b) describe the event that the device is not operative at that random time.
4. Suppose that, in a particular geographical area, of people aged 55 and older, 13.4% suffer from Parkinson's disease, 11.3% suffer from Alzheimer's disease, and 2.26% suffer from both of these completely separate neurodegenerative illnesses. What percentage of the people from this age group in this geographic area have neither Parkinson's disease nor Alzheimer's disease?
5. For an experiment, E and F are two events with $P(E) = 0.4$ and $P(E^c F^c) = 0.35$. Calculate $P(E^c F)$.
6. Last semester, all first-year students of a college took calculus, biology, and English. If 18% of them received an A in calculus, 10% received an A in both calculus and biology, 13% received an A in both English and calculus, and 7% received an A in all three courses, what is the probability that a randomly selected first-year student from this college, last semester, received an A in calculus, but not in biology and not in English?
7. For an experiment with sample space $S = (0, 2)$, for $n \geq 1$, let $E_n = (1 - 1/n, 1 + 1/n)$ and $P(E_n) = (2n + 1)/3n$. For this experiment, find the probability that the event $\{1\}$ occurs. Note that this is *not* the experiment of choosing a point at random from the interval $(0, 2)$, as defined in [Section 1.7](#).
8. Five customers enter a wireless corporate store to purchase smartphones. If the probability that at least three of them purchase an Android smartphone is 0.54, what is the probability that at most two of them buy such a phone?
Hint: For $0 \leq i \leq 5$, let A_i be the event that exactly i of these customers buy an Android smartphone. Note that $\sum_{i=0}^5 P(A_i) = 1$.
9. In some towns of a country, the harmful substances lead and asbestos fibers find their way into drinking water. In a study, it was found that, in 13% of those towns, the drinking water supplies have neither lead nor asbestos fibers, in 32% of them the drinking water supplies have lead, and in 43% the drinking water supplies have asbestos fibers. In what percentage of the towns are the drinking water supplies contaminated with exactly one of these two impurities?

10. Suppose that on a certain week, only one-third of the travelers visiting Paris took a trip to the Eiffel Tower's top, one-half visited the Louvre Museum, and one-third took a tour of Notre Dame Cathedral. If none of the tourists visited all three sites, and for each pair of the sites, one-fourth of the travelers visited that pair, find the fraction of this particular group of tourists who did not visit any of the three sites.





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Chapter 2

Combinatorial Methods

2.1 INTRODUCTION

The study of probability includes many applications, such as games of chance, occupancy and order problems, and sampling procedures. In some of such applications, we deal with finite sample spaces in which all sample points are equally likely to occur. Theorem 1.3 shows that, in such cases, the probability of an event A is evaluated simply by dividing the number of points of A by the total number of sample points. Therefore, some probability problems can be solved simply by counting the total number of sample points and the number of ways that an event can occur.

In this chapter we study a few rules that enable us to count systematically. A branch of mathematics, **combinatorial analysis**, deals with methods of counting: a very broad field with applications in virtually every branch of applied and pure mathematics. Besides probability and statistics, it is used in information theory, coding and decoding, linear programming, transportation problems, industrial planning, scheduling production, group theory, foundations of geometry, and other fields. Combinatorial analysis, as a formal branch of mathematics, began with Tartaglia in the sixteenth century. After Tartaglia, Pascal, Fermat, Chevalier Antoine de Méré (1607–1684), James Bernoulli, Gottfried Leibniz, and Leonhard Euler (1707–1783) made contributions to this field. The mathematical development of the twentieth century accelerated development by combinatorial analysis.

2.2 COUNTING PRINCIPLES

Suppose that there are n routes from town A to town B , and m routes from B to a third town, C . If we decide to go from A to C via B , then for each route that we choose from A to B , we have m choices from B to C . Therefore, altogether we have nm choices to go from A to C via B . This simple example motivates the following principle, which is the basis of this chapter.

Theorem 2.1 (Counting Principle) *If the set E contains n elements and the set F contains m elements, there are nm ways in which we can choose, first, an element of E and then an element of F .*

Proof: Let $E = \{a_1, a_2, \dots, a_n\}$ and $F = \{b_1, b_2, \dots, b_m\}$, then the following rectangular array, which consists of nm elements, contains all possible ways that we can choose, first, an

element of E and then an element of F .

$$\begin{array}{ccccccc} (a_1, b_1), & (a_1, b_2), & \dots, & (a_1, b_m) \\ (a_2, b_1), & (a_2, b_2), & \dots, & (a_2, b_m) \\ \vdots & & & \\ (a_n, b_1), & (a_n, b_2), & \dots, & (a_n, b_m) \end{array} \quad \blacklozenge$$

Now suppose that a fourth town, D , is connected to C by ℓ routes. If we decide to go from A to D , passing through C after B , then for each pair of routes that we choose from A to C , there are ℓ possibilities from C to D . Therefore, by the counting principle, the total number of ways we can go from A to D via B and C is the number of ways we can go from A to C through B times ℓ , that is, $nm\ell$. This concept motivates a generalization of the counting principle.

Theorem 2.2 (Generalized Counting Principle) *Let $E_1, E_2, \dots, E_k$ be sets with $n_1, n_2, \dots, n_k$ elements, respectively. Then there are $n_1 \times n_2 \times n_3 \times \dots \times n_k$ ways in which we can, first, choose an element of E_1 , then an element of E_2 , then an element of $E_3, \dots$, and finally an element of E_k .*

In probability, this theorem is used whenever we want to compute the total number of possible outcomes when k experiments are performed. Suppose that the first experiment has n_1 possible outcomes, the second experiment has n_2 possible outcomes, $\dots$, and the k th experiment has n_k possible outcomes. If we define E_i to be the set of all possible outcomes of the i th experiment, then the total number of possible outcomes coincides with the number of ways that we can, first, choose an element of E_1 , then an element of E_2 , then an element of $E_3, \dots$, and finally an element of E_k ; that is, $n_1 \times n_2 \times \dots \times n_k$.

Example 2.1 How many outcomes are there if we throw five dice?

Solution: Let $E_i, 1 \leq i \leq 5$, be the set of all possible outcomes of the i th die. Then $E_i = \{1, 2, 3, 4, 5, 6\}$. The number of the outcomes of throwing five dice equals the number of ways we can, first, choose an element of E_1 , then an element of $E_2, \dots$, and finally an element of E_5 . Thus we get $6 \times 6 \times 6 \times 6 \times 6 = 6^5$. $\blacklozenge$

Remark 2.1 Consider experiments such as flipping a fair coin several times, tossing a number of fair dice, drawing a number of cards from an ordinary deck of 52 cards at random and with replacement, and drawing a number of balls from an urn at random and with replacement. In [Section 3.5](#), discussing the concept of *independence*, we will show that all the possible outcomes in such experiments are equiprobable. Until then, however, in all the problems dealing with these kinds of experiments, we *assume*, without explicitly so stating in each case, that the sample points of the sample space of the experiment under consideration are all equally likely. $\blacklozenge$

Example 2.2 In tossing four fair dice, what is the probability of at least one 3?

Solution: Let A be the event of at least one 3. Then A^c is the event of no 3 in tossing the four dice. $N(A^c)$ and N , the number of sample points of A^c and the total number of sample points,

respectively, are given by $5 \times 5 \times 5 \times 5 = 5^4$ and $6 \times 6 \times 6 \times 6 = 6^4$. Therefore, $P(A^c) = N(A^c)/N = 5^4/6^4$. Hence $P(A) = 1 - P(A^c) = 1 - 625/1296 = 671/1296 \approx 0.52$. ♦

Example 2.3 Virginia wants to give her son, Brian, 14 different baseball cards within a 7-day period. If Virginia gives Brian cards no more than once a day, in how many ways can this be done?

Solution: Each of the baseball cards can be given on 7 different days. Therefore, in $7 \times 7 \times \cdots \times 7 = 7^{14} \approx 6.78 \times 10^{11}$ ways Virginia can give the cards to Brian. ♦

In experiments such as drawing cards from an ordinary deck of 52 cards or drawing balls from an urn, if after each draw the card or the ball is put into the deck or into the urn, respectively, we say that cards or balls are drawn **with replacement**. If they are not returned into the deck or into the urn, we say that they are drawn **without replacement**.

Example 2.4 A box contains 7 identical balls numbered 1 through 7. Three balls are drawn, one by one, at random, and *with replacement*, and their numbers are recorded. What is the probability that (a) all three outcomes are odd; (b) exactly one outcome is odd?

Solution: (a) Since the draws are with replacement, there are 7 possibilities for each draw. Thus the sample space has $7 \times 7 \times 7 = 343$ points. In $4 \times 4 \times 4 = 64$ ways, all three outcomes are odd. That is, each ball drawn has one of the numbers 1, 3, 5, or 7. So the probability that all three outcomes are odd is $64/343 \approx 0.187$.

(b) In $4 \times 3 \times 3 = 36$ ways, the first outcome is odd and the second and third outcomes are even (2, 4, or 6). In $3 \times 4 \times 3 = 36$ ways, the first and third outcomes are even, but the second outcome is odd. Similarly, in $3 \times 3 \times 4 = 36$ ways, the first two outcomes are even and the third one is odd. So the probability that exactly one outcome is odd is $(3 \times 36)/343 \approx 0.315$. ♦

Example 2.5 A box contains 7 identical balls numbered 1 through 7. Three balls are drawn, one by one, at random, and *without replacement*, and their numbers are recorded. What is the probability that (a) all three outcomes are odd; (b) exactly one outcome is odd?

Solution: (a) Since the draws are without replacement, there are 7 possibilities for the first draw, 6 possibilities for the second draw, and 5 possibilities for the third draw. Thus the sample space has $7 \times 6 \times 5 = 210$ points. Clearly, in $4 \times 3 \times 2 = 24$ ways, all three outcomes are odd. That is, each ball drawn has one of the numbers 1, 3, 5, or 7. So the probability that all three outcomes are odd is $24/210 \approx 0.114$.

(b) In $4 \times 3 \times 2 = 24$ ways, the first outcome is odd, and the second and third outcomes are even. In $3 \times 4 \times 2 = 24$ ways, the first and third outcomes are even, but the second outcome is odd. Similarly, in $3 \times 2 \times 4 = 24$ ways, the first two outcomes are even and the third one is odd. So the probability that exactly one outcome is odd is $(3 \times 24)/210 \approx 0.34$. ♦

Example 2.6 Rose has invited n friends to her birthday party. If they all attend, and each one shakes hands with everyone else at the party exactly once, what is the number of handshakes?

Solution 1: There are $n + 1$ people at the party and each of them shakes hands with the other n people. This is a total of $(n + 1)n$ handshakes, but that is an overcount since it counts “A shakes hands with B” as one handshake and “B shakes hand with A” as a second. Since each handshake is counted exactly twice, the actual number of handshakes is $(n + 1)n/2$.

Solution 2: Suppose that guests arrive one at a time. Rose will shake hands with all the n guests. The first guest to appear will shake hands with Rose and all the remaining guests. Since we have already counted his or her handshake with Rose, there will be $n - 1$ additional handshakes. The second guest will also shake hands with $n - 1$ fellow guests and Rose. However, we have already counted his or her handshakes with Rose and the first guest. So this will add $n - 2$ additional handshakes. Similarly, the third guest will add $n - 3$ additional handshakes, and so on. Therefore, the total number of handshakes will be $n + (n - 1) + (n - 2) + \cdots + 3 + 2 + 1$. Comparing solutions 1 and 2, we have the well-known relation

$$1 + 2 + 3 \cdots + (n - 2) + (n - 1) + n = \frac{n(n + 1)}{2}. \quad \blacklozenge$$

Example 2.7 At a state university in Maryland, there is hardly enough space for students to park their cars in their own lots. Jack, a student who parks in the faculty parking lot every day, noticed that none of the last 10 tickets he got was issued on a Monday or on a Friday. Is it wise for Jack to conclude that the campus police do not patrol the faculty parking lot on Mondays and on Fridays? Assume that police give no tickets on weekends.

Solution: Suppose that the answer is negative and the campus police patrol the parking lot randomly; that is, the parking lot is patrolled every day with the same probability. Let A be the event that out of 10 tickets given on random days, none is issued on a Monday or on a Friday. If $P(A)$ is very small, we can conclude that the campus police do not patrol the parking lot on these two days. Otherwise, we conclude that what happened is accidental and police patrol the parking lot randomly. To find $P(A)$, note that since each ticket has five possible days of being issued, there are 5^{10} possible ways for all tickets to have been issued. Of these, in only 3^{10} ways no ticket is issued on a Monday or on a Friday. Thus $P(A) = 3^{10}/5^{10} \approx 0.006$, a rather small probability. Therefore, it is reasonable to assume that the campus police do not patrol the parking lot on these two days. $\blacklozenge$

Example 2.8 (Standard Birthday Problem) What is the probability that at least two students of a class of size n have the same birthday? Compute the numerical values of such probabilities for $n = 23, 30, 50$, and 60 . Assume that the birth rates are constant throughout the year and that each year has 365 days.

Solution: There are 365 possibilities for the birthdays of each of the n students. Therefore, the sample space has 365^n points. In $365 \times 364 \times 363 \times \cdots \times [365 - (n - 1)]$ ways the birthdays of no two of the n students coincide. Hence $P(n)$, the probability that no two students have the same birthday, is

$$P(n) = \frac{365 \times 364 \times 363 \times \cdots \times [365 - (n - 1)]}{365^n},$$

and therefore the desired probability is $1 - P(n)$. For $n = 23, 30, 50$, and 60 the answers are $0.507, 0.706, 0.970$, and 0.995 , respectively. ♦

Example 2.9 Terra was born on July 4. Assuming that the birth rates are constant throughout the year and each year has 365 days, for Terra to have a 50% chance of meeting at least one person with her birthday, how many random people does she need to meet?

Solution: Let n be the minimum number of people that Terra needs to meet to have a 50% chance of meeting at least one person with her birthday. There are 365 possibilities for the birthdays of each of the n random people she needs to meet. Therefore, the sample space has 365^n points. Of these, in 364^n cases none of those people was born on July 4. So the probability that none shares a birthday with Terra is $364^n/365^n$. Thus the probability is $1 - (364^n/365^n)$ that at least one of the n people was born on July 4. We must have

$$1 - \frac{364^n}{365^n} \geq 0.5,$$

or

$$\left(\frac{364}{365}\right)^n \leq 0.5.$$

This gives $n \ln(364/365) \leq \ln(0.5)$, which implies that $n \geq 252.65$. So Terra needs to meet 253 random people to have a 50% chance of meeting at least one person with her birthday, July 4. ♦

Remark 2.2 In probability and statistics studies, birthday problems similar to Examples 2.8 and 2.9 have been very popular since 1939, when introduced by von Mises. This is probably because when solving such problems, numerical values obtained are often surprising. Persi Diaconis and Frederick Mosteller, two Harvard professors, have mentioned that they “find the utility of birthday problems impressive as a tool for thinking about coincidences.” Diaconis and Mosteller have illustrated basic statistical techniques for studying the fascinating, curious, and complicated “**Theory of Coincidences**” in the December 1989 issue of the *Journal of the American Statistical Association*. In their study, they have used birthday problems “as examples which make the point that in *many problems our intuitive grasp of the odds is far off*.” Throughout this book, where appropriate, we will present some interesting versions of these problems, but now that we have cited “coincidence,” let us examine this concept, just a little, in conjunction with the birthday problems.

To be certain that, among a group, two individuals share the same birthday, by the pigeon-hole principle, we need 367 people in a room since, including February 29, there are exactly 366 birthdays. To have a 50% chance of meeting someone with *your* own birthday, you need to meet with 253 random people (see Example 2.9). These and similar facts might mislead our intuition and astonish us with birthday coincidences such as the ones that require only 23 random people in a room for having a chance exceeding 50%, and 50 random people for a chance exceeding 97% that at least two individuals share a birthday.

Note that the probability that among, say, 23 or 50 random people at least one has *your* birthday is very low, 0.061 and 0.128, respectively. However, the probability that among a group of 23 random people or among a group of 50 random people, at least *two individuals* share a birthday exceeds 0.50 and 0.97, respectively. Similarly, if you purchase a lottery ticket,

the probability that *you* win the jackpot is negligible (see Example 2.22). But the probability that, in a lottery, *someone* wins the jackpot is high. Such observations show that the probability of coincidences when events are defined more broadly or more generically can be quite high. Clearly, billions of events occur everyday all over the world, and there are billions of ways for possible coincidences. No one can ever list all the possibilities, but inevitably some of them will occur, and when they do, we recognize them, and they confound our intuition. We find them astonishing because before they occur, their chances of occurrence are extremely small, one in billions. To elaborate this further, let us examine a simple case. Consider a boy who was born in Khulna, Bangladesh on February 2 of a given year, and a girl who was born on July 9 of the same year in Fort Wayne, Indiana. What are the chances for these infants to grow up both becoming interested in mathematics, both continue their education in math to the highest level and each earn a Ph.D., and then both end up teaching at a specific university in Massachusetts in the same department? Obviously, the probability of such a sequence of coincidences is negligible. However, two professors who were born in those towns on February 2 and July 9 of 1969, in fact, both ended up as professors of mathematics at the same university in Massachusetts. At American universities, such events, which have negligible probabilities, occur all the time. No surprises there since we actually know that each person's life is formed by an extremely long series of events, each with an infinitesimal probability. We end this remark with a few words by Diaconis and Mosteller in the abstract of their aforementioned paper.

... when enormous numbers of events and people and their interactions cumulate over time, almost any outrageous event is bound to occur. These sources account for much of the force of synchronicity. ♦

Number of Subsets of a Set

Let A be a set. The set of all subsets of A is called the **power set** of A . As an important application of the generalized counting principle, we now prove that the power set of a set with n elements has 2^n elements. This important fact has lots of good applications.

Theorem 2.3 *A set with n elements has 2^n subsets.*

Proof: Let $A = \{a_1, a_2, a_3, \dots, a_n\}$ be a set with n elements. Then there is a one-to-one correspondence between the subsets of A and the sequences of 0s and 1s of length n : To a subset B of A we associate a sequence $b_1b_2b_3 \cdots b_n$, where $b_i = 0$ if $a_i \notin B$, and $b_i = 1$ if $a_i \in B$. For example, if $n = 3$, we associate to the empty subset of A the sequence 000, to $\{a_2, a_3\}$ the sequence 011, and to $\{a_1\}$ the sequence 100. Now, by the generalized counting principle, the number of sequences of 0s and 1s of length n is $2 \times 2 \times 2 \times \cdots \times 2 = 2^n$. Thus the number of subsets of A is also 2^n . ♦

Example 2.10 A restaurant advertises that it offers over 1000 varieties of pizza. If, at the restaurant, it is possible to have on a pizza any combination of pepperoni, mushrooms, sausage, green peppers, onions, anchovies, salami, bacon, olives, and ground beef, is the restaurant's advertisement true?

Solution: Any combination of the 10 ingredients that the restaurant offers can be put on a pizza. Thus the number of different types of pizza that it is possible to make is equal to the number of subsets of the set {pepperoni, mushrooms, sausage, green peppers, onions, anchovies, salami, bacon, olives, ground beef}, which is $2^{10} = 1024$. Therefore, the restaurant's advertisement is true. Note that the empty subset of the set of ingredients corresponds to a plain cheese pizza. ♦

Tree Diagrams

We used the tree diagram in Example 1.19, even though we didn't explicitly mention it. Tree diagrams are useful pictorial representations that break down a complex counting problem into smaller, more tractable ones. They are used in situations where the number of possible ways an experiment can be performed is finite. The following examples illustrate how tree diagrams are constructed and why they are useful. A great advantage of tree diagrams is that they systematically identify all possible cases.

Example 2.11 Bill and John keep playing chess until one of them wins two games in a row or three games altogether. In what percent of all possible cases does the game end because Bill wins three games without winning two in a row?

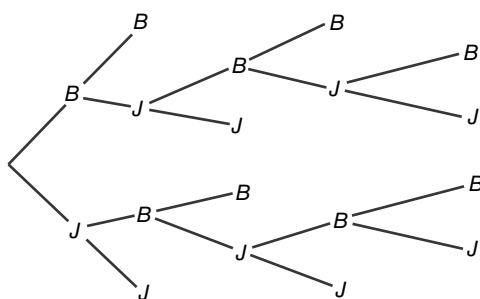


Figure 2.1 Tree diagram of Example 2.11.

Solution: The tree diagram of Figure 2.1 illustrates all possible cases. The total number of possible cases is equal to the number of the endpoints of the branches, which is 10. The number of cases in which Bill wins three games without winning two in a row, as seen from the figure, is one. So the answer is 10%. Note that the probability of this event is not 0.10 because not all of the branches of the tree are equiprobable. ♦

Example 2.12 Cheyenne has \$4. She decides to bet \$1 on the flip of a fair coin four times. What is the probability that (a) she breaks even; (b) she wins money?

Solution: The tree diagram shown in Figure 2.2 on the following page illustrates the different potential outcomes for Cheyenne. The diagram has 16 endpoints, showing that the sample space has 16 elements. In six of these 16 cases, Cheyenne breaks even and in five cases she wins money, so the desired probabilities are 6/16 and 5/16, respectively. ♦

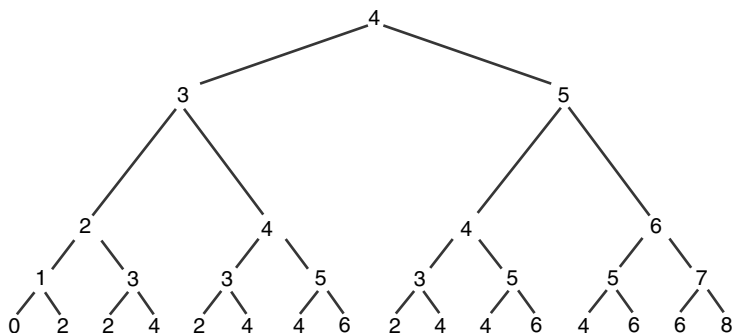


Figure 2.2 Tree diagram of Example 2.12.

EXERCISES

A

- How many six-digit numbers are there? How many of them contain the digit 5? Note that the first digit of an n -digit number is nonzero.
- How many different five-letter codes can be made using a , b , c , d , and e ? How many of them start with ab ?
- The population of a town is 20,000. If each resident has three initials, is it true that at least two people have the same initials?
- In how many different ways can 15 offices be painted with four different colors?
- In flipping a fair coin 23 times, what is the probability of all heads or all tails?
- In how many ways can we draw five cards from an ordinary deck of 52 cards (a) with replacement; (b) without replacement?
- In a word game, there are 6 tiles in a bag, each bearing one of the letters A, C, E, G, L, N. Zanya draws 8 tiles, one at a time, randomly, writes down the letter each time, and then returns the tile to the bag. What is the probability that, in the order drawn, the letters form the word ELEGANCE?
- Two fair dice are thrown. What is the probability that the outcome is a 6 and an odd number?
- From an ordinary deck of 52 cards, 4 cards are selected at random and placed on a table face down. A boy, who would like to be a magician, guesses the denomination and suit of all four cards randomly. What is the probability that he guesses at least one card correctly?

10. Mr. Smith has 12 shirts, eight pairs of slacks, eight ties, and four jackets. Suppose that four shirts, three pairs of slacks, two ties, and two jackets are blue. (a) What is the probability that an all-blue outfit is the result of a random selection? (b) What is the probability that he wears at least one blue item tomorrow?
11. A multiple-choice test has 15 questions, each having four possible answers, of which only one is correct. If the questions are answered at random, what is the probability of getting all of them right?
12. Suppose that in a state, license plates have three letters followed by three numbers, in a way that no letter or number is repeated in a single plate. Determine the number of possible license plates for this state.
13. A library has 800,000 books, and the librarian wants to encode each by using a code word consisting of three letters followed by two numbers. Are there enough code words to encode all of these books with different code words?
14. How many $n \times m$ arrays (matrices) with entries 0 or 1 are there?
15. How many divisors does 55,125 have?
Hint: $55,125 = 3^2 5^3 7^2$.
16. A delicatessen has advertised that it offers over 500 varieties of sandwiches. If at this deli it is possible to have any combination of salami, turkey, bologna, corned beef, ham, and cheese on French bread with the possible additions of lettuce, tomato, and mayonnaise, is the deli's advertisement true? Assume that a sandwich necessarily has bread and at least one type of meat or cheese.
17. For an international traveler, the only Automated Teller Machine (ATM) personal identification numbers (PINs) acceptable are those that do not begin with 0 and those that do not consist of identical digits. If PINs allowed can have 4, 5, or 6 digits, what is the probability that a 4, 5, or 6 digit number selected at random can be used as a PIN?
18. How many four-digit numbers can be formed by using only the digits 2, 4, 6, 8, and 9? How many of these have some digit repeated?
19. In a mental health clinic there are 12 patients. A therapist invites all these patients to join her for group therapy. How many possible groups could she get?
20. Suppose that four cards are drawn successively from an ordinary deck of 52 cards, with replacement and at random. What is the probability of drawing at least one king?
21. A campus telephone extension has four digits. How many different extensions with no repeated digits exist? Of these, (a) how many do not start with a 0; (b) how many do not have 01 as the first two digits?
22. There are N types of drugs sold to reduce acid indigestion. A random sample of n drugs is taken with replacement. What is the probability that brand A is included?
23. From an ordinary deck of 52 cards, 15 cards are drawn randomly and without replacement. What is the probability that the first four cards are all clubs?
24. A fair die is tossed eight times. What is the probability that the eighth outcome is not a repetition?

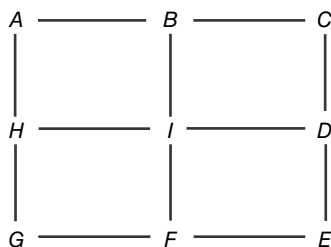


Figure 2.3 Islands and connecting bridges of Exercise 25.

25. A salesperson covers islands A, B, . . . , I. These islands are connected by the bridges shown in the Figure 2.3. While on an island, the salesperson takes one of the possible bridges at random and goes to another one. She does her business on this new island and then takes a bridge at random to go to the next one. She continues this until she reaches an island for the second time on the same day. She stays there overnight and then continues her trips the next day. If she starts her trip from island I tomorrow, in what percent of all possible trips will she end up staying overnight again at island I?
26. In North America, in the World Series, which is the most important baseball event each year, the American League champion team plays with the National League champion team a series of games. The first team to win four games will be the winner of the World Series championship and is awarded the Commissioner's Trophy. Assuming that it is equally likely for the teams to win, find the probability that the series (a) lasts 4 games; (b) lasts 5 games.
27. To log into their computer accounts, Alec and Mildred must create passwords that begin with a letter followed by 5 to 7 letters or numbers. If Mildred creates a password herself, but Alec uses a random password creation tool, what is the probability that they end up with the same password?

B

28. In a large town, Kennedy Avenue is a long north-south avenue with many intersections. A drunken man is wandering along the avenue and does not really know which way he is going. He is currently at an intersection O somewhere in the middle of the avenue. Suppose that, at the end of each block, he either goes north with probability $1/2$, or he goes south with probability $1/2$. Draw a tree diagram to find the probability that, after walking four blocks, (a) he is back at intersection O ; (b) he is only one block away from intersection O .
29. An integer is selected at random from the set $\{1, 2, \dots, 1000000\}$. What is the probability that it contains the digit 5?
30. How many divisors does a natural number N have?
Hint: A natural number N can be written as $p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$, where $p_1, p_2, \dots, p_k$ are distinct primes.
31. In tossing four fair dice, what is the probability of tossing, at most, one 3?

32. A delicatessen advertises that it offers over 3000 varieties of sandwiches. If at this deli it is possible to have any combination of salami, turkey, bologna, corned beef, and ham with or without Swiss and/or American cheese on French, white, or whole wheat bread, and possible additions of lettuce, tomato, and mayonnaise, is the deli's advertisement true? Assume that a sandwich necessarily has bread and at least one type of meat or cheese.
33. One of the five elevators in a building leaves the basement with eight passengers and stops at all of the remaining 11 floors. If it is equally likely that a passenger gets off at any of these 11 floors, what is the probability that no two of these eight passengers will get off at the same floor?
34. The elevator of a four-floor building leaves the first floor with six passengers and stops at all of the remaining three floors. If it is equally likely that a passenger gets off at any of these three floors, what is the probability that, at each stop of the elevator, at least one passenger departs?
35. A number is selected randomly from the set $\{0000, 0001, 0002, \dots, 9999\}$. What is the probability that the sum of the first two digits of the number selected is equal to the sum of its last two digits?
36. What is the probability that a random r -digit number ($r \geq 3$) contains at least one 0, at least one 1, and at least one 2?



Self-Quiz on Section 2.2

Time allotted: 20 Minutes

Each problem is worth 2.5 points.

1. In 2010, three unrelated faculty members, Professors Rodriguez, Bucs, and Beineke ended up sitting on three adjacent seats in a row at the spring commencement ceremony of a university, where faculty occupied the chairs randomly. These professors somehow got into a discussion about birthday problems, and they were astonished when they learned that all three of them had the same birthday. Calculate the probability of such a coincidence. Assume that the birth rates are constant throughout the year and that each year has 365 days.
2. The chair of the industry-academic partnership of a town invites all 12 members of the board and their spouses to his house for a Christmas party. If a board member may attend without his spouse, but not vice versa, how many different groups can the chair get?
3. In a pharmacy, there are 8 unrelated people standing in line to pick up their prescriptions. Before the pharmacist gives the patients their prescription medications, following the pharmacy's policy, she asks them the month and day of their birthdays. What is the probability that at least two of the last three of these 8 people have the same birthday? Assume that the birth rates are constant throughout the year and that each year has 365 days.

4. Cyrus and 27 other students are taking a course in probability this semester. If their professor chooses eight students at random and with replacement to ask them eight different questions, what is the probability that one of them is Cyrus?



2.3 PERMUTATIONS

To count the number of outcomes of an experiment or the number of possible ways an event can occur, it is often useful to look for special patterns. Sometimes patterns help us develop techniques for counting. Two simple cases in which patterns enable us to count easily are **permutations** and **combinations**. We study these two patterns in this section and the next.

Definition 2.1 *An ordered arrangement of r objects from a set A containing n objects ($0 < r \leq n$) is called an **r -element permutation** of A , or a permutation of the elements of A taken r at a time. The number of r -element permutations of a set containing n objects is denoted by ${}_nP_r$.*

By this definition, if three people, Brown, Smith, and Jones, are to be scheduled for job interviews, any possible order for the interviews is a three-element permutation of the set {Brown, Smith, Jones}.

If, for example, $A = \{a, b, c, d\}$, then ab is a two-element permutation of A , acd is a three-element permutation of A , and $adcb$ is a four-element permutation of A . The order in which objects are arranged is important. For example, ab and ba are considered different two-element permutations, abc and cba are distinct three-element permutations, and $abcd$ and $cbad$ are different four-element permutations.

To compute ${}_nP_r$, the number of permutations of a set A containing n elements taken r at a time ($1 \leq r \leq n$), we use the generalized counting principle: Since A has n elements, the number of choices for the first object in the r -element permutation is n . For the second object, the number of choices is the remaining $n - 1$ elements of A . For the third one, the number of choices is the remaining $n - 2$, $\dots$, and, finally, for the r th object the number of choices is $n - (r - 1) = n - r + 1$. Hence

$${}_nP_r = n(n - 1)(n - 2) \cdots (n - r + 1). \quad (2.1)$$

An n -element permutation of a set with n objects is simply called a **permutation**. The number of permutations of a set containing n elements, ${}_nP_n$, is evaluated from (2.1) by putting $r = n$.

$${}_nP_n = n(n - 1)(n - 2) \cdots (n - n + 1) = n!. \quad (2.2)$$

The formula $n!$ (the number of permutations of a set of n objects) has been well known for a long time. Although it first appeared in the works of Persian and Arab mathematicians in the twelfth century, there are indications that the mathematicians of India were aware of this rule a few hundred years before Christ. However, the surprise notation $!$ used for “factorial” was introduced by Christian Kramp in 1808. He chose this symbol perhaps because $n!$ gets *surprisingly* large even for small numbers. For example, we have that $18! \approx 6.402373 \times 10^{15}$,

a number that, according to Karl Smith,[†] is greater than six times “the number of all words ever printed.”

The Inconceivable Enormity of 52! Comparing with the number of all words ever printed, we immediately realize that the number of ways an ordinary deck of 52 cards can be arranged, $52! \approx 8.07 \times 10^{67}$, is beyond comprehension or imagination. The vastness of this number dwarfs even astronomical figures. To illustrate its enormity, let us start with some familiar comparisons. The time light takes to travel from the Moon to the Earth is about 1.28 seconds, whereas the distance from the Earth to the Sun is covered in approximately 8 minutes and 19 seconds at the speed of light.

Now, consider this: the number of arrangements of a deck of cards is roughly 1000 times the estimated volume of the Milky Way in *cubic inches*, which is approximately 9.76×10^{64} . Note that the volume of the Milky Way is often approximated by the volume of a disk with a 100,000 light-year diameter and a thickness of 1000 light years.

To provide a more tactile perspective, if we were to represent this using dice, and assume each side of an ordinary die is half an inch, then the Milky Way would be able to accommodate approximately $9.76 \times 10^{64} / (1/8) \approx 7.8 \times 10^{65}$ regular dice. Let us further imagine a fictional galaxy we will call *Megantic*. This hypothetical galaxy has a volume that is 103.46 times greater than the Milky Way. Fittingly, it would hold around $103.46 \times 7.8 \times 10^{65} \approx 8.07 \times 10^{67} \approx 52!$ ordinary dice.

Envision the vastness of the Megantic galaxy being filled with identical dice, all white, save for a single red die. The odds of randomly drawing that one red die from this immense galaxy parallels the probability of shuffling a standard deck of 52 cards and achieving a specific pre-determined arrangement. This analogy underscores a compelling point: the likelihood of randomly shuffling a deck of 52 cards and reproducing an arrangement that has ever existed in history is virtually zero. ♦

There is a popular alternative for relation (2.1). It is obtained by multiplying both sides of (2.1) by $(n - r)! = (n - r)(n - r - 1) \cdots 3 \cdot 2 \cdot 1$. We get

$$\begin{aligned} {}_nP_r \cdot (n - r)! \\ = [n(n - 1)(n - 2) \cdots (n - r + 1)] \cdot [(n - r)(n - r - 1) \cdots 3 \cdot 2 \cdot 1]. \end{aligned}$$

This gives ${}_nP_r \cdot (n - r)! = n!$. Therefore,

The number of r -element permutations of a set containing n objects is given by

$${}_nP_r = \frac{n!}{(n - r)!}. \quad (2.3)$$

Note that for $r = n$, this relation implies that ${}_nP_n = n!/0!$. But by (2.2), ${}_nP_n = n!$. Therefore, for $r = n$, to make (2.3) consistent with (2.2), we define $0! = 1$.

Example 2.13 Three people, Brown, Smith, and Jones, must be scheduled for job interviews. In how many different orders can this be done?

[†]Karl J. Smith, *The Nature of Mathematics*, 12th ed., Brooks/Cole, Belmont, Calif., 2012, p. 39.

Solution: The number of different orders is equal to the number of permutations of the set {Brown, Smith, Jones}. So there are $3! = 6$ possible orders for the interviews. ♦

Example 2.14 Suppose that two anthropology, four computer science, three statistics, three biology, and five music books are put on a bookshelf with a random arrangement. What is the probability that the books of the same subject are together?

Solution: Let A be the event that all the books of the same subject are together. Then $P(A) = N(A)/N$, where $N(A)$ is the number of arrangements in which the books dealing with the same subject are together and N is the total number of possible arrangements. Since there are 17 books and each of their arrangements is a permutation of the set of these books, $N = 17!$. To calculate $N(A)$, note that there are $2! \times 4! \times 3! \times 3! \times 5!$ arrangements in which anthropology books are first, computer science books are next, then statistics books, after that biology, and finally, music. Also, there are the same number of arrangements for each possible ordering of the subjects. Since the subjects can be ordered in $5!$ ways, $N(A) = 5! \times 2! \times 4! \times 3! \times 3! \times 5!$. Hence

$$P(A) = \frac{5! \times 2! \times 4! \times 3! \times 3! \times 5!}{17!} \approx 6.996 \times 10^{-8}. \quad \blacklozenge$$

Example 2.15 If five boys and five girls sit in a row in a random order, what is the probability that no two children of the same sex sit together?

Solution: There are $10!$ ways for 10 persons to sit in a row. In order that no two of the same sex sit together, boys must occupy positions 1, 3, 5, 7, 9, and girls positions 2, 4, 6, 8, 10, or vice versa. In each case there are $5! \times 5!$ possibilities. Therefore, the desired probability is equal to

$$\frac{2 \times 5! \times 5!}{10!} \approx 0.008. \quad \blacklozenge$$

We showed that the number of permutations of a set of n objects is $n!$. This formula is valid only if all of the objects of the set are distinguishable from each other. Otherwise, the number of permutations is different. For example, there are $8!$ permutations of the eight letters *STANFORD* because all of these letters are distinguishable from each other. But the number of permutations of the 8 letters *BERKELEY* is less than $8!$ since the second, the fifth, and the seventh letters in *BERKELEY* are indistinguishable. If in any permutation of these letters we change the positions of these three indistinguishable E 's with each other, no new permutations would be generated. Let us calculate the number of permutations of the letters *BERKELEY*. Suppose that there are x such permutations and we consider any particular one of them, say *BYERELEK*. If we label the E 's: $BYE_1RE_2LE_3K$ so that all of the letters are distinguishable, then by arranging E 's among themselves we get $3!$ new permutations, namely,

$$\begin{array}{ll} BYE_1RE_2LE_3K & BYE_2RE_3LE_1K \\ BYE_1RE_3LE_2K & BYE_3RE_1LE_2K \\ BYE_2RE_1LE_3K & BYE_3RE_2LE_1K \end{array}$$

which are otherwise all the same. Therefore, if all the letters were different, then for each one of the x permutations we would have $3!$ times as many. That is, the total number of permutations would have been $x \times 3!$. But since eight different letters generate exactly $8!$ permutations, we must have $x \times 3! = 8!$. This gives $x = 8!/3!$. We have shown that *the number of distinguishable permutations of the letters BERKELEY is $8!/3!$* . This sort of reasoning leads us to the following general theorem.

Theorem 2.4 *The number of distinguishable permutations of n objects of k different types, where n_1 are alike, n_2 are alike, $\dots$, n_k are alike and $n = n_1 + n_2 + \dots + n_k$, is*

$$\frac{n!}{n_1! \times n_2! \times \dots \times n_k!}.$$

Example 2.16 How many different 10-letter codes can be made using three a's, four b's, and three c's?

Solution: By Theorem 2.4, the number of such codes is $10!/(3! \times 4! \times 3!) = 4200$. ♦

Example 2.17 In how many ways can we paint 11 offices so that four of them will be painted green, three yellow, two white, and the remaining two pink?

Solution: Let “ggypgwpygwy” represent the situation in which the first office is painted green, the second office is painted green, the third one yellow, and so on, with similar representations for other cases. Then the answer is equal to the number of distinguishable permutations of “gggggyyywwpp,” which by Theorem 2.4 is $11!/(4! \times 3! \times 2! \times 2!) = 69,300$. ♦

Example 2.18 A fair coin is flipped 10 times. What is the probability of obtaining exactly three heads?

Solution: The set of all sequences of H (heads) and T (tails) of length 10 forms the sample space and contains 2^{10} elements. Of all these, those with three H, and seven T, are desirable. But the number of distinguishable sequences with three H's and seven T's is equal to

$$\frac{10!}{3! \times 7!}.$$

Therefore, the probability of exactly three heads is

$$\left(\frac{10!}{3! \times 7!}\right)/2^{10} \approx 0.12. \quad \blacklozenge$$

EXERCISES

A

1. In the popular TV show *Who Wants to Be a Millionaire*, contestants are asked to sort four items in accordance with some norm: for example, landmarks in geographical order, movies in the order of date of release, singers in the order of date of birth. What is the probability that a contestant can get the correct answer solely by guessing?
2. In New York City, a subway train arrives at the platform, and 12 people enter a subway car that has only 6 empty seats. In how many ways can these passengers be seated?
3. How many permutations of the set $\{a, b, c, d, e\}$ begin with a and end with e ?
4. How many different messages can be sent by five dashes and three dots?
5. Cynthia's husband is away Monday through Friday, and she does not know how to cook. If every evening she dines at one of her 8 favorite restaurants randomly, and during that period she dines at a restaurant at most once, what is the probability that, one evening she will eat at Wilbraham Pizzeria, one of her favorites?
6. A three-year old child, Sheridan, has 8 square tiles, each bearing one of the letters in her name. If she arranges the tiles randomly in a row all facing in the correct direction, what is the probability that they show her name?
7. Robert has eight guests, two of whom are Jim and John. If the guests will arrive in a random order, what is the probability that John will not arrive right after Jim?
8. Let A be the set of all sequences of 0s, 1s, and 2s of length 12.
 - (a) How many elements are there in A ?
 - (b) How many elements of A have exactly six 0s and six 1s?
 - (c) How many elements of A have exactly three 0s, four 1s, and five 2s?
9. Professor Haste is somewhat familiar with six languages. To translate texts from one language into another directly, how many one-way dictionaries does he need?
10. In an exhibition, 20 cars of the same style that are distinguishable only by their colors, are to be parked in a row, all facing a certain window. If four of the cars are blue, three are black, five are yellow, and eight are white, how many choices are there?
11. There are 25 hard metal bunk beds in an open bay barracks at a military base. In how many ways can newly recruited soldiers be assigned to the 50 beds in the barracks? If the sergeant divides the soldiers into two groups of 25 each and asks one group to occupy the lower bunks and the other the upper bunks of the bunk beds, in how many ways can this be done?
12. At various yard sales, a woman has acquired five forks, of which no two are alike. The same applies to her four knives and seven spoons. In how many different ways can

three place settings be chosen if each place setting consists of exactly one fork, one knife, and one spoon? Assume that the arrangement of the place settings on the table is unimportant.

13. In a conference, Dr. Richman's lecture is related to Dr. Chollet's and should not precede it. If there are six more speakers, how many schedules could be arranged?
Warning: Dr. Richman's lecture is not necessarily scheduled right after Dr. Chollet's lecture.
14. A dancing contest has 11 competitors, of whom three are Americans, two are Mexicans, three are Russians, and three are Italians. If the contest result lists only the nationality of the dancers, how many outcomes are possible?
15. Six fair dice are tossed. What is the probability that at least two of them show the same face?
16. (a) Find the number of distinguishable permutations of the letters *MISSISSIPPI*. (b) In how many of these permutations *P*'s are together? (c) In how many *I*'s are together? (d) In how many *P*'s are together, and *I*'s are together? (e) In a random order of the letters *MISSISSIPPI*, what is the probability that all *S*'s are together?
17. A fair die is tossed eight times. What is the probability of exactly two 3s, three 1s, and three 6s?
18. The letters in the word *SUPERCALIFRAGILISTICEXPALIDOCIOUS* are arranged randomly. (a) How many of the distinguishable arrangements begin with G and end with X? (b) What is the probability that the outcome begins with G and ends with X?
19. In drawing nine cards with replacement from an ordinary deck of 52 cards, what is the probability of three aces of spades, three queens of hearts, and three kings of clubs?
20. A university wants to plant four American elm, six maple, five northern red oak, and four tulip trees in a row along the road behind its dining hall. If the trees will be planted equally spaced, and the trees of the same type are identical, how many distinguishable possibilities for arranging the trees are there?
21. A construction company has acquired a vast lot, and on the side facing the main road, they plan to construct a total of 16 houses. The houses will be divided into three distinct styles: four modern farmhouses, three Cape Cod styles, five colonials, and four craftsmans . Each house style will be identical, and the houses will be constructed with equal spacing between them. Determine the number of distinguishable arrangements possible for these houses.
22. At an American Red Cross Blood Donation Center, five donors with type O, six with type A, one with type AB, and two with type B blood stand in a line randomly. What is the probability that all persons with type A blood stand together?
23. In a Napa Valley winery, guests are invited to a tasting room, and, for each of them, an affable staff member pours 6 types of wine, which are sold at different prices, in 6 small 3-ounce stylish glasses, at random and one at a time. What is the probability that the first three glasses of wine they serve a visitor (a) are, in order, the three most expensive of the 6 brands he or she tastes; (b) are the most expensive brands, but not necessarily in order of their prices?

24. At a party, n men and m women put their drinks on a table and go out on the floor to dance. When they return, none of them recognizes his or her drink, so everyone takes a drink at random. What is the probability that each man selects his own drink?
25. There are 20 chairs in a room numbered 1 through 20. If eight girls and 12 boys sit on these chairs at random, what is the probability that the thirteenth chair is occupied by a boy?
26. Five fair dice are tossed. What is the probability of getting a sum of 27?
27. Even with all the new advanced technological methods to send signals, some ships still use flags for that purpose. Each flag has its own significance as a signal, and each arrangement of two or more flags conveys a special message. If a ship has 8 flags and 4 flagpoles, how many signals can it send?
28. There are 12 students in a class. What is the probability that their birthdays fall in 12 different months? Assume that all months have the same probability of including the birthday of a randomly selected person.
29. If we put five math, six biology, eight history, and three literature books on a bookshelf at random, what is the probability that all the math books are together?
30. One of the five elevators in a building starts with seven passengers and stops at nine floors. Assuming that it is equally likely that a passenger gets off at any of these nine floors, find the probability that at least two of these passengers will get off at the same floor.
31. Five boys and five girls sit at random in a row. What is the probability that the boys are together and the girls are together?
32. If n balls are randomly placed into n cells, what is the probability that each cell will be occupied?
33. A town has six parks. On a Saturday, six classmates, who are unaware of each other's decision, choose a park at random and go there at the same time. What is the probability that at least two of them go to the same park? Convince yourself that this exercise is the same as Exercise 15, only expressed in a different context.
34. A list of all permutations of $abcdef$ is put in alphabetical order. What is the 601st entry in the list?
Hint: The first $5!$ entries all begin with a .
35. A club of 136 members is in the process of choosing a president, a vice president, a secretary, and a treasurer. If two of the members are not on speaking terms and do not serve together, in how many ways can these four people be chosen?

B

36. Let S and T be finite sets with n and m elements, respectively.
 - (a) How many functions $f: S \rightarrow T$ can be defined?
 - (b) If $m \geq n$, how many injective (one-to-one) functions $f: S \rightarrow T$ can be defined?

- (c) If $m = n$, how many surjective (onto) functions $f: S \rightarrow T$ can be defined?
37. A fair die is tossed eight times. What is the probability of exactly two 3s, exactly three 1s, and exactly two 6s?
 38. Suppose that 20 sticks are broken, each into one long and one short part. By pairing them randomly, the 40 parts are then used to make 20 new sticks. (a) What is the probability that long parts are all paired with short ones? (b) What is the probability that the new sticks are exactly the same as the old ones?
 39. At a party, 15 married couples are seated at random at a round table. What is the probability that all men are sitting next to their wives? Suppose that of these married couples, five husbands and their wives are older than 50 and the remaining husbands and wives are all younger than 50. What is the probability that all men over 50 are sitting next to their wives? Note that when people are sitting around a round table, only their seats relative to each other matters. The exact position of a person is not important.
 40. A box contains five blue and eight red balls. Jim and Jack start drawing balls from the box, respectively, one at a time, at random, and without replacement until a blue ball is drawn. What is the probability that Jack draws the blue ball?



Self-Quiz on Section 2.3

Time allotted: 20 Minutes

Each problem is worth 2.5 points.

1. At a university, all phone numbers begin with 782. If the remaining four digits are equally likely to be $0, 1, \dots, 9$, what is the probability that a randomly selected faculty member has a phone number consisting of 7 distinct digits?
2. In an Extrasensory Perception (ESP) experiment, five people are each asked to think of a card within the suit of hearts in an ordinary deck of 52 cards. What is the probability that (a) no two think of the same card? (b) At least two of them think of the same card?
3. One of Isabella's iPod playlists has 70 songs, 15 of which are by Lionel Richie. She sets the ipod to play the songs on that playlist. However, before doing that, Isabella unchecks and then rechecks the shuffle button of her iPod, and the iPod begins to play the playlist's songs in a random order without repeating a song. What is the probability that the first Lionel Richie song played is the 7th song played?
4. A psychologist specializes in dissocial personality disorder and, for each patient, she lists in order of strength, the 4 strongest traits of impulsivity, recklessness, deceitfulness, exploitative behavior, low conscientiousness, irresponsible behavior, and high negative emotionality. Suppose that, among patients, these traits are found in equal proportions.
 - (a) How many different possibilities are there for the list of a random patient?

- (b) What is the probability that the list of a random patient includes impulsivity and high negative emotionality?



2.4 COMBINATIONS

In many combinatorial problems, unlike permutations, the order in which elements are arranged is immaterial. For example, suppose that in a contest there are 10 semifinalists and we want to count the number of possible ways that three contestants enter the finals. If we argue that there are $10 \times 9 \times 8$ such possibilities, we are wrong since the contestants cannot be ordered. If A , B , and C are three of the semifinalists, then ABC , BCA , ACB , BAC , CAB , and CBA are all the same event and have the same meaning: “ A , B , and C are the finalists.” The technique known as **combinations** is used to deal with such problems.

Definition 2.2 *An unordered arrangement of r objects from a set A containing n objects ($r \leq n$) is called an **r -element combination** of A , or a combination of the elements of A taken r at a time.*

Therefore, two combinations are different only if they differ in composition. Let x be the number of r -element combinations of a set A of n objects. If all the permutations of each r -element combination are found, then all the r -element permutations of A are found. Since for each r -element combination of A there are $r!$ permutations and the total number of r -element permutations is ${}_nP_r$, we have

$$x \cdot r! = {}_nP_r.$$

Hence $x \cdot r! = n!/(n-r)!$, so $x = n!/[(n-r)! r!]$. Therefore, we have shown that

The number of r -element combinations of n objects is given by

$${}_nC_r = \frac{n!}{(n-r)! r!}.$$

Historically, a formula equivalent to $n!/[(n-r)! r!]$ turned up in the works of the Indian mathematician Bhaskara II (1114–1185) in the middle of the twelfth century. Bhaskara II used his formula to calculate the number of possible medicinal preparations using six ingredients. Therefore, the rule for calculation of the number of r -element combinations of n objects has been known for a long time.

It is worthwhile to observe that ${}_nC_r$ is the number of subsets of size r that can be constructed from a set of size n . By Theorem 2.3, a set with n elements has 2^n subsets. Therefore, of these 2^n subsets, the number of those that have exactly r elements is ${}_nC_r$.

Notation: By the symbol $\binom{n}{r}$ (read: n choose r) we mean the number of all r -element combinations of n objects. Therefore, for $r \leq n$,

$$\binom{n}{r} = \frac{n!}{r! (n-r)!}.$$

Observe that $\binom{n}{0} = \binom{n}{n} = 1$ and $\binom{n}{1} = \binom{n}{n-1} = n$. Also, for any $0 \leq r \leq n$,

$$\binom{n}{r} = \binom{n}{n-r}$$

and

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}. \quad (2.4)$$

These relations can be proved algebraically or verified combinatorially. Let us prove (2.4) by a combinatorial argument. Consider a set of $n+1$ objects, $\{a_1, a_2, \dots, a_n, a_{n+1}\}$. There are $\binom{n+1}{r}$ r -element combinations of this set. Now we separate these r -element combinations into two disjoint classes: one class consisting of all r -element combinations of $\{a_1, a_2, \dots, a_n\}$ and another consisting of all $(r-1)$ -element combinations of $\{a_1, a_2, \dots, a_n\}$ attached to a_{n+1} . The latter class contains $\binom{n}{r-1}$ elements and the former contains $\binom{n}{r}$ elements, showing that (2.4) is valid.

Example 2.19 In how many ways can two mathematics and three biology books be selected from eight mathematics and six biology books?

Solution: There are $\binom{8}{2}$ possible ways to select two mathematics books and $\binom{6}{3}$ possible ways to select three biology books. Therefore, by the counting principle,

$$\binom{8}{2} \times \binom{6}{3} = \frac{8!}{2! 6!} \times \frac{6!}{3! 3!} = 560$$

is the total number of ways in which two mathematics and three biology books can be selected. $\blacklozenge$

Example 2.20 A random sample of 45 instructors from different state universities were selected randomly and asked whether they are happy with their teaching loads. The responses of 32 were negative. If Drs. Smith, Brown, and Jones were among those questioned, what is the probability that all three of them gave negative responses?

Solution: There are $\binom{45}{32}$ different possible groups with negative responses. If three of them are Drs. Smith, Brown, and Jones, the other 29 are from the remaining 42 faculty members

questioned. Hence the desired probability is

$$\frac{\binom{42}{29}}{\binom{45}{32}} \approx 0.35. \quad \blacklozenge$$

Example 2.21 In a small town, 11 of the 25 schoolteachers are pro-life, eight are pro-choice, and the rest are indifferent. A random sample of five schoolteachers is selected for an interview. What is the probability that (a) all of them are pro-choice; (b) all of them have the same opinion?

Solution:

- (a) There are $\binom{25}{5}$ different ways to select random samples of size 5 out of 25 teachers.

Of these, only $\binom{8}{5}$ are all pro-choice. Hence the desired probability is

$$\frac{\binom{8}{5}}{\binom{25}{5}} \approx 0.0011.$$

- (b) By an argument similar to part (a), the desired probability equals

$$\frac{\binom{11}{5} + \binom{8}{5} + \binom{6}{5}}{\binom{25}{5}} \approx 0.0099. \quad \blacklozenge$$

Example 2.22 In a state lottery, players pick six different integers between 1 and 49, order of selection being irrelevant. The lottery commission then randomly selects six of these as the *winning numbers*. A player wins the grand prize if all six numbers that he or she has selected match the winning numbers. He or she wins the second prize if exactly five, and the third prize if exactly four of the six numbers chosen match with the winning ones. Find the probability that a certain choice of a bettor wins the grand, the second, and the third prizes, respectively.

Solution: The probability of winning the grand prize is

$$\frac{1}{\binom{49}{6}} = \frac{1}{13,983,816}.$$

The probability of winning the second prize is

$$\frac{\binom{6}{5}\binom{43}{1}}{\binom{49}{6}} = \frac{258}{13,983,816} \approx \frac{1}{54,200},$$

and the probability of winning the third prize is

$$\frac{\binom{6}{4}\binom{43}{2}}{\binom{49}{6}} = \frac{13,545}{13,983,816} \approx \frac{1}{1032}. \quad \blacklozenge$$

Example 2.23 From an ordinary deck of 52 cards, seven cards are drawn at random and without replacement. What is the probability that at least one of the cards is a king?

Solution: In $\binom{52}{7}$ ways seven cards can be selected from an ordinary deck of 52 cards. In $\binom{48}{7}$ of these, none of the cards selected is a king. Therefore, the desired probability is

$$P(\text{at least one king}) = 1 - P(\text{no kings}) = 1 - \frac{\binom{48}{7}}{\binom{52}{7}} = 0.4496.$$

Warning: A common mistake is to calculate this and similar probabilities as follows: To make sure that there is at least one king among the seven cards drawn, we will first choose a king; there are $\binom{4}{1}$ possibilities. Then we choose the remaining six cards from the remaining 51 cards; there are $\binom{51}{6}$ possibilities for this. Thus the answer is

$$\frac{\binom{4}{1}\binom{51}{6}}{\binom{52}{7}} = 0.5385.$$

This solution is wrong because it counts some of the possible outcomes several times. For example, the hand $K_H, 5_C, 6_D, 7_H, K_D, J_C$, and 9_S is counted twice: once when K_H is selected as the first card from the kings and $5_C, 6_D, 7_H, K_D, J_C$, and 9_S from the remaining 51, and once when K_D is selected as the first card from the kings and $K_H, 5_C, 6_D, 7_H, J_C$, and 9_S from the remaining 51 cards. ♦

Example 2.24 What is the probability that a poker hand is a full house? A poker hand consists of five randomly selected cards from an ordinary deck of 52 cards. It is a full house if three cards are of one denomination and two cards are of another denomination: for example, three queens and two 4s.

Solution: The number of different poker hands is $\binom{52}{5}$. To count the number of full houses, let us call a hand of type (Q,4) if it has three queens and two 4s, with similar representations for other types of full houses. Observe that (Q,4) and (4,Q) are different full houses, and types such as (Q,Q) and (K,K) do not exist. Hence there are 13×12 different types of full houses. Since for every particular type, say (4,Q), there are $\binom{4}{3}$ ways to select three 4s and $\binom{4}{2}$ ways to select two Q's, the desired probability equals

$$\frac{13 \times 12 \times \binom{4}{3} \times \binom{4}{2}}{\binom{52}{5}} \approx 0.0014. \quad \blacklozenge$$

Example 2.25 Show that the number of different ways n indistinguishable objects can be placed into k distinguishable cells is

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}.$$

Solution: Let the n indistinguishable objects be represented by n identical oranges, and the k distinguishable cells be represented by k people. We want to count the number of different ways that n identical oranges can be divided among k people. To do this, add $k-1$ identical apples to the oranges. Then take the $n+k-1$ apples and oranges and line them up in some random order. Give all of the oranges preceding the first apple to the first person, all of the oranges between the first and second apples to the second person, all of the oranges between the second and third apples to the third person, and so on. Note that if, for example, an apple appears in the beginning of the line, then the first person does not receive any oranges. Similarly, if two apples appear next to each other, say, at the i th and $(i+1)$ st positions, then the $(i+1)$ st person does not receive any oranges. This process establishes a one-to-one correspondence between the ways n identical oranges can be divided among k people, and the number of distinguishable

permutations of $n + k - 1$ apples and oranges, of which the n oranges are identical and the $k - 1$ apples are identical. By Theorem 2.4, the answer to this problem is

$$\frac{(n + k - 1)!}{n! (k - 1)!} = \binom{n + k - 1}{n} = \binom{n + k - 1}{k - 1}. \quad \blacklozenge$$

Example 2.26 Let n be a positive integer, and let $x_1 + x_2 + \cdots + x_k = n$ be a given equation. A vector $(x_1, x_2, \dots, x_k)$ satisfying $x_1 + x_2 + \cdots + x_k = n$ is said to be a *nonnegative integer solution* of the equation if for each i , $1 \leq i \leq k$, x_i is a nonnegative integer. It is said to be a *positive integer solution* of the equation if for each i , $1 \leq i \leq k$, x_i is a positive integer.

- (a) How many distinct nonnegative integer solutions does the equation $x_1 + x_2 + \cdots + x_k = n$ have?
- (b) How many distinct positive integer solutions does the equation $x_1 + x_2 + \cdots + x_k = n$ have?

Solution:

- (a) If we think of $x_1, x_2, \dots, x_k$ as k cells, then the problem reduces to that of dividing n identical objects (namely, n 1s) into k cells. Hence, by Example 2.25, the answer is

$$\binom{n + k - 1}{n} = \binom{n + k - 1}{k - 1}.$$

- (b) For $1 \leq i \leq k$, let $y_i = x_i - 1$. Then, for each positive integer solution $(x_1, x_2, \dots, x_k)$ of $x_1 + x_2 + \cdots + x_k = n$, there is exactly one nonnegative integer solution $(y_1, y_2, \dots, y_k)$ of $y_1 + y_2 + \cdots + y_k = n - k$, and conversely. Therefore, the number of positive integer solutions of $x_1 + x_2 + \cdots + x_k = n$ is equal to the number of nonnegative integer solutions of $y_1 + y_2 + \cdots + y_k = n - k$, which, by part (a), is

$$\binom{(n - k) + k - 1}{n - k} = \binom{n - 1}{n - k} = \binom{n - 1}{k - 1}. \quad \blacklozenge$$

In Example 2.26, we have established the following:

For positive integers n and k , the number of distinct **nonnegative** integer solutions of the equation $x_1 + x_2 + \cdots + x_k = n$ is given by

$$\binom{n + k - 1}{k - 1}.$$

The number of distinct **positive** integer solutions of this equation is $\binom{n - 1}{k - 1}$.

Example 2.27 A stockholder is considering whether to invest \$150,000 in five new stocks, each in multiples of \$10,000. In how many ways can he invest in these stocks if (a) he is determined to buy at least \$10,000 of each stock; (b) he does not necessarily select to invest in all five stocks?

Solution: For $1 \leq i \leq 5$, suppose that the stockholder decides to buy $\$(10000)x_i$ of stock i . We have

$$10000x_1 + 10000x_2 + \cdots + 10000x_5 = 150000.$$

So the answer to (a) is the number of positive integer solutions of the equation $x_1 + x_2 + \cdots + x_5 = 15$, which, by Example 2.26, is $\binom{15-1}{5-1} = \binom{14}{4} = 1001$. The answer to part (b) is the number of nonnegative integer solutions of the equation $x_1 + x_2 + \cdots + x_5 = 15$, which, by Example 2.26, is $\binom{15+5-1}{5-1} = \binom{19}{4} = 3876$. ♦

Example 2.28 An absentminded professor wrote n letters and sealed them in envelopes before writing the addresses on the envelopes. Then he wrote the n addresses on the envelopes at random. What is the probability that at least one letter was addressed correctly?

Solution: The total number of ways that one can write n addresses on n envelopes is $n!$; thus the sample space contains $n!$ points. Now we calculate the number of outcomes in which at least one envelope is addressed correctly. To do this, let E_i be the event that the i th letter is addressed correctly, then $E_1 \cup E_2 \cup \cdots \cup E_n$ is the event that at least one letter is addressed correctly. To calculate $P(E_1 \cup E_2 \cup \cdots \cup E_n)$, we use the inclusion-exclusion principle. To do so we must calculate the probabilities of all possible intersections of the events from $E_1, \dots, E_n$, add the probabilities that are obtained by intersecting an odd number of the events, and subtract all the probabilities that are obtained by intersecting an even number of the events. Therefore, we need to know the number of elements of E_i 's, $E_i E_j$'s, $E_i E_j E_k$'s, and so on. Now E_i contains $(n-1)!$ points, because when the i th letter is addressed correctly, there are $(n-1)!$ ways to address the remaining $n-1$ envelopes. So $P(E_i) = (n-1)!/n!$. Similarly, $E_i E_j$ contains $(n-2)!$ points, because if the i th and j th envelopes are addressed correctly, the remaining $n-2$ envelopes can be addressed in $(n-2)!$ ways. Thus $P(E_i E_j) = (n-2)!/n!$. Similarly, $P(E_i E_j E_k) = (n-3)!/n!$, and so on. Now in computing $P(E_1 \cup E_2 \cup \cdots \cup E_n)$, there are n terms of the form $P(E_i)$, $\binom{n}{2}$ terms of the form $P(E_i E_j)$, $\binom{n}{3}$ terms of the form $P(E_i E_j E_k)$, and so on. Hence

$$\begin{aligned} P(E_1 \cup E_2 \cup \cdots \cup E_n) &= n \frac{(n-1)!}{n!} - \binom{n}{2} \frac{(n-2)!}{n!} + \cdots + \\ &\quad (-1)^{n-2} \binom{n}{n-1} \frac{[n-(n-1)]!}{n!} + (-1)^{n-1} \binom{n}{n} \frac{1}{n!}. \end{aligned}$$

This expression simplifies to

$$P(E_1 \cup E_2 \cup \cdots \cup E_n) = 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \cdots + \frac{(-1)^{n-1}}{n!}.$$

Remark: Note that since $e^x = \sum_{n=0}^{\infty} (x^n/n!)$, if $n \rightarrow \infty$, then

$$\begin{aligned} P\left(\bigcup_{i=1}^{\infty} E_i\right) &= 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \cdots + \frac{(-1)^{n-1}}{n!} + \cdots \\ &= 1 - \left(1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots + \frac{(-1)^n}{n!} + \cdots\right) \\ &= 1 - \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} = 1 - \frac{1}{e} \approx 0.632. \end{aligned}$$

Hence even if the number of the envelopes is very large, there is still a very good chance for at least one envelope to be addressed correctly. ♦

Remark 2.3 Example 2.28 and its solution have a significant interpretation in the theory of **coincidences**. They indicate that in any random permutation of the integers $1, 2, \dots, n$, regardless of how large n is, the probability of having at least one integer remain unaltered is approximately 0.632. For instance, consider a college where, on exam days, the n students of a calculus class are randomly assigned seats marked from 1 to n . In a given semester, the probability that at least one student is assigned the same seat in both the midterm and final exams is approximately 0.632. In other words, the probability of at least one coincidence, which leaves at least one student wondering, is 0.632. ♦

Binomial Expansion

One of the most important applications of combinatorics is that the formula for the r -element combinations of n objects enables us to find an algebraic expansion for $(x + y)^n$.

Theorem 2.5 (Binomial Expansion) For any integer $n \geq 0$,

$$(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^{n-i} y^i.$$

Proof: By looking at some special cases, such as

$$(x + y)^2 = (x + y)(x + y) = x^2 + xy + yx + y^2$$

and

$$\begin{aligned} (x + y)^3 &= (x + y)(x + y)(x + y) \\ &= x^3 + x^2y + yx^2 + xy^2 + yx^2 + xy^2 + y^2x + y^3, \end{aligned}$$

it should become clear that when we carry out the multiplication

$$(x + y)^n = \underbrace{(x + y)(x + y) \cdots (x + y)}_{n \text{ times}},$$

we obtain only terms of the form $x^{n-i}y^i$, $0 \leq i \leq n$. Therefore, all we have to do is to find out how many times the term $x^{n-i}y^i$ appears, $0 \leq i \leq n$. This is seen to be $\binom{n}{n-i} = \binom{n}{i}$ because $x^{n-i}y^i$ emerges only whenever the x 's of $n-i$ of the n factors of $(x+y)$ are multiplied by the y 's of the remaining i factors of $(x+y)$. Hence

$$\begin{aligned} (x + y)^n &= \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \cdots \\ &\quad + \binom{n}{n-1}xy^{n-1} + \binom{n}{n}y^n. \quad \blacklozenge \end{aligned}$$

Remark 2.4 Note that, by Theorem 2.5, the expansion of $(x + y)^n$ has $n + 1$ terms, and the i th term is

$$\binom{n}{i-1}x^{n-i+1}y^{i-1}, \quad 1 \leq i \leq n + 1. \quad \blacklozenge$$

Example 2.29 What is the coefficient of x^2y^3 in the expansion of $(2x + 3y)^5$?

Solution: Let $u = 2x$ and $v = 3y$, then $(2x + 3y)^5 = (u + v)^5$. The coefficient of u^2v^3 in the expansion of $(u + v)^5$ is $\binom{5}{3}$ and $u^2v^3 = (2^2 \cdot 3^3)x^2y^3$; therefore, the coefficient of x^2y^3 in the expansion of $(2x + 3y)^5$ is $\binom{5}{3}(2^2 \cdot 3^3) = 1080$. $\blacklozenge$

Example 2.30 Evaluate the sum $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \cdots + \binom{n}{n}$.

Solution: A set containing n elements has $\binom{n}{i}$, $0 \leq i \leq n$, subsets with i elements. So the given expression is the total number of the subsets of a set of n elements, and therefore it equals 2^n . A second way to see this is to note that, by the binomial expansion, the given expression equals

$$\sum_{i=0}^n \binom{n}{i} 1^{n-i} 1^i = (1 + 1)^n = 2^n. \quad \blacklozenge$$

Example 2.31 Evaluate the sum $\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}$.

Solution:

$$i\binom{n}{i} = i \cdot \frac{n!}{i!(n-i)!} = \frac{n \cdot (n-1)!}{(i-1)!(n-i)!} = n\binom{n-1}{i-1}.$$

So

$$\begin{aligned} & \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} \\ &= n \left[\binom{n-1}{0} + \binom{n-1}{1} + \binom{n-1}{2} + \cdots + \binom{n-1}{n-1} \right] = n \cdot 2^{n-1}, \end{aligned}$$

by Example 2.30. $\blacklozenge$

Example 2.32 Prove that $\binom{2n}{n} = \sum_{i=0}^n \binom{n}{i}^2$.

Solution: We show this by using a combinatorial argument. For an analytic proof, see Exercise 65. Let $A = \{a_1, a_2, \dots, a_n\}$ and $B = \{b_1, b_2, \dots, b_n\}$ be two disjoint sets. The number of subsets of $A \cup B$ with n elements is $\binom{2n}{n}$. On the other hand, any subset of $A \cup B$ with n elements is the union of a subset of A with i elements and a subset of B with $n-i$ elements for some $0 \leq i \leq n$. Since for each i there are $\binom{n}{i}\binom{n}{n-i}$ such subsets, we have that the total number of subsets of $A \cup B$ with n elements is $\sum_{i=0}^n \binom{n}{i}\binom{n}{n-i}$. But since $\binom{n}{n-i} = \binom{n}{i}$, we have the identity. $\blacklozenge$

Example 2.33 In this example, we present an *intuitive* proof for the inclusion-exclusion principle explained in [Section 1.4](#):

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i=1}^n P(A_i) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n P(A_i A_j) + \sum_{i=1}^{n-2} \sum_{j=i+1}^{n-1} \sum_{k=j+1}^n P(A_i A_j A_k) \\ &\quad - \cdots + (-1)^{n-1} P(A_1 A_2 \cdots A_n). \end{aligned}$$

Let S be the sample space. If an outcome $\omega \notin \bigcup_{i=1}^n A_i$, then $\omega \notin A_i$, $1 \leq i \leq n$. Therefore, the probability of ω is not added to either side of the preceding equation. Suppose that $\omega \in \bigcup_{i=1}^n A_i$, then ω belongs to k of the A_i 's for some k , $1 \leq k \leq n$. Now, to the left side of the equation the probability of ω is added exactly once. We will show that the same is true on the right. Clearly, the first term on the right side contains the probability of ω exactly $k = \binom{k}{1}$ times. The second term subtracts this probability $\binom{k}{2}$ times, the third term adds it back, this time $\binom{k}{3}$ times, and so on until the k th term. Depending on whether k is even or odd, the k th term either subtracts or adds

the probability of ω exactly $\binom{k}{k}$ times. The remaining terms do not contain the probability of ω . Therefore, the total number of times the probability of ω is added to the right side is

$$\begin{aligned} & \binom{k}{1} - \binom{k}{2} + \binom{k}{3} - \cdots (-1)^{k-1} \binom{k}{k} \\ &= - \sum_{i=1}^k \binom{k}{i} 1^{k-i} (-1)^i = \binom{k}{0} - \sum_{i=0}^k \binom{k}{i} 1^{k-i} (-1)^i \\ &= 1 - (1-1)^k = 1, \end{aligned}$$

where the next-to-the-last equation follows from the binomial expansion. This establishes the inclusion-exclusion principle. ♦

Example 2.34 Suppose that we want to distribute n distinguishable balls into k distinguishable cells so that n_1 balls are distributed into the first cell, n_2 balls into the second cell, $\dots$, n_k balls into the k th cell, where $n_1 + n_2 + n_3 + \cdots + n_k = n$. To count the number of ways that this is possible, note that we have $\binom{n}{n_1}$ choices to distribute n_1 balls into the first cell; for each choice of n_1 balls in the first cell, we then have $\binom{n-n_1}{n_2}$ choices to distribute n_2 balls into the second cell; for each choice of n_1 balls in the first cell and n_2 balls in the second cell, we have $\binom{n-n_1-n_2}{n_3}$ choices to distribute n_3 balls into the third cell; and so on. Hence by the generalized counting principle the total number of such distributions is

$$\begin{aligned} & \binom{n}{n_1} \binom{n-n_1}{n_2} \binom{n-n_1-n_2}{n_3} \cdots \binom{n-n_1-n_2-\cdots-n_{k-1}}{n_k} \\ &= \frac{n!}{(n-n_1)! n_1!} \times \frac{(n-n_1)!}{(n-n_1-n_2)! n_2!} \times \frac{(n-n_1-n_2)!}{(n-n_1-n_2-n_3)! n_3!} \times \cdots \\ & \times \frac{(n-n_1-n_2-n_3-\cdots-n_{k-1})!}{(n-n_1-n_2-n_3-\cdots-n_{k-1}-n_k)! n_k!} = \frac{n!}{n_1! n_2! n_3! \cdots n_k!} \end{aligned}$$

because $n_1 + n_2 + n_3 + \cdots + n_k = n$ and $(n-n_1-n_2-\cdots-n_k)! = 0! = 1$. ♦

As an application of Example 2.34, we state a generalization of the binomial expansion (Theorem 2.5) and leave its proof as an exercise (see Exercise 44).

Theorem 2.6 (Multinomial Expansion) *In the expansion of*

$$(x_1 + x_2 + \cdots + x_k)^n,$$

the coefficient of the term $x_1^{n_1} x_2^{n_2} x_3^{n_3} \cdots x_k^{n_k}$, $n_1 + n_2 + \cdots + n_k = n$, is

$$\frac{n!}{n_1! n_2! \cdots n_k!}.$$

Therefore,

$$(x_1 + x_2 + \cdots + x_k)^n = \sum_{n_1+n_2+\cdots+n_k=n} \frac{n!}{n_1! n_2! \cdots n_k!} x_1^{n_1} x_2^{n_2} x_3^{n_3} \cdots x_k^{n_k}.$$

Note that the sum is taken over all nonnegative integers $n_1, n_2, \dots, n_k$ such that $n_1 + n_2 + \dots + n_k = n$.

Pascal's Triangle

At the intersection of mathematics and art, lies an intriguing construct called **Pascal's Triangle**. It is a testament to the endless wonders of mathematics. Whether a mathematics enthusiast or just someone intrigued by patterns, diving deep into this triangle promises a treasure trove of discoveries. We now introduce this captivating triangular arrangement of numbers, which is not just a pretty face; it conceals a plethora of patterns and mathematical wonders.

For small values of n , the coefficients of $(x + y)^n$ can be obtained from Pascal's triangle. To construct this triangle, consider the following relations:

$$(x + y)^0 = \binom{0}{0} x^0 y^0$$

$$(x + y)^1 = \binom{1}{0} x^1 y^0 + \binom{1}{1} x^0 y^1$$

$$(x + y)^2 = \binom{2}{0} x^2 y^0 + \binom{2}{1} x^1 y^1 + \binom{2}{2} x^0 y^2$$

$$(x + y)^3 = \binom{3}{0} x^3 y^0 + \binom{3}{1} x^2 y^1 + \binom{3}{2} x^1 y^2 + \binom{3}{3} x^0 y^3$$

and so on. From these expressions, we can construct a triangular table where the i th row provides the coefficients of $(x + y)^{i-1}$.

$$\begin{array}{ccccccc}
 & & & & \binom{0}{0} & & & & \\
 & & & & & & \binom{1}{1} & & \\
 & & & \binom{1}{0} & & & & & \\
 & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & \\
 & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & \\
 \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4}
 \end{array}$$

and so forth.

In Pascal's Triangle, every row starts and ends with the number 1. For any number in the $(n + 1)$ st row that is not 1, its form is $\binom{n+1}{r}$. The two adjacent numbers directly above this number are $\binom{n}{r-1}$ and $\binom{n}{r}$. Given the relation (2.3):

$$\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r},$$

each entry in the $(n + 1)$ st row, other than the first and last, is the sum of the two immediate numbers above it. As an illustration: $\binom{4}{3} = \binom{3}{2} + \binom{3}{3}$. Using this principle, we can construct the triangle numerically. The first row contains just a single 1, the second row has two 1s, and starting from the third row, both the first and last numbers are 1, with the internal numbers determined by adding the two numbers directly above them.

					1								
					1		1						
				1		2		1					
			1		3		3		1				
		1		4		6		4		1			
	1		5		10		10		5		1		
	1	6		15		20		15		6		1	
1		7	21		35		35		21		7		1

and the rows continuing as such.

For instance, the 7th row of Pascal's triangle provides the coefficients for the expansion of $(x + y)^6$. Thus, the expansion is:

$$(x + y)^6 = x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6.$$

The binomial coefficients, denoted as nCr , have a rich historical background. As early as the late eleventh century, Chinese mathematicians were already familiar with them. Omar Khyyām, Omar, the renowned Persian mathematician and poet, made significant contributions in the twelfth century, and Nasir ad-Din Tusi built upon this knowledge in the thirteenth century. In Europe, during the sixteenth century, the works of German mathematician Michael Stifel (1487–1567), as well as Italians Girolamo Cardano and Niccolò Tartaglia, shed light on the binomial coefficients.

Pertaining to the structure known as Pascal's Triangle, its knowledge in China predates its introduction in Europe. Chu Shih-Chieh made reference to this triangle in his works. Moreover, the configuration was already recognized as the Yang Hui triangle in China, named after another eminent mathematician, before Chu Shih-Chieh's acknowledgment. Blaise Pascal, in the seventeenth century, didn't "discover" this triangle, but explored it in great depth. He did offer a comprehensive examination of its structure and associated properties, notably in relation to the budding domain of probability. Consequently, in the Western context, it's often referred to as Pascal's Triangle.

EXERCISES

A

1. Jim has 20 friends. If he decides to invite six of them to his birthday party, how many choices does he have?
2. Each state of the 50 in the United States has two senators. In how many ways may a majority be achieved in the U.S. Senate? Ignore the possibility of absence or abstention. Assume that all senators are present and voting.
3. A panel consists of 20 men and 25 women. How many choices do we have to obtain a jury of six men and six women from this panel?
4. There are 24 eggs in a rectangular egg carton arranged in 4 rows of 6 eggs each. Three eggs are selected at random from the carton to make an omelet. What is the probability that the eggs are selected from the corners of the carton?
5. From an ordinary deck of 52 cards, five are drawn randomly. What is the probability of drawing exactly three face cards?
6. As part of an English department's requirements, students must take 5 courses out of a list of 9, of which 3 are advanced courses in Shakespeare. If it is required that at least one of these 5 courses be a Shakespeare course, how many choices does an English student have for this particular requirement?
7. A random sample of n elements is taken from a population of size N without replacement. What is the probability that a fixed element of the population is included? Simplify your answer.
8. In a box of 12 fuses, there are 3 that are defective. If a quality control engineer tests 4 of these fuses at random, what is the probability that he does not find any of the defective fuses?
9. Judy puts one piece of fruit in her child's lunch bag every day. If she has three oranges and two apples for the next five days, in how many ways can she do this?
10. Ann puts at most one piece of fruit in her child's lunch bag every day. If she has only three oranges and two apples for the next eight lunches of her child, in how many ways can she do this?
11. Lili has 20 friends. Among them are Kevin and Gerry, who are husband and wife. Lili wants to invite six of her friends to her birthday party. If neither Kevin nor Gerry will go to a party without the other, how many choices does Lili have?
12. In a binary code, each code word has 6 bits, each of which is 0 or 1. What is the probability that a random code word (a) has three 0s and three 1s? (b) Begins with two 0s? (c) Ends in three 1s?

13. In a game, there are 12 tiles in a bag, each bearing one of the numbers 1 through 12. Lida draws 6 tiles at random. What is the probability that the largest number she picks out is 9?
14. In front of Ilaria's office there is a parking lot with 13 parking spots in a row. When cars arrive at this lot, they park randomly at one of the empty spots. Ilaria parks her car in the only empty spot that is left. Then she goes to her office. On her return she finds that there are seven empty spots. If she has not parked her car at either end of the parking area, what is the probability that both of the parking spaces that are next to Ilaria's car are empty?
15. In addition to many scientific experiments that astronauts should operate while on the space station, they must make sure that the station is in perfect condition. Suppose that there are 13 astronauts on board, and on a specific day, three astronauts are needed to replace broken equipment, five to check other equipment, and five to clean. In how many ways can these tasks be divided?
16. Find the coefficient of x^9 in the expansion of $(2 + x)^{12}$.
17. Find the coefficient of x^3y^4 in the expansion of $(2x - 4y)^7$.
18. Find the 9th term of the binomial expansion of $(1 + \sqrt{x})^{13}$.
19. Find the 4th term of the binomial expansion of $\left(x + \frac{5}{x}\right)^9$.
20. A team consisting of three boys and four girls must be formed from a group of nine boys and eight girls. If two of the girls are feuding and refuse to play on the same team, how many possibilities do we have?
21. There are 23 girls and 19 boys lined up randomly for a play in a kindergarten classroom. What is the probability that the 12th child in the line is a girl?
22. A fair coin is tossed 10 times. What is the probability of (a) five heads; (b) at least five heads?
23. If five numbers are selected at random from the set $\{1, 2, 3, \dots, 20\}$, what is the probability that their minimum is larger than 5?
24. In North America, in the World Series, which is the most important baseball event each year, the American League champion team plays a series of games with the National League champion team. The first team to win four games will be the winner of the World Series championship and is awarded the Commissioner's Trophy. Assuming that it is equally likely for the teams to win, find the probability that the series (a) lasts 6 games; (b) lasts 7 games.
25. A soccer coach assigns his players randomly, 1 as a goalie, 4 as defenders, 3 as midfielders, and 3 as forwards. (a) How many choices does he have? (b) If only one player is experienced enough to play as a goalie, what is the probability that in the random division, he ends up to be the goalie?
26. From a faculty of six professors, six associate professors, ten assistant professors, and

twelve instructors, a committee of size six is formed randomly. What is the probability that (a) there are exactly two professors on the committee; (b) all committee members are of the same rank?

27. In a medical study, 25 patients with a duodenal ulcer have volunteered to be treated by an experimental drug. The researcher has decided to treat 18 of the patients, randomly selected, by the new drug and to give the remaining 7 a placebo as controls. If the duodenal ulcers of 17 of these patients are caused by the bacterium *H. pylori*, and the rest by nonsteroidal anti-inflammatory drugs such as aspirin, what is the probability that the control group (a) includes at least one patient whose duodenal ulcer is caused by the bacterium *H. pylori*? (b) Includes at least two such patients?
28. A lake contains 200 trout; 50 of them are caught randomly, tagged, and returned. If, again, we catch 50 trout at random, what is the probability of getting exactly five tagged trout?
29. A **round-robin tournament** is a competition in which each participant plays against every other participant exactly once. In a round-robin tournament with n contestants, what is the total number of possible outcomes? An outcome lists out the winner and the loser in each competition.
30. A round-robin tournament, as defined in the previous exercise, is a competition in which each participant plays against every other participant exactly once. Using a round-robin tournament with $n + 1$ contestants, give a combinatorial argument for the following identity:

$$1 + 2 + \cdots + n = \binom{n+1}{2}.$$

31. Find the values of $\sum_{i=0}^n 2^i \binom{n}{i}$ and $\sum_{i=0}^n x^i \binom{n}{i}$.
32. A fair die is tossed six times. What is the probability of getting exactly two 6s?
33. Suppose that 12 married couples take part in a contest. If 12 persons each win a prize, what is the probability that from every couple one of them is a winner? Assume that all of the $\binom{24}{12}$ possible sets of winners are equally probable.
34. How many all capital, different words of any length, meaningful or meaningless, can we create from the letters of “JUSTICE” without repeating a letter?
35. A young businessman is obsessed with buying life insurance policies to protect his family. He decides to allocate \$7200 a year toward all or some of the Term, Universal, Indexed Universal, Survivorship Universal, and Variable Universal life insurance premiums. If he decides to pay premiums in units of \$600 for each policy, how many choices does he have (a) if he insists in spending the entire \$7200 toward life insurance premiums; (b) if he does not necessarily want to spend the whole amount?
36. There are 12 nuts and 12 bolts in a box. If the contents of the box are divided between two handymen, what is the probability that each handyman will get six nuts and six bolts?

37. Poker hands are classified into the following 10 nonoverlapping categories in increasing order of likelihood. Calculate the probability of the occurrence of each class separately. Recall that a poker hand consists of five cards selected randomly from an ordinary deck of 52 cards.

Royal flush: The 10, jack, queen, king, and ace of the same suit.

Straight flush: All cards in the same suit, with consecutive denominations except for the royal flush.

Four of a kind: Four cards of one denomination and one card of a second denomination: for example, four 8s and a jack.

Full house: Three cards of one denomination and two cards of a second denomination: for example, three 4s and two queens.

Flush: Five cards all in one suit but not a straight or royal flush.

Straight: Cards of distinct consecutive denominations, not all in one suit: for example, 3 of hearts, 4 of hearts, 5 of spades, 6 of hearts, and 7 of clubs.

Three of a kind: Three cards of one denomination, a fourth card of a second denomination, and a fifth card of a third denomination.

Two pairs: Two cards from one denomination, another two from a second denomination, and the fifth card from a third denomination.

One pair: Two cards from one denomination, with the third, fourth, and fifth cards from a second, third, and fourth denomination, respectively: for example, two 8s, a king, a 5, and a 4.

None of the above.

38. A history professor who teaches three sections of the same course every semester decides to make several tests and use them for the next 10 years (20 semesters) as final exams. The professor has two policies: (1) not to give the same test to more than one class in a semester, and (2) not to repeat the same combination of three tests for any two semesters. Determine the minimum number of different tests that the professor should prepare.
39. A four-digit number is selected at random. What is the probability that its ones place is less than its tens place, its tens place is less than its hundreds place, and its hundreds place is less than its thousands place? Note that the first digit of an n -digit number is nonzero.
40. Using Theorem 2.6, expand $(x + y + z)^2$.
41. What is the coefficient of $x^2y^3z^2$ in the expansion of $(2x - y + 3z)^7$?
42. What is the coefficient of x^3y^7 in the expansion of $(2x - y + 3)^{13}$?
43. An ordinary deck of 52 cards is dealt, 13 each, to four players at random. What is the probability that each player receives 13 cards of the same suit?

44. Using induction, binomial expansion, and the identity

$$\binom{n}{n_1} \frac{(n - n_1)!}{n_2! n_3! \cdots n_k!} = \frac{n!}{n_1! n_2! \cdots n_k!},$$

prove the formula of multinomial expansion (Theorem 2.6).

45. A staircase is to be constructed between M and N (see Figure 2.4). The distances from M to L , and from L to N , are 5 and 2 meters, respectively. If the height of a step is 25 centimeters and its width can be any integer multiple of 50 centimeters, how many different choices do we have?

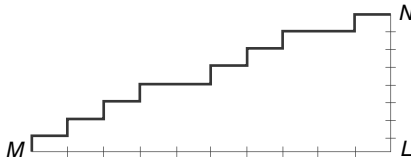


Figure 2.4 Staircase of Exercise 45.

46. Each state of the 50 in the United States has two senators. What is the probability that in a random committee of 50 senators (a) Maryland is represented; (b) all states are represented?
47. (**Newton-Pepys problem**) In 1693, Samuel Pepys, an English Naval administrator and member of the Parliament, corresponded with Isaac Newton concerning the following wager he intended to make: Which of the following trials is more likely than the other two?
- (a) At least one 6 when 6 dice are tossed.
 - (b) At least two 6s when 12 dice are tossed.
 - (c) At least three 6s when 18 dice are tossed.

Isaac Newton calculated the probabilities of these events correctly and provided Samuel Pepys with correct answers. He also justified his answers intuitively; however, his justifications were invalid. Calculate the probability of the events above posed by Samuel Pepys.

B

48. Prove the binomial expansion formula by induction.

Hint: Use the identity $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$.

49. A class contains 30 students. What is the probability that there are 6 months each containing the birthdays of two students, and 6 months each containing the birthdays of three students? Assume that all months have the same probability of including the birthday of a randomly selected person.
50. In a closet there are 10 pairs of shoes. If six shoes are selected at random, what is the probability of (a) no complete pairs; (b) exactly one complete pair; (c) exactly two complete pairs; (d) exactly three complete pairs?
51. An ordinary deck of 52 cards is divided into two equal sets randomly. What is the probability that each set contains exactly 13 red cards?
52. In a box, there are 12 balls, identical in every way, except that they are numbered 1 through 12. We draw 6 balls at random and without replacement one by one. What is the probability that the numbers on the balls appear in increasing order but not necessarily consecutive?
- Hint:* There is a one-to-one correspondence between all the possible ways that the numbers on the balls drawn are in increasing order and the set of subsets of $\{1, 2, \dots, 12\}$ with exactly 6 elements.
53. A train consists of n cars. Each of m passengers ($m > n$) will choose a car at random to ride in. What is the probability that (a) there will be at least one passenger in each car; (b) exactly r ($r < n$) cars remain unoccupied?
54. Suppose that n indistinguishable balls are placed at random into n distinguishable cells. What is the probability that exactly one cell remains empty?
55. At a departmental party, Dr. James P. Coughlin distributes 7 baby photos of 7 of his colleagues among the guests with a list of the names of those colleagues. Each guest is invited to identify the babies in the photos. What is the probability that a participant can identify exactly 3 of the 7 photos correctly by purely random guessing?

56. Show that

$$\binom{n}{0} + \binom{n+1}{1} + \cdots + \binom{n+r}{r} = \binom{n+r+1}{r}.$$

Hint: $\binom{n}{r} = \binom{n+1}{r} - \binom{n}{r-1}.$

57. Prove that

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^k \binom{n}{k} + \cdots + (-1)^n \binom{n}{n} = 0.$$

58. By a combinatorial argument, prove that for $r \leq n$ and $r \leq m$,

$$\binom{n+m}{r} = \binom{m}{0} \binom{n}{r} + \binom{m}{1} \binom{n}{r-1} + \cdots + \binom{m}{r} \binom{n}{0}.$$

59. Evaluate the following sum:

$$\binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \cdots + \frac{1}{n+1}\binom{n}{n}.$$

60. Suppose that five points are selected at random from the interval $(0, 1)$. What is the probability that exactly two of them are between 0 and $1/4$?

Hint: For any point there are four equally likely possibilities: to fall into $(0, 1/4)$, $[1/4, 1/2)$, $[1/2, 3/4)$, and $[3/4, 1)$.

61. A lake has N trout, and t of them are caught at random, tagged, and returned. We catch n trout at a later time randomly and observe that m of them are tagged.

- (a) Find P_N , the probability of what we observed actually happen.
- (b) To estimate the number of trout in the lake, statisticians find the value of N that maximizes P_N . Such a value is called the **maximum likelihood estimator** of N . Show that the maximum of P_N is $[nt/m]$, where by $[nt/m]$ we mean the greatest integer less than or equal to nt/m . That is, prove that the maximum likelihood estimator of the number of trout in the lake is $[nt/m]$.

Hint: Investigate for what values of N the probability P_N is increasing and for what values it is decreasing.

62. Let n be a positive integer. A random sample of four elements is taken from the set $\{0, 1, 2, \dots, n\}$, one at a time and *with replacement*. What is the probability that the sum of the first two elements is equal to the sum of the last two elements?

63. For a given position with n applicants, m applicants are equally qualified and $n - m$ applicants are not qualified at all. Assume that a recruitment process is considered to be fair if the probability that a qualified applicant is hired is $1/m$, and the probability that an unqualified applicant is hired is 0. One fair recruitment process is to interview all applicants, identify the qualified ones, and then hire one of the m qualified applicants randomly. A second fair process, which is more efficient, is to interview applicants in a random order and employ the first qualified applicant encountered. For the fairness of the second recruitment process,

- (a) present an *intuitive* argument;
- (b) give a rigorous combinatorial proof.

64. For $k \geq 1$, $n \geq k$, how many distinct positive integer vectors $(x_1, x_2, \dots, x_k)$ satisfy the inequality $x_1 + x_2 + \cdots + x_k \geq n$?

65. Using the binomial theorem, calculate the coefficient of x^n in the expansion of $(1+x)^{2n} = (1+x)^n(1+x)^n$ to prove that

$$\binom{2n}{n} = \sum_{i=0}^n \binom{n}{i}^2.$$

For a combinatorial proof of this relation, see Example 2.32.

- 66.** An absentminded professor wrote n letters and sealed them in envelopes without writing the addresses on them. Then he wrote the n addresses on the envelopes at random. What is the probability that exactly k of the envelopes were addressed correctly?
Hint: Consider a particular set of k letters. Let M be the total number of ways that only these k letters can be addressed correctly. The desired probability is the quantity $\binom{n}{k} M/n!$; using Example 2.28, argue that M satisfies $\sum_{i=2}^{n-k} (-1)^i / i! = M/(n-k)!$.
- 67.** A fair coin is tossed n times. Calculate the probability of getting no successive heads.
Hint: Let x_i be the number of sequences of H's and T's of length i with no successive H's. Show that x_i satisfies $x_i = x_{i-1} + x_{i-2}$, $i \geq 2$, where $x_0 = 1$ and $x_1 = 2$. The answer is $x_n/2^n$. Note that $\{x_i\}_{i=1}^{\infty}$ is a Fibonacci-type sequence.
- 68.** What is the probability that the birthdays of at least two students of a class of size n are at most k days apart? Assume that the birthrates are constants throughout the year and that each year has 365 days.



Self-Quiz on Section 2.4

Time allotted: 20 Minutes

Each problem is worth 2.5 points.

- For an experiment, Sheri, a neuroscientist, needs to select 5 rats from each of the 5 groups of rats being studied. If each group has 12 rats, how many choices does Sheri have?
- A broadcasting company recruits college students to help with their Olympic coverage. If they choose 4 athletes randomly from their 7 finalists, what is the probability that (a) the 4 tallest of the finalists are selected? (b) Exactly 3 of the 4 athletes selected are the tallest of the finalists? Assume that no two of the finalists are of the same exact height.
- There are 8 chairs in a row, on the upper deck of a boat, attached to the floor by screws and nails. A sociologist observed that from the first 4 passengers who sat in that row no two sat next to each other. Based on this observation, can she conclude that, when possible, passengers avoid taking seats next to occupied ones?
Hint: Suppose that passengers occupied the seats at random and calculate the probability of what the sociologist observed.
- For a chess tournament, 16 individuals are to be divided into 8 groups of 2 each. In how many ways can this be done?



2.5 STIRLING'S FORMULA

To estimate $n!$ for large values of n , the following formula is used. Discovered in 1730 by James Stirling (1692–1770), it appeared in his book *Methodus Differentialis* (Bowyer, London, 1730). Also, independently, a slightly less general version of the formula was discovered by De Moivre, who used it to prove the central limit theorem.

Theorem 2.7 (Stirling's Formula)

$$n! \sim \sqrt{2\pi n} n^n e^{-n},$$

where the sign $\sim$ means $\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi n} n^n e^{-n}} = 1$.

Stirling's formula, which usually gives excellent approximations in numerical computations, is also often used to prove theoretical problems. Note that although the ratio $R(n) = n! / (\sqrt{2\pi n} n^n e^{-n})$ becomes 1 at ∞ , it is still close to 1, even for very small values of n . The following table shows this fact.

n	$n!$	$\sqrt{2\pi n} n^n e^{-n}$	$R(n)$
1	1	0.922	1.084
2	2	1.919	1.042
5	120	118.019	1.017
8	40,320	39,902.396	1.010
10	3,628,800	3,598,695.618	1.008
12	479,001,600	475,687,486.474	1.007

Be aware that, even though $\lim_{n \rightarrow \infty} R(n)$ is 1, the difference between $n!$ and $\sqrt{2\pi n} n^n e^{-n}$ increases as n gets larger. In fact, it goes to ∞ as $n \rightarrow \infty$.

Example 2.35 Approximate the value of $[2^n (n!)^2] / (2n)!$ for large n .

Solution: By Stirling's formula, we have

$$\frac{2^n (n!)^2}{(2n)!} \sim \frac{2^n 2\pi n (n^n)^2 e^{-2n}}{\sqrt{4\pi n} (2n)^{2n} e^{-2n}} = \frac{\sqrt{\pi n}}{2^n}. \quad \blacklozenge$$

EXERCISE

1. Use Stirling's formula to approximate $\binom{2n}{n} \frac{1}{2^{2n}}$ and $\frac{[(2n)!]^3}{[(4n)! (n!)^2]}$ for large n .



2.6 COMBINATORIAL SIMULATIONS IN R

In R, the command **factorial(n)** calculates $n!$, while **choose(n,k)** determines $\binom{n}{k}$. Factorials increase at a rapid rate. In R, the highest factorial it can directly compute without yielding **Inf** (infinity) is typically **factorial(170)**. Attempting **factorial(171)** results in **Inf**. While this limitation suffices for many applications, those requiring higher precision or calculations involving larger factorials might consider using certain specialized packages. However, we should be cautious as operations with enormous numbers can be memory-intensive. Also, when using large factorials in mathematical operations, we must always be mindful of issues related to numerical stability.

Example 2.36 During a commencement ceremony, a clerk arranges the diplomas in the order in which the students will walk onto the stage. However, the last n diplomas are accidentally shuffled into a random order. Write an R program to simulate and approximate the probability that at least one of the final n graduates receives their correct diploma from the dean.

Solution A straightforward R program efficiently simulates the problem and provides an excellent approximation for the desired probability. In the program:

1. **num**: Represents the number of times the simulation will run to estimate the probability.
2. **k <- 0**: Initializes a counter to track how often at least one graduate receives their correct diploma during the simulations.
3. **1:n** vector: Denotes the correct order of the diplomas.
4. **sample(n)**: Generates a randomly shuffled version of the last n diplomas, simulating their accidental mixing.
5. The key command, **if(any(sample(n) == 1:n)) {k <- k+1}**, checks if any graduates in the shuffled order receive their correct diploma:
6. **any(sample(n) == 1:n)** portion: Tests if there is any position where the shuffled and actual orders match, which would indicate a graduate receiving the correct diploma.
7. If a match is detected, the counter k increments by one.

After all the simulations, the approximate probability, p , that at least one of the last n graduates receives his or her correct diploma is computed. This is done by dividing the number of successful trials, k , by the total number of trials, **num**. Here is the program that simulates this problem:

```
> k <- 0
> n <- 10
> num <- 15000
> for(i in 1:num) {
+   if(any(sample(n)==1:n)) {k <- k+1}
```

```
+ }
> p <- k/num
> print(p)
[1] 0.6284
```

This problem recounts an actual event experienced by the author of this book, involving the last 10 diplomas given to graduating students as they walked onto the stage. While the scenario mirrors the *absentminded professor* problem from Example 2.28, it appears here in an entirely different context. For $n = 10$, the precise solution from that example, rounded to eight decimal places, is 0.63212054. The result from the aforementioned simulation closely approximates this. When we executed the program with `num=50000`, the result was 0.63284, which closely aligns with the exact solution. ♦

Example 2.37 Using R simulation, determine the estimated probability that in a class of 87 students, at least four share the same birthday. Assume a consistent birth rate throughout the year with each year having 365 days.

Solution: We will start by crafting a function `t` that accepts two parameters: `n=87`, representing the class size, and `num=100000`, indicating the number of simulation iterations for estimating the probability. When invoked, this function runs the simulation `num` times, assessing in each iteration whether at least four students out of the n share the same birthday. The result is an estimated probability derived from the cumulative outcomes.

The R command `table`, when applied to a vector, enumerates the occurrences of each unique value. During each simulation iteration, the program randomly designates birthdays to all 87 students, forming a vector `v`. Each entry, `v[j]` for $1 \leq j \leq 87$, denotes the birthday of the j th student. Therefore, the j th entry of `table(v)` provides the count of students celebrating their birthday on the same day as the j th student.

Across the 100,000 simulations, 100,000 `table(v)` instances are produced. The program identifies and tallies instances where `table(v)` contains at least one value greater than or equal to 4. This count, when divided by `num`, yields the desired probability. Below is the program that simulates the given scenario:

```
> t <- function(n=87, num=100000) {
+ k <- 0
+ for(i in 1:num) {
+ v <- sample(1:365, n, replace=TRUE)
+ counts <- table(v)
+ if(any(counts >= 4)) {
+ k <- k+1
+ }
+ }
+ p <- k/num
+ return(p)
```

```

+}
> p <- t()
> print(p)
[1] 0.03649

```

Below is a streamlined version of this program, eliminating the use of functions for simplicity.

```

> num <- 15000
> n <- 87
> k <- 0
> for(i in 1:15000){
+ v <- sample(365,n,replace=TRUE)
+ counts <- table(v)
+ if(any(counts>=4)){k <- k+1}
+}
> p <- k/num
> print(p)
[1] 0.03573333

```



EXERCISES

Warning: In some combinatorial probability problems, the desired quantity is very small. For this reason, in simulations, the number of experiments should be very large to get good approximations.

1. Assume there are 18 customers in a queue at a box office. Half of them, i.e., nine customers, have \$10 bills, while the other nine have \$20 bills. Each ticket is priced at \$10. Initially, the box office does not have any change. Use an R program to simulate this scenario and estimate the likelihood that none of the customers has to wait to receive change.
2. A cereal company randomly inserts one of its 20 different prizes into each cereal box.
 - (a) Julie purchased 50 cereal boxes from the company. Develop a simulation in R to estimate the likelihood that she obtains all 20 prizes.
 - (b) Assuming Jim aims for a minimum 50% probability of obtaining all 20 prizes, design an R simulation to determine the estimated least number of cereal boxes he needs to purchase.
3. Nine students stood in a particular order and were photographed. A year later, these same students randomly arranged themselves for another photograph. Using an R simulation, determine the approximate probability that in the second photo, none of the students stood adjacent to the same person they stood next to in the first photograph.

4. Employ a simulation in R to estimate the likelihood that, out of 50 students in a class, precisely four share a common birthday while the remaining 46 have unique birthdays. Assume an even distribution of birthdays across the 365 days of the year.
5. Consider a regular 10-gon. Randomly select two pairs of its vertices and draw lines connecting them. Use R simulation to determine the approximate likelihood that these lines do not intersect.

Hint: The precise probability value sought is

$$\frac{\frac{1}{11} \binom{20}{10}}{\binom{20}{2} \binom{18}{2}} \approx 0.58.$$

6. Over the past 10 days, an executive penned seven letters daily, and his secretary readied seven corresponding envelopes. Each day's envelopes were bundled in a packet, maintaining chronological order within the bundle. However, the letters were never placed inside the envelopes. Today, the secretary discovers the mix-up: all 70 letters and the 10 envelope packets are jumbled. If she randomly picks envelope packets and inserts arbitrarily selected letters, what is the estimated probability, determined via a simulation in R, that no letter matches its intended envelope? It's worth noting that while envelopes within a packet remain in their daily sequence, the packets themselves are not in order.
Answer: Roughly 0.53416. For a detailed analysis of this topic, refer to the article by Steve Fisk in the April 1988 issue of *Mathematics Magazine*. ■

CHAPTER 2 SUMMARY

► **Generalized Counting Principle:** Let $E_1, E_2, \dots, E_k$ be sets with $n_1, n_2, \dots, n_k$ elements, respectively. Then there are $n_1 \times n_2 \times n_3 \times \dots \times n_k$ ways in which we can, first, choose an element of E_1 , then an element of E_2 , then an element of $E_3, \dots$, and finally an element of E_k .

► **Number of Subsets of a Set:** A set with n elements has 2^n subsets.

► **r -Element Permutations:** An ordered arrangement of r objects from a set A containing n objects ($0 < r \leq n$) is called an r -element permutation of A , or a permutation of the elements of A taken r at a time. The number of r -element permutations of a set containing n objects is denoted by ${}_nP_r$. We have that ${}_nP_r = \frac{n!}{(n-r)!}$. An n -element permutation of a set with n objects is simply called a *permutation*. The number of permutations of a set containing n elements, ${}_nP_n$, is $n!$.

► The number of distinguishable permutations of n objects of k different types, where n_1 are alike, n_2 are alike, $\dots$, n_k are alike and $n = n_1 + n_2 + \dots + n_k$, is

$$\frac{n!}{n_1! \times n_2! \times \dots \times n_k!}.$$

► **r -Element Combinations:** An unordered arrangement of r objects from a set A containing n objects ($r \leq n$) is called an r -element combination of A , or a combination of the elements of A taken r at a time. The number of r -element permutations of a set containing n objects is denoted by ${}_nC_r$ or $\binom{n}{r}$. We have that ${}_nC_r \equiv \binom{n}{r} = \frac{n!}{(n-r)!r!}$.

► **Binomial Expansion** For any integer $n \geq 0$,

$$(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^{n-i} y^i.$$

► **Multinomial Expansion** In the expansion of

$$(x_1 + x_2 + \dots + x_k)^n,$$

the coefficient of the term $x_1^{n_1} x_2^{n_2} x_3^{n_3} \dots x_k^{n_k}$, $n_1 + n_2 + \dots + n_k = n$, is

$$\frac{n!}{n_1! n_2! \dots n_k!}.$$

Therefore,

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{n_1+n_2+\dots+n_k=n} \frac{n!}{n_1! n_2! \dots n_k!} x_1^{n_1} x_2^{n_2} x_3^{n_3} \dots x_k^{n_k}.$$

Note that the sum is taken over all nonnegative integers $n_1, n_2, \dots, n_k$ such that $n_1 + n_2 + \dots + n_k = n$.

REVIEW PROBLEMS

1. Albert goes to the grocery store to buy fruit. There are seven different varieties of fruit, and Albert is determined to buy no more than one of any variety. How many different orders can he place?
2. In how many ways can we divide 9 toys among 3 children evenly?

3. Virginia has 1 one-dollar bill, 1 two-dollar bill, 1 five-dollar bill, 1 ten-dollar bill, and 1 twenty-dollar bill. She decides to give some money to her son Brian without asking for change. How many choices does she have?
4. To enhance children's STEM skills, a teacher gives each student 12 Lego bricks but asks them to use only 8 of them to construct a toy. If no two of the 12 bricks are identical, how many choices does each child have for his or her 8 Lego bricks?
5. If four fair dice are tossed, what is the probability that they will show four different faces?
6. From the 10 points that are placed on a circumference, two are selected randomly. What is the probability that they are adjacent?
7. A father buys nine different toys for his four children. In how many ways can he give one child three toys and the remaining three children two toys each?
8. A window dresser has decided to display five different dresses in a circular arrangement. How many choices does she have?
9. A student club has 50 members, of which 9 have volunteered themselves to serve on the executive committee. One of the volunteers, Ruth, is known to be a bully. If the faculty members in charge of the club choose the executive committee members at random from the volunteers, what is the probability that Ruth is included?
10. In a Napa Valley winery, guests are invited to a tasting room, and, for each of them, an affable staff member pours 6 types of wine, which are sold at different prices, in 6 small 3-ounce stylish glasses, at random and one at a time. What is the probability that the first glass of wine they serve a visitor is the most expensive brand and the last glass is the least expensive one?
11. A list of all permutations of 13579 is put in increasing order. What is the 100th number in the list?
Hint: The first $4! = 24$ numbers all begin with 1.
12. At a departmental party, Dr. James P. Coughlin distributes 4 baby photos of 4 of his colleagues among the guests with a list of the names of those colleagues. Each guest is invited to identify the babies in the photos. What is the probability that a participant can identify exactly 2 of the 4 photos correctly by purely random guessing?
13. In a soccer tryout, 25 players compete. The coach has decided to cut 8 of the players, assign 1 as a goalie, 4 as defenders, 3 as midfielders, 3 as forwards, and the remaining 6 as substitutes. How many choices does he have?
14. After each lecture, a professor assigns 10 problems. However, he grades only 4 of them randomly. If Salina's solution to only one of the recently assignment problems is wrong, what is the probability that she still gets a perfect score?
15. Judy has three sets of classics in literature, each set having four volumes. In how many ways can she put them on a bookshelf so that books of each set are not separated?
16. Korina is returning from a trip to the United Kingdom and is packing 12 wrapped chocolate bars of which 3 are Aero Caramels, 4 are Chomps, 3 are Duncans, and 2 are Freddos.

The bars of the same brand are indistinguishable. In how many distinguishable ways can Korina pack these chocolate bars?

17. Suppose that 30 lawn mowers, of which seven have defects, are sold to a hardware store. If the store manager inspects six of the lawn mowers randomly, what is the probability that he finds at least one defective lawn mower?
18. One of Zoey's iPod playlists has 12 songs, 5 by Beyoncé, 3 by Adele, and 4 by Céline Dion. She sets the ipod to play the songs of that playlist. However, before doing that Zoey unchecks and then rechecks the shuffle button of her iPod, and the iPod plays the playlist's songs in a random order, without repeating a song. What is the probability that all songs by Beyoncé are played back to back?
19. In how many ways can 23 identical refrigerators be allocated among four stores so that one store gets eight refrigerators, another four, a third store five, and the last one six refrigerators?
20. From a group of 8 male and 16 female potential jurors, a jury of 12 is selected at random. What is the probability that all males are included?
21. In how many arrangements of the letters *BERKELEY* are all three *E*'s adjacent?
22. Some watches feature a hand that displays the seconds. Despite the primary purpose of a second hand being to show the time accurately, the experience reveals that at any given moment, it appears to display a number between 1 and 60 at random. Nurses often wear watches with second hands to count respiratory and heart rates. Suppose, at a conference, 25 nurses are wearing watches with second hands. Find the probability that, at an instant, the second hands of at least two nurses' watches display the same number.
23. Show that for $n = 1, 2, 3, \dots$, $\binom{2n}{n} < 4^n$.
24. Bill and John play in a backgammon tournament. A player is the winner if he wins three games in a row or four games altogether. In what percent of all possible cases does the tournament end because John wins four games without winning three in a row?
25. For his calculus final exam, Dr. Channing, whose classes have a reputation for being very easy, gives his students 20 problems with their complete solutions, and tells them that he will choose 10 of them randomly for the final exam. Students who solve all 10 problems correctly will get an A, those who solve 9 correctly will get a B, and those who solve only 8 correctly will get a C. Dr. Channing told them that he does not give partial credit, and whoever solves fewer than 8 problems correctly, will get an F. Dolly, who has not studied calculus all semester, has no idea how to do any of the problems. However, she finds enough time to memorize only the solutions to 12 of the problems. What is the probability that she gets A, B, C and F, respectively?
26. For a card game tournament, 16 individuals must be divided into 4 groups of 4 each. In how many ways can this be done?
27. In a small town, both of the accidents that occurred during the week of June 8, 1988, were on Friday the 13th. Is this a good excuse for a superstitious person to argue that Friday the 13th's are inauspicious?

28. How many eight-digit numbers without two identical successive digits are there?
29. A broadcasting company recruits college students to help with their Olympic coverage. Suppose that three of their finalists are female athletes, and the remaining four are male athletes. If they choose four of them randomly, what is the probability that (a) two of them are female and two are male? (b) Two of them are the tallest females and two of them are the tallest males of the finalists? Assume that no two of the finalists are of the same height.
30. A **palindrome** is a sequence of characters that reads the same forward or backward. For example, *rotator*, *madam*, *Hannah*, the German name *Otto*, and an Indian language, *Malayalam*, are palindromes. So are the following expressions: “Put up,” “Madam I’m Adam,” “Was it a cat I saw?” and these two sentences in Latin concerning St. Martin, Bishop of Tours: “Signa te, signa; temere me tangis et angis. Roma tibi subito motibus ibit amor.” (Respectively: Cross, cross yourself; you annoy and vex me needlessly. Through my exertions, Rome, your desire, will soon be near.) Determine the number of palindromes containing 11 characters that can be made with (a) no letters repeating more than twice; (b) one letter repeating three times and no other letter more than twice (the chemical term *detartrated* is one such palindrome); (c) one letter repeating three times, two each repeating two times, and one repeating four times.
Historical Remark: It is said that the palindrome was invented by a Greek poet named Sotades in the third century B.C. It is also said that Sotades was drowned by order of a king of the Macedonian dynasty, the reigning Ptolemy, who found him a real bore.
31. In a lottery, players pick 5 different integers between 1 and 47, and the order of selection is irrelevant. The lottery commission then randomly selects 5 of these as the *winning numbers*. A player wins the grand prize if all 5 numbers that he or she has selected match the winning numbers. Nicole was born on April 12, 1946. Based on her birthday, she has purchased all tickets that include the numbers 4, 12, and 46. What is the probability that Nicole wins the grand prize?
32. Suppose that four women and two men enter a restaurant and sit at random around a table that has four chairs on one side and another four on the other side. What is the probability that the men are not all sitting on one side?
33. If five Americans, five Italians, and five Mexicans sit randomly at a round table, what is the probability that the persons of the same nationality sit together?
34. In a bridge game, each of the four players gets 13 random cards. What is the probability that every player has an ace?
35. An urn contains 15 white and 15 black balls. Suppose that 15 persons each draw two balls blindfolded from the urn without replacement. What is the probability that each of them draws one white ball and one black ball?
36. In a lottery the tickets are numbered 1 through N . A person purchases n ($1 \leq n \leq N$) tickets at random. What is the probability that the ticket numbers are consecutive? (This is a special case of a problem posed by Euler in 1763.)
37. The chair of an academic department needs to form 3 search committees each consisting of 5 tenured full professors. If the department has 18 such faculty, how many choices

does the chair have if he decides that no one can serve on more than one search committee? Note that a set of 5 professors serving on one search committee is distinguishable from the same professors serving on another search committee.

38. An ordinary deck of 52 cards is dealt, 13 each, at random among A, B, C, and D. What is the probability that (a) A and B together get two aces; (b) A gets all the face cards; (c) A gets five hearts and B gets the remaining eight hearts?
39. To test if a computer program works properly, we run it with 12 different data sets, using four computers, each running three data sets. If the data sets are distributed randomly among different computers, how many possibilities are there?
40. A four-digit number is selected at random. What is the probability that its ones place is greater than its tens place, its tens place is greater than its hundreds place, and its hundreds place is greater than its thousands place? Note that the first digit of an n -digit number is nonzero.
41. From the set of integers $\{1, 2, 3, \dots, 100000\}$ a number is selected at random. What is the probability that the sum of its digits is 8?

Hint: Establish a one-to-one correspondence between the set of integers from $\{1, 2, \dots, 100000\}$ the sum of whose digits is 8, and the set of possible ways 8 identical objects can be placed into 5 distinguishable cells. Then use Example 2.25.



Self-Test on Chapter 2

Time allotted: 120 Minutes

Each problem is worth 10 points.

1. What is the probability that exactly two students of a class of size 23 have the same birthday? Assume that the birth rates are constant throughout the year and each year has 365 days.
2. In a ternary code, each code has 7 bits, each of which is 0, 1, or 2. What is the probability that a random code ends in 2 and has exactly three 0s?
3. The card game, bridge, featuring two teams of two players each, is played with an ordinary deck of 52 cards. The cards are dealt among the players, 13 each, randomly. What is the probability that exactly 3 of the aces are in the hands of one team?
4. Possible passwords for a computer account are strings of length 6 of numbers, capital letters, and small letters, with no repetition allowed (all capital letters are distinguishable from all small letters. So a string such as AaBz8b has no repetition). What is the probability that a password generated for this computer, by a random password creation tool, has 2 digits, 2 capital letters, and 2 small letters?

5. Each state of the 50 in the United States has two senators. For each state, the senator with greater longevity in the chamber is the *senior senator*, and the other one is the *junior senator* of that state. Since no state has both senators up for election in the same year, every state always has a senior senator and a junior senator. If a committee of 12 U.S. senators is selected at random, find the probability that (a) no two senators are from the same state; (b) there are 6 senior senators and 6 junior senators, no two from the same state, on the committee.
6. There are 15 trees in a row of which 5 adjacent ones are infected by a fungal disease. Is this logical evidence for an arborist to conclude that the fungus spread from one tree to next through root-to-root contact?
Hint: Suppose that the disease strikes randomly, say, through airborne spores, and calculate the probability that five adjacent trees become infected.
7. Give a combinatorial proof for the relation

$$\binom{n}{k} \binom{k}{1} = \binom{n}{1} \binom{n-1}{k-1}.$$

Using this relation, show that

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}.$$

8. There are 48 students to be randomly distributed among 3 different jewelry classes, 16 students per class. If 3 of the students are visually impaired, find the probability that (a) each class gets one of the them; (b) all 3 end up in the same class.
9. Let $\ell < k < m < n$ be positive integers. There will be n new movies released next month. Suppose that a prominent movie critic, Mr. Wilbert, whose reviews are syndicated to more than 200 newspapers, is required to review exactly m movies next month for publication.
- (a) How many choices does he have?
 - (b) If k of the new releases are specified, and Mr. Wilbert is required to review them as part of his m reviews, how many choices does he have?
 - (c) If k of the new releases are specified, and Mr. Wilbert is required to review at least ℓ of them as part of his m reviews, how many choices does he have?
10. From a faculty of six professors, six associate professors, ten assistant professors, and twelve instructors, a committee of size six is formed randomly. What is the probability that there is at least one person from each rank on the committee?

Hint: Be careful, the answer is not

$$\frac{\binom{6}{1} \binom{6}{1} \binom{10}{1} \binom{12}{1} \binom{30}{2}}{\binom{34}{6}} = 1.397.$$

To find the correct answer, use the inclusion-exclusion principle explained in [Section 1.4](#): Let A_1 , A_2 , A_3 , and A_4 be the events that there is no professor, no associate professor, no assistant professor, and no instructor in the committee, respectively. The desired probability is

$$P(A_1^c A_2^c A_3^c A_4^c) = 1 - P(A_1 \cup A_2 \cup A_3 \cup A_4).$$



Chapter 3

Conditional Probability and Independence

3.1 CONDITIONAL PROBABILITY

To introduce the notion of conditional probability, let us first examine the following question: Suppose that all of the first-year students of an engineering college took calculus and discrete math last semester. Suppose that 70% of the students passed calculus, 55% passed discrete math, and 45% passed both. If a randomly selected first-year student is found to have passed calculus last semester, what is the probability that he or she also passed discrete math last semester? To answer this question, let A and B be the events that the randomly selected student passed discrete math and calculus last semester, respectively. Note that the quantity we are asked to find is not $P(A)$, which is 0.55; it would have been if we were not aware that B has occurred. Knowing that B has occurred changes the chances of the occurrence of A . To find the desired probability, denoted by the symbol $P(A | B)$ [read: probability of A given B] and called the **conditional probability** of A given B , let n be the number of all of the first-year students in the engineering college. Then $(0.7)n$ is the number of first-year students who passed calculus, and $(0.45)n$ is the number of those who passed both calculus and discrete math. Therefore, of the $(0.7)n$ first-year students who passed calculus, $(0.45)n$ of them passed discrete math as well. It is given that the randomly selected student is one of the $(0.7)n$ who passed calculus; we also want to find the probability that he or she is one of the $(0.45)n$ who passed discrete math. This is obviously equal to $(0.45)n/(0.7)n = 0.45/0.7$. Hence

$$P(A | B) = \frac{0.45}{0.7}.$$

But 0.45 is $P(AB)$ and 0.7 is $P(B)$. Therefore, this example suggests that

$$P(A | B) = \frac{P(AB)}{P(B)}. \quad (3.1)$$

As another example, suppose that a bus arrives at a station every day at a random time between 1:00 P.M. and 1:30 P.M. Suppose that at 1:10 the bus has not arrived and we want to calculate the probability that it will arrive in not more than 10 minutes from now. Let A be the event that the bus will arrive between 1:10 and 1:20 and B be the event that it will arrive

between 1:10 and 1:30. Then the desired probability is

$$P(A | B) = \frac{20 - 10}{30 - 10} = \frac{\frac{20 - 10}{30 - 0}}{\frac{30 - 10}{30 - 0}} = \frac{P(AB)}{P(B)}. \quad (3.2)$$

The relation $P(A | B) = P(AB)/P(B)$ discovered in (3.1) and (3.2) can also be verified for other types of conditional probability problems. It is compatible with our intuition and is a logical generalization of the existing relation between the conditional relative frequency of the event A with respect to the condition B and the ratio of the relative frequencies of AB and the relative frequency of B . For these reasons it is taken as a definition.

Definition 3.1 If $P(B) > 0$, the **conditional probability** of A given B , denoted by $P(A | B)$, is

$$P(A | B) = \frac{P(AB)}{P(B)}. \quad (3.3)$$

If $P(B) = 0$, formula (3.3) makes no sense, so the conditional probability is defined only for $P(B) > 0$. This formula may be expressed in words by saying that the conditional probability of A given that B has occurred is the ratio of the probability of joint occurrence of A and B and the probability of B . It is important to note that (3.3) is neither an axiom nor a theorem. It is a definition. As explained previously, this definition is fully justified and is not made arbitrarily. Indeed, historically, it was used implicitly even before it was formally introduced by De Moivre in the book *The Doctrine of Chance*.

To illustrate the conditional probability of A given B using a Venn diagram, recognize that once we know B has taken place, the sample space of potential outcomes is no longer S ; it becomes B . As depicted in Figure 3.1, the event A transitions to AB , since if B has happened, A can only occur concurrently with B . Hence, the relative frequency of event A conditional on B corresponds to the relative frequency of AB divided by that of B . This insight informs Definition 3.1.

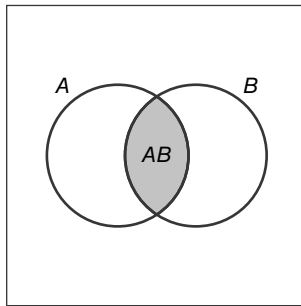


Figure 3.1 Venn diagram to illustrate the conditional probability of A given B .

Example 3.1 In a certain region of Russia, the probability that a person lives at least 80 years is 0.75, and the probability that he or she lives at least 90 years is 0.63. What is the probability that a randomly selected 80-year-old person from this region will survive to become 90?

Solution: Let A and B be the events that the person selected survives to become 90 and 80 years old, respectively. We are interested in $P(A | B)$. By definition,

$$P(A | B) = \frac{P(AB)}{P(B)}.$$

Now, $P(AB) = P(A)$ because “at least 80” and “at least 90” overlap with “at least 90,” which is A . Hence

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{P(A)}{P(B)} = \frac{0.63}{0.75} = 0.84. \quad \blacklozenge$$

Example 3.2 From the set of all families with two children, a *family* is selected at random and is found to have a girl. What is the probability that the other child of the family is a girl? Assume that in a two-child family all sex distributions are equally probable.

Solution: Let B and A be the events that the family has a girl and the family has two girls, respectively. We are interested in $P(A | B)$. Now, in a family with two children there are four equally likely possibilities: (boy, boy), (girl, girl), (boy, girl), (girl, boy), where by, say, (girl, boy), we mean that the older child is a girl and the younger is a boy. Thus $P(B) = 3/4$, $P(AB) = 1/4$. Hence

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{1/4}{3/4} = \frac{1}{3}. \quad \blacklozenge$$

Example 3.3 From the set of all families with two children, a *child* is selected at random and is found to be a girl. What is the probability that the second child of this girl’s family is also a girl? Assume that in a two-child family all sex distributions are equally probable.

Solution: Let B be the event that the child selected at random is a girl, and let A be the event that the second child of her family is also a girl. We want to calculate $P(A | B)$. Now the set of all possibilities is as follows: The child is a girl with a sister, the child is a girl with a brother, the child is a boy with a sister, and the child is a boy with a brother. Thus $P(B) = 2/4$ and $P(AB) = 1/4$. Hence

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{1/4}{2/4} = \frac{1}{2}. \quad \blacklozenge$$

Remark 3.1 There is a major difference between Examples 3.2 and 3.3. In Example 3.2, a family is selected and found to have a girl. Therefore, we have three equally likely possibilities: (girl, girl), (boy, girl), and (girl, boy), of which only one is desirable: (girl, girl). So the required probability is $1/3$. In Example 3.3, since a child rather than a family is selected,

families with (girl, girl), (boy, girl), and (girl, boy) are not equally likely to be chosen. In fact, the probability that a family with two girls is selected equals twice the probability that a family with one girl is selected. That is,

$$P(\text{girl, girl}) = 2P(\text{girl, boy}) = 2P(\text{boy, girl}).$$

This and the fact that the sum of the probabilities of (girl, girl), (girl, boy), (boy, girl) is 1 give

$$P(\text{girl, girl}) + \frac{1}{2}P(\text{girl, girl}) + \frac{1}{2}P(\text{girl, girl}) = 1,$$

which implies that

$$P(\text{girl, girl}) = \frac{1}{2}. \quad \blacklozenge$$

Example 3.4 An English class consists of 10 Koreans, 5 Italians, and 15 Hispanics. A paper is found belonging to one of the students of this class. If the name on the paper is not Korean, what is the probability that it is Italian? Assume that names completely identify ethnic groups.

Solution: Let A be the event that the name on the paper is Italian, and let B be the event that it is not Korean. To calculate $P(A | B)$, note that $AB = A$ because “the name being Italian,” and “the name not being Korean” overlap with “the name being Italian,” which is A . Hence $P(AB) = P(A) = 5/30$. Therefore,

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{5/30}{20/30} = \frac{1}{4}. \quad \blacklozenge$$

Example 3.5 (Cromwell’s Rule) The British leading Bayesian statistician and decision theorist, Dennis Lindley (1923–2013), stated a rule in subjective probability that we should refrain from using probabilities 0 and 1 except when assigning probability to logically true or false statements. He called this *Cromwell’s Rule*. Oliver Cromwell (1599–1658), an English military and political leader, in 1650, in a letter to the synod of the Church of Scotland wrote[†] “I beseech you, in the bowels of Christ, think it possible that you may be mistaken.” This well-known phrase was the reason for Lindley naming his precept, Cromwell’s rule. The reason for Cromwell’s rule is that if, for an event A , $P(A) = 1$, then for any event B with $P(B) > 0$, $P(A | B) = 1$. That is, **if you are certain that A is absolutely true, then no amount of evidence to the contrary will change your mind.** To prove this fact, note that, by Theorem 1.7,

$$P(B) = P(BA) + P(BA^c).$$

However, $BA^c \subseteq A^c$. So $0 \leq P(BA^c) \leq P(A^c) = 0$, and hence $P(BA^c) = 0$. This implies that $P(B) = P(BA)$ and $P(A | B) = \frac{P(AB)}{P(B)} = 1$. $\blacklozenge$

[†]Carlyle, Thomas, ed. (1855). *Oliver Cromwell’s letters and speeches*. 1. New York: Harper. p. 448.

Example 3.6 We draw eight cards at random from an ordinary deck of 52 cards. Given that three of them are spades, what is the probability that the remaining five are also spades?

Solution: Let B be the event that at least three of them are spades, and let A denote the event that all of the eight cards selected are spades. Then $P(A \mid B)$ is the desired quantity and is calculated from $P(A \mid B) = P(AB)/P(B)$ as follows:

$$P(AB) = \frac{\binom{13}{8}}{\binom{52}{8}}$$

since AB is the event that all eight cards selected are spades.

$$P(B) = \sum_{x=3}^8 \frac{\binom{13}{x} \binom{39}{8-x}}{\binom{52}{8}}$$

because at least three of the cards selected are spades if exactly three of them are spades, or exactly four of them are spades, $\dots$, or all eight of them are spades. Hence

$$P(A \mid B) = \frac{P(AB)}{P(B)} = \frac{\frac{\binom{13}{8}}{\binom{52}{8}}}{\sum_{x=3}^8 \frac{\binom{13}{x} \binom{39}{8-x}}{\binom{52}{8}}} \approx 5.44 \times 10^{-6}. \quad \blacklozenge$$

An important feature of conditional probabilities (stated in the following theorem) is that they satisfy the same axioms that ordinary probabilities satisfy. This enables us to use the theorems that are true for probabilities for conditional probabilities as well.

Theorem 3.1 Let S be the sample space of an experiment, and let B be an event of S with $P(B) > 0$. Then

- (a) $P(A \mid B) \geq 0$ for any event A of S .
- (b) $P(S \mid B) = 1$.

(c) If $A_1, A_2, \dots$ is a sequence of mutually exclusive events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i \mid B\right) = \sum_{i=1}^{\infty} P(A_i \mid B).$$

The proof of this theorem is left as an exercise.

Reduction of Sample Space

Let B be an event of a sample space S with $P(B) > 0$. For a subset A of B , define $Q(A) = P(A \mid B)$. Then Q is a function from the set of subsets of B to $[0, 1]$. Clearly, $Q(A) \geq 0$, $Q(B) = P(B \mid B) = 1$ and, by Theorem 3.1, if $A_1, A_2, \dots$ is a sequence of mutually exclusive subsets of B , then

$$Q\left(\bigcup_{i=1}^{\infty} A_i\right) = P\left(\bigcup_{i=1}^{\infty} A_i \mid B\right) = \sum_{i=1}^{\infty} P(A_i \mid B) = \sum_{i=1}^{\infty} Q(A_i).$$

Thus Q satisfies the axioms that probabilities do, and hence it is a probability function. Note that while P is defined for all subsets of S , the probability function Q is defined only for subsets of B . Therefore, for Q , the sample space is reduced from S to B . This reduction of sample space is sometimes very helpful in calculating conditional probabilities. Suppose that we are interested in $P(E \mid B)$, where $E \subseteq S$. One way to calculate this quantity is to reduce S to B and find $Q(EB)$. It is usually easier to compute the unconditional probability $Q(EB)$ rather than the conditional probability $P(E \mid B)$. Examples follow.

Example 3.7 A child mixes 10 good and three dead batteries. To find the dead batteries, his father tests them one-by-one and without replacement. If the first four batteries tested are all good, what is the probability that the fifth one is dead?

Solution: Using the information that the first four batteries tested are all good, we rephrase the problem in the reduced sample space: Six good and three dead batteries are mixed. A battery is selected at random: What is the probability that it is dead? The solution to this trivial question is $3/9 = 1/3$. Note that without reducing the sample space, the solution of this problem is not so easy. ♦

Example 3.8 A farmer decides to test four fertilizers for his soybean fields. He buys 32 bags of fertilizers, eight bags from each kind, and tries them randomly on 32 plots, eight plots from each of fields A, B, C, and D, one bag per plot. If from type I fertilizer one bag is tried on field A and three on field B, what is the probability that two bags of type I fertilizer are tried on field D?

Solution: Using the information that one and three bags of type I fertilizer are tried on fields A and B, respectively, we rephrase the problem in the reduced sample space: 16 bags of fertilizers (of which four are type I) are tried randomly, eight on field C and eight on field D. What is the

probability that two bags of type I fertilizer are tried on field D? The solution to this easier unconditional version of the problem is

$$\frac{\binom{4}{2} \binom{12}{6}}{\binom{16}{8}} \approx 0.43. \quad \blacklozenge$$

Example 3.9 Monty Hall Problem The television game show *Let's Make a Deal*, which debuted in 1963, featured Monty Hall (1921–2017) as its host for its first 30 years. Picture this: Olivia is a contestant on a show similar to *Let's Make a Deal*, hosted by Monty Hall. She is presented with three doors. Behind one is a car (the prize she desires), and behind the other two are goats. Olivia picks a door. Monty, being privy to what is behind each door, proceeds to open another door, revealing a goat. He then offers Olivia a choice: stay with her initial pick or switch to the remaining unopened door. The crux of the Monty Hall problem is this: Is Olivia better off sticking with her first choice, switching, or is the outcome unaffected by her decision?

Solution: We will show that the conditional probability that Olivia wins, given that she always changes her original choice is $2/3$. Therefore, she should definitely change her choice. To show this, suppose that the prize is behind Door 1, and Olivia always changes her choice. These two assumptions reduce the sample space. The elements of the reduced sample space can be described by 3-tuples (x, y, z) , where x is the door Olivia guesses first, y is the door Monty opens, and z is the door Olivia switches to. For example, $(3, 2, 1)$ represents a game in which Olivia guesses Door 3, Monty pulls back Door 2, and Olivia switches to Door 1. Therefore, given that the prize is behind Door 1 and Olivia changes her choice, the reduced sample space S is

$$S = \{(1, 2, 3), (1, 3, 2), (2, 3, 1), (3, 2, 1)\}.$$

Note that the event that Olivia guesses Door 2 is $\{(2, 3, 1)\}$, the event that she guesses Door 3 is $\{(3, 2, 1)\}$, whereas the event that she guesses Door 1 is $\{(1, 2, 3), (1, 3, 2)\}$. Given that the prize is behind Door 1 and Olivia always switches her original choice, the event that she wins is $\{(2, 3, 1), (3, 2, 1)\}$. Since she guesses a door with probability $1/3$,

$$P(\{(2, 3, 1)\}) = P(\{(3, 2, 1)\}) = P(\{(1, 2, 3), (1, 3, 2)\}) = \frac{1}{3}.$$

This shows that no matter what values are assigned to $P(\{(1, 2, 3)\})$ and $P(\{(1, 3, 2)\})$, as long as their sum is $1/3$, the conditional probability that Olivia wins, given that she always changes her original choice, is

$$P(\{(2, 3, 1)\}) + P(\{(3, 2, 1)\}) = \frac{2}{3}. \quad \blacklozenge$$

By Theorem 3.1, the function Q defined above by $Q(A) = P(A \mid B)$ is a probability function. Therefore, $Q(A) = P(A \mid B)$ satisfies the theorems stated for P . In particular, for all choices of B , $P(B) > 0$,

1. $P(\emptyset \mid B) = 0$.
2. $P(A^c \mid B) = 1 - P(A \mid B)$.
3. If $C \subseteq A$, then $P(AC^c \mid B) = P(A - C \mid B) = P(A \mid B) - P(C \mid B)$.
4. If $C \subseteq A$, then $P(C \mid B) \leq P(A \mid B)$.
5. $P(A \cup C \mid B) = P(A \mid B) + P(C \mid B) - P(AC \mid B)$.
6. $P(A \mid B) = P(AC \mid B) + P(AC^c \mid B)$.
7. To calculate $P(A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_n \mid B)$, we calculate conditional probabilities of all possible intersections of events from $\{A_1, A_2, \dots, A_n\}$, given B , add the conditional probabilities obtained by intersecting an odd number of the events, and subtract the conditional probabilities obtained by intersecting an even number of events.
8. For any increasing or decreasing sequences of events $\{A_n, n \geq 1\}$,

$$\lim_{n \rightarrow \infty} P(A_n \mid B) = P\left(\lim_{n \rightarrow \infty} A_n \mid B\right).$$

EXERCISES

A

1. Suppose that 15% of the population of a country are unemployed women, and a total of 25% are unemployed. What percent of the unemployed are women?
2. Suppose that 41% of Americans have blood type A, and 4% have blood type AB. If in the blood of a randomly selected American soldier the A antigen is found, what is the probability that his blood type is A? The A antigen is found only in blood types A and AB.
3. In a technical college all students are required to take calculus and physics. Statistics show that 32% of the students of this college get A's in calculus, and 20% of them get A's in both calculus and physics. Gino, a randomly selected student of this college, has passed calculus with an A. What is the probability that he got an A in physics?
4. Suppose that two fair dice have been tossed and the total of their top faces is found to be divisible by 5. What is the probability that both of them have landed 5?
5. A bus arrives at a station every day at a random time between 1:00 P.M. and 1:30 P.M. A person arrives at this station at 1:00 and waits for the bus. If at 1:15 the bus has not yet arrived, what is the probability that the person will have to wait at least an additional 5 minutes?
6. Prove that $P(A \mid B) > P(A)$ if and only if $P(B \mid A) > P(B)$. In probability, if for two events A and B , $P(A \mid B) > P(A)$, we say that A and B are **positively correlated**. If $P(A \mid B) < P(A)$, A and B are said to be **negatively correlated**.

7. A spinner is mounted on a wheel of unit circumference (radius $1/2\pi$). Arcs A , B , and C of lengths $1/3$, $1/2$, and $1/6$, respectively, are marked off on the wheel's perimeter (see Figure 3.2). The spinner is flicked and we know that it is not pointing toward C . What is the probability that it points toward A ?

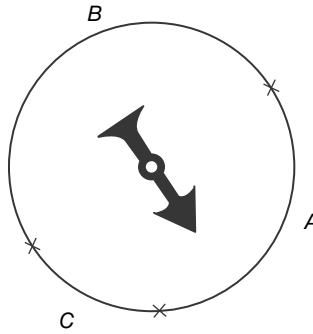


Figure 3.2 Spinner of Exercise 7.

8. In throwing two fair dice, what is the probability of a sum of 5 if they land on different numbers?
9. In a small lake, it is estimated that there are approximately 105 fish, of which 40 are trout and 65 are carp. A fisherman caught eight fish; what is the probability that exactly two of them are trout if we know that at least three of them are not?
10. From 100 cards numbered $00, 01, \dots, 99$, one card is drawn. Suppose that α and β are the sum and the product, respectively, of the digits of the card selected. Calculate $P(\{\alpha = i \mid \beta = 0\})$, $i = 0, 1, 2, 3, \dots, 18$.
11. From families with three children, a family is selected at random and found to have a boy. What is the probability that the boy has (a) an older brother and a younger sister; (b) an older brother; (c) a brother and a sister? Assume that in a three-child family all gender distributions have equal probabilities.
12. In a study of the records of 831 women who died in 2016, an actuary observed that 185 of them died of cancer. Furthermore, she discovered that the mothers of 257 of the 831 women suffered from cancer, and from the mothers of all who died of cancer 93 had died due to cancer. Find the probability that a randomly selected woman from this group died of cancer if her mother did not have cancer.
13. Show that if $P(A) = 1$, then $P(B \mid A) = P(B)$.
14. Prove that if $P(A) = a$ and $P(B) = b$, then $P(A \mid B) \geq (a + b - 1)/b$.
15. Prove Theorem 3.1.
16. Prove that if $P(E \mid F) \geq P(G \mid F)$ and $P(E \mid F^c) \geq P(G \mid F^c)$, then $P(E) \geq P(G)$.
17. Suppose that 28 crayons, of which four are red, are divided randomly among Jack,

Marty, Sharon, and Martha (seven each). If Sharon has exactly one red crayon, what is the probability that Marty has the remaining three?

18. Adam and three of his friends are playing bridge. (a) If, holding a certain hand, Adam announces that he has a king, what is the probability that he has at least one more king? (b) If, for some other hand, Adam announces that he has the king of diamonds, what is the probability that he has at least one more king? Compare parts (a) and (b) and explain why the answers are not the same.
19. An actuary studying the insurance preferences of homeowners in a region that is prone to earthquakes, hurricanes, and floods has discovered that, for each of these perils, the probability is 0.2 that a homeowner, selected randomly, has purchased coverage only for that peril. Moreover, she has observed that, for any two of the perils, the probability is 0.14 that the homeowner has purchased coverage for exactly those two perils. If one-third of the homeowners who have purchased a policy that covers earthquakes and hurricanes also purchased flood insurance, what percentage of homeowners did not purchase a policy that covered any of these perils?

B

20. A number is selected at random from the set $\{1, 2, \dots, 10,000\}$ and is observed to be odd. What is the probability that it is (a) divisible by 3; (b) divisible by neither 3 nor 5?
21. A retired person chooses randomly one of the six parks of his town everyday and goes there for hiking. We are told that he was seen in one of these parks, Oregon Ridge, once during the last 10 days. What is the probability that during this period he has hiked in this park two or more times?
22. A big urn contains 1000 red chips, numbered 1 through 1000, and 1750 blue chips, numbered 1 through 1750. A chip is removed at random, and its number is found to be divisible by 3. What is the probability that its number is also divisible by 5?
23. There are three types of animals in a laboratory: 15 type I, 13 type II, and 12 type III. Animals of type I react to a particular stimulus in 5 seconds, animals of types II and III react to the same stimulus in 4.5 and 6.2 seconds, respectively. A psychologist selects 10 of these animals at random and finds that exactly four of them react to this stimulus in 6.2 seconds. What is the probability that at least two of them react to the same stimulus in 4.5 seconds?
24. In an international school, 60 students, of whom 15 are Korean, 20 are French, eight are Greek, and the rest are Chinese, are divided randomly into four classes of 15 each. If there are a total of eight French and six Korean students in classes A and B, what is the probability that class C has four of the remaining 12 French and three of the remaining nine Korean students?
25. From the set of all families with two children, a *family* is selected at random and is found to have a girl called Mary. We want to know the probability that both children of the family are girls. By Example 3.2, the probability should apparently be $1/3$ because presumably knowing the name of the girl should not make a difference. However, if we ask a friend of that family whether Mary is the older or the younger child of the family,

upon receiving the answer, we can conclude that the probability of the other child being a girl is $1/2$. Therefore, there is no need to ask the friend, and we conclude already that the probability is $1/2$! What is the flaw in this argument? Explain. ■

Self-Quiz on Section 3.1

Time allotted: 20 Minutes

Each problem is worth 2.5 points.

1. In a bridge game, played with a normal deck of 52 cards, hearts is the designated trump suit. If the first 4 cards a player is dealt are ace, 5, 9, and the king, all of hearts, what is the probability that none of the remaining 9 cards the player is dealt are hearts?
Hint: Reduce the sample space.
2. For a certain loaded die, the probabilities of the possible outcomes are given by the following table:

Outcome	1	2	3	4	5	6
Probability	0.27	0.15	0.17	0.2	0.05	0.16

If the die is tossed and the outcome is an odd number, what is the probability that it is 1?

3. The theaters of a town are showing seven comedies and nine dramas. Marlon has seen five of the movies. If the first three movies he has seen are dramas, what is the probability that the last two are comedies? Assume that Marlon chooses the shows at random and sees each movie at most once.
Hint: Reduce the sample space.
4. For events E and F , suppose that $P(EF) = 0.23$, $P(E \cup F) = 0.67$, and $P(E | F) = 0.46$. Find $P(F | E)$. ■

Often, when we think of a collection of events, we have a tendency to think about them in either temporal or logical sequence. So, if, for example, a sequence of events $A_1, A_2, \dots, A_n$ occur in time or in some logical order, we can usually immediately write down the probabilities $P(A_1), P(A_2 | A_1), \dots, P(A_n | A_1 A_2 \cdots A_{n-1})$ without much computation. However, we may be interested in probabilities of intersection of events, or probabilities of events unconditional on the rest, or probabilities of earlier events, given later events. In the next three sections, we will develop techniques for calculating such probabilities.

Suppose that, in a random phenomenon, events occur in either temporal or logical sequence. In [Section 3.2](#) (The Multiplication Rule), we will discuss the probabilities of the intersection of events in terms of conditional probabilities of later events given earlier events. In [Section 3.3](#) (Law of Total Probability), we will find the probabilities of unconditional events in

terms of conditional probabilities of later events given the earlier ones. In [Section 3.4](#) (Bayes' Formula), we will discuss a formula relating probabilities of earlier events, given later events, to the conditional probabilities of later events given the earlier ones.

3.2 THE MULTIPLICATION RULE

The relation

$$P(A | B) = \frac{P(AB)}{P(B)}$$

is also useful for calculating $P(AB)$. If we multiply both sides of this relation by $P(B)$ [note that $P(B) > 0$], we get

$$P(AB) = P(B)P(A | B), \quad (3.4)$$

which means that the probability of the joint occurrence of A and B is the product of the probability of B and the conditional probability of A given that B has occurred. If $P(A) > 0$, then by letting $A = B$ and $B = A$ in (3.4), we obtain

$$P(BA) = P(A)P(B | A).$$

Since $P(BA) = P(AB)$, this relation gives

$$P(AB) = P(A)P(B | A). \quad (3.5)$$

Thus, to calculate $P(AB)$, depending on which of the quantities $P(A | B)$ and $P(B | A)$ is known, we may use (3.4) or (3.5), respectively. The following example clarifies the usefulness of these relations.

Example 3.10 Suppose that five good fuses and two defective ones have been mixed up. To find the defective fuses, we test them one-by-one, at random and without replacement. What is the probability that we are lucky and find both of the defective fuses in the first two tests?

Solution: Let D_1 and D_2 be the events of finding a defective fuse in the first and second tests, respectively. We are interested in $P(D_1 D_2)$. Using (3.5), we get

$$P(D_1 D_2) = P(D_1)P(D_2 | D_1) = \frac{2}{7} \times \frac{1}{6} = \frac{1}{21}. \quad \blacklozenge$$

Relation (3.5) can be generalized for calculating the probability of the joint occurrence of several events. For example, if $P(AB) > 0$, then

$$P(ABC) = P(A)P(B | A)P(C | AB). \quad (3.6)$$

To see this, note that $P(AB) > 0$ implies that $P(A) > 0$. Therefore,

$$P(A)P(B | A)P(C | AB) = P(A) \frac{P(AB)}{P(A)} \frac{P(ABC)}{P(AB)} = P(ABC).$$

The following theorem will generalize (3.5) and (3.6) to n events and can be shown in the same way as (3.6) is shown.

Theorem 3.2 (The Multiplication Rule) *If $P(A_1 A_2 A_3 \cdots A_{n-1}) > 0$, then*

$$\begin{aligned} P(A_1 A_2 A_3 \cdots A_{n-1} A_n) \\ = P(A_1)P(A_2 | A_1)P(A_3 | A_1 A_2) \cdots P(A_n | A_1 A_2 A_3 \cdots A_{n-1}). \end{aligned}$$

Example 3.11 A consulting firm is awarded 43% of the contracts it bids on. Suppose that Nordulf works for a division of the firm that gets to do 15% of the projects contracted for. If Nordulf directs 35% of the projects submitted to his division, what percentage of all bids submitted by the firm will result in contracts for projects directed by Nordulf?

Solution: Let A_1 be the event that the firm is awarded a contract for some randomly selected bid. Let A_2 be the event that the contract will be sent to Nordulf's division. Let A_3 be the event that Nordulf will direct the project. The desired probability is calculated as follows:

$$P(A_1 A_2 A_3) = P(A_1)P(A_2 | A_1)P(A_3 | A_1 A_2) = (0.43)(0.15)(0.35) = 0.0226.$$

Hence 2.26% of all bids submitted by the firm will be directed by Nordulf. ♦

Example 3.12 Suppose that five good and two defective fuses have been mixed up. To find the defective ones, we test them one by one, at random and without replacement. What is the probability that we find both of the defective fuses in exactly three tests?

Solution: Let D_1, D_2 , and D_3 be the events that the first, second, and third fuses tested are defective, respectively. Let G_1, G_2 , and G_3 be the events that the first, second, and third fuses tested are good, respectively. We are interested in $P(G_1 D_2 D_3 \cup D_1 G_2 D_3)$, which we calculate using the multiplication rule:

$$\begin{aligned} P(G_1 D_2 D_3 \cup D_1 G_2 D_3) \\ = P(G_1 D_2 D_3) + P(D_1 G_2 D_3) \\ = P(G_1)P(D_2 | G_1)P(D_3 | G_1 D_2) + P(D_1)P(G_2 | D_1)P(D_3 | D_1 G_2) \\ = \frac{5}{7} \times \frac{2}{6} \times \frac{1}{5} + \frac{2}{7} \times \frac{5}{6} \times \frac{1}{5} \approx 0.095. \quad \blacklozenge \end{aligned}$$

EXERCISES

A

1. In a trial, the judge is 65% sure that Susan has committed a crime. Robert is a witness who knows whether Susan is innocent or guilty. However, Robert is Susan's friend and will lie with probability 0.25 if Susan is guilty. He will tell the truth if she is innocent. What is the probability that Robert will commit perjury?

2. There are 14 marketing firms hiring new graduates. Kate randomly found the recruitment ads of six of these firms and sent them her resume. If three of these marketing firms are in Maryland, what is the probability that Kate did not apply to a marketing firm in Maryland?
3. In a game of cards, two cards of the same color and denomination form a pair. For example, 8 of hearts and 8 of diamonds is one pair, king of spades and king of clubs is another. If six cards are selected at random and without replacement, what is the probability that there will be no pairs?
4. If eight defective and 12 nondefective items are inspected one by one, at random and without replacement, what is the probability that (a) the first four items inspected are defective; (b) from the first three items at least two are defective?
5. An actuary works for an auto insurance company whose customers all have insured at least one car. She discovers that of all the customers who insure more than one car, 70% of them insure at least one SUV. Furthermore, she observes that 47% of all the customers insure at least one SUV. If 65% of all customers insure more than one car, what percentage of the customers insure exactly one non-SUV?
6. There are five boys and six girls in a class. For an oral exam, their teacher calls them one by one and randomly. (a) What is the probability that the boys and the girls alternate? (b) What is the probability that the boys are called first? Compare the answers to parts (a) and (b).
7. In a lottery scratch-off game, each ticket has 10 coated circles in the middle and one coated rectangle in the lower left corner. Underneath the coats of 4 of the circles, there is a dollar sign, "\$," and underneath the remaining 6 circles is blank. A winning ticket is the one with only those circles that contain the dollar sign scratched. The remaining circles and the rectangle must be left untouched. The winning amount is written underneath the coat of the rectangle, which will be scratched off by the lottery agent to reveal the amount won. Harlan buys one such ticket and scratches off four of the circles at random. Find the probability that (a) he wins; (b) he wins if the first two circles he scratches off have the dollar sign underneath.
8. An urn contains five white and three red chips. Each time we draw a chip, we look at its color. If it is red, we replace it along with two new red chips, and if it is white, we replace it along with three new white chips. What is the probability that, in successive drawing of chips, the colors of the first four chips alternate?
9. The law school of a university admits all applicants who have at least a 3.5 undergraduate GPA and a Law School Admissions Test (LSAT) score of 154 or higher. Suppose that of all students with a 3.5 or higher GPA, only 45% score at least 154 the first time they take the LSAT test; of all those students who scored less than 154 the first time, 58% will score 154 or higher the second time they take the test; and of all who scored lower than 154 both the first and second times, 75% will score 154 or higher the third time they take the test. Ladona has achieved the minimum GPA required by this law school. What is the probability that she does not need to take the test more than three times to earn at least the minimum LSAT score required by the law school?

10. In the card game, bridge, played with an ordinary deck of 52 cards, all cards are dealt among four players, 13 each, randomly. What is the probability that each player gets one ace?

Hint: Let A_1 be the event that the ace of hearts is dealt to one of the four players. Let A_2 be the event that the ace of hearts and ace of diamonds are dealt to two different players. Let A_3 be the event that the ace of hearts, ace of diamonds, and ace of spades are dealt to three different players, and let A_4 be the event that each player gets exactly one of the aces. Note that $A_4 \subseteq A_3 \subseteq A_2 \subseteq A_1$, and the desired probability is $P(A_4) = P(A_1 A_2 A_3 A_4)$.

11. Suppose that 75% of all people with credit records improve their credit ratings within 3 years. Suppose that 18% of the population at large have poor credit records, and of those only 30% will improve their credit ratings within 3 years. What percentage of the people who will improve their credit records within the next 3 years are the ones who currently have good credit ratings?

B

12. Cards are drawn at random from an ordinary deck of 52, one by one and without replacement. What is the probability that no heart is drawn before the ace of spades is drawn?

13. From an ordinary deck of 52 cards, cards are drawn one by one, at random and without replacement. What is the probability that the fourth heart is drawn on the tenth draw?

Hint: Let F denote the event that in the first nine draws there are exactly three hearts, and E be the event that the tenth draw is a heart. Use the multiplication rule: $P(FE) = P(F)P(E | F)$.

14. In a series of games, the winning number of the n th game, $n = 1, 2, 3, \dots$, is a number selected at random from the set of integers $\{1, 2, \dots, n + 2\}$. Don bets on 1 in each game and says that he will quit as soon as he wins. What is the probability that he has to play indefinitely?

Hint: Let A_n be the event that Don loses the first n games. To calculate the desired probability, $P(\bigcap_{i=1}^{\infty} A_i)$, use Theorem 1.8.



Self-Quiz on Section 3.2

Time allotted: 15 Minutes

Each problem is worth 5 points.

1. An urn contains 6 blue and 4 red balls. Three balls are drawn at random and without replacement. What is the probability that the balls drawn are alternatively of different colors?

2. On a given day, the first item produced by a manufacturer is defective with probability p and non-defective with probability $1 - p$. However, whether or not an item produced afterward is defective depends only on the item that was produced right before it. It is defective with probability p_1 if the previous item produced is non-defective, and it is defective with probability p_2 if the previous item produced is defective. Find the probability that, on a randomly selected working day, the first four items produced by this manufacturer are all non-defective.

■

3.3 LAW OF TOTAL PROBABILITY

Sometimes it is not possible to calculate directly the probability of the occurrence of an event A , but it is possible to find $P(A | B)$ and $P(A | B^c)$ for some event B . In such cases, the following theorem, which is conceptually rich and has widespread applications, is used. It states that $P(A)$ is the weighted average of the probability of A given that B has occurred and probability of A given that it has not occurred.

Theorem 3.3 (Law of Total Probability) *Let B be an event with $P(B) > 0$ and $P(B^c) > 0$. Then for any event A ,*

$$P(A) = P(A | B)P(B) + P(A | B^c)P(B^c).$$

Proof: By Theorem 1.7,

$$P(A) = P(AB) + P(AB^c). \quad (3.7)$$

Now $P(B) > 0$ and $P(B^c) > 0$. These imply that $P(AB) = P(A | B)P(B)$ and $P(AB^c) = P(A | B^c)P(B^c)$. Putting these in (3.7), we have proved the theorem. ♦

Example 3.13 An insurance company rents 35% of the cars for its customers from agency I and 65% from agency II. If 8% of the cars of agency I and 5% of the cars of agency II break down during the rental periods, what is the probability that a car rented by this insurance company breaks down?

Solution: Let A be the event that a car rented by this insurance company breaks down. Let I and II be the events that it is rented from agencies I and II, respectively. Then by the law of total probability,

$$\begin{aligned} P(A) &= P(A | I)P(I) + P(A | II)P(II) \\ &= (0.08)(0.35) + (0.05)(0.65) = 0.0605. \end{aligned}$$

Tree diagrams facilitate solutions to this kind of problem. Let B and B^c stand for breakdown and not breakdown during the rental period, respectively. Then, as Figure 3.3 shows, to find the probability that a car breaks down, all we need to do is to compute, by multiplication, the probability of each path that leads to a point B and then add them up. So, as seen from the tree, the probability that the car breaks down is $(0.35)(0.08) + (0.65)(0.05) = 0.06$, and the probability that it does not break down is $(0.35)(0.92) + (0.65)(0.95) = 0.94$.

♦

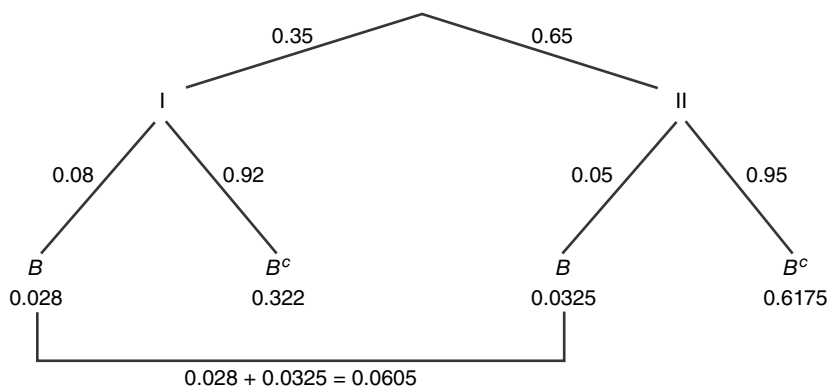


Figure 3.3 Tree diagram of Example 3.13.

Example 3.14 In a trial, the judge is 65% sure that Susan has committed a crime. Julie and Robert are two witnesses who know whether Susan is innocent or guilty. However, Robert is Susan’s friend and will lie with probability 0.25 if Susan is guilty. He will tell the truth if she is innocent. Julie is Susan’s enemy and will lie with probability 0.30 if Susan is innocent. She will tell the truth if Susan is guilty. What is the probability that, in the course of the trial, Robert and Julie will give conflicting testimony?

Solution: Let I be the event that Susan is innocent. Let C be the event that Robert and Julie will give conflicting testimony. By the law of total probability,

$$\begin{aligned} P(C) &= P(C | I)P(I) + P(C | I^c)P(I^c) \\ &= (0.30)(.35) + (0.25)(.65) = 0.2675. \quad \blacklozenge \end{aligned}$$

Example 3.15 (Gambler’s Ruin Problem) Two gamblers play the game of “heads or tails,” in which each time a fair coin lands heads up player A wins \$1 from B, and each time it lands tails up, player B wins \$1 from A. Suppose that player A initially has a dollars and player B has b dollars. If they continue to play this game successively, what is the probability that (a) A will be ruined; (b) the game goes forever with nobody winning?

Solution:

- (a) Let E be the event that A will be ruined if he or she starts with i dollars, and let $p_i = P(E)$. Our aim is to calculate p_a . To do so, we define F to be the event that A wins the first game. Then

$$P(E) = P(E | F)P(F) + P(E | F^c)P(F^c).$$

In this formula, $P(E | F)$ is the probability that A will be ruined, given that he wins the first game; so $P(E | F)$ is the probability that A will be ruined if his capital is $i + 1$; that is, $P(E | F) = p_{i+1}$. Similarly, $P(E | F^c) = p_{i-1}$. Hence

$$p_i = p_{i+1} \cdot \frac{1}{2} + p_{i-1} \cdot \frac{1}{2}. \quad (3.8)$$

Now $p_0 = 1$ because if A starts with 0 dollars, he or she is already ruined. Also, if the capital of A reaches $a + b$, then B is ruined; thus $p_{a+b} = 0$. Therefore, we have to solve the system of recursive equations (3.8), subject to the boundary conditions $p_0 = 1$ and $p_{a+b} = 0$. To do so, note that (3.8) implies that

$$p_{i+1} - p_i = p_i - p_{i-1}.$$

Hence, letting $p_1 - p_0 = \alpha$, we get

$$p_i - p_{i-1} = p_{i-1} - p_{i-2} = p_{i-2} - p_{i-3} = \cdots = p_2 - p_1 = p_1 - p_0 = \alpha.$$

Thus

$$\begin{aligned} p_1 &= p_0 + \alpha \\ p_2 &= p_1 + \alpha = p_0 + \alpha + \alpha = p_0 + 2\alpha \\ p_3 &= p_2 + \alpha = p_0 + 2\alpha + \alpha = p_0 + 3\alpha \\ &\vdots \\ p_i &= p_0 + i\alpha \\ &\vdots \end{aligned}$$

Now $p_0 = 1$ gives $p_i = 1 + i\alpha$. But $p_{a+b} = 0$; thus $0 = 1 + (a + b)\alpha$. This gives $\alpha = -1/(a + b)$; therefore,

$$p_i = 1 - \frac{i}{a + b} = \frac{a + b - i}{a + b}.$$

In particular, $p_a = b/(a + b)$. That is, the probability that A will be ruined is $b/(a + b)$.

- (b) The same method can be used with obvious modifications to calculate q_i , the probability that B is ruined if he or she starts with i dollars. The result is

$$q_i = \frac{a + b - i}{a + b}.$$

Since B starts with b dollars, he or she will be ruined with probability $q_b = a/(a + b)$. Thus the probability that the game goes on forever with nobody winning is $1 - (q_b + p_a)$. But $1 - (q_b + p_a) = 1 - a/(a + b) - b/(a + b) = 0$. Therefore, if this game is played successively, eventually either A is ruined or B is ruined. ♦

Remark 3.2 If the game is not fair and on each play gambler A wins \$1 from B with probability p , $0 < p < 1$, $p \neq 1/2$, and loses \$1 to B with probability $q = 1 - p$, relation (3.8) becomes

$$p_i = pp_{i+1} + qp_{i-1},$$

but $p_i = (p + q)p_i$, so $(p + q)p_i = pp_{i+1} + qp_{i-1}$. This gives

$$q(p_i - p_{i-1}) = p(p_{i+1} - p_i),$$

which reduces to

$$p_{i+1} - p_i = \frac{q}{p}(p_i - p_{i-1}).$$

Using this relation and following the same line of argument lead us to

$$p_i = \left[\left(\frac{q}{p} \right)^{i-1} + \left(\frac{q}{p} \right)^{i-2} + \cdots + \left(\frac{q}{p} \right) + 1 \right] \alpha + p_0,$$

where $\alpha = p_1 - p_0$. Using $p_0 = 1$, we obtain

$$p_i = \frac{(q/p)^i - 1}{(q/p) - 1} \alpha + 1.$$

After finding α from $p_{a+b} = 0$ and substituting in this equation, we get

$$p_i = \frac{(q/p)^i - (q/p)^{a+b}}{1 - (q/p)^{a+b}} = \frac{1 - (p/q)^{a+b-i}}{1 - (p/q)^{a+b}},$$

where the last equality is obtained by multiplying the numerator and denominator of the function by $(p/q)^{a+b}$. In particular,

$$p_a = \frac{1 - (p/q)^b}{1 - (p/q)^{a+b}},$$

and, similarly,

$$q_b = \frac{1 - (q/p)^a}{1 - (q/p)^{a+b}}.$$

In this case also, $p_a + q_b = 1$, meaning that eventually either A or B will be ruined and the game does not go on forever. For comparison, suppose that A and B both start with \$10. If they play a fair game, the probability that A will be ruined is 1/2, and the probability that B will be ruined is also 1/2. If they play an unfair game with $p = 3/4$, $q = 1/4$, then p_{10} , the probability that A will be ruined, is almost 0.00002. ♦

To generalize Theorem 3.3, we will now state a definition.

Definition 3.2 Let $\{B_1, B_2, \dots, B_n\}$ be a set of nonempty subsets of the sample space S of an experiment. If the events $B_1, B_2, \dots, B_n$ are mutually exclusive and $\bigcup_{i=1}^n B_i = S$, the set $\{B_1, B_2, \dots, B_n\}$ is called a **partition** of S .

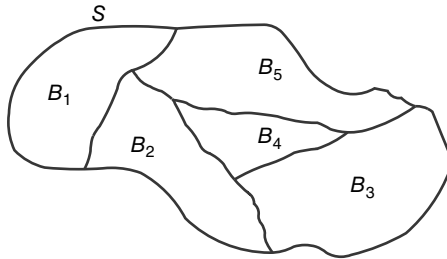


Figure 3.4 Partition of the given sample space S .

For example, if S of Figure 3.4 is the sample space of an experiment, then B_1, B_2, B_3, B_4 , and B_5 of the same figure form a partition of S . As another example, consider the experiment of drawing a card from an ordinary deck of 52 cards. Let B_1, B_2, B_3 , and B_4 denote the events that the card is a spade, a club, a diamond, and a heart, respectively. Then $\{B_1, B_2, B_3, B_4\}$ is a partition of the sample space of this experiment. If $A_i, 1 \leq i \leq 10$, denotes the event that the value of the card drawn is i , and A_{11}, A_{12} , and A_{13} are the events of jack, queen, and king, respectively, then $\{A_1, A_2, \dots, A_{13}\}$ is another partition of the same sample space. Note that

For an experiment with sample space S , for any event A , A and A^c both nonempty, the set $\{A, A^c\}$ is a partition.

As observed, Theorem 3.3 is used whenever it is not possible to calculate $P(A)$ directly, but it is possible to find $P(A | B)$ and $P(A | B^c)$ for some event B . In many situations, it is neither possible to find $P(A)$ directly, nor possible to find a single event B that enables us to use

$$P(A) = P(A | B)P(B) + P(A | B^c)P(B^c).$$

In such situations Theorem 3.4, which is a generalization of Theorem 3.3 and is also called the **law of total probability**, might be applicable.

Theorem 3.4 (Law of Total Probability) *If $\{B_1, B_2, \dots, B_n\}$ is a partition of the sample space of an experiment and $P(B_i) > 0$ for $i = 1, 2, \dots, n$, then for any event A of S ,*

$$\begin{aligned} P(A) &= P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + \dots + P(A | B_n)P(B_n) \\ &= \sum_{i=1}^n P(A | B_i)P(B_i). \end{aligned}$$

More generally, let $\{B_1, B_2, \dots\}$ be a sequence of mutually exclusive events of S such that $\bigcup_{i=1}^{\infty} B_i = S$. Suppose that, for all $i \geq 1$, $P(B_i) > 0$. Then for any event A of S ,

$$P(A) = \sum_{i=1}^{\infty} P(A | B_i)P(B_i).$$

Proof: Since $B_1, B_2, \dots, B_n$ are mutually exclusive, $B_i B_j = \emptyset$ for $i \neq j$, it follows that $(AB_i)(AB_j) = \emptyset$ for $i \neq j$. Hence $\{AB_1, AB_2, \dots, AB_n\}$ is a set of mutually exclusive events. Now

$$S = B_1 \cup B_2 \cup \dots \cup B_n$$

gives

$$A = AS = AB_1 \cup AB_2 \cup \dots \cup AB_n;$$

(see Figure 3.5) therefore,

$$P(A) = P(AB_1) + P(AB_2) + \dots + P(AB_n).$$

But $P(AB_i) = P(A | B_i)P(B_i)$ for $i = 1, 2, \dots, n$, so

$$P(A) = P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + \dots + P(A | B_n)P(B_n).$$

The proof of the more general case is similar. ♦

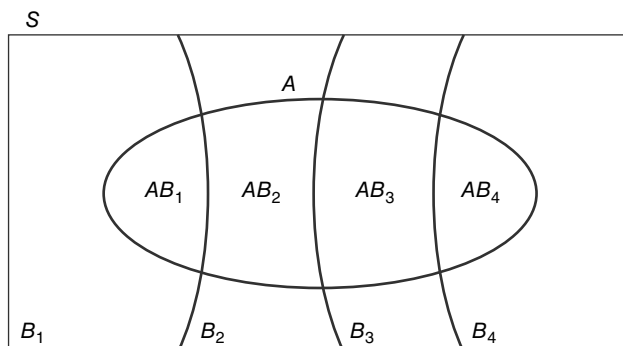


Figure 3.5 Venn diagram to illustrate $A = AB_1 \cup AB_2 \cup \cdots \cup AB_n$.

When using Theorem 3.4, one should be very careful to choose $B_1, B_2, \dots$, so that they form a partition of the sample space.

Example 3.16 Suppose that 80% of the seniors, 70% of the juniors, 50% of the sophomores, and 30% of the freshmen of a college use the library of their campus frequently. If 30% of all students are freshmen, 25% are sophomores, 25% are juniors, and 20% are seniors, what percent of all students use the library frequently?

Solution: Let U be the event that a randomly selected student is using the library frequently. Let F, O, J , and E be the events that he or she is a first-year student, sophomore, junior, or senior, respectively. Then $\{F, O, J, E\}$ is a partition of the sample space. Thus

$$\begin{aligned} P(U) &= P(U | F)P(F) + P(U | O)P(O) + P(U | J)P(J) + P(U | E)P(E) \\ &= (0.30)(0.30) + (0.50)(0.25) + (0.70)(0.25) + (0.80)(0.20) = 0.55. \end{aligned}$$

Therefore, 55% of these students use the campus library frequently. The same calculation can be carried out readily from the tree diagram of Figure 3.6, where U means they use the library frequently and N means that they do not. ♦

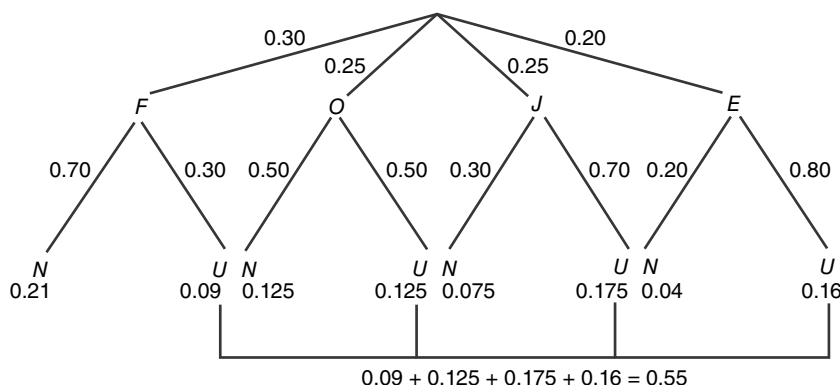


Figure 3.6 Tree diagram of Example 3.16.

Example 3.17 Suppose that the only parasite living in an aquatic habitat is a single-celled organism, which after a second, with equal probabilities, either splits into two organisms, remains as is, or dies. Suppose that in subsequent seconds, all the living parasites in the habitat follow the same behavior as the original parasite, independently of each other. What is the probability that eventually the aquatic habitat will be clean with no parasites?

Solution: The aquatic habitat will eventually be clean if ultimately the parasite population dies out. Let A be the event that this happens. Let T , R , and D be the events that, after a second, the parasite splits into two organisms, remains the same, and dies, respectively. By the law of total probability,

$$\begin{aligned} P(A) &= P(A | T)P(T) + P(A | R)P(R) + P(A | D)P(D) \\ &= [P(A)]^2 \cdot \frac{1}{3} + P(A) \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}. \end{aligned}$$

This gives $[P(A)]^2 - 2P(A) + 1 = 0$, or equivalently $[P(A) - 1]^2 = 0$. So $P(A) = 1$, and thus eventually the aquatic habitat will be clean with no parasite. ♦

Example 3.18 An urn contains 10 white and 12 red chips. Two chips are drawn at random and, without looking at their colors, are discarded. What is the probability that a third chip drawn is red?

Solution: For $i \geq 1$, let R_i be the event that the i th chip drawn is red and W_i be the event that it is white. Intuitively, it should be clear that the two discarded chips provide no information, so $P(R_3) = 12/22$, the same as if it were the first chip drawn from the urn. To prove this mathematically, note that $\{R_2W_1, W_2R_1, R_2R_1, W_2W_1\}$ is a partition of the sample space; therefore,

$$\begin{aligned} P(R_3) &= P(R_3 | R_2W_1)P(R_2W_1) + P(R_3 | W_2R_1)P(W_2R_1) \\ &\quad + P(R_3 | R_2R_1)P(R_2R_1) + P(R_3 | W_2W_1)P(W_2W_1). \end{aligned} \quad (3.9)$$

Now

$$P(R_2W_1) = P(R_2 | W_1)P(W_1) = \frac{12}{21} \times \frac{10}{22} = \frac{20}{77},$$

$$P(W_2R_1) = P(W_2 | R_1)P(R_1) = \frac{10}{21} \times \frac{12}{22} = \frac{20}{77},$$

$$P(R_2R_1) = P(R_2 | R_1)P(R_1) = \frac{11}{21} \times \frac{12}{22} = \frac{22}{77},$$

and

$$P(W_2W_1) = P(W_2 | W_1)P(W_1) = \frac{9}{21} \times \frac{10}{22} = \frac{15}{77}.$$

Substituting these values in (3.9), we get

$$P(R_3) = \frac{11}{20} \times \frac{20}{77} + \frac{11}{20} \times \frac{20}{77} + \frac{10}{20} \times \frac{22}{77} + \frac{12}{20} \times \frac{15}{77} = \frac{12}{22}. \quad \blacklozenge$$

EXERCISES

A

1. If 5% of men and 0.25% of women are color blind, what is the probability that a randomly selected person is color blind?
2. Suppose that 40% of the students of a campus are women. If 20% of the women and 16% of the men of this campus are A students, what percent of all of them are A students?
3. A random number is selected from the interval $(0, 1]$. Which one of the following is a partition of the sample space of this experiment? Which one is not?

(a) $\{(0, 1/2), (1/2, 1]\}$,

(b) $\{(0, 1/3], (1/3, 1]\}$,

(c) $\{(0, 2/5], [2/5, 1]\}$,

(d) $\left\{\left(\frac{i-1}{n}, \frac{i}{n}\right] : n \geq 1 \text{ is given and } 1 \leq i \leq n\right\}$.
4. Jim has three cars of different models: A, B, and C. The probabilities that models A, B, and C use over 3 gallons of gasoline from Jim's house to his work are 0.25, 0.32, and 0.53, respectively. On a certain day, all three of Jim's cars have 3 gallons of gasoline each. Jim chooses one of his cars at random, and without paying attention to the amount of gasoline in the car drives it toward his office. What is the probability that he makes it to the office?
5. One of the cards of an ordinary deck of 52 cards is lost. What is the probability that a random card drawn from this deck is a spade?
6. Two cards from an ordinary deck of 52 cards are missing. What is the probability that a random card drawn from this deck is a spade?
7. Of the patients in a hospital, 20% of those with, and 35% of those without myocardial infarction have had strokes. If 40% of the patients have had myocardial infarction, what percent of the patients have had strokes?
8. Suppose that 37% of a community are at least 45 years old. If 80% of the time a person who is 45 or older tells the truth, and 65% of the time a person below 45 tells the truth, what is the probability that a randomly selected person answers a question truthfully?
9. A person has six guns. The probability of hitting a target when these guns are properly aimed and fired is 0.6, 0.5, 0.7, 0.9, 0.7, and 0.8, respectively. What is the probability of hitting a target if a gun is selected at random, properly aimed, and fired?

10. A factory produces its entire output with three machines. Machines I, II, and III produce 50%, 30%, and 20% of the output, but 4%, 2%, and 4% of their outputs are defective, respectively. What fraction of the total output is defective?
11. When traveling from Springfield, Massachusetts, to Baltimore, Maryland, 30% of the time Charles takes I-91 south and then I-95 south, and 70% of the time he takes I-91 south, then I-84 west followed by I-81 south and I-83 south. Choosing the first option takes Charles a random time between 5 hours and 30 minutes and 10 hours depending upon the traffic. However, choosing the second option takes a random time between 6 hours and 30 minutes and 8 hours. The second option is a longer drive but has much less traffic. What is the probability that on his next trip from Springfield to Baltimore, Charles travels no more than 7 hours and 15 minutes?
12. Solve the following problem, from the “Ask Marilyn” column of *Parade Magazine*, October 29, 2000.

I recently returned from a trip to China, where the government is so concerned about population growth that it has instituted strict laws about family size. In the cities, a couple is permitted to have only one child. In the countryside, where sons traditionally have been valued, if the first child is a son, the couple may have no more children. But if the first child is a daughter, the couple may have another child. Regardless of the sex of the second child, no more are permitted. How will this policy affect the mix of males and females?

To pose the question mathematically, what is the probability that a randomly selected child from the countryside is a boy?

13. At a gas station, 85% of the customers use 87 octane gasoline, 5% use 91 octane, and 10% use 93 octane. Suppose that, 70%, 85%, and 95% of 87 octane users, 91 octane users, and 93 octane users fill their tanks, respectively. What is the probability that the next customer entering this station to put gas in his or her vehicle will fill the tank?
14. Suppose that five coins, of which exactly three are gold, are distributed among five persons, one each, at random, and one by one. Are the chances of getting a gold coin equal for all participants? Why or why not?
15. In a town, $\frac{7}{9}$ th of the men and $\frac{3}{5}$ th of the women are married. In that town, what fraction of the adults are married? Assume that all married adults are the residents of the town.
16. A child gets lost in the Disneyland at the Epcot Center in Florida. The father of the child believes that the probability of his being lost in the east wing of the center is 0.75 and in the west wing is 0.25. The security department sends an officer to the east and an officer to the west to look for the child. If the probability that a security officer who is looking in the correct wing finds the child is 0.4, find the probability that the child is found.
17. An actuary has discovered that, in her company, 65% of those who have only income protection insurance and 80% of those who have only legal expense insurance will renew their policies next year. Furthermore, she has observed that 87% of those who have both

of these policies will renew at least one of them next year. If 58% of policyholders of this company have income protection insurance, 43% have legal expense insurance, and 23% have both policies, what percentage of the policy holders will renew at least one policy next year?

18. A number is selected at random from the set $\{1, 2, \dots, 20\}$. Then a second number is selected randomly between 1 and the first number selected. What is the probability that the second number is 5?
19. Meteorologist 1 forecasts a 0.8 probability of rain for tomorrow. Meteorologist 2 gives a 0.65 probability for rain on the same day. Assuming both meteorologists are equally trustworthy and only one of their forecasts is conducted accurately. Demonstrate that, under the assumptions of this problem, the combined estimate for the likelihood of rain tomorrow, based on the two predictions, is the average of 0.8 and 0.65, resulting in 0.725.
20. Suppose that there exist N families on the earth and that the maximum number of children a family has is c . Let α_j ($0 \leq j \leq c$, $\sum_{j=0}^c \alpha_j = 1$) be the fraction of families with j children. Find the fraction of all children in the world who are the k th born of their families ($k = 1, 2, \dots, c$).
21. Let B be an event of a sample space S with $P(B) > 0$. For a subset A of S , define $Q(A) = P(A | B)$. By Theorem 3.1 we know that Q is a probability function. For E and F , events of S [$P(FB) > 0$], show that $Q(E | F) = P(E | FB)$.

B

22. Suppose that 40% of the students on a campus, who are married to students on the same campus, are female. Moreover, suppose that 30% of those who are married, but not to students at this campus, are also female. If one-third of the married students on this campus are married to other students on this campus, what is the probability that a randomly selected married student from this campus is a woman?

Hint: Let M , C , and F denote the events that the random student is married, is married to a student on the same campus and is female. For any event A , let $Q(A) = P(A | M)$. Then, by Theorem 3.1, Q satisfies the same axioms that probabilities satisfy. Applying Theorem 3.3 to Q and, using the result of Exercise 21, we obtain

$$P(F | M) = P(F | MC)P(C | M) + P(F | MC^c)P(C^c | M).$$

23. Suppose that the probability that a new seed planted in a specific farm germinates is equal to the proportion of all planted seeds that germinated in that farm previously. Suppose that the first seed planted in the farm germinated, but the second seed planted did not germinate. For positive integers n and k ($k < n$), what is the probability that of the first n seeds planted in the farm exactly k germinated?
24. For $n \geq 1$, let $E_1, E_2, \dots, E_n$ be events of a sample space. Find an n -element partition of $\bigcup_{i=1}^n E_i$. That is, find a set of mutually exclusive events $\{F_1, F_2, \dots, F_n\}$ that satisfies $\bigcup_{i=1}^n F_i = \bigcup_{i=1}^n E_i$.

25. Suppose that 10 good and three dead batteries are mixed up. Jack tests them one by one, at random and without replacement. But before testing the fifth battery he realizes that he does not remember whether the first one tested is good or is dead. All he remembers is that the last three that were tested were all good. What is the probability that the first one is also good?
26. A box contains 18 tennis balls, of which eight are new. Suppose that three balls are selected randomly, played with, and after play are returned to the box. If another three balls are selected for play a second time, what is the probability that they are all new?
27. From families with three children, a *child* is selected at random and found to be a girl. What is the probability that she has an older sister? Assume that in a three-child family all sex distributions are equally probable.
Hint: Let G be the event that the randomly selected child is a girl, A be the event that she has an older sister, and O , M , and Y be the events that she is the oldest, the middle, and the youngest child of the family, respectively. For any subset B of the sample space let $Q(B) = P(B \mid G)$, then apply Theorem 3.3 to Q . (See also Exercises 21 and 22.)
28. Suppose that three numbers are selected one by one, at random and without replacement from the set of numbers $\{1, 2, 3, \dots, n\}$. What is the probability that the third number falls between the first two if the first number is smaller than the second?



Self-Quiz on Section 3.3

Time allotted: 20 Minutes

- At a kindergarten, there is a box full of balls. Each child draws a ball at random to play and is not allowed to return it to the box to draw another one. When it is Natalie's turn to draw a ball, there are 3 red and 7 blue balls left in the box. Natalie loves to play with a red ball and so does Nicole, who will draw right after Natalie. Does Natalie have an advantage over Nicole for drawing a red ball? (3 points)
- There are three dice in a small box. The first die is unbiased; the second one is loaded, and, when tossed, the probability of obtaining 6 is $3/8$, and the probability of obtaining each of the other faces is $1/8$. The third die is also loaded, but the probability of obtaining 6 when tossed is $2/7$, and the probability of each of the other five faces is $1/7$. A die is selected at random and tossed. What is the probability that the outcome is a 6?
(3 points)
- Seventy percent of all students of a college participate in a study abroad program. If 60% of the students of this college are male and 55% of the male students attend a study abroad program, what percentage of female students participate in a study abroad program? (4 points)



3.4 BAYES' FORMULA

To introduce Bayes' formula, let us first examine the following problem. In a bolt factory, 30, 50, and 20% of production is manufactured by machines I, II, and III, respectively. If 4, 5, and 3% of the output of these respective machines is defective, what is the probability that a randomly selected bolt that is found to be defective is manufactured by machine III? To solve this problem, let A be the event that a random bolt is defective and B_3 be the event that it is manufactured by machine III. We are asked to find $P(B_3 | A)$. Now

$$P(B_3 | A) = \frac{P(B_3 A)}{P(A)}. \quad (3.10)$$

So, to calculate $P(B_3 | A)$, we need to know the quantities $P(B_3 A)$ and $P(A)$. But neither of these is given. However, since $P(A | B_3)$ and $P(B_3)$ are known we use the relation

$$P(B_3 A) = P(A | B_3)P(B_3) \quad (3.11)$$

to find $P(B_3 A)$. To calculate $P(A)$, we use the law of total probability. Let B_1 and B_2 be the events that the bolt is manufactured by machines I and II, respectively. Then $\{B_1, B_2, B_3\}$ is a partition of the sample space; hence

$$P(A) = P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + P(A | B_3)P(B_3). \quad (3.12)$$

Substituting (3.11) and (3.12) in (3.10), we arrive at a formula, called **Bayes' formula**, which enables us to calculate $P(B_3 | A)$ readily:

$$\begin{aligned} P(B_3 | A) &= \frac{P(B_3 A)}{P(A)} \\ &= \frac{P(A | B_3)P(B_3)}{P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + P(A | B_3)P(B_3)} \end{aligned} \quad (3.13)$$

$$= \frac{(0.03)(0.20)}{(0.04)(0.30) + (0.05)(0.50) + (0.03)(0.20)} \approx 0.14. \quad (3.14)$$

Relation (3.13) is a particular case of the general form of Bayes' formula (Theorem 3.5).

We will now explain how a tree diagram is used to write relation (3.14). [Figure 3.7](#), displayed on the next page, depicts a tree diagram for this problem, where D represents “defective” and N represents “not defective.” To find the desired probability all we need do is find (by multiplication) the probability of the required path, the path from III to D , and divide it by the sum of the probabilities of the paths that lead to D 's.

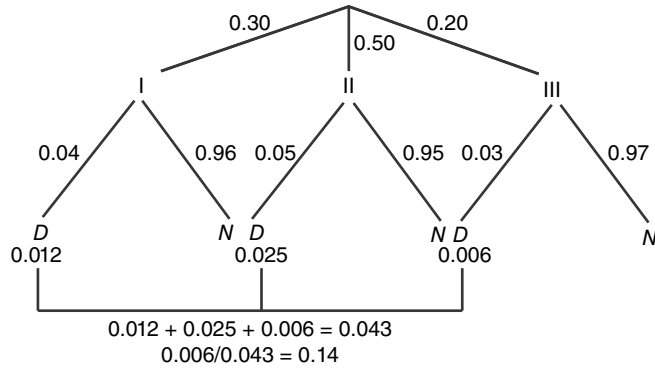


Figure 3.7 Tree diagram for relation (3.14).

In general, modifying the argument from which (3.13) was deduced, we arrive at the following theorem:

Theorem 3.5 (Bayes' Theorem) *Let $\{B_1, B_2, \dots, B_n\}$ be a partition of the sample space S of an experiment. If for $i = 1, 2, \dots, n$, $P(B_i) > 0$, then for any event A of S with $P(A) > 0$,*

$$P(B_k | A) = \frac{P(A | B_k)P(B_k)}{P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + \dots + P(A | B_n)P(B_n)}.$$

In statistical applications of Bayes' theorem, $B_1, B_2, \dots, B_n$ are called **hypotheses**, $P(B_i)$ is called the **prior** probability of B_i , and the conditional probability $P(B_i | A)$ is called the **posterior** probability of B_i after the occurrence of A .

Note that, for any event B of S , B and B^c both nonempty, the set $\{B, B^c\}$ is a partition of S . Thus, by Theorem 3.5,

If $P(B) > 0$ and $P(B^c) > 0$, then for any event A of S with $P(A) > 0$,

$$P(B | A) = \frac{P(A | B)P(B)}{P(A | B)P(B) + P(A | B^c)P(B^c)}.$$

Similarly,

$$P(B^c | A) = \frac{P(A | B^c)P(B^c)}{P(A | B)P(B) + P(A | B^c)P(B^c)}.$$

These are the simplest forms of Bayes' formula. They are used whenever the quantities $P(A | B)$, $P(A | B^c)$, and $P(B)$ are given or can be calculated. The typical situation is that A happens logically or temporally after B , so that the probabilities $P(B)$ and $P(A | B)$ can be readily computed. Bayes' formula is applicable when we know the probability of the more recent event, given that the earlier event has occurred, $P(A | B)$, and we wish to calculate the probability of the earlier event, given that the more recent event has occurred, $P(B | A)$.

In practice, Bayes' formula is used when we know the effect of a cause and we wish to make some inference about the cause.

Theorem 3.5, in its present form, is due to Laplace, who named it after Thomas Bayes (1701–1761). Bayes, a prominent English philosopher and an ordained minister, did a comprehensive study of the calculation of $P(B | A)$ in terms of $P(A | B)$. His work was continued by other mathematicians such as Laplace and Gauss.

We will now present several examples concerning Bayes' theorem. To emphasize the convenience of tree diagrams, in Example 3.24, we will use a tree diagram as well.

Example 3.19 In a study conducted 3 years ago, 82% of the people in a randomly selected sample were found to have “good” financial credit ratings, while the remaining 18% were found to have “bad” financial credit ratings. Current records of the people from that sample show that 30% of those with bad credit ratings have since improved their ratings to good, while 15% of those with good credit ratings have since changed to having a bad credit rating. What percentage of people with good credit ratings now had bad ratings 3 years ago?

Solution: Let G be the event that a randomly selected person from the sample has a good credit rating now. Let B be the event that he or she had a bad credit rating 3 years ago. The desired quantity is $P(B | G)$. By Bayes' formula,

$$\begin{aligned} P(B | G) &= \frac{P(G | B)P(B)}{P(G | B)P(B) + P(G | B^c)P(B^c)} \\ &= \frac{(0.30)(.18)}{(0.30)(.18) + (.85)(.82)} = 0.072, \end{aligned}$$

where $P(G | B^c) = 0.85$, because the probability is $1 - 0.15 = 0.85$ that a person with good credit rating 3 years ago has a good credit rating now. Therefore, 7.2% of people with good credit ratings now had bad ratings 3 years ago. ♦

Example 3.20 During a double homicide murder trial, based on circumstantial evidence alone, the jury becomes 15% certain that a suspect is guilty. DNA samples recovered from the murder scene are then compared with DNA samples extracted from the suspect. Given the size and conditions of the recovered samples, a forensic scientist estimates that the probability of the sample having come from someone other than the suspect is 10^{-9} . With this new information, how certain should the jury be of the suspect's guilt?

Solution: Let G and I be the events that the suspect is guilty and innocent, respectively. Let D be the event that the recovered DNA samples from the murder scene match with the DNA samples extracted from the suspect. Since $\{G, I\}$ is a partition of the sample space, we can use Bayes' formula to calculate $P(G | D)$, the probability that the suspect is the murderer in view of the new evidence.

$$\begin{aligned} P(G | D) &= \frac{P(D | G)P(G)}{P(D | G)P(G) + P(D | I)P(I)} \\ &= \frac{1(.15)}{1(.15) + 10^{-9}(.85)} = 0.9999999943. \end{aligned}$$

This shows that $P(I \mid D) = 1 - P(G \mid D)$ is approximately 5.67×10^{-9} , leaving no reasonable doubt for the innocence of the suspect.

In some trials, prosecutors have argued that if $P(D \mid I)$ is “small enough,” then there is no reasonable doubt for the guilt of the defendant. Such an argument, called the *prosecutor’s fallacy*, probably stems from confusing $P(D \mid I)$ for $P(I \mid D)$. One should pay attention to the fact that $P(D \mid I)$ is infinitesimal regardless of the suspect’s guilt or innocence. To elaborate this further, note that in this example, $P(I \mid D)$ is approximately 5.67 times larger than $P(D \mid I)$, which is 10^{-9} . This is because even without the DNA evidence, there is a 15% chance that the suspect is guilty.

We will now present a situation, which clearly demonstrates that $P(D \mid I)$ should not be viewed as the probability of guilt in evaluating reasonable doubt. Suppose that the double homicide in this example occurred in California, and there is no suspect identified. Suppose that a DNA data bank identifies a person in South Dakota whose DNA matches what was recovered at the California crime scene. Furthermore, suppose that the forensic scientist estimates that the probability of the sample recovered at the crime scene having come from someone other than the person in South Dakota is still 10^{-9} . If there is no evidence that this person ever traveled to California, or had any motive for committing the double homicide, it is doubtful that he or she would be indicted. In such a case $P(D \mid I)$ is still 10^{-9} , but this quantity hardly constitutes the probability of guilt for the person in South Dakota. The argument that since $P(D \mid I)$ is infinitesimal there is no reasonable doubt for the guilt of the person from South Dakota does not seem convincing at all. In such a case, the quantity $P(I \mid D)$, which can be viewed as the probability of guilt, cannot even be estimated if nothing beyond DNA evidence exists. ♦

Example 3.21 Consider a classic application of Bayes’ theorem in the realm of **medical diagnosis**: breast cancer screening for women. Here are the specifics: (i) In a certain population, 1% of women are diagnosed with breast cancer. (ii) When a woman has breast cancer, there is a 90% chance the diagnostic test will accurately detect it (true positive). (iii) Conversely, for women without breast cancer, the diagnostic test mistakenly indicates its presence in 9% of cases (false positive). Given these statistics, if a woman receives a positive test result, what is the probability she actually has breast cancer?

Solution: In a study, a random group of doctors was asked about the likelihood of a patient having cancer given a positive test result. “The most popular answer from a sample of doctors given these numbers ranged from 80 to 90 percent.”[†] This indicates that, upon receiving a positive test result, most doctors provided a specific probability between 80% and 90% as the likelihood of the patient having breast cancer.

Let us determine the true probability that a woman with a positive test result actually has breast cancer. Let A be the event that a woman has breast cancer, and let B be the event that she tests positive. By applying Bayes’ theorem, we can determine the desired probability as:

$$\begin{aligned} P(A \mid B) &= \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid A^c)P(A^c)} \\ &= \frac{(0.90)(0.01)}{(0.90)(0.01) + (0.09)(0.99)} \approx 0.09217. \end{aligned}$$

[†] See Steven Pinker, *Rationality*, VIKING, an imprint of Penguin Random House LLC, 2021, p. 150.

This reveals that the true probability of a woman with a positive result actually having cancer is 0.0917, or 9.17%, a number significantly lower than the doctors' interpretations and assessments. This example highlights that even the instincts of medical professionals, to whom we entrust our well-being, can sometimes misinterpret probabilistic outcomes of medical tests. While doctors might estimate close to a 90% probability that a woman has cancer based on a test result, the reality may be that there is a 90% chance she does not. Imagine the emotional distress and turmoil a patient would undergo upon learning of such a misjudged diagnosis.

To intuitively understand this problem, imagine a population of 1000 women. If 1% of these women have breast cancer, that means there are 10 women with the disease and 990 without it. If all of them undergo testing, we would expect the following: (i) Of the women with breast cancer, $(0.90)(10) = 9$ will test positive (true positives). (ii) Of the women without breast cancer, $(0.09)(990) \approx 89$ will still test positive (false positives). Thus, out of the 98 individuals who receive positive test results, only 9 genuinely have cancer. This translates to a probability of $9/98 \approx 0.0918$ or 9.18%. So, in this scenario, a woman with a positive test result has a 9.18% chance of actually having cancer. ♦

Example 3.22 On the basis of reconnaissance reports, Colonel Smith decides that the probability of an enemy attack against the left is 0.20, against the center is 0.50, and against the right is 0.30. A flurry of enemy radio traffic occurs in preparation for the attack. Since deception is normal as a prelude to battle, Colonel Brown, having intercepted the radio traffic, tells General Quick that if the enemy wanted to attack on the left, the probability is 0.20 that he would have sent this particular radio traffic. He tells the general that the corresponding probabilities for an attack on the center or the right are 0.70 and 0.10, respectively. How should General Quick use these two equally reliable staff members' views to get the best probability profile for the forthcoming attack?

Solution: Let A be the event that the attack would be against the left, B be the event that the attack would be against the center, and C be the event that the attack would be against the right. Let Δ be the event that this particular flurry of radio traffic occurs. Colonel Brown has provided information of *conditional* probabilities of a particular flurry of radio traffic given that the enemy is preparing to attack against the left, the center, and the right. However, Colonel Smith has presented *unconditional* probabilities, on the basis of reconnaissance reports, for the enemy attacking against the left, the center, and the right. Because of these, the general should take the opinion of Colonel Smith as prior probabilities for A , B , and C . That is, $P(A) = 0.20$, $P(B) = 0.50$, and $P(C) = 0.30$. Then he should calculate $P(A | \Delta)$, $P(B | \Delta)$, and $P(C | \Delta)$ based on Colonel Brown's view, using Bayes' theorem, as follows.

$$\begin{aligned} P(A | \Delta) &= \frac{P(\Delta | A)P(A)}{P(\Delta | A)P(A) + P(\Delta | B)P(B) + P(\Delta | C)P(C)} \\ &= \frac{(0.2)(0.2)}{(0.2)(0.2) + (0.7)(0.5) + (0.1)(0.3)} = \frac{(0.2)(0.2)}{0.42} \approx 0.095. \end{aligned}$$

Similarly,

$$P(B | \Delta) = \frac{(0.7)(0.5)}{0.42} \approx 0.83 \text{ and } P(C | \Delta) = \frac{(0.1)(0.3)}{0.42} \approx 0.071. \quad \blacklozenge$$

Example 3.23 In logic, the contrapositive of “ A implies B ” is defined to be not B implies not A . We know that a statement and its contrapositive are equivalent. Show the analogue of this fact in probability. That is, show that if A and B are two events with probabilities that are neither 0 nor 1, then $P(B | A) = 1$ implies that $P(A^c | B^c) = 1$.

Proof: Note that $P(B^c | A) = 1 - P(B | A) = 0$. So by Bayes’ theorem,

$$P(A^c | B^c) = \frac{P(B^c | A^c)P(A^c)}{P(B^c | A^c)P(A^c) + P(B^c | A)P(A)} = \frac{P(B^c | A^c)P(A^c)}{P(B^c | A^c)P(A^c)} = 1. \quad \blacklozenge$$

Example 3.24 A box contains seven red and 13 blue balls. Two balls are selected at random and are discarded without their colors being seen. If a third ball is drawn randomly and observed to be red, what is the probability that both of the discarded balls were blue?

Solution: Let BB , BR , and RR be the events that the discarded balls are blue and blue, blue and red, red and red, respectively. Also, let R be the event that the third ball drawn is red. Since $\{BB, BR, RR\}$ is a partition of sample space, Bayes’ formula can be used to calculate $P(BB | R)$.

$$P(BB | R) = \frac{P(R | BB)P(BB)}{P(R | BB)P(BB) + P(R | BR)P(BR) + P(R | RR)P(RR)}.$$

Now

$$P(BB) = \frac{13}{20} \times \frac{12}{19} = \frac{39}{95}, \quad P(RR) = \frac{7}{20} \times \frac{6}{19} = \frac{21}{190},$$

and

$$P(BR) = \frac{13}{20} \times \frac{7}{19} + \frac{7}{20} \times \frac{13}{19} = \frac{91}{190},$$

where the last equation follows since BR is the union of two disjoint events: namely, the first ball discarded was blue, the second was red, and vice versa. Thus

$$P(BB | R) = \frac{\frac{7}{18} \times \frac{39}{95}}{\frac{7}{18} \times \frac{39}{95} + \frac{6}{18} \times \frac{91}{190} + \frac{5}{18} \times \frac{21}{190}} \approx 0.46.$$

This result can be found from the tree diagram of [Figure 3.8](#) as well. Alternatively, reducing sample space, given that the third ball was red, there are 13 blue and six red balls which could have been discarded. Thus

$$P(BB | R) = \frac{13}{19} \times \frac{12}{18} \approx 0.46. \quad \blacklozenge$$

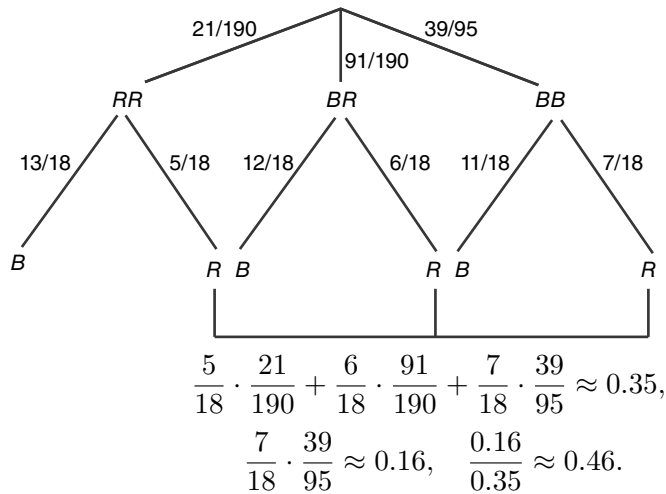


Figure 3.8 Tree diagram for Example 3.24.

EXERCISES

A

1. In transmitting dot and dash signals, a communication system changes $1/4$ of the dots to dashes and $1/3$ of the dashes to dots. If 40% of the signals transmitted are dots and 60% are dashes, what is the probability that a dot received was actually a transmitted dot?
2. On a multiple-choice exam with four choices for each question, a student either knows the answer to a question or marks it at random. If the probability that he or she knows the answers is $2/3$, what is the probability that an answer that was marked correctly was not marked randomly?
3. Suppose that 20% of the e-mails Derek receives are spam. Suppose that 12% of the spam e-mails and 0.05% of the non-spam e-mails are concerning the e-mail storage in Derek's email account. Derek has just received an e-mail concerning his e-mail storage. What is the probability that it is spam?
4. A judge is 65% sure that a suspect has committed a crime. During the course of the trial, a witness convinces the judge that there is an 85% chance that the criminal is left-handed. If 23% of the population is left-handed and the suspect is also left-handed, with this new information, how certain should the judge be of the guilt of the suspect?
5. When Professor Wagoner teaches calculus, he only grades 25% of the students' exam papers, randomly selected. The remaining papers are graded by his teaching assistants (TA's). Suppose that 78% of the papers graded by Professor Wagoner and 86% of the papers graded by the TA's get a passing grade. If Harris, a calculus student of Professor Wagoner, got a passing grade on his last exam, what is the probability that his exam paper was graded by a TA?

6. When traveling from Springfield, Massachusetts, to Baltimore, Maryland, 30% of the time Charles takes I-91 south and then I-95 south, and 70% of the time he takes I-91 south, then I-84 west followed by I-81 south and I-83 south. Choosing the first option takes Charles a random time between 5 hours and 30 minutes and 10 hours depending upon the traffic. However, choosing the second option takes a random time between 6 hours and 30 minutes and 8 hours. The second option is a longer drive but has much less traffic. We know that, on his last trip, it took Charles less than 7 hours and 15 minutes to travel from Springfield to Baltimore. What is the probability that he chose the second option?
7. In a trial, the judge is 65% sure that Susan has committed a crime. Julie and Robert are two witnesses who know whether Susan is innocent or guilty. However, Robert is Susan's friend and will lie with probability 0.25 if Susan is guilty. He will tell the truth if she is innocent. Julie is Susan's enemy and will lie with probability 0.30 if Susan is innocent. She will tell the truth if Susan is guilty. What is the probability that Susan is guilty if Robert and Julie give conflicting testimony?
8. Suppose that 5% of the men and 2% of the women working for a corporation make over \$120,000 a year. If 30% of the employees of the corporation are women, what percent of those who make over \$120,000 a year are women?
9. There are three dice in a small box. The first die is unbiased; the second one is loaded, and, when tossed, the probability of obtaining 6 is $\frac{3}{8}$, and the probability of obtaining each of the other faces is $\frac{1}{8}$. The third die is also loaded, but the probability of obtaining 6 when tossed is $\frac{2}{7}$, and the probability of each of the other five faces is $\frac{1}{7}$. A die was selected from the box at random and was tossed. If the outcome was a 6, for $i = 1, 2, 3$, find the probability that the die selected and tossed was the i th die.
10. A stack of cards consists of six red and five blue cards. A second stack of cards consists of nine red cards. A stack is selected at random and three of its cards are drawn. If all of them are red, what is the probability that the first stack was selected?
11. Suppose that currently it is a bull market, and in the stock market, share prices are rising. Furthermore, suppose that the Nasdaq Composite index closes at higher points 86% of the trading days. Iniko is a Nasdaq financial expert, and when she predicts that the Nasdaq Composite index closes at a lower or equal point the next trading day, her forecast will turn out to be true 90% of the time. However, her prediction is confirmed only 75% of the time that the Nasdaq Composite index closes at a higher point the next trading day. Tomorrow is a trading day, and Iniko has predicted that the Nasdaq Composite index will close at a higher point than it did last time. What is the probability that her prediction will turn out to be true?
12. Based on an insurance company's evaluation, with probability 0.85, Whitney is a safe driver, and with probability 0.15 she is not. Suppose that a safe driver avoids at-fault accidents in the next year with probability 0.87, and an unsafe driver avoids at-fault accidents in that period with probability 0.54. If next year Whitney gets involved in a car accident that is her fault, how should the insurance company revise the probability of Whitney being a safe driver?

13. An actuary has calculated that, for the age group 16–25, the probability is 0.08 that a driver whose car is insured by her company gets involved in a car accident within a year. For the age groups 26–35, 36–65, and 66–100, the corresponding probabilities are 0.03, 0.02, and 0.04, respectively. Suppose that the percentages of the company's insured drivers belonging to various age groups are given by the following table:

Age Group	16–25	26–35	36–65	66–100
Percentage of Insured Drivers	10%	15%	45%	30%

What proportion of the company's insured drivers who get involved in an accident within a year belong to the age group 16–25?

14. A certain cancer is found in one person in 5000. If a person does have the disease, in 92% of the cases the diagnostic procedure will show that he or she actually has it. If a person does not have the disease, the diagnostic procedure in one out of 500 cases gives a false positive result. Determine the probability that a person with a positive test result has the cancer.
15. At a grocery store, an absent-minded, honest professor, Dexter, hands the playful cashier, Hans, a dollar bill. Hans puts the bill in the cash register drawer and gives back Dexter \$2.75 in change. Dexter, who really does not remember whether he handed him a \$10 bill or a \$20 bill, claims that he gave Hans a \$20 bill and demands an additional \$10 back. Hans is an honest man as well, and each time dollar bills are given to him, he examines them and notes their amounts before putting them in the drawer. However, 8% of the time, due to his playfulness, he does not remember correctly how much money was handed to him. If the cash register has 13 \$20 bills and 18 \$10 bills, what is the probability that Dexter actually handed Hans a \$20 bill?
16. Urns I, II, and III contain three pennies and four dimes, two pennies and five dimes, three pennies and one dime, respectively. One coin is selected at random from each urn. If two of the three coins are dimes, what is the probability that the coin selected from urn I is a dime?
17. In a study it was discovered that 25% of the paintings of a certain gallery are not original. A collector in 15% of the cases makes a mistake in judging if a painting is authentic or a copy. If she buys a piece thinking that it is original, what is the probability that it is not?
18. There are three identical cards that differ only in color. Both sides of one are black, both sides of the second one are red, and one side of the third card is black and its other side is red. These cards are mixed up and one of them is selected at random. If the upper side of this card is red, what is the probability that its other side is black?
19. With probability of $1/6$ there are i defective fuses among 1000 fuses ($i = 0, 1, 2, 3, 4, 5$). If among 100 fuses selected at random, none was defective, what is the probability of no defective fuses at all?

20. Solve the following problem, asked of Marilyn Vos Savant in the “Ask Marilyn” column of *Parade Magazine*, February 18, 1996.

Say I have a wallet that contains either a \$2 bill or a \$20 bill (with equal likelihood), but I don't know which one. I add a \$2 bill. Later, I reach into my wallet (without looking) and remove a bill. It's a \$2 bill. There's one bill remaining in the wallet. What are the chances that it's a \$2 bill?

21. Some studies have shown that 1 in 177 Americans have celiac disease. Nevertheless, in general, for ordinary physicians, it is hard to diagnose it and, as a result, only 20% of the patients that have it are actually diagnosed with the disease. The good news is that of those who do not have celiac disease, 98% test negative. Hoyt is tested positive for this disease. However, to make sure, he is tested for a second time, and the result is positive again. Considering the results of both tests, what is the probability that Hoyt has celiac disease?

B

22. There are two stables on a farm, one that houses 20 horses and 13 mules, the other with 25 horses and eight mules. Without any pattern, animals occasionally leave their stables and then return to their stables. Suppose that during a period when all the animals are in their stables, a horse comes out of a stable and then returns. What is the probability that the next animal coming out of the same stable will also be a horse?
23. An urn contains five red and three blue chips. Suppose that four of these chips are selected at random and transferred to a second urn, which was originally empty. If a random chip from this second urn is blue, what is the probability that two red and two blue chips were transferred from the first urn to the second urn?
24. There are three dice in a small box. The first die is unbiased; the second one is loaded, and, when tossed, the probability of obtaining 6 is $\frac{3}{8}$, and the probability of obtaining each of the other faces is $\frac{1}{8}$. The third die is also loaded, but the probability of obtaining 6 when tossed is $\frac{2}{7}$, and the probability of each of the other five faces is $\frac{1}{7}$. A die was selected from the box at random and was tossed, and the outcome was 6. If the die is tossed once more, what is the probability that the outcome is 6 again?
25. The advantage of a certain blood test is that 90% of the time it is positive for patients having a certain disease. Its disadvantage is that 25% of the time it is also positive in healthy people. In a certain location 30% of the people have the disease, and anybody with a positive blood test is given a drug that cures the disease. If 20% of the time the drug produces a characteristic rash, what is the probability that a person from this location who has the rash had the disease in the first place?



Self-Quiz on Section 3.4

Time allotted: 20 Minutes

1. In a small town in Massachusetts, of the 60 students who took the SAT the last time that it was offered, 17 had attended a SAT test prep course. If taking this course increases the chances of a student to score 1200 or higher from 20% to 35%, what is the probability that a student who scored higher than 1200 had attended the prep course? (3 points)
2. A box has 10 coins of which 3 are gold. Eileen selects a coin at random first. Then Bernice draws a coin randomly and finds that it is gold. What is the probability that Eileen's coin is also gold? (3 point)
3. At a gas station, 85% of the customers use 87 octane gasoline, 5% use 91 octane, and 10% use 93 octane. Suppose that, 70%, 85%, and 95% of 87 octane users, 91 octane users, and 93 octane users fill their tanks, respectively. At this station, a customer just filled his tank. What is the probability that he used 93 octane gasoline? (4 points)



3.5 INDEPENDENCE

Let A and B be two events of a sample space S , and assume that $P(A) > 0$ and $P(B) > 0$. We have seen that, in general, the conditional probability of A given B is not equal to the probability of A . However, if it is, that is, if $P(A | B) = P(A)$, we say that A is **independent** of B . This means that if A is independent of B , knowledge regarding the occurrence of B does not change the chance of the occurrence of A . The relation $P(A | B) = P(A)$ is equivalent to the relations $P(AB)/P(B) = P(A)$, $P(AB) = P(A)P(B)$, $P(BA)/P(A) = P(B)$, and $P(B | A) = P(B)$. The equivalence of the first and last of these relations implies that if A is independent of B , then B is independent of A . In other words, if knowledge regarding the occurrence of B does not change the chance of occurrence of A , then knowledge regarding the occurrence of A does not change the chance of occurrence of B . Hence independence is a symmetric relation on the set of all events of a sample space. As a result of this property, instead of making the definitions “ A is independent of B ” and “ B is independent of A ,” we simply define the concept of the “independence of A and B .” To do so, we take $P(AB) = P(A)P(B)$ as the definition. We do this because a symmetrical definition relating A and B does not readily follow from either of the other relations given [i.e., $P(A | B) = P(A)$ or $P(B | A) = P(B)$]. Moreover, these relations require either that $P(B) > 0$ or $P(A) > 0$, whereas our definition does not.

Definition 3.3 Two events A and B are called **independent** if

$$P(AB) = P(A)P(B).$$

If two events are not independent, they are called **dependent**. If A and B are independent, we say that $\{A, B\}$ is an independent set of events.

Note that in this definition we did not require $P(A)$ or $P(B)$ to be strictly positive. Hence by this definition *any* event A with $P(A) = 0$ or 1 is independent of *every* event B (see Exercise 16).

Example 3.25 In the experiment of tossing a fair coin twice, let A and B be the events of getting heads on the first and second tosses, respectively. Intuitively, it is clear that A and B are independent. To prove this mathematically, note that $P(A) = 1/2$ and $P(B) = 1/2$. But since the sample space of this experiment consists of the four equally probable events: HH, HT, TH, and TT, we have $P(AB) = P(HH) = 1/4$. Hence $P(AB) = P(A)P(B)$ is valid, implying the independence of A and B . It is interesting to know that Jean Le Rond d'Alembert (1717–1783), a French mathematician, had argued that, since in the experiment of tossing a fair coin twice the possible number of heads is 0, 1, and 2, the probability of no heads, one heads, and two heads, each is $1/3$. ♦

Example 3.26 In the experiment of drawing a card from an ordinary deck of 52 cards, let A and B be the events of getting a heart and an ace, respectively. Whether A and B are independent cannot be answered easily on the basis of intuition alone. However, using the defining formula, $P(AB) = P(A)P(B)$, this can be answered at once since $P(AB) = 1/52$, $P(A) = 1/4$, $P(B) = 1/13$, and $1/52 = 1/4 \times 1/13$. Hence A and B are independent events. ♦

Example 3.27 An urn contains five red and seven blue balls. Suppose that two balls are selected at random and with replacement. Let A and B be the events that the first and the second balls are red, respectively. Then, using the counting principle, Theorem 2.1, we get $P(AB) = (5 \times 5)/(12 \times 12)$. Now $P(AB) = P(A)P(B)$ since $P(A) = 5/12$ and $P(B) = 5/12$. Thus A and B are independent. If we do the same experiment without replacement, then $P(B | A) = 4/11$ while

$$\begin{aligned} P(B) &= P(B | A)P(A) + P(B | A^c)P(A^c) \\ &= \frac{4}{11} \times \frac{5}{12} + \frac{5}{11} \times \frac{7}{12} = \frac{5}{12}, \end{aligned}$$

which might be quite surprising to some. But it is true. If no information is given on the outcome of the first draw, there is no reason for the probability of second ball being red to differ from $5/12$. Thus $P(B | A) \neq P(B)$, implying that A and B are dependent. ♦

Example 3.28 In the experiment of selecting a random number from the set of natural numbers $\{1, 2, 3, \dots, 100\}$, let A , B , and C denote the events that they are divisible by 2, 3, and 5, respectively. Clearly, $P(A) = 1/2$, $P(B) = 33/100$, $P(C) = 1/5$, $P(AB) = 16/100$, and $P(AC) = 1/10$. Hence A and B are dependent while A and C are independent. Note that if the random number is selected from $\{1, 2, 3, \dots, 300\}$, then each of $\{A, B\}$, $\{A, C\}$, and $\{B, C\}$ is an independent set of events. This is because 300 is divisible by 2, 3, and 5, but 100 is not divisible by 3. ♦

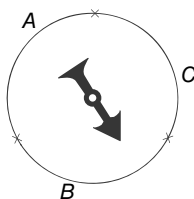


Figure 3.9 Spinner of Example 3.29.

Example 3.29 A spinner is mounted on a wheel. Arcs A , B , and C , of equal length, are marked off on the wheel's perimeter (see Figure 3.9). In a game of chance, the spinner is flicked, and depending on whether it stops on A , B , or C , the player wins 1, 2, or 3 points, respectively. Suppose that a player plays this game twice. Let E denote the event that he wins 1 point in the first game and any number of points in the second. Let F be the event that he wins a total of 3 points in both games, and G be the event that he wins a total of 4 points in both games. The sample space of this experiment has $3 \times 3 = 9$ elements, $E = \{(1, 1), (1, 2), (1, 3)\}$, $F = \{(1, 2), (2, 1)\}$, and $G = \{(1, 3), (2, 2), (3, 1)\}$. Therefore, $P(E) = 1/3$, $P(G) = 1/3$, $P(F) = 2/9$, $P(FE) = 1/9$, and $P(GE) = 1/9$. These show that E and G are independent, whereas E and F are not. To justify these intuitively, note that if we are interested in a total of 3 points, then getting 1 point in the first game is good luck, since obtaining 3 points in the first game makes it impossible to win a sum of 3. However, if we are interested in a sum of 4 points, it does not matter what we win in the first game. We have the same chance of obtaining a sum of 4 if the first game results in any of the numbers 1, 2, or 3. ♦

We now prove that if A and B are independent events, so are A and B^c .

Theorem 3.6 *If A and B are independent, then A and B^c are independent as well.*

Proof: By Theorem 1.7,

$$P(A) = P(AB) + P(AB^c).$$

Therefore,

$$\begin{aligned} P(AB^c) &= P(A) - P(AB) = P(A) - P(A)P(B) \\ &= P(A)[1 - P(B)] = P(A)P(B^c). \quad \blacklozenge \end{aligned}$$

Corollary *If A and B are independent, then A^c and B^c are independent as well.*

Proof: A and B are independent, so by Theorem 3.6 the events A and B^c are independent. Now, using the same theorem again, we have that A^c and B^c are independent. ♦

Thus, if A and B are independent, knowledge about the occurrence or nonoccurrence of A does not change the chances of the occurrence or nonoccurrence of B , and vice versa.

Remark 3.3 Dependence of Mutually Exclusive Events *If A and B are mutually exclusive events and $P(A) > 0$, $P(B) > 0$, then they are dependent. This is because, if we are given that one has occurred, the chance of the occurrence of the other one is zero. That is, the occurrence of one of them precludes the occurrence of the other. For example, let A be the event that the next*

president of the United States is a Democrat and B be the event that he or she is a Republican. Then A and B are mutually exclusive; hence they are dependent. If A occurs, that is, if the next president is a Democrat, the probability that B occurs, that is, he or she is a Republican is zero, and vice versa. ♦

The following example shows that if A is independent of B and if A is independent of C , then A is not necessarily independent of BC or of $B \cup C$.

Example 3.30 Dennis arrives at his office every day at a random time between 8:00 A.M. and 9:00 A.M. Let A be the event that Dennis arrives at his office tomorrow between 8:15 A.M. and 8:45 A.M. Let B be the event that he arrives between 8:30 A.M. and 9:00 A.M., and let C be the event that he arrives either between 8:15 A.M. and 8:30 A.M. or between 8:45 A.M. and 9:00 A.M. Then AB , AC , BC , and $B \cup C$ are the events that Dennis arrives at his office between 8:30 and 8:45, 8:15 and 8:30, 8:45 and 9:00, and 8:15 and 9:00, respectively. Thus $P(A) = P(B) = P(C) = 1/2$ and $P(AB) = P(AC) = 1/4$. So $P(AB) = P(A)P(B)$ and $P(AC) = P(A)P(C)$; that is, A is independent of B and it is independent of C . However, since BC and A are mutually exclusive, they are dependent. Also, $P(A | B \cup C) = 2/3 \neq P(A)$. Thus $B \cup C$ and A are dependent as well. ♦

Example 3.31 (Jailer's Paradox) The jailer of a prison in which Alex, Ben, and Tim are held is the only person, other than the judge, who knows which of these three prisoners is condemned to death, and which *two* will be freed. The prisoners know that exactly two of them will go free; they do not know which two. Alex has written a letter to his fiancée. Just in case he is not one of the two who will be freed, he wants to give the letter to a prisoner who goes free to deliver. So Alex asks the jailer to tell him the name of one of the two prisoners who will go free. The jailer refuses to give that information to Alex. To begin with, he is not allowed to tell Alex whether Alex goes free or not. Putting Alex aside, he argues that, if he reveals the name of a prisoner who will go free, then the probability of Alex dying increases from $1/3$ to $1/2$. He does not want to do that.

As Paul Frederick Zweifel (1929–2017) notes in the June 1986 issue of *Mathematics Magazine*, page 156, “this seems intuitively suspect, since the jailer is providing no new information to Alex, so why should his probability of dying change?” Zweifel is correct. Just revealing to Alex that Ben goes free, or just revealing to him that Tim goes free is not the type of information that changes the probability of Alex dying. What changes the probability of Alex dying is telling him whether both Ben and Tim are going free or exactly one of them is going free.

To explain this paradox, we will show that, under suitable conditions, if the jailer tells Alex that Tim goes free, still the probability is $1/3$ that Alex dies. Telling Alex that Tim is going free reveals to Alex that the probability of Ben dying is $2/3$. Similarly, telling Alex that Ben is going free reveals to Alex that the probability of Tim dying is $2/3$. To show these facts, let A , B , and T be the events that “Alex dies,” “Ben dies,” and “Tim dies.” Let

- ω_1 = Tim dies, and the jailer tells Alex that Ben goes free
- ω_2 = Ben dies, and the jailer tells Alex that Tim goes free
- ω_3 = Alex dies, and the jailer tells Alex that Ben goes free
- ω_4 = Alex dies, and the jailer tells Alex that Tim goes free.

The sample space of all possible episodes is $S = \{\omega_1, \omega_2, \omega_3, \omega_4\}$. Now, if Tim dies, with probability 1, the jailer will tell Alex that Ben goes free. Therefore, ω_1 occurs if and only if Tim dies. This implies that $P(\omega_1) = 1/3$. Similarly, if Ben dies, with probability 1, the jailer will tell Alex that Tim

goes free. Therefore, ω_2 occurs if and only if Ben dies. This shows that $P(\omega_2) = 1/3$. To assign probabilities to ω_3 and ω_4 , we will make two assumptions: (1) If Alex is the one who is scheduled to die, then the event that the jailer will tell Alex that Ben goes free is *independent* of the event that the jailer will tell Alex that Tim goes free, and (2) if Alex is the one who is scheduled to die, then the probability that the jailer will tell Alex that Ben goes free is $1/2$; hence the probability that he will tell Alex that Tim goes free is $1/2$ as well. Under these conditions, $P(\omega_3) = P(\omega_4) = 1/6$. Let J be the event that “the jailer tells Alex that Tim goes free;” then

$$P(A | J) = \frac{P(AJ)}{P(J)} = \frac{P(\omega_4)}{P(\omega_2) + P(\omega_4)} = \frac{\frac{1}{6}}{\frac{1}{3} + \frac{1}{6}} = \frac{1}{3},$$

This shows that if the jailer reveals no information about the fate of Alex, telling Alex the name of one prisoner who goes free does not change the probability of Alex dying; it remains to be $1/3$.

Note that the decision as which of Ben, Tim, or Alex is condemned to death, and which two will be freed, has been made by the judge. The jailer has no option to control Alex’s fate. With probability 1, the jailer knows which one of the three dies and which two will go free. It is Alex who doesn’t know any of these probabilities. Alex can only analyze these probabilities based on the information he receives from the jailer. If Alex is dying, and the jailer disobeys the conditions (1) and (2), then he could decide to tell Alex, with arbitrary probabilities, which of Ben or Tim goes free. In such a case, $P(A | J)$ will no longer equal $1/3$. It will vary depending on the probabilities of J and J^c . If the jailer insists on not giving extra information to Alex, he should obey conditions (1) and (2). Furthermore, the jailer should tell Alex what his rules are when he reveals information. Alex can only analyze the probability of dying based on the full information he receives from the jailer. If the jailer does not reveal his rules to Alex, there is no way for Alex to know whether the probability that he is the unlucky prisoner has changed or not.

Zweifel analyzes this paradox by using Bayes’ formula:

$$\begin{aligned} P(A | J) &= \frac{P(J | A)P(A)}{P(J | A)P(A) + P(J | B)P(B) + P(J | T)P(T)} \\ &= \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2} \times \frac{1}{3} + 1 \times \frac{1}{3} + 0 \times \frac{1}{3}} = \frac{1}{3}. \end{aligned}$$

Similarly, if D is the event that “the jailer tells Alex that Ben goes free,” then $P(A | D) = 1/3$. In his explanation of this paradox Zweifel writes

Aside from the formal application of Bayes’ theorem, one would like to understand this “paradox” from an intuitive point of view. The crucial point which can perhaps make the situation clear is the discrepancy between $P(J | A)$ and $P(J | B)$ noted above. Bridge players call this the “Principle of Restricted Choice.” The probability of a restricted choice is obviously greater than that of a free choice, and a common error made by those who attempt to solve such problems intuitively is to overlook this point. In the case of the jailer’s paradox, if the jailer says “Tim will go free” this is twice as likely to occur when Ben is scheduled to die (restricted choice; jailer *must* say “Tim”) as when Alex is scheduled to die (free choice; jailer could say either “Tim” or “Ben”).

The Jailer's paradox and its resolution have been recognized well before 1990. However, when presented under a new guise as the Monty Hall Problem (refer to Example 3.9), its popularity soared. Its widespread recognition can be attributed to a 1990 Parade Magazine column where Marilyn vos Savant addressed a reader's inquiry about the problem. The response elicited disbelief from many, including some with doctoral degrees, sparking considerable debate and discussion. Today, the Monty Hall problem serves as a quintessential example in probability and statistics courses, highlighting conditional probability and strategic thinking. This narrative recalls De Moivre's remark in the dedication of his work, *The Doctrine of Chance*:

Some of the Problems about Chance having a great appearance of Simplicity, the Mind is easily drawn into a belief, that their Solution may be attained by the mere Strength of natural good Sense; which generally proving otherwise and the Mistakes occasioned thereby being not unfrequent, 'tis presumed that a Book of this Kind, which teaches to distinguish Truth from what seems so nearly to resemble it, will be looked upon as a help to good Reasoning. ♦

We now extend the concept of independence to three events: A , B , and C are called **independent** if knowledge about the occurrence of any of them, or the joint occurrence of any two of them, does not change the chances of the occurrence of the remaining events. That is, A , B , and C are independent if $\{A, B\}$, $\{A, C\}$, $\{B, C\}$, $\{A, BC\}$, $\{B, AC\}$, and $\{C, AB\}$ are all independent sets of events. Hence A , B , and C are independent if

$$\begin{aligned}P(AB) &= P(A)P(B), \\P(AC) &= P(A)P(C), \\P(BC) &= P(B)P(C), \\P(A(BC)) &= P(A)P(BC), \\P(B(AC)) &= P(B)P(AC), \\P(C(AB)) &= P(C)P(AB).\end{aligned}$$

Now note that these relations can be reduced since the first three and the relation $P(ABC) = P(A)P(B)P(C)$ imply the last three relations. Hence the definition of the independence of three events can be shortened as follows.

Definition 3.4 *The events A , B , and C are called **independent** if*

$$\begin{aligned}P(AB) &= P(A)P(B), \\P(AC) &= P(A)P(C), \\P(BC) &= P(B)P(C), \\P(ABC) &= P(A)P(B)P(C).\end{aligned}$$

If A , B , and C are independent events, we say that $\{A, B, C\}$ is an independent set of events.

The following example demonstrates that $P(ABC) = P(A)P(B)P(C)$, in general, does not imply that $\{A, B, C\}$ is a set of independent events.

Example 3.32 Let an experiment consist of throwing a die twice. Let A be the event that in the second throw the die lands 1, 2, or 5; B the event that in the second throw it lands 4, 5 or 6; and C

the event that the sum of the two outcomes is 9. Then $P(A) = P(B) = 1/2$, $P(C) = 1/9$, and

$$\begin{aligned} P(AB) &= \frac{1}{6} \neq \frac{1}{4} = P(A)P(B), \\ P(AC) &= \frac{1}{36} \neq \frac{1}{18} = P(A)P(C), \\ P(BC) &= \frac{1}{12} \neq \frac{1}{18} = P(B)P(C), \end{aligned}$$

while

$$P(ABC) = \frac{1}{36} = P(A)P(B)P(C).$$

Thus the validity of $P(ABC) = P(A)P(B)P(C)$ is not sufficient for the independence of A , B , and C . ♦

If A , B , and C are three events and the occurrence of any of them does not change the chances of the occurrence of the remaining two, we say that A , B , and C are **pairwise independent**. Thus $\{A, B, C\}$ forms a set of pairwise independent events if $P(AB) = P(A)P(B)$, $P(AC) = P(A)P(C)$, and $P(BC) = P(B)P(C)$. The difference between pairwise independent events and independent events is that, in the former, knowledge about the joint occurrence of any two of them may change the chances of the occurrence of the remaining one, but in the latter it would not. The following example illuminates the difference between pairwise independence and independence.

Example 3.33 A regular tetrahedron is a body that has four faces and, if it is tossed, the probability that it lands on any face is $1/4$. Suppose that one face of a regular tetrahedron has three colors: red, green, and blue. The other three faces each have only one color: red, blue, and green, respectively. We throw the tetrahedron once and let R , G , and B be the events that the face on which it lands contains red, green, and blue, respectively. Then $P(R | G) = 1/2 = P(R)$, $P(R | B) = 1/2 = P(R)$, and $P(B | G) = 1/2 = P(B)$. Thus the events R , B , and G are pairwise independent. However, R , B , and G are not independent events since $P(R | GB) = 1 \neq P(R)$. ♦

The independence of more than three events may be defined in a similar manner. A set of n events $A_1, A_2, \dots, A_n$ is said to be **independent** if knowledge about the occurrence of any of them or the joint occurrence of any number of them does not change the chances of the occurrence of the remaining events. If we write this definition in terms of formulas, we get many equations. Similar to the case of three events, if we reduce the number of these equations to the minimum number that can be used to have all of the formulas satisfied, we reach the following definition.

Definition 3.5 The set of events $\{A_1, A_2, \dots, A_n\}$ is called **independent** if for every subset $\{A_{i_1}, A_{i_2}, \dots, A_{i_k}\}$, $k \geq 2$, of $\{A_1, A_2, \dots, A_n\}$,

$$P(A_{i_1} A_{i_2} \cdots A_{i_k}) = P(A_{i_1})P(A_{i_2}) \cdots P(A_{i_k}). \quad (3.15)$$

This definition is not in fact limited to finite sets and is extended to infinite sets of events (countable or uncountable) in the obvious way. For example, the sequence of events $\{A_i\}_{i=1}^{\infty}$ is called independent if for any of its subsets $\{A_{i_1}, A_{i_2}, \dots, A_{i_k}\}$, $k \geq 2$, (3.15) is valid.

By Definition 3.5, events $A_1, A_2, \dots, A_n$ are independent if, for all combinations $1 \leq i < j < k < \dots \leq n$, the relations

$$\begin{aligned} P(A_i A_j) &= P(A_i)P(A_j), \\ P(A_i A_j A_k) &= P(A_i)P(A_j)P(A_k), \\ &\vdots \\ P(A_1 A_2 \cdots A_n) &= P(A_1)P(A_2) \cdots P(A_n) \end{aligned}$$

are valid. Now we see that the first line stands for $\binom{n}{2}$ equations, the second line stands for $\binom{n}{3}$ equations, $\dots$, and the last line stands for $\binom{n}{n}$ equations. Therefore, $A_1, A_2, \dots, A_n$ are independent if all of the above $\binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n}$ relations are satisfied. Note that, by the binomial expansion (see Theorem 2.5 and Example 2.30),

$$\binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n} = (1+1)^n - \binom{n}{0} - \binom{n}{1} = 2^n - n - 1.$$

Thus the number of these equations is $2^n - n - 1$. Although these equations seem to be cumbersome to check, it usually turns out that they are obvious and checking is not necessary.

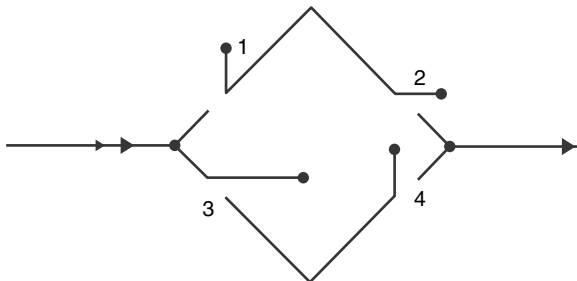


Figure 3.10 Electric circuit of Example 3.34.

Example 3.34 Figure 3.10 shows an electric circuit in which each of the switches located at 1, 2, 3, and 4 is independently closed or open with probabilities p and $1 - p$, respectively. If a signal is fed to the input, what is the probability that it is transmitted to the output?

Solution: Let E_i be the event that the switch at location i is closed, $1 \leq i \leq 4$. A signal fed to the input will be transmitted to the output if at least one of the events $E_1 E_2$ and $E_3 E_4$ occurs. Hence the desired probability is

$$\begin{aligned} P(E_1 E_2 \cup E_3 E_4) &= P(E_1 E_2) + P(E_3 E_4) - P(E_1 E_2 E_3 E_4) \\ &= p^2 + p^2 - p^4 = p^2(2 - p^2). \quad \blacklozenge \end{aligned}$$

We will now adopt a different approach to solve Example 2.8, which addresses the standard birthday problem.

Example 3.35 What is the minimum number of individuals needed in a room to ensure that the probability of at least two of them sharing the same birthday exceeds 0.5? Assume that each person's birthday is independent of the others', implying, for instance, that there are no twins or triplets present.

Solution: Let us assume that k represents the minimum number of individuals required. To identify matching birthdays, we have each person in the room verify their birth date with every other individual. This leads to $n = \binom{k}{2}$ comparisons. For each comparison, the probability of a birthday match is $1/365$. Given the independence of birthdays among individuals in the room, the probability of no matches across all n comparisons is:

$$\left(1 - \frac{1}{365}\right)^n = \left(\frac{364}{365}\right)^n.$$

Therefore, the probability of finding at least one match in the comparisons is $1 - \left(\frac{364}{365}\right)^n$. We aim to determine the value of n such that this probability exceeds 0.5. Specifically, we are looking for the smallest n that satisfies:

$$1 - \left(\frac{364}{365}\right)^n \geq 0.5 \quad \text{or} \quad \left(\frac{364}{365}\right)^n \leq 0.50 \quad \text{or} \quad n \ln\left(\frac{364}{365}\right) \leq \ln 0.5.$$

Upon solving the inequality, we find that $n \geq 252.65$. This means a minimum of 253 comparisons is required to achieve a probability of 0.5 or higher for at least one birthday match. Given that $n = 253$, this implies $\binom{k}{2} = 253$ or $\frac{k!}{2!(k-2)!} = 253$. This can be simplified to $\frac{k(k-1)}{2} = 253$, resulting in the equation $k^2 - k - 506 = 0$. The positive solution of this quadratic equation is $k = 23$. Therefore, a minimum of 23 people must be present in a room to ensure that at least two of them share the same birthday with a probability of 0.5 or greater. ♦

Example 3.36 We draw cards, one at a time, at random and successively from an ordinary deck of 52 cards with replacement. What is the probability that an ace appears before a face card?

Solution: We will elucidate two distinct techniques that can be employed to solve problems of this nature. For a third approach, refer to Exercise 31 in Section 11.3, available on the Instructor and Student Resources website linked to this textbook.

Technique 1: Let E be the event of an ace appearing before a face card. Let A , F , and B be the events of ace, face card, and neither in the first experiment, respectively. Then, by the law of total probability,

$$P(E) = P(E | A)P(A) + P(E | F)P(F) + P(E | B)P(B).$$

Thus

$$P(E) = 1 \times \frac{4}{52} + 0 \times \frac{12}{52} + P(E | B) \times \frac{36}{52}. \quad (3.16)$$

Now note that since the outcomes of successive experiments are all independent of each other, when the second experiment begins, the whole probability process starts all over again. Therefore, if in the first experiment neither a face card nor an ace are drawn, the probability of E before doing the

first experiment and after it would be the same; that is, $P(E | B) = P(E)$. Thus Equation (3.16) gives

$$P(E) = \frac{4}{52} + P(E) \times \frac{36}{52}.$$

Solving this equation for $P(E)$, we obtain $P(E) = 1/4$, a quantity expected because the number of face cards is three times the number of aces (see also Example 3.37).

Technique 2: Let A_n be the event that no face card or ace appears on the first $(n - 1)$ drawings, and the n th draw is an ace. Then the event of “an ace before a face card” is $\bigcup_{n=1}^{\infty} A_n$. Now $\{A_n, n \geq 1\}$ forms a sequence of mutually exclusive events because, if $n \neq m$, simultaneous occurrence of A_n and A_m is the impossible event that an ace appears for the first time in the n th and m th draws. Hence

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

To compute $P(A_n)$, note that $P(\text{an ace on any draw}) = 1/13$ and $P(\text{no face card and no ace in any trial}) = 9/13$. By the independence of trials we obtain

$$P(A_n) = \left(\frac{9}{13}\right)^{n-1} \left(\frac{1}{13}\right).$$

Therefore,

$$\begin{aligned} P\left(\bigcup_{n=1}^{\infty} A_n\right) &= \sum_{n=1}^{\infty} \left(\frac{9}{13}\right)^{n-1} \left(\frac{1}{13}\right) = \frac{1}{13} \sum_{n=1}^{\infty} \left(\frac{9}{13}\right)^{n-1} \\ &= \frac{1}{13} \cdot \frac{1}{1 - 9/13} = \frac{1}{4}. \end{aligned}$$

Here $\sum$ is calculated from the geometric series theorem: For $a \neq 0$, $|r| < 1$, the geometric series $\sum_{n=m}^{\infty} ar^n$ converges to $ar^m/(1 - r)$.

It is interesting to observe that, in this problem, if cards are drawn *without replacement*, even though the trials are no longer independent, the answer would still be the same. For $1 \leq n \leq 37$, let E_n be the event that no face card or ace appears on the first $n - 1$ drawings; let F_n be the event that the n th draw is an ace. Then the event of “an ace appearing before a face card” is $\bigcup_{n=1}^{37} E_n F_n$. Clearly, $\{E_n F_n, 1 \leq n \leq 37\}$ forms a sequence of mutually exclusive events. Hence

$$\begin{aligned} P\left(\bigcup_{n=1}^{37} E_n F_n\right) &= \sum_{n=1}^{37} P(E_n F_n) = \sum_{n=1}^{37} P(E_n) P(F_n | E_n) \\ &= \sum_{n=1}^{37} \frac{\binom{36}{n-1}}{\binom{52}{n-1}} \times \frac{4}{52 - (n-1)} = \frac{1}{4}. \quad \blacklozenge \end{aligned}$$

Example 3.37 Let S be the sample space of a repeatable experiment. Let A and B be mutually exclusive events of S . Prove that, in independent trials of this experiment, the event A occurs before the event B with probability $P(A)/[P(A) + P(B)]$.

Proof: Let $P(A) = p$ and $P(B) = q$. Let E_n be the event that none of A and B occurs in the first $n - 1$ trials and the outcome of the n th experiment is A . The desired probability is

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n) = \sum_{n=1}^{\infty} (1 - p - q)^{n-1} p = \frac{p}{1 - (1 - p - q)} = \frac{p}{p + q}. \quad \blacklozenge$$

Example 3.38 Show that in successive tosses of a fair die indefinitely, the probability of obtaining no 6 is 0.

Solution: For $n \geq 1$, let E_n be the event of at least one 6 in the first n tosses of the die. Clearly,

$$E_1 \subseteq E_2 \subseteq \cdots \subseteq E_n \subseteq E_{n+1} \subseteq \cdots.$$

Therefore, E_n 's form an increasing sequence of events. Note that $\lim_{n \rightarrow \infty} E_n = \bigcup_{i=1}^{\infty} E_i$ is the event that in successive tosses of the die indefinitely, eventually a 6 will occur. By the Continuity of the Probability Function (Theorem 1.8), we have

$$P\left(\lim_{n \rightarrow \infty} E_n\right) = \lim_{n \rightarrow \infty} P(E_n) = \lim_{n \rightarrow \infty} \left[1 - \left(\frac{5}{6}\right)^n\right] = 1 - \lim_{n \rightarrow \infty} \left(\frac{5}{6}\right)^n = 1 - 0 = 1.$$

This shows that, with probability 1, eventually a 6 will occur. Therefore, the probability of no 6 ever is 0. $\blacklozenge$

Example 3.39 Adam tosses a fair coin $n + 1$ times, Andrew tosses the same coin n times. What is the probability that Adam gets more heads than Andrew?

Solution: Let H_1 and H_2 be the number of heads obtained by Adam and Andrew, respectively. Also, let T_1 and T_2 be the number of tails obtained by Adam and Andrew, respectively. Since the coin is fair,

$$P(H_1 > H_2) = P(T_1 > T_2).$$

But

$$P(T_1 > T_2) = P(n + 1 - H_1 > n - H_2) = P(H_1 \leq H_2).$$

Therefore, $P(H_1 > H_2) = P(H_1 \leq H_2)$. So

$$P(H_1 > H_2) + P(H_1 \leq H_2) = 1$$

implies that

$$P(H_1 > H_2) = P(H_1 \leq H_2) = \frac{1}{2}.$$

Note that a combinatorial solution to this problem is neither elegant nor easy to handle:

$$\begin{aligned}
 P(H_1 > H_2) &= \sum_{i=0}^n P(H_1 > H_2 \mid H_2 = i)P(H_2 = i) \\
 &= \sum_{i=0}^n \sum_{j=i+1}^{n+1} P(H_1 = j)P(H_2 = i) \\
 &= \sum_{i=0}^n \sum_{j=i+1}^{n+1} \frac{(n+1)!}{j!(n+1-j)!} \frac{n!}{i!(n-i)!} \\
 &= \frac{1}{2^{2n+1}} \sum_{i=0}^n \sum_{j=i+1}^{n+1} \binom{n+1}{j} \binom{n}{i}.
 \end{aligned}$$

However, comparing these two solutions, we obtain the following interesting identity.

$$\sum_{i=0}^n \sum_{j=i+1}^{n+1} \binom{n+1}{j} \binom{n}{i} = 2^{2n}. \quad \blacklozenge$$

EXERCISES

A

1. Jean le Rond d'Alembert, a French mathematician, believed that in successive flips of a fair coin, after a long run of heads, a tail is more likely. Do you agree with d'Alembert on this? Explain.
2. Clark and Anthony are two old friends. Let A be the event that Clark will attend Anthony's funeral. Let B be the event that Anthony will attend Clark's funeral. Are A and B independent? Why or why not?
3. In a certain country, the probability that a fighter plane returns from a mission without mishap is $49/50$, *independent* of other missions. In a conversation, Mia concluded that any pilot who flew 49 consecutive missions without mishap should be returned home before the fiftieth mission. But, on considering the matter, Jim concluded that the probability of a randomly selected pilot being able to fly 49 consecutive missions safely is $(49/50)^{49} = 0.3716017$. In other words, the odds are almost two to one against an ordinary pilot performing the feat that this pilot has already performed. Hence the pilot would seem to be more skillful than most and thus has a better chance of surviving the 50th mission. Who is right, Mia, Jim, or neither of them? Explain.
4. A fair die is rolled twice. Let A denote the event that the sum of the outcomes is odd, and B denote the event that it lands 2 on the first toss. Are A and B independent? Why or why not?

5. The only information revealed concerning the three children of the new mayor of a large town is that their names are Daly, Emmett, and Karina. Is the event that Daly is younger than Emmett independent of the event that Karina is younger than Emmett?
6. Cal Ripken is a distinguished American baseball player who played 21 seasons in Major League Baseball for the Baltimore Orioles (1981–2001). Among his many achievements, the most important one is that he holds the record for playing in 2632 *consecutive* games over more than 16 years. Suppose that, independently of other games, the probability is 0.005 (5 in 1000) that a baseball player cannot play a game due to illness, injury, family emergency, or other emergencies. What is the probability that a baseball player breaks Cal Ripken's record?
7. An urn has three red and five blue balls. Suppose that eight balls are selected at random and with replacement. What is the probability that the first three are red and the rest are blue balls?
8. Suppose that two points are selected at random and independently from the interval $(0, 1)$. What is the probability that the first one is less than $3/4$, and the second one is greater than $1/4$?
9. According to a recent mortality table, the probability that a 35-year-old U.S. citizen will live to age 65 is 0.725. (a) What is the probability that John and Jim, two 35-year-old Americans who are not relatives, both live to age 65? (b) What is the probability that neither John nor Jim lives to that age?
10. The Italian mathematician Giorlamo Cardano once wrote that if the odds in favor of an event are 3 to 1, then the odds in favor of the occurrence of that event in two consecutive independent experiments are 9 to 1. (He squared 3 and 1 to obtain 9 to 1.) Was Cardano correct?
11. Consider the four “unfolded” dice in Figure 3.11 designed by Stanford professor Bradley Efron. Clearly, none of these dice is an ordinary die with sides numbered 1 through 6. A game consists of two players each choosing one of these four dice and rolling it. The player rolling a larger number is the winner. Show that if all four dice are fair, it is twice as likely for die *A* to beat die *B*, twice as likely for die *B* to beat die *C*, twice as likely for die *C* to beat die *D*, and *surprisingly*, it is twice as likely for die *D* to beat die *A*.

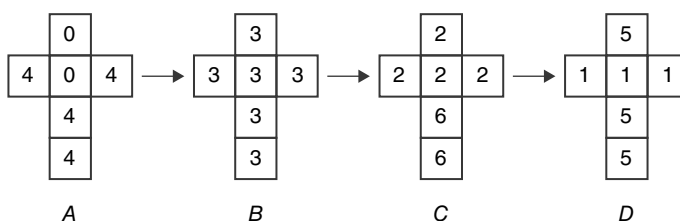


Figure 3.11 Efron dice.

12. **(Chevalier de Méré's Paradox[†])** In the seventeenth century in France there were two popular games, one to obtain at least one 6 in four throws of a fair die and the other to bet

[†]Some scholars consider this problem to be the inception of modern probability theory.

on at least one double 6 in 24 throws of two fair dice. French nobleman and mathematician Chevalier de Méré argued that the probabilities of a 6 in one throw of a fair die and a double 6 in one throw of two fair dice are $1/6$ and $1/36$, respectively. Therefore, the probability of at least one 6 in four throws of a fair die and at least one double 6 in 24 throws of two fair dice are $4 \times 1/6 = 2/3$ and $24 \times 1/36 = 2/3$, respectively. However, even then experience had convinced gamblers that the probability of winning the first game was higher than winning the second game, a contradiction. Explain why de Méré's argument is false, and calculate the correct probabilities of winning in these two games.

13. In data communications, a message transmitted from one end is subject to various sources of distortion and may be received erroneously at the other end. Suppose that a message of 64 bits (a bit is the smallest unit of information and is either 1 or 0) is transmitted through a medium. If each bit is received incorrectly with probability 0.0001 independently of the other bits, what is the probability that the message received is free of error?
14. Find an example in which $P(AB) < P(A)P(B)$.
15. Show that if an event A is independent of itself, then $P(A) = 0$ or 1 .
16. (a) Show that if $P(A) = 1$, then $P(AB) = P(B)$.
(b) Prove that *any* event A with $P(A) = 0$ or $P(A) = 1$ is independent of *every* event B .
17. Show that if A and B are independent and $A \subseteq B$, then either $P(A) = 0$ or $P(B) = 1$.
18. Suppose that 55% of the customers of a shoestore buy black shoes. Find the probability that at least one of the next six customers who purchase a pair of shoes from this store will buy black shoes. Assume that these customers decide independently.
19. An actuary studying the insurance preferences of homeowners in California discovers that the event that a homeowner purchases earthquake coverage is independent of the event that he or she purchases flood insurance. Furthermore, the actuary observes that a homeowner is five times as likely to purchase earthquake coverage as flood insurance. If the probability is 0.19 that a homeowner purchases both earthquake and flood coverage, what is the probability that a homeowner purchases neither earthquake nor flood insurance?
20. Three missiles are fired at a target and hit it independently, with probabilities 0.7, 0.8, and 0.9, respectively. What is the probability that the target is hit?
21. In the ball bearing manufacturing process, for each item being made, two types of defects will occur, independently of each other, with probabilities 0.03 and 0.05, respectively. What is the probability that a randomly selected ball bearing has at least one defect?
22. In his book, *Probability 1*, published by *Harcourt Brace and Company*, 1998, Amir Aczel estimates that the probability of life for any one given star in the known universe is 0.00000000000005 independently of life for any other star. Assuming that there are 100 billion galaxies in the universe and each galaxy has 300 billion stars, what is the probability of life on at least one other star in the known universe? How does this probability change if there were only a billion galaxies, each having 10 billion stars?
23. In a tire factory, the quality control inspector examines a randomly chosen sample of 15 tires. When more than one defective tire is found, production is halted, the existing tires are recycled, and production is then resumed. The purpose of this process is to ensure that the defect rate is no higher than 6%. Occasionally, by a stroke of bad luck, even if the defect rate

is no higher than 6%, more than one defective tire occurs among the 15 chosen. Determine the fraction of the time that this happens.

24. In the experiment of rolling two fair dice successively, what is the probability that a sum of 5 appears before a sum of 7?

Hint: See Example 3.36.

25. An experiment consists of first tossing a fair coin and then drawing a card randomly from an ordinary deck of 52 cards with replacement. If we perform this experiment successively, what is the probability of obtaining heads on the coin before an ace from the cards?

Hint: See Example 3.36.

26. In a community of M men and W women, m men and w women smoke, where $m \leq M$ and $w \leq W$. If a person is selected at random and A and B are the events that the person is a man and smokes, respectively, under what conditions are A and B independent?

27. Prove that if A , B , and C are independent, then A and $B \cup C$ are independent. Also show that $A - B$ and C are independent.

28. A fair die is rolled six times. If on the i th roll, $1 \leq i \leq 6$, the outcome is i , we say that a *match* has occurred. What is the probability that at least one match occurs?

29. There are n cards in a box numbered 1 through n . We draw cards successively and at random with replacement. If the i th draw is the card numbered i , we say that a match has occurred. (a) What is the probability of at least one match in n trials? (b) What happens if n increases without bound?

30. In a certain county, 15% of patients suffering heart attacks are younger than 40, 20% are between 40 and 50, 30% are between 50 and 60, and 35% are above 60. On a certain day, 10 unrelated patients suffering heart attacks are transferred to a county hospital. If among them there is at least one patient younger than 40, what is the probability that there are two or more patients younger than 40?

31. When a loaded die is rolled, the outcome is 6 with probability p . Find the probability that in successive rolls of the die, all outcomes are 6s.

32. If the events A and B are independent and the events B and C are independent, is it true that the events A and C are also independent? Why or why not?

33. From the set of all families with three children a *family* is selected at random. Let A be the event that “the family has children of both sexes” and B be the event that “there is at most one girl in the family.” Are A and B independent? Answer the same question for families with two children and families with four children. Assume that for any family size all sex distributions have equal probabilities.

34. An event occurs at least once in four independent trials with probability 0.59. What is the probability of its occurrence in one trial?

35. Let $\{A_1, A_2, \dots, A_n\}$ be an independent set of events and $P(A_i) = p_i$, $1 \leq i \leq n$.

(a) What is the probability that at least one of the events $A_1, A_2, \dots, A_n$ occurs?

(b) What is the probability that none of the events $A_1, A_2, \dots, A_n$ occurs?

36. From the set of all families with two children, a family is selected at random and is found

to have a girl who was born in January. What is the probability that the other child of the family is a girl? Assume that in a two-child family all sex distributions are equally probable, gender of a child is independent of the month he or she is born, and all months are equally likely to be a birth month of a randomly selected child.

37. Figure 3.12 shows an electric circuit in which each of the switches located at 1, 2, 3, 4, 5, and 6 is independently closed or open with probabilities p and $1 - p$, respectively. If a signal is fed to the input, what is the probability that it is transmitted to the output?

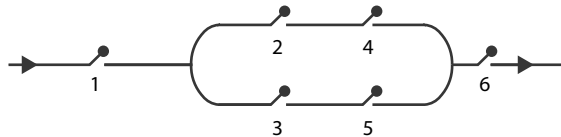


Figure 3.12 Electric circuit of Exercise 37.

38. Cards are drawn at random and with replacement from an ordinary deck of 52 cards, successively and indefinitely. Using Theorem 1.8, show that the probability that the ace of hearts never occurs is 0.

B

39. An urn contains two red and four white balls. Balls are drawn from the urn successively, at random and with replacement. What is the probability that exactly three whites occur in the first five trials?
40. A fair coin is tossed n times. Show that the events “at least two heads” and “one or two tails” are independent if $n = 3$ but dependent if $n = 4$.
41. A fair coin is flipped indefinitely. What is the probability of (a) at least one head in the first n flips; (b) exactly k heads in the first n flips; (c) getting heads in all of the flips indefinitely?
42. If two fair dice are tossed six times, what is the probability that the sixth sum obtained is not a repetition?
43. Suppose that an airplane passenger whose itinerary requires a change of airplanes in Ankara, Turkey, has a 4% chance of losing each piece of his or her luggage independently. Suppose that the probability of losing each piece of luggage in this way is 5% at Da Vinci airport in Rome, 5% at Kennedy airport in New York, and 4% at O’Hare airport in Chicago. Dr. Keating travels from Bombay to San Francisco with one piece of luggage in the baggage compartment. He changes airplanes in Ankara, Rome, New York, and Chicago.
- (a) What is the probability that his luggage does not reach his destination with him?
 - (b) If the luggage does not reach his destination with him, what is the probability that it was lost at Da Vinci airport in Rome?
44. Some watches feature a hand that displays the seconds. Despite the primary purpose of a second hand being to show the time accurately, the experience reveals that at any given moment, it appears to display a number between 1 and 60 at random, independently of what the second hands of other watches show. Nurses often wear watches with second hands to

count respiratory and heart rates. Suppose that n nurses are attending a conference, each wearing watches with second hands. Find the minimum value of n for which the probability is at least $\frac{1}{2}$ that, at an instant, the second hands of at least two nurses' watches display the same number.

45. **(The Game of Craps)** In the game of craps, the player rolls two unbiased dice. If the sum of the outcomes is 2, 3, or 12, she loses. If it is 7 or 11, she wins. However, if the sum is one of the numbers 4, 5, 6, 8, 9, or 10, she will continue rolling repeatedly until either she rolls the sum that she initially obtained, in which case she wins, or she rolls a 7, in which case she loses. What is the probability that in a game of craps, a player will win?
46. An urn contains nine red and one blue balls. A second urn contains one red and five blue balls. One ball is removed from each urn at random and without replacement, and all of the remaining balls are put into a third urn. If we draw two balls randomly from the third urn, what is the probability that one of them is red and the other one is blue?
47. From a population of people with unrelated birthdays, 30 people are selected at random. What is the probability that exactly four people of this group have the same birthday and that all the others have different birthdays (exactly 27 birthdays altogether)? Assume that the birthrates are constant throughout the first 365 days of a year but that on the 366th day it is one-fourth that of the other days.
48. In a contest, contestants A , B , and C are each asked, in turn, a general scientific question. If a contestant gives a wrong answer to a question, he drops out of the game. The remaining two will continue to compete until one of them drops out. The last person remaining is the winner. Suppose that a contestant knows the answer to a question independently of the other contestants, with probability p . Let CAB represent the event that C drops out first, A next, and B wins, with similar representations for other cases. Calculate and compare the probabilities of ABC , BCA , and CAB .

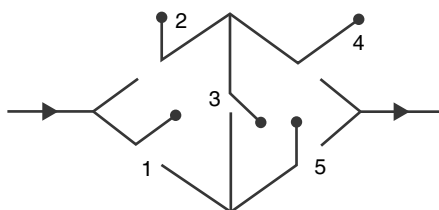


Figure 3.13 Electric circuit of Exercise 49.

49. [Figure 3.13](#) shows an electric circuit in which each of the switches located at 1, 2, 3, 4, and 5 is independently closed or open with probabilities p and $1 - p$, respectively. If a signal is fed to the input, what is the probability that it is transmitted to the output?
50. Hemophilia is a hereditary disease. If a mother has it, then with probability $1/2$, any of her sons independently will inherit it. Otherwise, none of the sons becomes hemophilic. Julie is the mother of two sons, and from her family's medical history it is known that, with the probability $1/4$, she is hemophilic. What is the probability that (a) her first son is hemophilic; (b) her second son is hemophilic; (c) none of her sons are hemophilic?

Hint: Let H , H_1 , and H_2 denote the events that the mother, the first son, and the

second son are hemophilic, respectively. Note that H_1 and H_2 are **conditionally independent** given H . That is, if we are given that the mother is hemophilic, knowledge about one son being hemophilic does not change the chance of the other son being hemophilic. However, H_1 and H_2 are not independent. This is because if we know that one son is hemophilic, the mother is hemophilic and therefore with probability $1/2$ the other son is also hemophilic.

- 51. (Laplace's Law of Succession)** Suppose that $n + 1$ urns are numbered 0 through n , and the i th urn contains i red and $n - i$ white balls, $0 \leq i \leq n$. An urn is selected at random, and then the balls it contains are removed one by one, at random, and with replacement. If the first m balls are all red, what is the probability that the $(m + 1)$ st ball is also red?

Hint: Let U_i be the event that the i th urn is selected, R_m the event that the first m balls drawn are all red, and R the event that the $(m + 1)$ st ball drawn is red. Note that R and R_m are conditionally independent given U_i ; that is, given that the i th urn is selected, R and R_m are independent. Hence

$$P(R \mid R_m U_i) = P(R \mid U_i) = \frac{i}{n}.$$

To find $P(R \mid R_m)$, use

$$P(R \mid R_m) = \sum_{i=0}^n P(R \mid R_m U_i) P(U_i \mid R_m),$$

where $P(U_i \mid R_m)$ is obtained from Bayes' theorem. The final answer is

$$P(R \mid R_m) = \frac{\sum_{i=0}^n \left(\frac{i}{n}\right)^{m+1}}{\sum_{k=0}^n \left(\frac{k}{n}\right)^m}.$$

Note that if we consider the function $f(x) = x^{m+1}$ on the interval $[0, 1]$, the right Riemann sum of this function on the partition $0 = x_0 < x_1 < x_2 < \cdots < x_n = 1$, $x_i = i/n$ is

$$\sum_{i=1}^n f(x_i)(x_i - x_{i-1}) = \sum_{i=1}^n x_i^{m+1} \frac{1}{n} = \frac{1}{n} \sum_{i=0}^n \left(\frac{i}{n}\right)^{m+1}.$$

Therefore, for large n , $(1/n) \sum_{i=0}^n (i/n)^{m+1}$ is approximately equal to $\int_0^1 x^{m+1} dx = 1/(m+2)$. Similarly, $(1/n) \sum_{k=0}^n (k/n)^m$ is approximately $\int_0^1 x^m dx = 1/(m+1)$. Hence

$$P(R \mid R_m) = \frac{\frac{1}{n} \sum_{i=0}^n \left(\frac{i}{n}\right)^{m+1}}{\frac{1}{n} \sum_{k=0}^n \left(\frac{k}{n}\right)^m} \approx \frac{1/(m+2)}{1/(m+1)} = \frac{m+1}{m+2}.$$

Laplace designed and solved this problem for philosophical reasons. He used it to argue that the sun will rise tomorrow with probability $(m+1)/(m+2)$ if we know that it has risen in the m preceding days. Therefore, according to this, as time passes by, the probability that the sun

rises again becomes higher, and the event of the sun rising becomes more and more certain day after day. It is clear that, to argue this way, we should accept the problematic assumption that the phenomenon of the sun rising is “random,” and the model of this problem is identical to its random behavior.



Self-Quiz on Section 3.5

Time allotted: 20 Minutes

Each problem is worth 2.5 points.

1. Suppose that the Dow-Jones Industrial Average (DJIA) rises 52% of trading days and falls 48% of those days. What is the probability that, in the next 5 trading days, the DJIA does not rise on any two consecutive trading days and does not fall on any two consecutive trading days?
2. A specific type of missile fired at a target hits it with probability 0.6. Find the minimum number of such missiles to be fired to have a probability of at least 0.95 of hitting the target.
3. In a ball bearing manufacturing process, after an item is manufactured, two inspectors will inspect it for defects. Suppose that each inspector finds a defect, independently of the other inspector, with probability 0.93. What is the probability that a defect is found by only one inspector?
4. Suppose that 48% of Dr. Darabi’s patients visit his office for a dental cleaning, 18% visit his office for orthodontic work, 12% for an extraction, 10% for a root canal, and the rest for other dental issues. If, for a person, the need for orthodontic work, an extraction, and a root canal are independent of each other, what is the probability that a random patient who just left Dr. Darabi’s office needed at least one of these three dental services?



3.6 R SIMULATION FOR CONDITIONAL PROBABILITIES

Let S represent the sample space of an experiment, and A and B be two events within S , where $P(B) > 0$. To approximate $P(A \mid B)$ through computer simulation, there are two primary approaches:

1. **Reduction Method:** We can restrict the sample space to event B and compute $P(AB)$ within this reduced space. To estimate $P(AB)$, conduct the experiment n times (for a sufficiently large n) within this restricted sample space. Let m be the count of occurrences where both A and B happen. Then, the estimate is given by m/n .
2. **Formula Method:** Alternatively, we can use the relation $P(A \mid B) = P(AB)/P(B)$. Here, we run the experiment numerous times and count n as the instances when B occurs and m as the instances when both A and B occur. The desired probability is then estimated as m/n .

Both methods provide a means to harness computational power in estimating conditional probabilities.

Example 3.40 Consider a deck of 52 standard playing cards. From this deck, we randomly draw a set of 10 cards. Using an R simulation, determine the probability that, in a given set of cards, exactly four are hearts, and among the remaining cards, at least two are spades.

Solution: The subsequent R program calculates the intended probability. Each line of the code will be elucidated immediately after its presentation.

```
> deck <- c(rep("H", 13), rep("S", 13), rep("D", 13), rep("C", 13))
> t <- function() {
+ d <- sample(deck, 10)
+ if(sum(d == "H") == 4) {
+ if(sum(d == "S") >= 2) {
+ return(TRUE) }
+ }
+ return(FALSE)
+ }
> num <- 100000
> results <- replicate(num, t())
> p <- sum(results)/num
> print(p)
[1] 0.09673
```

Now, we will break down the program and discuss each line in detail:

> deck=c(rep("H", 13), rep("S", 13), rep("D", 13), rep("C", 13)): Here, we are creating the deck vector. The **rep()** function is used to repeat values. So, **rep("H", 13)** creates a vector with the value "H" repeated 13 times, representing the 13 hearts in a standard deck. We do this for each suit (Hearts, Spades, Diamonds, Clubs) to create a full deck of 52 cards.

>t <- function() {: This starts the definition of a new function called **t**. This function will simulate drawing 10 cards from the deck and then checking if the conditions (exactly four hearts and at least two spades) are met.

d <- sample(deck, 10): The **sample()** function is used to randomly draw 10 cards from our deck. These drawn cards are stored in **d** vector.

if(sum(d == "H") == 4) {: Here, we are checking if exactly four of the drawn cards are hearts. The expression **d == "H"** creates a logical vector of **TRUE** and **FALSE** values. Each **TRUE** indicates that the card is a heart. By summing this logical vector with the **sum()** function, we get the number of hearts in **d**. We then check if this count is exactly 4.

if(sum(d == "S") >= 2) {: If the previous condition is met (four hearts), this block checks if there are at least two spades among the drawn cards. If so, the function returns **TRUE**, indicating that this particular simulation met the desired conditions.

return(FALSE); If either of the conditions (four hearts or at least two spades) is not met, the function returns FALSE.

num <- 100000: This line sets the number of simulations to 100,000. This means we will repeat our card-drawing scenario 100,000 times to get an approximate probability.

results <- replicate(num, t()): The **replicate()** command is used to run the **t** function 100,000 times (as specified by **num**). The **results** (a series of **TRUE** and **FALSE** values) are stored in the **results** vector.

p <- sum(results)/num: Finally, we calculate the approximate probability by dividing the number of **TRUE** values in the **results** vector by the total number of simulations (**num**). The result, **p**, gives the estimated probability that our conditions are met. ♦

In the following example we will use an R code **setdiff()**, which returns the difference between two sets. In other words, it returns all elements in the first set that are not in the second set. For example, the command **setdiff(c(1,2,3,7), c(1,3))** will return 2 and 7.

Example 3.41 In this example, we will address the **Monty Hall Problem** using an R simulation. The problem, as detailed in Example 3.9, is as follows: Olivia is a contestant on a show similar to *Let's Make a Deal*, hosted by Monty Hall. She is presented with three doors. Behind one is a car (the prize she desires), and behind the other two are goats. Olivia picks a door. Monty, being privy to what is behind each door, proceeds to open another door, revealing a goat. He then offers Olivia a choice: stay with her initial pick or switch to the remaining unopened door. The crux of the Monty Hall problem is this: Is Olivia better off sticking with her first choice, switching, or is the outcome unaffected by her decision?

Solution: The following R program is designed to compute the desired probability. We will provide explanations for each line of code right after presenting it. In the program **i** is Olivia's initial choice, **k** is the door Monty opens, **ns** is the result of the simulation for the strategy of no switching.

```
> t <- function(sw=FALSE){
+   doors <- c("goat", "goat" , "car")
+   if(doors[i]=="car"){
+     k <- sample(setdiff(1:3, i),1)
+   }
+   else{ g <- (1:3)[doors=="goat"]
+     k <- setdiff(g,i)[1]
+   }
+   i <- sample(1:3, 1)
+   if(doors[i]=="car"){
+     k <- sample(setdiff(1:3, i),1)
+   }
+   else{ g <- (1:3)[doors=="goat"]
+     k <- setdiff(g,i)[1]
+   }
+   if(sw){
```

```

+ fc <- setdiff(1:3, c(i.k))[1]
+ } else {
+ fc <- i
+ }
+ return(doors[fc])
+ }
> ns <- replicate(100000, t(sw=FALSE))
> pns <- mean(ns == "car")
> print(pns)
[1] 0.33277

```

As highlighted in Example 3.9, the exact answer is $1/3$. The simulation yielding a result of 0.33277 provides a close approximation to $1/3$.

Let us proceed to break down and discuss each segment of the program in detail:

t <- function(sw=FALSE) { This line starts the definition of a function called **t**. The function has an argument **sw** which is set to **FALSE**. By default, the simulation operates under the scenario where Olivia retains her original choice. If the simulation is executed with **sw=TRUE**, it implies that she opts to switch doors after Monty unveils a goat.

doors <- c("goat", "goat", "car") This line creates a vector **doors** which represents the three doors and what is behind them: two goats and one car.

i <- sample(1:3, 1) Here, the function randomly selects a door for the contestant using the **sample()** function. The door is chosen from 1:3 (representing the three doors) and only one door is chosen, hence the, 1.

```

+ if(doors[i]=="car"){
+ k <- sample(setdiff(1:3, i), 1)
+ else{ g <- (1:3)[doors=="goat"]
+ k <- setdiff(g, i)[1]}

```

This block of code determines which door Monty will open. If Olivia's initial choice hides the car, Monty chooses randomly from the two remaining doors, both of which hide goats. If her initial choice is a goat, Monty opens the other door with a goat.

```

if(sw){
fc <- setdiff(1:3, c(i.k))[1]
} else {
fc <- i}

```

This block decides the Olivia's final choice. If she chooses to switch (**sw** is **TRUE**), she changes her choice to the door that remains unopened. If she does not switch, she stays with her initial choice.

return(doors[fc]) The function then returns the result of Olivia's final choice, which is either "goat" or "car".

ns <- replicate(100000, t(sw=FALSE)) Here, the simulation is run 100,000 times for the strategy of not switching. The result of this simulation is stored in vector **ns**.

`pns <- mean(ns == "car")` This calculates the proportion of times the contestant wins the car.

`print(pns)` Prints the probability of winning the car if her strategy is not switching. ♦

EXERCISES

1. From a standard deck of 52 cards, A, B, C, and D each receive 13 cards. Given that A and B collectively have six hearts and five spades, use an R simulation to estimate the probability that C possesses two of the leftover seven hearts and three of the eight remaining spades.
2. From a group of families each having five children, a *family* is randomly chosen and found to include a boy. Utilizing R simulation, determine an approximate probability that this family has three boys and two girls, with the third child being a boy. Assume that all gender combinations are equally likely for a five-child family.
3. **(Laplace's Law of Succession)** Consider u urns, labeled from 0 to $u - 1$. The i th urn holds i red balls and $u - 1 - i$ white balls, where $0 \leq i \leq u - 1$. An urn is chosen randomly, and its balls are drawn sequentially, at random, and replaced after each draw. Given that the first m draws are all red, develop an R simulation to approximate the probability that the $(m + 1)$ th draw is also a red ball.
4. Referring to the previous exercise (Laplace's law of succession), assume that of the first m balls drawn, r are red and $m - r$ are white. Employ an R simulation to estimate the probability that the $(m + 1)$ st draw yields a red ball.
5. Targets A and B are positioned on a wall. Given that the probabilities of striking A and B with a missile are 0.3 and 0.4 respectively for each shot, utilize an R simulation to approximate the probability of hitting target B, given that target A was not struck. (For reference, the precise probability is $4/7$, so the simulation's result should be in the vicinity of $4/7$ or approximately 0.571.)



CHAPTER 3 SUMMARY

► **Conditional Probability** If $P(B) > 0$, the conditional probability of A given B , denoted by $P(A | B)$, is defined to be $P(A | B) = \frac{P(AB)}{P(B)}$.

► **Reduction of Sample Space** Let B be an event of a sample space S with $P(B) > 0$. For a subset A of B , define $Q(A) = P(A | B)$. Then Q is a function from the set of subsets of B to $[0, 1]$ and it satisfies the axioms that probabilities do, and hence it is a probability function. Note that while P is defined for all subsets of S , the probability function Q is defined only for subsets of B . Therefore, for Q , the sample space is reduced from S to B . This reduction of sample space is sometimes very helpful in calculating conditional probabilities. Suppose that we are interested in

$P(E | B)$, where $E \subseteq S$. One way to calculate this quantity is to reduce S to B and find $Q(EB)$. It is usually easier to compute the unconditional probability $Q(EB)$ rather than the conditional probability $P(E | B)$.

► **The Multiplication Rule** If $P(B) > 0$, then $P(AB) = P(B)P(A | B)$. In general, if $P(A_1 A_2 A_3 \cdots A_{n-1}) > 0$, then

$$\begin{aligned} P(A_1 A_2 A_3 \cdots A_{n-1} A_n) \\ = P(A_1)P(A_2 | A_1)P(A_3 | A_1 A_2) \cdots P(A_n | A_1 A_2 A_3 \cdots A_{n-1}). \end{aligned}$$

► **Partition of a Sample Space** Let $\{B_1, B_2, \dots, B_n\}$ be a set of nonempty subsets of the sample space S of an experiment. If the events $B_1, B_2, \dots, B_n$ are mutually exclusive and $\bigcup_{i=1}^n B_i = S$, the set $\{B_1, B_2, \dots, B_n\}$ is called a partition of S .

► **Law of Total Probability** Let B be an event with $P(B) > 0$ and $P(B^c) > 0$. Then for any event A , we have that

$$P(A) = P(A | B)P(B) + P(A | B^c)P(B^c).$$

In general, if $\{B_1, B_2, \dots, B_n\}$ is a partition of the sample space of an experiment and $P(B_i) > 0$ for $i = 1, 2, \dots, n$, then for any event A of S ,

$$\begin{aligned} P(A) &= P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + \cdots + P(A | B_n)P(B_n) \\ &= \sum_{i=1}^n P(A | B_i)P(B_i). \end{aligned}$$

Even more generally, let $\{B_1, B_2, \dots\}$ be a sequence of mutually exclusive events of S such that $\bigcup_{i=1}^{\infty} B_i = S$. Suppose that, for all $i \geq 1$, $P(B_i) > 0$. Then for any event A of S ,

$$P(A) = \sum_{i=1}^{\infty} P(A | B_i)P(B_i).$$

► **Bayes' Formula** If $P(B) > 0$ and $P(B^c) > 0$, then for any event A of S with $P(A) > 0$,

$$P(B | A) = \frac{P(A | B)P(B)}{P(A | B)P(B) + P(A | B^c)P(B^c)}.$$

In general, Let $\{B_1, B_2, \dots, B_n\}$ be a partition of the sample space S of an experiment. If for $i = 1, 2, \dots, n$, $P(B_i) > 0$, then for any event A of S with $P(A) > 0$,

$$P(B_k | A) = \frac{P(A | B_k)P(B_k)}{P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + \cdots + P(A | B_n)P(B_n)}.$$

► **Independence of Events** Two events A and B are called independent if $P(AB) = P(A)P(B)$. If A and B are independent, and $P(B) > 0$, then $P(A | B) = P(A)$; that is, if A is independent

of B , knowledge regarding the occurrence of B does not change the chance of the occurrence of A . If two events are not independent, they are called dependent. If A and B are independent, then A and B^c are independent as well. As a result, If A and B are independent, then so are A^c and B^c . In general, the set of events $\{A_1, A_2, \dots, A_n\}$ is called independent if for every subset $\{A_{i_1}, A_{i_2}, \dots, A_{i_k}\}$, $k \geq 2$, of $\{A_1, A_2, \dots, A_n\}$,

$$P(A_{i_1} A_{i_2} \cdots A_{i_k}) = P(A_{i_1})P(A_{i_2}) \cdots P(A_{i_k}).$$

REVIEW PROBLEMS

- Two persons arrive at a train station, independently of each other, at random times between 1:00 P.M. and 1:30 P.M. What is the probability that one will arrive between 1:00 P.M. and 1:12 P.M., and the other between 1:17 P.M. and 1:30 P.M.?
- From the students of a college that does not offer graduate programs, a student is selected at random. Let E_1 , E_2 , E_3 , and E_4 denote the events that the student is a first-year student, sophomore, junior, and senior, respectively. Let A be the event that the randomly selected student's grade point average (GPA) is A^+ , A , or A^- . Define the events B , C , D similarly. Let F be the event that the student's GPA is F. Which of the following sets of events is a partition of the sample space of this experiment? Which of them is not?

(a) $\{E_1, E_2, E_3, E_4\}$, (b) $\{A, B, C, D\}$, (c) $\{E_1, E_3, B, C\}$, (d) $\{A, B, C, D, F\}$.

- A polygraph operator detects innocent suspects as being guilty 3% of the time. If during a crime investigation six innocent suspects are examined by the operator, what is the probability that at least one of them is detected as guilty?
- Suppose that 5% of men and 0.25% of women are colorblind. In Belavia, only 42% of the persons 65 and older are male. What is the probability that a randomly selected person from this age group is colorblind?
- In statistical surveys where individuals are selected randomly and are asked questions, experience has shown that only 48% of those under 25 years of age, 67% between 25 and 50, and 89% above 50 will respond. A social scientist is about to send a questionnaire to a group of randomly selected people. If 30% of the population are younger than 25 and 17% are older than 50, what percent will answer her questionnaire?
- Let $n > 1$ be an integer. Suppose that a fair coin is tossed independently and repeatedly. Find the probability of tails on the first toss or heads on the n th toss.
- Every morning, Galya flips a fair coin to decide whether she wants to take the bus to school or she wants to drive there. No matter what her means of transport, if she leaves her house at or before 7:00 a.m., she gets to school on time. However, if she leaves 15 minutes later, at 7:15 a.m., then the probability is 0.80 to arrive at school on time if she drives her own car, and the probability is 0.62 to get to school on time if she takes the bus. On a certain day, Galya left her house at 7:15 and got to school on time. What is the probability that she took the bus on that day?

8. Diseases D_1 , D_2 , and D_3 cause symptom A with probabilities 0.5, 0.7, and 0.8, respectively. If 5% of a population have disease D_1 , 2% have disease D_2 , and 3.5% have disease D_3 , what percent of the population have symptom A? Assume that the only possible causes of symptom A are D_1 , D_2 , and D_3 and that no one carries more than one of these three diseases.
9. Professor Stern has three cars. The probability that on a given day car 1 is operative is 0.95, that car 2 is operative is 0.97, and that car 3 is operative is 0.85. If Professor Stern's cars operate independently, find the probability that on next Thanksgiving day (a) all three of his cars are operative; (b) at least one of his cars is operative; (c) at most two of his cars are operative; (d) none of his cars is operative.
10. A bus traveling from Baltimore to New York breaks down at a random location. If the bus was seen running at Wilmington, what is the probability that the breakdown occurred after passing through Philadelphia? The distances from New York, Philadelphia, and Wilmington to Baltimore are, respectively, 199, 96, and 67 miles.
11. Stacy and George are playing the heads or tails game with a fair coin. The coin is flipped repeatedly until either the fifth heads or the fifth tails appear. If the fifth heads occurs first, Stacy wins the game. Otherwise, George is the winner. Suppose that after the fifth flip, three heads and two tails have occurred. What is the probability that Stacy wins this game?
12. Roads A, B, and C are the only escape routes from a state prison. Prison records show that, of the prisoners who tried to escape, 30% used road A, 50% used road B, and 20% used road C. These records also show that 80% of those who tried to escape via A, 75% of those who tried to escape via B, and 92% of those who tried to escape via C were captured. What is the probability that a prisoner who succeeded in escaping used road C?
13. From an ordinary deck of 52 cards, 10 cards are drawn at random. If exactly four of them are hearts, what is the probability of at least one spade being among them?
14. A fair die is thrown twice. If the second outcome is 6, what is the probability that the first one is 6 as well?
15. Suppose that 10 dice are thrown and we are told that among them at least one has landed 6. What is the probability that there are two or more sixes?
16. Urns I and II contain three pennies and four dimes, and two pennies and five dimes, respectively. One coin is selected at random from each urn. If exactly one of them is a dime, what is the probability that the coin selected from urn I is the dime?
17. There are 49 unbiased dice and one loaded die in a basket. For the loaded die, when tossed the probability of obtaining 6 is $3/8$, and the probability of obtaining each of the other five faces is $1/8$. A die is selected at random and tossed five times. If all five times it lands on 6, what is the probability that the die selected at random is the loaded one?
18. Six fair dice are tossed independently. Find the probability that the number of 1s minus the number of 2s will be 3.
19. An experiment consists of first tossing an unbiased coin and then rolling a fair die. If we perform this experiment successively, what is the probability of obtaining a heads on the coin before a 1 or 2 on the die?
20. Suppose that the Dow-Jones Industrial Average (DJIA) rises 52% of trading days and falls

48% of those days. Elmer is a stock market expert, and his forecasts will turn out to be true 68% of the time when he predicts that the DJIA will rise the next trading day. However, when he predicts that it will fall the next trading day, only 59% of the time his forecasts will come true. Tomorrow is a trading day, and Elmer has predicted that the DJIA will fall. What is the probability that Elmer's prediction will turn out to be wrong, and the DJIA will rise tomorrow?

21. Urn I contains 25 white and 20 black balls. Urn II contains 15 white and 10 black balls. An urn is selected at random and one of its balls is drawn randomly and observed to be black and then returned to the same urn. If a second ball is drawn at random from this urn, what is the probability that it is black?
22. An urn contains nine red and one blue balls. A second urn contains one red and five blue balls. One ball is removed from each urn at random and without replacement, and all of the remaining balls are put into a third urn. What is the probability that a ball drawn randomly from the third urn is blue?
23. A fair coin is tossed. If the outcome is heads, a red hat is placed on Lorna's head. If it is tails, a blue hat is placed on her head. Lorna cannot see the hat. She is asked to guess the color of her hat. Is there a strategy that maximizes Lorna's chances of guessing correctly?
Hint: Suppose that Lorna chooses the color red with probability α and blue with probability $1 - \alpha$. Find the probability that she guesses correctly.
24. A child is lost at Epcot Center in Florida. The father of the child believes that the probability of his being lost in the east wing of the center is 0.75, and in the west wing 0.25. The security department sends three officers to the east wing and two to the west to look for the child. Suppose that an officer who is looking in the correct wing (east or west) finds the child, independently of the other officers, with probability 0.4. Find the probability that the child is found.
25. Solve the following problem, asked of Marilyn Vos Savant in the "Ask Marilyn" column of *Parade Magazine*, August 9, 1992.

Three of us couples are going to Lava Hot Springs next weekend. We're staying two nights, and we've rented two studios, because each holds a maximum of only four people. One couple will get their own studio on Friday, a different couple on Saturday, and one couple will be out of luck. We'll draw straws to see which are the two lucky couples. I told my wife we should just draw once, and the loser would be the couple out of luck both nights. I figure we'll have a two-out-of-three ($66\frac{2}{3}\%$) chance of winning one of the nights to ourselves. But she contends that we should draw straws *twice*—first on Friday and then, for the remaining two couples only, on Saturday—reasoning that a one-in-three ($33\frac{1}{3}\%$) chance for Friday and a one-in-two (50%) chance for Saturday will give us better odds. Which way should we go?

26. A student at a certain university will pass the oral Ph.D. qualifying examination if at least two of the three examiners pass her or him. Past experience shows that (a) 15% of the students who take the qualifying exam are not prepared, and (b) each examiner will independently pass 85% of the prepared and 20% of the unprepared students. Kevin took his Ph.D. qualifying exam with Professors Smith, Brown, and Rose. What is the probability that Professor

Rose has passed Kevin if we know that neither Professor Brown nor Professor Smith has passed him? Let S , B , and R be the respective events that Professors Smith, Brown, and Rose have passed Kevin. Are these three events independent? Are they conditionally independent given that Kevin is prepared? (See Exercises 50 and 51, [Section 3.5](#).)



Self-Test on Chapter 3

Time allotted: 120 Minutes

Each problem is worth 10 points.

1. Currently, negotiations on complicated border disputes are going on between Chernarus and Carpathia. Suppose that a summit between the foreign ministers of the two countries will resolve all the border issues with probability $1/4$. Furthermore, suppose that, for $i \geq 2$, if the previous $i - 1$ summits are unsuccessful, the i th summit will be successful with probability $1 - (1/i)$. What is the probability that no more than 4 summits are necessary to successfully resolve the border disputes between Chernarus and Carpathia?
2. Suppose that a system of five components is functional if at least one of the components A_1 , A_2 , and A_3 and both components B_1 and B_2 are operable. Suppose that a component is operable, independently of other components, with probability 0.85. Find the probability that the system functions.
3. In Helsinki, Finland, Hunter is the fourth in line at a station to board a city tour minibus having 16 passenger seats. A minibus with all empty passenger seats arrives at the station. Suppose that each passenger boarding the minibus selects one of the unoccupied seats at random. If Hunter's favorite seat is the window seat in the front row opposite from the minibus driver, what is the probability that when he boards the minibus his favorite seat is unoccupied?
4. Tasha has two dice, one is unbiased, but the other one is loaded so that the probability of 6 is twice the probability of any of the other five faces, which are all equally likely. Tasha picks one of the two dice randomly and tosses it three times. If all three times it lands on 6, what is the probability that the die Tasha picked is the unbiased one?
5. Suppose that before death, an individual in a population of living organisms gives birth to 0, 1, or 2 new individuals, independently of other organisms with probabilities $1/4$, $1/2$, and $1/4$, respectively. What is the probability that the second generation offspring of such a living organism consists of at least one individual?
6. In a metropolis, consecutive traffic lights are coordinated so that a driver who finds the first light green will find the second light also green with probability 0.75. Similarly, a driver who finds the first traffic light yellow or red will also find the second traffic light yellow or red with probability 0.75. Suppose that 50% of the time a traffic light is green. Find the probability that a driver has to stop at least once for the red or yellow lights when passing through two consecutive traffic lights.

7. In a lottery scratch-off game, each ticket has 10 coated circles in the middle and one coated rectangle in the lower left corner. Underneath the coats of 4 of the circles, there is a dollar sign, "\$," and underneath the remaining 6 circles is blank. A winning ticket is the one with only those circles that contain the dollar sign scratched. The remaining circles and the rectangle must be left untouched. The winning amount is written underneath the coat of the rectangle, which will be scratched off by the lottery agent to reveal the amount won. Harlan plays one such lottery scratch-off game and loses. What is the probability that the first circle he scratched off which was blank underneath was the third circle?
8. Suppose that of the individuals who have been exposed to the dust product of the mineral asbestos, 3.3 per 1000 people develop mesothelioma. Suppose that Keith, a mine worker exposed to such a dust product, was tested for mesothelioma, and the test result came back positive. If 92% of the time the test is accurate, what is the probability that Keith has mesothelioma?
9. Vincent is a patient with the life threatening blood cancer leukemia, and he is in need of a bone marrow transplant. He asks n people whether or not they are willing to donate bone marrow to him if they are a close bone marrow match for him. Suppose that each person's response, independently of others, is positive with probability p_1 and negative with probability $1 - p_1$. Furthermore, suppose that, independently of the others, the probability is p_2 that a person tested is a close match. (a) Find the probability that Vincent finds at least one close bone marrow match among these n people. (b) Suppose that the probability is 1 in 10,000 that two unrelated persons are compatible; that is, have a close bone marrow match. If 70% of the people asked would respond positively to Vincent's request, and he asks 1000 people, what is the probability that he will find a match? What if he asks 100,000 people?
10. Starting with Hillary, two players, Hillary and Donald, take turns and roll a fair die. The one who rolls a 6 first is the winner. What is the probability that Hillary will win?
Hint: For $i \geq 1$, let A_i be the event that the first 6 is rolled on Hillary's i th turn. Let H be the event that Hillary will win. Write $P(H)$ in terms of $P(A_i)$'s. Note that for $|r| < 1$, $m \geq 1$, $\sum_{i=m}^{\infty} r^i = r^m / (1 - r)$.

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Chapter 4

Distribution Functions and Discrete Random Variables

4.1 RANDOM VARIABLES

In real-world problems we are often faced with one or more quantities that do not have fixed values. The values of such quantities depend on random actions, and they usually change from one experiment to another. For example, the number of babies born in a certain hospital each day is not a fixed quantity. It is a complicated function of many random factors that vary from one day to another. So are the following quantities: the arrival time of a bus at a station, the sum of the outcomes of two dice when thrown, the amount of rainfall in Seattle during a given year, the number of earthquakes that occur in California per month, and the weight of grains of wheat grown on a certain plot of land (it varies from one grain to another). In probability, quantities introduced in these diverse examples are called **random variables**. The numerical values of random variables are unknown. They depend on random elements occurring at the time of the experiment and over which we have no control. For example, if in rolling two fair dice, X is the sum, then X can only assume the values 2, 3, 4, $\dots$, 12 with the following probabilities:

$$P(X = 2) = P(\{(1, 1)\}) = 1/36,$$

$$P(X = 3) = P(\{(1, 2), (2, 1)\}) = 2/36,$$

$$P(X = 4) = P(\{(1, 3), (2, 2), (3, 1)\}) = 3/36,$$

and, similarly,

Sum, i	5	6	7	8	9	10	11	12
$P(X = i)$	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

Clearly, $\{2, 3, 4, \dots, 12\}$ is the set of possible values of X . Since $X \in \{2, 3, 4, \dots, 12\}$, we should have $\sum_{i=2}^{12} P(X = i) = 1$, which is readily verified. The numerical value of a random variable depends on the outcome of the experiment. In this example, for instance, if the outcome is $(2, 3)$, then X is 5, and if it is $(5, 6)$, then X is 11. X is not defined for points that do not belong to S , the sample space of the experiment. Thus X is a real-valued function

on S . However, not all real-valued functions on S are considered to be random variables. For theoretical reasons, it is necessary that the inverse image of an interval in $\mathbb{R}$ be an event of S , which motivates the following definition.

Definition 4.1 *Let S be the sample space of an experiment. A real-valued function $X: S \rightarrow \mathbb{R}$ is called a **random variable** of the experiment if, for each interval $I \subseteq \mathbb{R}$, $\{s: X(s) \in I\}$ is an event.*

★ **Remark 4.1** Let S be the sample space of an experiment and let $\mathcal{F}$ be the σ -field of all events. A random variable X is a function from S to $\mathbb{R}$ so that, for each real number x , the set $\{\omega \in S: X(\omega) \leq x\}$ is an event. That is, $\{\omega \in S: X(\omega) \leq x\} \in \mathcal{F}$. ♦

In probability, the set $\{s: X(s) \in I\}$ is often abbreviated as $\{X \in I\}$, or simply as $X \in I$. For all practical purposes, random variables can be thought of as functions from the sample space S to the real numbers, $\mathbb{R}$.

Example 4.1 Suppose that three cards are drawn from an ordinary deck of 52 cards, one by one, at random and with replacement. Let X be the number of spades drawn, then X is a random variable. If an outcome of spades is denoted by s , and other outcomes are represented by t , then X is a real-valued function defined on the sample space

$$S = \{(s, s, s), (t, s, s), (s, t, s), (s, s, t), (s, t, t), (t, s, t), (t, t, s), (t, t, t)\},$$

by $X(s, s, s) = 3$, $X(s, t, s) = 2$, $X(s, s, t) = 2$, $X(s, t, t) = 1$, and so on. Now we must determine the values that X assumes and the probabilities that are associated with them. Clearly, X can take the values 0, 1, 2, and 3. The probabilities associated with these values are calculated as follows:

$$P(X = 0) = P(\{(t, t, t)\}) = \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} = \frac{27}{64},$$

$$\begin{aligned} P(X = 1) &= P(\{(s, t, t), (t, s, t), (t, t, s)\}) \\ &= \left(\frac{1}{4} \times \frac{3}{4} \times \frac{3}{4}\right) + \left(\frac{3}{4} \times \frac{1}{4} \times \frac{3}{4}\right) + \left(\frac{3}{4} \times \frac{3}{4} \times \frac{1}{4}\right) = \frac{27}{64}, \end{aligned}$$

$$\begin{aligned} P(X = 2) &= P(\{(s, s, t), (s, t, s), (t, s, s)\}) \\ &= \left(\frac{1}{4} \times \frac{1}{4} \times \frac{3}{4}\right) + \left(\frac{1}{4} \times \frac{3}{4} \times \frac{1}{4}\right) + \left(\frac{3}{4} \times \frac{1}{4} \times \frac{1}{4}\right) = \frac{9}{64}, \end{aligned}$$

$$P(X = 3) = P(\{(s, s, s)\}) = \frac{1}{64}.$$

If the cards are drawn without replacement, the probabilities associated with the values 0, 1, 2, and 3 are:

$$P(X = 0) = \frac{\binom{39}{3}}{\binom{52}{3}},$$

$$P(X = 1) = \frac{\binom{13}{1}\binom{39}{2}}{\binom{52}{3}},$$

$$P(X = 2) = \frac{\binom{13}{2}\binom{39}{1}}{\binom{52}{3}},$$

$$P(X = 3) = \frac{\binom{13}{3}}{\binom{52}{3}}.$$

Therefore,

$$P(X = i) = \frac{\binom{13}{i}\binom{39}{3-i}}{\binom{52}{3}}, \quad i = 0, 1, 2, 3. \quad \blacklozenge$$

Example 4.2 A bus stops at a station every day at some random time between 11:00 A.M. and 11:30 A.M. If X is the actual arrival time of the bus, X is a random variable. It is a function defined on the sample space $S = \{t: 11 < t < 11\frac{1}{2}\}$ by $X(t) = t$. As we know from [Section 1.7](#), $P(X = t) = 0$ for any $t \in S$, and $P(X \in (\alpha, \beta)) = \frac{\beta - \alpha}{11\frac{1}{2} - 11} = 2(\beta - \alpha)$ for any subinterval (α, β) of $(11, 11\frac{1}{2})$. $\blacklozenge$

Example 4.3 In the United States, the number of twin births is approximately 1 in 90. Let X be the number of births in a certain hospital until the first twins are born. X is a random variable. Denote twin births by T and single births by N . Then X is a real-valued function defined on the sample space

$$S = \{T, NT, NNT, NNNT, \dots\} \quad \text{by} \quad X(\underbrace{NNN \cdots N}_{i-1}T) = i.$$

The set of all possible values of X is $\{1, 2, 3, \dots\}$ and

$$P(X = i) = P(\underbrace{NNN \cdots N}_{i-1}T) = \left(\frac{89}{90}\right)^{i-1} \left(\frac{1}{90}\right). \quad \blacklozenge$$

Example 4.4 In a certain country, the draft-status priorities of eligible men are determined according to their birthdays. Suppose that numbers 1 to 31 are assigned to men with birthdays on January 1 to January 31, numbers 32 to 60 to men with birthdays on February 1 to February 29, numbers 61 to 91 to men with birthdays on March 1 to March 31, $\dots$, and finally,

numbers 336 to 366 to those with birthdays on December 1 to December 31. Then numbers are selected at random, one by one, and without replacement, from 1 to 366 until all of them are chosen. Those with birthdays corresponding to the first number drawn would have the highest draft priority, those with birthdays corresponding to the second number drawn would have the second-highest priority, and so on. Let X be the largest of the first 10 numbers selected. Then X is a random variable that assumes the values 10, 11, 12, $\dots$, 366. The event $X = i$ occurs if the largest number among the first 10 is i , that is, if one of the first 10 numbers is i and the other 9 are from 1 through $i - 1$. Thus

$$P(X = i) = \frac{\binom{i-1}{9}}{\binom{366}{10}}, \quad i = 10, 11, 12, \dots, 366.$$

As an application, let us calculate $P(X \geq 336)$, the probability that, among individuals having one of the 10 highest draft priorities, there are some with birthdays in December. We have that

$$P(X \geq 336) = \sum_{i=336}^{366} \frac{\binom{i-1}{9}}{\binom{366}{10}} \approx 0.592. \quad \blacklozenge$$

Remark: On December 1, 1969 the Selective Service headquarters in Washington, D.C. determined the draft priorities of 19-year-old males by using a method very similar to that of Example 4.4. $\blacklozenge$

We will now explain how, from given random variables, new ones can be formed. Let X and Y be two random variables over the same sample space S , then $X: S \rightarrow \mathbb{R}$ and $Y: S \rightarrow \mathbb{R}$ are real-valued functions having the same domain. Therefore, we can form the functions $X + Y$; $X - Y$; $aX + bY$, where a and b are constants; XY ; and X/Y , where $Y \neq 0$. Since the domain of these real-valued functions is S , they are also random variables defined on S . Hence the sum, the difference, linear combinations, the product, and the quotients (if they exist) of random variables are themselves random variables. Similarly, if $f: \mathbb{R} \rightarrow \mathbb{R}$ is an ordinary real-valued function, the composition of f and X , $f \circ X: S \rightarrow \mathbb{R}$, is also a random variable. For example, let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$, then $f \circ X: S \rightarrow \mathbb{R}$ is X^2 . Hence X^2 is a random variable as well. Similarly, functions such as $\sin X$, $\cos X^2$, e^X , and $X^3 - 2X$ are random variables. So are the following functions: $X^2 + Y^2$; $(X^2 + Y^2)/(2Y + 1)$, $Y \neq -1/2$; $\sin X + \cos Y$; $\sqrt{X^2 + Y^2}$; and so on. Functions of random variables appear naturally in probability problems. As an example, suppose that we choose a point at random from the unit disk in the plane. If X and Y are the x and the y coordinates of the point, X and Y are random variables. The distance of (X, Y) from the origin is $\sqrt{X^2 + Y^2}$, a random variable that is a function of both X and Y .

Example 4.5 The diameter of a flat metal disk manufactured by a factory is a random number between 4 and 4.5. What is the probability that the area of such a flat disk chosen at random is at least 4.41π ?

Solution: Let D be the diameter of the metal disk selected at random. D is a random variable, and the area of the metal disk, $\pi(D/2)^2$, which is a function of D , is also a random variable. We are interested in the probability of the event $\pi D^2/4 > 4.41\pi$, which is

$$P\left(\frac{\pi D^2}{4} > 4.41\pi\right) = P(D^2 > 17.64) = P(D > 4.2).$$

Now to calculate $P(D > 4.2)$, note that, since the length of D is a random number in the interval $(4, 4.5)$, the probability that it falls into the subinterval $(4.2, 4.5)$ is $(4.5 - 4.2)/(4.5 - 4) = 3/5$. Hence $P(\pi D^2/4 > 4.41\pi) = 3/5$. ♦

Example 4.6 A random number is selected from the interval $(0, \pi/2)$. What is the probability that its sine is greater than its cosine?

Solution: Let the number selected be X , then X is a random variable and therefore $\sin X$ and $\cos X$, which are functions of X , are also random variables. We are interested in the probability of the event $\sin X > \cos X$:

$$P(\sin X > \cos X) = P(\tan X > 1) = P\left(X > \frac{\pi}{4}\right) = \frac{\frac{\pi}{2} - \frac{\pi}{4}}{\frac{\pi}{2} - 0} = \frac{1}{2},$$

where the first equality holds since in the interval $(0, \pi/2)$, $\cos X > 0$, and the second equality holds since, in this interval, $\tan X$ is strictly increasing. ♦

4.2 DISTRIBUTION FUNCTIONS

Random variables are often used for the calculation of the probabilities of events. For example, in the experiment of throwing two dice, if we are interested in a sum of at least 8, we define X to be the sum and calculate $P(X > 8)$. Other examples are the following:

1. If a bus arrives at a random time between 10:00 A.M. and 10:30 A.M. at a station, and X is the arrival time, then $X < 10\frac{1}{6}$ is the event that the bus arrives before 10:10 A.M.
2. If X is the price of gold per troy ounce on a random day, then $X \leq 400$ is the event that the price of gold remains at or below \$400 per troy ounce.
3. If X is the number of votes that the next Democratic presidential candidate will get, then $X \geq 5 \times 10^7$ is the event that he or she will get at least 50 million votes.
4. If X is the number of heads in 100 tosses of a coin, then $40 < X \leq 60$ is the event that the number of heads is at least 41 and at most 60.

Usually, when dealing with a random variable X , for constants a and b ($b < a$), computation of one or several of the probabilities $P(X = a)$, $P(X < a)$, $P(X \leq a)$, $P(X > b)$, $P(X \geq b)$, $P(b \leq X \leq a)$, $P(b < X \leq a)$, $P(b \leq X < a)$, and $P(b < X < a)$ is our ultimate goal. For this reason we calculate $P(X \leq t)$ for all $t \in (-\infty, +\infty)$. As we will show shortly, if $P(X \leq t)$ is known for all $t \in \mathbb{R}$, then for any a and b , all of the probabilities that are mentioned above can be calculated. In fact, since the real-valued function $P(X \leq t)$ characterizes X , it tells us almost everything about X . This function is called the **distribution function** of X .

Definition 4.2 *If X is a random variable, then the function F defined on $(-\infty, +\infty)$ by $F(t) = P(X \leq t)$ is called the **distribution function** of X .*

Since F “accumulates” all of the probabilities of the values of X up to and including t , sometimes it is called the **cumulative distribution function** of X . The most important properties of the distribution functions are as follows:

1. **F is nondecreasing**; that is, if $t < u$, then $F(t) \leq F(u)$. To see this, note that the occurrence of the event $\{X \leq t\}$ implies the occurrence of the event $\{X \leq u\}$. Thus $\{X \leq t\} \subseteq \{X \leq u\}$ and hence $P(X \leq t) \leq P(X \leq u)$. That is, $F(t) \leq F(u)$.
2. **$\lim_{t \rightarrow \infty} F(t) = 1$** . To prove this, it suffices to show that for any increasing sequence $\{t_n\}$ of real numbers that converges to ∞ , $\lim_{n \rightarrow \infty} F(t_n) = 1$. This follows from the continuity property of the probability function (see Theorem 1.8). The events $\{X \leq t_n\}$ form an increasing sequence that converges to the event $\bigcup_{n=1}^{\infty} \{X \leq t_n\} = \{X < \infty\}$; that is, $\lim_{n \rightarrow \infty} \{X \leq t_n\} = \{X < \infty\}$. Hence

$$\lim_{n \rightarrow \infty} P(X \leq t_n) = P\left(\bigcup_{n=1}^{\infty} \{X \leq t_n\}\right) = P(X < \infty) = 1,$$

which means that

$$\lim_{n \rightarrow \infty} F(t_n) = 1.$$

3. **$\lim_{t \rightarrow -\infty} F(t) = 0$** . The proof of this is similar to the proof that $\lim_{t \rightarrow \infty} F(t) = 1$.
4. **F is right continuous**. That is, for every $t \in \mathbb{R}$, $F(t+) = F(t)$. This means that if t_n is a decreasing sequence of real numbers converging to t , then

$$\lim_{n \rightarrow \infty} F(t_n) = F(t).$$

To prove this, note that since t_n decreases to t , the events $\{X \leq t_n\}$ form a decreasing sequence that converges to the event $\bigcap_{n=1}^{\infty} \{X \leq t_n\} = \{X \leq t\}$. Thus, by the continuity property of the probability function,

$$\lim_{n \rightarrow \infty} P(X \leq t_n) = P\left(\bigcap_{n=1}^{\infty} \{X \leq t_n\}\right) = P(X \leq t),$$

which means that

$$\lim_{n \rightarrow \infty} F(t_n) = F(t).$$

As mentioned previously, by means of F , the distribution function of a random variable X , a wide range of probabilistic questions concerning X can be answered. Here are some examples.

1. To calculate $P(X > a)$, note that $P(X > a) = 1 - P(X \leq a)$, thus

$$P(X > a) = 1 - F(a).$$

2. To calculate $P(a < X \leq b)$, $b > a$, note that $\{a < X \leq b\} = \{X \leq b\} - \{X \leq a\}$ and $\{X \leq a\} \subseteq \{X \leq b\}$. Hence, by Theorem 1.5,

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a).$$

3. To calculate $P(X < a)$, note that the sequence of the events $\{X \leq a - 1/n\}$ is an increasing sequence that converges to $\bigcup_{n=1}^{\infty} \{X \leq a - 1/n\} = \{X < a\}$. Therefore, by the continuity property of the probability function (Theorem 1.8),

$$\lim_{n \rightarrow \infty} P\left(X \leq a - \frac{1}{n}\right) = P\left(\bigcup_{n=1}^{\infty} \left\{X \leq a - \frac{1}{n}\right\}\right) = P(X < a),$$

which means that

$$P(X < a) = \lim_{n \rightarrow \infty} F\left(a - \frac{1}{n}\right).$$

Hence $P(X < a)$ is the left-hand limit of the function F as $x \rightarrow a$; that is,

$$P(X < a) = F(a-).$$

4. To calculate $P(X \geq a)$, note that $P(X \geq a) = 1 - P(X < a)$. Thus

$$P(X \geq a) = 1 - F(a-).$$

5. Since $\{X = a\} = \{X \leq a\} - \{X < a\}$ and $\{X < a\} \subseteq \{X \leq a\}$, we can write

$$P(X = a) = P(X \leq a) - P(X < a) = F(a) - F(a-).$$

Note that since F is right continuous, $F(a)$ is the right-hand limit of F . This implies the following important fact:

Let F be the distribution function of a random variable X ; $P(X = a)$ is the difference between the right- and left-hand limits of F at a . If the function F is continuous at a , these limits are the same and equal to $F(a)$. Hence $P(X = a) = 0$. Otherwise, F has a *jump* at a , and the magnitude of the jump, $F(a) - F(a-)$, is the probability that $X = a$.

As in cases 1 to 5, we can establish similar cases to obtain the following table.

Event concerning X	Probability of the event in terms of F	Event concerning X	Probability of the event in terms of F
$X \leq a$	$F(a)$	$a < X \leq b$	$F(b) - F(a)$
$X > a$	$1 - F(a)$	$a < X < b$	$F(b-) - F(a)$
$X < a$	$F(a-)$	$a \leq X \leq b$	$F(b) - F(a-)$
$X \geq a$	$1 - F(a-)$	$a \leq X < b$	$F(b-) - F(a-)$
$X = a$	$F(a) - F(a-)$		

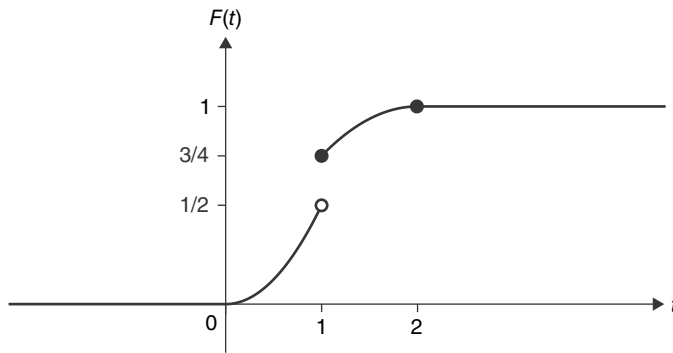


Figure 4.1 Distribution function of Example 4.7.

Example 4.7 The sales of a convenience store on a randomly selected day are X thousand dollars, where X is a random variable with a distribution function illustrated in Figure 4.1:

$$F(t) = \begin{cases} 0 & t < 0 \\ (1/2)t^2 & 0 \leq t < 1 \\ k(4t - t^2) & 1 \leq t < 2 \\ 1 & t \geq 2. \end{cases}$$

Suppose that this convenience store's total sales on any given day are less than \$2000.

- Find the value of k .
- Let A and B be the events that tomorrow the store's total sales are between 500 and 1500 dollars, and over 1000 dollars, respectively. Find $P(A)$ and $P(B)$.
- Are A and B independent events?

Solution: (a) Since $X < 2$, we have that $P(X < 2) = 1$, so $F(2-) = 1$. This gives $k(8 - 4) = 1$, so $k = 1/4$.

$$\begin{aligned}
 \text{(b)} \quad P(A) &= P\left(\frac{1}{2} \leq X \leq \frac{3}{2}\right) = F\left(\frac{3}{2}\right) - F\left(\frac{1}{2} -\right) \\
 &= F\left(\frac{3}{2}\right) - F\left(\frac{1}{2}\right) = \frac{15}{16} - \frac{1}{8} = \frac{13}{16}, \\
 P(B) &= P(X > 1) = 1 - F(1) = 1 - \frac{3}{4} = \frac{1}{4}.
 \end{aligned}$$

$$\text{(c)} \quad P(AB) = P\left(1 < X \leq \frac{3}{2}\right) = F\left(\frac{3}{2}\right) - F(1) = \frac{15}{16} - \frac{3}{4} = \frac{3}{16}. \text{ Since } P(AB) \neq P(A)P(B), A \text{ and } B \text{ are not independent. } \blacklozenge$$

Example 4.8 The distribution function of a random variable X is given by

$$F(x) = \begin{cases} 0 & x < 0 \\ x/4 & 0 \leq x < 1 \\ 1/2 & 1 \leq x < 2 \\ \frac{1}{12}x + \frac{1}{2} & 2 \leq x < 3 \\ 1 & x \geq 3, \end{cases}$$

where the graph of F is shown in Figure 4.2. Compute the following quantities:

- (a) $P(X < 2)$; (b) $P(X = 2)$; (c) $P(1 \leq X < 3)$;
 (d) $P(X > 3/2)$; (e) $P(X = 5/2)$; (f) $P(2 < X \leq 7)$.

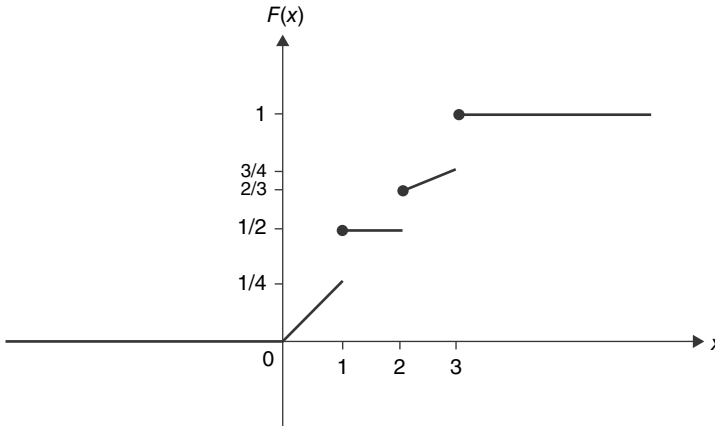


Figure 4.2 Distribution function of Example 4.8.

Solution: (a) $P(X < 2) = F(2-) = 1/2$.

(b) $P(X = 2) = F(2) - F(2-) = (2/12 + 1/2) - 1/2 = 1/6$.

- (c) $P(1 \leq X < 3) = P(X < 3) - P(X < 1) = F(3-) - F(1-) = (3/12 + 1/2) - 1/4 = 1/2.$
- (d) $P(X > 3/2) = 1 - F(3/2) = 1 - 1/2 = 1/2.$
- (e) $P(X = 5/2) = 0$ since F is continuous at $5/2$ and has no jumps.
- (f) $P(2 < X \leq 7) = F(7) - F(2) = 1 - (2/12 + 1/2) = 1/3. \quad \blacklozenge$

Example 4.9 For the experiment of flipping a fair coin twice, let X be the number of tails and calculate $F(t)$, the distribution function of X , and then sketch its graph.

Solution: Since X assumes only the values 0, 1, and 2, we have $F(t) = P(X \leq t) = 0$, if $t < 0$. If $0 \leq t < 1$, then

$$F(t) = P(X \leq t) = P(X = 0) = P(\{HH\}) = 1/4.$$

If $1 \leq t < 2$, then

$$F(t) = P(X \leq t) = P(X = 0 \text{ or } X = 1) = P(\{HH, HT, TH\}) = 3/4,$$

and if $t \geq 2$, then $P(X \leq t) = 1$. Hence

$$F(t) = \begin{cases} 0 & t < 0 \\ 1/4 & 0 \leq t < 1 \\ 3/4 & 1 \leq t < 2 \\ 1 & t \geq 2. \end{cases}$$

Figure 4.3 shows the graph of F . $\blacklozenge$

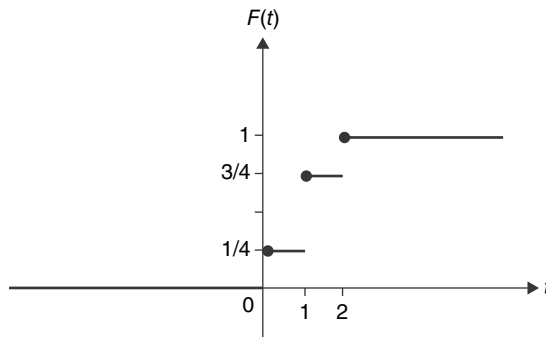


Figure 4.3 Distribution function of Example 4.9.

Example 4.10 Suppose that a bus arrives at a station every day between 10:00 A.M. and 10:30 A.M., at random. Let X be the arrival time; find the distribution function of X and sketch its graph.

Solution: The bus arrives at the station at random, between 10 and $10\frac{1}{2}$, so if $t \leq 10$, $F(t) = P(X \leq t) = 0$. Now if $t \in (10, 10\frac{1}{2})$, then

$$F(t) = P(X \leq t) = \frac{t - 10}{10\frac{1}{2} - 10} = 2(t - 10),$$

and if $t \geq 10\frac{1}{2}$, then $F(t) = P(X \leq t) = 1$. Thus

$$F(t) = \begin{cases} 0 & t < 10 \\ 2(t - 10) & 10 \leq t < 10\frac{1}{2} \\ 1 & t \geq 10\frac{1}{2}. \end{cases}$$

The graph of F is shown in Figure 4.4. ♦

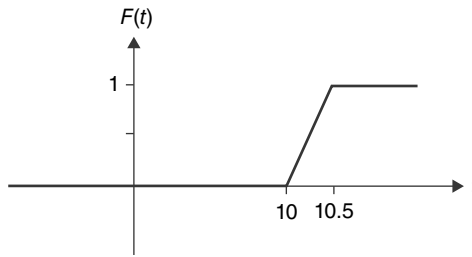


Figure 4.4 Distribution function of Example 4.10.

Remark 4.2 Suppose that F is a right-continuous, nondecreasing function on $(-\infty, \infty)$ that satisfies $\lim_{t \rightarrow \infty} F(t) = 1$ and $\lim_{t \rightarrow -\infty} F(t) = 0$. It can be shown that there exists a sample space S with a probability function and a random variable X over S such that the distribution function of X is F . Therefore, a function is a distribution function if it satisfies the conditions specified in this remark. ♦

EXERCISES

A

- Two fair dice are rolled and the absolute value of the difference of the outcomes is denoted by X . What are the possible values of X , and the probabilities associated with them?
- From an urn that contains five red, five white, and five blue chips, we draw two chips at random. For each blue chip we win \$1, for each white chip we win \$2, but for each red chip we lose \$3. If X represents the amount that we either win or we lose, what are the possible values of X and probabilities associated with them?

3. In a society of population N , the probability is p that a person has a certain rare disease independently of others. Let X be the number of people who should be tested until a person with the disease is found, $X = 0$ if no one with the disease is found. What are the possible values of X ? Determine the probabilities associated with these values.
4. The side measurement of a plastic die, manufactured by factory A, is a random number between 1 and $1\frac{1}{4}$ centimeters. What is the probability that the volume of a randomly selected die manufactured by this company is greater than 1.424? Assume that the die will always be a cube.
5. F , the distribution function of a random variable X , is given by

$$F(t) = \begin{cases} 0 & t < -1 \\ (1/4)t + 1/4 & -1 \leq t < 0 \\ 1/2 & 0 \leq t < 1 \\ (1/12)t + 7/12 & 1 \leq t < 2 \\ 1 & t \geq 2. \end{cases}$$

- (a) Sketch the graph of F .
- (b) Calculate the following quantities: $P(X < 1)$, $P(X = 1)$, $P(1 \leq X < 2)$, $P(X > 1/2)$, $P(X = 3/2)$, and $P(1 < X \leq 6)$.
6. From families with three children a family is chosen at random. Let X be the number of girls in the family. Calculate and sketch the distribution function of X . Assume that in a three-child family all gender distributions are equally probable.
7. A grocery store sells X hundred kilograms of rice every day, where the distribution of the random variable X is of the following form:

$$F(x) = \begin{cases} 0 & x < 0 \\ kx^2 & 0 \leq x < 3 \\ k(-x^2 + 12x - 3) & 3 \leq x < 6 \\ 1 & x \geq 6. \end{cases}$$

Suppose that this grocery store's total sales of rice do not reach 600 kilograms on any given day.

- (a) Find the value of k .
- (b) What is the probability that the store sells between 200 and 400 kilograms of rice next Thursday?
- (c) What is the probability that the store sells over 300 kilograms of rice next Thursday?
- (d) We are given that the store sold at least 300 kilograms of rice last Friday. What is the probability that it did not sell more than 400 kilograms on that day?

8. Let X be a random variable with distribution function F . For p ($0 < p < 1$), Q_p is said to be a **quantile** of order p if

$$F(Q_p-) \leq p \leq F(Q_p).$$

In a specific country, the annual rate of change in the price of oil per gallon follows the distribution function described below:

$$F(x) = \frac{1}{1 + e^{-x}}, \quad -\infty < x < \infty.$$

Find $Q_{0.50}$, called the **median** of F ; $Q_{0.25}$, called the **first quartile** of F ; and $Q_{0.75}$, called the **third quartile** of F . Interpret these quantities.

9. A random variable X is called **symmetric about 0** if for all $x \in \mathbb{R}$,

$$P(X \geq x) = P(X \leq -x).$$

Prove that if X is symmetric about 0, then for all $t > 0$ its distribution function F satisfies the following relations:

- (a) $P(|X| \leq t) = 2F(t) - 1$.
- (b) $P(|X| > t) = 2[1 - F(t)]$.
- (c) $P(X = t) = F(t) + F(-t) - 1$.

10. Prove that if a random variable X is symmetric about 0 (refer to Exercise 9), then the probability distribution functions of X and $-X$ are identical.
11. The probability distribution function of a random variable X is given by

$$F(t) = \begin{cases} 0 & \text{if } t < -1 \\ \frac{1}{2}(t+1) & \text{if } -1 \leq t \leq 1 \\ 1 & \text{if } t \geq 1. \end{cases}$$

- (a) Sketch the graph of F .
- (b) Show that X and $-X$ have the same distribution, implying that X is symmetric about 0 (refer to Exercise 10).

12. Determine if the following is a distribution function.

$$F(t) = \begin{cases} 1 - \frac{1}{\pi}e^{-t} & \text{if } t \geq 0 \\ 0 & \text{if } t < 0. \end{cases}$$

13. Determine if the following is a distribution function.

$$F(t) = \begin{cases} \frac{t}{1+t} & \text{if } t \geq 0 \\ 0 & \text{if } t < 0. \end{cases}$$

14. Determine if the following is a distribution function.

$$F(t) = \begin{cases} (1/2)e^t & t < 0 \\ 1 - (3/4)e^{-t} & t \geq 0. \end{cases}$$

15. In the U.S., for fellowship, the Casualty Actuarial Society requires passing a series of nine rigorous exams taken in order plus certain other learning objectives. Suppose that the probability is p_1 for a student to pass exam 1, and, for $2 \leq i \leq 9$, if the student has passed exam $i - 1$, then the probability is p_i for him or her to pass exam i . Let $X = 0$, if the student passes all of the exams, one after another, without failing any of them. Otherwise, let X be the first exam that he or she fails. Find the probability mass function of X .

Hint: Use Theorem 3.2, the multiplication rule.

16. Airline A has commuter flights every 45 minutes from San Francisco airport to Fresno. A passenger who wants to take one of these flights arrives at the airport at a random time. Suppose that X is the waiting time for this passenger; find the distribution function of X . Assume that seats are always available for these flights.

B

17. A scientific calculator can generate two-digit random numbers. That is, it can choose a number at random from the set $\{00, 01, 02, \dots, 99\}$. To obtain a random number from the set $\{4, 5, \dots, 18\}$, show that we have to keep generating two-digits random numbers until we obtain one between 4 and 18.
18. In a small town there are 40 taxis, numbered 1 to 40. Three taxis arrive at random at a station to pick up passengers. What is the probability that the number of at least one of the taxis is less than 5?
19. Let X be a randomly selected point from the interval $(0, 3)$. What is the probability that $X^2 - 5X + 6 > 0$?
20. Let X be a random point selected from the interval $(0, 1)$. Calculate F , the distribution function of $Y = X/(1 + X)$, and sketch its graph.
21. In the United States, the number of twin births is approximately 1 in 90. At a certain hospital let X be the number of births until the first twins are born. Find the first quartile, the median, and the third quartile of X . See Exercise 8 for the definitions of these quantities.
22. Let the time until a new car breaks down be denoted by X , and let

$$Y = \begin{cases} X & \text{if } X \leq 5 \\ 5 & \text{if } X > 5. \end{cases}$$

Then Y is the life of the car, if it lasts less than 5 years and is 5 if it lasts longer than 5 years. Calculate the distribution function of Y in terms of F , the distribution function of X .



Self-Quiz on Section 4.2

Time allotted: 20 Minutes

1. For what value of k , if any, is the function $p(n) = k/n$, $n = 1, 2, 3, \dots$, a probability mass function? (3 points)
2. Suppose that the probability distribution function of a random variable X is given by

$$F(t) = \begin{cases} 0 & t < 1 \\ 1 - \frac{1}{t^2} & t \geq 1. \end{cases}$$

Find $P(X > 3)$ and $P(X > 5 \mid X > 3)$. (3 points)

3. There are 50 students enrolled in a class, and they arrive one at a time, independently of each other. Suppose that the X th student is the first one who shares his birthday with another student already present in the classroom. Find the probability mass function of X . Assume that the birth rates are constant throughout the year and that each year has 365 days. (4 points)



4.3 DISCRETE RANDOM VARIABLES

In [Section 4.1](#), we observed that the set of possible values of a random variable might be finite, infinite but countable, or uncountable. For example, let X , Y , and Z be three random variables representing the respective number of tails in flipping a coin twice, the number of flips of until the first heads, and the amount of next year's rainfall. Then the sets of possible values for X , Y , and Z are the finite set $\{0, 1, 2\}$, the countable set $\{1, 2, 3, 4, \dots\}$, and the uncountable set $\{x: x \geq 0\}$, respectively. Whenever the set of possible values that a random variable X can assume is at most countable, X is called **discrete**. Therefore, X is discrete if either the set of its possible values is finite or it is countably infinite. To each discrete random variable, a real-valued function $p: \mathbb{R} \rightarrow \mathbb{R}$, defined by $p(x) = P(X = x)$, is assigned and is called the **probability mass function** of X . (It is also called the **probability function** of X or the **discrete probability function** of X .) Since the set of values of X is countable, $p(x)$ is positive at most for a countable set. It is zero elsewhere; that is, if possible values of X are $x_1, x_2, x_3, \dots$, then $p(x_i) \geq 0$ ($i = 1, 2, 3, \dots$) and $p(x) = 0$ if $x \notin \{x_1, x_2, x_3, \dots\}$. Now, clearly, the occurrence of the event $\{x_1, x_2, x_3, \dots\}$ is certain. Therefore, we have that $\sum_{i=1}^{\infty} P(X = x_i) = 1$ or, equivalently, $\sum_{i=1}^{\infty} p(x_i) = 1$.

Definition 4.3 *The probability mass function p of a random variable X whose set of possible values is $\{x_1, x_2, x_3, \dots\}$ is a function from $\mathbb{R}$ to $\mathbb{R}$ that satisfies the following properties.*

- (a) $p(x) = 0$ if $x \notin \{x_1, x_2, x_3, \dots\}$.
- (b) $p(x_i) = P(X = x_i)$ and hence $p(x_i) \geq 0$ ($i = 1, 2, 3, \dots$).
- (c) $\sum_{i=1}^{\infty} p(x_i) = 1$.

Because of this definition, if, for a set $\{x_1, x_2, x_3, \dots\}$, there exists a function $p: \mathbb{R} \rightarrow \mathbb{R}$ such that $p(x_i) \geq 0$ ($i = 1, 2, 3, \dots$), $p(x) = 0$, $x \notin \{x_1, x_2, x_3, \dots\}$, and $\sum_{i=1}^{\infty} p(x_i) = 1$, then p is called a **probability mass function**.

The probability mass function of a random variable is often demonstrated geometrically by a set of vertical lines connecting the points $(x_i, 0)$ and $(x_i, p(x_i))$. For example, if X is the number of heads in two flips of a fair coin, then $X = 0, 1, 2$ with $p(0) = 1/4$, $p(1) = 1/2$, and $p(2) = 1/4$. Hence the graphical representation of p is as shown in Figure 4.5.

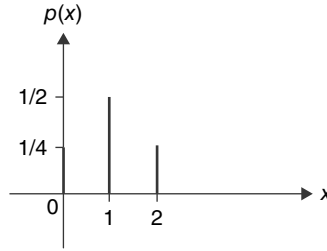


Figure 4.5 Graph of the number of heads in two flips of a fair coin.

The distribution function F of a discrete random variable X , with the set of possible values $\{x_1, x_2, x_3, \dots\}$, is a step function. Assuming that $x_1 < x_2 < x_3 < \dots$, we have that if $t < x_1$, then

$$F(t) = 0;$$

if $x_1 \leq t < x_2$, then

$$F(t) = P(X \leq t) = P(X = x_1) = p(x_1);$$

if $x_2 \leq t < x_3$, then

$$F(t) = P(X \leq t) = P(X = x_1 \text{ or } X = x_2) = p(x_1) + p(x_2);$$

and in general, if $x_{n-1} \leq t < x_n$, then

$$F(t) = \sum_{i=1}^{n-1} p(x_i).$$

Thus F is constant in the intervals $[x_{n-1}, x_n)$ with jumps at $x_1, x_2, x_3, \dots$. The magnitude of the jump at x_i is $p(x_i)$. For a typical representation of the distribution function of a discrete random variable, refer to Figure 4.6.

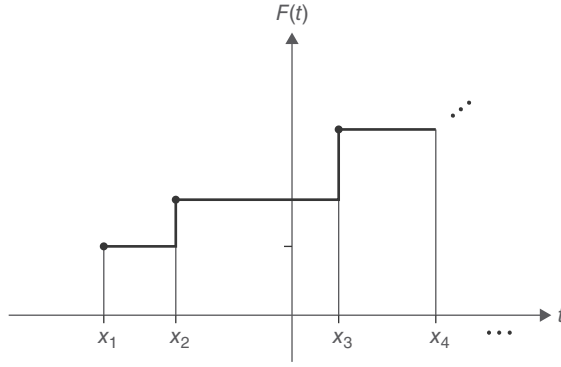


Figure 4.6 Typical distribution function for a discrete random variable.

Example 4.11 Can a function of the form

$$p(x) = \begin{cases} c\left(\frac{2}{3}\right)^x & x = 1, 2, 3, \dots \\ 0 & \text{elsewhere} \end{cases}$$

be a probability mass function?

Solution: A probability mass function should have three properties: (1) $p(x)$ must be zero at all points except on a finite or a countable set. Clearly, this property is satisfied. (2) $p(x)$ should be nonnegative. This is satisfied if and only if $c \geq 0$. (3) $\sum p(x_i) = 1$. This condition is satisfied if and only if $\sum_{i=1}^{\infty} c(2/3)^i = 1$. This happens precisely when

$$c = \frac{1}{\sum_{i=1}^{\infty} (2/3)^i} = \frac{1}{\frac{2/3}{1 - 2/3}} = \frac{1}{2},$$

where the second equality follows from the geometric series theorem. Thus only for $c = 1/2$, a function of the given form is a probability mass function. ♦

Example 4.12 In the experiment of rolling a balanced die twice, let X be the maximum of the two numbers obtained. Determine and sketch the probability mass function and the distribution function of X .

Solution: The possible values of X are 1, 2, 3, 4, 5, and 6. The sample space of this experiment consists of 36 equally likely outcomes. Hence the probability of any of them is $1/36$. Thus

$$p(1) = P(X = 1) = P(\{(1, 1)\}) = 1/36,$$

$$p(2) = P(X = 2) = P(\{(1, 2), (2, 2), (2, 1)\}) = 3/36,$$

$$p(3) = P(X = 3) = P(\{(1, 3), (2, 3), (3, 3), (3, 2), (3, 1)\}) = 5/36.$$

Similarly, $p(4) = 7/36$, $p(5) = 9/36$, and $p(6) = 11/36$; $p(x) = 0$ for $x \notin \{1, 2, 3, 4, 5, 6\}$. The graphical representation of p is displayed in [Figure 4.7](#) on the following page.

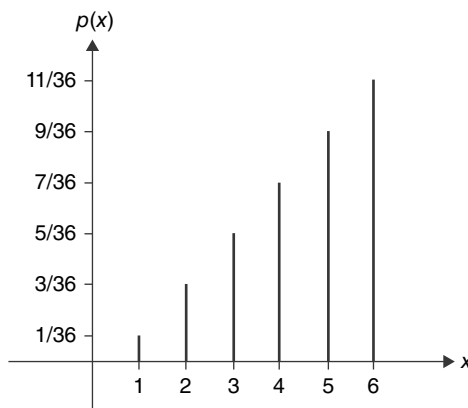


Figure 4.7 Probability mass function of Example 4.12.

The distribution function of X , F , is as follows (its graph is shown in [Figure 4.8](#)):

$$F(x) = \begin{cases} 0 & x < 1 \\ 1/36 & 1 \leq x < 2 \\ 4/36 & 2 \leq x < 3 \\ 9/36 & 3 \leq x < 4 \\ 16/36 & 4 \leq x < 5 \\ 25/36 & 5 \leq x < 6 \\ 1 & x \geq 6. \end{cases} \quad \blacklozenge$$

Example 4.13 Let X be the number of births in a hospital until the first girl is born. Determine the probability mass function and the distribution function of X . Assume that the probability is $1/2$ that a baby born is a girl.

Solution: Let p be the probability mass function of X and F be its distribution function. The random variable X can assume any positive integer i . $p(i) = P(X = i)$, and $X = i$ occurs if the first $i - 1$ births are all boys and the i th birth is a girl. Thus

$$p(i) = \begin{cases} \left(\frac{1}{2}\right)^{i-1} \left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^i & i = 1, 2, 3, \dots, \\ 0 & i \neq 1, 2, 3, \dots \end{cases}$$

The graphical representation of p is shown in [Figure 4.9](#).

To determine $F(t)$, note that for $t < 1$, $F(t) = 0$; for $1 \leq t < 2$, $F(t) = 1/2$; for $2 \leq t < 3$, $F(t) = 1/2 + 1/4 = 3/4$; for $3 \leq t < 4$, $F(t) = 1/2 + 1/4 + 1/8 = 7/8$; and in general for $n - 1 \leq t < n$,

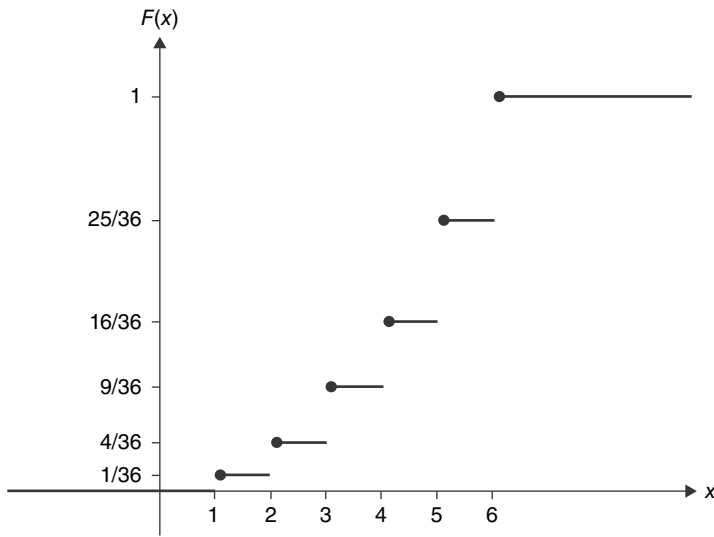


Figure 4.8 Distribution function of Example 4.12.

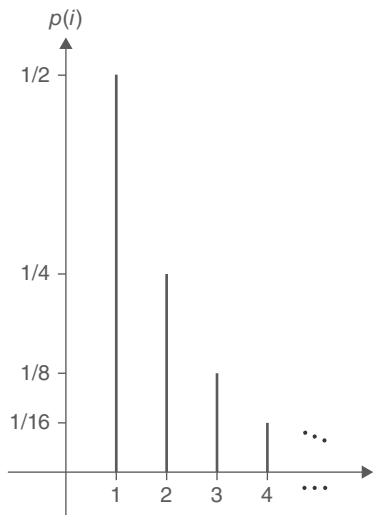


Figure 4.9 Probability mass function of Example 4.13.

$$\begin{aligned}
 F(t) &= \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^{n-1}} = \sum_{i=1}^{n-1} \left(\frac{1}{2}\right)^i \\
 &= \frac{1 - (1/2)^n}{1 - 1/2} - 1 = 1 - \left(\frac{1}{2}\right)^{n-1},
 \end{aligned}$$

by the partial sum formula for geometric series. Thus

$$F(t) = \begin{cases} 0 & t < 1 \\ 1 - (1/2)^{n-1} & n-1 \leq t < n, \quad n = 2, 3, 4, \dots \end{cases}$$

The graph for F is depicted in Figure 4.10. ♦

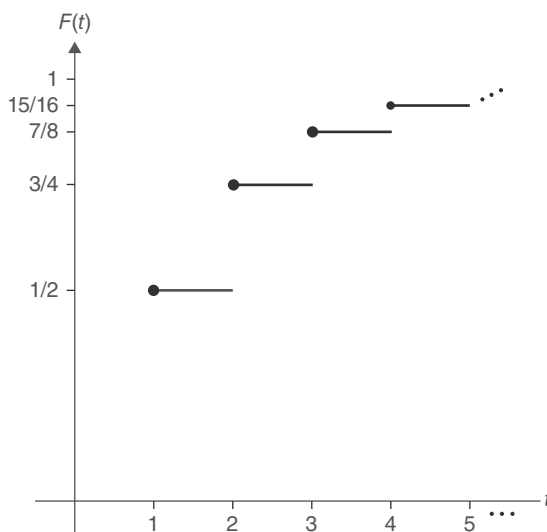


Figure 4.10 Distribution function of Example 4.13.

EXERCISES

A

1. Let $p(x) = x/15$, $x = 1, 2, 3, 4, 5$ be probability mass function of a random variable X . Determine F , the distribution function of X , and sketch its graph.
2. In the experiment of rolling a balanced die twice, let X be the minimum of the two numbers obtained. Determine the probability mass function and the distribution function of X and sketch their graphs.
3. In the experiment of rolling a balanced die twice, let X be the sum of the two numbers obtained. Determine the probability mass function of X .

4. The distribution function of a random variable X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < -2 \\ 1/2 & \text{if } -2 \leq x < 2 \\ 3/5 & \text{if } 2 \leq x < 4 \\ 8/9 & \text{if } 4 \leq x < 6 \\ 1 & \text{if } x \geq 6. \end{cases}$$

Determine the probability mass function of X and sketch its graph.

5. Let X be the number of random numbers selected from $\{0, 1, 2, \dots, 9\}$ independently until 0 is chosen. Find the probability mass functions of X and $Y = 2X + 1$.
6. A value i is said to be the **mode** of a discrete random variable X if it maximizes $p(x)$, the probability mass function of X . Find the modes of random variables X and Y with probability mass functions

$$p(x) = \left(\frac{1}{2}\right)^x, \quad x = 1, 2, 3, \dots,$$

and

$$q(y) = \frac{4!}{y!(4-y)!} \left(\frac{1}{4}\right)^y \left(\frac{3}{4}\right)^{4-y}, \quad y = 0, 1, 2, 3, 4,$$

respectively.

7. For each of the following, determine the value(s) of k for which p is a probability mass function. Note that in parts (d) and (e), n is a positive integer.
- (a) $p(x) = kx, x = 1, 2, 3, 4, 5$.
 - (b) $p(x) = k(1+x)^2, x = -2, 0, 1, 2$.
 - (c) $p(x) = k(1/9)^x, x = 1, 2, 3, \dots$.
 - (d) $p(x) = kx, x = 1, 2, 3, \dots, n$.
 - (e) $p(x) = kx^2, x = 1, 2, 3, \dots, n$.

Hint: Recall that

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

8. From 18 potential women jurors and 28 potential men jurors, a jury of 12 is chosen at random. Let X be the number of women selected. Find the probability mass function of X .
9. Let $p(x) = 3/4(1/4)^x, x = 0, 1, 2, 3, \dots$, be probability mass function of a random variable X . Find F , the distribution function of X , and sketch its graph.

10. In successive rolls of a fair die, let X be the number of rolls until the first 6 appears. Determine the probability mass function and the distribution function of X .
11. Let X be the number of claims filed by a randomly selected customer under a certain homeowner's insurance policy during a ten-year period. Suppose that an actuary has estimated that p , the probability mass function of X , satisfies

$$p(n+1) = 0.32p(n), \quad n \geq 0.$$

What is the probability that a policy holder files at least two claims during the next 10 years?

12. Suppose that the number of claims received by an insurance company in a given week is independent of the number of claims received in any other week. An actuary has calculated that the probability mass function of the number of claims received in a random week is

$$p(n) = \left(\frac{1}{5}\right)\left(\frac{4}{5}\right)^n, \quad n \geq 0.$$

Find the probability that the company will receive exactly 8 claims during the next two weeks.

13. A *binary digit* or *bit* is a zero or one. A computer assembly language can generate independent random bits. Let X be the number of independent random bits to be generated until both 0 and 1 are obtained. Find the probability mass function of X .

B

14. Every Sunday, Bob calls Liz to see if she will play tennis with him on that day. If Liz has not played tennis with Bob since i Sundays ago, the probability that she will say yes to him is i/k , $k \geq 2$, $i = 1, 2, \dots, k$. Therefore, if, for example, Liz does not play tennis with Bob for $k-1$ consecutive Sundays, then she will play with him next Sunday with probability 1. Let Z be the number of weeks it takes Liz to play again with Bob since they last played. Find the probability mass function of Z .
15. Let X be the number of vowels (not necessarily distinct) among the first five letters of a random arrangement of the following expression.

ELIZABETH TAYLOR

Find the probability mass function of X . Count the letter Y as a consonant.

16. From a drawer that contains 10 pair of gloves, six gloves are selected randomly. Let X be the number of pairs of gloves obtained. Find the probability mass function of X .
17. A fair die is tossed successively. Let X denote the number of tosses until each of the six possible outcomes occurs at least once. Find the probability mass function of X .
Hint: For $1 \leq i \leq 6$, let E_i be the event that the outcome i does not occur during the first n tosses of the die. First calculate $P(X > n)$ by writing the event $X > n$ in terms of $E_1, E_2, \dots, E_6$.

18. To an engineering class containing 23 male and three female students, there are 13 work stations available. To assign each work station to two students, the professor forms 13 teams one at a time, each consisting of two randomly selected students. In this process, let X be the total number of students selected when the first team consisting of a male and a female appears. Find the probability mass function of X .



Self-Quiz on Section 4.3

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. The number of claims filed with a car insurance company, per week, is a random variable with probability mass function

$$p(x) = \frac{46}{21(x+2)(x+3)}, \quad x = 0, 1, 2, \dots, 20.$$

- (a) Find the probability of at least one such claim next week.
- (b) If we are given that there were no more than 5 claims filed with the company two weeks ago, find the probability that there were at least 4 claims filed.
2. We choose 13 numbers at random and without replacement from the set $\{1, 2, \dots, 100\}$. Let X be the median of the numbers selected. Find the probability mass function of X . Note that the median of the 13 numbers selected is the number in the middle when they are put in order. For example, if the numbers chosen are 7, 12, 13, 25, 41, 48, 53, 59, 67, 77, 84, 90, and 96, then their median is 53.



4.4 EXPECTATIONS OF DISCRETE RANDOM VARIABLES

To clarify the concept of expectation, consider a casino game in which the probability of losing \$1 per game is 0.6, and the probabilities of winning \$1, \$2, and \$3 per game are 0.3, 0.08, and 0.02, respectively. The gain or loss of a gambler who plays this game only a few times depends on his luck more than anything else. For example, in one play of the game, a lucky gambler might win \$3, but he has a 60% chance of losing \$1. However, if a gambler decides to play the game a large number of times, his loss or gain depends more on the number of plays than on his luck. A calculating player argues that if he plays the game n times, for a large n , then in approximately $(0.6)n$ games he will lose \$1 per game, and in approximately $(0.3)n$, $(0.08)n$, and $(0.02)n$ games he will win \$1, \$2, and \$3, respectively. Therefore, his total gain is

$$(0.6)n \cdot (-1) + (0.3)n \cdot 1 + (0.08)n \cdot 2 + (0.02)n \cdot 3 = (-0.08)n. \quad (4.1)$$

This gives an average of \$ -0.08 , or about 8 cents of loss per game. The more the gambler plays, the less luck interferes and the closer his loss comes to \$0.08 per game. If X is the random variable denoting the gain in one play, then the number -0.08 is called the **expected value** of X . We write $E(X) = -0.08$. $E(X)$ is the average value of X . That is, if we play the game n times and find the average of the values of X , then as $n \rightarrow \infty$, $E(X)$ is obtained. Since, for this game, $E(X) < 0$, we have that, on the average, the more we play, the more we lose. If for some game $E(X) = 0$, then in long run the player neither loses nor wins. Such games are called **fair**. In this example, X is a discrete random variable with the set of possible values $\{-1, 1, 2, 3\}$. The probability mass function of X , $p(x)$, is given by

i	-1	1	2	3
$p(i) = P(X = i)$	0.6	0.3	0.08	0.02

and $p(x) = 0$ if $x \notin \{-1, 1, 2, 3\}$. Dividing both sides of (4.1) by n , we obtain

$$(0.6) \cdot (-1) + (0.3) \cdot 1 + (0.08) \cdot 2 + (0.02) \cdot 3 = -0.08.$$

Hence

$$-1 \cdot p(-1) + 1 \cdot p(1) + 2 \cdot p(2) + 3 \cdot p(3) = -0.08,$$

a relation showing that the expected value of X can be calculated directly by summing up the product of possible values of X by their probabilities. This and similar examples motivate the following general definition, which was first used casually by Pascal but introduced formally by Huygens in the late seventeenth century.

Definition 4.4 The **expected value** of a discrete random variable X with the set of possible values A and probability mass function $p(x)$ is defined by

$$E(X) = \sum_{x \in A} xp(x).$$

We say that $E(X)$ exists if this sum converges absolutely.

The expected value of a random variable X is also called the **mean**, or the **mathematical expectation**, or simply the **expectation** of X . It is also occasionally denoted by $E[X]$, EX , μ_X , or μ .

Note that if each value x of X is weighted by $p(x) = P(X = x)$, then $\sum_{x \in A} xp(x)$ is nothing but the weighted average of X . Similarly, suppose a mass is uniformly distributed along a rod. Let $x_1, x_2, \dots, x_5$ be specific points on the rod. We position five kettlebells with masses $m_1, m_2, \dots, m_5$, $\sum_{i=1}^5 m_i = 1$, at these points randomly. Define X as a random variable whose possible values are $x_1, x_2, \dots, x_5$ with probability $P(X = x_i) = m_i$. The expected value $E(X) = \sum_{i=1}^5 x_i m_i$ represents the center of mass of this configuration. To balance the rod with all the kettlebells, the fulcrum should be positioned at $E(X)$ (see Figure 4.11).

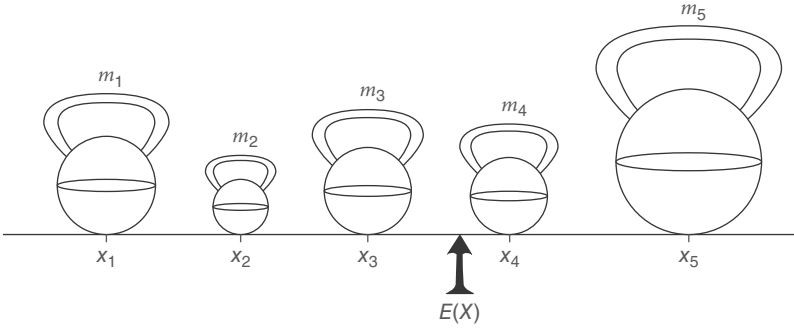


Figure 4.11 Expected value of X as the center of mass.

Here are some examples to illuminate the notion of expected value, a fundamental concept in probability and statistics.

Example 4.14 We flip a fair coin twice and let X be the number of heads obtained. What is the expected value of X ?

Solution: The possible values of X are 0, 1, and 2, and the probability mass function of X is given by $p(0) = P(X = 0) = 1/4$, $p(1) = P(X = 1) = 1/2$, $p(2) = P(X = 2) = 1/4$, and $p(x) = 0$ if $x \notin \{0, 1, 2\}$. Thus

$$E(X) = 0 \cdot p(0) + 1 \cdot p(1) + 2 \cdot p(2) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1.$$

Therefore, we can expect an average of one head in every two flips. ♦

Example 4.15 We write the numbers $a_1, a_2, \dots, a_n$ on n identical balls and mix them in a box. What is the expected value of a ball selected at random?

Solution: The set of possible values of X , the numbers written on the balls selected, is $\{a_1, a_2, \dots, a_n\}$. The probability mass function of X is given by

$$p(a_1) = p(a_2) = \dots = p(a_n) = 1/n,$$

and $p(x) = 0$ if $x \notin \{a_1, a_2, \dots, a_n\}$. Thus

$$\begin{aligned} E(X) &= \sum_{x \in \{a_1, a_2, \dots, a_n\}} xp(x) = a_1 \cdot \frac{1}{n} + a_2 \cdot \frac{1}{n} + \dots + a_n \cdot \frac{1}{n} \\ &= \frac{a_1 + a_2 + \dots + a_n}{n}. \end{aligned}$$

Therefore, as expected, $E(X)$ coincides with the average of the values $a_1, a_2, \dots, a_n$. That is, if for a large number of times we draw balls at random and with replacement, record their values, and then find their average, the result obtained is the arithmetic mean of the numbers $a_1, a_2, \dots, a_n$. ♦

Example 4.16 A college mathematics department sends 8 to 12 professors to the annual meeting of the American Mathematical Society, which lasts five days. The hotel at which the conference is held offers a bargain rate of a dollars per day per person if reservations are made 45 or more days in advance, but charges a cancellation fee of $2a$ dollars per person. The department is not certain how many professors will go. However, from past experience it is known that the probability of the attendance of i professors is $1/5$ for $i = 8, 9, 10, 11$, and 12 . If the regular rate of the hotel is $2a$ dollars per day per person, should the department make any reservations? If so, how many?

Solution: For $i = 8, 9, 10, 11$, and 12 , let X_i be the total cost in dollars if the department makes reservations for i professors. We should compute $E(X_i)$ for $i = 8, 9, 10, 11$, and 12 . If $E(X_j)$ is the smallest of all, the department should make reservations for j professors. To calculate $E(X_8)$, note that X_8 only assumes the values $40a, 50a, 60a, 70a$, and $80a$, which correspond to the cases that 8, 9, 10, 11, and 12 professors attend, respectively, while reservations are made only for eight professors. Since the probability of any of these is $1/5$, we have

$$E(X_8) = (40a)\frac{1}{5} + (50a)\frac{1}{5} + (60a)\frac{1}{5} + (70a)\frac{1}{5} + (80a)\frac{1}{5} = 60a.$$

Similarly,

$$E(X_9) = (42a)\frac{1}{5} + (45a)\frac{1}{5} + (55a)\frac{1}{5} + (65a)\frac{1}{5} + (75a)\frac{1}{5} = 56.4a,$$

$$E(X_{10}) = (44a)\frac{1}{5} + (47a)\frac{1}{5} + (50a)\frac{1}{5} + (60a)\frac{1}{5} + (70a)\frac{1}{5} = 54.2a,$$

$$E(X_{11}) = (46a)\frac{1}{5} + (49a)\frac{1}{5} + (52a)\frac{1}{5} + (55a)\frac{1}{5} + (65a)\frac{1}{5} = 53.4a,$$

$$E(X_{12}) = (48a)\frac{1}{5} + (51a)\frac{1}{5} + (54a)\frac{1}{5} + (57a)\frac{1}{5} + (60a)\frac{1}{5} = 54a.$$

We see that X_{11} has the smallest expected value. Thus making 11 reservations is the most reasonable policy. ♦

Example 4.17 In the lottery of a certain state, players pick six different integers between 1 and 49, the order of selection being irrelevant. The lottery commission then selects six of these numbers at random as the winning numbers. A player wins the grand prize of \$1,200,000 if all six numbers that he has selected match the winning numbers. He wins the second and third prizes of \$800 and \$35, respectively, if exactly five and four of his six selected numbers match the winning numbers. What is the expected value of the amount a player wins in one game?

Solution: Let X be the amount that a player wins in one game. Then the possible values of X are 1,200,000; 800; 35; and 0. The probabilities associated with these values are

$$P(X = 1200000) = \frac{1}{\binom{49}{6}} \approx 0.000000072,$$

$$P(X = 800) = \frac{\binom{6}{5} \binom{43}{1}}{\binom{49}{6}} \approx 0.000018, \quad P(X = 35) = \frac{\binom{6}{4} \binom{43}{2}}{\binom{49}{6}} \approx 0.00097,$$

and

$$P(X = 0) = 1 - 0.000000072 - 0.000018 - 0.00097 = 0.999011928.$$

Therefore,

$$\begin{aligned} E(X) &\approx 1200000(0.000000072) + 800(0.000018) + 35(0.00097) \\ &\quad + 0(0.999011928) \approx 0.13. \end{aligned}$$

This shows that on the average players will win 13 cents per game. If the cost per game is one dollar, then, on the average, a player will lose 87 cents per game. Therefore, a player who participates in 10,000 games over several years should anticipate an expected value of total losses amounting to approximately \$8700. ♦

Let X be a discrete random variable with a set of possible values A and probability mass function p . We say that $E(X)$ exists if the sum $\sum_{x \in A} xp(x)$ converges, that is, if $\sum_{x \in A} xp(x) < \infty$. We now present two examples of random variables whose mathematical expectations do not exist.

Example 4.18 (St. Petersburg Paradox) In a game, the player flips a fair coin successively until he gets a heads. If this occurs on the k th flip, the player wins 2^k dollars. Therefore, if the outcome of the first flip is heads, the player wins \$2. If the outcome of the first flip is tails but that of the second flip is heads, he wins \$4. If the outcomes of the first two are tails but the third one heads, he will win \$8, and so on. The question is, to play this game, how much should a person, who is willing to play a fair game, pay? To answer this question, let X be the amount of money the player wins. Then X is a random variable with the set of possible values $\{2, 4, 8, \dots, 2^k, \dots\}$ and

$$P(X = 2^k) = \left(\frac{1}{2}\right)^k, \quad k = 1, 2, 3, \dots$$

Therefore,

$$E(X) = \sum_{k=1}^{\infty} 2^k \left(\frac{1}{2}\right)^k = \sum_{k=1}^{\infty} 1 = 1 + 1 + 1 + \dots = \infty.$$

This result shows that the game remains unfair even if a person pays the largest possible amount to play it. In other words, this is a game in which one always wins no matter how expensive it is to play. To see what the flaw is, note that theoretically this game is not feasible to play because it requires an enormous amount of money. In practice, however, the probability that a gambler wins $\$2^k$ for a large k is close to zero. Even for small values of k , winning is highly unlikely. For example, to win $\$2^{30} = 1073741824$, you should get 29 tails in a row followed

by a head. The chance of this happening is 1 in 1,073,741,824, much less than 1 in a billion.



The following example serves to illuminate further the concept of expected value. At the same time, it shows the inadequacy of this concept as a central measure.

Example 4.19 Let X_0 be the amount of rain that will fall in the United States on the next Christmas day. For $n > 0$, let X_n be the amount of rain that will fall in the United States on Christmas n years later. Let N be the smallest number of years that elapse before we get a Christmas rainfall greater than X_0 . Suppose that $P(X_i = X_j) = 0$ if $i \neq j$, the events concerning the amount of rain on Christmas days of different years are all independent, and the X_n 's are identically distributed. Find the expected value of N .

Solution: Since N is the first value for n for which $X_n > X_0$,

$$\begin{aligned} P(N > n) &= P(X_0 > X_1, X_0 > X_2, \dots, X_0 > X_n) \\ &= P(\max(X_0, X_1, X_2, \dots, X_n) = X_0) = \frac{1}{n+1}; \end{aligned}$$

where the last equality follows from the symmetry, there is no more reason for the maximum to be at X_0 than there is for it to be at X_i , $0 \leq i \leq n$. Therefore,

$$P(N = n) = P(N > n-1) - P(N > n) = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}.$$

From this it follows that

$$E(N) = \sum_{n=1}^{\infty} nP(N = n) = \sum_{n=1}^{\infty} \frac{n}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n+1} = \infty.$$

Note that $P(N > n-1) = 1/n$ gives the probability that, in the United States, we will have to wait more than, say, 3 years for a Christmas rainfall that is greater than X_0 is only 1/4, and the probability that we must wait more than 9 years is only 1/10. Even with such low probabilities, on average, it will still take infinitely many years before we will have more rain on a Christmas day than we will have on next Christmas day. ◆

Example 4.20 The tanks of a country's army are numbered 1 to N . In a war this country loses n random tanks to the enemy, who discovers that the captured tanks are numbered. If $X_1, X_2, \dots, X_n$ are the numbers of the captured tanks, what is $E(\max X_i)$? How can the enemy use $E(\max X_i)$ to find an estimate of N , the total number of this country's tanks?

Solution: Let $Y = \max X_i$, then

$$P(Y = k) = \frac{\binom{k-1}{n-1}}{\binom{N}{n}} \quad \text{for } k = n, n+1, n+2, \dots, N,$$

because if the maximum of X_i 's is k , the numbers of the remaining $n - 1$ tanks are from 1 to $k - 1$. Now

$$\begin{aligned}
 E(Y) &= \sum_{k=n}^N k P(Y = k) = \sum_{k=n}^N \frac{k \binom{k-1}{n-1}}{\binom{N}{n}} \\
 &= \frac{1}{\binom{N}{n}} \sum_{k=n}^N \frac{k (k-1)!}{(n-1)! (k-n)!} = \frac{n}{\binom{N}{n}} \sum_{k=n}^N \frac{k!}{n! (k-n)!} \\
 &= \frac{n}{\binom{N}{n}} \sum_{k=n}^N \binom{k}{n}. \tag{4.2}
 \end{aligned}$$

To calculate $\sum_{k=n}^N \binom{k}{n}$, note that $\binom{k}{n}$ is the coefficient of x^n in the polynomial $(1+x)^k$.

Therefore, $\sum_{k=n}^N \binom{k}{n}$ is the coefficient of x^n in the polynomial $\sum_{k=n}^N (1+x)^k$. Since

$$\begin{aligned}
 \sum_{k=n}^N (1+x)^k &= (1+x)^n \sum_{k=0}^{N-n} (1+x)^k = (1+x)^n \frac{(1+x)^{N-n+1} - 1}{(1+x) - 1} \\
 &= \frac{1}{x} [(1+x)^{N+1} - (1+x)^n],
 \end{aligned}$$

and the coefficient of x^n in the polynomial $\frac{1}{x} [(1+x)^{N+1} - (1+x)^n]$ is $\binom{N+1}{n+1}$, we have

$$\sum_{k=n}^N \binom{k}{n} = \binom{N+1}{n+1}.$$

Substituting this result in (4.2), we obtain

$$E(Y) = \frac{n \binom{N+1}{n+1}}{\binom{N}{n}} = \frac{n \frac{(N+1)!}{(n+1)! (N-n)!}}{\frac{N!}{n! (N-n)!}} = \frac{n(N+1)}{n+1}.$$

To estimate N , the total number of this country's tanks, we solve $E(Y) = \frac{n(N+1)}{n+1}$ for N .

We obtain $N = \frac{n+1}{n} E(Y) - 1$. Therefore, if, for example, the enemy captures 12 tanks

and the maximum of the numbers of the tanks captured is 117, then assuming that $E(Y)$ is approximately equal to the value of Y observed, we get $N \approx (13/12) \times 117 - 1 \approx 126$. $\blacklozenge$

For more examples, including Pólya's Urn Model, please refer to Section 4A in the Supplementary Resources for [Chapter 4](#) on the Instructor and Student Resources website associated with this book.

We now discuss some elementary properties of the expectation of a discrete random variable. Further properties of expectations are discussed in subsequent chapters.

Theorem 4.1 *If X is a constant random variable, that is, if $P(X = c) = 1$ for a constant c , then $E(X) = c$.*

Proof: There is only one possible value for X and that is c ; hence $E(X) = c \cdot P(X = c) = c \cdot 1 = c$. $\blacklozenge$

Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a real-valued function and X be a discrete random variable with set of possible values A and probability mass function $p(x)$. Similar to $E(X) = \sum_{x \in A} xp(x)$, there is the important relation $E[g(X)] = \sum_{x \in A} g(x)p(x)$, known as the **law of the unconscious statistician**, which we now prove. This relation enables us to calculate the expected value of the random variable $g(X)$ without deriving its probability mass function. It implies that, for example,

$$E(X^2) = \sum_{x \in A} x^2 p(x),$$

$$E(X^2 - 2X + 4) = \sum_{x \in A} (x^2 - 2x + 4)p(x),$$

$$E(X \cos X) = \sum_{x \in A} (x \cos x)p(x),$$

$$E(e^X) = \sum_{x \in A} e^x p(x).$$

Theorem 4.2 *Let X be a discrete random variable with set of possible values A and probability mass function $p(x)$, and let g be a real-valued function. Then $g(X)$ is a random variable with*

$$E[g(X)] = \sum_{x \in A} g(x)p(x).$$

Proof: Let S be the sample space. We are given that $g: \mathbb{R} \rightarrow \mathbb{R}$ is a real-valued function and $X, : S \rightarrow A \subseteq \mathbb{R}$ is a random variable with the set of possible values A . As we know, $g(X)$, the composition of g and X , is a function from S to the set $g(A) = \{g(x) : x \in A\}$.

Hence $g(X)$ is a random variable with the possible set of values $g(A)$. Now, by the definition of expectation,

$$E[g(X)] = \sum_{z \in g(A)} zP\{g(X) = z\}.$$

Let $g^{-1}(\{z\}) = \{x: g(x) = z\}$, and notice that we are not claiming that g has an inverse function. We are simply considering the set $\{x: g(x) = z\}$, which is called *the inverse image* of z and is denoted by $g^{-1}(\{z\})$. Now

$$P(g(X) = z) = P(X \in g^{-1}(\{z\})) = \sum_{\{x: x \in g^{-1}(\{z\})\}} P(X = x) = \sum_{\{x: g(x)=z\}} p(x).$$

Thus

$$\begin{aligned} E[g(X)] &= \sum_{z \in g(A)} zP(g(X) = z) = \sum_{z \in g(A)} z \sum_{\{x: g(x)=z\}} p(x) \\ &= \sum_{z \in g(A)} \sum_{\{x: g(x)=z\}} zp(x) = \sum_{z \in g(A)} \sum_{\{x: g(x)=z\}} g(x)p(x) \\ &= \sum_{x \in A} g(x)p(x), \end{aligned}$$

where the last equality follows from the fact that the sum over A can be performed in two stages: We can first sum over all x with $g(x) = z$, and then over all z . ♦

Corollary *Let X be a discrete random variable; $g_1, g_2, \dots, g_n$ be real-valued functions, and let $\alpha_1, \alpha_2, \dots, \alpha_n$ be real numbers. Then*

$$\begin{aligned} E[\alpha_1 g_1(X) + \alpha_2 g_2(X) + \dots + \alpha_n g_n(X)] \\ = \alpha_1 E[g_1(X)] + \alpha_2 E[g_2(X)] + \dots + \alpha_n E[g_n(X)]. \end{aligned}$$

Proof: Let the set of possible values of X be A , and its probability mass function be $p(x)$. Then, by Theorem 4.2,

$$\begin{aligned} E[\alpha_1 g_1(X) + \alpha_2 g_2(X) + \dots + \alpha_n g_n(X)] \\ &= \sum_{x \in A} [\alpha_1 g_1(x) + \alpha_2 g_2(x) + \dots + \alpha_n g_n(x)]p(x) \\ &= \alpha_1 \sum_{x \in A} g_1(x)p(x) + \alpha_2 \sum_{x \in A} g_2(x)p(x) + \dots + \alpha_n \sum_{x \in A} g_n(x)p(x) \\ &= \alpha_1 E[g_1(X)] + \alpha_2 E[g_2(X)] + \dots + \alpha_n E[g_n(X)]. \quad \blacklozenge \end{aligned}$$

By this corollary, for example, we have relations such as the following:

$$E(2X^3 + 5X^2 + 7X + 4) = 2E(X^3) + 5E(X^2) + 7E(X) + 4$$

$$E(e^X + 2 \sin X + \log X) = E(e^X) + 2E(\sin X) + E(\log X).$$

Moreover, this corollary implies that $E(X)$ is linear. That is, if $\alpha, \beta \in \mathbb{R}$, then

$$E(\alpha X + \beta) = \alpha E(X) + \beta.$$

Example 4.21 The probability mass function of a discrete random variable X is given by

$$p(x) = \begin{cases} x/15 & x = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise.} \end{cases}$$

What is the expected value of $X(6 - X)$?

Solution: By Theorem 4.2,

$$E[X(6 - X)] = 5 \cdot \frac{1}{15} + 8 \cdot \frac{2}{15} + 9 \cdot \frac{3}{15} + 8 \cdot \frac{4}{15} + 5 \cdot \frac{5}{15} = 7. \quad \blacklozenge$$

Example 4.22 A box contains 10 disks of radii 1, 2, ..., and 10, respectively. What is the expected value of the area of a disk selected at random from this box?

Solution: Let the radius of the disk be R , then R is a random variable with the probability mass function $p(x) = 1/10$ if $x = 1, 2, \dots, 10$, and $p(x) = 0$ otherwise. $E(\pi R^2)$, the desired quantity is calculated as follows:

$$E(\pi R^2) = \pi E(R^2) = \pi \left(\sum_{i=1}^{10} i^2 \frac{1}{10} \right) = 38.5\pi. \quad \blacklozenge$$

EXERCISES

A

1. There is a story about Charles Dickens (1812–1870), the English novelist and one of the most popular writers in the history of literature. It is known that Dickens was interested in practical applications of mathematics. On the final day in March during a year in the second half of the nineteenth century, he was scheduled to leave London by train and travel about an hour to visit a very good friend. However, Mr. Dickens was aware of the fact that in England there were, on the average, two serious train accidents each month. Knowing that there had been only one serious accident so far during the month of March, Dickens thought that the probability of a serious train accident on the last day of March would be very high. Thus he called his friend and postponed his visit until the next day. He boarded the train on April 1, feeling much safer and believing that he had used his knowledge of mathematics correctly by leaving the next day. He *did* arrive safely! Is there a fallacy in Dickens' argument? Explain.

2. In a certain part of downtown Baltimore parking lots charge \$7 per day. A car that is illegally parked on the street will be fined \$25 if caught, and the chance of being caught is 60%. If money is the only concern of a commuter who must park in this location every day, should he park at a lot or park illegally?
3. Let X be a discrete random variable with probability mass function

x	-2	0	2	4
$p(x)$	1/3	1/4	1/4	1/6

Find $E[2(X - 1)(3 - X)]$.

4. In a lottery every week, 2,000,000 tickets are sold for \$1 apiece. If 4000 of these tickets pay off \$30 each, 500 pay off \$800 each, one ticket pays off \$1,200,000, and no ticket pays off more than one prize, what is the expected value of the winning amount for a player with a single ticket?
5. In a lottery, a player pays \$1 and selects four distinct numbers from 0 to 9. Then, from an urn containing 10 identical balls numbered from 0 to 9, four balls are drawn at random and without replacement. If the numbers of three or all four of these balls matches the player's numbers, he wins \$5 and \$10, respectively. Otherwise, he loses. On the average, how much money does the player *gain* per game? (Gain = win – loss.)
6. An urn contains five balls, two of which are marked \$1, two \$5, and one \$15. A game is played by paying \$10 for winning the sum of the amounts marked on two balls selected randomly from the urn. Is this a fair game?
7. Let X be a discrete random variable with the following probability mass function

x	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$p(x)$	0.2	0.4	0.3	0.1

Find $E(\cos X)$.

8. A box contains 20 fuses, of which five are defective. What is the expected number of defective items among three fuses selected randomly?
9. The demand for a certain weekly magazine at a newsstand is a random variable with probability mass function $p(i) = (10 - i)/18$, $i = 4, 5, 6, 7$. If the magazine sells for \$ a and costs \$ $2a/3$ to the owner, and the unsold magazines cannot be returned, how many magazines should be ordered every week to maximize the profit in the long run?
10. It is well known that $\sum_{x=1}^{\infty} 1/x^2 = \pi^2/6$.
 - (a) Show that $p(x) = 6/(\pi x)^2$, $x = 1, 2, 3, \dots$ is the probability mass function of a random variable X .
 - (b) Prove that $E(X)$ does not exist.
11. (a) Show that $p(x) = (|x| + 1)^2/27$, $x = -2, -1, 0, 1, 2$, is the probability mass function of a random variable X .

(b) Calculate $E(X)$, $E(|X|)$, and $E(2X^2 - 5X + 7)$.

12. A box contains 10 disks of radii 1, 2, ..., 10, respectively. What is the expected value of the circumference of a disk selected at random from this box?
13. The amount that an insurance policy pays for hospitalization is \$ a per day up to 4 days and $\$(a/2)$ thereafter. Let X be the number of days a randomly selected policy holder who needs hospitalization is hospitalized. If the probability mass function of X is given by

$$p(n) = \frac{8-n}{28}, \quad 1 \leq n \leq 7,$$

find the expected amount of money that the policy pays for each hospitalization.

14. The distribution function of a random variable X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < -3 \\ 3/8 & \text{if } -3 \leq x < 0 \\ 1/2 & \text{if } 0 \leq x < 3 \\ 3/4 & \text{if } 3 \leq x < 4 \\ 1 & \text{if } x \geq 4. \end{cases}$$

Calculate $E(X)$, $E(X^2 - 2|X|)$, and $E(X|X|)$.

15. If X is a random number selected from the first 10 positive integers, what is the expected value of $X(11 - X)$?
16. Let X be the number of different birthdays among four persons selected randomly. Find $E(X)$.
17. A newly married couple decides to continue having children until they have one of each sex. If the events of having a boy and a girl are independent and equiprobable, how many children should this couple expect?
Hint: Note that $\sum_{i=1}^{\infty} ir^i = r/(1-r)^2$, $|r| < 1$.

B

18. Suppose that there exist N families on the earth and that the maximum number of children a family has is c . For $j = 0, 1, 2, \dots, c$, let α_j be the fraction of families with j children ($\sum_{j=0}^c \alpha_j = 1$). A *child* is selected at random from the set of all children in the world. Let this child be the K th born of his or her family, then K is a random variable. Find $E(K)$.
19. An ordinary deck of 52 cards is well-shuffled, and then the cards are turned face up one by one until an ace appears. Find the expected number of cards that are face up.
20. Suppose that n random integers are selected from $\{1, 2, \dots, N\}$ with replacement. What is the expected value of the largest number selected? Show that for large N the answer is approximately $nN/(n+1)$.

21. (a) Show that

$$p(n) = \frac{1}{n(n+1)}, \quad n \geq 1,$$

is a probability mass function.

- (b) Let X be a random variable with probability mass function p given in part (a); find $E(X)$.
22. To an engineering class containing $2n - 3$ male and three female students, there are n work stations available. To assign each workstation to two students, the professor forms n teams one at a time, each consisting of two randomly selected students. In this process, let X be the number of students selected until a team of a male and a female is formed. Find the expected value of X .



Self-Quiz on Section 4.4

Time allotted: 20 Minutes

Each problem is worth 2.5 points.

1. Let X be a discrete random variable with probability mass function

x	0	1	3	7
$p(x)$	0.2	0.3	0.4	0.1

Find $E(2^X)$.

2. Two fair dice are tossed and the maximum of the outcomes is denoted by X . Find $E(X)$.
3. In a certain lottery, 15,000 tickets are sold. If there are 300 prizes of \$5 each, 50 prizes of \$50 each, and one grand prize of \$1,000, what is the fair price for each ticket?
4. Let F be the distribution function of a random variable X . Find $E(X)$.

$$F(t) = \begin{cases} 0 & t < 2 \\ 0.2 & 2 \leq t < 3 \\ 0.3 & 3 \leq t < 5 \\ 0.8 & 5 \leq t < 7 \\ 1 & t \geq 7 \end{cases}$$



4.5 VARIANCES AND MOMENTS OF DISCRETE RANDOM VARIABLES

Thus far, through many examples, we have explained the importance of mathematical expectation in detail. For instance, in Example 4.16, we have shown how expectation is applied in decision making. Also, in Example 4.17, concerning lottery, we showed that the expectation of the winning amount per game gives an excellent estimation for the total amount a player will win if he or she plays a large number of times. In these and many other situations, mathematical expectation is the only quantity one needs to calculate. However, very frequently we face situations in which the expectation by itself does not say much. In such cases more information should be extracted from the probability mass function. As an example, suppose that we are interested in measuring a certain quantity. Let X be the true value[†] of the quantity minus the value obtained by measurement. Then X is the error of measurement. It is a random variable with expected value zero, the reason being that in measuring a quantity a very large number of times, positive and negative errors of the same magnitudes occur with equal probabilities. Now consider an experiment in which a quantity is measured several times, and the average of the errors is obtained to be a number close to zero. Can we conclude that the measurements are very close to the true value and thus are accurate? The answer is no because they might differ from the true value by relatively large quantities but be scattered both in positive and negative directions, resulting in zero expectation. Thus in this and similar cases, expectation by itself does not give adequate information, so additional measures for decision making are needed. One such quantity is the **variance** of a random variable.

Variance measures the average magnitude of the fluctuations of a random variable from its expectation. This is particularly important because random variables fluctuate from their expectations. To mathematically define the variance of a random variable X , the first temptation is to consider the expectation of the difference of X from its expectation, that is, $E[X - E(X)]$. But the difficulty with this quantity is that the positive and negative deviations of X from $E(X)$ cancel each other, and we always get 0. This can be seen mathematically from the corollary of Theorem 4.2: Let $E(X) = \mu$, then

$$E[X - E(X)] = E(X - \mu) = E(X) - \mu = E(X) - E(X) = 0.$$

Hence $E[X - E(X)]$ is not an appropriate measure for the variance. However, if we consider $E(|X - E(X)|)$ instead, the problem of negative and positive deviations canceling each other disappears. Since this quantity is the true average magnitude of the fluctuations of X from $E(X)$, it seems that it is the best candidate for an expression for the variance of X . But mathematically, $E(|X - E(X)|)$ is difficult to handle; for this reason the quantity $E[(X - E(X))^2]$, analogous to Euclidean distance in geometry, is used instead and is called the **variance** of X . The square root of $E[(X - E(X))^2]$ is called the **standard deviation** of X .

Definition 4.5 Let X be a discrete random variable with a set of possible values A , probability mass function $p(x)$, and $E(X) = \mu$. Then σ_X and $\text{Var}(X)$, called the **standard deviation** and the **variance** of X , respectively, are defined by

$$\sigma_X = \sqrt{E[(X - \mu)^2]} \quad \text{and} \quad \text{Var}(X) = E[(X - \mu)^2].$$

[†]True value is a nebulous concept. Here we shall use it to mean the average of a *large* number of measurements.

Note that by this definition and Theorem 4.2,

$$\text{Var}(X) = E[(X - \mu)^2] = \sum_{x \in A} (x - \mu)^2 p(x).$$

Let X be a discrete random variable with the set of possible values A and probability mass function $p(x)$. Suppose that the prediction of the value of X is in order, and if the value t is predicted for X , then based on the error $X - t$, a penalty is charged. To minimize the penalty, it seems reasonable to minimize $E[(X - t)^2]$. But

$$E[(X - t)^2] = \sum_{x \in A} (x - t)^2 p(x).$$

Assuming that this series converges [i.e., $E(X^2) < \infty$], we differentiate it to find the minimum value of $E[(X - t)^2]$:

$$\frac{d}{dt} E[(X - t)^2] = \frac{d}{dt} \sum_{x \in A} (x - t)^2 p(x) = \sum_{x \in A} -2(x - t)p(x) = 0.$$

This gives

$$\sum_{x \in A} xp(x) = t \sum_{x \in A} p(x) = t.$$

Therefore, $E[(X - t)^2]$ is a minimum for $t = \sum_{x \in A} xp(x) = E(X)$ and the minimum value is $E[(X - E(X))^2] = \text{Var}(X)$. So the smaller that $\text{Var}(X)$ is, the better $E(X)$ predicts X . We have that

$$\text{Var}(X) = \min_t E[(X - t)^2].$$

Earlier we mentioned that, if we think of a unit mass distributed along the real line at the points of A so that the mass at $x \in A$ is $p(x) = P(X = x)$, then $E(X)$ is the center of gravity. As we know, since center of gravity does not provide any information about how the mass is distributed around this center, the concept of moment of inertia is introduced. Moment of inertia is a measure of dispersion (spread) of the mass distribution about the center of gravity. $E(X)$ is analogous to the center of gravity, and it too does not provide any information about the distribution of X about this center of location. However, variance, the analog of the moment of inertia, measures the dispersion, or spread, of a distribution about its expectation.

Related Historical Remark: In 1900, the Wright brothers were looking for a private location with consistent wind to test their gliders. The data they got from the U.S. Weather Bureau indicated that Kill Devil Hill, near Kitty Hawk, North Carolina, had, on average, suitable winds, so they chose that location for their tests. However, the wind was not consistent. There were many calm days and many days with strong winds that were not suitable for their tests. The summary of the data was obtained by averaging undesirable extreme wind conditions. The Weather Bureau statistics were misleading because they failed to utilize the standard deviation. If the Wright brothers had been provided with the standard deviation of the wind speed at Kitty Hawk, they would not have chosen that location for their tests. ♦

Example 4.23 Karen is interested in two games, Keno and Bolita. To play Bolita, she buys a ticket for \$1, draws a ball at random from a box of 100 balls numbered 1 to 100. If the ball drawn matches the number on her ticket, she wins \$75; otherwise, she loses. To play Keno, Karen bets \$1 on a single number that has a 25% chance to win. If she wins, they will return her dollar plus two dollars more; otherwise, they keep the dollar. Let B and K be the amounts that Karen gains in one play of Bolita and Keno, respectively. Then

$$\begin{aligned} E(B) &= (74)(0.01) + (-1)(0.99) = -0.25, \\ E(K) &= (2)(0.25) + (-1)(0.75) = -0.25. \end{aligned}$$

Therefore, in the long run, it does not matter which of the two games Karen plays. Her gain would be about the same. However, by virtue of

$$\text{Var}(B) = E[(B - \mu)^2] = (74 + 0.25)^2(0.01) + (-1 + 0.25)^2(0.99) = 55.69$$

and

$$\text{Var}(K) = E[(K - \mu)^2] = (2 + 0.25)^2(0.25) + (-1 + 0.25)^2(0.75) = 1.6875,$$

we can say that in Bolita, on average, the deviation of the gain from the expectation is much higher than in Keno. In other words, the risk with Keno is far less than the risk with Bolita. In Bolita, the probability of winning is very small, but the amount one might win is high. In Keno, players win more often but in smaller amounts. ♦

The following theorem states another useful formula for $\text{Var}(X)$.

Theorem 4.3 $\text{Var}(X) = E(X^2) - [E(X)]^2$.

Proof: By the definition of variance,

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] = E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) - 2\mu E(X) + \mu^2 \\ &= E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2 \\ &= E(X^2) - [E(X)]^2. \quad \blacklozenge \end{aligned}$$

One immediate application of this formula is that, since $\text{Var}(X) \geq 0$, for any discrete random variable X ,

$$[E(X)]^2 \leq E(X^2).$$

The formula $\text{Var}(X) = E(X^2) - [E(X)]^2$ is usually a better alternative for computing the variance of X . Here is an example.

Example 4.24 What is the variance of the random variable X , the outcome of rolling a fair die?

Solution: The probability mass function of X is given by $p(x) = 1/6$; $x = 1, 2, 3, 4, 5, 6$, and $p(x) = 0$, otherwise. Hence

$$E(X) = \sum_{x=1}^6 xp(x) = \frac{1}{6} \sum_{x=1}^6 x = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{7}{2},$$

$$E(X^2) = \sum_{x=1}^6 x^2 p(x) = \frac{1}{6} \sum_{x=1}^6 x^2 = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6}.$$

Thus

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}. \quad \blacklozenge$$

Suppose that a random variable X is constant, then $E(X) = X$ and the deviations of X from $E(X)$ are 0. Therefore, the average deviation of X from $E(X)$ is also 0. We have the following theorem.

Theorem 4.4 *Let X be a discrete random variable with the set of possible values A , and mean μ . Then $\text{Var}(X) = 0$ if and only if X is a constant with probability 1.*

Proof: We will show that $\text{Var}(X) = 0$ implies that $X = \mu$ with probability 1. Suppose not, then there exists some $k \neq \mu$ such that $p(k) = P(X = k) > 0$. But then

$$\text{Var}(X) = (k - \mu)^2 p(k) + \sum_{x \in A - \{k\}} (x - \mu)^2 p(x) > 0,$$

which is a contradiction to $\text{Var}(X) = 0$. Conversely, if X is a constant c with probability 1, then $X = E(X) = c = \mu$ with probability 1. This implies that $\text{Var}(X) = E[(X - \mu)^2] = 0$. $\blacklozenge$

For constants a and b , a linear relation similar to $E(aX + b) = aE(X) + b$ does not exist for variance and for standard deviation. However, other important relations exist and are given by the following theorem.

Theorem 4.5 *Let X be a discrete random variable, then for constants a and b we have that*

$$\text{Var}(aX + b) = a^2 \text{Var}(X),$$

$$\sigma_{aX+b} = |a| \sigma_X.$$

Proof: To see this, note that

$$\begin{aligned}
 \text{Var}(aX + b) &= E[(aX + b) - E(aX + b)]^2 \\
 &= E[(aX + b) - (aE(X) + b)]^2 \\
 &= E[a(X - E(X))]^2 \\
 &= E[a^2(X - E(X))^2] \\
 &= a^2 E[(X - E(X))^2] \\
 &= a^2 \text{Var}(X).
 \end{aligned}$$

Taking the square roots of both sides of this relation, we find that $\sigma_{aX+b} = |a|\sigma_X$. ♦

Example 4.25 Suppose that, for a discrete random variable X , $E(X) = 2$ and $E[X(X - 4)] = 5$. Find the variance and the standard deviation of $-4X + 12$.

Solution: By the Corollary of Theorem 4.2, $E(X^2 - 4X) = 5$ implies that

$$E(X^2) - 4E(X) = 5.$$

Substituting $E(X)$ in this relation gives $E(X^2) = 13$. Hence, by Theorem 4.3,

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - [E(X)]^2 = 13 - 4 = 9, \\
 \sigma_X &= \sqrt{9} = 3.
 \end{aligned}$$

By Theorem 4.5,

$$\begin{aligned}
 \text{Var}(-4X + 12) &= 16 \text{Var}(X) = 16 \times 9 = 144, \\
 \sigma_{-4X+12} &= |-4|\sigma_X = 4 \times 3 = 12. \quad \blacklozenge
 \end{aligned}$$

Optional

As we know, variance measures the dispersion, or spread, of a distribution about its expectation. One way to find out which one of the two given random variables X and Y is more dispersed, or spread, about an arbitrary point ω is to see which one is more *concentrated* about ω . The following is a mathematical definition to this concept.

Definition 4.6 Let X and Y be two random variables and ω be a given point. If for all $t > 0$,

$$P(|Y - \omega| \leq t) \leq P(|X - \omega| \leq t),$$

then we say that X is more **concentrated** about ω than is Y .

A useful consequence of this definition is the following theorem, the proof of which we leave as an exercise. This theorem should be intuitively clear.

Theorem 4.6 Suppose that X and Y are two random variables with $E(X) = E(Y) = \mu$. If X is more concentrated about μ than is Y , then $\text{Var}(X) \leq \text{Var}(Y)$.

MOMENTS

Let X be a random variable with expected value μ . Let c be a constant, $n \geq 0$ be an integer, and $r > 0$ be any real number, integral or not. The expected value of X , $E(X)$, is also called the **first moment** of X . In practice, expected values of some important functions of X have also numerical and theoretical significance. Some of these functions are $g(X) = X^n$, $|X|^n$, $X - c$, $(X - c)^n$, and $(X - \mu)^n$. Provided that $E(|g(X)|) < \infty$, $E[g(X)]$ in each of these cases is defined as follows.

$E[g(X)]$	Definition
$E(X^n)$	The n th moment of X
$E(X ^r)$	The r th absolute moment of X
$E(X - c)$	The first moment of X about c
$E[(X - c)^n]$	The n th moment of X about c
$E[(X - \mu)^n]$	The n th central moment of X

★ **Remark 4.3** Let X be a discrete random variable with probability mass function $p(x)$ and set of possible values A . Let n be a positive integer. It is important to know that if $E(X^{n+1})$ exists, then $E(X^n)$ also exists. That is, *the existence of higher moments implies the existence of lower moments*. In particular, this implies that if $E(X^2)$ exists, then $E(X)$ and, hence, $\text{Var}(X)$ exist. To prove this fact, note that, by definition, $E(X^{n+1})$ exists if $\sum_{x \in A} |x|^{n+1} p(x) < \infty$. Let $B = \{x \in A: |x| < 1\}$, then $B^c = \{x \in A: |x| \geq 1\}$. We have

$$\begin{aligned} \sum_{x \in B} |x|^n p(x) &\leq \sum_{x \in B} p(x) \leq \sum_{x \in A} p(x) = 1; \\ \sum_{x \in B^c} |x|^n p(x) &\leq \sum_{x \in B^c} |x|^{n+1} p(x) \leq \sum_{x \in A} |x|^{n+1} p(x) < \infty. \end{aligned}$$

By these inequalities,

$$\sum_{x \in A} |x|^n p(x) = \sum_{x \in B} |x|^n p(x) + \sum_{x \in B^c} |x|^n p(x) \leq 1 + \sum_{x \in A} |x|^{n+1} p(x) < \infty,$$

showing that $E(X^n)$ also exists. ♦

EXERCISES

A

1. Mr. Jones is about to purchase a business. There are two businesses available. The first has a daily expected profit of \$150 with standard deviation \$30, and the second has a daily expected profit of \$150 with standard deviation \$55. If Mr. Jones is interested in a business with a steady income, which should he choose?
2. The temperature of a material is measured by two devices. Using the first device, the expected temperature is t with standard deviation 0.8; using the second device, the expected temperature is t with standard deviation 0.3. Which device measures the temperature more precisely?
3. Find the variance of X , the random variable with probability mass function

$$p(x) = \begin{cases} (|x - 3| + 1)/28 & x = -3, -2, -1, 0, 1, 2, 3 \\ 0 & \text{otherwise.} \end{cases}$$

4. In the inventory of a multinational office supply retailing corporation, there are 7,200 80-sheet smooth paper Renee pads made of acid free and ink-friendly paper in France. The retailing corporation sells these pads only in packs of 12. Orders received by the retailing corporation are from stationary stores, and the probability mass function of the number of packs of pads per order is

x	3	4	5	10
$p(x)$	0.4	0.1	0.3	0.2

Find the expected value and the standard deviation of the number of pads left in the inventory of the office supply corporation after the next order is shipped.

5. Find the variance and the standard deviation of a random variable X with distribution function

$$F(x) = \begin{cases} 0 & x < -3 \\ 3/8 & -3 \leq x < 0 \\ 3/4 & 0 \leq x < 6 \\ 1 & x \geq 6. \end{cases}$$

6. Let X be a discrete random variable with the set of possible values $\{x_1, x_2, \dots, x_n\}$; X is called a **discrete uniform random variable** if

$$P(X = x_i) = \frac{1}{n}, \quad 1 \leq i \leq n.$$

- (a) Find $E(X)$ and $\text{Var}(X)$ for the special case, where $x_i = i$, $1 \leq i \leq n$. Note that $\sum_{i=1}^n i = n(n+1)/2$ and $\sum_{i=1}^n i^2 = n(n+1)(2n+1)/6$.
- (b) Let X be a random integer from the set $\{1, 2, \dots, N\}$. Find $E(X)$, $\text{Var}(X)$, and σ_X .
7. Let X be the number of claims received, within a year, for an auto insurance policy offered by an insurance company. Let p , the probability mass function of X , be given by

n	25	30	35	40	45	50	55
$p(n)$	0.20	0.30	0.15	0.10	0.10	0.10	0.05

What percentage of the number of claims are within one standard deviation of the mean?

8. What are the expected number, the variance, and the standard deviation of the number of spades in a poker hand? (A poker hand is a set of five cards that are randomly selected from an ordinary deck of 52 cards.)
9. Suppose that X is a discrete random variable with $E(X) = 1$ and $E[X(X-2)] = 3$. Find $\text{Var}(-3X + 5)$.
10. In a game, Emily gives Harry three well-balanced quarters to flip. Harry will get to keep all the ones that will land heads. He will return those landing tails. However, if all three coins land tails, Harry must pay Emily two dollars. Find the expected value and the variance of Harry's net gain.
11. Let X be a discrete random variable. Show that if $E(X^2) = 0$, then $X = 0$ with probability 1.
12. Let X be a random variable defined by

$$P(X = -1) = P(X = 1) = 1/2.$$

Let Y be a random variable defined by

$$P(Y = -10) = P(Y = 10) = 1/2.$$

Which one of X and Y is more concentrated about 0 and why?

B

13. A drunken man has n keys, one of which opens the door to his office. He tries the keys at random, one by one, and independently. Compute the mean and the variance of the number of trials required to open the door if the wrong keys (a) are not eliminated; (b) are eliminated.
14. For $n = 1, 2, 3, \dots$, let $x_n = (-1)^n \sqrt{n}$. Let X be a discrete random variable with the set of possible values $A = \{x_n : n = 1, 2, 3, \dots\}$ and probability mass function

$$p(x_n) = P(X = x_n) = \frac{6}{(\pi n)^2}.$$

Show that even though $\sum_{n=1}^{\infty} x_n^3 p(x_n) < \infty$, $E(X^3)$ does not exist.

15. Let X be a discrete random variable; let $0 < s < r$. Show that if the r th absolute moment of X exists, then the absolute moment of order s of X also exists.
16. Let X and Y be two discrete random variables with the identical set of possible values $A = \{a, b, c\}$, where a, b , and c are three *different* real numbers. Show that if $E(X) = E(Y)$ and $\text{Var}(X) = \text{Var}(Y)$, then X and Y are identically distributed. That is,

$$P(X = t) = P(Y = t) \quad \text{for } t = a, b, c.$$

17. Let X and Y be two discrete random variables with the identical set of possible values $A = \{a_1, a_2, \dots, a_n\}$, where $a_1, a_2, \dots, a_n$ are n *different* real numbers. Show that if

$$E(X^r) = E(Y^r), \quad r = 1, 2, \dots, n-1,$$

then X and Y are identically distributed. That is,

$$P(X = t) = P(Y = t) \quad \text{for } t = a_1, a_2, \dots, a_n.$$



Self-Quiz on Section 4.5

Time allotted: 20 Minutes

Each problem is worth 5 points.

- Two fair dice are tossed. Let X be the sum of the outcomes. Find $\text{Var}(X)$ and σ_X .
- Let X be a random variable with $E(X) = 3$ and $E[(X - 3)(4 - X)] = -15$. Find $\text{Var}(-3X + 8)$.



4.6 STANDARDIZED RANDOM VARIABLES

Let X be a random variable with mean μ and standard deviation σ . The random variable $X^* = (X - \mu)/\sigma$ is called the **standardized** X . We have that

$$E(X^*) = E\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma}E(X) - \frac{\mu}{\sigma} = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0,$$

$$\text{Var}(X^*) = \text{Var}\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma^2}\text{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1.$$

When standardizing a random variable X , we change the origin to μ and the scale to the units of standard deviation. The value that is obtained for X^* is independent of the units in which X is measured. It is the number of standard deviation units by which X differs from $E(X)$. For example, let X be a random variable with mean 10 feet and standard deviation 2 feet. Suppose

that in a random observation we obtain $X = 16$, then $X^* = (16 - 10)/2 = 3$. This shows that the distance of X from its mean is 3 standard deviation units regardless of the scale of measurement. That is, if the same quantities are measured, say, in inches (12 inches = 1 foot), then we will get the same standardized value:

$$X^* = \frac{16 \times 12 - 10 \times 12}{2 \times 12} = 3.$$

Standardization is particularly useful if two or more random variables with different distributions must be compared. Suppose that, for example, a student's grade in a probability test is 72 and that her grade in a history test is 85. At first glance these grades suggest that the student is doing much better in the history course than in the probability course. However, this might not be true—the relative grade of the student in probability might be better than that in history. To illustrate, suppose that the mean and standard deviation of all grades in the history test are 82 and 7, respectively, while these quantities in the probability test are 68 and 4. If we convert the student's grades to their standard deviation units, we find that her standard scores on the probability and history tests are given by $(72 - 68)/4 = 1$ and $(85 - 82)/7 = 0.43$, respectively. These show that her grade in probability is 1 and in history is 0.43 standard deviation unit higher than their respective averages. Therefore, she is doing relatively better in the probability course than in the history course. This comparison is most useful when only the means and standard deviations of the random variables being studied are known. If the distribution functions of these random variables are given, better comparisons might be possible.

We now prove that, for a random variable X , the standardized X , denoted by X^* , is independent of the units in which X is measured. To do so, let X_1 be the observed value of X when a different scale of measurement is used. Then for some $\alpha > 0$, we have that $X_1 = \alpha X + \beta$, and

$$\begin{aligned} X_1^* &= \frac{X_1 - E(X_1)}{\sigma_{X_1}} = \frac{(\alpha X + \beta) - [\alpha E(X) + \beta]}{\sigma_{\alpha X + \beta}} \\ &= \frac{\alpha[X - E(X)]}{\alpha \sigma_X} = \frac{X - E(X)}{\sigma_X} = X^*. \end{aligned}$$

EXERCISES

1. Mr. Norton owns two appliance stores. In store 1 the number of TV sets sold by a salesperson is, on average, 13 per week with a standard deviation of five. In store 2 the number of TV sets sold by a salesperson is, on average, seven with a standard deviation of four. Mr. Norton has a position open for a person to sell TV sets. There are two applicants. Mr. Norton asked one of them to work in store 1 and the other in store 2, each for one week. The salesperson in store 1 sold 10 sets, and the salesperson in store 2 sold six sets. Based on this information, which person should Mr. Norton hire?

2. The mean and standard deviation in midterm tests of a probability course are 72 and 12, respectively. These quantities for final tests are 68 and 15. What final grade is comparable to Velma's 82 in the midterm.



4.7 JOINT PROBABILITY MASS FUNCTIONS

Thus far we have studied probability mass functions of single discrete random variables. We now consider two or more random variables that are defined simultaneously on the same sample space.

Definition 4.7 Let X and Y be two discrete random variables defined on the same sample space. Let the sets of possible values of X and Y be A and B , respectively. The function

$$p(x, y) = P(X = x, Y = y)$$

is called the **joint probability mass function** of X and Y .

Joint probability distribution, also referred to as **bivariate distribution**, is a broader term encompassing both discrete and continuous cases, while the specific terms *joint probability mass function of two random variables* and *joint probability density functions of two random variables* (discussed in Chapter 5) are used when referring to discrete and continuous distributions, respectively.

Note that $p(x, y) \geq 0$. If $x \notin A$ or $y \notin B$, then $p(x, y) = 0$. Also,

$$\sum_{x \in A} \sum_{y \in B} p(x, y) = 1. \quad (4.3)$$

Let X and Y have joint probability mass function $p(x, y)$. Let p_X be the probability mass function of X . Then

$$\begin{aligned} p_X(x) &= P(X = x) = P(X = x, Y \in B) \\ &= \sum_{y \in B} P(X = x, Y = y) = \sum_{y \in B} p(x, y). \end{aligned}$$

Similarly, p_Y , the probability mass function of Y , is given by

$$p_Y(y) = \sum_{x \in A} p(x, y).$$

These relations motivate the following definition.

Definition 4.8 Let X and Y have joint probability mass function $p(x, y)$. Let A be the set of possible values of X and B be the set of possible values of Y . Then the functions $p_X(x) = \sum_{y \in B} p(x, y)$ and $p_Y(y) = \sum_{x \in A} p(x, y)$ are called, respectively, the **marginal probability mass functions** of X and Y .

Example 4.26 Let X and Y be discrete random variables with the set of possible values $\{2, 3, 4, 5\}$ and $\{1, 2, 3\}$, respectively. Let $p(x, y)$ their probability mass function be give by the following table:

x	y		
	1	2	3
2	1/6	0	1/12
3	0	1/6	1/12
4	1/12	1/6	0
5	1/12	1/12	1/12

- (a) Find the probability that $X \geq 4$ and $Y \leq 2$.
 (b) Sketch the graph of the joint probability mass function of X and Y .

Solution:

- (a) We need to find the probability that X is 4 or 5, and Y is 1 or 2:

$$\begin{aligned}
 P(X \geq 4 \text{ and } Y \leq 2) &= p(4, 1) + p(4, 2) + p(5, 1) + p(5, 2) \\
 &= \frac{1}{12} + \frac{1}{6} + \frac{1}{12} + \frac{1}{12} = \frac{5}{12}.
 \end{aligned}$$

- (b) The graph of the joint probability mass function of X and Y is given by [Figure 4.12](#).

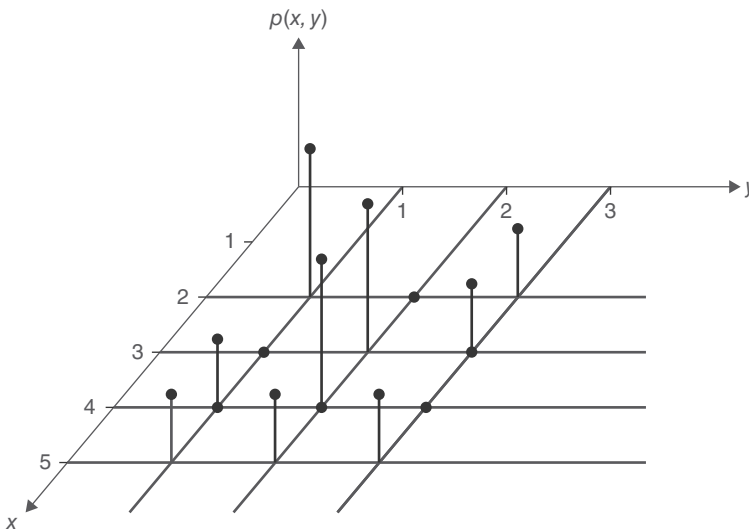


Figure 4.12 Joint probability mass function of Example 4.26.

Example 4.27 A small college has 90 male and 30 female professors. An ad hoc committee of five is selected at random to write the vision and mission of the college. Let X and Y be the number of men and women on this committee, respectively.

- (a) Find the joint probability mass function of X and Y .
- (b) Find p_X and p_Y , the marginal probability mass functions of X and Y .

Solution:

- (a) The set of possible values for both X and Y is $\{0, 1, 2, 3, 4, 5\}$. The joint probability mass function of X and Y , $p(x, y)$, is given by

$$p(x, y) = \begin{cases} \frac{\binom{90}{x} \binom{30}{y}}{\binom{120}{5}} & \text{if } x, y \in \{0, 1, 2, 3, 4, 5\}, \quad x + y = 5 \\ 0 & \text{otherwise.} \end{cases}$$

- (b) To find p_X and p_Y , the marginal probability mass functions of X and Y , respectively, note that $p_X(x) = \sum_{y=0}^5 p(x, y)$, $p_Y(y) = \sum_{x=0}^5 p(x, y)$. Since $p(x, y) = 0$ if $x + y \neq 5$, $\sum_{y=0}^5 p(x, y) = p(x, 5 - x)$ and $\sum_{x=0}^5 p(x, y) = p(5 - y, y)$. Therefore,

$$p_X(x) = \frac{\binom{90}{x} \binom{30}{5-x}}{\binom{120}{5}}, \quad x \in \{0, 1, 2, 3, 4, 5\},$$

$$p_Y(y) = \frac{\binom{90}{5-y} \binom{30}{y}}{\binom{120}{5}}, \quad y \in \{0, 1, 2, 3, 4, 5\}.$$

The probability mass functions p_X and p_Y are known functions. They are hypergeometric distributions studied in [Chapter 6](#). ♦

Example 4.28 Roll a balanced die and let the outcome be X . Then toss a fair coin X times and let Y denote the number of tails. What is the joint probability mass function of X and Y and the marginal probability mass functions of X and Y ?

Solution: Let $p(x, y)$ be the joint probability mass function of X and Y . Clearly, $X \in \{1, 2, 3, 4, 5, 6\}$ and $Y \in \{0, 1, 2, 3, 4, 5, 6\}$. Now if $X = 1$, then $Y = 0$ or 1 ; we have

$$p(1, 0) = P(X = 1, Y = 0) = P(X = 1)P(Y = 0 | X = 1) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12},$$

$$p(1, 1) = P(X = 1, Y = 1) = P(X = 1)P(Y = 1 | X = 1) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}.$$

If $X = 2$, then $y = 0, 1$, or 2 , where

$$p(2, 0) = P(X = 2, Y = 0) = P(X = 2)P(Y = 0 | X = 2) = \frac{1}{6} \cdot \frac{1}{4} = \frac{1}{24}.$$

Similarly, $p(2, 1) = 1/12$, $p(2, 2) = 1/24$. If $X = 3$, then $y = 0, 1, 2$, or 3 , and

$$\begin{aligned} p(3, 0) &= P(X = 3, Y = 0) = P(X = 3)P(Y = 0 | X = 3) \\ &= \frac{1}{6} \binom{3}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^3 = \frac{1}{48}, \end{aligned}$$

$$\begin{aligned} p(3, 1) &= P(X = 3, Y = 1) = P(X = 3)P(Y = 1 | X = 3) \\ &= \frac{1}{6} \binom{3}{1} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^1 = \frac{3}{48}. \end{aligned}$$

Similarly, $p(3, 2) = 3/48$, $p(3, 3) = 1/48$. Similar calculations will yield the following table for $p(x, y)$.

x	y							$p_X(x)$
	0	1	2	3	4	5	6	
1	1/12	1/12	0	0	0	0	0	1/6
2	1/24	2/24	1/24	0	0	0	0	1/6
3	1/48	3/48	3/48	1/48	0	0	0	1/6
4	1/96	4/96	6/96	4/96	1/96	0	0	1/6
5	1/192	5/192	10/192	10/192	5/192	1/192	0	1/6
6	1/384	6/384	15/384	20/384	15/384	6/384	1/384	1/6
$p_Y(y)$	63/384	120/384	99/384	64/384	29/384	8/384	1/384	

Note that $p_X(x) = P(X = x)$ and $p_Y(y) = P(Y = y)$, the probability mass functions of X and Y , are obtained by summing up the rows and the columns of this table, respectively.

Let X and Y be discrete random variables with joint probability mass function $p(x, y)$. Let the sets of possible values of X and Y be A and B , respectively. To find $E(X)$ and $E(Y)$, first we calculate p_X and p_Y , the marginal probability mass functions of X and Y , respectively. Then we will use the following formulas.

$$E(X) = \sum_{x \in A} xp_X(x); \quad E(Y) = \sum_{y \in B} yp_Y(y).$$

Example 4.29 Let the joint probability mass function of random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{70} x(x + y) & \text{if } x = 1, 2, 3, \quad y = 3, 4 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X)$ and $E(Y)$.

Solution: To find $E(X)$ and $E(Y)$, first we need to calculate $p_X(x)$ and $p_Y(y)$, the marginal probability mass functions of X and Y , respectively. Note that

$$\begin{aligned} p_X(x) &= p(x, 3) + p(x, 4) \\ &= \frac{1}{70}x(x + 3) + \frac{1}{70}x(x + 4) = \frac{1}{35}x^2 + \frac{1}{10}x, \quad x = 1, 2, 3; \\ p_Y(y) &= p(1, y) + p(2, y) + p(3, y) \\ &= \frac{1}{70}(1 + y) + \frac{2}{70}(2 + y) + \frac{3}{70}(3 + y) = \frac{1}{5} + \frac{3}{35}y, \quad y = 3, 4. \end{aligned}$$

Therefore,

$$\begin{aligned} E(X) &= \sum_{x=1}^3 xp_X(x) = \sum_{x=1}^3 x \left(\frac{1}{35}x^2 + \frac{1}{10}x \right) = \frac{17}{7} \approx 2.43; \\ E(Y) &= \sum_{y=3}^4 yp_Y(y) = \sum_{y=3}^4 y \left(\frac{1}{5} + \frac{3}{35}y \right) = \frac{124}{35} \approx 3.54. \quad \blacklozenge \end{aligned}$$

We will now state the following generalization of Theorem 4.2 from one dimension to two. As an immediate application of this important generalization, we will show that the expected value of the sum of two random variables is equal to the sum of their expected values.

Theorem 4.7 Let $p(x, y)$ be the joint probability mass function of discrete random variables X and Y . Let A and B be the set of possible values of X and Y , respectively. If h is a function of two variables from $\mathbb{R}^2$ to $\mathbb{R}$, then $h(X, Y)$ is a discrete random variable with the expected value given by

$$E[h(X, Y)] = \sum_{x \in A} \sum_{y \in B} h(x, y)p(x, y),$$

provided that the sum is absolutely convergent.

Letting $h(x, y) = x + y$ in this theorem, it follows that

$$\begin{aligned} E(X + Y) &= \sum_{x \in A} \sum_{y \in B} (x + y)p(x, y) \\ &= \sum_{x \in A} \sum_{y \in B} xp(x, y) + \sum_{x \in A} \sum_{y \in B} yp(x, y) \\ &= E(X) + E(Y). \end{aligned}$$

We have shown the following important corollary.

Corollary For discrete random variables X and Y ,

$$E(X + Y) = E(X) + E(Y).$$

Example 4.30 Let the joint probability mass function of the discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{2}{11} \left(\frac{x}{y} \right) & \text{if } x = 1, 2, \quad y = 1, 2, 3 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(XY)$.

Solution: By Theorem 4.7,

$$\begin{aligned} E(XY) &= \sum_{x=1}^2 \sum_{y=1}^3 xy \cdot \frac{2}{11} \left(\frac{x}{y} \right) \\ &= \frac{2}{11} \sum_{x=1}^2 \sum_{y=1}^3 x^2 = \frac{2}{11} \sum_{x=1}^2 3x^2 = \frac{30}{11} \approx 2.73. \quad \blacklozenge \end{aligned}$$

The following definition generalizes the concept of joint probability mass function of two discrete random variables to $n > 2$ discrete random variables.

Definition 4.9 Let $X_1, X_2, \dots, X_n$ be discrete random variables defined on the same sample space, with sets of possible values $A_1, A_2, \dots, A_n$, respectively. The function

$$p(x_1, x_2, \dots, x_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

is called the **joint probability mass function** of $X_1, X_2, \dots, X_n$.

Similar to the case of two random variables, **Joint probability distribution**, also referred to as **multivariate distribution**, is a broader term encompassing both discrete and continuous

cases, while the specific terms *joint probability mass function of several random variables* and *joint probability density functions of several random variables* (discussed in [Chapter 5](#)) are used when referring to discrete and continuous distributions, respectively.

Note that

- (a) $p(x_1, x_2, \dots, x_n) \geq 0$.
- (b) If for some i , $1 \leq i \leq n$, $x_i \notin A_i$, then $p(x_1, x_2, \dots, x_n) = 0$.
- (c) $\sum_{x_i \in A_i, 1 \leq i \leq n} p(x_1, x_2, \dots, x_n) = 1$.

Moreover, if the joint probability mass function of random variables $X_1, X_2, \dots, X_n$, $p(x_1, x_2, \dots, x_n)$, is given, then for $1 \leq i \leq n$, the **marginal probability mass function** of X_i , p_{X_i} , can be found from $p(x_1, x_2, \dots, x_n)$ by

$$\begin{aligned} p_{X_i}(x_i) &= P(X_i = x_i) = P(X_i = x_i; X_j \in A_j, 1 \leq j \leq n, j \neq i) \\ &= \sum_{x_j \in A_j, j \neq i} p(x_1, x_2, \dots, x_n). \end{aligned} \quad (4.4)$$

The following is a generalization of Theorem 4.7 from dimension 2 to n .

Theorem 4.8 *Let $p(x_1, x_2, \dots, x_n)$ be the joint probability mass function of discrete random variables $X_1, X_2, \dots, X_n$. For $1 \leq i \leq n$, let A_i be the set of possible values of X_i . If h is a function of n variables from $\mathbb{R}^n$ to $\mathbb{R}$, then $Y = h(X_1, X_2, \dots, X_n)$ is a discrete random variable with expected value given by*

$$E(Y) = \sum_{x_n \in A_n} \cdots \sum_{x_1 \in A_1} h(x_1, x_2, \dots, x_n) p(x_1, x_2, \dots, x_n),$$

provided that the sum is finite.

Example 4.31 Let

$$p(x, y, z) = k(x^2 + y^2 + yz), \quad x = 0, 1, 2; \quad y = 2, 3; \quad z = 3, 4.$$

- (a) For what value of k is $p(x, y, z)$ a joint probability mass function?
- (b) Suppose that, for the value of k found in part (a), $p(x, y, z)$ is the joint probability mass function of random variables X , Y , and Z . Find $p_Z(z)$.
- (c) Find $E(XZ)$.

Solution:

- (a) We must have

$$k \sum_{x=0}^2 \sum_{y=2}^3 \sum_{z=3}^4 (x^2 + y^2 + yz) = 1.$$

This implies that $k = 1/203$.

(b)

$$\begin{aligned}
 p_Z(z) &= \frac{1}{203} \sum_{x=0}^2 \sum_{y=2}^3 (x^2 + y^2 + yz) \\
 &= \frac{15}{203}z + \frac{7}{29}, \quad z = 3, 4.
 \end{aligned}$$

(c) By Theorem 4.8,

$$E(XZ) = \frac{1}{203} \sum_{x=0}^2 \sum_{y=2}^3 \sum_{z=3}^4 xz(x^2 + y^2 + yz) = \frac{774}{203} \approx 3.81. \quad \blacklozenge$$

Now, we present the linearity property of expectation for discrete random variables. In Theorem 4.8, for real numbers $\alpha_1, \alpha_2, \dots, \alpha_n$, let $Y = h(X_1, X_2, \dots, X_n) = \sum_{i=1}^n \alpha_i X_i$. We obtain the following theorem.

Theorem 4.9 For random variables $X_1, X_2, \dots, X_n$ defined on the same sample space, and for real numbers $\alpha_1, \alpha_2, \dots, \alpha_n$,

$$E\left(\sum_{i=1}^n \alpha_i X_i\right) = \sum_{i=1}^n \alpha_i E(X_i).$$

In Theorem 4.9, for $1 \leq i \leq n$, letting $\alpha_i = 1$, we have the following important corollary.

Corollary Let $X_1, X_2, \dots, X_n$ be random variables on the same sample space. Then

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n).$$

Example 4.32 A die is rolled 15 times. What is the expected value of the sum of the outcomes?

Solution: Let X be the sum of the outcomes, and for $i = 1, 2, \dots, 15$; let X_i be the outcome of the i th roll. Then $X = X_1 + X_2 + \dots + X_{15}$. Thus

$$E(X) = E(X_1) + E(X_2) + \dots + E(X_{15}).$$

Now since for all values of i , X_i is 1, 2, 3, 4, 5, and 6, with probability of $1/6$ for all six values,

$$E(X_i) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{7}{2}.$$

Hence $E(X) = 15(7/2) = 52.5$. $\blacklozenge$

Example 4.32 and the following examples show the robust applicability of Theorem 4.9 and its corollary. Note that in Example 4.32, while the formula

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

enables us to compute $E(X)$ readily, computing $E(X)$ directly from probability mass function of X is not easy. This is because calculation of the probability mass function of X is time-consuming and cumbersome.

Example 4.33 A well-shuffled ordinary deck of 52 cards is divided randomly into four piles of 13 each. Counting jack, queen, and king as 11, 12, and 13, respectively, we say that a match occurs in a pile if the j th card is j . What is the expected value of the total number of matches in all four piles?

Solution: Let X_i , $i = 1, 2, 3, 4$ be the number of matches in the i th pile. $X = X_1 + X_2 + X_3 + X_4$ is the total number of matches, and $E(X) = E(X_1) + E(X_2) + E(X_3) + E(X_4)$. To calculate $E(X_i)$, $1 \leq i \leq 4$, let A_{ij} be the event that the j th card in the i th pile is j ($1 \leq i \leq 4$, $1 \leq j \leq 13$). Then by defining

$$X_{ij} = \begin{cases} 1 & \text{if } A_{ij} \text{ occurs} \\ 0 & \text{otherwise,} \end{cases}$$

we have that $X_i = \sum_{j=1}^{13} X_{ij}$. Now $P(A_{ij}) = 4/52 = 1/13$ implies that

$$E(X_{ij}) = 1 \cdot P(A_{ij}) + 0 \cdot P(A_{ij}^c) = P(A_{ij}) = \frac{1}{13}.$$

Hence

$$E(X_i) = E\left(\sum_{j=1}^{13} X_{ij}\right) = \sum_{j=1}^{13} E(X_{ij}) = \sum_{j=1}^{13} \frac{1}{13} = 1.$$

Thus on average there is one match in every pile. From this we get

$$E(X) = E(X_1) + E(X_2) + E(X_3) + E(X_4) = 1 + 1 + 1 + 1 = 4,$$

showing that on average there are a total of four matches. ♦

Example 4.34 Dr. Windler's secretary accidentally threw a patient's file into the wastebasket. A few minutes later, the janitor cleaned the entire clinic, dumped the wastebasket containing the patient's file randomly into one of the seven garbage cans outside the clinic, and left. Determine the expected number of cans that Dr. Windler should empty to find the file.

Solution: Let X be the number of garbage cans that Dr. Windler should empty to find the patient's file. For $i = 1, 2, \dots, 7$, let $X_i = 1$ if the patient's file is in the i th garbage can that Dr. Windler will empty, and $X_i = 0$, otherwise. Then

$$X = 1 \cdot X_1 + 2 \cdot X_2 + \cdots + 7 \cdot X_7,$$

and, therefore,

$$\begin{aligned} E(X) &= 1 \cdot E(X_1) + 2 \cdot E(X_2) + \cdots + 7 \cdot E(X_7) \\ &= 1 \cdot \frac{1}{7} + 2 \cdot \frac{1}{7} + \cdots + 7 \cdot \frac{1}{7} = 4. \end{aligned} \quad \blacklozenge$$

Example 4.35 A box contains nine light bulbs, of which two are defective. What is the expected value of the number of light bulbs that one will have to test (at random and without replacement) to find both defective bulbs?

Solution: For $i = 1, 2, \dots, 8$ and $j > i$, let $X_{ij} = j$ if the i th and j th light bulbs to be examined are defective, and $X_{ij} = 0$ otherwise. Then $\sum_{i=1}^8 \sum_{j=i+1}^9 X_{ij}$ is the number of light bulbs to be examined. Therefore,

$$\begin{aligned} E(X) &= \sum_{i=1}^8 \sum_{j=i+1}^9 E(X_{ij}) = \sum_{i=1}^8 \sum_{j=i+1}^9 j \frac{1}{\binom{9}{2}} \\ &= \frac{1}{36} \sum_{i=1}^8 \sum_{j=i+1}^9 j = \frac{1}{36} \sum_{i=1}^8 \frac{90 - i^2 - i}{2} \approx 6.67, \end{aligned}$$

where the next-to-last equality follows from

$$\sum_{j=i+1}^9 j = \sum_{j=1}^9 j - \sum_{j=1}^i j = \frac{9 \times 10}{2} - \frac{i(i+1)}{2} = \frac{90 - i^2 - i}{2}. \quad \blacklozenge$$

Example 4.36 Exactly n married couples are living in a small town. What is the expected number of intact couples after m deaths occur among the couples? Assume that the deaths occur at random, there are no divorces, and there are no new marriages.

Solution: Let X be the number of intact couples after m deaths, and for $i = 1, 2, \dots, n$ define

$$X_i = \begin{cases} 1 & \text{if the } i\text{th couple is left intact} \\ 0 & \text{otherwise.} \end{cases}$$

Then $X = X_1 + X_2 + \dots + X_n$, and hence

$$E(X) = E(X_1) + E(X_2) + \dots + E(X_n),$$

where

$$E(X_i) = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) = P(X_i = 1).$$

The event $\{X_i = 1\}$ occurs if the i th couple is left intact, and thus all of the m deaths are among the remaining $n - 1$ couples. Knowing that the deaths occur at random among these individuals, we can write

$$P(X_i = 1) = \frac{\binom{2n-2}{m}}{\binom{2n}{m}} = \frac{\frac{(2n-2)!}{m!(2n-m-2)!}}{\frac{(2n)!}{(2n-m)!m!}} = \frac{(2n-m)(2n-m-1)}{2n(2n-1)}.$$

Thus the desired quantity is equal to

$$\begin{aligned} E(X) &= E(X_1) + E(X_2) + \cdots + E(X_n) = n \frac{(2n-m)(2n-m-1)}{2n(2n-1)} \\ &= \frac{(2n-m)(2n-m-1)}{2(2n-1)}. \end{aligned}$$

To have a numerical feeling for this interesting example, let $n = 1000$. Then the following table shows the effect of the number of deaths on the expected value of the number of intact couples.

m	100	300	600	900	1200	1500	1800
$E(X)$	902.48	722.44	489.89	302.38	159.88	62.41	9.95

This example was posed by Daniel Bernoulli (1700–1782). ♦

Remark 4.4 For $n = \infty$, Theorem 4.9 is not necessarily true. That is, $E(\sum_{i=1}^{\infty} X_i)$ might not be equal to $\sum_{i=1}^{\infty} E(X_i)$. To show this, we give a counterexample. For $i = 1, 2, 3, \dots$, let

$$Y_i = \begin{cases} i & \text{with probability } 1/i \\ 0 & \text{otherwise,} \end{cases}$$

and $X_i = Y_{i+1} - Y_i$. Then since $E(Y_i) = 1$ for $i = 1, 2, \dots$, $E(X_i) = E(Y_{i+1}) - E(Y_i) = 0$. Hence $\sum_{i=1}^{\infty} E(X_i) = 0$. However, since $\sum_{i=1}^{\infty} X_i = -Y_1$, we have that $E(\sum_{i=1}^{\infty} X_i) = E(-Y_1) = -1$. Therefore, $E(\sum_{i=1}^{\infty} X_i) \neq \sum_{i=1}^{\infty} E(X_i)$. ♦

It can be shown that, in general,

If $\sum_{i=1}^{\infty} E(|X_i|) < \infty$ or if, for all i , the random variables $X_1, X_1, \dots$ are nonnegative [that is, $P(X_i \geq 0) = 1$ for $i \geq 1$], then

$$E\left(\sum_{i=1}^{\infty} X_i\right) = \sum_{i=1}^{\infty} E(X_i). \quad (4.5)$$

As an application of (4.5), we prove the following important theorem.

Theorem 4.10 Let N be a discrete random variable with set of possible values $\{1, 2, 3, \dots\}$. Then

$$E(N) = \sum_{i=1}^{\infty} P(N \geq i).$$

Proof: For $i \geq 1$, let

$$X_i = \begin{cases} 1 & \text{if } N \geq i \\ 0 & \text{otherwise;} \end{cases}$$

then

$$\sum_{i=1}^{\infty} X_i = \sum_{i=1}^N X_i + \sum_{i=N+1}^{\infty} X_i = \sum_{i=1}^N 1 + \sum_{i=N+1}^{\infty} 0 = N.$$

Since $P(X_i \geq 0) = 1$, for $i \geq 1$,

$$E(N) = E\left(\sum_{i=1}^{\infty} X_i\right) = \sum_{i=1}^{\infty} E(X_i) = \sum_{i=1}^{\infty} P(N \geq i). \quad \blacklozenge$$

EXERCISES

A

1. Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} k\left(\frac{x}{y}\right) & \text{if } x = 1, 2, \quad y = 1, 2 \\ 0 & \text{otherwise.} \end{cases}$$

Determine (a) the value of the constant k , (b) the marginal probability mass functions of X and Y , (c) $P(X > 1 \mid Y = 1)$, (d) $E(X)$ and $E(Y)$.

2. Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} c(x + y) & \text{if } x = 1, 2, 3, \quad y = 1, 2 \\ 0 & \text{otherwise.} \end{cases}$$

Determine (a) the value of the constant c , (b) the marginal probability mass functions of X and Y , (c) $P(X \geq 2 \mid Y = 1)$, (d) $E(X)$ and $E(Y)$.

3. Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} k(x^2 + y^2) & \text{if } (x, y) = (1, 1), (1, 3), (2, 3) \\ 0 & \text{otherwise.} \end{cases}$$

Determine (a) the value of the constant k , (b) the marginal probability mass functions of X and Y , and (c) $E(X)$ and $E(Y)$.

4. Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{25}(x^2 + y^2) & \text{if } x = 1, 2, \quad y = 0, 1, 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find $P(X > Y)$, $P(X + Y \leq 2)$, and $P(X + Y = 2)$.

5. Thieves stole four animals at random from a farm that had seven sheep, eight goats, and five burros. Calculate the joint probability mass function of the number of sheep and goats stolen.
6. Two dice are rolled. The sum of the outcomes is denoted by X and the absolute value of their difference by Y . Calculate the joint probability mass function of X and Y and the marginal probability mass functions of X and Y .
7. In a community 30% of the adults are Republicans, 50% are Democrats, and the rest are independent. For a randomly selected person, let

$$X = \begin{cases} 1 & \text{if he or she is a Republican} \\ 0 & \text{otherwise,} \end{cases}$$

$$Y = \begin{cases} 1 & \text{if he or she is a Democrat} \\ 0 & \text{otherwise.} \end{cases}$$

Calculate the joint probability mass function of X and Y .

8. In an area prone to flood, an insurance company covers losses only up to three separate floods per year. Let X be the total number of floods in a random year and Y be the number of the floods that each cause over \$74 million in insured damage. Suppose that, for some constant c , the joint probability mass function of X and Y is

$$p(x, y) = \begin{cases} c(3x + 2y) & 0 \leq x \leq 3, 0 \leq y \leq x \\ 0 & \text{otherwise.} \end{cases}$$

Find the expected value of the number of floods, in a random year, that cause \$74 million or less in insured damage.

9. From an ordinary deck of 52 cards, 13 cards are selected at random. Calculate the joint probability mass function of the numbers of hearts, diamonds, clubs, and spades selected.
10. Let $p(x, y, z) = (xyz)/162$, $x = 4, 5$, $y = 1, 2, 3$, and $z = 1, 2$, be the joint probability mass function of the random variables X , Y , Z . Find $E(YZ)$.
11. An absentminded professor wrote n letters and sealed them in envelopes without writing the addresses on the envelopes. Having forgotten which letter he had put in which envelope, he wrote the n addresses on the envelopes at random. What is the expected

number of the letters addressed correctly?

Hint: For $i = 1, 2, \dots, n$, let

$$X_i = \begin{cases} 1 & \text{if the } i\text{th letter is addressed correctly} \\ 0 & \text{otherwise.} \end{cases}$$

Calculate $E(X_1 + X_2 + \dots + X_n)$.

- 12.** A cultural society is arranging a party for its members. The cost of a band to play music, the amount that the caterer will charge, the rent of a hall to give the party, and other expenses (in dollars) are uniform random variables over the intervals (1300, 1800), (1800, 2000), (800, 1200), and (400, 700), respectively. If the number of party guests is a random integer from $(150, 200]$, what is the least amount that the society should charge each participant to have no loss, on average?
- 13.** Shiante does not remember in which one of her 11 disk storage wallets she stored the last 4K Ultra HD disc that she purchased. Determine the expected number of the wallets that she should search to find it. Assume that, in her search, she chooses the wallets randomly.
- Hint:* For $1 \leq i \leq 11$, let $X_i = 1$ if the 4K Ultra HD is stored in the i th wallet that Shiante searches, and $X_i = 0$, otherwise. The number of wallets that she should search to find the disc can be written in terms of the X_i 's.

B

- 14.** Solve the following problem posed by Michael Khoury, U.S. Mathematics Olympiad Member, in "The Problem Solving Competition," Oklahoma Publishing Company and the American Society for Communication of Mathematics, February 1999.

Bob is teaching a class with n students. There are n desks in the classroom, numbered from 1 to n . Bob has prepared a seating chart, but the students have already seated themselves randomly. Bob calls off the name of the person who belongs in seat 1. This person vacates the seat he or she is currently occupying and takes his or her rightful seat. If this displaces a person already in the seat, that person stands at the front of the room until he or she is assigned a seat. Bob does this for each seat in turn. After k ($1 \leq k < n$) names have been called, what is the expected number of students standing at the front of the room?

- 15.** A coin is tossed n times ($n > 4$). What is the expected number of exactly three consecutive heads?

Hint: Let E_1 be the event that the first three outcomes are heads and the fourth outcome is tails. For $2 \leq i \leq n-3$, let E_i be the event that the outcome $(i-1)$ is tails, the outcomes i , $(i+1)$, and $(i+2)$ are heads, and the outcome $(i+3)$ is tails. Let E_{n-2} be the event that the outcome $(n-3)$ is tails, and the last three outcomes are heads. Let

$$X_i = \begin{cases} 1 & \text{if } E_i \text{ occurs} \\ 0 & \text{otherwise.} \end{cases}$$

Then calculate the expected value of an appropriate sum of X_i 's.

16. Suppose that 80 balls are placed into 40 boxes at random and independently. What is the expected number of the empty boxes?
17. There are 25 students in a probability class. What is the expected number of birthdays that belong only to one student? Assume that the birthrates are constant throughout the year and that each year has 365 days.
Hint: Let $X_i = 1$ if the birthday of the i th student is not the birthday of any other student, and $X_i = 0$, otherwise. Find $E(X_1 + X_2 + \cdots + X_{25})$.
18. There are 25 students in a probability class. What is the expected number of the days of the year that are birthdays of at least two students? Assume that the birthrates are constant throughout the year and that each year has 365 days.
19. From an ordinary deck of 52 cards, cards are drawn at random, one by one, and *without* replacement until a heart is drawn. What is the expected value of the number of cards drawn?
Hint: See Exercise 13, [Section 3.2](#).
20. From an urn that contains a large number of red and blue chips, mixed in equal proportions, 10 chips are removed one by one and at random. The chips that are removed before the first red chip are returned to the urn. The first red chip, together with all those that follow, is placed in another urn that is initially empty. Calculate the expected number of the chips in the second urn.



Self-Quiz on Section 4.7

Time allotted: 25 Minutes

Each problem is worth 5 points.

1. In a commencement ceremony, for the dean of a college to present the diplomas of the graduates, a clerk piles the diplomas in the order that the students will walk on the stage. However, the clerk mixes the last 10 diplomas in some random order accidentally. Find the expected number of the last 10 graduates walking on the stage who will receive their own diplomas from the dean of the college.
2. From an ordinary deck of 52 cards, seven cards are drawn at random and without replacement. Let X and Y be the number of hearts and the number of spades drawn, respectively.
 - (a) Find the joint probability mass function of X and Y .
 - (b) Calculate $P(X \geq Y)$.



4.8 INDEPENDENCE OF DISCRETE RANDOM VARIABLES

Two random variables X and Y are called **independent** if, for arbitrary subsets A and B of real numbers, the events $\{X \in A\}$ and $\{Y \in B\}$ are independent, that is, if

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B). \quad (4.6)$$

If X and Y are discrete random variables with sets of possible values E and F , respectively, and joint probability mass function $p(x, y)$, then the definition of independence is satisfied if for all $x \in E$ and $y \in F$, the events $\{X = x\}$ and $\{Y = y\}$ are independent; that is,

$$P(X = x, Y = y) = P(X = x)P(Y = y). \quad (4.7)$$

Relation (4.7) shows that X and Y are independent if and only if their joint probability mass function is the product of the marginal probability mass functions of X and Y . Therefore, we have the following theorem.

Theorem 4.11 *Let X and Y be two discrete random variables defined on the same sample space. If $p(x, y)$ is the joint probability mass function of X and Y , then X and Y are independent if and only if for all real numbers x and y ,*

$$p(x, y) = p_X(x)p_Y(y).$$

Let X and Y be discrete *independent* random variables with sets of possible values A and B , respectively. Then (4.7) implies that for all $x \in A$ and $y \in B$,

$$P(X = x \mid Y = y) = P(X = x)$$

and

$$P(Y = y \mid X = x) = P(Y = y).$$

Hence X and Y are *independent if knowing the value of one of them does not change the probability mass function of the other*.

We now prove that functions of independent random variables are also independent.

Theorem 4.12 *Let X and Y be independent random variables. Then for the real-valued functions $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$, $g(X)$ and $h(Y)$ are also independent random variables.*

Proof: To show that $g(X)$ and $h(Y)$ are independent, by (4.6) it suffices to prove that, for any two real numbers a and b ,

$$P(g(X) \leq a, h(Y) \leq b) = P(g(X) \leq a)P(h(Y) \leq b).$$

Let $A = \{x: g(x) \leq a\}$ and $B = \{y: h(y) \leq b\}$. Clearly, $x \in A$ if and only if $g(x) \leq a$, and $y \in B$ if and only if $h(y) \leq b$. Therefore,

$$\begin{aligned} P(g(X) \leq a, h(Y) \leq b) &= P(X \in A, Y \in B) = P(X \in A)P(Y \in B) \\ &= P(g(X) \leq a)P(h(Y) \leq b). \quad \blacklozenge \end{aligned}$$

By this theorem, if X and Y are independent random variables, then sets such as $\{X^2, Y\}$, $\{\sin X, e^Y\}$, $\{X^2 - 2X, Y^3 + 3Y\}$ are sets of independent random variables.

Another important property of independent random variables is that the expected value of their product is equal to the product of their expected values. To prove this, we use Theorem 4.8.

Theorem 4.13 *Let X and Y be independent random variables. Then for all real-valued functions $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$,*

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)],$$

where, as usual, we assume that $E[g(X)]$ and $E[h(Y)]$ are finite.

Solution: Let A be the set of possible values of X , and B be the set of possible values for Y . Let $p(x, y)$ be the joint probability mass function of X and Y . Then

$$\begin{aligned} E[g(X)h(Y)] &= \sum_{x \in A} \sum_{y \in B} g(x)h(y)p(x, y) = \sum_{x \in A} \sum_{y \in B} g(x)h(y)p_X(x)p_Y(y) \\ &= \sum_{x \in A} \left[g(x)p_X(x) \sum_{y \in B} h(y)p_Y(y) \right] = \sum_{x \in A} g(x)p_X(x)E[h(Y)] \\ &= E[h(Y)] \sum_{x \in A} g(x)p_X(x) = E[h(Y)]E[g(X)]. \quad \blacklozenge \end{aligned}$$

By this theorem, if X and Y are independent, then $E(XY) = E(X)E(Y)$, and relations such as the following are valid.

$$\begin{aligned} E(X^2|Y|) &= E(X^2)E(|Y|), \\ E[(\sin X)e^Y] &= E(\sin X)E(e^Y), \\ E[(X^2 - 2X)(Y^3 + 3Y)] &= E(X^2 - 2X)E(Y^3 + 3Y). \end{aligned}$$

However, the converse of Theorem 4.13 is not necessarily true. That is, two random variables X and Y might be dependent while $E(XY) = E(X)E(Y)$. Here is an example:

Example 4.37 Let X be a random variable with the set of possible values $\{-1, 0, 1\}$ and probability mass function $p(-1) = p(0) = p(1) = 1/3$. Letting $Y = X^2$, we have

$$\begin{aligned} E(X) &= -1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = 0, \\ E(Y) &= E(X^2) = (-1)^2 \cdot \frac{1}{3} + 0^2 \cdot \frac{1}{3} + (1)^2 \cdot \frac{1}{3} = \frac{2}{3}, \\ E(XY) &= E(X^3) = (-1)^3 \cdot \frac{1}{3} + 0^3 \cdot \frac{1}{3} + (1)^3 \cdot \frac{1}{3} = 0. \end{aligned}$$

Thus $E(XY) = E(X)E(Y)$ while, clearly, X and Y are dependent. ♦

We now introduce the concept of independence of random variables, extending it from two variables to any number of random variables.

Suppose that $X_1, X_2, \dots, X_n$ are random variables (not necessarily discrete) on a sample space. We say that they are **independent** if, for arbitrary subsets $A_1, A_2, \dots, A_n$ of real numbers,

$$P(X_1 \in A_1, X_2 \in A_2, \dots, X_n \in A_n) = P(X_1 \in A_1)P(X_2 \in A_2) \cdots P(X_n \in A_n).$$

If $X_1, X_2, \dots, X_n$ are discrete, the definition of independence reduces to the following condition:

$$P(X_1 = x_1, \dots, X_n = x_n) = P(X_1 = x_1) \cdots P(X_n = x_n) \quad (4.8)$$

for any set of points, $x_i, i = 1, 2, \dots, n$.

The following theorem is the generalization of Theorem 4.11. It follows from (4.8).

Theorem 4.14 *Let $X_1, X_2, \dots, X_n$ be jointly discrete random variables with the joint probability mass function $p(x_1, x_2, \dots, x_n)$. Then $X_1, X_2, \dots, X_n$ are independent if and only if $p(x_1, x_2, \dots, x_n)$ is the product of their marginal probability mass functions $p_{X_1}(x_1), p_{X_2}(x_2), \dots, p_{X_n}(x_n)$.*

Remark 4.5 Let X, Y , and Z be independent random variables. Then, by definition, for arbitrary subsets A_1, A_2 , and A_3 of $\mathbb{R}$,

$$P(X \in A_1, Y \in A_2, Z \in A_3) = P(X \in A_1)P(Y \in A_2)P(Z \in A_3). \quad (4.9)$$

Now, if in (4.9) we let $A_2 = \mathbb{R}$, then since the event $Y \in \mathbb{R}$ is certain and has probability 1, we get

$$P(X \in A_1, Z \in A_3) = P(X \in A_1)P(Z \in A_3).$$

This shows that X and Z are independent random variables. In the same way it can be shown that $\{X, Y\}$, and $\{Y, Z\}$ are also independent sets. Similarly, if $\{X_1, X_2, \dots, X_n\}$ is a sequence of independent random variables, its subsets are also independent sets of random variables. This observation motivates the following definition for the concept of independence of any collection of random variables. ♦

Definition 4.10 *A collection of random variables is called **independent** if all of its finite subcollections are independent.*

Sometimes independence of a collection of random variables is self-evident and requires no checking. For example, let E be an experiment with the sample space S . Let X be a random variable defined on S . If the experiment E is repeated n times independently and on the i th experiment X is called X_i , the sequence $\{X_1, X_2, \dots, X_n\}$ is an independent sequence of random variables.

It is also important to know that the result of Theorem 4.12 is true for any number of random variables:

If $\{X_1, X_2, \dots\}$ is a sequence of independent random variables and for $i = 1, 2, \dots$, $g_i: \mathbb{R} \rightarrow \mathbb{R}$ is a real-valued function, then the sequence $\{g_1(X_1), g_2(X_2), \dots\}$ is also an independent sequence of random variables.

Using Theorems 4.8 and 4.14, yields the following important theorem.

Theorem 4.15 *Let $X_1, X_2, \dots, X_n$ be independent discrete random variables defined on the same sample space, with sets of possible values $A_1, A_2, \dots, A_n$, respectively. Let the joint probability mass function of these random variables be given by $p(x_1, x_2, \dots, x_n)$. Then*

$$E(X_1 X_2 \cdots X_n) = E(X_1)E(X_2) \cdots E(X_n).$$

Proof: By Theorems 4.8 and 4.14,

$$\begin{aligned} E(X_1 X_2 \cdots X_n) &= \sum_{x_1 \in A_1} \sum_{x_2 \in A_2} \cdots \sum_{x_n \in A_n} x_1 x_2 \cdots x_n p(x_1, x_2, \dots, x_n) \\ &= \sum_{x_1 \in A_1} \sum_{x_2 \in A_2} \cdots \sum_{x_n \in A_n} x_1 x_2 \cdots x_n p_{X_1}(x_1) p_{X_2}(x_2) \cdots p_{X_n}(x_n) \\ &= \sum_{x_1 \in A_1} x_1 p_{X_1}(x_1) \sum_{x_2 \in A_2} x_2 p_{X_2}(x_2) \cdots \sum_{x_n \in A_n} x_n p_{X_n}(x_n) \\ &= E(X_1)E(X_2) \cdots E(X_n). \end{aligned}$$

Remark 4.6 In probability, **Reliability Theory** refers to the likelihood that a system or component will perform its intended function without failure for a specified period or under specific conditions. It is a fundamental concept in the field of reliability engineering, which aims to study and enhance the dependability and performance of systems over time. One key aspect of reliability is to determine the conditions under which a system functions and the probability of its successful functioning. For a discussion on the Reliability of Systems, please refer to Section 4C in the Supplementary Resources for [Chapter 4](#) on the Instructor and Student Resources website associated with this book.

EXERCISES

1. Let the joint probability mass function of random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{25}(x^2 + y^2) & \text{if } x = 1, 2, \quad y = 0, 1, 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Are X and Y independent? Why or why not?

2. Let the joint probability mass function of random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{7}x^2y & \text{if } (x, y) = (1, 1), (1, 2), (2, 1) \\ 0 & \text{elsewhere.} \end{cases}$$

Are X and Y independent? Why or why not?

3. Let X and Y be independent random variables each having the probability mass function

$$p(x) = \frac{1}{2} \left(\frac{2}{3} \right)^x, \quad x = 1, 2, 3, \dots$$

Find $P(X = 1, Y = 3)$ and $P(X + Y = 3)$.

4. From an ordinary deck of 52 cards, eight cards are drawn at random and without replacement. Let X and Y be the number of clubs and spades, respectively. Are X and Y independent?
5. What is the probability that there are exactly two girls among the first seven and exactly four girls among the first 15 babies born in a hospital in a given week? Assume that the events that a child born is a girl or is a boy are equiprobable.
6. Suppose that X and Y are discrete random variables with joint probability mass function $p(x, y)$. Show that if there are two functions g and h for which $p(x, y) = g(x)h(y)$, then X and Y are independent random variables.
7. The joint probability mass function $p(x, y)$ of the random variables X and Y is given by the following table. Determine if X and Y are independent.

x	y			
	0	1	2	3
0	0.1681	0.1804	0.0574	0.0041
1	0.1804	0.1936	0.0616	0.0044
2	0.0574	0.0616	0.0196	0.0014
3	0.0041	0.0044	0.0014	0.0001

8. From the set of families with two children a *family* is selected at random. Let $X_1 = 1$ if the first child of the family is a girl; $X_2 = 1$ if the second child of the family is a girl; and $X_3 = 1$ if the family has exactly one boy. For $i = 1, 2, 3$, let $X_i = 0$ in other cases. Determine if X_1 , X_2 , and X_3 are independent. Assume that in a family the probability that a child is a girl is independent of the gender of the other children and is $1/2$.
9. Let X , Y , and Z be three independent random variables such that $E(X) = E(Y) = E(Z) = 0$, and $\text{Var}(X) = \text{Var}(Y) = \text{Var}(Z) = 1$. Calculate $E[X^2(Y + 5Z)^2]$.
10. Let $\{X_1, X_2, \dots, X_n\}$ be a set of independent random variables with $P(X_j = i) = p_i$ ($1 \leq j \leq n$ and $i \geq 1$). Let $h_k = \sum_{i=k}^{\infty} p_i$. Using Theorem 4.10,

prove that

$$E[\min(X_1, X_2, \dots, X_n)] = \sum_{k=1}^{\infty} h_k^n.$$

11. Let X and Y be two independent random integers from the set $\{0, 1, 2, \dots, n\}$. Find $P(X = Y)$ and $P(X \leq Y)$.
12. Let X and Y be jointly distributed discrete random variables with the marginal probability density functions given by

$$p_X(x) = \begin{cases} 1/3 & \text{if } x = 0 \\ 2/3 & \text{if } x = 1, \end{cases} \quad p_Y(y) = \begin{cases} 1/4 & \text{if } y = 0 \\ 3/4 & \text{if } y = 1 \end{cases}$$

Let $p(x, y)$ be the joint probability mass function of X and Y . Show that if $p(1, 1) = p_X(1)p_Y(1)$, then X and Y are independent. ■

Self-Quiz on Section 4.8

Time allotted: 25 Minutes

Each problem is worth 5 points.

1. Let the joint probability density function of the discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} 2/8 & x = 0, y = 0 \\ 2/8 & x = 0, y = 2 \\ 1/8 & x = 0, y = 3 \\ 1/8 & x = 1, y = 0 \\ 2/8 & x = 1, y = 1 \end{cases}$$

Are X and Y independent random variables? Why or why not?

2. Suppose that the number of claims received by an insurance company in a given week is independent of the number of claims received in any other week. An actuary has calculated that the probability mass function of the number of claims received in a random week is

$$p(n) = \left(\frac{1}{5}\right)\left(\frac{4}{5}\right)^n, \quad n \geq 0.$$

Find the probability that the company will receive exactly 8 claims during the next two weeks. ■

4.9 SIMULATING DISCRETE RANDOM VARIABLES IN R

The method illustrated in the subsequent example, used to generate 500 independent **realizations** from a discrete random variable with the same distribution as X , is generally applicable to all discrete random variables. That is, we are drawing 500 *samples* from one random variable with a distribution identical to X . Therefore, a **realization** (or observation or sample) from a random variable is an actual observed value that the random variable takes on during a particular instance or experiment.

Example 4.38 A fair coin is tossed three times. Let X represent the number of heads obtained. The sample space for this experiment is denoted as

$$S = \{\text{HHT, HTH, HTT, HHH, THT, TTH, THH, TTT}\}.$$

The probability mass function of X is given by

$$\begin{aligned} p(0) &= P(X = 0) = 1/8, & p(1) &= P(X = 1) = 3/8, \\ p(2) &= P(X = 2) = 3/8, & p(3) &= P(X = 3) = 1/8. \end{aligned}$$

We now define the vector $\mathbf{v}$ to represent the possible values of X , and the vector $\mathbf{p}$ to represent the probabilities associated with those values:

$$\begin{aligned} \mathbf{v} &= \mathbf{c}(0, 1, 2, 3), \\ \mathbf{p} &= \mathbf{c}(1/8, 3/8, 3/8, 1/8). \end{aligned}$$

The command below produces 500 independent realizations from the desired random variable X .

```
sample(v, 500, prob=p, replace=TRUE) .    ♦
```

Example 4.39 A fair die is rolled 8 times. Let X represent the sum of the outcomes. Determine an approximate value for $P(35 \leq X \leq 37)$ using R simulation.

Solution: The R code below provides the desired approximation:

```
v <- replicate(100000, sum(sample(6, 8, replace=TRUE)))
m <- sum(v==35)+sum(v==36)+sum(v==37)
p <- m/100000
print(p)
[1] 0.06582
```

In this R code, the command `sample(6, 8, replace=TRUE)` samples 8 times (with replacement) from the numbers 1 through 6. This is analogous to rolling a fair 6-sided die 8 times. After sampling (rolling the die) 8 times, the command

```
sum(sample(6, 8, replace=TRUE))
```


adds up the results. The sum can be anywhere between 8 (if you roll a 1 all eight times) and 48 (if we roll a 6 all eight times). The command

```
replicate(100000, sum(sample(6, 8, replace=TRUE)))
```

repeats the process just described a total of 100,000 times. The output **v** will be a vector of length 100,000 where each entry represents the sum of 8 rolls of a fair 6-sided die. This command essentially simulates the distribution of the sum of 8 dice rolls over 100,000 trials.

The command **v == 35** creates a logical vector of the same length as **v** where each element is either **TRUE** (if the corresponding element in **v** is **35**) or **FALSE** (if it is not). **sum(v == 35)** sums up the logical vector of length 100,000 with **TRUE** and **FALSE** entries. In R, **TRUE** is equivalent to 1 and **FALSE** is equivalent to 0. So, summing up the logical vector gives the count of **TRUE** values, which in this context, is the number of times the value 35 appears in the vector **v**. The terms, **sum(v==36)** and **sum(v==37)**, are calculated similarly. In summary, **m** will give us the count of how many times the numbers 35, 36, and 37 are present in the vector **v**. The approximation we aim to compute is given by **p=m/100000**. ♦

Simulating Expected Values in R

Consider a random variable X defined over the sample space S . To approximate $E(X)$ through simulation, we conduct the experiment (over which X is defined) independently for a large number n of times. In each iteration, we determine the value of X . After all iterations, we sum up the n obtained values and divide the result by n .

Example 4.40 Suppose we have 60 cards numbered from 1 to 60. We shuffle them until they are in a random order. In the shuffled arrangement, if the card in the i th position is card number i , we say that a match has occurred. Let X be the number of matches. Use R simulation to determine $E(X)$.

Solution: Let us shuffle the cards thoroughly 40,000 times, counting the number of matches in each shuffle. Let **v** be a vector of length 40,000, where **v[i]** represents the number of matches in the i th shuffle. The R code provided below computes **ev**, the expected value of X .

```
n <- 60
v <- replicate(40000, sum(sample(n) == (1:n)))
ev <- mean(v)
> print(ev)
[1] 1.0064
```

The command **mean(v)** calculates **sum(v)/40000**. We executed the program using 100,000 simulations and obtained a result of 0.99905. The exact answer is 1. ♦

Before delving into the next example, let us craft an R command that, for a given vector **v**, removes all entries with duplicates, retaining only those values that appear precisely once. Such a command is

```
v[!duplicated(v) & !duplicated(v, fromLast=TRUE) ]
```

To explain this command, note that **duplicated(v)** returns a logical vector identifying duplicates based on their first occurrence. **duplicated(v, fromLast=TRUE)** returns a logical vector identifying duplicates based on their last occurrence. Using the **!** operator negates the logical vectors. Combining both with the **&** operator ensures that only values that are not identified as duplicates from either the beginning or the end of the vector are retained. Here is a step-by-step breakdown of how this command functions:

```
> v <- c(1,2,3,1,4,3,5)
> duplicated(v)
[1] FALSE FALSE FALSE TRUE FALSE TRUE FALSE
> duplicated(v, fromLast=TRUE)
[1] TRUE FALSE TRUE FALSE FALSE FALSE FALSE
> !duplicated(v)
[1] TRUE TRUE TRUE FALSE TRUE FALSE TRUE
> !duplicated(v, fromLast=TRUE)
[1] FALSE TRUE FALSE TRUE TRUE TRUE TRUE
> v[!duplicated(v) & !duplicated(v, fromLast=TRUE) ]
[1] 2 4 5
```

It is also worth noting that, in R, the **unique()** function returns a vector with duplicate values removed, preserving the order of their first occurrence. Here is how this code functions:

```
unique(c(1,2,3,1,4,3,5))
[1] 1 2 3 4 5
```

Example 4.41 In a class of 87 random students, what is the expected value of X , representing the number of students whose birthdays are unique and not shared with any of their classmates?

Solution: To simulate the birthday problem with 87 students and determine the expected value of X —the number of students with unique, non-shared birthdays—we will adopt the following approach:

- Generate random birthdays for the 87 students.
- Use the command **v[!duplicated(v) & !duplicated(v, fromLast=TRUE)]** to isolate only the unique birthdays.
- Repeat the two steps above for a substantial number of iterations, say 30,000, to thoroughly simulate the scenario and ascertain the expected value.

For the simulation, let **num** represent the number of iterations and **n** denote the student count. In each iteration, **bd** will be our vector containing the 87 randomly generated birthdays, and **ubd** will represent the vector of unique birthdays.

Initializing our storage with the **numeric(num)** command, we create a numeric vector named **results** of length **num** (which we have set to 30,000 in this case). Initially, this vector's elements are all zeroes. This initialization serves to predefine a space where the non-shared birthday count from each simulation can be stored. As we run through the loop, each element in the **results** vector gets updated with the count of unique birthdays from that respective iteration.

```
> num <- 30000
> n <- 87
> results <- numeric(num)
> for(i in 1:num) {
+ bd <- sample(1:365,n,replace=TRUE)
+ ubd <- bd[!duplicated(bd) & !duplicated(bd,fromLast=TRUE)]
+ results[i] <- length(ubd)
+ }
> ev <- mean(results)
> print(ev)
[1] 68.68277 ◆
```

Example 4.42 A fair coin is tossed repeatedly until four consecutive heads appear for the first time. Create an R program to simulate and estimate the expected number of trials required to achieve this outcome.

Let H represent the outcome of heads. We provide an R code that simulates the coin-flipping experiment until four consecutive heads, denoted as HHHH, are obtained. This is repeated **num** times. Within the code:

- **j** is the counter for the number of experiments, starting at 1. After each completed experiment, **j** increments by 1. When **j=num**, the final experiment is executed.
- Random numbers from the interval $(0, 1)$ are generated to simulate coin flips. An outcome within $(0, 0.5)$ corresponds to heads; otherwise, it is tails.
- **i** tracks the consecutive occurrences of heads. It starts at 0. With each heads result, **i** increments by 1; but upon a tails outcome, **i** resets to 0. An experiment concludes when **i** reaches 4 for the first time.
- **k** counts the number of trials to achieve the first HHHH sequence.
- **m** aggregates all the **k** values across the **num** experiments.
- **a** calculates the average number of trials to reach the first HHHH, with the formula $a=m/\text{num}$.

An R simulation for this is presented next.

```
> num <- 30000
> m <- 0
> j <- 0
> while(j <= num){
+ k <- 0
+ i <- 0
+ while(i<4){
+ k <- k+1
+ r <- runif(1)
+ if(r < 0.5){
+ i <- i+1}
+ else{
+ i <- 0}}
+ m <- m+k
+ j <- j+1}
> a <- m/num
> print(a)
[1] 30.12043
```

It can be shown that the exact solution to this problem is 30 (see Section 4B “Pattern Appearance” in the Supplementary Resources for [Chapter 4](#) on the Instructor and Student Resources website associated with this book). Thus, this simulation result is quite accurate. ♦

CHAPTER 4 SUMMARY

► **Random Variable** Let S be the sample space of an experiment. A real-valued function $X: S \rightarrow \mathbb{R}$ is called a *random variable* of the experiment if, for each interval $I \subseteq \mathbb{R}$, $\{s: X(s) \in I\}$ is an event.

► **Distribution Function** If X is a random variable, then the function F defined on $(-\infty, +\infty)$ by $F(t) = P(X \leq t)$ is called the *distribution function* of X . The most important properties of a distribution function F are as follows: (i) F is nondecreasing; (ii) $\lim_{t \rightarrow \infty} F(t) = 1$; (iii) $\lim_{t \rightarrow -\infty} F(t) = 0$; (iv) F is right continuous. That is, for every $t \in \mathbb{R}$, $F(t+) = F(t)$. For a point $a \in \mathbb{R}$, $P(X = a) = F(a) - F(a-)$. Thus $P(X = a)$ is the difference between the right- and left-hand limits of F at a . If the function F is continuous at a , these limits are the same and equal to $F(a)$. Hence $P(X = a) = 0$. Otherwise, F has a *jump* at a , and the magnitude of the jump, $F(a) - F(a-)$, is the probability that $X = a$.

► **Discrete Random Variable** Whenever the set of possible values that a random variable X can assume is at most countable, X is called *discrete*. Therefore, X is discrete if either the set of its possible values is finite or it is countably infinite. To each discrete random variable, a real-valued function $p: \mathbb{R} \rightarrow \mathbb{R}$, defined by $p(x) = P(X = x)$, is assigned and is called the *probability mass function* of X . Since the set of values of X is countable, $p(x)$ is positive at most for a countable set. It is zero elsewhere; that is, if possible values of X are $x_1, x_2, x_3, \dots$, then $p(x_i) \geq 0$ ($i = 1, 2, 3, \dots$) and $p(x) = 0$ if $x \notin \{x_1, x_2, x_3, \dots\}$. Now, clearly, the occurrence of the event $\{x_1, x_2, x_3, \dots\}$ is certain. Therefore, $\sum_{i=1}^{\infty} P(X = x_i) = 1$ or, equivalently, $\sum_{i=1}^{\infty} p(x_i) = 1$.

► The distribution function F of a discrete random variable X , with the set of possible values $\{x_1, x_2, x_3, \dots\}$, is a step function; that is, F is constant in the intervals $[x_{n-1}, x_n)$ with jumps at $x_1, x_2, x_3, \dots$. The magnitude of the jump at x_i is $p(x_i)$.

► **The Expected Value of a Discrete Random Variable** The expected value of a discrete random variable X with the set of possible values A and probability mass function $p(x)$ is defined by

$$E(X) = \sum_{x \in A} xp(x).$$

We say that $E(X)$ exists if this sum converges absolutely. Note that if each value x of X is weighted by $p(x) = P(X = x)$, then $\sum_{x \in A} xp(x)$ is the weighted average of X .

If g is a real-valued function, then $g(X)$ is a random variable with

$$E[g(X)] = \sum_{x \in A} g(x)p(x).$$

► If X is a constant random variable; that is, if $P(X = c) = 1$ for a constant c , then $E(X) = c$.

► Let X be a discrete random variable; $g_1, g_2, \dots, g_n$ be real-valued functions, and let $\alpha_1, \alpha_2, \dots, \alpha_n$ be real numbers. Then

$$\begin{aligned} E[\alpha_1 g_1(X) + \alpha_2 g_2(X) + \dots + \alpha_n g_n(X)] \\ = \alpha_1 E[g_1(X)] + \alpha_2 E[g_2(X)] + \dots + \alpha_n E[g_n(X)]. \end{aligned}$$

This implies that $E(X)$ is linear. That is, if $\alpha, \beta \in \mathbb{R}$, then

$$E(\alpha X + \beta) = \alpha E(X) + \beta.$$

► **The Standard Deviation and Variance of a Discrete Random Variable** Let X be a discrete random variable with a set of possible values A , probability mass function $p(x)$,

and $E(X) = \mu$. Then σ_X and $\text{Var}(X)$, called the *standard deviation* and the *variance* of X , respectively, are defined by

$$\sigma_X = \sqrt{E[(X - \mu)^2]} \quad \text{and} \quad \text{Var}(X) = E[(X - \mu)^2].$$

Variance measures the dispersion, or spread, of a distribution about its expectation. The following formula is a better alternative for computing the variance of X .

$$\text{Var}(X) = E(X^2) - [E(X)]^2.$$

This relation implies that $[E(X)]^2 \leq E(X^2)$.

- $\text{Var}(X) = 0$ if and only if X is a constant with probability 1.
- For constants a and b ,

$$\begin{aligned} \text{Var}(aX + b) &= a^2 \text{Var}(X), \\ \sigma_{aX+b} &= |a| \sigma_X. \end{aligned}$$

► **Standardized Random Variables** Let X be a random variable with mean μ and standard deviation σ . The random variable $X^* = (X - \mu)/\sigma$ is called the *standardized X* . We have that $E(X^*) = 0$ and $\text{Var}(X^*) = 1$. When standardizing a random variable X , we change the origin to μ and the scale to the units of standard deviation. The value that is obtained for X^* is independent of the units in which X is measured. It is the number of standard deviation units by which X differs from $E(X)$. Standardization is particularly useful if two or more random variables with different distributions must be compared.

► **Joint Probability Mass Functions** Let X and Y be two discrete random variables defined on the same sample space. Let the sets of possible values of X and Y be A and B , respectively. The function

$$p(x, y) = P(X = x, Y = y)$$

is called the *joint probability mass function of X and Y* . The functions $p_X(x) = \sum_{y \in B} p(x, y)$ and $p_Y(y) = \sum_{x \in A} p(x, y)$ are called, respectively, the *marginal probability mass functions* of X and Y .

► If h is an ordinary function of two variables from $\mathbb{R}^2$ to $\mathbb{R}$, then $h(X, Y)$ is a discrete random variable with the expected value given by

$$E[h(X, Y)] = \sum_{x \in A} \sum_{y \in B} h(x, y) p(x, y),$$

provided that the sum is absolutely convergent. By this relation, $E(X + Y) = E(X) + E(Y)$.

► **Independence of Discrete Random Variables** Two discrete random variables X and Y are called *independent* if, for arbitrary subsets A and B of real numbers, the events $\{X \in A\}$

and $\{Y \in B\}$ are independent, that is, if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$. Two discrete random variables are independent if and only if their joint probability mass function is the product of their marginal probability mass functions.

Let X and Y be independent random variables and $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$ be real-valued functions, then $g(X)$ and $h(Y)$ are also independent random variables. Furthermore,

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)],$$

In particular, $E(XY) = E(X)E(Y)$. However, two random variables X and Y might be dependent while $E(XY) = E(X)E(Y)$.

► **Independence of $n > 2$ Discrete Random Variables** Suppose that $X_1, X_2, \dots, X_n$ are n random variables (not necessarily discrete) on a sample space. We say that they are *independent* if, for arbitrary subsets $A_1, A_2, \dots, A_n$ of real numbers,

$$P(X_1 \in A_1, X_2 \in A_2, \dots, X_n \in A_n) = P(X_1 \in A_1)P(X_2 \in A_2) \cdots P(X_n \in A_n).$$

► If $\{X_1, X_2, \dots\}$ is a sequence of independent random variables and for $i = 1, 2, \dots$, $g_i: \mathbb{R} \rightarrow \mathbb{R}$ is a real-valued function, then the sequence $\{g_1(X_1), g_2(X_2), \dots\}$ is also an independent sequence of random variables.

► **Joint Probability Mass Functions** Let $X_1, X_2, \dots, X_n$ be discrete random variables defined on the same sample space, with sets of possible values $A_1, A_2, \dots, A_n$, respectively. The function

$$p(x_1, x_2, \dots, x_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

is called the *joint probability mass function* of $X_1, X_2, \dots, X_n$. Clearly,

- (a) $p(x_1, x_2, \dots, x_n) \geq 0$.
- (b) If for some i , $1 \leq i \leq n$, $x_i \notin A_i$, then $p(x_1, x_2, \dots, x_n) = 0$.
- (c) $\sum_{x_i \in A_i, 1 \leq i \leq n} p(x_1, x_2, \dots, x_n) = 1$.

Moreover, for $1 \leq i \leq n$, the *marginal probability mass function* of X_i , p_{X_i} , is $p_{X_i}(x_i) = \sum_{x_j \in A_j, j \neq i} p(x_1, x_2, \dots, x_n)$.

► Let $X_1, X_2, \dots, X_n$ be n discrete random variables. For $1 \leq i \leq n$, let A_i be the set of possible values of X_i . If h is a function of n variables from $\mathbb{R}^n$ to $\mathbb{R}$, then $Y = h(X_1, X_2, \dots, X_n)$ is a discrete random variable with expected value given by

$$E(Y) = \sum_{x_n \in A_n} \cdots \sum_{x_1 \in A_1} h(x_1, x_2, \dots, x_n) p(x_1, x_2, \dots, x_n),$$

provided that the sum is finite. This fact can be used to prove that the expected value of the product of several independent discrete random variables is equal to the product of their expected values.

► **Expected Values of Sums of Random Variables** For discrete random variables $X_1, X_2, \dots, X_n$ defined on the same sample space, and for real numbers $\alpha_1, \alpha_2, \dots, \alpha_n$,

$$E\left(\sum_{i=1}^n \alpha_i X_i\right) = \sum_{i=1}^n \alpha_i E(X_i).$$

In particular,

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n).$$

In general, If $\sum_{i=1}^{\infty} E(|X_i|) < \infty$ or if, for all i , the random variables $X_1, X_2, \dots$ are nonnegative [that is, $P(X_i \geq 0) = 1$ for $i \geq 1$], then $E\left(\sum_{i=1}^{\infty} X_i\right) = \sum_{i=1}^{\infty} E(X_i)$.

► Let N be a discrete random variable with set of possible values $\{1, 2, 3, \dots\}$. Then

$$E(N) = \sum_{i=1}^{\infty} P(N \geq i).$$

REVIEW PROBLEMS

1. An urn contains 10 chips numbered from 0 to 9. Two chips are drawn at random and without replacement. What is the probability mass function of their total?
2. A word is selected at random from the following poem of Persian poet and mathematician Omar Khayyām (1048–1131), translated by English poet Edward Fitzgerald (1808–1883). Find the expected value of the length of the word.

The moving finger writes and, having writ,
Moves on; nor all your Piety nor Wit
Shall lure it back to cancel half a line,
Nor all your tears wash out a word of it.

3. A statistical survey shows that only 2% of secretaries know how to use the highly sophisticated word processor language T_EX. If a certain mathematics department prefers to hire a secretary who knows T_EX, what is the least number of applicants that should be interviewed so as to have at least a 50% chance of finding one such secretary?
4. An electronic system fails if both of its components fail. Let X be the time (in hours) until the system fails. Experience has shown that

$$P(X > t) = \left(1 + \frac{t}{200}\right)e^{-t/200}, \quad t \geq 0.$$

What is the probability that the system lasts at least 200 but not more than 300 hours?

5. A professor has prepared 30 exams of which 8 are difficult, 12 are reasonable, and 10 are easy. The exams are mixed up, and the professor selects four of them at random to give to four sections of the course he is teaching. How many sections would be expected to get a difficult test?
6. The annual amount of rainfall (in centimeters) in a certain area is a random variable with the distribution function

$$F(x) = \begin{cases} 0 & x < 5 \\ 1 - (5/x^2) & x \geq 5. \end{cases}$$

What is the probability that next year it will rain (a) at least 6 centimeters; (b) at most 9 centimeters; (c) at least 2 and at most 7 centimeters?

7. Let X be the amount (in fluid ounces) of soft drink in a randomly chosen bottle from company A , and Y be the amount of soft drink in a randomly chosen bottle from company B . A study has shown that the probability distributions of X and Y are as follows:

x	15.85	15.9	16	16.1	16.2
$P(X = x)$	0.15	0.21	0.35	0.15	0.14
$P(Y = x)$	0.14	0.05	0.64	0.08	0.09

Find $E(X)$, $E(Y)$, $\text{Var}(X)$, and $\text{Var}(Y)$ and interpret them.

8. The fasting blood-glucose levels of 30 children are as follows.

58 62 80 58 64 76 80 80 80 58
 62 64 76 76 58 64 62 80 58 58
 80 64 58 62 76 62 64 80 62 76

Let X be the fasting blood-glucose level of a child chosen randomly from this group. Find the distribution function of X .

9. Experience shows that X , the number of customers entering a post office, during any period of length t , is a random variable the probability mass function of which is of the form

$$p(i) = k \frac{(2t)^i}{i!}, \quad i = 0, 1, 2, \dots$$

- (a) Determine the value of k .
- (b) Compute $P(X < 4)$ and $P(X > 1)$.

10. From the set of families with three children a *family* is selected at random, and the number of its boys is denoted by the random variable X . Find the probability mass function and the distribution functions of X . Assume that in a three-child family all gender distributions are equally probable.
11. If the motor of a certain commercial grade dishwasher fails within the first year of purchase, the insurance company pays \$2500 for its replacement or repair. Thereafter, each year, the company's payment for motor failure will be \$500 less than the previous year until the sixth year. If the motor fails during or after the sixth year, the company offers no payments. Suppose that if such a motor has not failed by the first day of a year, the probability that it fails during that year is 0.06. Find the expected value of the insurance company's payment for each such dishwasher motor.
12. A fair die is tossed twice. The sum of the outcomes is denoted by X and the largest value by Y . (a) Calculate the joint probability mass function of X and Y ; (b) find the marginal probability mass functions of X and Y ; (c) find $E(X)$ and $E(Y)$.
13. Calculate the probability mass function of the number of spades in a random bridge hand that includes exactly four hearts.
14. Suppose that three cards are drawn at random from an ordinary deck of 52 cards. If X and Y are the numbers of diamonds and clubs, respectively, calculate the joint probability mass function of X and Y .
15. Calculate the probability mass function of the number of spades in a random bridge hand that includes exactly four hearts and three clubs.
16. An urn contains 100 chips of which 20 are blue, 30 are red, and 50 are green. Suppose that 20 chips are drawn at random and without replacement. Let B , R , and G be the number of blue, red, and green chips, respectively. Calculate the joint probability mass function of B , R , and G .
17. Let X be the smallest number obtained in rolling a balanced die n times. Calculate the probability distribution function and the probability mass function of X .
18. From a leap year calendar, a month is selected at random. Let X be the number of days in that month. Find $E(X)$ and σ_X . If the month is selected from a non-leap year, on average, how many days does that month have?
19. Let x be a nonnegative real number. By $[x]$, we mean the greatest integer less than or equal to x . For example, $[1.2] = 1$, $[5.8] = 5$, and $[0.8] = 0$. In a box, there are n identical balls numbered from 1 to n . Let X be the number on a ball that is drawn at random from the box. Find F , the distribution function of X by expressing $F(t) = P(X \leq t)$ in terms of $[t]$ for $1 \leq t < n$.
20. From an ordinary deck of 52 cards, cards are drawn successively and with replacement. Let X and Y denote the number of spades in the first 10 cards and in the second 15 cards, respectively. Calculate the joint probability mass function of X and Y .
21. There are prizes in 10% of the boxes of a certain type of cereal. Let X be the number of boxes of such cereal that Kim should buy to find a prize. Let Y be the number of additional boxes of such cereal that she should purchase to find another prize. Calculate

the joint probability mass function of X and Y .

22. In the inventory of a pharmacy, in a carton, there are 40 boxes of painkillers of which 15 are brand A, 10 are brand B, 6 are brand C, 4 are brand D, and 5 are brand E. An assistant pharmacist chooses 20 of these boxes randomly to move them inside the store for the over-the-counter sale. Let X , Y , Z , U , and V be the respective numbers of the boxes of the aforementioned brands selected. Find the joint probability mass function of X , Y , Z , U , and V .



Self-Test on Chapter 4

Time allotted: 120 Minutes

Each problem is worth 10 points.

1. A large company has w women and m men in retirement age. If a random set of n , $n \leq w + m$, of these employees decides to retire next year, what is the probability mass function of the number of women who will retire next year?
2. In the front yard of a doctor's clinic, there are exactly six parking spaces next to each other in a row. Suppose that at a time when all six parking spaces are occupied, two of the cars leave at random and their spots remain empty. Let X be the number of cars between the empty spots. Find p , the probability mass function, and F , the distribution function, of X .
3. Let X be the number of bagels purchased by a random customer from the Longmeadow Bagel Shop. Let F be the distribution function of X , and suppose that $F(0) = 4/33$, $F(1) = 16/33$, $F(2) = 26/33$, and $F(3) = 30/33$. If $P(X = 4) = P(X > 4)$, for $i = 0, 1, 2, 3, 4$, find $P(X = i)$.
4. The season finale of a Belavian reality television series, "Belavia's Got Talent," was held in an oval auditorium which has 25 rows of seats. The first row has 10 seats, and each succeeding row has 2 more seats than the previous row. All the seats in the auditorium were occupied, and before doing some magic tricks, a magician, who was a finalist, picked someone from the audience at random to help him with the performance. Let X be the row number of the seat of the person picked from the audience by the magician. Find the probability mass function of X .
5. At a department store, summer polo shirts are sold from April 1st until the end of September. Suppose that for each polo shirt the department store sells the net profit is \$15.00, and for each shirt unsold by the end of September the net loss is \$10.00. Also suppose that, at the department store, for each i , $100 \leq i \leq 150$, each year, the demand for polo shirts is i with probability $1/51$. Determine the expected net profit of the department store from selling polo shirts if it stocks 130 shirts each year.

6. Under what conditions on α , β , γ , and θ is the following a distribution function?

$$F(t) = \begin{cases} 1 - \alpha & t < 0 \\ \beta & 0 \leq t < 1 \\ \beta - \gamma & 1 \leq t < 2 \\ \theta & t \geq 2. \end{cases}$$

7. The joint probability mass function of X and Y is given by the following table.

	x		
	1	2	3
y			
2	0.05	0.25	0.15
4	0.14	0.10	0.17
6	0.10	0.02	0.02

- (a) Find $P(XY \leq 6)$.
 (b) Find $E(X)$ and $E(Y)$.

8. A fair coin is tossed n times by Adam and n times by Andrew. What is the probability that they get the same number of heads?
9. A jury of 12 people is randomly selected from a group of eight Afro-American, seven Hispanic, three Native American, and 20 white potential jurors. Let A , H , N , and W be the number of Afro-American, Hispanic, Native American, and white jurors selected, respectively. Calculate the joint probability mass function of A , H , N , W and the marginal probability mass function of A .
10. Hooroo Jackson, a professor of cinema studies, has 12 Blu-ray movies inside his briefcase. The subject of his current lesson is method acting, and he plans to show his students clips from each of the three Marlon Brando movies that he has in his briefcase. If Professor Jackson withdraws movies one by one, at random, and without replacement from his briefcase, what is the expected number of the movies that he has to withdraw until all three Brando movies are pulled out?

Hint: For $i = 1, 2, \dots, 10$, $j = i + 1, i + 2, \dots, 11$, and $k = j + 1, j + 2, \dots, 12$, let $X_{ijk} = k$ if the i th, the j th, and the k th movies withdrawn are Marlon Brando movies. Let $X_{ijk} = 0$, otherwise. Then $X = \sum_{i=1}^{10} \sum_{j=i+1}^{11} \sum_{k=j+1}^{12} X_{ijk}$ is the number of movies Professor Jackson withdraws until all Brando movies are pulled out. Also note that $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ and $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$.





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Chapter 5

Continuous Random Variables

5.1 PROBABILITY DENSITY FUNCTIONS

As discussed in [Section 4.2](#), the distribution function of a random variable X is a function F from $(-\infty, +\infty)$ to $\mathbb{R}$ defined by $F(t) = P(X \leq t)$. From the definition of F we deduced that it is nondecreasing, right continuous, and satisfies $\lim_{t \rightarrow \infty} F(t) = 1$ and $\lim_{t \rightarrow -\infty} F(t) = 0$. Furthermore, we showed that, for discrete random variables, distributions are step functions. We also proved that if X is a discrete random variable with set of possible values $\{x_1, x_2, \dots\}$, probability mass function p , and distribution function F , then F has jump discontinuities at $x_1, x_2, \dots$, where the magnitude of the jump at x_i is $p(x_i)$ and for $x_{n-1} \leq t < x_n$,

$$F(t) = P(X \leq t) = \sum_{i=1}^{n-1} p(x_i). \quad (5.1)$$

In the case of discrete random variables a very small change in t may cause relatively large changes in the values of F . For example, if t changes from $x_{n-1} - \varepsilon$ to x_n , $\varepsilon > 0$ being arbitrarily small, then $F(t)$ changes from $\sum_{i=1}^{n-1} p(x_i)$ to $\sum_{i=1}^n p(x_i)$, a change of magnitude $p(x_n)$, which might be large. In cases such as the lifetime of a random light bulb, the arrival time of a train at a station, and the weight of a random watermelon grown in a certain field, where the set of possible values of X is uncountable, small changes in x produce correspondingly small changes in the distribution of X . In such cases we expect that F , the distribution function of X , will be a continuous function. Random variables that have continuous distributions can be studied under general conditions. However, for practical reasons and mathematical simplicity, we restrict ourselves to a class of random variables that are called **absolutely continuous** and are defined as follows:

Definition 5.1 *Let X be a random variable. Suppose that there exists a nonnegative real-valued function $f: \mathbb{R} \rightarrow [0, \infty)$ such that for any subset of real numbers A that can be constructed from intervals by a countable number of set operations,*

$$P(X \in A) = \int_A f(x) dx. \quad (5.2)$$

*Then X is called **absolutely continuous** or, in this book, for simplicity, **continuous**. Therefore, whenever we say that X is continuous, we mean that it is absolutely continuous and hence satisfies (5.2). The function f is called the **probability density function**, or simply the **density function** of X .*

Let f be the density function of a random variable X with distribution function F . Some immediate properties of f are as follows:

(a) $F(t) = \int_{-\infty}^t f(x) dx.$

Let $A = (-\infty, t]$, then using (5.2), we can write

$$F(t) = P(X \leq t) = P(X \in A) = \int_A f(x) dx = \int_{-\infty}^t f(x) dx.$$

Comparing (a) with (5.1), we see that f , the probability density function of a continuous random variable X , can be regarded as an extension of the idea of p , the probability mass function of a discrete random variable. As we study f further, the resemblance between f and p becomes clearer. For example, recall that if X is a discrete random variable with the set of possible values A and probability mass function p , then $\sum_{x \in A} p(x) = 1$. In the continuous case, the relation analogous to this is the following:

(b) $\int_{-\infty}^{\infty} f(x) dx = 1.$

This is true by (a) and the fact that $F(\infty) = 1$ [i.e., $\lim_{t \rightarrow \infty} F(t) = 1$]. It also follows from (5.2) with $A = \mathbb{R}$. Because of this property, in general, if a function $g: \mathbb{R} \rightarrow [0, \infty)$ satisfies $\int_{-\infty}^{\infty} g(x) dx = 1$, we say that g is a probability density function, or simply a density function.

(c) **If f is continuous, then $F'(x) = f(x)$.** This follows from (a) and the fundamental theorem of calculus: $\frac{d}{dt} \int_a^t f(x) dx = f(t)$. Note that even if f is not continuous, still $F'(x) = f(x)$ for every x at which f is continuous. *Therefore, the distribution functions of continuous random variables are integrals of their derivatives.*

(d) **For real numbers $a \leq b$, $P(a \leq X \leq b) = \int_a^b f(t) dt$.**

Let $A = [a, b]$ in (5.2), then

$$P(a \leq X \leq b) = P(X \in A) = \int_A f(x) dx = \int_a^b f(x) dx.$$

Property (d) states that the probability of X being between a and b is equal to the area under the graph of f from a to b . Letting $a = b$ in (d), we obtain

$$P(X = a) = \int_a^a f(x) dx = 0.$$

This means that for any real number a , $P(X = a) = 0$. That is, the probability that a continuous random variable assumes a certain value is 0. From $P(X = a) = 0$, $\forall a \in \mathbb{R}$, we have that the probability of X lying in an interval does not depend on the endpoints of the interval. Therefore,

(e) $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b) = \int_a^b f(t) dt.$

Another implication of $P(X = a) = 0, \forall a \in \mathbb{R}$, is that *the value of the density function f at no point represents a probability*. The probabilistic significance of f is that its integral over any subset of real numbers B gives the probability that X lies in B . In particular,

The area over an interval I under the graph of f represents the probability that the random variable X will belong to I . The area under f to the left of a given point t is $F(t)$, the value of the distribution function of X at t .

As an example, let X be a continuous random variable with probability density function f the graph of which is sketched in Figure 5.1. Then the shaded area under f is the probability that X is between a and b . Similarly, let Y be a continuous random variable with probability density function g , as depicted in Figure 5.2. The shaded region beneath the curve of g represents the probability that Y is greater than or equal to 1.5.

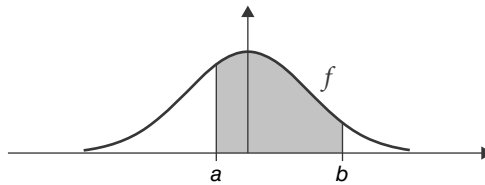


Figure 5.1 The shaded area under f is the probability that $a < X < b$.

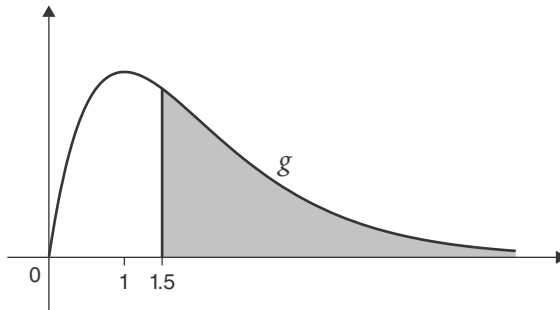


Figure 5.2 The shaded area under g is the probability that $Y \geq 1.5$.

Intuitively, $f(a)$ is a measure that determines how likely it is for X to be close to a . To see this, note that if $\varepsilon > 0$ is very small, then $P(a - \varepsilon < X < a + \varepsilon)$ is the probability that X is close to a . Now

$$P(a - \varepsilon < X < a + \varepsilon) = \int_{a-\varepsilon}^{a+\varepsilon} f(t) dt$$

is the area under $f(t)$ from $a - \varepsilon$ to $a + \varepsilon$. This area is almost equal to the area of a rectangle with sides of lengths $(a + \varepsilon) - (a - \varepsilon) = 2\varepsilon$ and $f(a)$. Thus

$$P(a - \varepsilon < X < a + \varepsilon) = \int_{a-\varepsilon}^{a+\varepsilon} f(t) dt \approx 2\varepsilon f(a). \quad (5.3)$$

Similarly, for any other real number b in the domain of f ,

$$P(b - \varepsilon < X < b + \varepsilon) \approx 2\varepsilon f(b). \quad (5.4)$$

Relations (5.3) and (5.4) show that for small fixed $\varepsilon > 0$, if $f(a) < f(b)$, then

$$P(a - \varepsilon < X < a + \varepsilon) < P(b - \varepsilon < X < b + \varepsilon).$$

That is, if $f(a) < f(b)$, the probability that X is close to b is higher than the probability that X is close to a . Thus the larger $f(x)$, the higher the probability will be that X is close to x .

Example 5.1 Experience has shown that while walking in a certain park, the time X , in minutes, between seeing two people smoking has a density function of the form

$$f(x) = \begin{cases} \lambda x e^{-x} & x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Calculate the value of λ .
- (b) Find the probability distribution function of X .
- (c) What is the probability that Jeff, who has just seen a person smoking, will see another person smoking in 2 to 5 minutes? In at least 7 minutes?

Solution:

- (a) To determine λ , we use the property $\int_{-\infty}^{\infty} f(x) dx = 1$:

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} \lambda x e^{-x} dx = \lambda \int_0^{\infty} x e^{-x} dx.$$

Now, by integration by parts,

$$\int x e^{-x} dx = -(x + 1)e^{-x}.$$

Thus

$$\lambda \left[-(x + 1)e^{-x} \right]_0^{\infty} = 1. \quad (5.5)$$

But as $x \rightarrow \infty$, using l'Hôpital's rule, we get

$$\lim_{x \rightarrow \infty} (x + 1)e^{-x} = \lim_{x \rightarrow \infty} \frac{x + 1}{e^x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0.$$

Therefore, (5.5) implies that $\lambda[0 - (-1)] = 1$ or $\lambda = 1$.

- (b) To find F , the distribution function of X , note that $F(t) = 0$ if $t < 0$. For $t \geq 0$,

$$F(t) = \int_{-\infty}^t f(x) dx = \left[-(x+1)e^{-x} \right]_0^t = -(t+1)e^{-t} + 1.$$

Thus

$$F(t) = \begin{cases} 0 & \text{if } t < 0 \\ -(t+1)e^{-t} + 1 & \text{if } t \geq 0. \end{cases}$$

- (c) The desired probabilities $P(2 < X < 5)$ and $P(X \geq 7)$ are calculated as follows:

$$\begin{aligned} P(2 < X < 5) &= P(2 < X \leq 5) = P(X \leq 5) - P(X \leq 2) = F(5) - F(2) \\ &= (1 - 6e^{-5}) - (1 - 3e^{-2}) = 3e^{-2} - 6e^{-5} \approx 0.37. \end{aligned}$$

$$\begin{aligned} P(X \geq 7) &= 1 - P(X < 7) = 1 - P(X \leq 7) = 1 - F(7) \\ &= 8e^{-7} \approx 0.007. \end{aligned}$$

Note: To calculate these quantities, we can use $P(2 < X < 5) = \int_2^5 f(t) dt$ and $P(X \geq 7) = \int_7^{\infty} f(t) dt$ as well. ♦

Example 5.2

- (a) Sketch the graph of the function

$$f(x) = \begin{cases} \frac{1}{2} - \frac{1}{4}|x-3| & 1 \leq x \leq 5 \\ 0 & \text{otherwise,} \end{cases}$$

and show that it is the probability density function of a random variable X .

- (b) Find F , the distribution function of X , and sketch its graph.
 (c) Show that F is continuous.

Solution:

- (a) Note that

$$f(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{2} + \frac{1}{4}(x-3) & 1 \leq x < 3 \\ \frac{1}{2} - \frac{1}{4}(x-3) & 3 \leq x < 5 \\ 0 & x \geq 5. \end{cases}$$

Therefore, the graph of f is as shown in [Figure 5.3](#) on the following page. Now since $f(x) \geq 0$ and the area under f from 1 to 5, being the area of the triangle ABC (as observed in [Figure 5.3](#)) is $(1/2)(4 \times 1/2) = 1$, f is a density function of some random variable X .

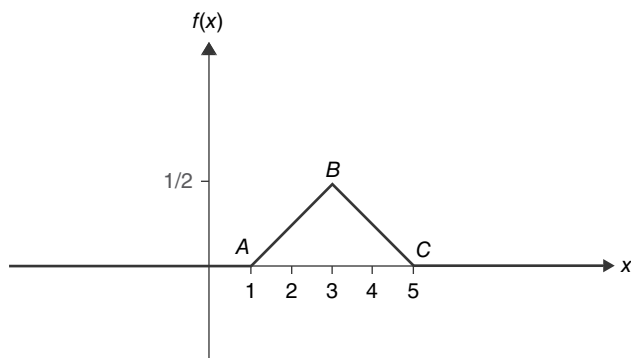


Figure 5.3 Probability density function of Example 5.2.

- (b) To calculate the distribution function of X , we use the formula $F(t) = \int_{-\infty}^t f(x) dx$. For $t < 1$,

$$F(t) = \int_{-\infty}^t 0 dx = 0;$$

for $1 \leq t < 3$,

$$F(t) = \int_{-\infty}^t f(x) dx = \int_1^t \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx = \frac{1}{8}t^2 - \frac{1}{4}t + \frac{1}{8};$$

for $3 \leq t < 5$,

$$\begin{aligned} F(t) &= \int_{-\infty}^t f(x) dx = \int_1^3 \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx + \int_3^t \left[\frac{1}{2} - \frac{1}{4}(x-3) \right] dx \\ &= \frac{1}{2} + \left(-\frac{1}{8}t^2 + \frac{5}{4}t - \frac{21}{8} \right) = -\frac{1}{8}t^2 + \frac{5}{4}t - \frac{17}{8}; \end{aligned}$$

and for $t \geq 5$,

$$\begin{aligned} F(t) &= \int_{-\infty}^t f(x) dx = \int_1^3 \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx + \int_3^5 \left[\frac{1}{2} - \frac{1}{4}(x-3) \right] dx \\ &= \frac{1}{2} + \frac{1}{2} = 1. \end{aligned}$$

Therefore,

$$F(t) = \begin{cases} 0 & t < 1 \\ \frac{1}{8}t^2 - \frac{1}{4}t + \frac{1}{8} & 1 \leq t < 3 \\ -\frac{1}{8}t^2 + \frac{5}{4}t - \frac{17}{8} & 3 \leq t < 5 \\ 1 & t \geq 5. \end{cases}$$

The graph of F is shown in [Figure 5.4](#).

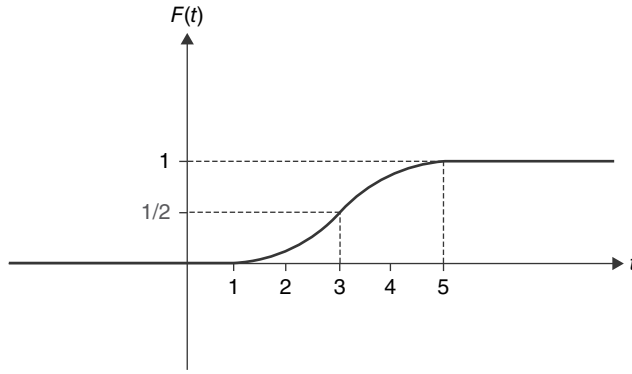


Figure 5.4 Distribution function of Example 5.2.

(c) F is continuous because

$$\lim_{t \rightarrow 1^-} F(t) = 0 = F(1),$$

$$\lim_{t \rightarrow 3^-} F(t) = \frac{1}{8}(3)^2 - \frac{1}{4}(3) + \frac{1}{8} = \frac{1}{2} = -\frac{1}{8}(3)^2 + \frac{5}{4}(3) - \frac{17}{8} = F(3),$$

$$\lim_{t \rightarrow 5^-} F(t) = -\frac{1}{8}(5)^2 + \frac{5}{4}(5) - \frac{17}{8} = 1 = F(5). \quad \blacklozenge$$

Remark 5.1 Under what conditions does a distribution function F represent the distribution of a continuous random variable? The answer is that if F is a continuous function and is differentiable everywhere except possibly at a finite number of points, then it represents the distribution of a continuous random variable. To show this, let $f(x) = F'(x)$.

Then, (i) $F(x) = \int_{-\infty}^x f(t) dt$; (ii) since F is a distribution function and hence nondecreasing, we must have $f(x) \geq 0$; (iii) $\lim_{t \rightarrow \infty} F(t) = 1$ implies that $\int_{-\infty}^{\infty} f(t) dt = 1$; (iv) for any interval $I = (a, b)$, $[a, b)$, $(a, b]$, or $[a, b]$,

$$P(X \in I) = F(b) - F(a) = \int_a^b f(x) dx,$$

and, as a result of (iv), if A is a subset of $\mathbb{R}$ that can be constructed from intervals by a countable number of set operations, then

$$P(X \in A) = \int_A f(x) dx. \quad \blacklozenge$$

EXERCISES

1. When a certain car breaks down, the time that it takes to fix it (in hours) is a random variable with the density function

$$f(x) = \begin{cases} ce^{-3x} & \text{if } 0 \leq x < \infty \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Calculate the value of c .
- (b) Find the probability that when this car breaks down, it takes at most 30 minutes to fix it.
2. The distribution function for the duration of a certain soap opera (in tens of hours) is

$$F(x) = \begin{cases} 1 - \frac{16}{x^2} & x \geq 4 \\ 0 & x < 4. \end{cases}$$

- (a) Calculate f , the probability density function of the soap opera.
- (b) Sketch the graphs of F and f .
- (c) What is the probability that the soap opera takes at most 50 hours? At least 60 hours? Between 50 and 70 hours? Between 10 and 35 hours?
3. The time it takes for a student to finish an aptitude test (in hours) has a density function of the form

$$f(x) = \begin{cases} c(x-1)(2-x) & \text{if } 1 < x < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Determine the constant c .
- (b) Calculate the distribution function of the time it takes for a randomly selected student to finish the aptitude test.
- (c) What is the probability that a student will finish the aptitude test in less than 75 minutes? Between $1\frac{1}{2}$ and 2 hours?
4. The lifetime of a tire selected randomly from a used tire shop is $10000X$ miles, where X is a random variable with the density function

$$f(x) = \begin{cases} 2/x^2 & \text{if } 1 < x < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) What percentage of the tires of this shop last fewer than 15,000 miles?
- (b) What percentage of those having lifetimes fewer than 15,000 miles last between 10,000 and 12,500 miles?

5. The probability density function of a random variable X is given by

$$f(x) = \begin{cases} \frac{c}{\sqrt{1-x^2}} & \text{if } -1 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Calculate the value of c .
 - (b) Find the probability distribution function of X .
6. At a grocery store in Idaho, the demand for honey, in kilograms, per week, is a random variable with probability density function

$$f(x) = \begin{cases} \frac{5}{6656} (4-x)(4+x)^3 & 0 < x < 4 \\ 0 & \text{otherwise.} \end{cases}$$

How many kilograms of honey must the grocery store carry per week so that the stock out probability for honey is at most 0.05?

7. Let X be a continuous random variable with the following probability density function

$$f(x) = \begin{cases} kx & \text{if } 0 \leq x < 4 \\ k(8-x) & \text{if } 4 \leq x < 8 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Determine the value of the constant k .
 - (b) Find the probability distribution function of X .
 - (c) Calculate the median and the 95th percentile of the random variable X . That is, find values of α and β such that $P(X \leq \alpha) = 0.5$ and $P(X \leq \beta) = 0.95$.
8. Let X be a continuous random variable with density and distribution functions f and F , respectively. Assuming that $\alpha \in \mathbb{R}$ is a point at which $P(X \leq \alpha) < 1$, prove that

$$h(x) = \begin{cases} \frac{f(x)}{1-F(\alpha)} & \text{if } x \geq \alpha \\ 0 & \text{if } x < \alpha \end{cases}$$

is also a probability density function.

9. An insurance company has sold a large number of homeowner insurance policies in an affluent area, where the insured value of homes are above \$750,000. An actuary has calculated that X , the insured value of a randomly selected home, in millions of dollars, is a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{81}{64x^5} & x > 3/4 \\ 0 & \text{otherwise.} \end{cases}$$

Of all homes that are insured for at least 1.5 million dollars, what percentage are insured for less than 2 million dollars?

10. Let X be a continuous random variable with density function f . We say that X is **symmetric about α** if for all x ,

$$P(X \geq \alpha + x) = P(X \leq \alpha - x).$$

- (a) Prove that X is symmetric about α if and only if for all x , we have $f(\alpha - x) = f(\alpha + x)$.
 (b) Show that X is symmetric about α if and only if $f(x) = f(2\alpha - x)$ for all x .
 (c) Let X be a continuous random variable with probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-(x-3)^2/2}, \quad x \in \mathbb{R},$$

and Y be a continuous random variable with probability density function

$$g(x) = \frac{1}{\pi[1 + (x - 1)^2]}, \quad x \in \mathbb{R}.$$

Find the points about which X and Y are symmetric.

11. The function

$$F(x) = \begin{cases} 0 & x < \pi \\ \cos x & \pi \leq x < 2\pi \\ 1 & x \geq 2\pi \end{cases}$$

is continuous, differentiable, nondecreasing, $F(-\infty) = 0$, and $F(\infty) = 1$. Is F the distribution function of a continuous random variable?

12. Suppose that the loss in a certain investment, in thousands of dollars, is a continuous random variable X that has a density function of the form

$$f(x) = \begin{cases} k(2x - 3x^2) & -1 < x < 0 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Calculate the value of k .
- (b) Find the probability that the loss is at most \$500.
13. Let X denote the lifetime of a radio, in years, manufactured by a certain company. The density function of X is given by

$$f(x) = \begin{cases} \frac{1}{15}e^{-x/15} & \text{if } 0 \leq x < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

What is the probability that, of eight such radios, at least four last more than 15 years?

14. Let X be a continuous random variable with probability density function

$$f(x) = \begin{cases} 1 - |x| & -1 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Sketch the graph of f .
- (b) Find the distribution function of X .
15. Prove that if f and g are two probability density functions, then for $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \beta = 1$, $\alpha f + \beta g$ is also a probability density function.
16. The distribution function of a random variable X is given by

$$F(x) = \alpha + \beta \arctan \frac{x}{2}, \quad -\infty < x < \infty.$$

Determine the constants α and β and the density function of X .



Self-Quiz on Section 5.1

Time allotted: 15 Minutes

Each problem is worth 5 points.

1. For what value of α is the following the probability density function of a random variable X ? For that value of α , find $P(1.5 < X < 2.5)$.

$$f(x) = \begin{cases} \alpha/x^4 & 1 < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

2. The distribution for the duration of a certain soap opera, in tens of hours, is

$$F(x) = \begin{cases} 1 - \frac{16}{x^2} & x \geq 4 \\ 0 & \text{otherwise.} \end{cases}$$

If the soap opera has been running for 110 hours, what is the probability that it will run for at least another 110 hours?



5.2 DENSITY FUNCTION OF A FUNCTION OF A RANDOM VARIABLE

In the theory and applications of probability, we are often faced with situations in which f , the density function of a random variable X , is known but the density function $h(X)$, for some function of X , is needed. In such cases there are two common methods for the evaluation of the density function of $h(X)$ that we will explain in this section. One method is to find the density of $h(X)$ by calculating its distribution function. Another method is to find it directly from the density function of X . We now explain the first method, called **the method of distribution functions**. To calculate the probability density function of $h(X)$, we calculate the distribution function of $h(X)$, $P(h(X) \leq t)$, by finding a set A for which $h(X) \leq t$ if and only if $X \in A$ and then computing $P(X \in A)$ from the distribution function of X . When $P(h(X) \leq t)$ is found, the density function of $h(X)$, if it exists, is obtained by differentiation. Some examples follow.

Example 5.3 Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} 2/x^2 & \text{if } 1 < x < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Find the distribution and the density functions of $Y = X^2$.

Solution: Let G and g be the distribution function and the density function of Y , respectively. By definition,

$$\begin{aligned} G(t) &= P(Y \leq t) = P(X^2 \leq t) = P(-\sqrt{t} \leq X \leq \sqrt{t}) \\ &= \begin{cases} 0 & t < 1 \\ P(1 \leq X \leq \sqrt{t}) & 1 \leq t < 4 \\ 1 & t \geq 4. \end{cases} \end{aligned}$$

Now

$$P(1 \leq X \leq \sqrt{t}) = \int_1^{\sqrt{t}} \frac{2}{x^2} dx = \left[-\frac{2}{x} \right]_1^{\sqrt{t}} = 2 - \frac{2}{\sqrt{t}}.$$

Therefore,

$$G(t) = \begin{cases} 0 & t < 1 \\ 2 - \frac{2}{\sqrt{t}} & 1 \leq t < 4 \\ 1 & t \geq 4. \end{cases}$$

The density function of Y , g , is found by differentiation of G :

$$g(t) = G'(t) = \begin{cases} \frac{1}{t\sqrt{t}} & \text{if } 1 \leq t \leq 4 \\ 0 & \text{elsewhere.} \quad \blacklozenge \end{cases}$$

Example 5.4 Let X be a continuous random variable with distribution function F and probability density function f . In terms of f , find the distribution and the density functions of $Y = X^3$.

Solution: Let G and g be the distribution function and the density function of Y . Then

$$G(t) = P(Y \leq t) = P(X^3 \leq t) = P(X \leq \sqrt[3]{t}) = F(\sqrt[3]{t}).$$

Hence

$$g(t) = G'(t) = \frac{1}{3\sqrt[3]{t^2}} F'(\sqrt[3]{t}) = \frac{1}{3\sqrt[3]{t^2}} f(\sqrt[3]{t}),$$

where g is defined at all points $t \neq 0$ at which f is defined. $\blacklozenge$

Example 5.5 The error of a measurement has the density function

$$f(x) = \begin{cases} 1/2 & \text{if } -1 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the distribution and the density functions of the magnitude of the error.

Solution: Let X be the error of the measurement. We want to find G , the distribution, and g , the density functions of $|X|$, the magnitude of the error. By definition,

$$G(t) = P(|X| \leq t) = P(-t \leq X \leq t) = \begin{cases} 0 & t < 0 \\ \int_{-t}^t \frac{1}{2} dx = t & 0 \leq t < 1 \\ 1 & t \geq 1. \end{cases}$$

The probability density function of Y , g , is obtained by differentiation of G :

$$g(t) = G'(t) = \begin{cases} 1 & \text{if } 0 \leq t \leq 1 \\ 0 & \text{elsewhere.} \quad \blacklozenge \end{cases}$$

In Examples 5.3, 5.4, and 5.5, the method used to find the density function of $h(X)$, a function of the random variable X , is to determine the distribution of $h(X)$ and then differentiate it. It is also possible to find the density function of $h(X)$ without obtaining its distribution function. We now explain this method, called the **method of transformations**.

Recall that a real-valued function $h: \mathbb{R} \rightarrow \mathbb{R}$ defined by $y = h(x)$ is *invertible* if and only if when solving $y = h(x)$ for x , we find that $x = h^{-1}(y)$ is itself a function. That is, $y = h(x)$ if and only if $x = h^{-1}(y)$, where $x = h^{-1}(y)$ is a function. For example, the function $y = h(x) = x^3$ is invertible because if we solve it for x , we obtain $x = h^{-1}(y) = \sqrt[3]{y}$, which is itself a function. The function $y = h(x) = x^2$ is not invertible if its domain is $(-\infty, \infty)$. When solving for x we obtain $x = \pm\sqrt{y}$, which is not a function. The function $y = x^2$ will be invertible if we restrict its domain to $[0, \infty)$, the set of nonnegative numbers. This follows since solving $y = x^2$ for x gives $x = \sqrt{y}$, which is a function. Other examples of invertible functions are $h(x) = e^x$, $x \in \mathbb{R}$; $h(x) = \ln x$, $x \in (0, \infty)$; and $h(x) = \sin x$, $x \in (0, \pi/2)$. Examples of functions that are not invertible are $h(x) = \sin x$, $x \in \mathbb{R}$; $h(x) = x^2 + 5$, $x \in \mathbb{R}$; and $h(x) = \exp(x^2)$, $x \in \mathbb{R}$.

In calculus, we have seen that a continuous function is invertible if and only if it is either strictly increasing or strictly decreasing. If it is strictly increasing, its inverse is also strictly increasing. If it is strictly decreasing, its inverse is strictly decreasing as well. We use these to prove the next theorem, which enables us to find the density function of $h(X)$ mainly by finding the inverse of h . For this application, all we need to know is how to find the inverse of a function, if it exists. Studying different criteria for invertibility does not serve our purpose.

Theorem 5.1 (Method of Transformations) *Let X be a continuous random variable with density function f_X and the set of possible values A . For the invertible function $h: A \rightarrow \mathbb{R}$, let $Y = h(X)$ be a random variable with the set of possible values $B = h(A) = \{h(a) : a \in A\}$. Suppose that the inverse of $y = h(x)$ is the function $x = h^{-1}(y)$, which is differentiable for all values of $y \in B$. Then f_Y , the density function of Y , is given by*

$$f_Y(y) = f_X(h^{-1}(y)) |(h^{-1})'(y)|, \quad y \in B.$$

Proof: Let F_X and F_Y be distribution functions of X and $Y = h(X)$, respectively. Differentiability of h^{-1} implies that it is continuous. Since a continuous invertible function is strictly monotone, h^{-1} is either strictly increasing or strictly decreasing. If it is strictly increasing, $(h^{-1})'(y) > 0$, and hence $(h^{-1})'(y) = |(h^{-1})'(y)|$. Moreover, in this case h is also strictly increasing, so

$$F_Y(y) = P(h(X) \leq y) = P(X \leq h^{-1}(y)) = F_X(h^{-1}(y)).$$

Differentiating this by chain rule, we obtain

$$F'_Y(y) = (h^{-1})'(y) F'_X(h^{-1}(y)) = |(h^{-1})'(y)| f_X(h^{-1}(y)),$$

which gives the theorem. If h^{-1} is strictly decreasing, $(h^{-1})'(y) < 0$ and hence $|(h^{-1})'(y)| = -(h^{-1})'(y)$. In this case h is also strictly decreasing and we get

$$F_Y(y) = P(h(X) \leq y) = P(X \geq h^{-1}(y)) = 1 - F_X(h^{-1}(y)).$$

Differentiating this by chain rule, we find that

$$F'_Y(y) = -(h^{-1})'(y)F'_X(h^{-1}(y)) = |(h^{-1})'(y)|f_X(h^{-1}(y)),$$

showing that the theorem is valid in this case as well. ♦

Example 5.6 Let X be a random variable with the density function

$$f_X(x) = \begin{cases} 2e^{-2x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Using the method of transformations, find the probability density function of $Y = \sqrt{X}$.

Solution: The set of possible values of X is $A = (0, \infty)$. Let $h: (0, \infty) \rightarrow \mathbb{R}$ be defined by $h(x) = \sqrt{x}$. We want to find the density function of $Y = h(X) = \sqrt{X}$. The set of possible values of $h(X)$ is $B = \{h(a): a \in A\} = (0, \infty)$. The function h is invertible with the inverse $x = h^{-1}(y) = y^2$, which is differentiable, and its derivative is $(h^{-1})'(y) = 2y$. Therefore, by Theorem 5.1,

$$f_Y(y) = f_X(h^{-1}(y))|(h^{-1})'(y)| = 2e^{-2y^2}|2y|, \quad y \in B = (0, \infty).$$

Since $y \in (0, \infty)$, $|2y| = 2y$ and we find that

$$f_Y(y) = \begin{cases} 4ye^{-2y^2} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

is the density function of $Y = \sqrt{X}$. ♦

Example 5.7 Let X be a continuous random variable with the probability density function

$$f_X(x) = \begin{cases} 4x^3 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Using the method of transformations, find the probability density function of the random variable $Y = 1 - 3X^2$.

Solution: The set of possible values of X is $A = (0, 1)$. Let $h: (0, 1) \rightarrow \mathbb{R}$ be defined by $h(x) = 1 - 3x^2$. We want to find the density function of $Y = h(X) = 1 - 3X^2$. The set of possible values of $h(X)$ is $B = \{h(a): a \in A\} = (-2, 1)$. Since the domain of the function h is $(0, 1)$, h is invertible, and its inverse is found by solving $1 - 3x^2 = y$ for x , which gives $x = \sqrt{(1 - y)/3}$. Therefore, the inverse of h is

$$x = h^{-1}(y) = \sqrt{(1 - y)/3},$$

which is differentiable, and its derivative is $(h^{-1})'(y) = -1/(2\sqrt{3(1-y)})$. Using Theorem 5.1, we find that

$$f_Y(y) = f_X(h^{-1}(y)) |(h^{-1})'(y)| = 4 \left(\sqrt{\frac{1-y}{3}} \right)^3 \left| -\frac{1}{2\sqrt{3(1-y)}} \right| = \frac{2}{9}(1-y)$$

is the density function of Y when $y \in (-2, 1)$. ♦

EXERCISES

A

1. Let X be a continuous random variable with the density function

$$f(x) = \begin{cases} 1/4 & \text{if } x \in (-2, 2) \\ 0 & \text{otherwise.} \end{cases}$$

Using the method of distribution functions, find the probability density functions of $Y = X^3$ and $Z = X^4$.

2. A manufacturer produces metal disks with radii being random variables with the identical probability density function

$$f(x) = \begin{cases} \frac{x^5}{431} - \frac{5x^4}{2586} & \text{if } 3 < x < 4 \\ 0 & \text{otherwise.} \end{cases}$$

What is the probability that the area of a random disk is greater than 39?

3. Let X be a continuous random variable with distribution function F and density function f . Calculate the density function of the random variable $Y = e^X$.
4. Let the density function of X be

$$f(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Using the method of transformations, find the density functions of $Y = X\sqrt{X}$ and $Z = e^{-X}$.

5. Let X be a continuous random variable with the density function

$$f(x) = \begin{cases} 3e^{-3x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Using the method of transformations, find the probability density function of the random variable $Y = \log_2 X$.

6. Let the probability density function of X be

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise,} \end{cases}$$

for some $\lambda > 0$. Using the method of distribution functions, calculate the probability density function of $Y = \sqrt[3]{X^2}$.

7. Let f be the probability density function of a random variable X . In terms of f , calculate the probability density function of X^2 .
8. Let X be a random variable with the density function

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

(X is called a **Cauchy random variable**.) Find the density function of $Z = \arctan X$.

B

9. Let X be a random variable with the probability density function given by

$$f(x) = \begin{cases} e^{-x} & \text{if } x \geq 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Let

$$Y = \begin{cases} X & \text{if } X \leq 1 \\ 1/X & \text{if } X > 1. \end{cases}$$

Find the probability density function of Y . ■

Self-Quiz on Section 5.2

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. Let X be a random point from the interval $(-\pi/2, \pi/2)$. Find the probability density function of $Y = \tan X$.
2. Let X be a random variable with probability density function

$$f(x) = \begin{cases} xe^{-x^2/2} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Using the method of transformations, find the probability density function of $Y = X^2$. Then calculate $E(Y)$. ■

5.3 EXPECTATIONS AND VARIANCES OF CONTINUOUS RANDOM VARIABLES

Expectations of Continuous Random Variables

Let X be a continuous random variable with probability density and distribution functions f and F , respectively. To define $E(X)$, the average or expected value of X , first suppose that X only takes values from the interval $[a, b]$, and divide $[a, b]$ into n subintervals of equal lengths. Let $h = (b - a)/n$, $x_0 = a$, $x_1 = a + h$, $x_2 = a + 2h$, $\dots$, $x_n = a + nh = b$. Then $a = x_0 < x_1 < x_2 < \dots < x_n = b$ is a partition of $[a, b]$. Since F is continuous, let us assume that it is differentiable on (a, b) . By the mean-value theorem (of calculus), there exists $t_i \in (x_{i-1}, x_i)$ such that

$$F(x_i) - F(x_{i-1}) = F'(t_i)(x_i - x_{i-1}), \quad 1 \leq i \leq n,$$

or, equivalently,

$$P(x_{i-1} < X \leq x_i) = f(t_i)h, \quad 1 \leq i \leq n. \quad (5.6)$$

If n is sufficiently large ($n \rightarrow \infty$), then h , the width of the intervals, is sufficiently small ($h \rightarrow 0$), and $f(x)$ does not vary appreciably over any subinterval of $(x_{i-1}, x_i]$, $1 \leq i \leq n$. Thus t_i is an approximate value of X in the interval $(x_{i-1}, x_i]$. Now

$$\sum_{i=1}^n t_i P(x_{i-1} < X \leq x_i)$$

finds the product of an approximate value of X when it is in $(x_{i-1}, x_i]$ and the probability that it is in $(x_{i-1}, x_i]$, and then sums over all these intervals. From the concept of expectation of a discrete random variable, it is clear that, as the lengths of these intervals get smaller and smaller, $\sum_{i=1}^n t_i P(x_{i-1} < X \leq x_i)$ gets closer and closer to the “average” value of X . So it is desirable to define $E(X)$ as

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n t_i P(x_{i-1} < X \leq x_i).$$

But by (5.6) this is the same as $\lim_{n \rightarrow \infty} \sum_{i=1}^n t_i f(t_i)h$, where this limit is equal to $\int_a^b x f(x) dx$ as known from calculus. If X is not restricted to an interval $[a, b]$, a definition for $E(X)$ is motivated in the same way, but at the end the limit is taken as $a \rightarrow -\infty$ and $b \rightarrow \infty$.

Definition 5.2 If X is a continuous random variable with probability density function f , the **expected value** of X is defined by

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx.$$

The expected value of X is also called the **mean**, or **mathematical expectation**, or simply the **expectation** of X , and as in the discrete case, sometimes it is denoted by EX , $E[X]$, μ , or μ_X .

To get a geometric feeling for mathematical expectation, consider a piece of cardboard of uniform density on which the graph of the density function f of a random variable X is drawn. Suppose that the cardboard is cut along the graph of f and we are asked to balance it on a given edge perpendicular to the x -axis. Then, to have it in equilibrium, we must balance the cardboard on the given edge at the point $x = E(X)$ on the x -axis.

Example 5.8 In a group of adult males, the difference between the uric acid value and 6, the standard value, is a random variable X with the following probability density function:

$$f(x) = \begin{cases} \frac{27}{490}(3x^2 - 2x) & \text{if } 2/3 < x < 3 \\ 0 & \text{elsewhere.} \end{cases}$$

Calculate the mean of these differences for the group.

Solution: By definition,

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x) dx = \int_{2/3}^3 \frac{27}{490}(3x^3 - 2x^2) dx \\ &= \frac{27}{490} \left[\frac{3}{4}x^4 - \frac{2}{3}x^3 \right]_{2/3}^3 = \frac{283}{120} = 2.36. \quad \blacklozenge \end{aligned}$$

Remark 5.2 If X is a continuous random variable with density function f , X is said to have a **finite expected value** if

$$\int_{-\infty}^{\infty} |x|f(x) dx < \infty;$$

that is, X has a finite expected value if the integral of $xf(x)$ converges absolutely. Otherwise, we say that the expected value of X is not finite. We now justify why the absolute convergence of the integral is required. Note that

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x) dx = \int_{-\infty}^0 xf(x) dx + \int_0^{\infty} xf(x) dx \\ &= - \int_{-\infty}^0 (-x)f(x) dx + \int_0^{\infty} xf(x) dx, \end{aligned}$$

where $\int_{-\infty}^0 (-x)f(x) dx \geq 0$ and $\int_0^{\infty} xf(x) dx \geq 0$. Thus $E(X)$ is well defined if the integrals $\int_{-\infty}^0 (-x)f(x) dx$ and $\int_0^{\infty} xf(x) dx$ are not both ∞ . Moreover, $E(X) < \infty$ if neither of these two integrals is $+\infty$. Hence a necessary and sufficient condition for $E(X)$ to

exist and to be finite is that $\int_{-\infty}^0 (-x)f(x) dx < \infty$ and $\int_0^{\infty} xf(x) dx < \infty$. Since both of these integrals are finite if and only if

$$\int_{-\infty}^{\infty} |x|f(x) dx = \int_{-\infty}^0 (-x)f(x) dx + \int_0^{\infty} xf(x) dx$$

is finite, we have that $E(X)$ is well defined and finite if and only if the integral $\int_{-\infty}^{\infty} xf(x) dx$ is absolutely convergent. ♦

Example 5.9 A random variable X with density function

$$f(x) = \frac{c}{1+x^2}, \quad -\infty < x < \infty,$$

is called a **Cauchy random variable**.

- (a) Find c .
- (b) Show that $E(X)$ does not exist.

Solution:

- (a) Since f is a density function $\int_{-\infty}^{\infty} f(x) dx = 1$. Thus

$$\int_{-\infty}^{\infty} \frac{c}{1+x^2} dx = c \int_{-\infty}^{\infty} \frac{dx}{1+x^2} = 1.$$

Now

$$\int \frac{dx}{1+x^2} = \arctan x.$$

Since the range of $\arctan x$ is $(-\pi/2, +\pi/2)$, we get

$$1 = c \int_{-\infty}^{\infty} \frac{dx}{1+x^2} = c \left[\arctan x \right]_{-\infty}^{\infty} = c \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right] = c\pi.$$

Thus $c = 1/\pi$.

- (b) To show that $E(X)$ does not exist, note that

$$\begin{aligned} \int_{-\infty}^{\infty} |x|f(x) dx &= \int_{-\infty}^{\infty} \frac{|x| dx}{\pi(1+x^2)} = 2 \int_0^{\infty} \frac{x dx}{\pi(1+x^2)} \\ &= \frac{1}{\pi} \left[\ln(1+x^2) \right]_0^{\infty} = \infty. \quad \blacklozenge \end{aligned}$$

Remark 5.3 In this book, unless otherwise specified, it is implicitly assumed that the expectation of a random variable is finite. ♦

The following theorem directly relates the distribution function of a random variable to its expectation. It enables us to find the expected value of a continuous random variable without calculating its probability density function. It also has important theoretical applications.

Theorem 5.2 For any continuous random variable X with probability distribution function F and density function f ,

$$E(X) = \int_0^{\infty} [1 - F(t)] dt - \int_0^{\infty} F(-t) dt.$$

Proof: Note that

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^0 x f(x) dx + \int_0^{\infty} x f(x) dx \\ &= - \int_{-\infty}^0 \left(\int_0^{-x} dt \right) f(x) dx + \int_0^{\infty} \left(\int_0^x dt \right) f(x) dx \\ &= - \int_0^{\infty} \left(\int_{-\infty}^{-t} f(x) dx \right) dt + \int_0^{\infty} \left(\int_t^{\infty} f(x) dx \right) dt, \end{aligned}$$

where the last equality is obtained by changing the order of integration. The theorem follows since $\int_{-\infty}^{-t} f(x) dx = F(-t)$ and $\int_t^{\infty} f(x) dx = P(X > t) = 1 - F(t)$. ♦

Remark 5.4 In the proof of this theorem we assumed that the random variable X is continuous. Even without this condition the theorem is still valid. Also note that, since $1 - F(t) = P(X > t)$, this theorem may be stated as follows.

For any random variable X ,

$$E(X) = \int_0^{\infty} P(X > t) dt - \int_0^{\infty} P(X \leq -t) dt.$$

In particular, if X is nonnegative, that is, $P(X < 0) = 0$, this theorem states that

$$E(X) = \int_0^{\infty} [1 - F(t)] dt = \int_0^{\infty} P(X > t) dt. \quad \blacklozenge$$

Note that for a continuous nonnegative random variable X , the relation

$$E(X) = \int_0^{\infty} P(X > t) dt.$$

is analogous to the result stated in Theorem 4.10: For a discrete random variable N with set of possible values $\{1, 2, 3, \dots\}$,

$$E(N) = \sum_{i=1}^{\infty} P(N \geq i).$$

As an important application of Theorem 5.2, we now prove the law of the unconscious statistician, Theorem 4.2, for continuous random variables.

Theorem 5.3 *Let X be a continuous random variable with probability density function $f(x)$, then for any function $h: \mathbb{R} \rightarrow \mathbb{R}$,*

$$E[h(X)] = \int_{-\infty}^{\infty} h(x)f(x) dx.$$

Proof: Let

$$h^{-1}(t, \infty) = \{x: h(x) \in (t, \infty)\} = \{x: h(x) > t\}$$

with similar representation for $h^{-1}(-\infty, -t)$. Notice that we are not claiming that h has an inverse function. We are simply considering the set $\{x: h(x) \in (t, \infty)\}$, which is called the *inverse image* of (t, ∞) and is denoted by $h^{-1}(t, \infty)$. By Theorem 5.2,

$$\begin{aligned} E[h(X)] &= \int_0^{\infty} P(h(X) > t) dt - \int_0^{\infty} P(h(X) \leq -t) dt \\ &= \int_0^{\infty} P(X \in h^{-1}(t, \infty)) dt - \int_0^{\infty} P(X \in h^{-1}(-\infty, -t]) dt \\ &= \int_0^{\infty} \left(\int_{\{x: x \in h^{-1}(t, \infty)\}} f(x) dx \right) dt \\ &\quad - \int_0^{\infty} \left(\int_{\{x: x \in h^{-1}(-\infty, -t]\}} f(x) dx \right) dt \\ &= \int_0^{\infty} \left(\int_{\{x: h(x) > t\}} f(x) dx \right) dt - \int_0^{\infty} \left(\int_{\{x: h(x) \leq -t\}} f(x) dx \right) dt. \end{aligned}$$

Now we change the order of integration for both of these double integrals. Since

$$\{(t, x): 0 < t < \infty, h(x) > t\} = \{(t, x): h(x) > 0, 0 < t < h(x)\},$$

and

$$\{(t, x): 0 < t < \infty, h(x) \leq -t\} = \{(t, x): h(x) < 0, 0 < t \leq -h(x)\},$$

we get

$$\begin{aligned}
 E[h(X)] &= \int_{\{x: h(x) > 0\}} \left(\int_0^{h(x)} dt \right) f(x) dx \\
 &\quad - \int_{\{x: h(x) < 0\}} \left(\int_0^{-h(x)} dt \right) f(x) dx \\
 &= \int_{\{x: h(x) > 0\}} h(x)f(x) dx + \int_{\{x: h(x) < 0\}} h(x)f(x) dx \\
 &= \int_{-\infty}^{\infty} h(x)f(x) dx.
 \end{aligned}$$

Note that the last equality follows because $\int_{\{x: h(x) = 0\}} h(x)f(x) dx = 0$. ♦

Corollary Let X be a continuous random variable with probability density function $f(x)$. Let $h_1, h_2, \dots, h_n$ be real-valued functions, and $\alpha_1, \alpha_2, \dots, \alpha_n$ be real numbers. Then

$$\begin{aligned}
 E[\alpha_1 h_1(X) + \alpha_2 h_2(X) + \dots + \alpha_n h_n(X)] \\
 = \alpha_1 E[h_1(X)] + \alpha_2 E[h_2(X)] + \dots + \alpha_n E[h_n(X)].
 \end{aligned}$$

Proof: In the discrete case, in the proof of the corollary of Theorem 4.2, replace $\sum$ by $\int$, and $p(x)$ by $f(x) dx$. ♦

Just as in the discrete case, by this corollary we can write that, for example,

$$E(3X^4 + \cos X + 3e^X + 7) = 3E(X^4) + E(\cos X) + 3E(e^X) + 7.$$

Moreover, this corollary implies that if α and β are constants, then

$$E(\alpha X + \beta) = \alpha E(X) + \beta.$$

Example 5.10 A point X is selected from the interval $(0, \pi/4)$ randomly. Calculate $E(\cos 2X)$ and $E(\cos^2 X)$.

Solution: First we calculate the distribution function of X . Clearly,

$$F(t) = P(X \leq t) = \begin{cases} 0 & t < 0 \\ \frac{t - 0}{\frac{\pi}{4} - 0} = \frac{4t}{\pi} & 0 \leq t < \frac{\pi}{4} \\ 1 & t \geq \frac{\pi}{4}. \end{cases}$$

Thus f , the probability density function of X , is

$$f(t) = \begin{cases} \frac{4}{\pi} & 0 < t < \frac{\pi}{4} \\ 0 & \text{otherwise.} \end{cases}$$

Now, by Theorem 5.3,

$$E(\cos 2X) = \int_0^{\pi/4} \frac{4}{\pi} \cos 2x \, dx = \left[\frac{2}{\pi} \sin 2x \right]_0^{\pi/4} = \frac{2}{\pi}.$$

To calculate $E(\cos^2 X)$, note that $\cos^2 X = (1 + \cos 2X)/2$. So by the corollary of Theorem 5.3,

$$E(\cos^2 X) = E\left(\frac{1}{2} + \frac{1}{2} \cos 2X\right) = \frac{1}{2} + \frac{1}{2}E(\cos 2X) = \frac{1}{2} + \frac{1}{2} \cdot \frac{2}{\pi} = \frac{1}{2} + \frac{1}{\pi}. \quad \blacklozenge$$

Variances of Continuous Random Variables

Definition 5.3 If X is a continuous random variable with $E(X) = \mu$, then $\text{Var}(X)$ and σ_X , called the **variance** and **standard deviation** of X , respectively, are defined by

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2], \\ \sigma_X &= \sqrt{E[(X - \mu)^2]}. \end{aligned}$$

Therefore, if f is the density function of X , then by Theorem 5.3,

$$\text{Var}(X) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) \, dx.$$

Also, as before, we have the following important relations whose proofs are analogous to those in the discrete case.

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2, \\ \text{Var}(aX + b) &= a^2 \text{Var}(X), \quad \sigma_{aX+b} = |a| \sigma_X, \quad a \text{ and } b \text{ being constants.} \end{aligned}$$

As discussed in 4, variance serves as an indicator of variability or dispersion. Typically, a higher variance denotes greater variability and fluctuations. For continuous random variables, this concept is clearly depicted in [Figure 5.5](#), where the wider dashed curve has a variance of 4 (standard deviation 2), and the solid curve has a variance of 1 (standard deviation 1).

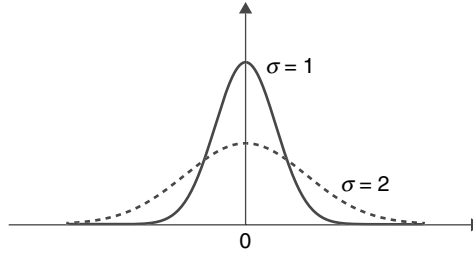


Figure 5.5 The greater the variance, the broader the curve.

Example 5.11 The time elapsed, in minutes, between the placement of an order of pizza and its delivery is random with the density function

$$f(x) = \begin{cases} 1/15 & \text{if } 25 < x < 40 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine the mean and standard deviation of the time it takes for the pizza shop to deliver pizza.
- (b) Suppose that it takes 12 minutes for the pizza shop to bake pizza. Determine the mean and standard deviation of the time it takes for the delivery person to deliver pizza.

Solution:

- (a) Let the time elapsed between the placement of an order and its delivery be X minutes. Then

$$E(X) = \int_{25}^{40} x \frac{1}{15} dx = 32.5,$$

$$E(X^2) = \int_{25}^{40} x^2 \frac{1}{15} dx = 1075.$$

Therefore, $\text{Var}(X) = 1075 - (32.5)^2 = 18.75$, and hence $\sigma_X = \sqrt{\text{Var}(X)} = 4.33$.

- (b) The time it takes for the delivery person to deliver pizza is $X - 12$. Therefore, the desired quantities are

$$E(X - 12) = E(X) - 12 = 32.5 - 12 = 20.5$$

$$\sigma_{X-12} = |1|\sigma_X = \sigma_X = 4.33. \quad \blacklozenge$$

Remark 5.5 For a continuous random variable X , the moments, absolute moments, moments about a constant c , and central moments are all defined in a manner similar to those of [Section 4.5](#) (the discrete case). $\blacklozenge$

★ **Remark 5.6** In Remark 4.3, we showed that, for a discrete random variable X and positive integer n , if $E(X^{n+1})$ exists, then $E(X^n)$ also exists. That is, the existence of higher moments of a discrete random variable implies the existence of its lower moments. In particular, the existence of $E(X^2)$ implies the existence of $E(X)$ and, hence, $\text{Var}(X)$. These facts are also true for continuous random variables, and their proofs are similar. Let X be a continuous random variable with probability density function $f(x)$. Let n be a positive integer. We will show that if $E(X^{n+1})$ exists, then $E(X^n)$ also exists. Clearly,

$$\begin{aligned}\int_{-1}^1 |x|^n f(x) dx &\leq \int_{-1}^1 f(x) dx \leq \int_{-\infty}^{\infty} f(x) dx = 1; \\ \int_{|x|>1} |x|^n f(x) dx &\leq \int_{|x|>1} |x|^{n+1} f(x) dx \leq \int_{-\infty}^{\infty} |x|^{n+1} f(x) dx.\end{aligned}$$

These inequalities yield

$$\int_{-\infty}^{\infty} |x|^n f(x) dx = \int_{-1}^1 |x|^n f(x) dx + \int_{|x|>1} |x|^n f(x) dx \leq 1 + \int_{-\infty}^{\infty} |x|^{n+1} f(x) dx.$$

Therefore, $\int_{-\infty}^{\infty} |x|^{n+1} f(x) dx < \infty$ implies that $\int_{-\infty}^{\infty} |x|^n f(x) dx < \infty$. By Remark 5.2, this shows that the existence of $E(X^{n+1})$ implies the existence of $E(X^n)$. ♦

★ **Remark 5.7 Random Variables that are neither Discrete nor Continuous**

Here we discuss one such random variable. Examples 4.7 and 4.8 present two more. At a bank, when customers arrive, they take their place in a waiting line for service if all of the servers are busy. If there is no queue and a server is free, then customers will be served immediately upon arrival. Otherwise, the customer waits in the line until his or her turn for service. Let W be the waiting time of a randomly selected customer. Then W is not a continuous random variable, since $P(W = 0) > 0$ and is not a discrete random variable, since its set of possible values is the interval $[0, \infty)$, a set, which is not countable. W can take any value in $[0, \infty)$. Suppose that the probability is p that a customer who arrives at the bank finds no queue and a free server. Furthermore, suppose that if the customer has to wait to be served, then for some $\lambda > 0$, his or her waiting time is a continuous random variable, X , with probability density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Clearly,

$$W = \begin{cases} 0 & \text{with probability } p \\ X & \text{with probability } 1 - p. \end{cases}$$

Random variables such as W are called **mixtures of discrete and continuous random variables**. In general, let

$$Z = \begin{cases} Y & \text{with probability of } p \\ X & \text{with probability } 1 - p. \end{cases}$$

Then, by the methods of [Section 9.1](#), we can show that

$$E(Z) = pE(Y) + (1 - p)E(X).$$

(See Exercise 4, [Section 9.1](#).) Applying this to W , we obtain

$$E(W) = p \cdot 0 + (1 - p)E(X) = (1 - p) \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx.$$

Using integration by parts with $u = x$ and $dv = \lambda e^{-\lambda x} dx$, we have

$$\int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx = \left[-x e^{-\lambda x} \right]_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx = 0 - \left[\frac{1}{\lambda} e^{-\lambda x} \right]_0^{\infty} = \frac{1}{\lambda}.$$

Therefore, $E(W) = \frac{1 - p}{\lambda}$.

Let us now calculate $P(W \leq 5)$; that is, the probability that a customer's waiting time at the bank is at most 5. By the law of total probability,

$$\begin{aligned} P(W \leq 5) &= P(W \leq 5 \mid W = 0)P(W = 0) + P(W \leq 5 \mid W > 0)P(W > 0) \\ &= 1 \cdot p + \left(\int_0^5 \lambda e^{-\lambda x} dx \right) \cdot (1 - p) \\ &= p + (1 - p)(1 - e^{-5\lambda}). \end{aligned}$$

It should be noted that there are also random variables that are neither discrete nor continuous, and they cannot be expressed as a mixture of discrete and continuous random variables. Discussion of such random variables is beyond the scope of this book. ♦

EXERCISES

A

1. The distribution function for the duration of a certain soap opera (in tens of hours) is

$$F(x) = \begin{cases} 1 - \frac{16}{x^2} & \text{if } x \geq 4 \\ 0 & \text{if } x < 4. \end{cases}$$

- (a) Find $E(X)$.
- (b) Show that $\text{Var}(X)$ does not exist.

2. The time it takes for a student to finish an aptitude test (in hours) has the density function

$$f(x) = \begin{cases} 6(x-1)(2-x) & \text{if } 1 < x < 2 \\ 0 & \text{otherwise.} \end{cases}$$

Determine the mean and standard deviation of the time it takes for a randomly selected student to finish the aptitude test.

3. The mean and standard deviation of the lifetime of a car muffler manufactured by company A are 5 and 2 years, respectively. These quantities for car mufflers manufactured by company B are, respectively, 4 years and 18 months. Brian buys one muffler from company A and one from company B. That of company A lasts 4 years and 3 months, and that of company B lasts 3 years and 9 months. Determine which of these mufflers has performed relatively better.

Hint: Find the standardized lifetimes of the mufflers and compare (see [Section 4.6](#)).

4. At a university, which does not qualify as a tax-exempt entity under federal and state laws, each year, without taxes, it costs an average of \$56,000 with a variance of 81 million dollars to repair and maintain all the dormitories. If all items associated with the maintenance and repair of the dormitories are subject to 8.875% tax, find the standard deviation of the total cost of maintaining and repairing the dormitories, in a random year, after all the taxes are paid.
5. An actuary of an insurance company has discovered that the company's total monthly claims, in hundred thousands of dollars, is a continuous random variable proportional to $\frac{1}{(1+x)(1+x^2)}$, for $0 < x < \infty$. Find the expected value of the total amount of next month's claims.

6. A random variable X has the density function

$$f(x) = \begin{cases} 3e^{-3x} & \text{if } 0 \leq x < \infty \\ 0 & \text{otherwise.} \end{cases}$$

Calculate $E(e^X)$.

7. Find the expected value of a random variable X with the density function

$$f(x) = \begin{cases} \frac{1}{\pi\sqrt{1-x^2}} & \text{if } -1 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

8. Let Y be a continuous random variable with probability distribution function

$$F(y) = \begin{cases} e^{-k(\alpha-y)/A} & -\infty < y \leq \alpha \\ 1 & y > \alpha, \end{cases}$$

where A , k , and α are positive constants. (Such distribution functions arise in the study of local computer network performance.) Find $E(Y)$.

9. Let the probability density function of tomorrow's Celsius temperature be h . In terms of h , calculate the corresponding probability density function and its expectation for Fahrenheit temperature.

Hint: Let C and F be tomorrow's temperature in Celsius and Fahrenheit, respectively. Then $F = 1.8C + 32$.

10. Find $E(\ln X)$ if X is a continuous random variable with probability density function

$$f(x) = \begin{cases} 2/x^2 & \text{if } 1 < x < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

11. A right triangle has a hypotenuse of length 9. If the probability density function of one side's length is given by

$$f(x) = \begin{cases} x/6 & \text{if } 2 < x < 4 \\ 0 & \text{otherwise,} \end{cases}$$

what is the expected value of the length of the other side?

12. Let X be a random variable with probability density function

$$f(x) = \frac{1}{2}e^{-|x|}, \quad -\infty < x < \infty.$$

Calculate $\text{Var}(X)$.

13. This exercise is the continuous counterpart to the discrete case presented in Exercise 11, [Section 4.5](#). Let X be a continuous random variable. Show that if $E(X^2) = 0$, then $X = 0$ with probability 1.
14. Let X be a continuous random variable with probability density function $f(x)$. Show that if $E|X| = 0$, then $X = 0$ with probability 1.

B

15. Let X be a random variable with the probability density function

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

Prove that $E(|X|^\alpha)$ converges if $0 < \alpha < 1$ and diverges if $\alpha \geq 1$.

16. Suppose that X , the interarrival time between two customers entering a certain postoffice, satisfies

$$P(X > t) = \alpha e^{-\lambda t} + \beta e^{-\mu t}, \quad t \geq 0,$$

where $\alpha + \beta = 1$, $\alpha \geq 0$, $\beta \geq 0$, $\lambda > 0$, $\mu > 0$. Calculate the expected value of X .

Hint: For a fast calculation, use Remark 5.4.

17. For $n \geq 1$, let X_n be a continuous random variable with the probability density function

$$f_n(x) = \begin{cases} \frac{c_n}{x^{n+1}} & \text{if } x \geq c_n \\ 0 & \text{otherwise.} \end{cases}$$

X_n 's are called **Pareto random variables** and are used to study income distributions.

- (a) Calculate c_n , $n \geq 1$.
 - (b) Find $E(X_n)$, $n \geq 1$.
 - (c) Determine the density function of $Z_n = \ln X_n$, $n \geq 1$.
 - (d) For what values of m does $E(X_n^{m+1})$ exist?
18. Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{\pi} x \sin x & \text{if } 0 < x < \pi \\ 0 & \text{otherwise.} \end{cases}$$

Prove that

$$E(X^{n+1}) + (n+1)(n+2)E(X^{n-1}) = \pi^{n+1}.$$

19. Let X be a continuous random variable with density function f . A number t is said to be the **median** of X if

$$P(X \leq t) = P(X \geq t) = \frac{1}{2}.$$

By Exercise 10, [Section 5.1](#), X is symmetric about α if and only if for all x we have $f(\alpha - x) = f(\alpha + x)$. Show that if X is symmetric about α , then

$$E(X) = \text{Median}(X) = \alpha.$$

20. Let X be a continuous random variable with probability density function $f(x)$. Determine the value of y for which $E(|X - y|)$ is minimum.
21. Let X be a nonnegative random variable with distribution function F . Define

$$I(t) = \begin{cases} 1 & \text{if } X > t \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Prove that $\int_0^\infty I(t) dt = E(X)$.
- (b) By calculating the expected value of both sides of part (a), prove that

$$E(X) = \int_0^\infty [1 - F(t)] dt.$$

This is a special case of Theorem 5.2.

(c) For $r > 0$, use part (b) to prove that

$$E(X^r) = r \int_0^\infty t^{r-1} [1 - F(t)] dt.$$

22. Let X be a continuous random variable. Prove that

$$\sum_{n=1}^{\infty} P(|X| \geq n) \leq E(|X|) \leq 1 + \sum_{n=1}^{\infty} P(|X| \geq n).$$

These important inequalities show that $E(|X|) < \infty$ if and only if the series $\sum_{n=1}^{\infty} P(|X| \geq n)$ converges.

Hint: By Exercise 21,

$$E(|X|) = \int_0^\infty P(|X| > t) dt = \sum_{n=0}^{\infty} \int_n^{n+1} P(|X| > t) dt.$$

Note that on the interval $[n, n+1)$,

$$P(|X| \geq n+1) < P(|X| > t) \leq P(|X| \geq n).$$

23. Let X be the random variable introduced in Exercise 16. Applying the results of Exercise 21, calculate $\text{Var}(X)$.

24. Suppose that X is the lifetime of a randomly selected fan used in certain types of diesel engines. Let Y be a randomly selected competing fan for the same type of diesel engines manufactured by another company. To compare the lifetimes X and Y , it is not sufficient to compare $E(X)$ and $E(Y)$. For example, $E(X) > E(Y)$ does not necessarily imply that the first manufacture's fan outlives the second manufacture's fan. Knowing $\text{Var}(X)$ and $\text{Var}(Y)$ will help, but variance is also a crude measure. One of the best tools for comparing random variables in such situations is *stochastic comparison*. Let X and Y be two random variables. We say that X is **stochastically larger** than Y , denoted by $X \geq_{st} Y$, if for all t ,

$$P(X > t) \geq P(Y > t).$$

Show that if $X \geq_{st} Y$, then $E(X) \geq E(Y)$, but not conversely.

Hint: Use Theorem 5.2.

25. Let X be a continuous random variable with probability density function f . Show that if $E(X)$ exists; that is, if $\int_{-\infty}^{\infty} |x|f(x) dx < \infty$, then

$$\lim_{x \rightarrow -\infty} xP(X \leq x) = \lim_{x \rightarrow \infty} xP(X > x) = 0.$$



Self-Quiz on Section 5.3

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. Suppose that the weekly profit of a certain store, in thousands of dollars, is a random variable X with probability density function

$$f(x) = \begin{cases} \frac{3}{14}(x + x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise.} \end{cases}$$

- (i) What is the expected weekly profit of this store?
 - (ii) What is the probability that for 8 consecutive weeks, every week, the store's profit is less than \$500? Assume that the profit in one week is independent of the profits in other weeks.
2. Let X be the value of the claims made, in 1 year, in millions of dollars, for partial losses from fires that damage homes but do not destroy them. Suppose that the probability density function of X is given by

$$f(x) = \begin{cases} \frac{1}{4}(2 - x)^3 & \text{if } 0 < x < 2 \\ 0 & \text{otherwise.} \end{cases}$$

- (i) Find $E(X)$, $\text{Var}(X)$, and σ_X .
- (ii) Calculate the probability that the claims made in a certain year exceed 1,200,000, if we know that they exceed 800,000.



5.4 JOINT PROBABILITY DENSITY FUNCTIONS

To define the concept of *joint probability density function* of two continuous random variables X and Y , recall from [Section 5.1](#) that a single random variable X is called continuous if there exists a nonnegative real-valued function $f: \mathbb{R} \rightarrow [0, \infty)$ such that for any subset A of real numbers that can be constructed from intervals by a countable number of set operations,

$$P(X \in A) = \int_A f(x) dx.$$

This definition is generalized in the following obvious way:

Definition 5.4 Two random variables X and Y , defined on the same sample space, have a continuous joint distribution if there exists a nonnegative function of two variables, $f(x, y)$ on $\mathbb{R} \times \mathbb{R}$, such that for any region R in the xy -plane that can be formed from rectangles by a countable number of set operations,

$$P((X, Y) \in R) = \iint_R f(x, y) dx dy. \quad (5.7)$$

The function $f(x, y)$ is called the **joint probability density function** of X and Y .

Note that if in (5.7) the region R is a plane curve, then $P((X, Y) \in R) = 0$. That is, the probability that (X, Y) lies on any curve (in particular, a circle, an ellipse, a straight line, etc.) is 0.

As explained in [Chapter 4, Joint probability distribution](#), also referred to as **bivariate distribution**, is a broader term encompassing both continuous and discrete cases, while the specific terms *joint probability density function of two random variables* and *joint probability mass functions of two random variables* are used when referring to continuous and discrete distributions, respectively.

Let $R = \{(x, y) : x \in A, y \in B\}$, where A and B are any subsets of real numbers that can be constructed from intervals by a countable number of set operations. Then (5.7) gives

$$P(X \in A, Y \in B) = \int_B \int_A f(x, y) dx dy. \quad (5.8)$$

Letting $A = (-\infty, \infty)$, $B = (-\infty, \infty)$, (5.8) implies the relation

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1,$$

which is the continuous analog of (4.3). The relation (5.8) also implies that

$$P(X = a, Y = b) = \int_b^b \int_a^a f(x, y) dx dy = 0.$$

Hence, for $a < b$ and $c < d$,

$$\begin{aligned} P(a < X \leq b, c \leq Y \leq d) &= P(a < X < b, c < Y < d) \\ &= P(a \leq X < b, c \leq Y < d) = \cdots \\ &= \int_c^d \left(\int_a^b f(x, y) dx \right) dy. \end{aligned}$$

Because, for real numbers a and b , $P(X = a, Y = b) = 0$, in general, $f(a, b)$ is not equal to $P(X = a, Y = b)$. At no point do the values of f represent probabilities. Intuitively, $f(a, b)$ is a measure that determines how likely it is that X is close to a and Y is close to b . To see this, note that if ε and δ are very small positive numbers, then

$$P(a - \varepsilon < X < a + \varepsilon, b - \delta < Y < b + \delta)$$

is the probability that X is close to a and Y is close to b . Now

$$P(a - \varepsilon < X < a + \varepsilon, b - \delta < Y < b + \delta) = \int_{b-\delta}^{b+\delta} \int_{a-\varepsilon}^{a+\varepsilon} f(x, y) dx dy$$

is the volume under the surface $z = f(x, y)$, above the region $(a - \varepsilon, a + \varepsilon) \times (b - \delta, b + \delta)$. For infinitesimal ε and δ , this volume is approximately equal to the volume of a rectangular parallelepiped with sides of lengths 2ε , 2δ , and height $f(a, b)$ [i.e., $(2\varepsilon)(2\delta)f(a, b) = 4\varepsilon\delta f(a, b)$]. Therefore,

$$P(a - \varepsilon < X < a + \varepsilon, b - \delta < Y < b + \delta) \approx 4\varepsilon\delta f(a, b).$$

Hence for fixed small values of ε and δ , we observe that a larger value of $f(a, b)$ makes $P(a - \varepsilon < X < a + \varepsilon, b - \delta < Y < b + \delta)$ larger. That is, the larger the value of $f(a, b)$, the higher the probability that X and Y are close to a and b , respectively. A sketch of a joint probability density function of two continuous random variables is shown in [Figure 5.6](#). Note that $f(x, y)$ is the height of the surface at each point (x, y) in its domain.

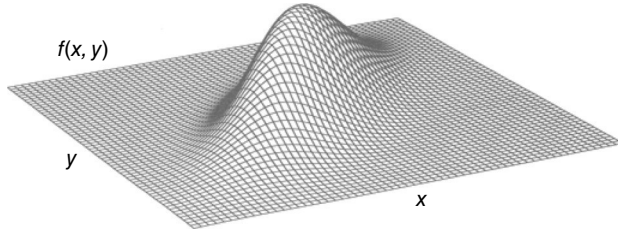


Figure 5.6 An example of a joint probability density function.

Let X and Y have joint probability density function $f(x, y)$. Let f_Y be the probability density function of Y . To find f_Y in terms of f , note that, on the one hand, for any subset B of $\mathbb{R}$,

$$P(Y \in B) = \int_B f_Y(y) dy, \quad (5.9)$$

and, on the other hand, using (5.8),

$$P(Y \in B) = P(X \in (-\infty, \infty), Y \in B) = \int_B \left(\int_{-\infty}^{\infty} f(x, y) dx \right) dy.$$

Comparing this with (5.9), we can write

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx. \quad (5.10)$$

Similarly,

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy. \quad (5.11)$$

Therefore, it is reasonable to make the following definition:

Definition 5.5 *Let X and Y have joint probability density function $f(x, y)$, then the functions f_X and f_Y , given by (5.11) and (5.10), are called, respectively, the **marginal probability density functions** of X and Y .*

Note that while from the joint probability density function of X and Y we can find the marginal probability density functions of X and Y , it is not possible, in general, to find the joint probability density function of two random variables from their marginals. This is because the marginal probability density function of a random variable X gives information about X without looking at the possible values of other random variables. However, sometimes with more information the marginal probability density functions enable us to find the joint probability density function of two random variables. An example of such a case, studied below in this section, is when X and Y are independent random variables.

Let X and Y be two random variables (discrete, continuous, or mixed). The **joint probability distribution function**, or *joint cumulative probability distribution function*, or simply the *joint distribution function* of X and Y , is defined by

$$F(t, u) = P(X \leq t, Y \leq u)$$

for all $-\infty < t, u < \infty$. The **marginal probability distribution function of X** , F_X , can be found from F as follows:

$$\begin{aligned} F_X(t) &= P(X \leq t) = P(X \leq t, Y < \infty) = P\left(\lim_{n \rightarrow \infty} \{X \leq t, Y \leq n\}\right) \\ &= \lim_{n \rightarrow \infty} P(X \leq t, Y \leq n) = \lim_{n \rightarrow \infty} F(t, n) \equiv F(t, \infty). \end{aligned}$$

To justify the fourth equality, note that the sequence of events $\{X \leq t, Y \leq n\}$, $n \geq 1$, is an increasing sequence. Therefore, by continuity of probability function (Theorem 1.8),

$$P\left(\lim_{n \rightarrow \infty} \{X \leq t, Y \leq n\}\right) = \lim_{n \rightarrow \infty} P(X \leq t, Y \leq n).$$

Similarly, F_Y , the **marginal probability distribution function of Y** , is

$$F_Y(u) = P(Y \leq u) = \lim_{n \rightarrow \infty} F(n, u) \equiv F(\infty, u).$$

Now suppose that the joint probability density function of X and Y is $f(x, y)$. Then

$$\begin{aligned} F(x, y) &= P(X \leq x, Y \leq y) = P(X \in (-\infty, x], Y \in (-\infty, y]) \\ &= \int_{-\infty}^y \int_{-\infty}^x f(t, u) dt du. \end{aligned} \tag{5.12}$$

Assuming that the partial derivatives of F exist, by differentiation of (5.12), we get

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y). \tag{5.13}$$

Moreover, from (5.12), we obtain

$$\begin{aligned} F_X(x) &= F(x, \infty) = \int_{-\infty}^{\infty} \left(\int_{-\infty}^x f(t, u) dt \right) du \\ &= \int_{-\infty}^x \left(\int_{-\infty}^{\infty} f(t, u) du \right) dt = \int_{-\infty}^x f_X(t) dt, \end{aligned}$$

and, similarly,

$$F_Y(y) = \int_{-\infty}^y f_Y(u) du.$$

These relations show that if X and Y have joint probability density function $f(x, y)$, then X and Y are continuous random variables with density functions f_X and f_Y , and distribution functions F_X and F_Y , respectively. Therefore, $F'_X(x) = f_X(x)$ and $F'_Y(y) = f_Y(y)$. We also have

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx; \quad E(Y) = \int_{-\infty}^{\infty} y f_Y(y) dy.$$

Example 5.12 The joint probability density function of random variables X and Y is given by

$$f(x, y) = \begin{cases} \lambda xy^2 & 0 \leq x \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine the value of λ .
- (b) Find the marginal probability density functions of X and Y .
- (c) Calculate $E(X)$ and $E(Y)$.

Solution:

- (a) To find λ , note that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$ gives

$$\int_0^1 \left(\int_x^1 \lambda xy^2 dy \right) dx = 1.$$

Therefore,

$$\begin{aligned} 1 &= \int_0^1 \left(\int_x^1 y^2 dy \right) \lambda x dx = \int_0^1 \left[\frac{1}{3} y^3 \right]_x^1 \lambda x dx \\ &= \lambda \int_0^1 \left(\frac{1}{3} - \frac{1}{3} x^3 \right) x dx = \frac{\lambda}{3} \int_0^1 (1 - x^3) x dx = \frac{\lambda}{3} \int_0^1 (x - x^4) dx \\ &= \frac{\lambda}{3} \left[\frac{1}{2} x^2 - \frac{1}{5} x^5 \right]_0^1 = \frac{\lambda}{10}, \end{aligned}$$

and hence $\lambda = 10$.

- (b) To find f_X and f_Y , the respective marginal probability density functions of X and Y , we use (5.11) and (5.10):

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_x^1 10xy^2 dy = \left[\frac{10}{3}xy^3 \right]_x^1 = \frac{10}{3}x(1 - x^3), \quad 0 \leq x \leq 1;$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^y 10xy^2 dx = \left[5x^2y^2 \right]_0^y = 5y^4, \quad 0 \leq y \leq 1.$$

- (c) To find $E(X)$ and $E(Y)$, we use the results obtained in part (b):

$$E(X) = \int_0^1 x \cdot \frac{10}{3}x(1 - x^3) dx = \frac{5}{9};$$

$$E(Y) = \int_0^1 y \cdot 5y^4 dy = \frac{5}{6}. \quad \blacklozenge$$

Example 5.13 For $\lambda > 0$, let

$$F(x, y) = \begin{cases} 1 - \lambda e^{-\lambda(x+y)} & \text{if } x > 0, \quad y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Determine if F is the joint probability distribution function of two random variables X and Y .

Solution: If F is the joint probability distribution function of two random variables X and Y , then $\frac{\partial^2}{\partial x \partial y} F(x, y)$ is the joint probability density function of X and Y . But

$$\frac{\partial^2}{\partial x \partial y} F(x, y) = \begin{cases} -\lambda^3 e^{-\lambda(x+y)} & \text{if } x > 0, \quad y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Since $\frac{\partial^2}{\partial x \partial y} F(x, y) < 0$, it cannot be a joint probability density function. Therefore, F is not a joint probability distribution function. $\blacklozenge$

Example 5.14 A circle of radius 1 is inscribed in a square with sides of length 2. A point is selected at random from the square. What is the probability that it is inside the circle? Note that by a point being selected at random from the square we mean that the point is selected in a way that all the subsets of equal areas of the square are equally likely to contain the point.

Solution: Let the square and the circle be situated in the coordinate system as shown in [Figure 5.7](#) on the following page. Let the coordinates of the point selected at random be (X, Y) , then X and Y are random variables. By definition, regions inside the square with equal areas are equally likely to contain (X, Y) . Hence, for all (a, b) inside the square, the probability that (X, Y) is “close” to (a, b) is the same. Let $f(x, y)$ be the joint probability density function of X and Y .

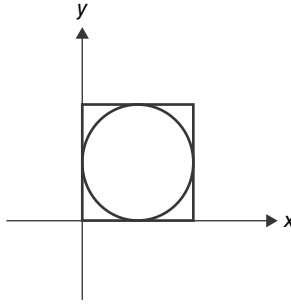


Figure 5.7 Geometric model of Example 5.14.

Since $f(x, y)$ is a measure that determines how likely it is that X is close to x and Y is close to y , $f(x, y)$ must be constant for the points inside the square, and 0 elsewhere. Therefore, for some $c > 0$,

$$f(x, y) = \begin{cases} c & \text{if } 0 < x < 2, \quad 0 < y < 2 \\ 0 & \text{otherwise,} \end{cases}$$

where $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$ gives

$$\int_0^2 \int_0^2 c dx dy = 1,$$

implying that $c = 1/4$. Now let R be the region inside the circle. Then the desired probability is

$$\iint_R f(x, y) dx dy = \iint_R \frac{1}{4} dx dy = \frac{\iint_R dx dy}{4}.$$

Note that $\iint_R dx dy$ is the area of the circle, and 4 is the area of the square. Thus the desired probability is

$$\frac{\text{area of the circle}}{\text{area of the square}} = \frac{\pi(1)^2}{4} = \frac{\pi}{4}. \quad \blacklozenge$$

What we showed in Example 5.14 is true, in general. Let S be a bounded region in the Euclidean plane and suppose that R is a region inside S . Fix a coordinate system, and let the coordinates of a point selected at random from S be (X, Y) . By an argument similar to that of

Example 5.14, we have that, for some $c > 0$, the joint probability density function of X and Y , $f(x, y)$, is given by

$$f(x, y) = \begin{cases} c & \text{if } (x, y) \in S \\ 0 & \text{otherwise,} \end{cases}$$

where $\iint_S f(x, y) dx dy = 1$. This gives $c \iint_S dx dy = 1$, or, equivalently, $c \times \text{area}(S) = 1$. Therefore, $c = 1/\text{area}(S)$ and hence

$$f(x, y) = \begin{cases} \frac{1}{\text{area}(S)} & \text{if } (x, y) \in S \\ 0 & \text{otherwise.} \end{cases} \quad (5.14)$$

Thus

$$P((X, Y) \in R) = \iint_R f(x, y) dx dy = \frac{1}{\text{area}(S)} \iint_R dx dy = \frac{\text{area}(R)}{\text{area}(S)}.$$

Based on these observations, we make the following definition.

Definition 5.6 Let S be a subset of the plane with area $A(S)$. A point is said to be **randomly selected** from S if for any subset R of S with area $A(R)$, the probability that R contains the point is $A(R)/A(S)$.

This definition is essential in the field of **geometric probability**. By the following examples, we will show how it can help to solve problems readily.

Example 5.15 A man invites his fiancée to a fine hotel for a Sunday brunch. They decide to meet in the lobby of the hotel between 11:30 A.M. and 12 noon. If they arrive at random times during this period, what is the probability that they will meet within 10 minutes?

Solution: Let X and Y be the minutes past 11:30 A.M. that the man and his fiancée arrive at the lobby, respectively. Let

$$S = \{(x, y) : 0 \leq x \leq 30, 0 \leq y \leq 30\}, \quad \text{and} \quad R = \{(x, y) \in S : |x - y| \leq 10\}.$$

Then the desired probability, $P(|X - Y| \leq 10)$, is given by

$$P(|X - Y| \leq 10) = \frac{\text{area of } R}{\text{area of } S} = \frac{\text{area of } R}{30 \times 30} = \frac{\text{area}(R)}{900}.$$

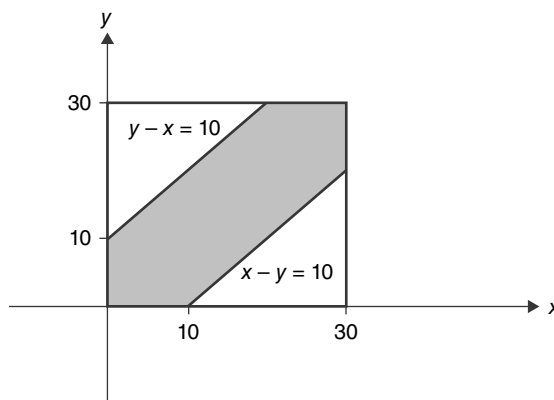


Figure 5.8 Geometric model of Example 5.15.

$R = \{(x, y) \in S: x - y \leq 10 \text{ and } y - x \leq 10\}$ is the shaded region of Figure 5.8, and its area is the area of the square minus the areas of the two unshaded triangles:
 $(30)(30) - 2(1/2 \times 20 \times 20) = 500$. Hence the desired probability is $500/900 = 5/9$. ♦

Example 5.16 A farmer decides to build a pen in the shape of a triangle for his chickens. He sends his son out to cut the lumber and the boy, without taking any thought as to the ultimate purpose, makes two cuts at two points selected at random. What are the chances that the resulting three pieces of lumber can be used to form a triangular pen?

Solution: Suppose that the length of the lumber is ℓ . Let A and B be the random points placed on the lumber; let the distances of A and B from the left end of the lumber be denoted by X and Y , respectively. If $X < Y$, the lumber is divided into three parts of lengths X , $Y - X$, and $\ell - Y$; otherwise, it is divided into three parts of lengths Y , $X - Y$, and $\ell - X$. Because of symmetry, we calculate the probability that $X < Y$, and X , $Y - X$, and $\ell - Y$ form a triangle. Then we multiply the result by 2 to obtain the desired probability. We know that three segments form a triangle if and only if the length of any one of them is less than the sum of the lengths of the remaining two. Therefore, we must have

$$\begin{aligned} X &< Y \\ X &< (Y - X) + (\ell - Y) \\ (Y - X) &< X + (\ell - Y) \\ \ell - Y &< X + (Y - X), \end{aligned}$$

or, equivalently,

$$X < Y, \quad X < \frac{\ell}{2}, \quad Y < X + \frac{\ell}{2}, \quad \text{and} \quad Y > \frac{\ell}{2}.$$

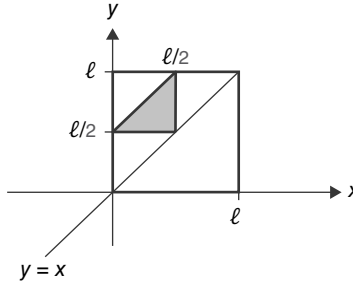


Figure 5.9 Geometric model of Example 5.16.

Since (X, Y) is a random point from the square $(0, \ell) \times (0, \ell)$, the probability that it satisfies these inequalities is the area of R , where

$$R = \left\{ (x, y) : x < y, x < \frac{\ell}{2}, y < x + \frac{\ell}{2}, y > \frac{\ell}{2} \right\},$$

divided by ℓ^2 , the area of the square. Note that R is the shaded region of Figure 5.9 and its area is $1/8$ of the area of the square. Thus the probability that (X, Y) lies in R is $1/8$ and hence the desired probability is $1/4$. ♦

We now state the continuous analog of Theorem 4.7, which is a generalization of Theorem 5.3 from one dimension to two. As an immediate application of this important generalization, We will demonstrate the continuous version of the fact that the expected value of the sum of two discrete random variables is equal to the sum of their expected values.

Theorem 5.4 Let $f(x, y)$ be the joint probability density function of random variables X and Y . If h is a function of two variables from $\mathbb{R}^2$ to $\mathbb{R}$, then $h(X, Y)$ is a random variable with the expected value given by

$$E[h(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dx dy,$$

provided that the integral is absolutely convergent.

Corollary For random variables X and Y ,

$$E(X + Y) = E(X) + E(Y).$$

Proof: In Theorem 5.4 let $h(x, y) = x + y$. Then

$$\begin{aligned} E(X + Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy \\ &= E(X) + E(Y). \quad \blacklozenge \end{aligned}$$

Example 5.17 Let X and Y have joint probability density function

$$f(x, y) = \begin{cases} \frac{3}{2}(x^2 + y^2) & \text{if } 0 < x < 1, \quad 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

The geometric representation of f is a surface, a portion of which is shown in [Figure 5.10](#) on the next page. Find $E(X^2 + Y^2)$.

Solution: By Theorem 5.4,

$$\begin{aligned} E(X^2 + Y^2) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x^2 + y^2) f(x, y) dx dy = \int_0^1 \int_0^1 \frac{3}{2}(x^2 + y^2)^2 dx dy \\ &= \frac{3}{2} \int_0^1 \int_0^1 (x^4 + 2x^2y^2 + y^4) dx dy = \frac{14}{15}. \quad \blacklozenge \end{aligned}$$

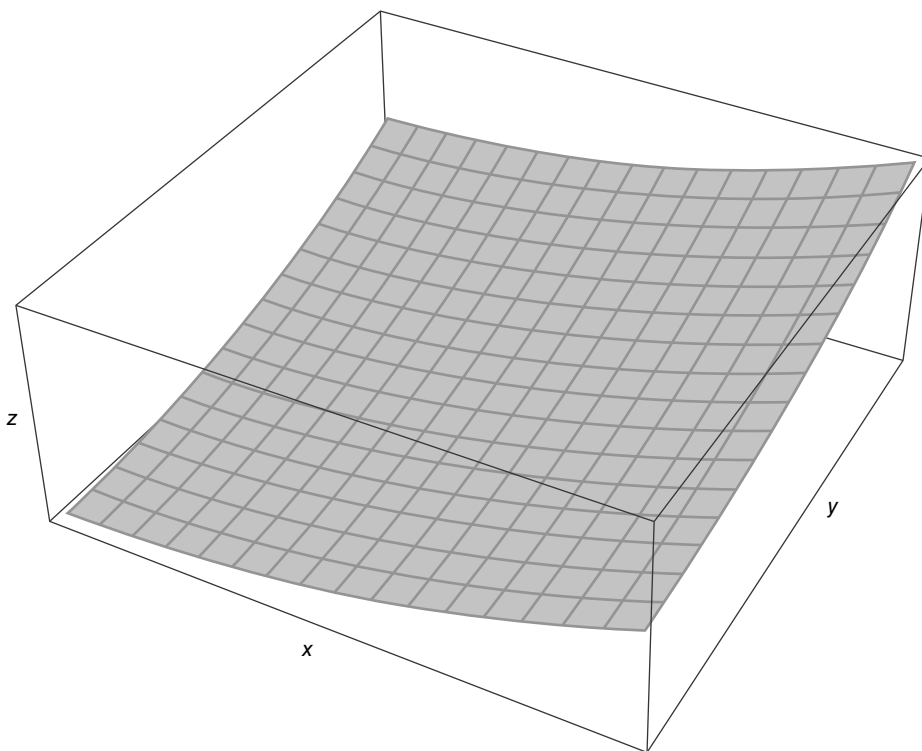


Figure 5.10 The surface representing f in Example 5.17.

We now proceed with the definition of the concept of a jointly continuous density function for more than two random variables.

Definition 5.7 Let $X_1, X_2, \dots, X_n$ be continuous random variables defined on the same sample space. We say that $X_1, X_2, \dots, X_n$ have a **continuous joint distribution** if there exists a nonnegative function of n variables, $f(x_1, x_2, \dots, x_n)$, on $\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R} \equiv \mathbb{R}^n$ such that for any region R in $\mathbb{R}^n$ that can be formed from n -dimensional rectangles by a countable number of set operations,

$$P((X_1, X_2, \dots, X_n) \in R) = \int \cdots \int_R f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n. \quad (5.15)$$

The function $f(x_1, x_2, \dots, x_n)$ is called the **joint probability density function** of $X_1, X_2, \dots, X_n$.

As explained in [Chapter 4](#), **Joint probability distribution**, also referred to as **multivariate distribution**, is a broader term encompassing both continuous and discrete cases. The specific terms *joint probability density function of several random variables* and *joint probability mass functions of several random variables* are used when referring to continuous and discrete distributions, respectively.

Let $R = \{(x_1, x_2, \dots, x_n) : x_i \in A_i, 1 \leq i \leq n\}$, where $A_i, 1 \leq i \leq n$, is any subset of real numbers that can be constructed from intervals by a countable number of set operations. Then (5.15) gives

$$\begin{aligned} P(X_1 \in A_1, X_2 \in A_2, \dots, X_n \in A_n) \\ = \int_{A_n} \int_{A_{n-1}} \cdots \int_{A_1} f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n. \end{aligned}$$

Letting $A_i = (-\infty, +\infty)$, $1 \leq i \leq n$, this implies that

$$\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n = 1.$$

Let f_{X_i} be the marginal probability density function of X_i , $1 \leq i \leq n$. Then

$$f_{X_i}(x_i) = \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{n-1 \text{ terms}} f(x_1, x_2, \dots, x_n) dx_1 \cdots dx_{i-1} dx_{i+1} \cdots dx_n. \quad (5.16)$$

Therefore, for instance,

$$f_{X_3}(x_3) = \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{n-1 \text{ terms}} f(x_1, x_2, \dots, x_n) dx_1 dx_2 dx_4 \cdots dx_n.$$

The following is the continuous analog of Theorem 4.8. It is also a generalization of Theorem 5.4 from dimension 2 to n .

Theorem 5.5 *Let $f(x_1, x_2, \dots, x_n)$ be the joint probability density function of random variables $X_1, X_2, \dots, X_n$. If h is a function of these n variables from $\mathbb{R}^n$ to $\mathbb{R}$, then $Y = h(X_1, X_2, \dots, X_n)$ is a random variable with expected value given by*

$$E(Y) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} h(x_1, x_2, \dots, x_n) f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n,$$

provided that the integral is absolutely convergent.

Example 5.18

- (a) Prove that the following is a joint probability density function.

$$f(x, y, z, t) = \begin{cases} \frac{1}{xyz} & \text{if } 0 < t \leq z \leq y \leq x \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

- (b) Suppose that f is the joint probability density function of random variables X, Y, Z , and T . Find $f_Z(z)$.

Solution:

- (a) Since $f(x, y, z, t) \geq 0$ and

$$\begin{aligned} \int_0^1 \int_0^x \int_0^y \int_0^z \frac{1}{xyz} dt dz dy dx &= \int_0^1 \int_0^x \int_0^y \frac{1}{xy} dz dy dx \\ &= \int_0^1 \int_0^x \frac{1}{x} dy dx = \int_0^1 dx = 1, \end{aligned}$$

f is a joint probability density function.

- (b) To find $f_Z(z)$, we have that for $0 < z \leq 1$,

$$\begin{aligned} f_Z(z) &= \int_z^1 \int_z^x \int_0^z \frac{1}{xyz} dt dy dx = \int_z^1 \int_z^x \frac{1}{xy} dy dx \\ &= \int_z^1 \left[\frac{1}{x} \ln y \right]_z^x dx = \int_z^1 \left(\frac{1}{x} \ln x - \frac{1}{x} \ln z \right) dx \\ &= \left[\frac{1}{2} (\ln x)^2 - (\ln x)(\ln z) \right]_z^1 = \frac{1}{2} (\ln z)^2. \end{aligned}$$

Thus

$$f_Z(z) = \begin{cases} \frac{1}{2} (\ln z)^2 & \text{if } 0 < z \leq 1 \\ 0 & \text{otherwise.} \quad \blacklozenge \end{cases}$$

Earlier, we showed how Definition 5.6 was justified. The following analogue of that definition in three dimensions is justified similarly.

Definition 5.8 *Let S be a subset of the three dimensional Euclidean space with volume $\text{Vol}(S) \neq 0$. A point is said to be **randomly selected** from S if for any subset Ω of S with volume $\text{Vol}(\Omega)$, the probability that Ω contains the point is $\text{Vol}(\Omega)/\text{Vol}(S)$.*

Let X , Y , and Z be the coordinates of the point randomly selected from S . Let $f(x, y, z)$ be the joint probability density function of X , Y , and Z . By an argument similar to the two dimensional case, which resulted in (5.14), we have that

$$f(x, y, z) = \begin{cases} \frac{1}{\text{Vol}(S)} & \text{if } (x, y, z) \in S \\ 0 & \text{otherwise.} \end{cases} \quad (5.17)$$

Let us now proceed with the proof of the linearity property of expectation for continuous random variables.

Theorem 5.6 *For random variables $X_1, X_2, \dots, X_n$ defined on the same sample space, and for real numbers $\alpha_1, \alpha_2, \dots, \alpha_n$,*

$$E\left(\sum_{i=1}^n \alpha_i X_i\right) = \sum_{i=1}^n \alpha_i E(X_i).$$

Proof: Let $f(x_1, x_2, \dots, x_n)$ be the joint probability density function of $X_1, X_2, \dots, X_n$, then, by Theorem 5.5,

$$\begin{aligned} E\left(\sum_{i=1}^n \alpha_i X_i\right) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left(\sum_{i=1}^n \alpha_i x_i\right) f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n \\ &= \sum_{i=1}^n \alpha_i \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} x_i f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n \\ &= \sum_{i=1}^n \alpha_i E(X_i). \quad \blacklozenge \end{aligned}$$

In Theorem 5.6, letting $\alpha_i = 1$, $1 \leq i \leq n$, we have the following important corollary.

Corollary *Let $X_1, X_2, \dots, X_n$ be random variables on the same sample space. Then*

$$E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n).$$

The following theorem, known as the *Cauchy-Schwarz Inequality*, holds significant importance in probability theory and mathematics. It provides valuable insights into the relationships

between random variables and finds wide-ranging applications in various mathematical and statistical contexts. This inequality serves as the foundation for other essential inequalities and offers bounds for the product of expectations, making it particularly useful when dealing with complex probability distributions.

Theorem 5.7 (Cauchy-Schwarz Inequality) *For random variables X and Y with finite variances,*

$$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}.$$

Proof: For any real number t , $(X - tY)^2 \geq 0$. Hence for all values of t ,

$$X^2 - 2XYt + t^2Y^2 \geq 0.$$

Since nonnegative random variables have nonnegative expected values,

$$E(X^2 - 2XYt + t^2Y^2) \geq 0,$$

which implies that

$$E(Y^2)t^2 - 2E(XY)t + E(X^2) \geq 0.$$

This inequality is valid for all real numbers t including the minimum of the function $g(t) = E(Y^2)t^2 - 2E(XY)t + E(X^2)$, which is obtained by solving $g'(t) = 0$ for t . Now $g'(t) = 0$ implies that $2E(Y^2)t - 2E(XY) = 0$ or $t = E(XY)/E(Y^2)$. Plugging this into the inequality above gives

$$E(Y^2) \cdot \frac{[E(XY)]^2}{[E(Y^2)]^2} - 2E(XY) \cdot \frac{E(XY)}{E(Y^2)} + E(X^2) \geq 0,$$

or

$$[E(XY)]^2 \leq E(X^2)E(Y^2).$$

Taking square root of both sides gives the Cauchy-Schwarz inequality:

$$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}. \quad \blacklozenge$$

Corollary *For a random variable X , $[E(X)]^2 \leq E(X^2)$.*

Proof: In Cauchy-Schwarz's inequality, let $Y = 1$, then

$$|E(X)| = |E(XY)| \leq \sqrt{E(X^2)E(1)} = \sqrt{E(X^2)};$$

thus $[E(X)]^2 \leq E(X^2)$. $\blacklozenge$

EXERCISES

A

1. Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} 2 & \text{if } 0 \leq y \leq x \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Calculate the marginal probability density functions of X and Y , respectively.
 (b) Find $E(X)$ and $E(Y)$.
 (c) Calculate $P(X < 1/2)$, $P(X < 2Y)$, and $P(X = Y)$.
2. Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} 8xy & \text{if } 0 \leq y \leq x \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Calculate the marginal probability density functions of X and Y , respectively.
 (b) Calculate $E(X)$ and $E(Y)$.
3. Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} \frac{1}{2}ye^{-x} & \text{if } x > 0, \quad 0 < y < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Find the marginal probability density functions of X and Y .

4. Let X and Y have the joint probability density function

$$f(x, y) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Calculate $P(X + Y \leq 1/2)$, $P(X - Y \leq 1/2)$, $P(XY \leq 1/4)$, and $P(X^2 + Y^2 \leq 1)$.

5. Let X be the proportion of customers of an insurance company who bundle their auto and home insurance policies. Let Y be the proportion of customers who insure at least their car with the insurance company. An actuary has discovered that for, $0 \leq x \leq y \leq 1$, the joint distribution function of X and Y is $F(x, y) = x(y^2 + xy - x^2)$. Find the expected value of the proportion of the customers of the insurance company who bundle their auto and home insurance policies.

6. Let F be the joint probability distribution function of random variables X and Y . For $x_1 < x_2$ and $y_1 < y_2$, in terms of F , calculate $P(x_1 < X \leq x_2, y_1 < Y \leq y_2)$.
7. Let F be the joint probability distribution function of the random variables X and Y . In terms of F , calculate $P(X > t, Y > u)$.
8. Let R be the bounded region between $y = x$ and $y = x^2$. A random point (X, Y) is selected from R .
 - (a) Find the joint probability density function of X and Y .
 - (b) Calculate the marginal probability density functions of X and Y .
 - (c) Find $E(X)$ and $E(Y)$.
9. A man invites his fiancée to an elegant hotel for a Sunday brunch. They decide to meet in the lobby of the hotel between 11:30 A.M. and 12 noon. If they arrive at random times during this period, what is the probability that the first to arrive has to wait at least 12 minutes?
10. A farmer makes cuts at two points selected at random on a piece of lumber of length ℓ . What is the expected value of the length of the middle piece?
11. On a line segment AB of length ℓ , two points C and D are placed at random and independently. What is the probability that C is closer to D than to A ?
12. Two points X and Y are selected at random and independently from the interval $(0, 1)$. Calculate $P(Y \leq X \text{ and } X^2 + Y^2 \leq 1)$.
13. For $\lambda > 0$, let

$$F(x, y) = \begin{cases} 1 - \lambda e^{-\lambda(x+y)} & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Determine if F is the joint probability distribution function of two random variables X and Y .

14. Let

$$F(x, y) = \begin{cases} 0 & x \leq 0 \text{ or } y \leq 0 \\ \frac{1}{2} xy(x+y) & 0 < x \leq 1, 0 < y \leq 1 \\ \frac{1}{2} x(x+1) & 0 \leq x \leq 1, y \geq 1 \\ \frac{1}{2} y(y+1) & x \geq 1, 0 < y \leq 1 \\ 1 & x \geq 1, y \geq 1. \end{cases}$$

Determine if F is the joint probability distribution function of two random variables X and Y .

15. Let X and Y have the joint probability density function

$$f(x, y) = \begin{cases} \frac{3}{2}(x^2 + y^2) & 0 < x < 1 \text{ and } 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the joint probability distribution function, the marginal distribution functions, and the marginal probability density functions of X and Y .

16. Let the joint probability density function of X , Y , and Z be given by

$$f(x, y, z) = \begin{cases} 6e^{-x-y-z} & \text{if } 0 < x < y < z < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X)$.

17. (a) Show that the following is a joint probability density function.

$$f(x, y, z) = \begin{cases} -\frac{\ln x}{xy} & \text{if } 0 < z \leq y \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

- (b) Suppose that f is the joint probability density function of X , Y , and Z . Find $f_Y(y)$.

18. A point is selected at random from the cube

$$\Omega = \{(x, y, z) : -a \leq x \leq a, -a \leq y \leq a, -a \leq z \leq a\}.$$

What is the probability that it is inside the sphere inscribed in the cube?

19. Is the following a joint probability density function?

$$f(x_1, x_2, \dots, x_n) = \begin{cases} e^{-x_n} & \text{if } 0 < x_1 < x_2 < \dots < x_n \\ 0 & \text{otherwise.} \end{cases}$$

20. Let the probability density function of a random variable X be given by

$$f(x) = \begin{cases} |x - 1| & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X^2 + X)$.

21. A calculator is able to generate random numbers from the interval $(0, 1)$. We need five random numbers from $(0, 2/5)$. Using this calculator, how many independent random numbers should we generate, on average, to find the five numbers needed?

22. Let the joint probability density function of random variables X and Y be

$$f(x, y) = \begin{cases} 2e^{-(x+2y)} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X)$, $E(Y)$, and $E(X^2 + Y^2)$.

23. Let $X_1, X_2, \dots, X_n$ be positive, identically distributed random variables. For $1 \leq i \leq n$, let

$$Z_i = \frac{X_i}{X_1 + X_2 + \dots + X_n}.$$

Show that $Z_1, Z_2, \dots, Z_n$ are also identically distributed and, for $1 \leq i \leq n$, find $E(Z_i)$.

24. Let X be the number of students in Dr. Lisa Kerr's English 101 who will fail the course next semester. Show that

$$P(X = 0) \leq \frac{\text{Var}(X)}{E(X^2)}.$$

Hint: Let

$$Y = \begin{cases} 1 & \text{if } X > 0 \\ 0 & \text{if } X = 0 \end{cases}$$

and apply the Cauchy-Schwarz inequality.

B

25. Suppose that h is the probability density function of a continuous random variable. Let the joint probability density function of two random variables X and Y be given by

$$f(x, y) = h(x)h(y), \quad x \in \mathbb{R}, \quad y \in \mathbb{R}.$$

Prove that $P(X \geq Y) = 1/2$.

26. Let X and Y be random variables with finite expectations. Show that if $P(X \leq Y) = 1$, then $E(X) \leq E(Y)$.
27. Let g and h be two probability density functions with probability distribution functions G and H , respectively. Show that for $-1 \leq \alpha \leq 1$, the function

$$f(x, y) = g(x)h(y)(1 + \alpha[2G(x) - 1][2H(y) - 1])$$

is a joint probability density function of two random variables. Moreover, prove that g and h are the marginal probability density functions of f .

Note: This exercise gives an infinite family of joint probability density functions all with the same marginal probability density function of X and of Y .

28. Three points M , N , and L are placed on a circle at random and independently. What is the probability that MNL is an acute angle?

29. Two numbers x and y are selected at random from the interval $(0, 1)$. For $i = 0, 1, 2$, determine the probability that the integer nearest to $x + y$ is i .

Note: This problem was given by Hilton and Pedersen in the paper “A Role for Untraditional Geometry in the Curriculum,” published in the December 1989 issue of *Kolloquium Mathematik-Didaktik der Universität Bayreuth*. It is not hard to solve, but it has interesting consequences in arithmetic.

30. A farmer who has two pieces of lumber of lengths a and b ($a < b$) decides to build a pen in the shape of a triangle for his chickens. He sends his son out to cut the lumber and the boy, without taking any thought as to the ultimate purpose, makes two cuts, one on each piece, at randomly selected points. He then chooses three of the resulting pieces at random and take them to his father. What are the chances that they can be used to form a triangular pen?
31. Two points are placed on a segment of length ℓ independently and at random to divide the line into three parts. What is the probability that the length of none of the three parts exceeds a given value α , $\ell/3 \leq \alpha \leq \ell$?

32. A point is selected at random and uniformly from the region

$$R = \{(x, y) : |x| + |y| \leq 1\}.$$

Find the probability density function of the x -coordinate of the point selected at random.

33. Let X and Y be continuous random variables with joint probability density function $f(x, y)$. Let $Z = Y/X$, $X \neq 0$. Prove that the probability density function of Z is given by

$$f_Z(z) = \int_{-\infty}^{\infty} |x| f(x, xz) dx.$$

34. Consider a disk centered at O with radius R . Suppose that $n \geq 3$ points $P_1, P_2, \dots, P_n$ are independently placed at random inside the disk. Find the probability that all these points are contained in a closed semicircular disk.

Hint: For each $1 \leq i \leq n$, let A_i be the endpoint of the *radius* through P_i and let B_i be the corresponding antipodal point. Let D_i be the closed semicircular disk OA_iB_i with negatively (clockwise) oriented boundary. Note that there is at most one D_i , $1 \leq i \leq n$, that contains all the P_i 's.

35. As Liu Wen from Hebei University of Technology in Tianjin, China, has noted in the April 2001 issue of *The American Mathematical Monthly*, in some reputable probability and statistics texts it has been asserted that “if a two-dimensional distribution function $F(x, y)$ has a continuous density of $f(x, y)$, then

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}. \quad (5.18)$$

Furthermore, some intermediate textbooks in probability and statistics even assert that at a point of continuity for $f(x, y)$, $F(x, y)$ is twice differentiable, and (5.18) holds at

that point.” Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} 1 + 2xe^y & \text{if } y \geq 0, (-1/2)e^{-y} \leq x \leq 0 \\ 1 - 2xe^y & \text{if } y \geq 0, 0 \leq x \leq (1/2)e^{-y} \\ 1 + 2xe^{-y} & \text{if } y \leq 0, (-1/2)e^y \leq x \leq 0 \\ 1 - 2xe^{-y} & \text{if } y \leq 0, 0 \leq x \leq (1/2)e^y \\ 0 & \text{otherwise.} \end{cases}$$

Show that even though f is continuous everywhere, the partial derivatives of its distribution function F do not exist at $(0, 0)$. This counterexample is constructed based on a general example given by Liu Wen in the aforementioned paper.

- 36.** Suppose that h is the probability density function of a continuous random variable. Let the joint probability density function of X , Y , and Z be

$$f(x, y, z) = h(x)h(y)h(z), \quad x, y, z \in \mathbb{R}.$$

Prove that $P(X < Y < Z) = 1/6$.

- 37.** Let X and Y be nonnegative random variables with an arbitrary joint probability distribution function. Let

$$I(x, y) = \begin{cases} 1 & \text{if } X > x, \quad Y > y \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Show that

$$\int_0^\infty \int_0^\infty I(x, y) dx dy = XY.$$

- (b) By calculating expected values of both sides of part (a), prove that

$$E(XY) = \int_0^\infty \int_0^\infty P(X > x, Y > y) dx dy.$$

Note that this is a generalization of the result explained in Remark 5.4.

- 38.** Under what condition does Cauchy-Schwarz's inequality become equality?



Self-Quiz on Section 5.4

Time allotted: 25 Minutes

Each problem is worth 5 points.

1. A device with two components functions only if both components function. Suppose that the joint probability density function of X and Y , the lifetimes of the components, in years, is

$$f(x, y) = \begin{cases} \frac{1}{60}(x + y) & \text{if } 0 < x < 3 \text{ and } 0 < y < 5 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the probability that the device functions for at least 2 years.
 (b) Find the marginal distribution function of X .
2. Let X be a car insurance company's annual losses under medical coverage insurance. Let Y be its annual losses under comprehensive and collisions losses. Suppose that the joint probability density function of X and Y , in millions of dollars, is

$$f(x, y) = \begin{cases} \frac{1}{432}(2x + 3y) & 1 < x < 5, 1 < y < 7 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the marginal probability density function of Y .
 (b) Find the expected value of the sum of the annual losses of the company under medical coverage and under comprehensive and collision.
3. Let X , Y , and Z be continuous random variables with the joint probability density function given by

$$f(x, y, z) = \begin{cases} 2y e^{-(2x+y+z)} & x > 0, y > 0, z > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find $P(X < Y < Z)$.



5.5 INDEPENDENCE OF CONTINUOUS RANDOM VARIABLES

Two random variables X and Y are called **independent** if, for arbitrary subsets A and B of real numbers, the events $\{X \in A\}$ and $\{Y \in B\}$ are independent, that is, if

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B). \quad (5.19)$$

Using the axioms of probability, we can prove that X and Y are independent if and only if for any two real numbers a and b ,

$$P(X \leq a, Y \leq b) = P(X \leq a)P(Y \leq b). \quad (5.20)$$

Hence (5.19) and (5.20) are equivalent.

Relation (5.20) states that X and Y are independent random variables if and only if their joint probability distribution function is the product of their marginal distribution functions. The following theorem states this fact.

Theorem 5.8 *Let X and Y be two random variables defined on the same sample space. If F is the joint probability distribution function of X and Y , then X and Y are independent if and only if for all real numbers t and u ,*

$$F(t, u) = F_X(t)F_Y(u).$$

For jointly continuous *independent* random variables X and Y having joint probability distribution function $F(x, y)$ and joint probability density function $f(x, y)$, Theorem 5.8 implies that, for any two real numbers x and y ,

$$F(x, y) = F_X(x)F_Y(y). \quad (5.21)$$

Differentiating (5.21) with respect to x , we get

$$\frac{\partial F}{\partial x} = f_X(x)F_Y(y),$$

which upon differentiation with respect to y yields

$$\frac{\partial^2 F}{\partial y \partial x} = f_X(x)f_Y(y).$$

By (5.13), $\frac{\partial^2 F}{\partial y \partial x} = f(x, y)$. Thus,

$$f(x, y) = f_X(x)f_Y(y). \quad (5.22)$$

We leave it as an exercise that, if (5.21) is valid, then X and Y are independent. We have the following theorem:

Theorem 5.9 *Let X and Y be jointly continuous random variables with joint probability density function $f(x, y)$. Then X and Y are independent if and only if $f(x, y)$ is the product of their marginal densities $f_X(x)$ and $f_Y(y)$.*

Example 5.19 A point is selected at random from the rectangle

$$R = \{(x, y) \in \mathbb{R}^2 : 0 < x < a, 0 < y < b\}.$$

Let X be the x -coordinate and Y be the y -coordinate of the point selected. Determine if X and Y are independent random variables.

Solution: We know that $f(x, y)$, the joint probability density function of X and Y , is given by

$$f(x, y) = \begin{cases} \frac{1}{\text{area}(R)} = \frac{1}{ab} & \text{if } (x, y) \in R \\ 0 & \text{elsewhere.} \end{cases}$$

Now the marginal density functions of X and Y , f_X and f_Y , are given by

$$f_X(x) = \int_0^b \frac{1}{ab} dy = \frac{1}{a}, \quad x \in (0, a),$$

$$f_Y(y) = \int_0^a \frac{1}{ab} dx = \frac{1}{b}, \quad y \in (0, b).$$

Therefore, $f(x, y) = f_X(x)f_Y(y)$, $\forall x, y \in \mathbb{R}$, and hence X and Y are independent. ♦

Example 5.20 Stores A and B , which belong to the same owner, are located in two different towns. If the probability density function of the weekly profit of each store, in thousands of dollars, is given by

$$f(x) = \begin{cases} x/4 & \text{if } 1 < x < 3 \\ 0 & \text{otherwise,} \end{cases}$$

and the profit of one store is independent of the other, what is the probability that next week one store makes at least \$500 more than the other store?

Solution: Let X and Y denote next week's profits of A and B , respectively. The desired probability is $P(X > Y + 1/2) + P(Y > X + 1/2)$. Since X and Y have the same probability density function, by symmetry, this sum equals $2P(X > Y + 1/2)$. To calculate this, we need to know $f(x, y)$, the joint probability density function of X and Y . Since X and Y are independent, $f(x, y) = f_X(x)f_Y(y)$, where

$$f_X(x) = \begin{cases} x/4 & \text{if } 1 < x < 3 \\ 0 & \text{otherwise,} \end{cases} \quad f_Y(y) = \begin{cases} y/4 & \text{if } 1 < y < 3 \\ 0 & \text{otherwise.} \end{cases}$$

Thus

$$f(x, y) = \begin{cases} xy/16 & \text{if } 1 < x < 3, 1 < y < 3 \\ 0 & \text{otherwise.} \end{cases}$$

The desired probability, as seen from Figure 5.11, is therefore equal to

$$\begin{aligned} 2P\left(X > Y + \frac{1}{2}\right) &= 2P\left((X, Y) \in \left\{(x, y) : \frac{3}{2} < x < 3, 1 < y < x - \frac{1}{2}\right\}\right) \\ &= 2 \int_{3/2}^3 \left(\int_1^{x-1/2} \frac{xy}{16} dy \right) dx = \frac{1}{8} \int_{3/2}^3 \left[\frac{xy^2}{2} \right]_1^{x-1/2} dx \\ &= \frac{1}{16} \int_{3/2}^3 x \left[\left(x - \frac{1}{2}\right)^2 - 1 \right] dx \\ &= \frac{1}{16} \int_{3/2}^3 \left(x^3 - x^2 - \frac{3}{4}x \right) dx \\ &= \frac{1}{16} \left[\frac{1}{4}x^4 - \frac{1}{3}x^3 - \frac{3}{8}x^2 \right]_{3/2}^3 = \frac{549}{1024} \approx 0.54. \quad \blacklozenge \end{aligned}$$

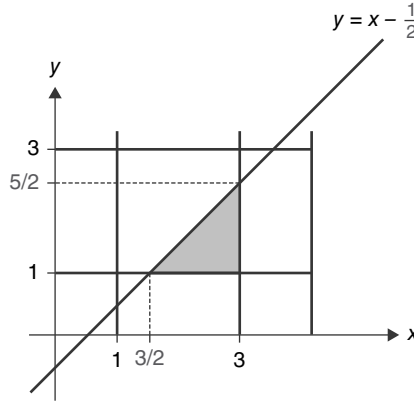


Figure 5.11 Figure of Example 5.20.

We now explain one of the most interesting problems of geometric probability, **Buffon's needle problem**. In 10, we will show how the solution of this problem and the Monte Carlo method can be used to find estimations for π by simulation. Georges Louis Buffon (1707–1784), who proposed and solved this famous problem, was a French naturalist who, in the eighteenth century, used probability to study natural phenomena. In addition to the needle problem, his studies of the distribution and expectation of the remaining lifetimes of human beings are famous among mathematicians. These works, together with many more, are published in his gigantic 44-volume, *Histoire Naturelle* (Natural History; 1749–1804).

Example 5.21 (Buffon's Needle Problem) A plane is ruled with parallel lines a distance d apart. A needle of length ℓ , $\ell < d$, is tossed at random onto the plane. What is the probability that the needle intersects one of the parallel lines?

Solution: Let X denote the distance from the center of the needle to the closest line, and let Θ denote the angle between the needle and the line. The position of the needle is completely determined by the coordinates X and Θ .

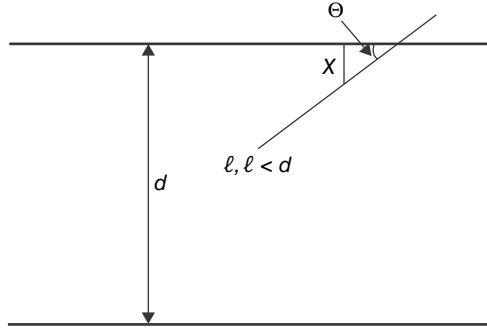


Figure 5.12 Buffon's needle problem.

As [Figure 5.12](#) shows, the needle intersects the line if and only if the length of the hypotenuse of the triangle, $X/\sin \Theta$, is less than $\ell/2$. Since X and Θ are independent uniform random variables over the intervals $(0, d/2)$ and $(0, \pi)$, respectively, the probability that the needle intersects a line is $\iint_R f(x, \theta) dx d\theta$, where $f(x, \theta)$ is the joint probability density function of X and Θ and is given by

$$f(x, \theta) = f_X(x)f_\Theta(\theta) = \begin{cases} \frac{2}{\pi d} & \text{if } 0 \leq \theta \leq \pi, 0 \leq x \leq \frac{d}{2}. \\ 0 & \text{elsewhere} \end{cases}$$

and

$$R = \left\{ (x, \theta) : \frac{x}{\sin \theta} < \frac{\ell}{2} \right\} = \left\{ (x, \theta) : x < \frac{\ell}{2} \sin \theta \right\}.$$

Thus the desired probability is

$$\iint_R f(x, \theta) dx d\theta = \int_0^\pi \int_0^{(\ell/2) \sin \theta} \frac{2}{\pi d} dx d\theta = \frac{\ell}{\pi d} \int_0^\pi \sin \theta d\theta = \frac{2\ell}{\pi d}. \quad \blacklozenge$$

Example 5.22 Prove that two random variables X and Y with the following joint probability density function are not independent.

$$f(x, y) = \begin{cases} 8xy & 0 \leq x \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Solution: To check the validity of (5.21), we first calculate f_X and f_Y .

$$f_X(x) = \int_x^1 8xy \, dy = 4x(1 - x^2), \quad 0 \leq x \leq 1,$$

$$f_Y(y) = \int_0^y 8xy \, dx = 4y^3, \quad 0 \leq y \leq 1.$$

Now since $f(x, y) \neq f_X(x)f_Y(y)$, X and Y are dependent. This is expected because of the range $0 \leq x \leq y \leq 1$, which is not a Cartesian product of one-dimensional regions. ♦

The results of Theorems 4.12 and 4.13 are valid for continuous random variables as well:

Let X and Y be independent continuous random variables and $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$ be real-valued functions, then $g(X)$ and $h(Y)$ are also independent random variables.

The proof of this fact is identical to the proof of Theorem 4.12. The continuous analog of Theorem 4.13 and its proof are as follows:

Let X and Y be independent continuous random variables. Then for all real-valued functions $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$,

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)],$$

where, as usual, we assume that $E[g(X)]$ and $E[h(Y)]$ are finite.

Proof: Let $f(x, y)$ be the joint probability density function of X and Y . Then

$$\begin{aligned} E[g(X)h(Y)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x)h(y)f(x, y) \, dx \, dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x)h(y)f_X(x)f_Y(y) \, dx \, dy \\ &= \int_{-\infty}^{\infty} h(y)f_Y(y) \left(\int_{-\infty}^{\infty} g(x)f_X(x) \, dx \right) dy \\ &= \left(\int_{-\infty}^{\infty} g(x)f_X(x) \, dx \right) \left(\int_{-\infty}^{\infty} h(y)f_Y(y) \, dy \right) \\ &= E[g(X)]E[h(Y)]. \quad \blacklozenge \end{aligned}$$

Again, by this theorem, if X and Y are independent, then $E(XY) = E(X)E(Y)$. As we know from the discrete case, the converse of this fact is not *necessarily* true. (Refer to Example 4.37.)

It is also essential to understand that the subsequent generalizations are valid facts.

If $\{X_1, X_2, \dots\}$ is a sequence of independent continuous random variables and for $i = 1, 2, \dots$, $g_i: \mathbb{R} \rightarrow \mathbb{R}$ is a real-valued function, then the sequence $\{g_1(X_1), g_2(X_2), \dots\}$ is also an independent sequence of random variables.

Let $X_1, X_2, \dots, X_n$ be independent continuous random variables defined on the same sample space, then

$$E(X_1 X_2 \cdots X_n) = E(X_1)E(X_2) \cdots E(X_n).$$

Random Sample

Definition 5.9 We say that n random variables $X_1, X_2, \dots, X_n$ form a **random sample of size n** , from a (continuous or discrete) distribution function F , if they are independent and, for $1 \leq i \leq n$, the distribution function of X_i is F . Therefore, elements of a random sample are independent and identically distributed.

To elucidate this definition, consider that a company manufactures light bulbs with a lifetime distribution represented by a random variable X . The probability density function of X is given by:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

This type of random variable, known as **exponential** with parameter λ , will be discussed in depth in 7. The expected value of X is:

$$E(X) = \int_0^{\infty} \lambda e^{-\lambda x} dx = \frac{1}{\lambda}.$$

To estimate $1/\lambda$, which represents the average lifetime of a light bulb, we randomly and independently select n light bulbs from those produced by the company, where n is a positive integer. For each bulb in this selection, from $1 \leq i \leq n$, let X_i represent its lifetime. Thus, the set $\{X_1, X_2, \dots, X_n\}$ constitutes a random sample of size n from the exponential distribution with parameter λ . It is important to note that for every $1 \leq i \leq n$ the X_i variables are independent and each follows an exponential distribution with parameter λ .

A logical estimate for $1/\lambda$ is the mean of the random sample $\{X_1, X_2, \dots, X_n\}$, denoted by $\bar{X}$:

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

To find this estimate, one simply measures the lifetime of each bulb in the random sample of size n and computes the average of the observed values. Later, in [Sections 7.3](#) and [10.4](#), methods to determine the appropriate sample size, n , will be discussed, ensuring that the estimation error remains within a predefined limit.

EXERCISES

A

1. Let X and Y be two independent random variables with distribution functions F and G , respectively. Find the distribution functions of $\max(X, Y)$ and $\min(X, Y)$.
2. Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} 2 & \text{if } 0 \leq y \leq x \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Are X and Y independent? Why or why not?

3. Suppose that the amount of cholesterol in a certain type of sandwich is $100X$ milligrams, where X is a random variable with the following density function:

$$f(x) = \begin{cases} \frac{2x+3}{18} & \text{if } 2 < x < 4 \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability that two such sandwiches made independently have the same amount of cholesterol.

4. Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} x^2 e^{-x(y+1)} & \text{if } x \geq 0, \quad y \geq 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Are X and Y independent? Why or why not?

5. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} 8xy & \text{if } 0 \leq x < y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Determine if $E(XY) = E(X)E(Y)$.

6. Let X and Y be two independent random variables with the same probability density function given by

$$f(x) = \begin{cases} e^{-x} & \text{if } 0 < x < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

Show that g , the probability density function of X/Y , is given by

$$g(t) = \begin{cases} \frac{1}{(1+t)^2} & \text{if } 0 < t < \infty \\ 0 & t \leq 0. \end{cases}$$

7. A point is selected at random from the disk

$$R = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

Let X be the x -coordinate and Y be the y -coordinate of the point selected. Determine if X and Y are independent random variables.

8. The lifetimes of mufflers manufactured by company A are random with the following density function:

$$f(x) = \begin{cases} \frac{1}{6}e^{-x/6} & \text{if } x > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

The lifetimes of mufflers manufactured by company B are random and independent of the lifetimes of the mufflers of company A , with the following density function:

$$g(y) = \begin{cases} \frac{2}{11}e^{-2y/11} & \text{if } y > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Elizabeth buys two mufflers, one from company A and the other one from company B and installs them on her cars at the same time. What is the probability that the muffler of company B outlasts that of company A ?

9. Let X , Y , and Z be jointly continuous with the following joint probability density function:

$$f(x, y, z) = \begin{cases} x^2 e^{-x(1+y+z)} & \text{if } x, y, z > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Are X , Y , and Z independent?

10. Let the joint probability distribution function of X , Y , and Z be given by

$$F(x, y, z) = (1 - e^{-\lambda_1 x})(1 - e^{-\lambda_2 y})(1 - e^{-\lambda_3 z}), \quad x, y, z > 0,$$

where $\lambda_1, \lambda_2, \lambda_3 > 0$.

- (a) Are X , Y , and Z independent?
- (b) Find the joint probability density function of X , Y , and Z .
- (c) Find $P(X < Y < Z)$.

11. Inside a circle of radius R , n points are selected at random and independently. Find the probability that the distance of the nearest point to the center is at least r .

B

12. Let E be an event; the random variable I_E , defined as follows, is called the **indicator** of E :

$$I_E = \begin{cases} 1 & \text{if } E \text{ occurs} \\ 0 & \text{otherwise.} \end{cases}$$

Show that A and B are independent events if and only if I_A and I_B are independent random variables.

13. Let B and C be two independent random variables both having the following density function:

$$f(x) = \begin{cases} \frac{3x^2}{26} & \text{if } 1 < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

What is the probability that the quadratic equation $X^2 + BX + C = 0$ has two real roots?

14. Let the joint probability density function of two random variables X and Y satisfy

$$f(x, y) = g(x)h(y), \quad -\infty < x < \infty, \quad -\infty < y < \infty,$$

where g and h are two functions from $\mathbb{R}$ to $\mathbb{R}$. Show that X and Y are independent.

15. Let $f(x, y)$ be the joint probability density function of two continuous random variables; f is called **circularly symmetrical** if it is a function of $\sqrt{x^2 + y^2}$, the distance of (x, y) from the origin; that is, if there exists a function φ so that $f(x, y) = \varphi(\sqrt{x^2 + y^2})$. Prove that if X and Y are independent random variables, their joint probability density function is circularly symmetrical if and only if they are both normal with mean 0 and equal variance.

Hint: Suppose that f is circularly symmetrical, then

$$f_X(x)f_Y(y) = \varphi(\sqrt{x^2 + y^2}).$$

Differentiating this relation with respect to x yields

$$\frac{\varphi'(\sqrt{x^2 + y^2})}{\varphi(\sqrt{x^2 + y^2})\sqrt{x^2 + y^2}} = \frac{f'_X(x)}{xf_X(x)}.$$

This implies that both sides are constants, so that

$$\frac{f'_X(x)}{xf_X(x)} = k$$

for some constant k . Solve this and use the fact that f_X is a probability density function to show that f_X is normal. Repeat the same procedure for f_Y .

16. Let F be a probability distribution function. Prove that the functions F^n and $1 - (1 - F)^n$ are also probability distribution functions.

Hint: Let $X_1, X_2, \dots, X_n$ be independent random variables each with the probability distribution function F . Find the probability distribution functions of the random variables $\max(X_1, X_2, \dots, X_n)$ and $\min(X_1, X_2, \dots, X_n)$.

17. A point is selected at random from the pyramid

$$V = \{(x, y, z) : x, y, z \geq 0, x + y + z \leq 1\}.$$

Letting (X, Y, Z) be its coordinates, determine if X , Y , and Z are independent.

Hint: Recall that the volume of a pyramid is $Bh/3$, where h is the height and B is the area of the base.

18. **(Roots of Quadratic Equations)** Three numbers A , B , and C are selected at random and independently from the interval $(0, 1)$. Determine the probability that the quadratic equation $Ax^2 + Bx + C = 0$ has real roots. In other words, what fraction of “all possible quadratic equations” with coefficients in $(0, 1)$ have real roots?

19. **(Roots of Cubic Equations)** Solve the following exercise posed by S. A. Patil and D. S. Hawkins, Tennessee Technological University, Cookeville, Tennessee, in *The College Mathematics Journal*, September 1992.

Let A , B , and C be independent random variables uniformly distributed on $[0, 1]$. What is the probability that all of the roots of the cubic equation $x^3 + Ax^2 + Bx + C = 0$ are real?

20. Let $\{X_1, X_2, \dots\}$ be a sequence of continuous, independent, and identically distributed random variables. Let

$$N = \min\{n : X_1 \geq X_2 \geq X_3 \geq \dots \geq X_{n-1}, X_{n-1} < X_n\}.$$

Find $E(N)$. ■

Self-Quiz on Section 5.5

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. The joint probability density function of the continuous random variables X and Y is given by

$$f(x, y) = \begin{cases} \frac{1}{2}y^3e^{-2x} & x \geq 0, 0 \leq y \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$

Are X and Y independent? Why or why not?

2. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} 1 & \text{if } |x| < y, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Show that X and Y are dependent but $E(XY) = E(X)E(Y)$. ■

CHAPTER 5 SUMMARY

► **Continuous Random Variables** Let X be a random variable. Suppose that there exists a nonnegative real-valued function $f: \mathbb{R} \rightarrow [0, \infty)$ such that for any subset of real numbers A that can be constructed from intervals by a countable number of set operations,

$$P(X \in A) = \int_A f(x) dx.$$

Then X is called *continuous*. The function f is called the *probability density function*, or simply the *density function* of X . Let F be the distribution function of X . Some immediate properties of the density function f are: (i) $F(t) = \int_{-\infty}^t f(x) dx$; (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$; (iii) If f is continuous, then $F'(x) = f(x)$; (iv) For real numbers $a \leq b$, $P(a \leq X \leq b) = \int_a^b f(t) dt$; (v) for any real number a , $P(X = a) = 0$; and (vi) $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b) = \int_a^b f(t) dt$.

► The area over an interval I under the graph of f represents the probability that the random variable X will belong to I . The area under f to the left of a given point t is $F(t)$, the value of the distribution function of X at t .

► **Density Function of a Function of a Random Variable** Suppose that f , the density function of a random variable X , is known but the density function $h(X)$, for some function of X , is needed. In such cases there are two common methods for the evaluation of the density function of $h(X)$. In a method called *the method of distribution functions*, to calculate the probability density function of $h(X)$, we calculate the distribution function of $h(X)$, $P(h(X) \leq t)$, by finding a set E for which $h(X) \leq t$ if and only if $X \in E$ and then computing $P(X \in E)$ from the distribution function of X . When $P(h(X) \leq t)$ is found, the density function of $h(X)$, if it exists, is obtained by differentiation.

In the second method, called *the method of transformations*, we find the density function of $h(X)$ without obtaining its distribution function. Let f_X be the density function of X , and let its set of possible values be A . For the invertible function $h: A \rightarrow \mathbb{R}$, let $Y = h(X)$ be a random variable with the set of possible values $B = h(A) = \{h(a): a \in A\}$. Suppose that the inverse of $y = h(x)$ is the function $x = h^{-1}(y)$, which is differentiable for all values of $y \in B$. Then f_Y , the density function of Y , is given by

$$f_Y(y) = f_X(h^{-1}(y)) |(h^{-1})'(y)|, \quad y \in B.$$

► **Expectations of Continuous Random Variables** If X is a continuous random variable with probability density function f , the *expected value* of X is defined by

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx.$$

X is said to have a *finite expected value* if $\int_{-\infty}^{\infty} |x|f(x) dx < \infty$; that is, X has a finite expected value if the integral of $xf(x)$ converges absolutely.

► For any continuous random variable X with distribution function F and probability density function f ,

$$E(X) = \int_0^{\infty} [1 - F(t)] dt - \int_0^{\infty} F(-t) dt,$$

$$E(X) = \int_0^{\infty} P(X > t) dt - \int_0^{\infty} P(X \leq -t) dt.$$

► If X is a nonnegative continuous random variable with distribution function F , that is, if $P(X < 0) = 0$, then

$$E(X) = \int_0^{\infty} [1 - F(t)] dt = \int_0^{\infty} P(X > t) dt.$$

► Let X be a continuous random variable with probability density function $f(x)$, then for any function $h: \mathbb{R} \rightarrow \mathbb{R}$,

$$E[h(X)] = \int_{-\infty}^{\infty} h(x)f(x) dx.$$

► Let $h_1, h_2, \dots, h_n$ be real-valued functions, and $\alpha_1, \alpha_2, \dots, \alpha_n$ be real numbers. Then

$$\begin{aligned} E[\alpha_1 h_1(X) + \alpha_2 h_2(X) + \dots + \alpha_n h_n(X)] \\ = \alpha_1 E[h_1(X)] + \alpha_2 E[h_2(X)] + \dots + \alpha_n E[h_n(X)]. \end{aligned}$$

This relation implies the *linearity property of continuous random variables*. That is, if α and β are constants, then $E(\alpha X + \beta) = \alpha E(X) + \beta$.

► **Variances of Continuous Random Variables** If X is a continuous random variable with $E(X) = \mu$, then $\text{Var}(X)$ and σ_X , called the *variance* and *standard deviation* of X , respectively, are defined by

$$\text{Var}(X) = E[(X - \mu)^2],$$

$$\sigma_X = \sqrt{E[(X - \mu)^2]}.$$

As in discrete case, we have the following relations:

$$\begin{aligned}\text{Var}(X) &= E(X^2) - [E(X)]^2, \\ \text{Var}(aX + b) &= a^2 \text{Var}(X), \quad \sigma_{aX+b} = |a| \sigma_X, \quad a \text{ and } b \text{ being constants.}\end{aligned}$$

► **Joint Probability Density Functions** Two random variables X and Y , defined on the same sample space, have a continuous joint distribution if there exists a nonnegative function of two variables, $f(x, y)$, on $\mathbb{R} \times \mathbb{R}$, such that for any region R in the xy -plane that can be formed from rectangles by a countable number of set operations,

$$P((X, Y) \in R) = \iint_R f(x, y) dx dy.$$

The function $f(x, y)$ is called the *joint probability density function* of X and Y . The functions f_X and f_Y , given by $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$ and $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$ are called, respectively, the *marginal probability density functions* of X and Y . We have $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$ and $E(Y) = \int_{-\infty}^{\infty} y f_Y(y) dy$. Furthermore, if h is an ordinary function of two variables from $\mathbb{R}^2$ to $\mathbb{R}$, then $h(X, Y)$ is a random variable with the expected value given by

$$E[h(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dx dy,$$

provided that the integral is absolutely convergent. By this relation,

$$E(X + Y) = E(X) + E(Y).$$

► **Expected Values of Sums of Random Variables** For random variables $X_1, X_2, \dots, X_n$ defined on the same sample space, and for real numbers $\alpha_1, \alpha_2, \dots, \alpha_n$,

$$E\left(\sum_{i=1}^n \alpha_i X_i\right) = \sum_{i=1}^n \alpha_i E(X_i).$$

In particular,

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n).$$

In general, If $\sum_{i=1}^{\infty} E(|X_i|) < \infty$ or if, for all i , the random variables $X_1, X_2, \dots$ are nonnegative [that is, $P(X_i \geq 0) = 1$ for $i \geq 1$], then $E\left(\sum_{i=1}^{\infty} X_i\right) = \sum_{i=1}^{\infty} E(X_i)$.

► **Cauchy-Schwarz Inequality** For random variables X and Y with finite variances,

$$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}.$$

► **Geometric Probability** Let S be a subset of the plane with area $A(S)$. A point is said to be *randomly selected* from S if for any subset R of S with area $A(R)$, the probability that R contains the point is $A(R)/A(S)$.

► **Independence of Random Variables** Two random variables X and Y are called *independent* if, for arbitrary subsets A and B of real numbers, the events $\{X \in A\}$ and $\{Y \in B\}$ are independent, that is, if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$. Two random variables are independent if and only if their joint distribution function is the product of their marginal distribution functions. Two discrete random variables are independent if and only if their joint probability mass function is the product of their marginal probability mass functions. Two continuous random variables are independent if and only if their joint probability density function is the product of their marginal probability density functions.

Let X and Y be independent random variables and $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$ be real-valued functions, then $g(X)$ and $h(Y)$ are also independent random variables. Furthermore,

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)],$$

In particular, $E(XY) = E(X)E(Y)$. However, two random variables X and Y might be dependent while $E(XY) = E(X)E(Y)$.

► **Independence of $n > 2$ Random Variables** Suppose that $X_1, X_2, \dots, X_n$ are random variables (discrete, continuous, or mixed) on a sample space. We say that they are *independent* if, for arbitrary subsets $A_1, A_2, \dots, A_n$ of real numbers,

$$P(X_1 \in A_1, X_2 \in A_2, \dots, X_n \in A_n) = P(X_1 \in A_1)P(X_2 \in A_2) \cdots P(X_n \in A_n).$$

► If $\{X_1, X_2, \dots\}$ is a sequence of independent random variables and for $i = 1, 2, \dots$, $g_i: \mathbb{R} \rightarrow \mathbb{R}$ is a real-valued function, then the sequence $\{g_1(X_1), g_2(X_2), \dots\}$ is also an independent sequence of random variables.

► **Joint Probability Density Functions** Let $X_1, X_2, \dots, X_n$ be continuous random variables defined on the same sample space. We say that $X_1, X_2, \dots, X_n$ have a *continuous joint distribution* if there exists a nonnegative function of n variables, $f(x_1, x_2, \dots, x_n)$, on $\mathbb{R}^n$ such that for any region R in $\mathbb{R}^n$ that can be formed from n -dimensional rectangles by a countable number of set operations,

$$P((X_1, X_2, \dots, X_n) \in R) = \int \cdots \int_R f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n.$$

The function $f(x_1, x_2, \dots, x_n)$ is called the *joint probability density function* of $X_1, X_2, \dots, X_n$. Let $R = \{(x_1, x_2, \dots, x_n): x_i \in A_i, 1 \leq i \leq n\}$, where $A_i, 1 \leq i \leq n$, is any subset of real numbers that can be constructed from intervals by a countable number of set operations. Then

$$\begin{aligned} P(X_1 \in A_1, X_2 \in A_2, \dots, X_n \in A_n) \\ = \int_{A_n} \int_{A_{n-1}} \cdots \int_{A_1} f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n. \end{aligned}$$

Let f_{X_i} be the marginal probability density function of X_i , $1 \leq i \leq n$. Then

$$f_{X_i}(x_i) = \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{n-1 \text{ terms}} f(x_1, x_2, \dots, x_n) dx_1 \cdots dx_{i-1} dx_{i+1} \cdots dx_n.$$

In general, to find the *joint probability density function marginalized* over a given set of k of these random variables, we integrate $f(x_1, x_2, \dots, x_n)$ over all possible values of the remaining $n-k$ random variables. The continuous random variables $X_1, X_2, \dots, X_n$ are independent if and only if $f(x_1, x_2, \dots, x_n)$ is the product of their marginal densities $f_{X_1}(x_1), f_{X_2}(x_2), \dots, f_{X_n}(x_n)$.

► If h is a function of n variables from $\mathbb{R}^n$ to $\mathbb{R}$, then $Y = h(X_1, X_2, \dots, X_n)$ is a random variable with expected value given by

$$E(Y) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} h(x_1, x_2, \dots, x_n) f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n,$$

provided that the integral is absolutely convergent. This fact can be used to prove that the expected value of the product of several independent continuous random variables is equal to the product of their expected values.

REVIEW PROBLEMS

1. There is a spring inside pendulum clocks that is an important component of its mechanism. Suppose that, for some c , the lifetime of such a spring, in tens of thousands of hours, is a random variable that has a probability density function of the form

$$f(x) = \begin{cases} \frac{c}{(1-x)^2} & 8 \leq x \leq 15 \\ 0 & \text{otherwise.} \end{cases}$$

- (i) Determine the constant c .
 - (ii) Find the probability that the spring of a randomly selected pendulum clock lasts at least 120,000 hours.
 - (iii) Calculate the distribution function of the spring of a randomly selected pendulum clock with the probability density function f given above.
2. Let F , the distribution of a random variable X , be defined by

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{1}{2} + \frac{\arcsin x}{\pi} & -1 \leq x < 1 \\ 1 & x \geq 1, \end{cases}$$

where $\arcsin x$ lies between $-\pi/2$ and $\pi/2$. Find f , the probability density function of X and $E(X)$.

3. Let X be a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{1}{2} \cos x & -\pi/2 \leq x \leq \pi/2 \\ 0 & \text{otherwise.} \end{cases}$$

Find $\text{Var}(X)$.

4. Each year, a business loses some money because the market prices of some merchandise in its inventory decline. Let X be the amount of loss in a random year, in thousands of dollars, and suppose that the probability density function of X is given by

$$f(x) = \begin{cases} \frac{1}{50}(10 - x) & 0 < x < 10 \\ 0 & \text{otherwise.} \end{cases}$$

Of all the years that the company loses at least \$4000 due to decline of market prices of some of its merchandise, in what percentage its total loss exceeds \$8000?

5. Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} 2/x^3 & \text{if } x > 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X)$ and $\text{Var}(X)$ if they exist.

6. Let X be a continuous random variable with density function

$$f(x) = 6x(1 - x), \quad 0 < x < 1.$$

What is the probability that X is within two standard deviations of the mean?

7. Let X be a random variable with density function

$$f(x) = \frac{e^{-|x|}}{2}, \quad -\infty < x < \infty.$$

Find $P(-2 < X < 1)$.

8. Let X be a continuous random variable with set of possible values $\{x: 0 < x < \alpha\}$ (where $\alpha < \infty$), distribution function F , and density function f . Using integration by parts, prove the following special case of Theorem 5.2.

$$E(X) = \int_0^\alpha [1 - F(t)] dt.$$

9. Prove or disprove: If $\sum_{i=1}^n \alpha_i = 1$, $\alpha_i \geq 0$, $\forall i$, and $\{f_i\}_{i=1}^n$ is a sequence of density functions, then $\sum_{i=1}^n \alpha_i f_i$ is a probability density function.
10. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} \frac{c}{x} & \text{if } 0 < y < x, 0 < x < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Determine the value of c .
- (b) Find the marginal probability density functions of X and Y .
11. Let X and Y have the joint probability density function below. Determine if $E(XY) = E(X)E(Y)$.

$$f(x, y) = \begin{cases} \frac{3}{4}x^2y + \frac{1}{4}y & \text{if } 0 < x < 1 \text{ and } 0 < y < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

12. Prove that the following cannot be the joint probability distribution function of two random variables X and Y .

$$F(x, y) = \begin{cases} 1 & \text{if } x + y \geq 1 \\ 0 & \text{if } x + y < 1. \end{cases}$$

13. Three concentric circles of radii r_1 , r_2 , and r_3 , $r_1 > r_2 > r_3$, are the boundaries of the regions that form a circular target. If a person fires a shot at random at the target, what is the probability that it lands in the middle region?
14. Let the joint probability distribution function of the lifetimes of two brands of lightbulb be given by

$$F(x, y) = \begin{cases} (1 - e^{-x^2})(1 - e^{-y^2}) & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability that one lightbulb lasts more than twice as long as the other.

15. For $\Omega = \{(x, y) : 0 < x + y < 1, 0 < x < 1, 0 < y < 1\}$, a region in the plane, let

$$f(x, y) = \begin{cases} 3(x + y) & \text{if } (x, y) \in \Omega \\ 0 & \text{otherwise} \end{cases}$$

be the joint probability density function of the random variables X and Y . Find the marginal probability density functions of X and Y , and $P(X + Y > 1/2)$.

16. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} cx(1-x) & \text{if } 0 \leq x \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine the value of c .
- (b) Determine if X and Y are independent.
17. A point is selected at random from the bounded region between the curves $y = x^2 - 1$ and $y = 1 - x^2$. Let X be the x -coordinate, and let Y be the y -coordinate of the point selected. Determine if X and Y are independent.
18. Let the probability density function of a random variable X be given by

$$f(x) = \begin{cases} 2x - 2 & \text{if } 1 < x < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X^3 + 2X - 7)$.

19. Let the joint probability density function of random variables X and Y be

$$f(x, y) = \begin{cases} \frac{3x^3 + xy}{3} & \text{if } 0 \leq x \leq 1, \quad 0 \leq y \leq 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X^2 + 2XY)$.

20. The joint probability density function of random variables X , Y , and Z is given by

$$f(x, y, z) = \begin{cases} c(x + y + 2z) & \text{if } 0 \leq x, y, z \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine the value of c .
- (b) Find $P(X < 1/3 \mid Y < 1/2, Z < 1/4)$.
21. A bar of length ℓ is broken into three pieces at two random spots. What is the probability that the length of at least one piece is less than $\ell/20$?
22. If F is the probability distribution function of a random variable X , is $G(x, y) = F(x) + F(y)$ a joint probability distribution function?
23. Suppose that n points are selected at random and independently inside the cube

$$\Omega = \{(x, y, z) : -a \leq x \leq a, -a \leq y \leq a, -a \leq z \leq a\}.$$

Find the probability that the distance of the nearest point to the center is at least r ($r < a$).

24. Does there exist a constant c for which the following is a density function?

$$f(x) = \begin{cases} \frac{c}{1+x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

25. Let X be a random variable with density function

$$f(x) = \begin{cases} 4x^3/15 & 1 \leq x \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find the density functions of $Y = e^X$, $Z = X^2$, and $W = (X - 1)^2$.

26. The probability density function of a continuous random variable X is

$$f(x) = \begin{cases} 30x^2(1-x)^2 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability density function of $Y = X^4$.



Self-Test on Chapter 5

Time allotted: 120 Minutes

Each problem is worth 10 points.

- Let X be a strictly positive and continuous random variable with distribution function F and density function f . Find the distribution and the density functions of $Y = \log_2 X$.
- An actuary has calculated that the probability density function of the cost to repair a vehicle after a certain type of car accident, in thousands of dollars, is

$$f(x) = \begin{cases} x^2/9 & \text{if } 0 < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

The actuary has also calculated that, for all auto insurance policies with the same deductible amount d , in 7.24% of the time, the cost of such a repair to the insurance company is less than \$1000. Calculate d .

- At a local hardware store, the demand, per week, for propane gas, primarily for barbecue propane tanks, in 100s of gallons, is a random variable with probability density function

$$f(x) = \begin{cases} \frac{1}{875} (6 - x^2)^2 & 1 \leq x \leq 6 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) How many gallons of propane gas must the hardware store carry, per week, so that the stock out probability for the gas is at most 0.05?
- (b) Suppose that on a Sunday morning, when the store opens, it has 590 gallons of propane gas. What is the expected value of its inventory of gas on the following Saturday evening when the store closes, given that no propane gas delivery has been made all week?

4. Let X be a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{1}{2} \cos x & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X)$, $\text{Var}(X)$, and $E(\cos X)$.

Hint: Note that $\cos^2 x = \frac{1 + \cos 2x}{2}$.

5. Let X be a random variable with probability density function

$$f(x) = \begin{cases} (2/3)x & \text{if } 1 < x < 2 \\ 0 & \text{otherwise.} \end{cases}$$

- (i) Determine the density function of e^X .
- (ii) Calculate $E(e^X)$.
6. A point is selected at random from the set $S = \{(x, y) : x^2 + y^2 \leq 4\}$.
- (a) What is the probability that it falls inside the set $E = \{(x, y) : x^2 + y^2 \leq 1\}$?
- (b) What is the probability that it falls inside the set $F = \{(x, y) : |x| + |y| < 2\}$?
7. Let

$$f(x, y) = \begin{cases} 60x^2y & x \geq 0, y \geq 0, x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

be the joint probability density function of continuous random variables X and Y . Are X and Y independent? Why or why not?

8. A certain admission test, given to prospective graduate applicants, consists of two sections, verbal reasoning and quantitative reasoning. Let $100X$ and $100Y$ be the respective verbal and quantitative scores of a randomly selected applicant with an undergraduate 3.0 GPA. Suppose that X and Y are independent and both have the following probability density function

$$f(x) = \begin{cases} \frac{8}{x^2} & 4 \leq x \leq 8 \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability that the total score, $100X + 100Y$, of the randomly selected applicant is at least 1300?

9. Let (X, Y, Z) be a point selected at random in the unit sphere

$$\{(x, y, z): x^2 + y^2 + z^2 \leq 1\};$$

that is, the sphere of radius 1 centered at the origin. [Note that the volume of a sphere with radius R is $(4/3)\pi R^3$.]

(a) Find f , the joint probability density function of X , Y and Z .

(c) Find the marginal probability density function of Z .

Hint: To find $f_Z(z)$, convert the Cartesian coordinates (x, y) to polar coordinates (r, θ) .

10. For what value of c is the following a joint probability density function of four random variables X , Y , Z , and T ? For that value of c find $P(X < Y < Z < T)$.

$$f(x, y, z, t) = \begin{cases} \frac{c}{(1 + x + y + z + t)^6} & x > 0, y > 0, z > 0, t > 0 \\ 0 & \text{otherwise.} \end{cases}$$



Chapter 6

Special Discrete Distributions

In this chapter we study some examples of discrete random variables. These random variables appear frequently in theory and applications of probability, statistics, and branches of science and engineering.

6.1 BERNOULLI AND BINOMIAL RANDOM VARIABLES

Bernoulli trials, named after the Swiss mathematician James Bernoulli, are perhaps the simplest type of random variable. They have only two possible outcomes. One outcome is usually called a **success**, denoted by s . The other outcome is called a **failure**, denoted by f . The experiment of flipping a coin is a Bernoulli trial. Its only outcomes are “heads” and “tails.” If we are interested in heads, we may call it a success; tails is then a failure. The experiment of tossing a die is a Bernoulli trial if, for example, we are interested in knowing whether the outcome is odd or even. An even outcome may be called a success, and hence an odd outcome a failure, or vice versa. If a fuse is inspected, it is either “defective” or it is “good.” So the experiment of inspecting fuses is a Bernoulli trial. A good fuse may be called a success, a defective fuse a failure.

The sample space of a Bernoulli trial contains two points, s and f . The random variable defined by $X(s) = 1$ and $X(f) = 0$ is called a **Bernoulli random variable**. Therefore, a Bernoulli random variable takes on the value 1 when the outcome of the Bernoulli trial is a success and 0 when it is a failure. If p is the probability of a success, then $1 - p$ (sometimes denoted q) is the probability of a failure. Hence the probability mass function of X is

$$p(x) = \begin{cases} 1 - p \equiv q & \text{if } x = 0 \\ p & \text{if } x = 1 \\ 0 & \text{otherwise.} \end{cases} \quad (6.1)$$

Note that the same symbol p is used for the probability mass function and the Bernoulli parameter. This duplication should not be confusing since the p 's used for the probability mass function often appear in the form $p(x)$.

An accurate mathematical definition for Bernoulli random variables is as follows.

Definition 6.1 *A random variable is called **Bernoulli** with parameter p if its probability mass function is given by equation (6.1).*

From (6.1) it follows that the expected value of a Bernoulli random variable X , with parameter p , is p , because

$$E(X) = 0 \cdot P(X = 0) + 1 \cdot P(X = 1) = P(X = 1) = p.$$

Also, since

$$E(X^2) = 0 \cdot P(X = 0) + 1 \cdot P(X = 1) = p,$$

we have

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = p - p^2 = p(1 - p).$$

We will now summarize what we have shown.

For a Bernoulli random variable X with parameter p , $0 < p < 1$,

$$E(X) = p, \quad \text{Var}(X) = p(1 - p), \quad \sigma_X = \sqrt{p(1 - p)}.$$

Example 6.1 If in a throw of a fair die the event of obtaining 4 or 6 is called a success, and the event of obtaining 1, 2, 3, or 5 is called a failure, then

$$X = \begin{cases} 1 & \text{if 4 or 6 is obtained} \\ 0 & \text{otherwise} \end{cases}$$

is a Bernoulli random variable with the parameter $p = 1/3$. Therefore, its probability mass function is

$$p(x) = \begin{cases} 2/3 & \text{if } x = 0 \\ 1/3 & \text{if } x = 1 \\ 0 & \text{elsewhere.} \end{cases}$$

The expected value of X is given by $E(X) = p = 1/3$, and its variance by $\text{Var}(X) = 1/3(1 - 1/3) = 2/9$. ♦

Let $X_1, X_2, X_3, \dots$ be a sequence of Bernoulli random variables. If, for all $j_i \in \{0, 1\}$, the sequence of events $\{X_1 = j_1\}, \{X_2 = j_2\}, \{X_3 = j_3\}, \dots$ are independent, we say that $\{X_1, X_2, X_3, \dots\}$ and the corresponding Bernoulli trials are independent.

Although Bernoulli trials are simple, if they are repeated independently, they may pose interesting and even sometimes complicated questions. Consider an experiment in which n Bernoulli trials are performed independently. The sample space of such an experiment, S , is the set of different sequences of length n with x ($x = 0, 1, \dots, n$) successes (s 's) and $(n - x)$ failures (f 's). For example, if, in an experiment, three Bernoulli trials are performed independently, then the sample space is

$$\{fff, sff, fsf, ffs, fss, sfs, ssf, sss\}.$$

If n Bernoulli trials all with probability of success p are performed independently, then X , the number of successes, is one of the most important random variables. It is called a **binomial with parameters n and p** . The set of possible values of X is $\{0, 1, 2, \dots, n\}$, it is defined on the set S described previously, and its probability mass function is given by the following theorem.

Theorem 6.1 *Let X be a binomial random variable with parameters n and p . Then $p(x)$, the probability mass function of X , is*

$$p(x) = P(X = x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & \text{if } x = 0, 1, 2, \dots, n \\ 0 & \text{elsewhere.} \end{cases} \quad (6.2)$$

Proof: Observe that the number of ways that, in n Bernoulli trials, x ($x = 0, 1, 2, \dots, n$) successes can occur is equal to the number of different sequences of length n with x successes (s 's) and $(n-x)$ failures (f 's). But the number of such sequences is $\binom{n}{x}$ because the number of distinguishable permutations of n objects of two different types, where x are alike and $n-x$ are alike is $\frac{n!}{x!(n-x)!} = \binom{n}{x}$ (see Theorem 2.4). Since by the independence of the trials the probability of each of these sequences is $p^x(1-p)^{n-x}$, we have $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$. Hence (6.2) follows. ♦

Definition 6.2 *The function $p(x)$ given by equation (6.2) is called the **binomial** probability mass function with parameters (n, p) .*

The reason for this name is that the binomial expansion theorem (Theorem 2.5) guarantees that p is a probability mass function:

$$\sum_{x=0}^n p(x) = \sum_{x=0}^n \binom{n}{x} p^x (1-p)^{n-x} = [p + (1-p)]^n = 1^n = 1.$$

Example 6.2 A restaurant serves 8 entrées of fish, 12 of beef, and 10 of poultry. If customers select from these entrées randomly, what is the probability that two of the next four customers order fish entrées?

Solution: Let X denote the number of fish entrées (successes) ordered by the next four customers. Then X is binomial with the parameters $(4, 8/30 = 4/15)$. Thus

$$P(X = 2) = \binom{4}{2} \left(\frac{4}{15}\right)^2 \left(\frac{11}{15}\right)^2 = 0.23. \quad \blacklozenge$$

Example 6.3 In a county hospital 10 babies, of whom six were boys, were born last Thursday. What is the probability that the first six births were all boys? Assume that the events that a child born is a girl or is a boy are equiprobable.

Solution: Let A be the event that the first six births were all boys and the last four all girls. Let X be the number of boys, then X is binomial with parameters 10 and $1/2$. The desired probability is

$$\begin{aligned} P(A \mid X = 6) &= \frac{P(A \text{ and } X = 6)}{P(X = 6)} = \frac{P(A)}{P(X = 6)} \\ &= \frac{\left(\frac{1}{2}\right)^{10}}{\binom{10}{6} \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^4} = \frac{1}{\binom{10}{6}} \approx 0.0048. \end{aligned}$$

Remark: We can also argue as follows: $bbbbbbgggg$, the event that the first six were all boys, and the remainder girls, is one sequence out of the $\binom{10}{6}$ possible sequences of the sexes of the births. Since each of these sequences has the same probability of occurrence, the answer is $1/\binom{10}{6}$. ♦

Example 6.4 In a small town, out of 12 accidents that occurred in June 1986, four happened on Friday the 13th. Is this a good reason for a superstitious person to argue that Friday the 13th is inauspicious?

Solution: Suppose the probability that each accident occurs on Friday the 13th is $1/30$, just as on any other day. Then the probability of at least four accidents on Friday the 13th is

$$1 - \sum_{i=0}^3 \binom{12}{i} \left(\frac{1}{30}\right)^i \left(\frac{29}{30}\right)^{12-i} \approx 0.000,493.$$

Since the probability of four or more of these accidents occurring on Friday the 13th is very small, this is a good reason for superstitious persons to argue that Friday the 13th is inauspicious. ♦

Example 6.5 Suppose that jury members decide independently and that each with probability p ($0 < p < 1$) makes the correct decision. If the decision of the majority is final, which is preferable: a three-person jury or a single juror?

Solution: Let X denote the number of persons who decide correctly among a three-person jury. Then X is a binomial random variable with parameters $(3, p)$. Hence the probability that

a three-person jury decides correctly is

$$\begin{aligned} P(X \geq 2) &= P(X = 2) + P(X = 3) = \binom{3}{2} p^2 (1-p) + \binom{3}{3} p^3 (1-p)^0 \\ &= 3p^2(1-p) + p^3 = 3p^2 - 2p^3. \end{aligned}$$

Since the probability is p that a single juror decides correctly, a three-person jury is preferable to a single juror if and only if

$$3p^2 - 2p^3 > p.$$

This is equivalent to $3p - 2p^2 > 1$, so $-2p^2 + 3p - 1 > 0$. But

$$-2p^2 + 3p - 1 = 2(1-p)(p - 1/2).$$

Since $1-p > 0$, $2(1-p)(p - 1/2) > 0$ if and only if $p > 1/2$. Hence a three-person jury is preferable if $p > 1/2$. If $p < 1/2$, the decision of a single juror is preferable. For $p = 1/2$ there is no difference. ♦

Example 6.6 Machuca's favorite bridge hands are those with at least two aces. Determine the number of times he should play in order to have a chance of 90% or more to get at least two favorite hands? Recall that a bridge hand consists of 13 randomly selected cards from an ordinary deck of 52 cards.

Solution: We call a bridge hand a "success" if it is Machuca's favorite. Let X be the number of successes in n independent bridge hands. Clearly, X is a binomial random variable with parameters (n, p) , where

$$p = 1 - \frac{\binom{48}{13}}{\binom{52}{13}} - \frac{\binom{4}{1} \binom{48}{12}}{\binom{52}{13}} \approx 0.257.$$

We want to determine n so that $P(X \geq 2) \geq 0.90$. Since

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) \geq 0.90,$$

we must have

$$P(X = 0) + P(X = 1) \leq 0.10.$$

Therefore, n is the smallest integer satisfying

$$\binom{n}{0} (0.257)^0 (1 - 0.257)^n + \binom{n}{1} (0.257)^1 (1 - 0.257)^{n-1} \leq 0.10.$$

This inequality reduces to

$$(0.743)^{n-1} (0.743 + 0.257n) \leq \frac{1}{10}.$$

By trial and error, we find that for $n = 13$, the left side is approximately 0.116 while for $n = 14$, it is 0.091. Therefore $n = 14$ is the answer.

Remark: Note that n the number of times Machuca should play in order to get at least two favorite hands with certainty (chance of 100%) satisfies the relation

$$(0.743)^{n-1}(0.743 + 0.257n) = 0.$$

This gives $(0.743)^{n-1} = 0$ or $n = \infty$. However, similar calculations show that, the number of times he should play to get at least two favorite hands with probability of at least 0.99 is only 23. ♦

Example 6.7 A Canadian pharmaceutical company has developed three types of Ebola virus vaccines. Suppose that vaccines I, II, and III are effective 92%, 88%, and 96% of the time, respectively. Assume that a person reacts to the Ebola vaccine independently of other people. The company sends to Guinea hundreds of qualified vaccine carrier containers containing 100 ml vaccine bottles each filled with one of the three types of the Ebola vaccines. Suppose that the bottles of 40% of the containers are filled with only the type I vaccine, and the bottles of 35% and 25% of the containers are filled only with type II and type III vaccines, respectively. In Guinea, a physician assistant takes one of the containers at random to a small village and using its contents, vaccinates its entire 200-people population. It is certain that sooner or later every member of the village will be exposed to the Ebola virus. What is the probability that at most 6 of the villagers will become sick with Ebola?

Solution: Let X be the number of the villagers who will become sick with Ebola. Let A_1 , A_2 , and A_3 , respectively, be the events that the physician assistant takes the container that has type I, type II, and type III vaccines. The desired probability is

$$\begin{aligned} P(X \leq 6) &= P(X \leq 6 \mid A_1)P(A_1) + P(X \leq 6 \mid A_2)P(A_2) + P(X \leq 6 \mid A_3)P(A_3) \\ &= \sum_{i=0}^6 \binom{200}{i} (0.08)^i (1 - 0.08)^{200-i} \cdot (0.40) + \\ &\quad \sum_{i=0}^6 \binom{200}{i} (0.12)^i (1 - 0.12)^{200-i} \cdot (0.35) + \\ &\quad \sum_{i=0}^6 \binom{200}{i} (0.04)^i (1 - 0.04)^{200-i} \cdot (0.25) \approx 0.0783. \quad \blacklozenge \end{aligned}$$

Example 6.8 Let p be the probability that a randomly chosen person is pro-life, and let X be the number of persons pro-life in a random sample of size n . Suppose that, in a particular random sample of n persons, k are pro-life. Show that $P(X = k)$ is maximum for $\hat{p} = k/n$. That is, $\hat{p}$ is the value of p that makes the outcome $X = k$ most probable.

Solution: By definition of X ,

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

This gives

$$\begin{aligned}\frac{d}{dp}P(X = k) &= \binom{n}{k} [kp^{k-1}(1-p)^{n-k} - (n-k)p^k(1-p)^{n-k-1}] \\ &= \binom{n}{k} p^{k-1}(1-p)^{n-k-1} [k(1-p) - (n-k)p].\end{aligned}$$

Letting $\frac{d}{dp}P(X = k) = 0$, we obtain $p = k/n$. Now since $\frac{d^2}{dp^2}P(X = k) < 0$, $\hat{p} = k/n$ is the maximum of $P(X = k)$, and hence it is an estimate of p that makes the outcome $x = k$ most probable. ♦

Let X be a binomial random variable with parameters (n, p) , $0 < p < 1$, and probability mass function $p(x)$. We will now find the value of X at which $p(x)$ is maximum. For any real number t , let $[t]$ denote the largest integer less than or equal to t . We will prove that $p(x)$ is maximum at $x = [(n+1)p]$. To do so, we note that

$$\begin{aligned}\frac{p(x)}{p(x-1)} &= \frac{\frac{n!}{(n-x)!x!} p^x(1-p)^{n-x}}{\frac{n!}{(n-x+1)!(x-1)!} p^{x-1}(1-p)^{n-x+1}} \\ &= \frac{(n-x+1)p}{x(1-p)} \\ &= \frac{(n+1)p - xp + x - x}{x(1-p)} = \frac{[(n+1)p - x] + x(1-p)}{x(1-p)} \\ &= \frac{(n+1)p - x}{x(1-p)} + 1.\end{aligned}\tag{6.3}$$

This equality shows that $p(x) > p(x-1)$ if and only if $(n+1)p - x > 0$, or, equivalently, if and only if $x < (n+1)p$. Hence as x changes from 0 to $[(n+1)p]$, $p(x)$ increases. As x changes from $[(n+1)p]$ to n , $p(x)$ decreases. The maximum value of $p(x)$ [the peak of the graphical representation of $p(x)$] occurs at $[(n+1)p]$.

What we have just shown is illustrated by the graphical representations of $p(x)$ in [Figure 6.1](#), displayed on the next page, for binomial random variables with parameters $(5, 1/2)$, $(10, 1/2)$, and $(20, 1/2)$.

Expectations and Variances of Binomial Random Variables

Let X be a binomial random variable with parameters (n, p) . Intuitively, we expect that the expected value of X will be np . For example, if we toss a fair coin 100 times, we expect that the average number of heads will be 50, which is $100 \cdot 1/2 = np$. Also, if we choose 10 fuses from a set with 30% defective fuses, we expect the average number of defective fuses to be

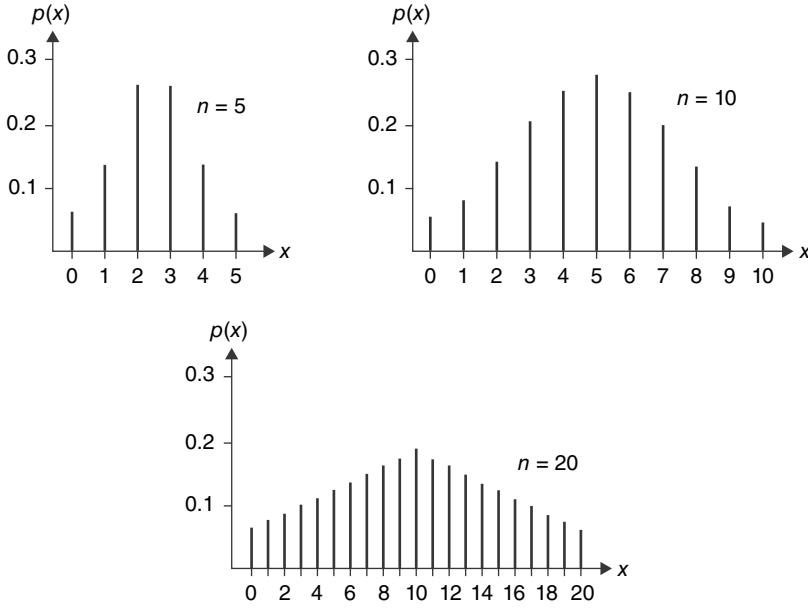


Figure 6.1 Examples of binomial probability mass functions.

$np = 10(0.30) = 3$. The formula $E(X) = np$ can be verified directly from the definition of mathematical expectation as follows:

$$\begin{aligned}
 E(X) &= \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=1}^n x \frac{n!}{x! (n-x)!} p^x (1-p)^{n-x} \\
 &= \sum_{x=1}^n \frac{n!}{(x-1)! (n-x)!} p^x (1-p)^{n-x} \\
 &= np \sum_{x=1}^n \frac{(n-1)!}{(x-1)! (n-x)!} p^{x-1} (1-p)^{n-x} \\
 &= np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} (1-p)^{n-x}.
 \end{aligned}$$

Letting $i = x - 1$ (by reindexing this sum), we obtain

$$E(X) = np \sum_{i=0}^{n-1} \binom{n-1}{i} p^i (1-p)^{(n-1)-i} = np[p + (1-p)]^{n-1} = np,$$

where the next-to-last equality follows from binomial expansion (Theorem 2.5). To calculate the variance of X , from a procedure similar to the one we used to compute $E(X)$, we obtain (see Exercise 31)

$$E(X^2) = \sum_{x=1}^n x^2 \binom{n}{x} p^x (1-p)^{n-x} = n^2 p^2 - np^2 + np.$$

Therefore,

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = -np^2 + np = np(1 - p).$$

We have established that

If X is a binomial random variable with parameters n and p , then

$$E(X) = np, \quad \text{Var}(X) = np(1 - p), \quad \sigma_X = \sqrt{np(1 - p)}.$$

For the computation of $E(X)$, while it is essential to grasp such techniques, there are more efficient shortcuts available to achieve the desired results. We have previously explored such shortcuts in Examples 4.32 through 4.36. In this context, we employ the same methodology by leveraging the Corollary of Theorem 4.9 to swiftly compute the expected value of a binomial random variable.

Example 6.9 Let X be a binomial random variable with parameters (n, p) . Recall that X is the number of successes in n independent Bernoulli trials. Thus, for $i = 1, 2, \dots, n$, letting

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{otherwise,} \end{cases}$$

we get

$$X = X_1 + X_2 + \cdots + X_n,$$

where X_i is a Bernoulli random variable for $i = 1, 2, \dots, n$. Now, since $\forall i, 1 \leq i \leq n$,

$$E(X_i) = 1 \cdot p + 0 \cdot (1 - p) = p,$$

(6.4) implies that

$$E(X) = E(X_1) + E(X_2) + \cdots + E(X_n) = np. \quad \blacklozenge$$

Example 6.10 (One-dimensional Random Walk) We will now examine an elementary example of a random walk. In Chapter 11 of this book, which is accessible on the Instructor and Student Resources website for this text, we will revisit this concept and explore its applications. Suppose that a particle is at 0 on the integer number line and suppose that at step 1, the particle will move to 1 with probability p , $0 < p < 1$, and will move to -1 with probability $1 - p$. Furthermore, if at step n , the particle is at i , then independently of the previous moves, it will move 1 unit to the right to $i + 1$ with probability p and will move 1 unit to the left to $i - 1$ with probability $1 - p$. Let X be the position of the particle after n moves. Find the probability mass function of X .

Solution: Let Y be the number of times the particle moves one unit to the right. Then Y is a binomial random variable with parameters n and p . If $Y = k$, after the n th step, the particle has moved k units to the right and $n - k$ units to the left, $k = 0, 1, \dots, n$. So after n

steps, it will be at $k - (n - k) = 2k - n$. This shows that the set of possible values of X is $E = \{-n, 2 - n, 4 - n, \dots, 2n - n\}$ and, for $i \in E$,

$$\begin{aligned} P(X = i) &= P(2Y - n = i) = P\left(Y = \frac{n+i}{2}\right) \\ &= \binom{n}{(n+i)/2} p^{(n+i)/2} (1-p)^{n-[(n+i)/2]} \\ &= \binom{n}{(n+i)/2} p^{(n+i)/2} (1-p)^{(n-i)/2}. \quad \blacklozenge \end{aligned}$$

Example 6.11 A town of 100,000 inhabitants is exposed to a contagious disease. If the probability that a person becomes infected is 0.04, what is the expected number of people who become infected?

Solution: Assuming that people become infected independently of each other, the number of those who will become infected is a binomial random variable with parameters 100,000 and 0.04. Thus the expected number of such people is $100000 \times 0.04 = 4000$. Note that here “getting infected” is implicitly defined to be a success for mathematical simplicity! Therefore, in the context of Bernoulli trials, what is called a success may be a failure in real life. $\blacklozenge$

Example 6.12 Two proofreaders, Ruby and Myra, read a book independently and found r and m misprints, respectively. Suppose that the probability that a misprint is noticed by Ruby is p and the probability that it is noticed by Myra is q , where these two probabilities are independent. If the number of misprints noticed by *both* Ruby and Myra is b , estimate the number of unnoticed misprints. [This problem was posed and solved by George Pólya (1888–1985) in the January 1976 issue of the *American Mathematical Monthly*.]

Solution: Let M be the total number of misprints in the book. The expected number of misprints that may be noticed by Ruby, Myra, and both of them are Mp , Mq , and Mpq , respectively. Assuming that the number of misprints found are approximately equal to the expected number, we have $Mp \approx r$, $Mq \approx m$, and $Mpq \approx b$. Therefore,

$$M = \frac{(Mp)(Mq)}{Mpq} \approx \frac{rm}{b}.$$

The number of unnoticed misprints is therefore estimated by

$$M - (r + m - b) \approx \frac{rm}{b} - (r + m - b) = \frac{(r - b)(m - b)}{b}. \quad \blacklozenge$$

Example 6.13 Suppose that 4% of the bicycle fenders, produced by a stamping machine from the strips of steel, need smoothing. What is the probability that, of the next 13 bicycle fenders stamped by this machine, two need smoothing and, of the next 20, three need smoothing?

Solution: Let X be the number of bicycle fenders among the first 13 that need smoothing. Let Y be the number of those among the next 7 that need smoothing. We want to calculate $P(X = 2, Y = 1)$. Since X and Y are independent binomial random variables with parameters $(13, 0.04)$ and $(7, 0.04)$, respectively, we can write

$$\begin{aligned} P(X = 2, Y = 1) &= P(X = 2)P(Y = 1) \\ &= \binom{13}{2}(0.04)^2(0.96)^{11} \binom{7}{1}(0.04)^1(0.96)^6 \approx 0.0175. \quad \blacklozenge \end{aligned}$$

The idea of Example 6.13 can be stated in general terms: Suppose that $n + m$ independent Bernoulli trials, each with parameter p , are performed. If X and Y are the number of successes in the first n and in the last m trials, then X and Y are binomial random variables with parameters (n, p) and (m, p) , respectively. Furthermore, they are independent because knowing the number of successes in the first n trials does not change the probability mass function of the number of successes in the last m trials. Hence

$$\begin{aligned} P(X = i, Y = j) &= P(X = i)P(Y = j) \\ &= \binom{n}{i} p^i (1 - p)^{n-i} \binom{m}{j} p^j (1 - p)^{m-j} \\ &= \binom{n}{i} \binom{m}{j} p^{i+j} (1 - p)^{(n+m)-(i+j)}. \end{aligned}$$

Theorem 6.2 *If X and Y are independent binomial random variables with parameters (n, p) and (m, p) , respectively, then $X + Y$ is a binomial random variable with parameters $(n + m, p)$.*

Proof: Suppose a Bernoulli trial with a probability of success p is independently performed $n + m$ times. Let X be the number of successes in the first n trials, and Y denote the number of successes in the subsequent m trials. It follows that $X + Y$ corresponds to the number of successes in the $n + m$ Bernoulli trials, and therefore, it follows a binomial distribution with parameters $(n + m, p)$. Formally, for $i = 1, 2, \dots, n + m$, letting

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{otherwise,} \end{cases}$$

we get $X = X_1 + X_2 + \dots + X_n$ and $Y = X_{n+1} + X_{n+2} + \dots + X_{n+m}$. where X_i is a Bernoulli random variable for $i = 1, 2, \dots, n + m$. Therefore, $X + Y = X_1 + X_2 + \dots + X_{n+m}$ is the sum of $n + m$ independent Bernoulli random variables, each with parameter p . This shows that $X + Y$ is binomial with parameters $(n + m, p)$. $\blacklozenge$

EXERCISES

A

1. From an ordinary deck of 52 cards, cards are drawn at random and with replacement. What is the probability that, of the first eight cards drawn, four are spades?
2. The probability that a randomly selected person is female is $1/2$. What is the expected number of the girls in the first-grade classes of an elementary school that has 64 first graders? What is the expected number of females in a family with six children?
3. A graduate class consists of six students. What is the probability that exactly three of them are born either in April or in October?
4. Let X be a Bernoulli random variable with parameter p , $0 < p < 1$. Find the probability mass function of $Y = 1 - X$, $E(Y)$, and $\text{Var}(Y)$.
5. In a state where license plates contain six digits, what is the probability that the license number of a randomly selected car has two 9s? Assume that each digit of the license number is randomly selected from $\{0, 1, \dots, 9\}$.
6. A box contains 30 balls numbered 1 through 30. Suppose that five balls are drawn at random, one at a time, with replacement. What is the probability that the numbers of two of them are prime?
7. A manufacturer of nails claims that only 3% of its nails are defective. A random sample of 24 nails is selected, and it is found that two of them are defective. Is it fair to reject the manufacturer's claim based on this observation?
8. Let X be a discrete random variable with probability mass function p given by

x	-1	0	1	other values
$p(x)$	$2/9$	$4/9$	$3/9$	0

Find the probability mass functions of the random variables $Y = |X|$ and $Z = X^2$.

9. Only 60% of certain kinds of seeds germinate when planted under normal conditions. Suppose that four such seeds are planted, and X denotes the number of those that will germinate. Find the probability mass functions of X and $Y = 2X + 1$.
10. Suppose that the Internal Revenue Service will audit 20% of income tax returns reporting an annual gross income of over \$80,000. What is the probability that of 15 such returns, at most four will be audited?
11. If two fair dice are rolled 10 times, what is the probability of at least one 6 (on either die) in exactly five of these 10 rolls?

12. From the interval $(0, 1)$, five points are selected at random and independently. What is the probability that (a) at least two of them are less than $1/3$; (b) the first decimal point of exactly two of them is 3?
13. Let X be a binomial random variable with parameters (n, p) and probability mass function $p(x)$. Prove that if $(n + 1)p$ is an integer, then $p(x)$ is maximum at two different points. Find both of these points.
14. On average, how many times should Ernie play poker in order to be dealt a straight flush (royal flush included)? (See Exercise 37 of [Section 2.4](#) for definitions of a royal and a straight flush.)
15. Suppose that each day the price of a stock moves up $1/8$ th of a point with probability $1/3$ and moves down $1/8$ th of a point with probability $2/3$. If the price fluctuations from one day to another are independent, what is the probability that after six days the stock has its original price?
16. Even though there are some differences between the taste of Max Cola and the taste of Golden Cola, most people cannot tell the difference. Pedro claims that in the absence of brand information, he can say whether a cup of cola is Max or Golden by tasting the drink. A statistician decided to test his claim. So she asked him to taste 6 cups of cola, 3 of which were Max and 3 of which were Golden, all in unmarked clear glasses, in a random order. If Pedro has no ability to distinguish Max Cola from Golden Cola, what is the probability that he could successfully identify the brand of four or more of the samples based on pure guesses? That is, by guessing the brand of each sample randomly and independently from other samples.
17. From the set $\{x: 0 \leq x \leq 1\}$, 100 independent numbers are selected at random and rounded to three decimal places. What is the probability that at least one of them is 0.345?
18. Dr. Willis is teaching two sections of Calculus I, each with 25 students. Suppose that each student will earn a passing grade, independently of other students, with probability of 0.82. What is the probability that in one of Dr. Willis's classes more than 22 students will earn a passing grade and in the other one less than 23?
19. A certain basketball player makes a foul shot with probability 0.45. Determine for what value of k the probability of k baskets in 10 shots is maximum, and find this maximum probability.
20. What are the expected value and variance of the number of full house hands in n poker hands? A poker hand consists of five randomly selected cards from an ordinary deck of 52 cards. It is a full house if three cards are of one denomination and two cards are of another denomination: for example, three queens and two 4s.
21. Let X be the number of sixes obtained when a balanced die is tossed five times. Find the probability mass function of $Y = (X - 3)^2$.
22. A certain rare blood type can be found in only 0.05% of people. If the population of a randomly selected group is 3000, what is the probability that at least two persons in the group have this rare blood type?

23. Edward's experience shows that 7% of the parcels he mails will not reach their destination. He has bought two books for \$20 each and wants to mail them to his brother. If he sends them in one parcel, the postage is \$5.20, but if he sends them in separate parcels, the postage is \$3.30 for each book. To minimize the expected value of his expenses (loss + postage), which way is preferable to send the books, as two separate parcels or as a single parcel?
24. Vincent is a patient with the life threatening blood cancer leukemia, and he is in need of a bone marrow transplant. He asks n people whether or not they are willing to donate bone marrow to him if they are a close bone marrow match for him. Suppose that each person's response, independently of others, is positive with probability p_1 and negative with probability $1 - p_1$. Furthermore, suppose that, independently of the others, the probability is p_2 that a person tested is a close match. Find the probability that Vincent finds at least one close bone marrow match among these n people.
Hint: For $0 \leq i \leq n$, let A_i be the event that i of the n individuals respond positively to Vincent's request. Let X be the number of close bone marrow matches among those who will be tested. To find $P(X \geq 1) = 1 - P(X = 0)$, calculate $P(X = 0)$ by using the Law of Total Probability, Theorem 3.4, applied to the partition $\{A_0, A_1, \dots, A_n\}$.
25. A woman and her husband want to have a 95% chance for at least one boy and at least one girl. What is the minimum number of children that they should plan to have? Assume that the events that a child is a girl and a boy are equiprobable and independent of the gender of other children born in the family.
26. A computer network consists of several stations connected by various media (usually cables). There are certain instances when no message is being transmitted. At such "suitable instances," each station will send a message with probability p independently of the other stations. However, if two or more stations send messages, a collision will corrupt the messages and they will be discarded. These messages will be retransmitted until they reach their destination. Suppose that the network consists of N stations.
- What is the probability that at a "suitable instance" a message is initiated by one of the stations and will go through without a collision?
 - Show that, to maximize the probability of a message going through with no collisions, exactly one message, on average, should be initiated at each "suitable instance."
 - Find the limit of the maximum probability obtained in (b) as the number of stations of the network grows to ∞ .
27. Consider the following problem posed by Michael Khoury, U.S. Math Olympiad Team Member, in "The Problem Solving Competition," Oklahoma Publishing Company and the American Society for the Communication of Mathematics, February 1999.
- Bob is teaching a class with n students. There are n desks in the classroom numbered from 1 to n . Bob has prepared a seating chart, but the students have already seated themselves randomly. Bob calls off the name of the person who belongs in seat 1. This person vacates the seat he or she is currently occupying and takes his or her rightful seat. If this

displaces a person already in the seat, that person stands at the front of the room until he or she is assigned a seat. Bob does this for each seat in turn.

Let X be the number of students standing at the front of the room after k , $1 \leq k < n$, names have been called. Call a person “success” if he or she is standing. Is X a binomial random variable? Why or why not?

28. Six brothers and sisters who are all either under 10 or in their early teens are having dinner with their parents and four grandparents. Their mother unintentionally feeds the entire family (including herself) a type of poisonous mushrooms that makes 20% of the adults and 30% of the children sick. What is the probability that more adults than children get sick?

B

29. In a community, a persons are pro-choice, b ($b < a$) are pro-life, and n ($n > a - b$) are undecided. Suppose that there will be a vote to determine the will of the majority with regard to legalizing abortion. If by then all of the undecided persons make up their minds, what is the probability that those who are pro-life will win? Assume that it is equally likely that an undecided person eventually votes pro-choice or pro-life.
30. A game often played in carnivals and gambling houses is called *chuck-a-luck*, where a player bets on any number 1 through 6. Then three fair dice are tossed. If one, two, or all three land the same number as the player's, then he or she receives one, two, or three times the original stake plus his or her original bet, respectively. Otherwise, the player loses his or her stake. Let X be the net gain of the player per unit of stake. First find the probability mass function of X , then determine the expected amount that the player will lose per unit of stake.
31. Let X be a binomial random variable with the parameters (n, p) . Prove that

$$E(X^2) = \sum_{x=1}^n x^2 \binom{n}{x} p^x (1-p)^{n-x} = n^2 p^2 - np^2 + np.$$

32. Suppose that an aircraft engine will fail in flight with probability $1 - p$ independently of the plane's other engines. Also suppose that a plane can complete the journey successfully if at least half of its engines do not fail.
- (a) Is it true that a four-engine plane is always preferable to a two-engine plane? Explain.
- (b) Is it true that a five-engine plane is always preferable to a three-engine plane? Explain.
33. The simplest error detection scheme used in data communication is **parity checking**. Usually messages sent consist of characters, each character consisting of a number of bits (a *bit* is the smallest unit of information and is either 1 or 0). In parity checking, a 1 or 0 is appended to the end of each character at the transmitter to make the total

number of 1s even. The receiver checks the number of 1s in every character received, and if the result is odd it signals an error. Suppose that each bit is received correctly with probability 0.999, independently of other bits. What is the probability that a 7-bit character is received in error, but the error is not detected by the parity check?

- 34.** In Exercise 33, suppose that a message consisting of six characters is transmitted. If each character consists of seven bits, what is the probability that the message is erroneously received, but none of the errors is detected by the parity check?
- 35.** How many games of poker occur until a preassigned player is dealt at least one straight flush with probability of at least $3/4$? (See Exercise 37 of [Section 2.4](#) for a definition of a straight flush.)
- 36.** The postoffice of a certain small town has only one clerk to wait on customers. The probability that a customer will be served in any given minute is 0.6, regardless of the time that the customer has already taken. The probability of a new customer arriving is 0.45, regardless of the number of customers already in line. The chances of two customers arriving during the same minute are negligible. Similarly, the chances of two customers being served in the same minute are negligible. Suppose that we start with exactly two customers: one at the postal window and one waiting on line. After 4 minutes, what is the probability that there will be exactly four customers: one at the window and three waiting on line?
- 37. (a)** What is the probability of an even number of successes in n independent Bernoulli trials?
Hint: Let r_n be the probability of an even number of successes in n Bernoulli trials. By conditioning on the first trial and using the law of total probability (Theorem 3.3), show that for $n \geq 1$,

$$r_n = p(1 - r_{n-1}) + (1 - p)r_{n-1}.$$

$$\text{Then prove that } r_n = \frac{1}{2}[1 + (1 - 2p)^n].$$

- (b)** Prove that

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} p^{2k} (1-p)^{n-2k} = \frac{1}{2}[1 + (1 - 2p)^n].$$

- 38.** An urn contains n balls whose colors, red or blue, are equally probable. [For example, the probability that all of the balls are red is $(1/2)^n$.] If in drawing k balls from the urn, successively with replacement and randomly, no red balls appear, what is the probability that the urn contains no red balls?
Hint: Use Bayes' theorem.
- 39.** While Rose always tells the truth, four of her friends, Albert, Brenda, Charles, and Donna, tell the truth randomly only in one out of three instances, independent of each other. Albert makes a statement. Brenda tells Charles that Albert's statement is the truth. Charles tells Donna that Brenda is right, and Donna says to Rose that Charles is telling

the truth and Rose agrees. What is the probability that Albert's statement is the truth? ■

Self-Quiz on Section 6.1

Time allotted: 10 Minutes

Each problem is worth 5 points.

1. What is the probability that at least two of the six members of a family are not born in the fall? Assume that all seasons have the same probability of containing the birthday of a person selected randomly.
2. Suppose that a prescription drug causes 7 side effects each with probability of 0.05 independently of the others. What is the probability that a patient taking this drug experiences at most 2 side effects. ■

6.2 POISSON RANDOM VARIABLES

Poisson as an Approximation to Binomial

From [Section 6.1](#), it should have become clear that in many natural phenomena and real-world problems, we are dealing with binomial random variables. Therefore, for possible values of parameters n , x , and p , we need to calculate $p(x) = \binom{n}{x} p^x (1-p)^{n-x}$. However, in many cases, direct calculation of $p(x)$ from this formula is not possible, because even for moderate values of n , $n!$ exceeds the largest integer that a computer can store. To overcome this difficulty, indirect methods for calculation of $p(x)$ are developed. For example, we may use the recursive formula

$$p(x) = \frac{(n-x+1)p}{x(1-p)} p(x-1),$$

obtained from (6.3), with the initial condition $p(0) = (1-p)^n$, to evaluate $p(x)$. An important study concerning calculation of $p(x)$ is the one by the French mathematician Simeon-Denis Poisson in 1837 in his book concerning the applications of probability to law. Poisson introduced the following procedure to obtain the formula that approximates the binomial probability mass function when the number of trials is large ($n \rightarrow \infty$), the probability of success is small ($p \rightarrow 0$), and the average number of successes remains a fixed quantity of moderate value ($np = \lambda$ for some constant λ).

Let X be a binomial random variable with parameters (n, p) , then

$$\begin{aligned} P(X = i) &= \binom{n}{i} p^i (1-p)^{n-i} = \frac{n!}{(n-i)! i!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^{n-i} \\ &= \frac{n(n-1)(n-2) \cdots (n-i+1)}{n^i} \frac{\lambda^i}{i!} \left(1 - \frac{\lambda}{n}\right)^n. \end{aligned}$$

Now, for large n and appreciable λ , $(1 - \lambda/n)^i$ is approximately 1, $(1 - \lambda/n)^n$ is approximately $e^{-\lambda}$ [from calculus we know that $\lim_{n \rightarrow \infty} (1 + x/n)^n = e^x$. Thus $\lim_{n \rightarrow \infty} (1 - \lambda/n)^n = e^{-\lambda}$], and $[n(n-1)(n-2) \cdots (n-i+1)]/n^i$ is approximately 1, because its numerator and denominator are both polynomials in n of degree i . Thus as $n \rightarrow \infty$,

$$P(X = i) \rightarrow \frac{e^{-\lambda} \lambda^i}{i!}.$$

Three years after his book was published, Poisson died. The importance of this approximation was left unknown until 1889, when the German-Russian mathematician L. V. Bortkiewicz demonstrated its significance for both in probability theory and its applications. Among other things, Bortkiewicz argued that since

$$\sum_{i=0}^{\infty} \frac{e^{-\lambda} \lambda^i}{i!} = e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{-\lambda} e^{\lambda} = 1,$$

Poisson's approximation by itself is a probability mass function. This and the introduction of the Poisson processes in the twentieth century made the Poisson probability mass function one of the three most important probability functions, the other two being the binomial and normal probability functions.

Definition 6.3 A discrete random variable X with possible values $0, 1, 2, 3, \dots$ is called **Poisson** with parameter λ , $\lambda > 0$, if

$$P(X = i) = \frac{e^{-\lambda} \lambda^i}{i!}, \quad i = 0, 1, 2, 3, \dots$$

Figure 6.2 depicts an example of a probability mass function of a Poisson random variable.

Under the conditions specified in our discussion, binomial probabilities can be approximated by Poisson probabilities. Such approximations are generally good if $p < 0.1$ and $np \leq 10$. If $np > 10$, it would be more appropriate to use normal approximation, discussed in Section 7.2.

Since a Poisson probability mass function is the limit of a binomial probability mass function, the expected value of a binomial random variable with parameters (n, p) is np , and

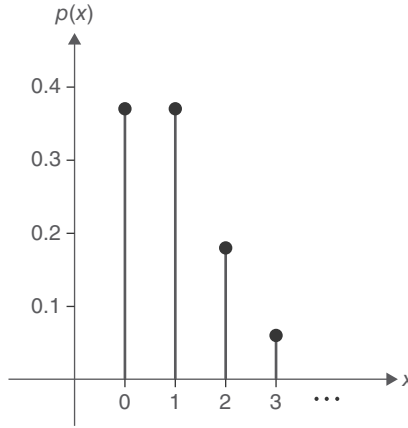


Figure 6.2 An example of a Poisson probability mass functions.

$np = \lambda$, it is reasonable to expect that the mean of a Poisson random variable with parameter λ is λ . To prove that this is the case, observe that

$$\begin{aligned} E(X) &= \sum_{i=0}^{\infty} iP(X=i) = \sum_{i=1}^{\infty} i \frac{e^{-\lambda} \lambda^i}{i!} \\ &= \lambda e^{-\lambda} \sum_{i=1}^{\infty} \frac{\lambda^{i-1}}{(i-1)!} = \lambda e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = \lambda e^{-\lambda} e^{\lambda} = \lambda. \end{aligned}$$

The variance of a Poisson random variable X with parameter λ is also λ . To see this, note that

$$\begin{aligned} E(X^2) &= \sum_{i=0}^{\infty} i^2 P(X=i) = \sum_{i=1}^{\infty} i^2 \frac{e^{-\lambda} \lambda^i}{i!} \\ &= \lambda e^{-\lambda} \sum_{i=1}^{\infty} \frac{i \lambda^{i-1}}{(i-1)!} = \lambda e^{-\lambda} \sum_{i=1}^{\infty} \frac{1}{(i-1)!} \frac{d}{d\lambda} (\lambda^i) \\ &= \lambda e^{-\lambda} \frac{d}{d\lambda} \left[\sum_{i=1}^{\infty} \frac{\lambda^i}{(i-1)!} \right] = \lambda e^{-\lambda} \frac{d}{d\lambda} \left[\lambda \sum_{i=1}^{\infty} \frac{\lambda^{i-1}}{(i-1)!} \right] \\ &= \lambda e^{-\lambda} \frac{d}{d\lambda} (\lambda e^{\lambda}) = \lambda e^{-\lambda} (e^{\lambda} + \lambda e^{\lambda}) = \lambda + \lambda^2, \end{aligned}$$

and hence

$$\text{Var}(X) = (\lambda + \lambda^2) - \lambda^2 = \lambda.$$

We have shown that

If X is a Poisson random variable with parameter λ , then

$$E(X) = \text{Var}(X) = \lambda, \quad \sigma_X = \sqrt{\lambda}.$$

Some examples of binomial random variables that obey Poisson's approximation are as follows:

1. Let X be the number of babies in a community who grow up to at least 190 centimeters. If a baby is called a success, provided that he or she grows up to the height of 190 or more centimeters, then X is a binomial random variable. Since n , the total number of babies, is large, p , the probability that a baby grows to the height of 190 centimeters or more, is small, and np , the average number of such babies, is appreciable, X is approximately a Poisson random variable.
2. Let X be the number of winning tickets among the Maryland lottery tickets sold in Baltimore during one week. Then, calling winning tickets successes, we have that X is a binomial random variable. Since n , the total number of tickets sold in Baltimore, is large, p , the probability that a ticket wins, is small, and the average number of winning tickets is appreciable, X is approximately a Poisson random variable.
3. Let X be the number of misprints on a document page typed by a secretary. Then X is a binomial random variable if a word is called a success, provided that it is misprinted! Since misprints are rare events, the number of words is large, and np , the average number of misprints, is of moderate value, X is approximately a Poisson random variable.

In the same way we can argue that random variables such as “the number of defective fuses produced by a factory in a given year,” “the number of inhabitants of a town who live at least 85 years,” “the number of machine failures per day in a plant,” and “the number of drivers arriving at a gas station per hour” are approximately Poisson random variables.

We will now give examples of problems that can be solved by Poisson's approximations. In the course of these examples, new Poisson random variables are also introduced.

Example 6.14 Every week the average number of wrong-number phone calls received by a certain mail-order house is seven. What is the probability that they will receive (a) two wrong calls tomorrow; (b) at least one wrong call tomorrow?

Solution: Assuming that the house receives a lot of calls, the number of wrong numbers received tomorrow, X , is approximately a Poisson random variable with $\lambda = E(X) = 1$. Thus

$$P(X = n) = \frac{e^{-1} \cdot (1)^n}{n!} = \frac{1}{e \cdot n!},$$

and hence the answers to (a) and (b) are, respectively,

$$P(X = 2) = \frac{1}{2e} \approx 0.18$$

and

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{1}{e} \approx 0.63. \quad \blacklozenge$$

Example 6.15 Suppose that, on average, in every three pages of a book there is one typographical error. If the number of typographical errors on a single page of the book is a Poisson random variable, what is the probability of at least one error on a specific page of the book?

Solution: Let X be the number of errors on the page we are interested in. Then X is a Poisson random variable with $E(X) = 1/3$. Hence $\lambda = E(X) = 1/3$, and thus

$$P(X = n) = \frac{(1/3)^n e^{-1/3}}{n!}.$$

Therefore,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-1/3} \approx 0.28. \quad \blacklozenge$$

Example 6.16 The atoms of a radioactive element are randomly disintegrating. If every gram of this element, on average, emits 3.9 alpha particles per second, what is the probability that during the next second the number of alpha particles emitted from 1 gram is (a) at most 6; (b) at least 2; (c) at least 3 and at most 6?

Solution: Every gram of radioactive material consists of a large number n of atoms. If for these atoms the event of disintegrating and emitting an alpha particle during the next second is called a success, then X , the number of alpha particles emitted during the next second, is a binomial random variable. Now $E(X) = 3.9$, so $np = 3.9$ and $p = 3.9/n$. Since n is very large, p is very small, and hence, to a very close approximation, X has a Poisson distribution with the appreciable parameter $\lambda = 3.9$. Therefore,

$$P(X = n) = \frac{(3.9)^n e^{-3.9}}{n!},$$

and hence (a), (b), and (c) are calculated as follows.

$$(a) \quad P(X \leq 6) = \sum_{n=0}^6 \frac{(3.9)^n e^{-3.9}}{n!} \approx 0.899.$$

$$(b) \quad P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 0.901.$$

$$(c) \quad P(3 \leq X \leq 6) = \sum_{n=3}^6 \frac{(3.9)^n e^{-3.9}}{n!} \approx 0.646. \quad \blacklozenge$$

Example 6.17 Suppose that n raisins are thoroughly mixed in dough. If we bake k raisin cookies of equal sizes from this mixture, what is the probability that a given cookie contains at least one raisin?

Solution: Since the raisins are thoroughly mixed in the dough, the probability that a given cookie contains any particular raisin is $p = 1/k$. If for the raisins the event of ending up in the given cookie is called a success, then X , the number of raisins in the given cookie, is a binomial random variable. For large values of k , $p = 1/k$ is small. If n is also large but n/k has a moderate value, it is reasonable to assume that X is approximately Poisson. Hence

$$P(X = i) = \frac{\lambda^i e^{-\lambda}}{i!},$$

where $\lambda = np = n/k$. Therefore, the probability that a given cookie contains at least one raisin is

$$P(X \neq 0) = 1 - P(X = 0) = 1 - e^{-n/k}. \quad \blacklozenge$$

Example 6.18 Let X and Y be independent Poisson random variables with parameters λ and μ , respectively.

(a) Show that

$$P(X + Y = n) = \sum_{i=0}^n P(X = i)P(Y = n - i).$$

(b) Use part (a) to prove that $X + Y$ is a Poisson random variable with parameter $\lambda + \mu$.

Solution:

(a) By the law of total probability (Theorem 3.4),

$$\begin{aligned} P(X + Y = n) &= \sum_{i=0}^n P(X + Y = n \mid X = i)P(X = i) \\ &= \sum_{i=0}^n P(X + Y = n, X = i) = \sum_{i=0}^n P(Y = n - i, X = i) \\ &= \sum_{i=0}^n P(X = i)P(Y = n - i). \end{aligned}$$

(b) By part (a),

$$\begin{aligned} P(X + Y = n) &= \sum_{i=0}^n \frac{e^{-\lambda} \lambda^i}{i!} \cdot \frac{e^{-\mu} \mu^{n-i}}{(n-i)!} = e^{-(\lambda+\mu)} \cdot \frac{1}{n!} \cdot \sum_{i=0}^n \binom{n}{i} \lambda^i \mu^{n-i} \\ &= \frac{e^{-(\lambda+\mu)} (\lambda + \mu)^n}{n!}, \end{aligned}$$

where the last equality follows from the binomial expansion (Theorem 2.5). $\blacklozenge$

For real-world problems, it is important to know that in most cases numerical examples show that, even for small values of n , the agreement between the binomial and the Poisson probabilities is surprisingly good (see Exercise 11).

Poisson Processes

Earlier, the Poisson distribution was introduced to approximate binomial distribution for very large n , very small p , and moderate np . On its own merit, the Poisson distribution appears in

connection with the study of sequences of random events occurring over time. Examples of such sequences of random events are accidents occurring at an intersection, β -particles emitted from a radioactive substance, customers entering a post office, and California earthquakes. Suppose that, starting from a time point labeled $t = 0$, we begin counting the number of events. Then for each value of t , we obtain a number denoted by $N(t)$, which is the number of events that have occurred during $[0, t]$. For example, in studying the number of β -particles emitted from a radioactive substance, $N(t)$ is the number of β -particles emitted by time t . Similarly, in studying the number of California earthquakes, $N(t)$ is the number of earthquakes that have occurred in the time interval $[0, t]$. Clearly, for each value of t , $N(t)$ is a discrete random variable with the set of possible values $\{0, 1, 2, \dots\}$. To study the distribution of $N(t)$, the number of events occurring in $[0, t]$, we make the following three simple and natural assumptions about the way in which events occur.

Stationarity: *For all $n \geq 0$, and for any two equal time intervals Δ_1 and Δ_2 , the probability of n events in Δ_1 is equal to the probability of n events in Δ_2 .*

Independent Increments: *For all $n \geq 0$, and for any time interval $(t, t + s)$, the probability of n events in $(t, t + s)$ is independent of how many events have occurred earlier or how they have occurred. In particular, suppose that the times $0 \leq t_1 < t_2 < \dots < t_k$ are given. For $1 \leq i < k - 1$, let A_i be the event that n_i events of the process occur in $[t_i, t_{i+1})$. The independent increments mean that $\{A_1, A_2, \dots, A_{k-1}\}$ is an independent set of events.*

Orderliness: *The occurrence of two or more events in a very small time interval is practically impossible. This condition is mathematically expressed by $\lim_{h \rightarrow 0} P(N(h) > 1)/h = 0$. This implies that as $h \rightarrow 0$, the probability of two or more events, $P(N(h) > 1)$, approaches 0 faster than h does. That is, if h is negligible, then $P(N(h) > 1)$ is even more negligible.*

Note that, by the stationarity property, the number of events in the interval $(t_1, t_2]$ has the same distribution as the number of events in $(t_1 + s, t_2 + s]$, $s \geq 0$. This means that the random variables $N(t_2) - N(t_1)$ and $N(t_2 + s) - N(t_1 + s)$ have the same probability mass function. In other words, the probability of occurrence of n events during the interval of time from t_1 to t_2 is a function of n and $t_2 - t_1$ and not of t_1 and t_2 independently. The number of events in $(t_i, t_{i+1}]$, $N(t_{i+1}) - N(t_i)$, is called the **increment** in $\{N(t)\}$ between t_i and t_{i+1} . That is why the second property in the preceding list is called independent-increments property. It is also noteworthy to mention that the combined concepts of stationarity and orderliness lead to the following result, which is proven in Section 11.2 of Chapter 11 within this book, accessible on the Instructor and Student Resources website for this text.

The simultaneous occurrence of two or more events is impossible. Therefore, under the aforementioned properties, events occur one at a time, not in pairs or groups.

It is for this reason that the third property is called the orderliness property.

Suppose that random events occur in time in a way that the conditions discussed above—stationarity, independent increments, and orderliness—are always satisfied. Then, if for *some* interval of length $t > 0$, $P(N(t) = 0) = 0$, we have that in *any* interval of length t at

least one event occurs. In such a case, it can be shown that in any interval of *arbitrary* length, with probability 1, at least one event occurs. Similarly, if for *some* interval of length $t > 0$, $P(N(t) = 0) = 1$, then in *any* interval of length t no event will occur. In such a case, it can be shown that in any interval of *arbitrary* length, with probability 1, no event occurs. To avoid these uninteresting, trivial cases, throughout the book, we assume that, for all $t > 0$,

$$0 < P(N(t) = 0) < 1.$$

We are now ready to state a celebrated theorem, for the validity of which we present a motivating argument. The theorem is presented again, with a rigorous proof, in Chapter 11 of this book, which is accessible on the Instructor and Student Resources website for this text.

Theorem 6.3 *If random events occur in time in a way that the preceding conditions—stationarity, independent increments, and orderliness—are always satisfied, $N(0) = 0$ and, for all $t > 0$, $0 < P(N(t) = 0) < 1$, then there exists a positive number λ such that*

$$P(N(t) = n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}.$$

That is, for all $t > 0$, $N(t)$ is a Poisson random variable with parameter λt . Hence $E[N(t)] = \lambda t$ and therefore $\lambda = E[N(1)]$.

A Motivating Argument: The property that under the stated conditions, $N(t)$ (the number of events that has occurred in $[0, t]$) is a Poisson random variable is not accidental. It is related to the fact that a Poisson random variable is approximately binomial for large n , small p , and moderate np . To see this, divide the interval $[0, t]$ into n subintervals of equal length. Then, as $n \rightarrow \infty$, the probability of two or more events in any of these subintervals is 0. Therefore, $N(t)$, the number of events in $[0, t]$, is equal to the number of subintervals in which an event occurs. If a subinterval in which an event occurs is called a success, then $N(t)$ is the number of successes. Moreover, the stationarity and independent-increments properties imply that $N(t)$ is a binomial random variable with parameters (n, p) , where p is the probability of success (i.e., the probability that an event occurs in a subinterval). Now let λ be the expected number of events in an interval of unit length. Because of stationarity, events occur at a uniform rate over the entire time period. Therefore, the expected number of events in any period of length t is λt . Hence, in particular, the expected number of the events in $[0, t]$ is λt . But by the formula for the expectation of binomial random variables, the expected number of the events in $[0, t]$ is np . Thus $np = \lambda t$ or $p = (\lambda t)/n$. Since n is extremely large ($n \rightarrow \infty$), we have that p is extremely small while $np = \lambda t$ is of moderate size. Therefore, $N(t)$ is a Poisson random variable with parameter λt . ♦

In the study of sequences of random events occurring in time, suppose that $N(0) = 0$ and, for all $t > 0$, $0 < P(N(t) = 0) < 1$. Furthermore, suppose that the events occur in a way that the preceding conditions—stationarity, independent increments, and orderliness—are always satisfied. We argued that for each value of t , the discrete random variable $N(t)$, the number of events in $[0, t]$ and hence in any other time interval of length t , is a Poisson random variable with parameter λt . Any process with this property is called a **Poisson process with rate λ** and is often denoted by $\{N(t), t \geq 0\}$.

Theorem 6.3 is remarkable as it reveals that the probability mass functions of random variables $N(t)$, where $t > 0$, can be characterized by just three straightforward and inherent physical conditions on $N(t)$. Moreover, λ , the only unknown parameter of the probability mass functions of $N(t)$'s, equals $E[N(1)]$. That is, it is the average number of events that occur in one unit of time. Hence in practice it can be measured readily. Historically, Theorem 6.3 stands as the culmination of a series of earlier theorems, representing the most sophisticated and refined form among them. It was discovered in 1955 by the Russian mathematician Alexander Khinchin (1894–1959). The first major work in this direction was done by Thornton Fry (1892–1992) in 1929. But before Fry, Albert Einstein (1879–1955) and Roman Smoluchowski (1910–1996) had also discovered important results in connection with their work on the theory of Brownian motion.

Example 6.19 (Standard Birthday Problem Revisited) Consider a class of size k of unrelated students. Assuming that the birth rates are constant throughout the year and each year has 365 days, with probability $p = 1/365$, each pair of the students has the same birthday. Let X be the number of such pairs. Argue that X is approximately a Poisson random variable and find the probability of at least one birthday match for $k = 23, 30, 50$, and 60 . Compare your results with the exact probabilities that were calculated up to three decimal places in Example 2.8.

Solution: There are $n = \binom{k}{2}$ pairs of birthdays. Call a pair success if its elements are identical and failure otherwise. Clearly X is a binomial random variable with parameters n and p . Since n is large and p is small, X is approximately a Poisson random variable with the appreciable parameter $\lambda = np = \binom{k}{2} \frac{1}{365}$. The probability of at least one birthday match is

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{e^{-\lambda} \cdot \lambda^0}{0!} = 1 - e^{-\lambda}.$$

The following table shows how good this approximation is

k	23	30	50	60
$P(X \geq 1)$ by Poisson approximation	0.500	0.696	0.965	0.992
$P(X \geq 1)$ obtained in Example 2.8	0.507	0.706	0.970	0.995

Using Poisson random variables, similarly to the method used in this exercise, we can easily find excellent approximations for the probability that two students have, say, birthdays within one day of each other or within two days of each other and so forth. For having birthdays within one day, in the solution above, all we need to do is to change p from $1/365$ to $3/365$. Similarly, for having birthdays within two days, it suffices to change p from $1/365$ to $5/365$. Such birthday problems are not easy to solve. Poisson approximation comes in handy and gives excellent results. ♦

Example 6.20 Suppose that children are born at a Poisson rate of five per day in a certain hospital. What is the probability that (a) at least two babies are born during the next six hours; (b) no babies are born during the next two days?

Solution: Let $N(t)$ denote the number of babies born at or prior to t . The assumption that $\{N(t), t \geq 0\}$ is a Poisson process is reasonable because it is stationary, it has independent increments, $N(0) = 0$, and simultaneous births are impossible. Thus $\{N(t), t \geq 0\}$ is a Poisson process. If we choose one day as time unit, then $\lambda = E[N(1)] = 5$. Therefore,

$$P(N(t) = n) = \frac{(5t)^n e^{-5t}}{n!}.$$

Hence the probability that at least two babies are born during the next six hours is

$$\begin{aligned} P(N(1/4) \geq 2) &= 1 - P(N(1/4) = 0) - P(N(1/4) = 1) \\ &= 1 - \frac{(5/4)^0 e^{-5/4}}{0!} - \frac{(5/4)^1 e^{-5/4}}{1!} \approx 0.36, \end{aligned}$$

where $1/4$ is used since 6 hours is $1/4$ of a day. The probability that no babies are born during the next two days is

$$P(N(2) = 0) = \frac{(10)^0 e^{-10}}{0!} \approx 4.54 \times 10^{-5}. \quad \blacklozenge$$

Example 6.21 Suppose that earthquakes occur in a certain region of California, in accordance with a Poisson process, at a rate of seven per year.

- (a) What is the probability of no earthquakes in 1 year?
- (b) What is the probability that in exactly 3 of the next 8 years no earthquakes will occur?

Solution:

- (a) Let $N(t)$ be the number of earthquakes in this region at or prior to t . We are given that $\{N(t), t \geq 0\}$ is a Poisson process. If we choose 1 year as the unit of time, then $\lambda = E[N(1)] = 7$. Thus

$$P(N(t) = n) = \frac{(7t)^n e^{-7t}}{n!}, \quad n = 0, 1, 2, 3, \dots$$

Let p be the probability of no earthquakes in 1 year, then

$$p = P(N(1) = 0) = e^{-7} \approx 0.00091.$$

- (b) Suppose that a year is called a success if during its course no earthquakes occur. Of the next 8 years, let X be the number of years in which no earthquakes will occur. Then X is a binomial random variable with parameters $(8, p)$. Thus

$$P(X = 3) \approx \binom{8}{3} (0.00091)^3 (1 - 0.00091)^5 \approx 4.2 \times 10^{-8}. \quad \blacklozenge$$

Example 6.22 A fisherman catches fish at a Poisson rate of two per hour from a large lake with lots of fish. Yesterday, he went fishing at 10:00 A.M. and caught just one fish by 10:30 and a total of three by noon. What is the probability that he can duplicate this feat tomorrow?

Solution: Label the time the fisherman starts fishing tomorrow at $t = 0$. Let $N(t)$ denote the total number of fish caught at or prior to t . Clearly, $N(0) = 0$. It is reasonable to assume that catching two or more fish simultaneously is impossible. It is also reasonable to assume that $\{N(t), t \geq 0\}$ is stationary and has independent increments. Thus the assumption that $\{N(t), t \geq 0\}$ is a Poisson process is well grounded. Choosing 1 hour as the unit of time, we have $\lambda = E[N(1)] = 2$. Thus

$$P(N(t) = n) = \frac{(2t)^n e^{-2t}}{n!}, \quad n = 0, 1, 2, \dots$$

We want to calculate the probability of the event

$$\{N(1/2) = 1 \text{ and } N(2) = 3\}.$$

But this event is the same as $\{N(1/2) = 1 \text{ and } N(2) - N(1/2) = 2\}$. Thus by the independent-increments property,

$$P(N(1/2) = 1 \text{ and } N(2) - N(1/2) = 2) = P(N(1/2) = 1) \cdot P(N(2) - N(1/2) = 2).$$

Since stationarity implies that

$$P(N(2) - N(1/2) = 2) = P(N(3/2) = 2),$$

the desired probability equals

$$P(N(1/2) = 1) \cdot P(N(3/2) = 2) = \frac{1^1 e^{-1}}{1!} \cdot \frac{3^2 e^{-3}}{2!} \approx 0.082. \quad \blacklozenge$$

Example 6.23 Let $N(t)$ be the number of earthquakes that occur at or prior to time t worldwide. Suppose that $\{N(t): t \geq 0\}$ is a Poisson process with rate λ and the probability that the magnitude of an earthquake on the Richter scale is 5 or more is p . Find the probability of k earthquakes of such magnitudes at or prior to t worldwide.

Solution: Let $X(t)$ be the number of earthquakes of magnitude 5 or more on the Richter scale at or prior to t worldwide. Since the sequence of events $\{N(t) = n\}$, $n = 0, 1, 2, \dots$ is mutually exclusive and $\bigcup_{n=0}^{\infty} \{N(t) = n\}$ is the sample space, by the law of total probability, Theorem 3.4,

$$P(X(t) = k) = \sum_{n=0}^{\infty} P(X(t) = k \mid N(t) = n) P(N(t) = n).$$

Now clearly, $P(X(t) = k \mid N(t) = n) = 0$ if $n < k$. If $n > k$, the conditional probability mass

function of $X(t)$ given that $N(t) = n$ is binomial with parameters n and p . Thus

$$\begin{aligned}
 P(X(t) = k) &= \sum_{n=k}^{\infty} \binom{n}{k} p^k (1-p)^{n-k} \frac{e^{-\lambda t} (\lambda t)^n}{n!} \\
 &= \sum_{n=k}^{\infty} \frac{n!}{k! (n-k)!} p^k (1-p)^{n-k} \frac{e^{-\lambda t} (\lambda t)^k (\lambda t)^{n-k}}{n!} \\
 &= \frac{e^{-\lambda t} (\lambda t p)^k}{k!} \sum_{n=k}^{\infty} \frac{1}{(n-k)!} [\lambda t (1-p)]^{n-k} \\
 &= \frac{e^{-\lambda t} (\lambda t p)^k}{k!} \sum_{j=0}^{\infty} \frac{1}{(j)!} [\lambda t (1-p)]^j \\
 &= \frac{e^{-\lambda t} (\lambda t p)^k}{k!} e^{\lambda t (1-p)} \\
 &= \frac{e^{-\lambda t p} (\lambda t p)^k}{k!}.
 \end{aligned}$$

It should be clear that $\{X(t) : t \geq 0\}$ is stationary, orderly, and possesses independent increment property. Therefore, $\{X(t) : t \geq 0\}$ is itself a Poisson process with parameter λp . ♦

Remark 6.1 We only considered sequences of random events that occur in *time*. However, the restriction to time is not necessary. Random events that occur on the real line, the plane, or in space and satisfy the stationarity and orderliness conditions and possess independent increments also form Poisson processes. For example, suppose that a wire manufacturing company produces a wire that has various fracture sites where the wire will fail in tension. Let $N(t)$ be the number of fracture sites in the first t meters of wire. The process $\{N(t), t \geq 0\}$ may be modeled as a Poisson process. As another example, suppose that in a certain region S , the numbers of trees that grow in nonoverlapping subregions are independent of each other, the distributions of the number of trees in subregions of equal area are identical, and the probability of two or more trees in a very small subregion is negligible. Let λ be the expected number of trees in a region of area 1 and $A(R)$ be the area of a region R . Then $N(R)$, the number of trees in a subregion R is a Poisson random variable with parameter $\lambda A(R)$, and the set $\{N(R), R \subseteq S\}$ is a two-dimensional Poisson process. ♦

EXERCISES

A

1. Jim buys 60 lottery tickets every week. If only 5% of the lottery tickets win, what is the probability that he wins next week?
2. Suppose that 3% of the families in a large city have an annual income of over \$60,000. What is the probability that, of 60 random families, at most three have an annual income of over \$60,000?

3. Suppose that 2.5% of the population of a border town are illegal immigrants. Find the probability that, in a theater of this town with 80 random viewers, there are at least two illegal immigrants.
4. By Example 2.24, the probability that a poker hand is a full house is 0.0014. What is the probability that in 500 random poker hands there are at least two full houses?
5. On a random day, the number of vacant rooms of a big hotel in New York City is 35, on average. What is the probability that next Saturday this hotel has at least 30 vacant rooms?
6. On average, there are three misprints in every 10 pages of a particular book. If every chapter of the book contains 35 pages, what is the probability that [Chapters 1](#) and [5](#) have 10 misprints each?
7. Suppose that X is a Poisson random variable with $P(X = 1) = P(X = 3)$. Find $P(X = 5)$.
8. Suppose that n raisins have been carefully mixed with a batch of dough. If we bake k ($k > 4$) raisin buns of equal size from this mixture, what is the probability that two out of four randomly selected buns contain no raisins?
Hint: Note that, by Example 6.17, the number of raisins in a given bun is approximately Poisson with parameter n/k .
9. The children in a small town all own slingshots. In a recent contest, 4% of them were such poor shots that they did not hit the target even once in 100 shots. If the number of times a randomly selected child has hit the target is approximately a Poisson random variable, determine the percentage of children who have hit the target at least twice.
10. At a large mall, on a major holiday, they give a gift to the first, and then to every 100th customer who enters the mall. If customers arrive at the mall according to a Poisson process with rate λ , find the probability mass function of X_t , the number of gifts given by time t .
11. The department of mathematics of a state university has 26 faculty members. For $i = 0, 1, 2, 3$, find p_i , the probability that i of them were born on Independence Day (a) using the binomial distribution; (b) using the Poisson distribution. Assume that the birth rates are constant throughout the year and that each year has 365 days.
12. Suppose that on a summer evening, shooting stars are observed at a Poisson rate of one every 12 minutes. What is the probability that three shooting stars are observed in 30 minutes?
13. Let X be the number of times, within the next 5 years, that a policyholder of an insurance company gets injured in a car accident. An actuary has shown that X is a Poisson random variable, and the probability that, within 5 years, a policyholder gets injured in exactly two accidents is $1/5$ th of the probability that he or she gets injured in exactly one accident. Find the mean and the standard deviation of X .
14. It is 6:00 P.M. now and, for $t > 0$, the number of orders that are placed, between now and t minutes past 6:00 P.M., through an electronic commerce company's website, is a Poisson process. Suppose that the first order placed through the company's website will

arrive within the next minute with probability 0.09. Find the expected number of all the orders that will be placed within the next hour.

15. Suppose that in Japan earthquakes occur at a Poisson rate of three per week. What is the probability that the next earthquake occurs after two weeks?
16. Shocks occur independently to a system at a Poisson rate of λ . The probability that the system survives a shock is p independently of other shocks. Find the probability that the system has survived all shocks by time t .
17. A pharmaceutical company has developed an Ebola vaccine, which is 98.5% of the time effective. If 5,000 people have received the vaccine, and sooner or later all of them will be exposed to the Ebola virus, what is the probability that at most 70 of them will become sick with Ebola? Assume that a person reacts to the Ebola vaccine independently of other people.
18. Suppose that, for a telephone subscriber, the number of wrong numbers is Poisson, at a rate of $\lambda = 1$ per week. A certain subscriber has not received any wrong numbers from Sunday through Friday. What is the probability that he receives no wrong numbers on Saturday either?
19. In a certain town, crimes occur at a Poisson rate of five per month. What is the probability of having exactly 2 months (not necessarily consecutive) with no crimes during the next year?
20. Accidents occur at an intersection at a Poisson rate of three per day. What is the probability that during January there are exactly three days (not necessarily consecutive) without any accidents?
21. Customers arrive at a bookstore at a Poisson rate of six per hour. Given that the store opens at 9:30 A.M., what is the probability that exactly one customer arrives by 10:00 A.M. and 10 customers by noon?
22. A company is located in a region that is prone to dust storms. The company has purchased an insurance policy that pays \$200,000 for each dust storm in a year except the first one. If the number of the dust storms per year is a Poisson random variable with mean 0.83, what is the expected amount of money that the company is paid in a random year under this insurance policy?
23. A wire manufacturing company has inspectors to examine the wire for fractures as it comes out of a machine. The number of fractures is distributed in accordance with a Poisson process, having one fracture on the average for every 60 meters of wire. One day an inspector has to take an emergency phone call and is missing from his post for 10 minutes. If the machine turns out 7 meters of wire per minute, what is the probability that the inspector will miss more than one fracture?
24. Let X be a Poisson random variable with parameter λ . Let

$$Y = \begin{cases} 0 & \text{if } X \text{ is zero or even} \\ 1 & \text{If } X \text{ is odd.} \end{cases}$$

Find the probability mass function of Y .

B

25. On a certain two-lane north-south highway, there is a T junction. Cars arrive at the junction according to a Poisson process, on the average four per minute. For cars to turn left onto the side street, the highway is widened by the addition of a left-turn lane that is long enough to accommodate three cars. If four or more cars are trying to turn left, the fourth car will effectively block north-bound traffic. At the junction for the left-turn lane there is a left-turn signal that allows cars to turn left for 1 minute and prohibits such turns for the next 3 minutes. The probability of a randomly selected car turning left at this T junction is 0.22. Suppose that during a green light for the left-turn lane all waiting cars were able to turn left. What is the probability that during the subsequent red light for the left-turn lane, the north-bound traffic will be blocked?
26. Suppose that, on the Richter scale, earthquakes of magnitude 5.5 or higher have probability 0.015 of damaging certain types of bridges. Suppose that such intense earthquakes occur at a Poisson rate of 1.5 per 10 years. If a bridge of this type is constructed to last at least 60 years, what is the probability that it will be undamaged by earthquakes for that period of time?
27. According to the United States Postal Service, <http://www.usps.gov>, May 15, 1998,

Dogs have caused problems for letter carriers for so long that the situation has become a cliché. In 1983, more than 7,000 letter carriers were bitten by dogs. . . . However, the 2,795 letter carriers who were bitten by dogs last year represent less than one-half of 1 percent of *all reported* dog-bite victims.

Suppose that during a year 94% of the letter carriers are not bitten by dogs. Assuming that dogs bite letter carriers randomly, what percentage of those who sustained one bite will be bitten again?

28. Suppose that in Maryland, on a certain day, N lottery tickets are sold and M win. To have a probability of at least α of winning on that day, approximately how many tickets should be purchased?
29. Balls numbered $1, 2, \dots$, and n are randomly placed into cells numbered $1, 2, \dots$, and n . Therefore, for $1 \leq i \leq n$ and $1 \leq j \leq n$, the probability that ball i is in cell j is $1/n$. For each i , $1 \leq i \leq n$, if ball i is in cell i , we say that a *match* has occurred at cell i .
 - (a) What is the probability of exactly k matches?
 - (b) Let $n \rightarrow \infty$. Show that the probability mass function of the number of matches is Poisson with mean 1.
30. Let $\{N(t), t \geq 0\}$ be a Poisson process. What is the probability of (a) an even number of events in $(t, t + \alpha)$; (b) an odd number of events in $(t, t + \alpha)$?

- 31.** Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ . Suppose that $N(t)$ is the total number of two types of events that have occurred in $[0, t]$. Let $N_1(t)$ and $N_2(t)$ be the total number of events of type 1 and events of type 2 that have occurred in $[0, t]$, respectively. If events of type 1 and type 2 occur independently with probabilities p and $1 - p$, respectively, prove that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are Poisson processes with respective rates λp and $\lambda(1 - p)$.

Hint: First calculate $P(N_1(t) = n \text{ and } N_2(t) = m)$ using the relation

$$\begin{aligned} P(N_1(t) = n \text{ and } N_2(t) = m) \\ = \sum_{i=0}^{\infty} P(N_1(t) = n \text{ and } N_2(t) = m \mid N(t) = i) P(N(t) = i). \end{aligned}$$

(This is true because of Theorem 3.4.) Then use the relation

$$P(N_1(t) = n) = \sum_{m=0}^{\infty} P(N_1(t) = n \text{ and } N_2(t) = m).$$

- 32.** Customers arrive at a grocery store at a Poisson rate of one per minute. If $2/3$ of the customers are female and $1/3$ are male, what is the probability that 15 females enter the store between 10:30 and 10:45?

Hint: Use the result of Exercise 31.

- 33.** In a forest, the number of trees that grow in a region of area R has a Poisson distribution with mean λR , where λ is a given positive number.

- (a) Find the probability that the distance from a certain tree to the nearest tree is more than d .
- (b) Find the probability that the distance from a certain tree to the n th nearest tree is more than d .

- 34.** Let X be a Poisson random variable with parameter λ . Show that the maximum of $P(X = i)$ occurs at $[\lambda]$, where $[\lambda]$ is the greatest integer less than or equal to λ .

Hint: Let p be the probability mass function of X . Prove that

$$p(i) = \frac{\lambda}{i} p(i - 1).$$

Use this to find the values of i at which p is increasing and the values of i at which it is decreasing.



Self-Quiz on Section 6.2

Time allotted: 15 Minutes

Each problem is worth 5 points.

1. Suppose that due to an exhaust leak in 0.3% of cars manufactured by a car company, their passenger side exhaust manifolds must be replaced. What is the probability that in the next 5,000 cars manufactured by this company, at least 12 but no more than 15 need such a repair?
2. Patients arrive at a pharmacy for flu shots at a Poisson rate of 3 per hour. Given that the pharmacy opens at 8:00 A.M., what is the probability that exactly 2 patients arrive by 9:00 A.M. and 12 patients by noon?



6.3 OTHER DISCRETE RANDOM VARIABLES

Geometric Random Variables

Consider an experiment in which independent Bernoulli trials are performed until the first success occurs. The sample space for such an experiment is

$$S = \{s, fs, ffs, fffs, \dots, ff \cdots fs, \dots\}.$$

Now, suppose that a sequence of independent Bernoulli trials, each with probability of success p , $0 < p < 1$, are performed. Let X be the number of experiments until the first success occurs. Then X is a discrete random variable called **geometric**. It is defined on S , its set of possible values is $\{1, 2, \dots\}$, and

$$P(X = n) = (1 - p)^{n-1}p, \quad n = 1, 2, 3, \dots$$

This equation follows since (a) the first $(n - 1)$ trials are all failures, (b) the n th trial is a success, and (c) the successive Bernoulli trials are all independent.

Let $p(x) = (1 - p)^{x-1}p$ for $x = 1, 2, 3, \dots$, and 0 elsewhere. Then, for all values of x in $\mathbf{R}$, $p(x) \geq 0$ and

$$\sum_{x=1}^{\infty} p(x) = \sum_{x=1}^{\infty} (1 - p)^{x-1}p = \frac{p}{1 - (1 - p)} = 1,$$

by the geometric series theorem. Hence $p(x)$ is a probability mass function.

Definition 6.4 The probability mass function

$$p(x) = \begin{cases} (1 - p)^{x-1}p & 0 < p < 1, \quad x = 1, 2, 3, \dots, \\ 0 & \text{elsewhere} \end{cases}$$

is called **geometric with parameter p** .

Figure 6.3 depicts the probability mass function of the geometric random variable with parameter $p = 1/2$.

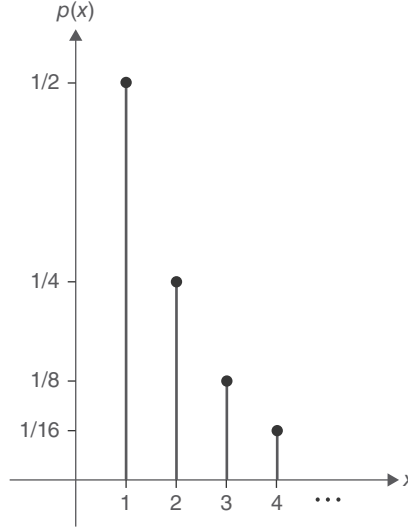


Figure 6.3 Plot of the probability mass function for a geometric random variable.

Let X be a geometric random variable with parameter p , then

$$\begin{aligned} E(X) &= \sum_{x=1}^{\infty} xp(1-p)^{x-1} = \frac{p}{1-p} \sum_{x=1}^{\infty} x(1-p)^x \\ &= \frac{p}{1-p} \frac{1-p}{[1-(1-p)]^2} = \frac{1}{p}, \end{aligned}$$

where the third equality follows from the relation $\sum_{x=1}^{\infty} xr^x = r/(1-r)^2$, $|r| < 1$. $E(X) = 1/p$ indicates that to get the first success, on average, $1/p$ independent Bernoulli trials are needed. The relation $\sum_{x=1}^{\infty} x^2 r^x = [r(r+1)]/(1-r)^3$, $|r| < 1$, implies that

$$E(X^2) = \sum_{x=1}^{\infty} x^2 p(1-p)^{x-1} = \frac{p}{1-p} \sum_{x=1}^{\infty} x^2 (1-p)^x = \frac{2-p}{p^2}.$$

Hence

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2-p}{p^2} - \left(\frac{1}{p}\right)^2 = \frac{1-p}{p^2}.$$

We have established the following formulas:

Let X be a geometric random variable with parameter p , $0 < p < 1$. Then

$$E(X) = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}, \quad \sigma_X = \frac{\sqrt{1-p}}{p}.$$

Example 6.24 From an ordinary deck of 52 cards we draw cards at random, *with replacement*, and successively until an ace is drawn. What is the probability that at least 10 draws are needed?

Solution: Let X be the number of draws until the first ace. The random variable X is geometric with the parameter $p = 1/13$. Thus

$$P(X = n) = \left(\frac{12}{13}\right)^{n-1} \left(\frac{1}{13}\right), \quad n = 1, 2, 3, \dots,$$

and so the probability that at least 10 draws are needed is

$$\begin{aligned} P(X \geq 10) &= \sum_{n=10}^{\infty} \left(\frac{12}{13}\right)^{n-1} \left(\frac{1}{13}\right) = \frac{1}{13} \sum_{n=10}^{\infty} \left(\frac{12}{13}\right)^{n-1} \\ &= \frac{1}{13} \cdot \frac{(12/13)^9}{1 - 12/13} = \left(\frac{12}{13}\right)^9 \approx 0.49. \end{aligned}$$

Remark: There is a shortcut to the solution of this problem: The probability that at least 10 draws are needed to get an ace is the same as the probability that in the first nine draws there are no aces. This is equal to $(12/13)^9 \approx 0.49$. ♦

Let X be a geometric random variable with parameter p , $0 < p < 1$. Then, for all positive integers n and m ,

$$P(X > n + m | X > m) = \frac{P(X > n + m)}{P(X > m)} = \frac{(1 - p)^{n+m}}{(1 - p)^m} = (1 - p)^n = P(X > n).$$

This is called the **memoryless property** of geometric random variables. It means that

In successive independent Bernoulli trials, the probability that the next n outcomes are all failures does not change if we are given that the previous m successive outcomes were all failures.

This is obvious by the independence of the trials. Interestingly enough, in the following sense, geometric random variable is the only memoryless discrete random variable.

Let X be a discrete random variable with the set of possible values $\{1, 2, 3, \dots\}$. If for all positive integers n and m ,

$$P(X > n + m | X > m) = P(X > n),$$

then X is a geometric random variable. That is, there exists a number p , $0 < p < 1$, such that

$$P(X = n) = p(1 - p)^{n-1}, \quad n \geq 1.$$

We leave the proof of this theorem as an exercise (see Exercise 26).

Example 6.25 A father asks his sons to cut their backyard lawn. Since he does not specify which of the three sons is to do the job, each boy tosses a coin to determine the odd person, who must then cut the lawn. In the case that all three get heads or tails, they continue tossing until they reach a decision. Let p be the probability of heads and $q = 1 - p$, the probability of tails.

- (a) Find the probability that they reach a decision in less than n tosses.
- (b) If $p = 1/2$, what is the minimum number of tosses required to reach a decision with probability 0.95?

Solution:

- (a) The probability that they reach a decision on a certain round of coin tossing is

$$\binom{3}{2}p^2q + \binom{3}{2}q^2p = 3pq(p + q) = 3pq.$$

The probability that they do not reach a decision on a certain round is $1 - 3pq$. Let X be the number of tosses until they reach a decision, then X is a geometric random variable with parameter $3pq$. Therefore,

$$P(X < n) = 1 - P(X \geq n) = 1 - (1 - 3pq)^{n-1},$$

where the second equality follows since $X \geq n$ if and only if none of the first $n - 1$ tosses results in a success.

- (b) We want to find the minimum n so that $P(X \leq n) \geq 0.95$. This gives $1 - P(X > n) \geq 0.95$ or $P(X > n) \leq 0.05$. But

$$P(X > n) = (1 - 3pq)^n = (1 - 3/4)^n = (1/4)^n.$$

Therefore, we must have $(1/4)^n \leq 0.05$, or $n \ln 1/4 \leq \ln 0.05$. This gives $n \geq 2.16$; hence the smallest n is 3. ♦

Negative Binomial Random Variables

Negative binomial random variables are generalizations of geometric random variables. Suppose that a sequence of independent Bernoulli trials, each with probability of success p , $0 < p < 1$, is performed. Let X be the number of experiments until the r th success occurs. Then X is a discrete random variable called a **negative binomial**. Its set of possible values is $\{r, r + 1, r + 2, r + 3, \dots\}$ and

$$P(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}, \quad n = r, r + 1, \dots \quad (6.4)$$

This equation follows since if the outcome of the n th trial is the r th success, then in the first $(n - 1)$ trials exactly $(r - 1)$ successes have occurred and the n th trial is a success. The probability of the former event is

$$\binom{n-1}{r-1} p^{r-1} (1-p)^{(n-1)-(r-1)} = \binom{n-1}{r-1} p^{r-1} (1-p)^{n-r},$$

and the probability of the latter is p . Therefore, by the independence of the trials, (6.4) follows.

Definition 6.5 *The probability mass function*

$$p(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad 0 < p < 1, \quad x = r, r+1, r+2, r+3, \dots,$$

is called **negative binomial** with parameters (r, p) .

Note that a negative binomial probability mass function with parameters $(1, p)$ is geometric. In Examples 6.29 and 8.8, we have demonstrated the following:

If X is a negative binomial random variable with parameters (r, p) , then

$$E(X) = \frac{r}{p}, \quad \text{Var}(X) = \frac{r(1-p)}{p^2}, \quad \sigma_X = \frac{\sqrt{r(1-p)}}{p}.$$

Example 6.26 Sharon and Ann play a series of backgammon games until one of them wins five games. Suppose that the games are independent and the probability that Sharon wins a game is 0.58.

- (a) Find the probability that the series ends in seven games.
- (b) If the series ends in seven games, what is the probability that Sharon wins?

Solution: (a) Let X be the number of games until Sharon wins five games. Let Y be the number of games until Ann wins five games. The random variables X and Y are negative binomial with parameters $(5, 0.58)$ and $(5, 0.42)$, respectively. The probability that the series ends in seven games is

$$\begin{aligned} P(X = 7) + P(Y = 7) &= \binom{6}{4} (0.58)^5 (0.42)^2 + \binom{6}{4} (0.42)^5 (0.58)^2 \\ &\approx 0.17 + 0.066 \approx 0.24. \end{aligned}$$

(b) Let A be the event that Sharon wins and B be the event that the series ends in seven games. Then the desired probability is

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{P(X = 7)}{P(X = 7) + P(Y = 7)} \approx \frac{0.17}{0.24} \approx 0.71. \quad \blacklozenge$$

The following example, given by Kaigh in January 1979 issue of *Mathematics Magazine*, is a modification of the gambler's ruin problem, Example 3.15.

Example 6.27 (Attrition Ruin Problem) Two gamblers play a game in which in each play gambler A beats B with probability p , $0 < p < 1$, and loses to B with probability $q = 1 - p$. Suppose that each play results in a forfeiture of \$1 for the loser and in no change for the winner. If player A initially has a dollars and player B has b dollars, what is the probability that B will be ruined?

Solution: Let E_i be the event that, in the first $b + i$ plays, B loses b times. Let A^* be the event that A wins. Then

$$P(A^*) = \sum_{i=0}^{a-1} P(E_i).$$

If every time that A wins is called a success, E_i is the event that the b th success occurs on the $(b + i)$ th play. Using the negative binomial distribution, we have

$$P(E_i) = \binom{i+b-1}{b-1} p^b q^i.$$

Therefore,

$$P(A^*) = \sum_{i=0}^{a-1} \binom{i+b-1}{b-1} p^b q^i.$$

As a numerical illustration, let $a = b = 4$, $p = 0.6$, and $q = 0.4$. We get $P(A^*) = 0.710208$ and $P(B^*) = 0.289792$ exactly. ♦

The following example is due to Hugo Steinhaus (1887–1972), who brought it up in a conference honoring Stefan Banach (1892–1945), a smoker and one of the greatest mathematicians of the twentieth century.

Example 6.28 (Banach Matchbox Problem) A smoking mathematician carries two matchboxes, one in his right pocket and one in his left pocket. Whenever he wants to smoke, he selects a pocket at random and takes a match from the box in that pocket. If each matchbox initially contains N matches, what is the probability that when the mathematician for the first time *discovers* that one box is empty, there are exactly m matches in the other box, $m = 0, 1, 2, \dots, N$?

Solution: Every time that the left pocket is selected we say that a success has occurred. When the mathematician discovers that the left box is empty, the right one contains m matches if and only if the $(N + 1)$ st success occurs on the $(N - m) + (N + 1) = (2N - m + 1)$ st trial. The probability of this event is

$$\binom{(2N - m + 1) - 1}{(N + 1) - 1} \left(\frac{1}{2}\right)^{N+1} \left(\frac{1}{2}\right)^{(2N - m + 1) - (N + 1)} = \binom{2N - m}{N} \left(\frac{1}{2}\right)^{2N - m + 1}.$$

By symmetry, when the mathematician discovers that the right box is empty, with probability $\binom{2N-m}{N} \left(\frac{1}{2}\right)^{2N-m+1}$, the left box contains m matches. Therefore, the desired probability is

$$2 \binom{2N-m}{N} \left(\frac{1}{2}\right)^{2N-m+1} = \binom{2N-m}{N} \left(\frac{1}{2}\right)^{2N-m}. \quad \blacklozenge$$

Example 6.29 Let X be a negative binomial random variable with parameters (r, p) . Then in a sequence of independent Bernoulli trials each with success probability p , X is the number of trials until the r th success. Let X_1 be the number of trials until the first success, X_2 be the number of additional trials to get the second success, X_3 the number of additional ones to obtain the third success, and so on. Then clearly

$$X = X_1 + X_2 + \cdots + X_r,$$

where for $i = 1, 2, \dots, r$, the random variable X_i is geometric with parameter p . This is because $P(X_i = n) = (1-p)^{n-1}p$ by the independence of the trials. Since $E(X_i) = 1/p$ ($i = 1, 2, \dots, r$),

$$E(X) = E(X_1) + E(X_2) + \cdots + E(X_r) = \frac{r}{p}.$$

This formula shows that, for example, in the experiment of throwing a fair die successively, on the average, it takes $5/(1/6) = 30$ trials to get five 6s. $\blacklozenge$

Hypergeometric Random Variables

Suppose that, from a box containing D defective and $N - D$ nondefective items, n are drawn at random and *without replacement*. Furthermore, suppose that the number of items drawn does not exceed the number of defective or the number of nondefective items. That is, suppose that $n \leq \min(D, N - D)$. Let X be the number of defective items drawn. Then X is a discrete random variable with the set of possible values $\{0, 1, \dots, n\}$, and a probability mass function

$$p(x) = P(X = x) = \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, 2, \dots, n.$$

Any random variable X with such a probability mass function is called a **hypergeometric random variable**. The fact that $p(x)$ is a probability mass function is readily verified. Clearly, $p(x) \geq 0, \forall x$. To prove that $\sum_{x=0}^n p(x) = 1$, note that this is equivalent to

$$\sum_{x=0}^n \binom{D}{x} \binom{N-D}{n-x} = \binom{N}{n},$$

which can be shown by a simple combinatorial argument. (See Exercise 58 of [Section 2.4](#); in that exercise let $m = D$, $n = N - D$, and $r = n$.)

Definition 6.6 Let N , D , and n be positive integers with $n \leq \min(D, N - D)$. Then

$$p(x) = P(X = x) = \begin{cases} \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}} & \text{if } x \in \{0, 1, 2, \dots, n\} \\ 0 & \text{elsewhere} \end{cases}$$

is said to be a **hypergeometric** probability mass function.

For the following important results, see Example 6.34, and Exercise 26, [Section 8.2](#).

For the hypergeometric random variable X , defined above,

$$E(X) = \frac{nD}{N}, \quad \text{Var}(X) = \frac{nD(N-D)}{N^2} \left(1 - \frac{n-1}{N-1}\right).$$

Note that if the experiment of drawing n items from a box containing D defective and $N - D$ nondefective items is performed *with replacement*, then X is binomial with parameters n and D/N . Hence

$$E(X) = \frac{nD}{N}, \quad \text{Var}(X) = n \frac{D}{N} \left(1 - \frac{D}{N}\right) = \frac{nD(N-D)}{N^2}.$$

These show that if items are drawn *with replacement*, then the expected value of X will not change, but the variance will increase. However, if n is much smaller than N , then, as the variance formulae confirm, drawing with replacement is a good approximation for drawing without replacement.

Example 6.30 In 500 independent calculations a scientist has made 25 errors. If a second scientist checks seven of these calculations randomly, what is the probability that he detects two errors? Assume that the second scientist will definitely find the error of a false calculation.

Solution: Let X be the number of errors found by the second scientist. Then X is hypergeometric with $N = 500$, $D = 25$, and $n = 7$. We are interested in $P(X = 2)$, which is given by

$$p(2) = \frac{\binom{25}{2} \binom{500-25}{7-2}}{\binom{500}{7}} \approx 0.04. \quad \blacklozenge$$

Example 6.31 In a community of $a + b$ potential voters, a are pro-choice and b ($b < a$) are pro-life. Suppose that a vote is taken to determine the will of the majority with regard to legalizing abortion. If n ($n < b$) random persons of these $a + b$ potential voters do not vote, what is the probability that the pro-life advocates will win?

Solution: Let X be the number of those who do not vote and are pro-choice. The pro-life advocates will win if and only if

$$a - X < b - (n - X).$$

But this is true if and only if $X > (a - b + n)/2$. Since X is a hypergeometric random variable, we have that

$$P\left(X > \frac{a - b + n}{2}\right) = \sum_{i=\lceil \frac{a-b+n}{2} \rceil + 1}^n P(X = i) = \sum_{i=\lceil \frac{a-b+n}{2} \rceil + 1}^n \frac{\binom{a}{i} \binom{b}{n-i}}{\binom{a+b}{n}},$$

where $\lceil \frac{a-b+n}{2} \rceil$ is the greatest integer less than or equal to $\frac{a-b+n}{2}$. ♦

Example 6.32 Every month, a large lot of 2000 fuses manufactured by a certain company is shipped to a major retail store. The policy of the store is to return a lot if more than 2% of its fuses are defective. Since 2% of 50 fuses is 1 fuse, the store selects 50 of the 2000 fuses at random and tests them all. If it finds more than one defective fuse among them, it will return the entire lot. Does this sampling strategy, with probability of 0.95 or higher, guarantee that the store's return policy is satisfied?

Solution: Let X be the number of defective items among the 50 randomly selected fuses from a lot. According to the store's sampling strategy, $P(X > 1)$ is the probability that the store will return the lot. Note that, based on the store's policy, the lot must contain at least 41 defective fuses for the store to return it. Suppose that it actually has exactly 41 such fuses. Then X is hypergeometric with parameters $N = 2000$, $D = 41$, and $n = 50$. In this case, the probability of returning the lot, based on the sampling strategy, is

$$\begin{aligned} P(X > 1) &= 1 - P(X = 0) - P(X = 1) \\ &= 1 - \frac{\binom{41}{0} \binom{2000-41}{50}}{\binom{2000}{50}} - \frac{\binom{41}{1} \binom{2000-41}{49}}{\binom{2000}{50}} \approx 0.274. \end{aligned}$$

Thus the probability of returning a lot that must be returned is too low, much lower than the required 0.95. This alone shows that the store's sampling strategy does not guarantee that its return policy is satisfied. Furthermore, by similar calculations, the following table shows that even if there are greater than 41 defective fuses in the lot, still the store's sampling strategy fails to guarantee that its policy is satisfied with the required probability.

Number of defective items in the lot	50	70	100	150
$P(X > 1)$	0.357	0.580	0.724	0.900

It is interesting to note that, with probability of 0.95 or higher, the store's sampling strategy works if its policy is to return a lot in which more than 9% of the fuses are defective. That is, a lot with at least 181 defective fuses. For exactly 181 such fuses, $P(X > 1) = 0.95$, and if there are more than 181 defective items in the lot, then $P(X > 1)$ is greater than 0.95. ♦

Example 6.33 Professors Davidson and Johnson from the University of Victoria in Canada gave the following problem to their students in a finite math course:

An urn contains N balls of which B are black and $N - B$ are white; n balls are chosen at random and without replacement from the urn. If X is the number of black balls chosen, find $P(X = i)$.

Using the hypergeometric formula, the solution to this problem is

$$P(X = i) = \frac{\binom{B}{i} \binom{N - B}{n - i}}{\binom{N}{n}}, \quad i \leq \min(n, B).$$

However, by mistake, some students interchanged B with n and came up with the following answer:

$$P(X = i) = \frac{\binom{n}{i} \binom{N - n}{B - i}}{\binom{N}{B}},$$

which can be verified immediately as the correct answer for all values of i . Explain if this is accidental or if there is a probabilistic reason for its validity.

Solution: In their paper “Interchanging Parameters of Hypergeometric Distributions,” *Mathematics Magazine*, December 1993, Volume 66, Davidson and Johnson indicate that what happened is not accidental and that there is a probabilistic reason for it. Their justification is elegant and is as follows:

We begin with an urn that contains N white balls. When Ms. Painter arrives, she picks B balls at random without replacement from the urn, then paints each of the drawn balls black with instant dry paint, and finally returns these B balls to the urn. When Mr. Carver arrives, he chooses n balls at random without replacement from the urn, then engraves each of the chosen balls with the letter C and, finally, returns these n balls to the urn. Let the random variable X denote the number of black balls chosen (i.e., painted and engraved) when both Painter and Carver have finished their jobs. Since the tasks of Painter

and Carver do not depend on which task is done first, the probability distribution of X is the same whether Painter does her job before or after Carver does his. If Painter goes first, Carver chooses n balls from an urn that contains B black balls and $N - B$ white balls, so

$$P(X = i) = \frac{\binom{B}{i} \binom{N-B}{n-i}}{\binom{N}{n}} \quad \text{for } i = 0, 1, 2, \dots, \min(n, B).$$

On the other hand, if Carver goes first, Painter draws B balls from an urn that contains n balls engraved with C and $N - n$ balls not engraved, so

$$P(X = i) = \frac{\binom{n}{i} \binom{N-n}{B-i}}{\binom{N}{B}} \quad \text{for } i = 0, 1, 2, \dots, \min(n, B).$$

Thus changing the order of who goes first provides an urn-drawing explanation of why the probability distribution of X remains the same when B is interchanged with n . ♦

Let X be a hypergeometric random variable with parameters n , D , and N . Calculation of $E(X)$ and $\text{Var}(X)$ directly from $p(x)$, the probability mass function of X , is tedious. By the methods developed in [Sections 4.7](#) and [8.2](#), calculation of these quantities is considerably easier. In [Section 4.7](#), we have established that that $E(X) = nD/N$; it can be shown that

$$\text{Var}(X) = \frac{nD(N-D)}{N^2} \left(1 - \frac{n-1}{N-1}\right).$$

(See Exercise 26, [Section 8.2](#).) Using these formulas, for example, we have that the average number of face cards in a bridge hand is $(13 \times 12)/52 = 3$ with variance

$$\frac{13 \times 12(52-12)}{52^2} \left(1 - \frac{13-1}{52-1}\right) = 1.765.$$

Example 6.34 Let X be a hypergeometric random variable with probability mass function

$$p(x) = P(X = x) = \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}},$$

$$n \leq \min(D, N-D), \quad x = 0, 1, 2, \dots, n.$$

Then X is the number of defective items among n items drawn at random and without replacement from an urn containing D defective and $N - D$ nondefective items. To calculate $E(X)$, let A_i be the event that the i th item drawn is defective. Also, for $i = 1, 2, \dots, n$, let

$$X_i = \begin{cases} 1 & \text{if } A_i \text{ occurs} \\ 0 & \text{otherwise;} \end{cases}$$

then $X = X_1 + X_2 + \dots + X_n$. Hence

$$E(X) = E(X_1) + E(X_2) + \dots + E(X_n),$$

where for $i = 1, 2, \dots, n$,

$$E(X_i) = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) = P(X_i = 1) = P(A_i) = \frac{D}{N}.$$

This follows since the i th item can be any of the N items with equal probabilities. Therefore,

$$E(X) = \frac{nD}{N}.$$

This shows that, for example, the expected number of spades in a random bridge hand is $(13 \times 13)/52 = 13/4 \approx 3.25$. ♦

Example 6.35 Dr. Shams has 23 hypertensive patients, of whom five do not use any medicine but try to lower their blood pressures by self-help: dieting, exercise, not smoking, relaxation, and so on. Of the remaining 18 patients, 10 use beta blockers and 8 use diuretics. A random sample of seven of all these patients is selected. Let X , Y , and Z be the number of the patients in the sample trying to lower their blood pressures by self-help, beta blockers, and diuretics, respectively. Find the joint probability mass function and the marginal probability mass functions of X , Y , and Z .

Solution: Let $p(x, y, z)$ be the joint probability mass function of X , Y , and Z . Then for $0 \leq x \leq 5$, $0 \leq y \leq 7$, $0 \leq z \leq 7$, $x + y + z = 7$,

$$p(x, y, z) = \frac{\binom{5}{x} \binom{10}{y} \binom{8}{z}}{\binom{23}{7}};$$

$p(x, y, z) = 0$, otherwise. The marginal probability mass function of X , p_X , is obtained as follows:

$$p_X(x) = \sum_{\substack{x+y+z=7 \\ 0 \leq y \leq 7 \\ 0 \leq z \leq 7}} p(x, y, z) = \sum_{y=0}^{7-x} \frac{\binom{5}{x} \binom{10}{y} \binom{8}{7-x-y}}{\binom{23}{7}}$$

$$= \frac{\binom{5}{x}}{\binom{23}{7}} \sum_{y=0}^{7-x} \binom{10}{y} \binom{8}{7-x-y}.$$

Now $\sum_{y=0}^{7-x} \binom{10}{y} \binom{8}{7-x-y}$ is the total number of the ways we can choose $7-x$ patients from 18 (in Exercise 58 of [Section 2.4](#), let $m = 10$, $n = 8$, and $r = 7-x$); hence it is equal to $\binom{18}{7-x}$. Therefore, as expected,

$$p_X(x) = \frac{\binom{5}{x} \binom{18}{7-x}}{\binom{23}{7}}, \quad x = 0, 1, 2, 3, 4, 5.$$

Similarly,

$$p_Y(y) = \frac{\binom{10}{y} \binom{13}{7-y}}{\binom{23}{7}}, \quad y = 0, 1, 2, \dots, 7,$$

$$p_Z(z) = \frac{\binom{8}{z} \binom{15}{7-z}}{\binom{23}{7}}, \quad z = 0, 1, 2, \dots, 7.$$

Remark: A joint probability mass function, such as $p(x, y, z)$ of this Example, is called *multivariate hypergeometric*, which will be thoroughly examined in the upcoming section. ♦

Binomial and Hypergeometric Relationships

If the n items that are selected at random from the D defective and $N - D$ nondefective items are chosen *with replacement* rather than *without replacement*, then X , the number of defective items, is a binomial random variable with parameters n and D/N . Thus

$$P(X = x) = \binom{n}{x} \left(\frac{D}{N}\right)^x \left(1 - \frac{D}{N}\right)^{n-x}, \quad x = 0, 1, 2, \dots, n.$$

Now, if N is very large, it is not that important whether a sample is taken with or without replacement. For example, if we have to choose 1000 persons from a population of 200,000,000, replacing persons that are selected back into the population will change the chance of other persons to be selected only insignificantly. Therefore, for large N , the binomial probability mass function is an excellent approximation for the hypergeometric probability mass function, which can be stated mathematically as follows.

$$\lim_{\substack{N \rightarrow \infty \\ D \rightarrow \infty \\ D/N \rightarrow p}} \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}} = \binom{n}{x} p^x (1-p)^{n-x}.$$

To prove this, note that, by Example 6.33,

$$\frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}} = \binom{n}{x} \frac{\binom{N-n}{D-x}}{\binom{N}{D}}.$$

Therefore,

$$\begin{aligned} \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}} &= \binom{n}{x} \frac{\binom{N-n}{D-x}}{\binom{N}{D}} = \binom{n}{x} \frac{\frac{(N-n)!}{(D-x)!(N-n-D+x)!}}{\frac{N!}{D!(N-D)!}} \\ &= \binom{n}{x} \frac{D!}{(D-x)!} \cdot \frac{(N-D)!}{(N-D-n+x)!} \cdot \frac{(N-n)!}{N!} \\ &= \binom{n}{x} \frac{D(D-1) \cdots (D-x+1)(N-D)(N-D-1) \cdots (N-D-n+x+1)}{N(N-1) \cdots (N-n+1)} \\ &= \binom{n}{x} \frac{\frac{D(D-1) \cdots (D-x+1)(N-D)(N-D-1) \cdots (N-D-n+x+1)}{N^n}}{\frac{N(N-1) \cdots (N-n+1)}{N^n}} \\ &= \binom{n}{x} \frac{\left(\frac{D}{N}\right) \left(\frac{D-1}{N}\right) \cdots \left(\frac{D-x+1}{N}\right) \left(\frac{N-D}{N}\right) \left(\frac{N-D-1}{N}\right) \cdots \left(\frac{N-D-n+x+1}{N}\right)}{\left(\frac{N}{N}\right) \left(\frac{N-1}{N}\right) \left(\frac{N-2}{N}\right) \cdots \left(\frac{N-n+1}{N}\right)} \\ &= \binom{n}{x} \frac{\frac{D}{N} \left(\frac{D}{N} - \frac{1}{N}\right) \cdots \left(\frac{D}{N} - \frac{x-1}{N}\right) \left(1 - \frac{D}{N}\right) \left(1 - \frac{D}{N} - \frac{1}{N}\right) \cdots \left(1 - \frac{D}{N} - \frac{n-x-1}{N}\right)}{\left(1 - \frac{1}{N}\right) \left(1 - \frac{2}{N}\right) \cdots \left(1 - \frac{n-1}{N}\right)} \end{aligned}$$

Now as $N \rightarrow \infty$, $D \rightarrow \infty$, and $D/N \rightarrow p$, the denominator approaches 1, and the numerator approaches $p^x(1-p)^{n-x}$. So $\frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}}$ approaches $\binom{n}{x} p^x(1-p)^{n-x}$.

In summary, when drawing a sample from a large population, without significantly impacting the proportion of successes in the population, the hypergeometric distribution behaves similarly to the binomial distribution with the same probability of success. This property allows the binomial distribution to be used as an approximation for the hypergeometric distribution in such scenarios.

The binomial and hypergeometric distributions share another significant connection. Firstly, we demonstrated that we can transition from the hypergeometric distribution to the binomial distribution by taking a limit. Secondly, the reverse process is also possible, where we can move from the binomial to the hypergeometric distribution through conditioning. To explain this, let X and Y be independent binomial random variables with parameters (n, p) and (m, p) , respectively. We now calculate the conditional probability $P(X = x \mid X + Y = k)$ using relations (3.1) and (3.5), and noting that, by Theorem 6.2, $X + Y$ is binomial with parameters $(n + m, p)$.

$$\begin{aligned}
 P(X = x \mid X + Y = k) &= \frac{P(X = x, X + Y = k)}{P(X + Y = k)} \\
 &= \frac{P(X + Y = k \mid X = x)P(X = x)}{P(X + Y = k)} \\
 &= \frac{P(Y = k - x)P(X = x)}{P(X + Y = k)} \\
 &= \frac{\binom{m}{k-x} p^{k-x} (1-p)^{m-k+x} \cdot \binom{n}{x} p^x (1-p)^{n-x}}{\binom{n+m}{k} p^k (1-p)^{n+m-k}} \\
 &= \frac{\binom{n}{x} \binom{m}{k-x}}{\binom{n+m}{k}}.
 \end{aligned}$$

This reveals that the conditional distribution of X given $X + Y = k$ is hypergeometric with parameters $n + m$, n , and k . To intuitively explain this relationship, consider a scenario where the probability, p , represents the likelihood of a first-year student at a college, independently of other students, taking a calculus course. Let's assume the college has n female and m male first-year students. Given that k first-year students have enrolled in calculus courses, the

probability that x of them are females no longer depends on p . Instead, it is expressed as follows:

$$\frac{\binom{n}{x} \binom{m}{k-x}}{\binom{n+m}{k}}.$$

This expression represents the probability that x out of k enrolled students are females, and it can be derived using the hypergeometric probability mass function. To illustrate this, imagine a box containing n identical blue balls (representing female students) and m identical red balls (representing male students) - distinguished only by their colors. When k balls are randomly drawn from the box, the above expression corresponds to the probability of x blue balls being drawn.

In summary, the conditional distribution's connection between the binomial and the hypergeometric distribution is elucidated through this intuitive explanation involving the probabilities of first-year students taking calculus courses and the analogy of drawing balls from a box with colored balls.

EXERCISES

A

1. Define a sample space for the experiment that, from a box containing two defective and five nondefective items, three items are drawn at random and without replacement.
2. Define a sample space for the experiment of performing independent Bernoulli trials until the second success occurs.
3. An absentminded professor does not remember which of his 12 keys will open his office door. If he tries them at random and with replacement:
 - (a) On average, how many keys should he try before his door opens?
 - (b) What is the probability that he opens his office door after only three tries?
4. The probability is p that Marty hits target M when he fires at it. The probability is q that Alvie hits target A when he fires at it. Marty and Alvie fire one shot each at their targets. If both of them hit their targets, they stop; otherwise, they will continue.
 - (a) What is the probability that they stop after each has fired r times?
 - (b) What is the expected value of the number of the times that each of them has fired before stopping?
5. Suppose that 20% of a group of people have hazel eyes. What is the probability that the eighth passenger boarding a plane is the third one having hazel eyes? Assume that passengers boarding the plane form a randomly chosen group.

6. A certain basketball player makes a foul shot with probability 0.45. What is the probability that (a) his first basket occurs on the sixth shot; (b) his first and second baskets occur on his fourth and eighth shots, respectively?
7. A store has 50 light bulbs available for sale. Of these, five are defective. A customer buys eight light bulbs randomly from this store. What is the probability that he finds exactly one defective light bulb among them?
8. The probability is p that a randomly chosen light bulb is defective. We screw a bulb into a lamp and switch on the current. If the bulb works, we stop; otherwise, we try another and continue until a good bulb is found. What is the probability that at least n bulbs are required?
9. Suppose that independent Bernoulli trials with parameter p are performed successively. Let N be the number of trials needed to get x successes, and X be the number of successes in the first n trials. Show that

$$P(N = n) = \frac{x}{n} P(X = x).$$

Remark: By this relation, in coin tossing, for example, we can state that the probability of getting a fifth head on the seventh toss is $5/7$ of the probability of five heads in seven tosses.

10. In rural Ireland, a century ago, the students had to form a line. The student at the front of the line would be asked to spell a word. If he spelled it correctly, he was allowed to sit down. If not, he received a whack on the hand with a switch and was sent to the end of the line. Suppose that a student could spell correctly 70% of the words in the lesson. What is the probability that the student would be able to sit down before receiving four whacks on the hand? Assume that the master chose the words to be spelled randomly and independently.
11. The digits after the decimal point of a random number between 0 and 1 are numbers selected at random, with replacement, independently, and successively from the set $\{0, 1, \dots, 9\}$. In a random number from $(0, 1)$, on the average, how many digits are there before the fifth 3?
12. On average, how many games of bridge are necessary before a player is dealt three aces? A bridge hand is 13 randomly selected cards from an ordinary deck of 52 cards.
13. Solve the Banach matchbox problem (Example 6.28) for the case where the matchbox in the right pocket contains M matches, the matchbox in his left pocket contains N matches, and $m \leq \min(M, N)$.
14. For certain software, independently of other users, the probability is 0.07 that a user encounters a fault. What are the chances that the 30th user is the 5th person encountering a fault?
15. In recent months, an electric utility company has received complaints from its customers who think that, due to their unusually high electricity bills, their electric meters are defective. The company begins to inspect the meters one by one. If only 3.6% of them are defective, what is the probability that it takes at least 50 inspections until the 5th defective electric meter is found?

16. Suppose that 15% of the population of a town are senior citizens. Let X be the number of nonsenior citizens who enter a mall before the tenth senior citizen arrives. Find the probability mass function of X . Assume that each customer who enters the mall is a random person from the entire population.
17. Florence is relocating to another town and intends to sell her collection of 100 Blu-ray movies. Unbeknownst to her, 10 of these Blu-rays are defective. Sharon is interested in buying the collection, but she will only do so if a random sample of 10 movies contains no more than one defective item. What is the probability that Sharon decides to make the purchase?
18. Let X be a geometric random variable with parameter p , $0 < p < 1$. Find $E\left(\frac{1}{1+X}\right)$.
19. In an annual charity drive, 35% of a population of 560 make contributions. If, in a statistical survey, 15 people are selected at random and without replacement, what is the probability that at least two persons have contributed?
20. The probability is p that a message sent over a communication channel is garbled. If the message is sent repeatedly until it is transmitted reliably, and if each time it takes 2 minutes to process it, what is the probability that the transmission of a message takes more than t minutes?
21. A vending machine contains cans of grapefruit juice that cost 75 cents each, but it is not working properly. The probability that it accepts a coin is 10%. Angela has a quarter and five dimes. Determine the probability that she should try the coins at least 50 times before she gets a can of grapefruit juice.
22. A computer network consists of several stations connected by various media (usually cables). There are certain instances when no message is being transmitted. At such “suitable instances,” each station will send a message with probability p , independently of the other stations. However, if two or more stations send messages, a collision will corrupt the messages, and they will be discarded. These messages will be retransmitted until they reach their destination. If the network consists of N stations, on average, how many times should a certain station transmit and retransmit a message until it reaches its destination?
23. A company puts five different types of prizes into their cereal boxes, one in each box and in equal proportions. If a customer decides to collect all five prizes, what is the expected number of the boxes of cereals that he or she should buy?

B

24. A fair coin is flipped repeatedly. What is the probability that the fifth tail occurs before the tenth head?
25. On average, how many independent games of poker are required until a preassigned player is dealt a straight? (See Exercise 37 of [Section 2.4](#) for a definition of a straight. The cards have distinct consecutive values that are not of the same suit: for example, 3 of hearts, 4 of hearts, 5 of spades, 6 of hearts, and 7 of hearts.)
26. Let X be a geometric random variable with parameter p , and n and m be nonnegative

integers.

- (a) For what values of n is $P(X = n)$ maximum?
- (b) What is the probability that X is even?
- (c) Show that the geometric is the only distribution on the positive integers with the memoryless property:

$$P(X > n + m \mid X > m) = P(X > n).$$

- 27. In data communication, messages are usually combinations of characters, and each character consists of a number of bits. A bit is the smallest unit of information and is either 1 or 0. Suppose that the length of a character (in bits) is a geometric random variable with parameter p . Furthermore, suppose that the lengths of characters are independent random variables. What is the distribution of the total number of the bits forming a message of k random characters?
- 28. Twelve hundred eggs, of which 200 are rotten, are distributed randomly in 100 cartons, each containing a dozen eggs. These cartons are then sold to a restaurant. How many cartons should we expect the chef of the restaurant to open before finding one without rotten eggs?
- 29. In the Banach matchbox problem, Example 6.28, find the probability that when the first matchbox is emptied (not found empty) there are exactly m matches in the other box.
- 30. In the Banach matchbox problem, Example 6.28, find the probability that the box which is emptied first is not the one that is first found empty.
- 31. Adam rolls a well-balanced die until he gets a 6. Andrew rolls the same die until he rolls an odd number. What is the probability that Andrew rolls the die more than Adam does?
- 32. To estimate the number of trout in a lake, we caught 50 trout, tagged and returned them. Later we caught 50 trout and found that four of them were tagged. From this experiment estimate n , the total number of trout in the lake.
Hint: Let p_n be the probability of four tagged trout among the 50 trout caught. Find the value of n that maximizes p_n .
- 33. Suppose that, from a box containing D defective and $N - D$ nondefective items, n ($n \leq D$) are drawn one by one, at random and *without replacement*.
 - (a) Find the probability that the k th item drawn is defective.
 - (b) If the $(k - 1)$ st item drawn is defective, what is the probability that the k th item is also defective?



Self-Quiz on Section 6.3

Time allotted: 15 Minutes

Each problem is worth 5 points.

1. A maximum of m ($m \geq 1$) independent Bernoulli trials, each with probability of success p , $0 < p < 1$, are performed successively. However, we stop the trials if the first success occurs before the m th trial. Let Z be the number of experiments performed. Find the probability mass function of Z .
2. At the Antonio Car dealership, the probability that during a month at least one consumer returns a car for warranty work is 0.45. Suppose that the numbers of cars returned for warranty work in different months are independent. Consider the period until the fifth month in which at least one car was returned for warranty work. What is the probability that, during that period, there were *exactly* 3 months with no car returned for warranty work?

Hint: Call a month “success” if, in that month, at least one car is returned for warranty work.



6.4 MULTINOMIAL DISTRIBUTIONS

Multinomial distribution is a generalization of a binomial. Suppose that, for $r \geq 2$, whenever an experiment is performed, one of the disjoint outcomes $A_1, A_2, \dots, A_r$ will occur. Let $P(A_i) = p_i$, $1 \leq i \leq r$. Then $p_1 + p_2 + \dots + p_r = 1$. If, in n independent performances of this experiment, X_i , $i = 1, 2, \dots, r$, denotes the number of times that A_i occurs, then $p(x_1, \dots, x_r)$, the joint probability mass function of $X_1, X_2, \dots, X_r$, is called **multinomial joint probability mass function**, and its distribution is said to be a **multinomial distribution**. For any set of nonnegative integers $\{x_1, x_2, \dots, x_r\}$ with $x_1 + x_2 + \dots + x_r = n$,

$$\begin{aligned} p(x_1, x_2, \dots, x_r) &= P(X_1 = x_1, X_2 = x_2, \dots, X_r = x_r) \\ &= \frac{n!}{x_1! x_2! \dots x_r!} p_1^{x_1} p_2^{x_2} \dots p_r^{x_r}. \end{aligned} \quad (6.5)$$

To prove this relation, recall that, by Theorem 2.4, the number of distinguishable permutations of n objects of r different types where x_1 are alike, x_2 are alike, $\dots$, x_r are alike ($n = x_1 + \dots + x_r$) is $n!/(x_1! x_2! \dots x_r!)$. Hence there are $n!/(x_1! x_2! \dots x_r!)$ sequences of $A_1, A_2, \dots, A_r$ in which the number of A_i 's is x_i , $i = 1, 2, \dots, r$. Relation (6.5) follows since the probability of the occurrence of any of these sequences is $p_1^{x_1} p_2^{x_2} \dots p_r^{x_r}$. By Theorem 2.6, $p(x_1, x_2, \dots, x_r)$'s given by (6.5) are the terms in the expansion of $(p_1 + p_2 + \dots + p_r)^n$. For this reason, the multinomial distribution sometimes is called the

polynomial distribution. The following relation guarantees that $p(x_1, x_2, \dots, x_r)$ is a joint probability mass function:

$$\begin{aligned} & \sum_{x_1+x_2+\dots+x_r=n} p(x_1, x_2, \dots, x_r) \\ &= \sum_{x_1+x_2+\dots+x_r=n} \frac{n!}{x_1! x_2! \dots x_r!} p_1^{x_1} p_2^{x_2} \dots p_r^{x_r} = (p_1 + p_2 + \dots + p_r)^n = 1. \end{aligned}$$

Note that, for $r = 2$, the multinomial distribution coincides with the binomial distribution. This is because, for $r = 2$, the experiment has only two possible outcomes, A_1 and A_2 . Hence it is a Bernoulli trial.

Example 6.36 In a certain town, at 8:00 P.M., 30% of the TV viewing audience watch the news, 25% watch a certain comedy, and the rest watch other programs. What is the probability that, in a statistical survey of seven randomly selected viewers, exactly three watch the news and at least two watch the comedy?

Solution: In the random sample, let X_1 , X_2 , and X_3 be the numbers of viewers who watch the news, the comedy, and other programs, respectively. Then the joint distribution of X_1 , X_2 , and X_3 is multinomial with $p_1 = 0.30$, $p_2 = 0.25$, and $p_3 = 0.45$. Therefore, for $i + j + k = 7$,

$$P(X_1 = i, X_2 = j, X_3 = k) = \frac{7!}{i! j! k!} (0.30)^i (0.25)^j (0.45)^k.$$

The desired probability equals

$$\begin{aligned} P(X_1 = 3, X_2 \geq 2) &= P(X_1 = 3, X_2 = 2, X_3 = 2) \\ &\quad + P(X_1 = 3, X_2 = 3, X_3 = 1) + P(X_1 = 3, X_2 = 4, X_3 = 0) \\ &= \frac{7!}{3! 2! 2!} (0.30)^3 (0.25)^2 (0.45)^2 + \frac{7!}{3! 3! 1!} (0.30)^3 (0.25)^3 (0.45)^1 \\ &\quad + \frac{7!}{3! 4! 0!} (0.30)^3 (0.25)^4 (0.45)^0 \approx 0.103. \quad \blacklozenge \end{aligned}$$

Example 6.37 A warehouse contains 500 TV sets, of which 25 are defective, 300 are in working condition but used, and the rest are brand new. What is the probability that, in a random sample of five TV sets from this warehouse, there are exactly one defective and exactly two brand new sets?

Solution: In the random sample, let X_1 be the number of defective TV sets, X_2 be the number of TV sets in working condition but used, and X_3 be the number of brand new sets. The desired probability is given by

$$P(X_1 = 1, X_2 = 2, X_3 = 2) = \frac{\binom{25}{1} \binom{300}{2} \binom{175}{2}}{\binom{500}{5}} \approx 0.067.$$

Note that, since the number of sets is large and the sample size is small, the joint distribution of X_1 , X_2 , and X_3 is *approximately* multinomial with $p_1 = 25/500 = 1/20$, $p_2 = 300/500 = 3/5$, and $p_3 = 175/500 = 7/20$. Hence

$$P(X_1 = 1, X_2 = 2, X_3 = 2) \approx \frac{5!}{1! 2! 2!} \left(\frac{1}{20}\right)^1 \left(\frac{3}{5}\right)^2 \left(\frac{7}{20}\right)^2 \approx 0.066. \quad \blacklozenge$$

Example 6.38 (Marginals of Multinomials) Let $X_1, X_2, \dots, X_r$ ($r \geq 2$) have the joint multinomial probability mass function $p(x_1, x_2, \dots, x_r)$ with parameters n and $p_1, p_2, \dots, p_r$. Find the marginal probability mass functions p_{X_1} .

Solution: From the definition of marginal probability mass functions,

$$\begin{aligned} p_{X_1}(x_1) &= \sum_{x_2+x_3+\dots+x_r=n-x_1} \frac{n!}{x_1! x_2! \dots x_r!} p_1^{x_1} p_2^{x_2} \dots p_r^{x_r} \\ &= \frac{n!}{x_1! (n-x_1)!} p_1^{x_1} \sum_{x_2+x_3+\dots+x_r=n-x_1} \frac{(n-x_1)!}{x_2! x_3! \dots x_r!} p_2^{x_2} p_3^{x_3} \dots p_r^{x_r} \\ &= \frac{n!}{x_1! (n-x_1)!} p_1^{x_1} (p_2 + p_3 + \dots + p_r)^{n-x_1} \\ &= \frac{n!}{x_1! (n-x_1)!} p_1^{x_1} (1-p_1)^{n-x_1}, \end{aligned}$$

where the next-to-last equality follows from multinomial expansion (Theorem 2.6) and the last inequality follows from $p_1 + p_2 + \dots + p_r = 1$. This shows that *the marginal probability mass function of X_1 is binomial with parameters n and p_1* . There is nothing distinctive about X_1 . A similar proof shows that, for $1 \leq i \leq r$, the marginal probability mass function of X_i is binomial with parameters n and p_i . $\blacklozenge$

EXERCISES

1. Light bulbs manufactured by a certain factory last a random time between 400 and 1200 hours. What is the probability that, of eight such bulbs, three burn out before 550 hours, two burn out after 800 hours, and three burn out after 550 but before 800 hours?
2. An urn contains 100 chips of which 20 are blue, 30 are red, and 50 are green. We draw 20 chips at random and with replacement. Let B , R , and G be the number of blue, red, and green chips, respectively. Calculate the joint probability mass function of B , R , and G .
3. Suppose that each day the price of a stock moves up $1/8$ of a point with probability $1/4$, remains the same with probability $1/3$, and moves down $1/8$ of a point with probability $5/12$. If the price fluctuations from one day to another are independent, what is the probability that after six days the stock has its original price?

4. At a certain college, 16% of the calculus students get A's, 34% B's, 34% C's, 14% D's, and 2% F's. What is the probability that, of 15 calculus students selected at random, five get B's, five C's, two D's, and at least two A's?
5. Of the drivers who are insured by a certain insurance company and get into at least one accident during a random year, 15% are low-risk drivers, 35% are moderate-risk drivers, and 50% are high-risk drivers. Suppose that drivers insured by this company get into an accident independently of each other. If an actuary randomly chooses five drivers insured by this company, who got into an accident one or more times last year, what is the probability that there are at least two more high-risk drivers among them than the low-risk and moderate-risk drivers combined?
6. Suppose that 50% of the watermelons grown on a farm are classified as large, 30% as medium, and 20% as small. Joanna buys five watermelons at random from this farm. What is the probability that (a) at least two of them are large; (b) two of them are large, two are medium, and one is small; (c) exactly two of them are medium if it is given that at least two of them are large?
7. Suppose that the ages of 30% of the teachers of a country are over 50, 20% are between 40 and 50, and 50% are below 40. In a random committee of 10 teachers from this country, two are above 50. What is the probability mass function of those who are below 40?
8. Customers enter a department store at the rate of three per minute, in accordance with a Poisson process. If 30% of them buy nothing, 20% pay cash, 40% use charge cards, and 10% write personal checks, what is the probability that in five operating minutes of the store, five customers use charge cards, two write personal checks, and three pay cash?



Self-Quiz on Section 6.4

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. Suppose that 40% of the students joining the mathematics department of a certain university major in Actuarial Science, 35% major in statistics and operation research, and 25% major in pure mathematics. There are 30 new students entering this department next fall. What is the probability that exactly 12 of them will major in actuarial sciences and either 7 or 8 students will major in pure mathematics?
2. Of the emails that arrive in Samantha's inbox, 45% are personal, 40% are work-related, and 15% are unsolicited commercial emails. Samantha logs into her email account and finds that she has 25 new emails. What is the probability that exactly 10 of them are work-related and at most 4 of them are unsolicited commercial emails?



6.5 SIMULATING SPECIAL DISCRETE DISTRIBUTIONS USING R

In R, the `help("distributions")` command provides a list of native distributions. Numerous additional distributions can be accessed via R packages. In this section, we will delve into simulating various discrete distributions using R. This discussion will be further extended in [Section 7.8](#), where we will explore the simulation of continuous random variables.

In R, when simulating different types of random variables, whether they are discrete or continuous, the convention is that commands beginning with **d** typically represent the probability mass function for discrete random variables and the probability density function for continuous random variables. Commands starting with **p** are used for probability distribution functions, while those starting with **r** are used to simulate random variables and generate realizations or samples.

Binomial Random Variable

In R, three commands are instrumental in simulating binomial random variables: **dbinom** represents the probability mass function, **pbinom** signifies the probability distribution function, and **rbinom** is used to produce realizations of binomial random variables. Suppose X is a binomial random variable with parameters $n = 10$ and a success probability $p = 0.57$. Then,

$$P(X = 6) = \binom{10}{6} (0.57)^6 (1 - 0.57)^4 = \mathbf{dbinom(6, 10, 0.57)} = 0.2462307.$$

$$P(X \leq 5) = \sum_{i=0}^5 \binom{10}{i} (0.57)^i (1 - 0.57)^{10-i} = \mathbf{pbinom(5, 10, 0.57)} = 0.4436094.$$

To determine **dbinom(6, 10, 0.57)** and **pbinom(5, 10, 0.57)**, we ran the following commands in R:

```
> dbinom(6, 10, 0.57)
[1] 0.2462307
> pbinom(5, 10, 0.57)
[1] 0.4436094
```

The command **rbinom(k, n, p)** generates k independent binomial random variables, each identically distributed with parameters n and p . For instance, **rbinom(50, 5, 0.333)** yields 50 independent instances (realizations) of a binomial random variable with parameters $n = 5$ and $p = 0.333$. In other words, it generates a sample of size 50 from this distribution. Here is an example:

Example 6.39 A box contains 30 balls, numbered from 1 to 30. If five balls are drawn randomly, one at a time, with replacement, what's the expected value of the count of balls marked with prime numbers? Additionally, what is the probability that two of these balls have prime numbers? Address these questions using simulation.

Solution: We will define a successful draw as one where the ball's number is prime. There are 10 balls labeled with prime numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. As a reminder, a prime number is greater than 1 and has no positive divisors other than 1 and itself. Let X denote the number of successes in five draws. X follows a binomial distribution with parameters $n = 5$ and $p = 10/30$, which simplifies to $p = 1/3$. The command `rbinom(5000, 5, 1/3)` simulates drawing five balls from the box 5000 times and, for each simulation, records the number of balls with prime numbers. To find the expected value of the count of balls with prime numbers, we compute the sum of these simulations and divide by 5000:

```
sum(rbinom(5000, 5, 1/3)) / 5000.
```

The probability that exactly two of the balls drawn are prime is found using the binomial probability mass function: `dbinom(2, 5, 1/3)`. Using R, the answers are:

```
> sum(rbinom(5000, 5, 1/3)) / 5000
[1] 1.6688
> pbinom(2, 5, 1/3)
[1] 0.329218      ◆
```

The provided code below utilizes R simulation to compute the expected value (mean), variance, and standard deviation of a binomial random variable. This is achieved using the commands `mean()`, `var()`, `sd()`, and the function `rbinom()` for generating random samples. Here is how it is done:

```
> n <- 16
> p <- 0.6
> v <- rbinom(3000, n, p)
> ev <- mean(v)
> va <- var(v)
> st <- sd(v)
> print(ev)
[1] 9.610667
> print(va)
[1] 3.957739
> print(st)
[1] 1.989407
```

To simulate a **Bernoulli distribution** with parameter p , we utilize a binomial distribution with $n = 1$.

Poisson Random Variable

Let X be a Poisson random variable with parameter λ . In R, the three commands associated with simulating Poisson random variables are:

- **dpois (n, λ)** for the probability mass function,
- **ppois (n, λ)** to represent the probability distribution function, and
- **rpois (k, λ)** to generate k samples of a Poisson random variable with parameter λ .

Let X be a Poisson random variable with parameter $\lambda = 3.9$, then

$$P(X = 5) = \frac{e^{-3.9}(3.9)^5}{5!} = \mathbf{dpois(5, 3.9)} = 0.1521925.$$

$$P(X \leq 6) = \sum_{i=0}^6 \frac{e^{-3.9}(3.9)^i}{i!} = \mathbf{ppois(6, 3.9)} = 0.899483.$$

Geometric Random Variable

Similar to the binomial and Poisson random variables, the three commands in R for simulating a geometric random variable are **dgeom**, **pgeom**, and **rgeom**. Let X be a geometric random variable with $p = 0.76$. The following two equations compute the probability that, in successive Bernoulli trials with a success probability of 0.76, the first success is on the 4th trial, and the probability that success occurs on the 4th trial or earlier.

$$P(X = 4) = \mathbf{dgeom(4, 0.76)} = 0.002521498.$$

$$P(X \leq 4) = \mathbf{pgeom(4, 0.76)} = 0.9992037.$$

The command **rgeom(30, 0.76)** produces a sample of 30 geometric random variables, each with parameter $p = 0.76$.

```
> rgeom(30, 0.76)
[1] 0 0 0 0 1 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 2 0 0 2 0 0 0
```

Running **rgeom(30, 0.76)** in RStudio or R and receiving a bunch of zeros, as we did, is not indicative of an error. The **rgeom()** function in R returns the number of trials needed to get the first success in a sequence of Bernoulli experiments. If the probability of success p is high, such as 0.76 in our case, it is very likely that success will occur on the first trial for many of the 30 simulated sequences, hence returning a value of 0 (since R's **rgeom()** function is zero-indexed). To clarify, in R's implementation of the geometric distribution, a result of 0 from **rgeom()** means the first success occurred on the first trial, a result of 1 means it occurred on the second trial, and so forth. So, if we are seeing many zeros, it is consistent with the high probability of success: many of the sequences have their first success on the very first trial.

Negative Binomial Random Variable

Let X represent the count of independent Bernoulli trials needed, each with probability p , to achieve the r th success. In this case, X follows a negative binomial distribution with parameters r and p . In R, the three functions to simulate aspects of this distribution are **dnbinom**, **pnbinom**, and **rnbinom**. If the parameters of X are $r = 6$ and $p = 0.43$, then

$$P(X = 12) = \text{dnbinom}(12, 6, 0.43) = 0.04601075.$$

$$\begin{aligned} P(6 < X \leq 12) &= P(X \leq 12) - P(X \leq 6) \\ &= \text{pnbinom}(12, 6, 0.43) - \text{pnbinom}(6, 6, 0.43) = 0.4406216. \end{aligned}$$

In R's implementation of the negative binomial distribution, the **rnbinom** function returns the number of failures before achieving the r th success. Therefore, an output of 3 from, say, **rnbinom(10, 6, 0.43)** means there were 3 failures before achieving the 6 successes. So, the total number of trials to achieve 6 successes is 3 (failures) + 6 (successes) = 9 trials. We should always remember to check the documentation and understand the conventions of the functions we are using.

Using the command **rnbinom(10, 6, 0.43)**, the following sequence was generated in a simulation:

```
rnbinom(10, 6, 0.43) [1] 12 1 16 6 7 8 7 5 14 11
```

Each value in this sequence represents the number of failures before achieving the sixth success. Thus, for a negative binomial random variable with parameters $r = 6$ and $p = 0.43$, the sixth success took place on these trials: 18 7 22 12 13 14 13 11 20 17. This set of values forms a sample of size 10 from the negative binomial distribution with the given parameters.

Hypergeometric Random Variable

From a box containing w white balls and b black balls, n balls are drawn at random without replacement, where $(n \leq \min(w, b))$. Let X denote the number of white balls drawn, then X is described as a hypergeometric random variable with parameters w , b , and n . In R, the corresponding commands for the hypergeometric distribution, analogous to those for binomial, Poisson, geometric, and negative binomial distributions, are **dhyper**, **phyper**, and **rhyper**. Thus, the probabilities can be computed as:

$$P(X = i) = \text{dhyper}(i, w, b, n)$$

$$P(X \leq i) = \text{bhyper}(i, w, b, n)$$

The code **rhyper(k, w, b, n)** produces a sample of size k by simulating the hypergeometric distribution with parameters w , b , and n . In the subsequent run:

```
> rhyper(15, 25, 18, 9)
[1] 7 5 6 3 3 7 7 5 4 5 6 6 5 5 4,
```

the output represents the number of white balls drawn in each of the 15 simulations. For example, in the first simulation, 7 white balls were drawn out of 9 total balls, and in the second simulation, 5 white balls were drawn, and so on.

Note that, in the context of the hypergeometric distribution, these draws are without replacement. So, if you draw a white ball in one of the trials, the number of remaining white balls decreases for subsequent draws within that trial. However, in this instant, each of the 15 simulations resets the conditions and starts fresh.

Multinomial Distribution

We will now delve into simulating a multinomial distribution for the case where r , the number of possible outcomes, is 4. We do this because the same set of commands and methodology can be applied regardless of the number of possible outcomes. Suppose that, whenever an experiment is performed, one of the disjoint outcomes A_1 , A_2 , A_3 , and A_4 will occur. Let $P(A_i) = p_i$, $1 \leq i \leq 4$. Then $p_1 + p_2 + p_3 + p_4 = 1$. If, in n independent iterations of this experiment, we denote by X_i , where $1 \leq i \leq 4$, the number of occurrences of event A_i , then $p(x_1, x_2, x_3, x_4)$ represents the joint probability mass function of the multinomial distribution for X_1, X_2, X_3, X_4 . Thus,

$$P(X_1 = x_1, X_2 = x_2, X_3 = x_3, X_4 = x_4) = \frac{n!}{x_1! x_2! x_3! x_4!} p_1^{x_1} p_2^{x_2} p_3^{x_3} p_4^{x_4}.$$

In R, we can use the code **dmultinom(x,n,prob)** to calculate the probability $P(X_1 = x_1, X_2 = x_2, X_3 = x_3, X_4 = x_4)$ for a multinomial distribution. In this code **x** is the vector **c(x₁, x₂, x₃, x₄)**, **n** is the number of experiments ($n = x_1 + x_2 + x_3 + x_4$) and **prob** is the vector **c(p₁, p₂, p₃, p₄)**.

In a particular case where, $p_1 = 0.2$, $p_2 = 0.4$, $p_3 = 0.1$ and $p_4 = 0.3$, the following code calculates $P(X_1 = 3, X_2 = 2, X_3 = 4, X_4 = 7)$.

```
x <- c(3, 2, 4, 7)
n <- sum(x)
prob <- c(0.2, 0.4, 0.1, 0.3)
p <- dmultinom(x, n, prob)
print(p)
[1] 0.0004035109
```

In R, the **rmultinom()** function is used to simulate random draws from a multinomial distribution. Here is how it works:

rmultinom(num, n, prob)

We denote the number of random samples we want to generate by **num**. We represent the number of trials or experiments in each simulation by **n**. The vector **prob** is representing the probabilities of each possible outcome. The output is a table where, for $1 \leq i \leq \text{num}$, column i represents the result of simulation i , and row i displays the count of occurrences of A_i across

all **num** simulations. To illustrate, let us draw a random sample of size 8 from the multinomial distribution defined above:

```
> num=8
> n=16
> prob=c(0.2, 0.4, 0.1, 0.3)
> results=rmultinom(num, n, prob)
> print(results)
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]
[1,]	2	5	2	7	3	5	4	2
[2,]	6	6	7	2	5	3	6	8
[3,]	2	2	3	1	1	1	2	2
[4,]	6	3	4	6	7	7	4	4

EXERCISES

Within this chapter, there is a range of exercises in almost all sections that provide excellent problems for simulation using R. Professors have the option to assign a selected number of these exercises to students, enabling them to create R simulations. Likewise, students can independently select exercises to practice simulation techniques with R.

CHAPTER 6 SUMMARY

► **Bernoulli Random Variables** Bernoulli trials are experiments that have only two possible outcomes. One outcome is usually called a *success*, denoted by s . The other outcome is called a *failure*, denoted by f . The random variable defined by $X(s) = 1$ and $X(f) = 0$ is called a *Bernoulli random variable*. Therefore, a Bernoulli random variable takes on the value 1 when the outcome of the Bernoulli trial is a success and 0 when it is a failure. If p is the probability of a success, then $1 - p$ (sometimes denoted q) is the probability of a failure. Hence the probability mass function of X is $p(x) = 1 - p \equiv q$ if $x = 0$, $p(x) = p$ if $x = 1$, and $p(x) = 0$, otherwise. We have that $E(X) = E(X^2) = p$, $\text{Var}(X) = p(1 - p)$, and $\sigma_X = \sqrt{p(1 - p)}$.

► **Binomial Random Variables** If n Bernoulli trials all with probability of success p are performed independently, then X , the number of successes, is said to be a *binomial random variable* with parameters n and p . The set of possible values of X is $\{0, 1, 2, \dots, n\}$, and its

probability mass function is $p(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$, if $x = 0, 1, 2, \dots, n$, and $p(x) = 0$, elsewhere. For a binomial random variable X with parameters n and p , $E(X) = np$, $\text{Var}(X) = np(1-p)$, and $\sigma_X = \sqrt{np(1-p)}$.

► **Poisson Random Variables** For a binomial random variable X with parameters n and p , when the number of trials is large ($n \rightarrow \infty$), the probability of success is small ($p \rightarrow 0$), and the average number of successes remains a fixed quantity of moderate value ($np = \lambda$ for some constant λ), then $P(X = i) \rightarrow \frac{e^{-\lambda} \lambda^i}{i!}$. This approximation is called *Poisson approximation to binomial*. Poisson's approximation by itself is a probability mass function, and a discrete random variable X with possible values $0, 1, 2, 3, \dots$ is called *Poisson* with parameter λ , $\lambda > 0$, if its probability mass function is given by

$$P(X = i) = \frac{e^{-\lambda} \lambda^i}{i!}, \quad i = 0, 1, 2, 3, \dots$$

For a Poisson random variable with parameter λ , $\lambda > 0$, $E(X) = \text{Var}(X) = \lambda$ and $\sigma_X = \sqrt{\lambda}$.

► **Poisson Processes** The Poisson distribution also appears in connection with the study of sequences of random events occurring over time. Suppose that, starting from a time point labeled $t = 0$, we begin counting the number of events. Then for each value of t , we obtain a number denoted by $N(t)$, which is the number of events that have occurred during $[0, t]$. If random events occur in time in a way that the conditions stationarity, independent increments, and orderliness are always satisfied, $N(0) = 0$ and, for all $t > 0$, $0 < P(N(t) = 0) < 1$, then there exists a positive number λ such that

$$P(N(t) = n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}.$$

That is, for all $t > 0$, $N(t)$ is a Poisson random variable with parameter λt . Hence $E[N(t)] = \lambda t$ and therefore $\lambda = E[N(1)]$.

► **Geometric Random Variables** Suppose that a sequence of independent Bernoulli trials, each with probability of success p , $0 < p < 1$, are performed. Let X be the number of experiments until the first success occurs. Then X is a discrete random variable called *geometric*. Its set of possible values is $\{1, 2, \dots\}$, and its probability mass function is given by $p(x) = (1-p)^{x-1} p$, $0 < p < 1$, $x = 1, 2, 3, \dots$, $p(x) = 0$, elsewhere. For X , a geometric random variable with parameter p , $0 < p < 1$, $E(X) = 1/p$, $\text{Var}(X) = (1-p)/p^2$, and $\sigma_X = \sqrt{1-p}/p$.

► **Negative Binomial Random Variables** Suppose that a sequence of independent Bernoulli trials, each with probability of success p , $0 < p < 1$, is performed. Let X be the number of experiments until the r th success occurs. Then X is a discrete random variable

called a *negative binomial* with parameters (r, p) . Clearly, $\{r, r+1, r+2, r+3, \dots\}$ is the set of possible values of X and its probability mass function is given by

$$p(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad 0 < p < 1, \quad x = r, r+1, r+2, r+3, \dots$$

For a negative binomial random variable X with parameters (r, p) ,

$$E(X) = \frac{r}{p}, \quad \text{Var}(X) = \frac{r(1-p)}{p^2}, \quad \sigma_X = \frac{\sqrt{r(1-p)}}{p}.$$

► **Hypergeometric Random Variables** Suppose that, from a box containing D defective and $N-D$ nondefective items, n are drawn at random and *without replacement*. Furthermore, suppose that the number of items drawn does not exceed the number of defective or the number of nondefective items. That is, suppose that $n \leq \min(D, N-D)$. Let X be the number of defective items drawn. Then X is a discrete random variable with the set of possible values $\{0, 1, \dots, n\}$, and a probability mass function

$$p(x) = P(X = x) = \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, 2, \dots, n.$$

Any random variable X with such a probability mass function is called a *hypergeometric random variable*. For the hypergeometric random variable X , defined above,

$$E(X) = \frac{nD}{N}, \quad \text{Var}(X) = \frac{nD(N-D)}{N^2} \left(1 - \frac{n-1}{N-1}\right).$$

► **Multinomial Distributions** Suppose that, whenever an experiment is performed, one of the disjoint outcomes $A_1, A_2, \dots, A_r$ will occur. Let $P(A_i) = p_i$, $1 \leq i \leq r$. Then $p_1 + p_2 + \dots + p_r = 1$. If, in n independent performances of this experiment, X_i , $i = 1, 2, \dots, r$, denotes the number of times that A_i occurs, then $p(x_1, \dots, x_r)$, the joint probability mass function of $X_1, X_2, \dots, X_r$, is called *multinomial joint probability mass function*, and its distribution is said to be a *multinomial distribution*. For any set of nonnegative integers $\{x_1, x_2, \dots, x_r\}$ with $x_1 + x_2 + \dots + x_r = n$,

$$\begin{aligned} p(x_1, x_2, \dots, x_r) &= P(X_1 = x_1, X_2 = x_2, \dots, X_r = x_r) \\ &= \frac{n!}{x_1! x_2! \dots x_r!} p_1^{x_1} p_2^{x_2} \dots p_r^{x_r}. \end{aligned}$$

Furthermore, for $1 \leq i \leq r$, the marginal probability mass function of X_i is binomial with parameters n and p_i .

REVIEW PROBLEMS

1. Of police academy applicants, only 25% will pass all the examinations. Suppose that 12 successful candidates are needed. What is the probability that, by examining 20 candidates, the academy finds all of the 12 persons needed?
2. The time between the arrival of two consecutive customers at a postoffice is 3 minutes, on average. Assuming that customers arrive in accordance with a Poisson process, find the probability that tomorrow during the lunch hour (between noon and 12:30 P.M.) fewer than seven customers arrive.
3. A restaurant serves 8 fish entrées, 12 beef, and 10 poultry. If customers select from these entrées randomly, what is the expected number of fish entrées ordered by the next four customers?
4. A university has n students, 70% of whom will finish an aptitude test in less than 25 minutes. If 12 students are selected at random, what is the probability that at most two of them will not finish in 25 minutes?
5. A doctor has five patients with migraine headaches. He prescribes for all five a drug that relieves the headaches of 82% of such patients. What is the probability that the medicine will not relieve the headaches of two of these patients?
6. In a community, the chance of a set of triplets is 1 in 1000 births. Determine the probability that the second set of triplets in this community occurs before the 2000th birth.
7. From a panel of prospective jurors, 12 are selected at random. If there are 200 men and 160 women on the panel, what is the probability that more than half of the jury selected are women?
8. Suppose that 10 trains arrive independently at a station every day, each at a random time between 10:00 A.M. and 11:00 A.M.. What is the expected number and the variance of those that arrive between 10:15 A.M. and 10:28 A.M.?
9. Suppose that a certain bank returns bad checks at a Poisson rate of three per day. What is the probability that this bank returns at most four bad checks during the next two days?
10. The policy of the quality control division of a certain corporation is to reject a shipment if more than 5% of its items are defective. A shipment of 500 items is received, 30 of them are randomly tested, and two have been found defective. Should that shipment be rejected?
11. A fair coin is tossed successively until a head occurs. If N is the number of tosses required, what are the expected value and the variance of N ?
12. What is the probability that the sixth toss of a die is the first 6?
13. A company produces liquid silicone rubber that is designed specifically for mold making. If in the molding process, defective items are produced at a rate of 0.6%, find the probability that, in a sample of 200 items, there are at most 3 defective ones.

14. In data communication, one method for error control makes the receiver send a positive acknowledgment for every message received with no detected error and a negative acknowledgment for all the others. Suppose that (i) acknowledgments are transmitted error free, (ii) a negative acknowledgment makes the sender transmit the same message over, (iii) there is no upper limit on the number of retransmissions of a message, and (iv) errors in messages are detected with probability p , independently of other messages. On average, how many times is a message retransmitted?
15. Past experience shows that 30% of the customers entering Harry's Clothing Store will make a purchase. Of the customers who make a purchase, 85% use credit cards. Let X be the number of the next six customers who enter the store, make a purchase, and use a credit card. Find the probability mass function, the expected value, and the variance of X .
16. Of the 28 professors in a certain department, 18 drive foreign and 10 drive domestic cars. If five of these professors are selected at random, what is the probability that at least three of them drive foreign cars?
17. A bowl contains 10 red and six blue chips. What is the expected number of blue chips among five randomly selected chips?
18. A certain type of seed when planted fails to germinate with probability 0.06. If 40 of such seeds are planted, what is the probability that at least 36 of them germinate?
19. Suppose that 6% of the claims received by an insurance company are for damages from vandalism. What is the probability that at least three of the 20 claims received on a certain day are for vandalism damages?
20. At the Antonio Car dealership, the probability that during a given month at least one consumer returns a car for warranty work is 0.45. Suppose that the numbers of cars returned for warranty work in different months are independent. Consider the period until the fifth month in which at least one car was returned for warranty work. What is the probability that, during that period, there were *at least* 3 months with no car returned for warranty work?
21. Passengers are making reservations for a particular flight on a small commuter plane 24 hours a day at a Poisson rate of 3 reservations per 8 hours. If 24 seats are available for the flight, what is the probability that by the end of the second day all the plane seats are reserved?
22. An insurance company claims that only 70% of the drivers regularly use seat belts. In a statistical survey, it was found that out of 20 randomly selected drivers, 12 regularly used seat belts. Is this sufficient evidence to conclude that the insurance company's claim is false?
23. From the set $\{x: 0 \leq x \leq 1\}$ numbers are selected at random and independently and rounded to three decimal places. What is the probability that 0.345 is obtained (a) for the first time on the 1000th selections; (b) for the third time on the 3000th selections?
24. The probability that a child of a certain family inherits a certain disease is 0.23 independently of other children inheriting the disease. If the family has five children and the

disease is detected in one child, what is the probability that exactly two more children have the disease as well?

25. A bowl contains w white and b blue chips. Chips are drawn at random and with replacement until a blue chip is drawn. What is the probability that (a) exactly n draws are required; (b) at least n draws are required?
26. Experience shows that 75% of certain kinds of seeds germinate when planted under normal conditions. Determine the minimum number of seeds to be planted so that the chance of at least five of them germinating is more than 90%.
27. Suppose that n babies were born at a county hospital last week. Also suppose that the probability of a baby having blonde hair is p . If k of these n babies are blondes, what is the probability that the i th baby born is blonde?
28. Show that if all three of n , N , and $D \rightarrow \infty$ so that $n/N \rightarrow 0$, D/N converges to a small number, and $nD/N \rightarrow \lambda$, then for all x ,

$$\frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}} \rightarrow \frac{e^{-\lambda} \lambda^x}{x!}.$$

This formula shows that the Poisson distribution is the limit of the hypergeometric distribution.

29. A farmer, plagued by insects, seeks to attract birds to his property by distributing seeds over a wide area. Let λ be the average number of seeds per unit area, and suppose that the seeds are distributed in a way that the probability of having any number of seeds in a given area depends only on the size of the area and not on its location. Argue that X , the number of seeds that fall on the area A , is approximately a Poisson random variable with parameter λA .
30. In a town there are n taxis. A woman takes one of these taxis every day at random and with replacement. On average, how long does it take before she can claim that she has been in every taxi in the town?
Hint: The final answer is in terms of $a_n = 1 + 1/2 + \cdots + 1/n$.
31. Slugger Bubble Gum Company markets its best-selling brand to young baseball fans by including pictures of current baseball stars in packages of its bubble gum. In the latest series, there are 20 players included, but there is no way of telling which player's picture is inside until the package of gum is purchased and opened. If a young fan wishes to collect all 20 pictures, how many packages must he buy, on average? Assume that there is no trading of one player's picture for another and that the number of cards printed for each player is the same.
32. A fair die is tossed 18 times. What is the probability that each face appears three times?
33. Alvie, a marksman, fires seven independent shots at a target. Suppose that the probabilities that he hits the bull's-eye, he hits the target but not the bull's-eye, and he misses

the target are 0.4, 0.35, and 0.25, respectively. What is the probability that he hits the bull's-eye three times, the target but not the bull's-eye two times, and misses the target two times?

- 34.** As discussed in [Section 7.7](#), if X represents the lifetime of a system, the function defined by $\bar{F}(t) = 1 - F(t) = P(X > t)$ is known as the *survival function* of X . For $t > 0$, $\bar{F}(t)$ signifies the probability that the system has survived for at least t units of time. Now consider a system composed of n components, with their lifetimes constituting an independent sequence of random variables. The system remains operational until one of its components fails. Let $F_1, F_2, \dots, F_n$ denote the distribution functions of the lifetimes of the system's components. The task is to derive the survival function of the system's lifetime, using $F_1, F_2, \dots, F_n$.
- 35.** When Kara throws a dart at a dartboard, X , the distance, in centimeters, between the impact point of the dart and O , the center of the dartboard, is a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{3}{500}x(10 - x) & 0 < x < 10 \\ 0 & \text{otherwise.} \end{cases}$$

If the bull's eye of the dartboard is a circle of diameter 6 centimeters centered at O , and the probability of hitting the bull's eye in one throw is independent of hitting it in other throws, what is the probability that Kara misses the bull's eye in exactly 7 of her next 10 throws?

- 36.** The lifetime (in hours) of a light bulb manufactured by a certain company is a random variable with probability density function

$$f(x) = \begin{cases} 0 & \text{if } x \leq 500 \\ \frac{5 \times 10^5}{x^3} & \text{if } x > 500. \end{cases}$$

Suppose that, for all nonnegative real numbers a and b , the event that any light bulb lasts at least a hours is independent of the event that any other light bulb lasts at least b hours. Find the probability that, of six such light bulbs selected at random, exactly two last over 1000 hours.



Self-Test on Chapter 6

Time allotted: 120 Minutes

Each problem is worth 10 points.

1. Fifty identical balls are numbered 1 through 50 and are mixed in a box. Five balls are drawn randomly, one by one, and without replacement. Let X be the number of balls drawn that are numbered at least 40. Let Y be the number on the 3rd ball drawn. Find the probability mass functions of X and Y .
2. A company has insurance coverage to pay for its indirect cost each time it closes down from a devastating calamity such as hurricane, tornado, flood, and snowstorm. Suppose that the policy pays \$50,000 for each calamity that causes the company to shut down temporarily, except for the first calamity, for which it pays nothing. Furthermore, suppose that such calamities occur at a Poisson rate of $4/3$ per year. Find the expected amount paid to the company each year under this policy?
3. For certain software, independently of other users, the probability is 0.07 that a user encounters a fault. Let X be the number of users who do not encounter a fault before the 12th user who encounters a fault. Find the probability mass function of X .
Hint: Note that $X + 12$ is a negative binomial random variable.
4. Ginny has begun a new website that aggregates reviews of movies and TV shows. Suppose that, all over the world, every day, 10 million people search for film review websites, and Ginny's site is one of the last sites out of hundreds that a search turns up. If independently of the others, the probability is 5×10^{-8} that a person visits Ginny's site via a search, find the probability that tomorrow at least 4 people visit her site by searching the Web.
5. To estimate the number of kangaroos living in a particular region of Western Victoria in Australia, Liam captured 25 kangaroos, marked them and then let them go free. After a few days he captured 30 kangaroos and observed that 5 of them were marked. Using this experiment, estimate n , the total number of kangaroos in the area.
Hint: Let p_n be the probability of 5 marked kangaroos among the 30 captured kangaroos. By analyzing p_n/p_{n-1} , find the value of n that makes p_n maximum.
6. There are two fair coins on a table. The first coin is flipped successively and independently until a heads appears. Then the second coin is flipped repeatedly and independently until a heads appears. Let X be the number of flips until the first coin lands heads, and let Y be the number of flips until the second coin lands heads. Find $P(X = Y)$.
7. Suppose that 20% of the physicians working for a certain hospital retire before age 65, 30% retire at ages 65–69, and the remaining 50% retire at age 70 or later. If physicians retire independently of each other, what is the probability that the median age of five randomly selected retired physicians from this hospital is below 65?

8. To reward improved product quality, a small manufacturing company that has only 27 employees, has developed a cash incentive program. Every year, the company pays out a \$7,000 bonus on top of the salaries of the individuals who have performed at or above certain thresholds. Suppose that, independently of other employees, each individual has an 8% chance of achieving that level of performance. Find the minimum amount of bonus money that the company should fund annually so that the probability is less than 1% that the amount is inadequate
9. Suppose that, independently of other tornado-force winds, with probability 0.063, such a wind even damages tornado-proof fortified rooms that are constructed with cinder blocks and are filled with mortar and rebar. Such a room is constructed in a town in the plains located in a Tornado alley. If in this town, tornado-force winds occur at a Poisson rate of 0.3 per year, what is the probability that this room can remain undamaged from tornadoes for at least 70 years?
Hint: Let X be the number of tornado-force winds during the next 70 years. Observe that X is a Poisson random variable with parameter $\lambda = (70)(0.3) = 21$.
10. A gambler has two dice, an unbiased one, and a loaded one which, when tossed, lands on 6 with probability of $4/9$ and lands on any of the other faces with probability of $1/9$. The gambler does not know which die is the loaded one. He chooses one of the two dice at random and tosses that die ten times in a row. If it lands on 6 exactly four times, what is the probability that it is the loaded one?





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Chapter 7

Special Continuous Distributions

In this chapter we study some examples of continuous random variables. These random variables appear frequently in theory and applications of probability, statistics, and branches of science and engineering.

7.1 UNIFORM RANDOM VARIABLES

In [Sections 1.6](#) and [1.7](#), we explained that in random selection of a point from an interval (a, b) , the probability of the occurrence of any particular point is zero. As a result, we stated that if $[\alpha, \beta] \subseteq (a, b)$, the events that the point falls in $[\alpha, \beta]$, (α, β) , $[\alpha, \beta)$, and $(\alpha, \beta]$ are all equiprobable. Moreover, we said that a point is randomly selected from an interval (a, b) if any two of its subintervals that have the same length are equally likely to include the point. We also mentioned that the probability associated with the event that the subinterval (α, β) includes the point is defined to be $(\beta - \alpha)/(b - a)$. Applications of these facts have been discussed throughout the book. Therefore, their significance should be clear by now. In particular, across various chapters, we have demonstrated that a fundamental aspect of computer simulations involves the selection of random points from within intervals. In this section, we introduce the concept of a uniform random variable. Then we study its properties and applications. As we will see now, uniform random variables are directly related to random selection of points from intervals.

Suppose that X is the value of the random point selected from an interval (a, b) . Then X is called a **uniform random variable** over (a, b) . Let F and f be probability distribution and density functions of X , respectively. Clearly,

$$F(t) = \begin{cases} 0 & t < a \\ \frac{t - a}{b - a} & a \leq t < b \\ 1 & t \geq b. \end{cases}$$

Therefore,

$$f(t) = F'(t) = \begin{cases} \frac{1}{b - a} & \text{if } a < t < b \\ 0 & \text{otherwise.} \end{cases} \quad (7.1)$$

Definition 7.1 A random variable X is said to be **uniformly distributed** over an interval (a, b) if its probability density function is given by (7.1).

Another way of reaching this definition is to note that $f(x)$ is a measure that determines how likely it is for X to be close to x . Since for all $x \in (a, b)$ the probability that X is close to x is the same, f should be a nonzero constant on (a, b) ; zero, elsewhere. Therefore,

$$f(x) = \begin{cases} c & \text{if } a < x < b \\ 0 & \text{elsewhere.} \end{cases}$$

Now $\int_a^b f(x) dx = 1$ implies that $\int_a^b c dx = c(b - a) = 1$. Thus $c = 1/(b - a)$.

Figure 7.1 represents the graphs of f and F , the density function and the distribution function of a uniform random variable over the interval (a, b) .

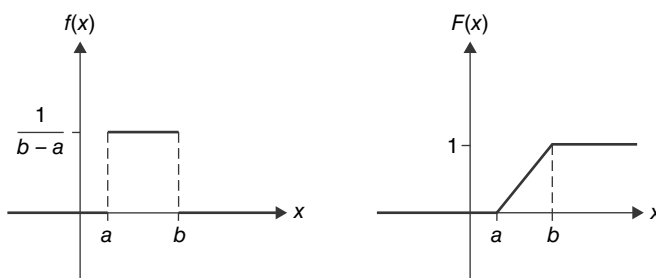


Figure 7.1 Density and distribution functions of a uniform random variable.

In random selections of a large number of points from (a, b) , we expect that the average of the values of the points will be approximately $(a + b)/2$, the midpoint of (a, b) . This can be shown by calculating $E(X)$, the expected value of a uniform random variable X over (a, b) .

$$\begin{aligned} E(X) &= \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{1}{2} x^2 \right]_a^b = \frac{1}{b-a} \left(\frac{1}{2} b^2 - \frac{1}{2} a^2 \right) \\ &= \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}. \end{aligned}$$

To find $\text{Var}(X)$, note that

$$E(X^2) = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{3} \frac{b^3 - a^3}{b-a} = \frac{1}{3} (a^2 + ab + b^2).$$

Hence

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{3} (a^2 + ab + b^2) - \left(\frac{a+b}{2} \right)^2 = \frac{(b-a)^2}{12}.$$

We have shown that

If X is uniformly distributed over an interval (a, b) , then

$$E(X) = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}, \quad \sigma_X = \frac{b-a}{\sqrt{12}}.$$

It is interesting to note that the expected value and the variance of a randomly selected integer Y from the set $\{1, 2, 3, \dots, N\}$ are very similar to $E(X)$ and $\text{Var}(X)$ obtained above. By Exercise 6 of Section 4.5, $E(Y) = (1+N)/2$ and $\text{Var}(Y) = (N^2-1)/12$.

Example 7.1 Starting at 5:00 A.M., every half hour there is a flight from San Francisco airport to Los Angeles International airport. Suppose that none of these planes is completely sold out and that they always have room for passengers. A person who wants to fly to L.A. arrives at the airport at a random time between 8:45 A.M. and 9:45 A.M. Find the probability that she waits (a) at most 10 minutes; (b) at least 15 minutes.

Solution: Let the passenger arrive at the airport X minutes past 8:45. Then X is a uniform random variable over the interval $(0, 60)$. Hence the density function of X is given by

$$f(x) = \begin{cases} 1/60 & \text{if } 0 < x < 60 \\ 0 & \text{elsewhere.} \end{cases}$$

Now the passenger waits at most 10 minutes if she arrives between 8:50 and 9:00 or 9:20 and 9:30; that is, if $5 < X < 15$ or $35 < X < 45$. So the answer to (a) is

$$P(5 < X < 15) + P(35 < X < 45) = \int_5^{15} \frac{1}{60} dx + \int_{35}^{45} \frac{1}{60} dx = \frac{1}{3}.$$

The passenger waits at least 15 minutes if she arrives between 9:00 and 9:15 or 9:30 and 9:45; that is, if $15 < X < 30$ or $45 < X < 60$. Thus the answer to (b) is

$$P(15 < X < 30) + P(45 < X < 60) = \int_{15}^{30} \frac{1}{60} dx + \int_{45}^{60} \frac{1}{60} dx = \frac{1}{2}. \quad \blacklozenge$$

Example 7.2 A person arrives at a bus station every day at 7:00 A.M. If a bus arrives at a random time between 7:00 A.M. and 7:30 A.M., what is the average time spent waiting?

Solution: If the bus arrives X minutes past 7:00 A.M., then X is a uniform random variable over the interval $(0, 30)$. Hence the average waiting time is

$$E(X) = \frac{0+30}{2} = 15 \text{ minutes.} \quad \blacklozenge$$

The uniform distribution is often used to solve elementary problems in **geometric probability**. In Chapter 5 we solved several such problems. Here we discuss only a famous problem introduced by the French mathematician Joseph Bertrand (1822–1900) in 1889. It is called **Bertrand's paradox**. Bertrand seriously doubted that probability could be defined on infinite sample spaces. To make his point, he posed this problem:

Example 7.3 What is the probability that a *random chord* of a circle is longer than a side of an equilateral triangle inscribed into the circle?

Solution: Since the exact meaning of a *random chord* is not given, we cannot solve this problem as stated. We interpret the expression *random chord* in three different ways and solve the problem in each case. Let the center of the circle be at C and its radius be r .

First interpretation: To draw a random chord, by considerations of symmetry, first choose a random point A on the circle and connect it to C , the center; then choose a random number d from $(0, r)$ and place M on AC so that $\overline{CM} = d$. Finally, from M draw a chord perpendicular to the radius AC (see Figure 7.2). Since $d < r/2$ if and only if the chord is longer than a side of an equilateral triangle inscribed into the circle and d is uniformly distributed over $(0, r)$, the desired probability is

$$P\left(d < \frac{r}{2}\right) = \frac{r/2}{r} = \frac{1}{2}.$$

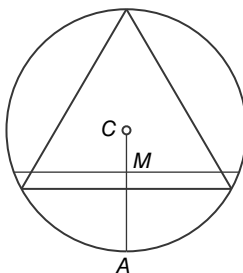


Figure 7.2 First interpretation of Example 7.3.

Second interpretation: To draw a random chord, by considerations of symmetry, first choose a random point A and then another random point D on the circle and connect AD . Let B and E be the points on the circle that make ABE an equilateral triangle. The random chord AD is longer than a side of ABE if and only if D lies on the arc BE . Since the length of the arc BE is one-third of the length of the whole circle and D is a random point on the circle, D lies on the arc BE with probability $1/3$. Thus the desired probability is $1/3$ as well (see Figure 7.3).

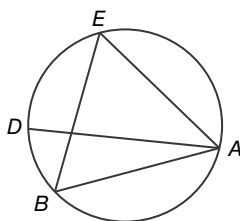


Figure 7.3 Second interpretation of Example 7.3.

Third interpretation: Since a chord is perpendicular to the radius connecting its midpoint to the center of the circle, every chord is uniquely determined by its midpoint. To draw a random chord, choose a random point M inside the circle, connect it to C , and draw a chord perpendicular to MC from M . It is clear that the chord is longer than a side of the equilateral triangle inscribed into the circle if and only if its midpoint M lies inside the circle centered at C with radius $r/2$ (see Figure 7.4). Since each choice of M uniquely determines one choice of a chord, the desired probability is the area of the small circle divided by the area of the original circle (a fact discussed in detail in Section 5.4). It is equal to

$$\frac{\pi(r/2)^2}{\pi r^2} = \frac{1}{4}.$$

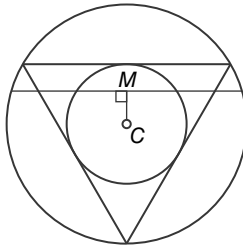


Figure 7.4 Third interpretation of Example 7.3.

As we showed, three different interpretations of random chord resulted in three different answers. Because of this, Bertrand's problem was once considered a paradox. At that time, one did not pay attention to the fact that the three interpretations correspond to three different experiments concerning the selection of a random chord. In this process we are dealing with three different probability functions defined on the same set of events.



EXERCISES

A

1. It takes a professor a random time between 20 and 27 minutes to walk from his home to school every day. If he has a class at 9:00 A.M. and he leaves home at 8:37 A.M., find the probability that he reaches his class on time.
2. Suppose that 15 points are selected at random and independently from the interval $(0, 1)$. How many of them can be expected to be greater than $3/4$?
3. The time at which a bus arrives at a station is uniform over an interval (a, b) with mean 2:00 P.M. and standard deviation $\sqrt{12}$ minutes. Determine the values of a and b .

4. Suppose that b is a random number from the interval $(-3, 3)$. What is the probability that the quadratic equation $x^2 + bx + 1 = 0$ has at least one real root?
5. The radius of a sphere is a random number between 2 and 4. What is the expected value of its volume? What is the probability that its volume is at most 36π ?
6. A point is selected at random on a line segment of length ℓ . What is the probability that the longer segment is at least twice as long as the shorter segment?
7. The loss for an insurance policy, in thousands of dollars, denoted by X , is a random amount between 0 and 10. The insurance company wants to determine the amount of deductible under this policy so that the expected value of the payment is $(0.81)E(X)$, which is 81% of the expected value of the loss if there was no deductible. What should the amount of the deductible be?
8. From the class of all triangles one is selected at random. What is the probability that it is obtuse?
Hint: The largest angle of a triangle is less than 180 degrees but greater than or equal to 60 degrees.
9. A farmer who has two pieces of lumber of lengths a and b ($a < b$) decides to build a pen in the shape of a triangle for his chickens. He sends his foolish son out to cut the longer piece and the boy, without taking any thought as to the ultimate purpose, makes a cut on the lumber of length b , at a point selected randomly. What are the chances that the two resulting pieces and the piece of length a can be used to form a triangular pen?
Hint: Three segments form a triangle if and only if the length of any one of them is less than the sum of the lengths of the remaining two.
10. Let θ be a random number between $-\pi/2$ and $\pi/2$. Find the probability density function of $X = \tan \theta$.
11. Let X be a random number from $[0, 1]$. Find the probability mass function of $[nX]$, the greatest integer less than or equal to nX .
12. Let X and Y be independent random points from the interval $(-1, 1)$. Find $E[\max(X, Y)]$.
13. Let X and Y be independent random points from the interval $(0, 1)$. Find the probability density function of the random variable XY .
14. Let X be a random number from $(0, 1)$. Find the probability density function of the random variable $Y = 1/X$.
15. Let X and Y be two independent uniformly distributed random variables over the intervals $(0, 1)$ and $(0, 2)$, respectively. Find the probability density function of X/Y .

B

16. Let X be a random number from $(0, 1)$. Find the density functions of the random variables $Y = -\ln(1 - X)$ and $Z = X^n$, $n \neq 0$.
17. Let X be a uniform random variable over the interval $(0, 1 + \theta)$, where $0 < \theta < 1$ is a given parameter. Find a function of X , say $g(X)$, so that $E[g(X)] = \theta^2$.

18. Let X be a continuous random variable with distribution function F . Prove that $F(X)$ is uniformly distributed over $(0, 1)$.
19. Let g be a nonnegative real-valued function on $\mathbb{R}$ that satisfies the relation $\int_{-\infty}^{\infty} g(t) dt = 1$. Show that if, for a random variable X , the random variable $Y = \int_{-\infty}^X g(t) dt$ is uniform, then g is the density function of X .
20. The sample space of an experiment is $S = (0, 1)$, and for every subset A of S , $P(A) = \int_A dx$. Let X be a random variable defined on S by $X(\omega) = 5\omega - 1$. Prove that X is a uniform random variable over the interval $(-1, 4)$.
21. Let Y be a random number from $(0, 1)$. Let X be the second digit of $\sqrt{Y}$. Prove that for $n = 0, 1, 2, \dots, 9$, $P(X = n)$ increases as n increases. This is remarkable because it shows that $P(X = n)$, $n = 1, 2, 3, \dots$ is not constant. That is, Y is uniform but X is not.
22. Let X and Y be two independent random points from the interval $(0, 1)$. Calculate the distribution function and the probability density function of $\max(X, Y)/\min(X, Y)$.
23. Let $X_1, X_2, \dots, X_n$ be n independent random numbers from the interval $(0, 1)$. Find $E\left(\max_{1 \leq i \leq n} X_i\right)$ and $E\left(\min_{1 \leq i \leq n} X_i\right)$.
24. Let $X_1, X_2, \dots, X_n$ be n independent random numbers from $(0, 1)$, and

$$Y_n = n \cdot \min(X_1, X_2, \dots, X_n).$$

Prove that

$$\lim_{n \rightarrow \infty} P(Y_n > x) = e^{-x}, \quad x \geq 0.$$



Self-Quiz on Section 7.1

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. A point is selected at random on a line segment of length ℓ . What is the probability that none of the two segments is smaller than $\ell/3$?
2. Suppose that X , the error of a measurement, is a random number between -1 and 1 . Find the expected value and the variance of $|X|$, the magnitude of the error.
Hint: By calculating the distribution function of $|X|$, show that it is a uniform random variable.



7.2 NORMAL RANDOM VARIABLES

Standard Normal Random Variables

In search of formulas to approximate binomial probabilities, Poisson was not alone. Other mathematicians had also realized the importance of such investigations. In 1718, before Poisson, De Moivre had discovered the following approximation, which is completely different from Poisson's.

De Moivre's Theorem *Let X be a binomial random variable with parameters n and $1/2$. Then for any numbers a and b , $a < b$,*

$$\lim_{n \rightarrow \infty} P\left(a < \frac{X - (1/2)n}{(1/2)\sqrt{n}} < b\right) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2} dt.$$

Note that in this formula $(1/2)n = E(X)$ and $(1/2)\sqrt{n} = \sigma_X$.

De Moivre's theorem was appreciated by Laplace, who recognized its importance. In 1812 he generalized it to binomial random variables with parameters n and p . He showed the following theorem, now called the De Moivre-Laplace theorem.

Theorem 7.1 (De Moivre-Laplace Theorem) *Let X be a binomial random variable with parameters n and p . Then for any numbers a and b , $a < b$,*

$$\lim_{n \rightarrow \infty} P\left(a < \frac{X - np}{\sqrt{np(1-p)}} < b\right) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2} dt.$$

Note that np and $\sqrt{np(1-p)}$ appearing in this formula are, respectively, $E(X)$ and σ_X .

Poisson's approximation, as discussed, is good when n is large, p is relatively small, and np is appreciable. The De Moivre-Laplace formula yields excellent approximations for values of n and p for which $np(1-p) \geq 10$.

To find his approximation, Poisson first considered a binomial random variable X with parameters (n, p) . Then he showed that if $\lambda = np$, $P(X = i)$ is approximately $(e^{-\lambda} \lambda^i)/i!$ for large n . Finally, he proved that $(e^{-\lambda} \lambda^i)/i!$ itself is a probability mass function. The same procedure can be followed for the De Moivre-Laplace approximation as well. By this theorem, if X is a binomial random variable with parameters (n, p) , the sequence of probabilities

$$P\left(\frac{X - np}{\sqrt{np(1-p)}} \leq t\right), \quad n = 1, 2, 3, 4, \dots,$$

converges to $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx$, where the function $\Phi(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx$ is a distribution function itself. To prove that Φ is a distribution function, note that it is increasing, continuous, and $\Phi(-\infty) = 0$. The proof of $\Phi(\infty) = 1$ is tricky. We use the following ingenious technique introduced by Gauss to show it. Let

$$I = \int_{-\infty}^{\infty} e^{-x^2/2} dx.$$

Then

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2/2} dy \right) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy.$$

To evaluate this double integral, we change the variables to polar coordinates. That is, we let $x = r \cos \theta$, $y = r \sin \theta$. We get $dx dy = r d\theta dr$ and

$$\begin{aligned} I^2 &= \int_0^{\infty} \int_0^{2\pi} e^{-r^2/2} r d\theta dr = \int_0^{\infty} e^{-r^2/2} r \left(\int_0^{2\pi} d\theta \right) dr \\ &= 2\pi \int_0^{\infty} r e^{-r^2/2} dr = 2\pi \left[-e^{-r^2/2} \right]_0^{\infty} = 2\pi. \end{aligned}$$

Thus

$$I = \int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi},$$

and hence

$$\Phi(\infty) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} dx = 1.$$

Therefore, Φ is a distribution function.

Definition 7.2 A random variable X is called **standard normal** if its distribution function is Φ , that is, if

$$P(X \leq t) = \Phi(t) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx.$$

By the fundamental theorem of calculus, f , the density function of a standard normal random variable, is given by

$$f(x) = \Phi'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

The standard normal density function is a bell-shaped curve that is symmetric about the y -axis (see Figure 7.5). Figure 7.6, presented on the following page, illustrates the probability distribution function of a standard normal random variable.

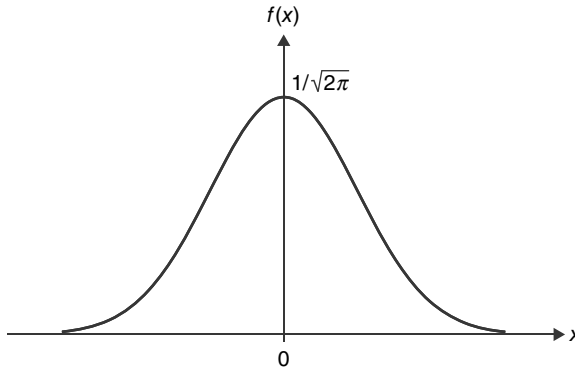


Figure 7.5 Graph of the standard normal density function.

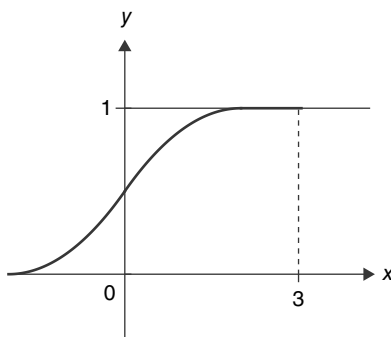


Figure 7.6 Graph of the standard normal distribution function.

Since Φ is the distribution function of the standard normal random variable, $\Phi(t)$ is the area under this curve from $-\infty$ to t . Because $\Phi(\infty) = 1$ and the curve is symmetric about the y -axis, $\Phi(0) = 1/2$. Moreover,

$$\Phi(-t) = 1 - \Phi(t).$$

To see this, note that

$$\Phi(-t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-t} e^{-x^2/2} dx.$$

Substituting $u = -x$, we obtain

$$\begin{aligned} \Phi(-t) &= \frac{-1}{\sqrt{2\pi}} \int_{\infty}^t e^{-u^2/2} du = \frac{1}{\sqrt{2\pi}} \int_t^{\infty} e^{-u^2/2} du \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2} du - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-u^2/2} du \\ &= 1 - \Phi(t). \end{aligned}$$

Example 7.4 Let Z be a standard normal random variable and α be a given constant. Find the real number x that maximizes $P(x < Z < x + \alpha)$.

Solution: It should be intuitively clear that $P(x < Z < x + \alpha)$ is maximum at $x = -\alpha/2$.

To show this mathematically, let $g(x) = P(x < Z < x + \alpha) = \frac{1}{\sqrt{2\pi}} \int_x^{x+\alpha} e^{-y^2/2} dy$. The number x that maximizes $P(x < Z < x + \alpha)$ is the root of $g'(x) = 0$. Therefore, by the fundamentals theorem of calculus, it is the solution of

$$g'(x) = \frac{1}{\sqrt{2\pi}} [e^{-(x+\alpha)^2/2} - e^{-x^2/2}] = 0,$$

which is $x = -\alpha/2$. ♦

Now that we have clarified the significance of the standard normal random variable, we will proceed to calculate its expectation and variance. The graph of the standard normal density function (Figure 7.5) implies that its expected value is zero. To prove this, let X be a standard normal random variable. Then

$$E(X) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-x^2/2} dx = 0,$$

because the integrand, $x e^{-x^2/2}$, is a finite odd function, and the integral is taken from $-\infty$ to $+\infty$. To calculate $\text{Var}(X)$, note that

$$E(X^2) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 e^{-x^2/2} dx.$$

Using integration by parts, we get (let $u = x$, $dv = x e^{-x^2/2} dx$)

$$\int_{-\infty}^{\infty} x x e^{-x^2/2} dx = \left[-x e^{-x^2/2} \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-x^2/2} dx = 0 + \sqrt{2\pi} = \sqrt{2\pi}.$$

Therefore, $E(X^2) = 1$, $\text{Var}(X) = E(X^2) - [E(X)]^2 = 1$, and $\sigma_X = \sqrt{\text{Var}(X)} = 1$. We have shown that

The expected value of a standard normal random variable is 0. Its standard deviation is 1.

Correction for Continuity

Thus far we have shown that the De Moivre-Laplace theorem approximates the distribution of a discrete random variable by that of a continuous one. But we have not demonstrated how this approximation works in practice. To do so, first we explain how, in general, a probability based on a discrete random variable is approximated by a probability based on a continuous one. Let X be a discrete random variable with probability mass function $p(x)$, and suppose that we want to find $P(i \leq X \leq j)$, $i < j$. Consider the **histogram** representing X on the following page in Figure 7.7, spanning from i to j . In that figure the base of each rectangle equals 1, and the height (and therefore the area) of the rectangle with the base midpoint k is $p(k)$, $i \leq k \leq j$. Thus the sum of the areas of all rectangles is $\sum_{k=i}^j p(k)$, which is the exact value of $P(i \leq X \leq j)$. Now, let us consider that $f(x)$, the density function of a continuous random variable, depicted in Figure 7.8 on the following page, provides a close approximation to $p(x)$. As illustrated in this figure, the probability $P(i \leq X \leq j)$, which is the sum of the areas of all rectangles in the figure, is approximately equal to the area under $f(x)$ from $i - 1/2$ to $j + 1/2$ rather than from i to j . That is,

$$P(i \leq X \leq j) \approx \int_{i-1/2}^{j+1/2} f(x) dx.$$

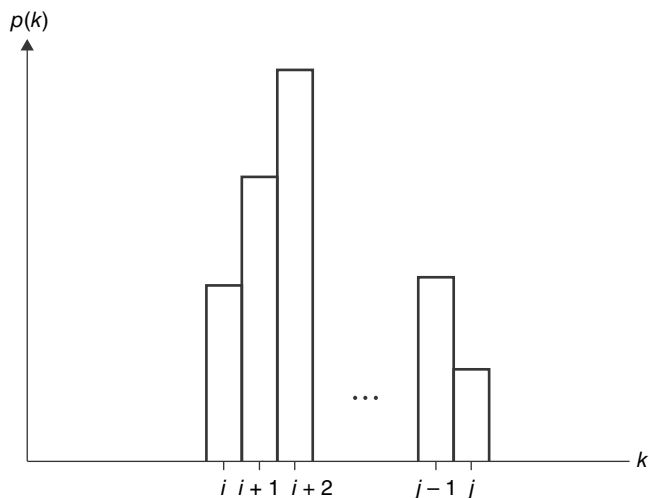


Figure 7.7 Histogram of X from i to j .

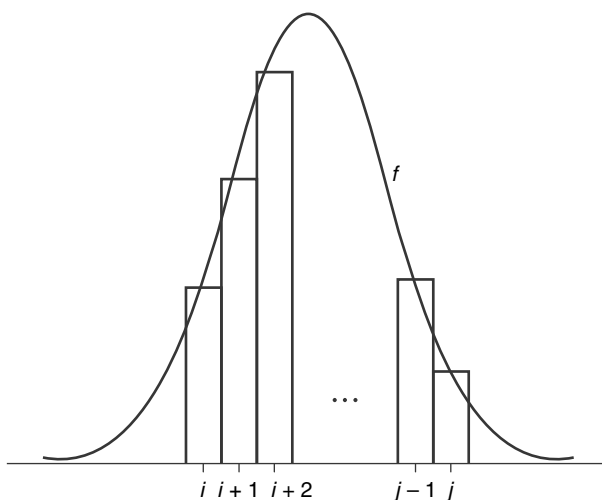


Figure 7.8 Histogram of X and the density function f .

This adjustment is called **correction for continuity** and is necessary for approximation of the distribution of a discrete random variable with that of a continuous one. Similarly, the following corrections for continuity are made to calculate the given probabilities.

$$P(X = k) \approx \int_{k-1/2}^{k+1/2} f(x) dx,$$

$$P(X \geq i) \approx \int_{i-1/2}^{\infty} f(x) dx,$$

$$P(X \leq j) \approx \int_{-\infty}^{j+1/2} f(x) dx.$$

In real-world problems, or even sometimes in theoretical ones, to apply the De Moivre-Laplace theorem, we need to calculate the numerical values of $\int_a^b e^{-x^2/2} dx$ for some real numbers a and b . Since $e^{-x^2/2}$ has no antiderivative in terms of elementary functions, such integrals are approximated by numerical techniques. Tables for such approximations have been available since 1799. Nowadays, most of the scientific calculators give excellent approximate values for these integrals. **Table 1** of the Appendix Tables gives the values of $\Phi(x) = (1/\sqrt{2\pi}) \int_{-\infty}^x e^{-y^2/2} dy$ for $x = -3.89$ to $x = 0$. For $x \leq -3.90$, $\Phi(x) \approx 0$. **Table 2** of the Appendix Tables gives the values of $\Phi(x)$ for $x = 0$ to $x = 3.89$. For $x > 3.89$, $\Phi(x) \approx 1$. Note that, using the relation $\Phi(x) = 1 - \Phi(-x)$, we can also use **Table 1** to find $\Phi(x)$ for $x > 0$ and **Table 2** to find $\Phi(x)$ for $x < 0$. Therefore, one of the **Tables 1** or **2** is sufficient to find $\Phi(x)$ for all values of x . However, for convenience, we have included a table for negative values of x and a separate table for positive values of x . The following example shows how the De Moivre-Laplace theorem is applied.

Example 7.5 Suppose that of all the clouds that are seeded with silver iodide, 58% show splendid growth. If 60 clouds are seeded with silver iodide, what is the probability that exactly 35 show splendid growth?

Solution: Let X be the number of clouds that show splendid growth. Then $E(X) = 60(0.58) = 34.80$ and $\sigma_X = \sqrt{60(0.58)(1 - 0.58)} = 3.82$. By correction for continuity and De Moivre-Laplace theorem,

$$\begin{aligned} P(X = 35) &\approx P(34.5 < X < 35.5) \\ &= P\left(\frac{34.5 - 34.80}{3.82} < \frac{X - 34.80}{3.82} < \frac{35.5 - 34.80}{3.82}\right) \\ &= P\left(-0.08 < \frac{X - 34.8}{3.82} < 0.18\right) \approx \frac{1}{\sqrt{2\pi}} \int_{-0.08}^{0.18} e^{-x^2/2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0.18} e^{-x^2/2} dx - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-0.08} e^{-x^2/2} dx \\ &= \Phi(0.18) - \Phi(-0.08) = 0.5714 - 0.4681 = 0.1033. \end{aligned}$$

The exact value of $P(X = 35)$ is $\binom{60}{35} (0.58)^{35} (0.42)^{25}$, which up to four decimal points equals 0.1039. Therefore, the answer obtained by the De Moivre-Laplace approximation is very close to the actual probability. ♦

★ **Remark 7.1** Using today's powerful scientific calculators, calculating the numerical value of $\Phi(x) = (1/\sqrt{2\pi}) \int_{-\infty}^x e^{-y^2/2} dy$ up to several decimal points of accuracy is almost as easy as addition or subtraction. However, both for theoretical and computational reasons, approximating $\Phi(x)$ by simpler functions has been the subject of many studies. One such approximation was given by A. K. Shah in his paper "A Simpler Approximation for Areas Under the Standard Normal Curve" (*The American Statistician*, **39**, 80, 1985). According to this

approximation, for $0 \leq x \leq 2.2$,

$$\Phi(x) \approx 0.5 + \frac{x(4.4 - x)}{10},$$

where the error of approximation is at most 0.005. That is,

$$\left| \Phi(x) - 0.5 - \frac{x(4.4 - x)}{10} \right| \leq 0.005. \quad \blacklozenge$$

Normal Random Variables with Varying Parameters

Although some natural phenomena follow a standard normal distribution, it cannot be used for data analysis due to the lack of parameters. To address this issue, mathematicians introduced a generalized density function that encompasses all normal random variables, with the standard normal distribution being a special case. This general normal distribution allows for various mean and standard deviation values, resulting in a broader family of distributions.

Definition 7.3 A random variable X is called **normal**, with parameters μ and σ , if its density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[\frac{-(x - \mu)^2}{2\sigma^2} \right], \quad -\infty < x < \infty.$$

f is a density function because by the change of variable $y = (x - \mu)/\sigma$:

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp \left[\frac{-(x - \mu)^2}{2\sigma^2} \right] dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-y^2/2} dy = \Phi(\infty) = 1.$$

If X is a normal random variable with parameters μ and σ , we write $X \sim N(\mu, \sigma^2)$.

One of the first applications of $N(\mu, \sigma^2)$ was given by Gauss in 1809. Gauss used $N(\mu, \sigma^2)$ to model the errors of observations in astronomy. For this reason, the normal distribution is sometimes called **the Gaussian distribution**. Larsen and Marx, in their *An Introduction to Mathematical Statistics and Its Applications* (Prentice Hall, Upper Saddle River, N.J., 1986), explain that $N(\mu, \sigma^2)$ was popularized by the Belgian scholar Lambert Quetelet (1796–1874), who used it successfully in data analysis in many situations.[†] The chest measurement of Scottish soldiers is one of the studies of $N(\mu, \sigma^2)$ by Quetelet. He measured X_i , $i = 1, 2, \dots, 5738$, the respective sizes of the chests of 5738 Scottish soldiers, and, using statistical methods, found that their average and standard deviation were 39.8 and 2.05 inches, respectively. Then he counted the number of soldiers who had a chest of size i , $i = 33, 34, \dots, 48$, and calculated the relative frequencies of sizes 33 through 48. He showed that for $i = 33, \dots, 48$ the relative frequency of the size i is very close to $P(i - 1/2 < X < i + 1/2)$, where X is $N(\mu, \sigma^2)$.

[†]An excellent treatise on the work of Quetelet concerning the applications of probability to the measurement of uncertainty in the social sciences can be found in [Chapter 5](#) of *The History of Statistics* by Stephen M. Stigler, The Belknap Press of Harvard University Press, 1986.

with $\mu = 39.8$ and $\sigma = 2.05$. Hence he concluded that the measure of the chest of a Scottish soldier is approximately a normal random variable with parameters 39.8 and 2.05.

The following lemma shows that, by a simple change of variable, $N(\mu, \sigma^2)$ can be transformed to $N(0, 1)$.

Lemma 7.1 *If $X \sim N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma$ is $N(0, 1)$. That is, if $X \sim N(\mu, \sigma^2)$, the standardized X is $N(0, 1)$.*

Proof: We show that the distribution function of Z is $(1/\sqrt{2\pi}) \int_{-\infty}^x e^{-y^2/2} dy$. Note that

$$\begin{aligned} P(Z \leq x) &= P\left(\frac{X - \mu}{\sigma} \leq x\right) = P(X \leq \sigma x + \mu) \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\sigma x + \mu} \exp\left[-\frac{(t - \mu)^2}{2\sigma^2}\right] dt. \end{aligned}$$

Let $y = (t - \mu)/\sigma$, then $dt = \sigma dy$ and we get

$$P(Z \leq x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} \sigma dy = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy. \quad \blacklozenge$$

Let $X \sim N(\mu, \sigma^2)$. Quetelet's analysis of the sizes of the chests of Scottish soldiers is an indication that μ and σ are the expected value and the standard deviation of X , respectively. To prove this, note that the random variable $Z = (X - \mu)/\sigma$ is $N(0, 1)$ and $X = \sigma Z + \mu$. Hence $E(X) = E(\sigma Z + \mu) = \sigma E(Z) + \mu = \mu$ and $\text{Var}(X) = \text{Var}(\sigma Z + \mu) = \sigma^2 \text{Var}(Z) = \sigma^2$. Therefore,

The parameters μ and σ that appear in the formula of the density function of X are its expected value and standard deviation, respectively.

The graph of $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(t - \mu)^2}{2\sigma^2}\right]$ is a bell-shaped curve symmetric about $x = \mu$ with the maximum at $(\mu, 1/\sigma\sqrt{2\pi})$ and inflection points at $\mu \pm \sigma$ (refer to [Figures 7.9](#) and [7.10](#) on the subsequent page).

The transformation $Z = (X - \mu)/\sigma$ enables us to use [Tables 1](#) and [2](#) of the Appendix Tables to calculate the probabilities concerning X . Some examples follow.

Example 7.6 Suppose that a Scottish soldier's chest size is normally distributed with mean 39.8 and standard deviation 2.05 inches, respectively. What is the probability that of 20 randomly selected Scottish soldiers, five have a chest of at least 40 inches?

Solution: Let p be the probability that a randomly selected Scottish soldier has a chest of 40 or more inches. If X is the normal random variable with mean 39.8 and standard deviation 2.05, then

$$\begin{aligned} p &= P(X \geq 40) = P\left(\frac{X - 39.8}{2.05} \geq \frac{40 - 39.8}{2.05}\right) = P\left(\frac{X - 39.8}{2.05} \geq 0.10\right) \\ &= P(Z \geq 0.10) = 1 - \Phi(0.1) \approx 1 - 0.5398 \approx 0.46. \end{aligned}$$

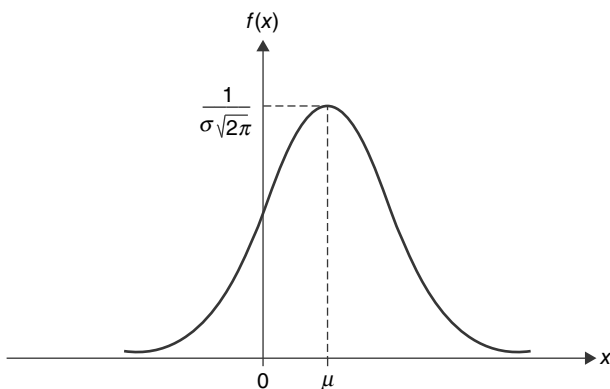


Figure 7.9 Density of $N(\mu, \sigma^2)$.

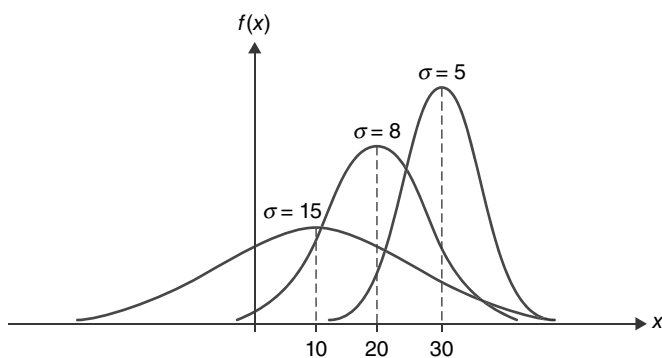


Figure 7.10 Different normal densities with specified parameters.

Therefore, the probability that of 20 randomly selected Scottish soldiers, five have a chest of at least 40 inches is

$$\binom{20}{5} (0.46)^5 (0.54)^{15} \approx 0.03. \quad \blacklozenge$$

Example 7.7 Let X , the grade of a randomly selected student in a test of a probability course, be a normal random variable. A professor is said to grade such a test *on the curve* if he finds the average μ and the standard deviation σ of the grades and then assigns letter grades according to the following table.

Range of the grade	$X \geq \mu + \sigma$	$\mu \leq X < \mu + \sigma$	$\mu - \sigma \leq X < \mu$	$\mu - 2\sigma \leq X < \mu - \sigma$	$X < \mu - 2\sigma$
Letter grade	A	B	C	D	F

Suppose that the professor of the probability course grades the test on the curve. Determine the percentage of the students who will get A, B, C, D, and F, respectively.

Solution: By the fact that $(X - \mu)/\sigma$ is standard normal,

$$P(X \geq \mu + \sigma) = P\left(\frac{X - \mu}{\sigma} \geq 1\right) = 1 - \Phi(1) \approx 0.1587,$$

$$P(\mu \leq X < \mu + \sigma) = P\left(0 \leq \frac{X - \mu}{\sigma} < 1\right) = \Phi(1) - \Phi(0) \approx 0.3413,$$

$$\begin{aligned} P(\mu - \sigma \leq X < \mu) &= P\left(-1 \leq \frac{X - \mu}{\sigma} < 0\right) = \Phi(0) - \Phi(-1) \\ &= 0.5 - 0.1587 \approx 0.3413, \end{aligned}$$

$$\begin{aligned} P(\mu - 2\sigma \leq X < \mu - \sigma) &= P\left(-2 \leq \frac{X - \mu}{\sigma} < -1\right) = \Phi(-1) - \Phi(-2) \\ &= 0.1587 - 0.0228 \approx 0.1359, \end{aligned}$$

$$P(X < \mu - 2\sigma) = P\left(\frac{X - \mu}{\sigma} < -2\right) = \Phi(-2) \approx 0.0228.$$

Therefore, approximately 16% should get A, 34% B, 34% C, 14% D, and 2% F. If an instructor grades a test on the curve, instead of calculating μ and σ , he or she may assign A to the top 16%, B to the next 34%, and so on. ♦

Example 7.8 The scores on an achievement test given to 100,000 students are normally distributed with mean 500 and standard deviation 100. What should the score of a student be to place him among the top 10% of all students?

Solution: Letting X be a normal random variable with mean 500 and standard deviation 100, we must find x so that $P(X \geq x) = 0.10$ or $P(X < x) = 0.90$. This gives

$$P\left(\frac{X - 500}{100} < \frac{x - 500}{100}\right) = 0.90.$$

Thus

$$\Phi\left(\frac{x - 500}{100}\right) = 0.90.$$

From Table 2 of the Appendix Tables, we have that $\Phi(1.28) \approx 0.8997$, implying that $(x - 500)/100 \approx 1.28$. This gives $x \approx 628$; therefore, a student should earn 628 or more to be among the top 10% of the students. ♦

Example 7.9 The annual rate of return for a share of a specific stock is a normal random variable with mean 10% and standard deviation 12%. Ms. Couture buys 100 shares of the stock at a price of \$60 per share. What is the probability that after a year her net profit from that investment is at least \$750? Ignore transaction costs and assume that there is no annual dividend.

Solution: Let r be the rate of return of this stock. The random variable r is normal with $\mu = 0.10$ and $\sigma = 0.12$. Let X be the price of the total shares of the stock that Ms. Couture buys this year. We are given that $X = 6000$. Let Y be the total value of the shares next year. The desired probability is

$$\begin{aligned}
 P(Y - X \geq 750) &= P\left(\frac{Y - X}{X} \geq \frac{750}{X}\right) = P\left(r \geq \frac{750}{6000}\right) \\
 &= P(r \geq 0.125) = P\left(Z \geq \frac{0.125 - 0.10}{0.12}\right) \\
 &= P(Z \geq 0.21) = 1 - P(Z < 0.21) \\
 &= 1 - \Phi(0.21) = 1 - 0.5832 = 0.4168. \quad \blacklozenge
 \end{aligned}$$

It is important to note that a large family of random variables in mathematics and statistics have distributions that are either normal or approximately normal. One of the reasons for this is the celebrated central limit theorem which we will study in [Section 7.3](#). To express that theorem, we will use Richard von Mises' words from his *Probability, Statistics and the Truth* (page 171, Dover Publications, 1981):

The normal curve represents the distribution in all cases where a final collective is formed by combination of a very large number of initial collectives, the attribute in the final collective being the sum of the results in the initial collectives. The original collectives are not necessarily simple alternatives as they are in Bernoulli's problem. It is not even necessary for them to have the same attributes or the same distributions. The only conditions are that a very great number of collectives are combined and that the attributes are mixed in such a way that the final attribute is the sum of all the original ones. Under these conditions the final distribution is always represented by a normal curve.

For example, consider the annual profit of an insurance company that sells thousands of policies each year. This is a normal random variable since the total profit is the sum of profits (losses being treated as negative profits) obtained from all policies sold. Some policies might be very complicated (i.e., those sold to the banks or oil companies). Moreover, the profit from each policy is itself a random variable which depends on many factors: the premium, risk involved, damages to be paid, quit claims, profit or loss from investments, the level of competition in the industry, and so on.

EXERCISES

A

1. Let $X \sim N(-2, 5)$. Find $P(|X| < 4)$.

2. Suppose that 90% of the patients with a certain disease can be cured with a certain drug. What is the approximate probability that, of 50 such patients, at least 45 can be cured with the drug?
3. A small college has 1095 students. What is the approximate probability that more than five students were born on Christmas day? Assume that the birthrates are constant throughout the year and that each year has 365 days.
4. Let $\Psi(x) = 2\Phi(x) - 1$. The function Ψ is called the **positive normal distribution**. Prove that if Z is standard normal, then $|Z|$ is positive normal.
5. Let X be a standard normal random variable. Calculate $E(X \cos X)$, $E(\sin X)$, and $E\left(\frac{X}{1 + X^2}\right)$.
6. The ages of subscribers to a certain newspaper are normally distributed with mean 35.5 years and standard deviation 4.8. What is the probability that the age of a random subscriber is (a) more than 35.5 years; (b) between 30 and 40 years?
7. The grades for a certain exam are normally distributed with mean 67 and variance 64. What percent of students get A(≥ 90), B($80 - 90$), C($70 - 80$), D($60 - 70$), and F(< 60)?
8. Suppose that the distribution of the diastolic blood pressure is normal for a randomly selected person in a certain population is normal with mean 80 mm Hg and standard deviation 7 mm Hg. If people with diastolic blood pressures 95 or above are considered hypertensive and people with diastolic blood pressures above 89 and below 95 are considered to have mild hypertension, what percent of that population have mild hypertension and what percent are hypertensive? Assume in that population no one has abnormal systolic blood pressure.
9. The length of an aluminum-coated steel sheet manufactured by a certain factory is approximately normal with mean 75 centimeters and standard deviation 1 centimeter. Find the probability that a randomly selected sheet manufactured by this factory is between 74.5 and 75.8 centimeters.
10. Suppose that the IQ of a randomly selected student from a university is normal with mean 110 and standard deviation 20. Determine the interval of values that is centered at the mean and includes 50% of the IQ's of the students at that university.
11. The amount of cereal in a box is normal with mean 16.5 ounces. If the packager is required to fill at least 90% of the cereal boxes with 16 or more ounces of cereal, what is the largest standard deviation for the amount of cereal in a box?
12. For a group of male workers, the number of red blood cells in a microliter is a normally distributed random variable with a mean of 5.4 million cells and a standard deviation of 0.31 million cells. Find the minimum number a for which the probability is 0.32 that the red blood cell count is greater than a . In other words, find a , the 68th percentile of the red blood cells counts.
13. Suppose that the scores on a certain manual dexterity test are normal with mean 12 and standard deviation 3. If eight randomly selected individuals take the test, what is the

probability that none will make a score less than 14?

14. A number t is said to be the **median** of a continuous random variable X if

$$P(X \leq t) = P(X \geq t) = 1/2.$$

Calculate the median of the normal random variable with parameters μ and σ^2 .

15. Let $X \sim N(\mu, \sigma^2)$. Prove that $P(|X - \mu| > k\sigma)$ does not depend on μ or σ .
16. Suppose that lifetimes of light bulbs produced by a certain company are normal random variables with mean 1000 hours and standard deviation 100 hours. Is this company correct when it claims that 95% of its light bulbs last at least 900 hours?
17. Suppose that lifetimes of light bulbs produced by a certain company are normal random variables with mean 1000 hours and standard deviation 100 hours. Suppose that lifetimes of light bulbs produced by a second company are normal random variables with mean 900 hours and standard deviation 150 hours. Howard buys one light bulb manufactured by the first company and one by the second company. What is the probability that at least one of them lasts 980 or more hours?
18. The annual rate of return for a share of a specific stock is a normal random variable with mean 0.12 and standard deviation of 0.06. The current price of the stock is \$35 per share. Mrs. Lovotti would like to purchase enough shares of this stock to make at least \$1000 profit with a probability of at least 90% in 1 year. Find the minimum number of shares that she should buy. Ignore transaction costs and assume that there are no annual dividends.
19. Find the expected value and the variance of a random variable with the probability density function

$$f(x) = \sqrt{\frac{2}{\pi}} e^{-2(x-1)^2}, \quad -\infty < x < \infty.$$

20. Let $X \sim N(\mu, \sigma^2)$. Find the distribution function of $|X - \mu|$ and its expected value.
21. Determine the value(s) of k for which the following is the probability density function of a normal random variable.

$$f(x) = \sqrt{k} e^{-k^2 x^2 - 2kx - 1}, \quad -\infty < x < \infty.$$

22. The viscosity of a brand of motor oil is normal with mean 37 and standard deviation 10. What is the lowest possible viscosity for a specimen that has viscosity higher than at least 90% of that brand of motor oil?
23. Let $\alpha \in (-\infty, \infty)$ and $Z \sim N(0, 1)$; find $E(e^{\alpha Z})$.
24. Let $X \sim N(0, \sigma^2)$. Calculate the density function of $Y = X^2$.
25. Let $X \sim N(\mu, \sigma^2)$. Calculate the density function of $Y = e^X$.
26. Let $X \sim N(0, 1)$. Calculate the density function of $Y = \sqrt{|X|}$.

B

27. Suppose that the odds are 1 to 5000 in favor of a customer of a particular bookstore buying a certain fiction bestseller. If 800 customers enter the store every day, how many copies of that bestseller should the store stock every month so that, with a probability of more than 98%, it does not run out of this book? For simplicity, assume that a month is 30 days.
28. To examine the accuracy of an algorithm that selects random numbers from the set $\{1, 2, \dots, 40\}$, 100,000 numbers are selected and there are 3500 ones. Given that the expected number of ones is 2500, is it fair to say that this algorithm is not accurate?
29. Prove that for some constant k , $f(x) = ka^{-x^2}$, $a \in (0, \infty)$, is a normal probability density function.
30. (a) Prove that for all $x > 0$,

$$\frac{1}{x\sqrt{2\pi}}\left(1 - \frac{1}{x^2}\right)e^{-x^2/2} < 1 - \Phi(x) < \frac{1}{x\sqrt{2\pi}}e^{-x^2/2}.$$

Hint: Integrate the following inequalities:

$$(1 - 3y^{-4})e^{-y^2/2} < e^{-y^2/2} < (1 + y^{-2})e^{-y^2/2}.$$

- (b) Use part (a) to prove that $1 - \Phi(x) \sim \frac{1}{x\sqrt{2\pi}}e^{-x^2/2}$. That is, as $x \rightarrow \infty$, the ratio of the two sides approaches 1.
31. Let Z be a standard normal random variable. Show that for $x > 0$,

$$\lim_{t \rightarrow \infty} P\left(Z > t + \frac{x}{t} \mid Z \geq t\right) = e^{-x}.$$

Hint: Use part (b) of Exercise 30.

32. The amount of soft drink in a bottle is a normal random variable. Suppose that in 7% of the bottles containing this soft drink there are less than 15.5 ounces, and in 10% of them there are more than 16.3 ounces. What are the mean and standard deviation of the amount of soft drink in a randomly selected bottle?
33. At an archaeological site 130 skeletons are found and their heights are measured and found to be approximately normal with mean 172 centimeters and variance 81 centimeters. At a nearby site, five skeletons are discovered and it is found that the heights of exactly three of them are above 185 centimeters. Based on this information is it reasonable to assume that the second group of skeletons belongs to the same family as the first group of skeletons?
34. In a forest, the number of trees that grow in a region of area R has a Poisson distribution with mean λR , where λ is a positive real number. Find the expected value of the distance from a certain tree to its nearest neighbor.

35. Let $I = \int_0^\infty e^{-x^2/2} dx$, then

$$I^2 = \int_0^\infty \left[\int_0^\infty e^{-(x^2+y^2)/2} dy \right] dx.$$

Let $y/x = s$ and change the order of integration to show that $I^2 = \pi/2$. This gives an alternative proof of the fact that Φ is a distribution function. The advantage of this method is that it avoids polar coordinates. ■

Self-Quiz on Section 7.2

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. The grades of students in a calculus-based probability course are normal with mean 72 and standard deviation 7. If 90, 80, 70, and 60 are the respective lowest, A, B, C, and D, what percent of students in this course get A's, B's, C's, D's, and F's?
2. Every day a factory produces 5000 light bulbs, of which 2500 are type I and 2500 are type II. If a sample of 40 light bulbs is selected at random to be examined for defects, what is the approximate probability that this sample contains at least 18 light bulbs of each type? ■

7.3 CENTRAL LIMIT THEOREM

The central limit theorem is a fundamental concept in statistics and probability theory. It allows us to approximate the distribution of a sum of independent random variables with a normal distribution, even if the individual variables themselves are not normally distributed. Moreover, the central limit theorem serves as the foundation for many statistical inference methods. For instance, when dealing with the means of samples from a population, we can utilize the normal distribution to make inferences about the population mean. Additionally, it aids in determining the appropriate sample size for conducting experiments or surveys. Furthermore, this theorem finds wide-ranging applications in various fields, including social sciences, engineering, finance, biology, and more. It provides a framework for understanding the behavior of sample statistics and is instrumental in designing experiments, analyzing data, and making predictions..

Let X be a binomial random variable with parameters n and p . In [Section 7.2](#), we discussed the De Moivre-Laplace theorem, which states that, for real numbers a and b , $a < b$,

$$\lim_{n \rightarrow \infty} P\left(a < \frac{X - np}{\sqrt{np(1-p)}} < b\right) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2} dt.$$

We mentioned that, for values of n and p for which $np(1-p) \geq 10$, this theorem can be used to approximate binomial by standard normal. Since this approximation is very useful, it is highly desirable to obtain similar ones for distributions other than binomial. However, this form of the De Moivre-Laplace theorem does not readily suggest any generalizations. Therefore, we restate it in an alternative way that can be generalized. To do so, note that if X is a binomial random variable with parameters n and p , then X is the sum of n independent Bernoulli random variables, each with parameter p . That is, $X = X_1 + X_2 + \cdots + X_n$, where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{if the } i\text{th trial is a failure.} \end{cases}$$

With this representation, the De Moivre-Laplace theorem is restated as follows:

Let $X_1, X_2, X_3, \dots$ be a sequence of independent Bernoulli random variables each with parameter p . Then $\forall i, E(X_i) = p, \text{Var}(X_i) = p(1-p)$, and

$$\lim_{n \rightarrow \infty} P\left(\frac{X_1 + X_2 + \cdots + X_n - np}{\sqrt{np(1-p)}} \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy.$$

It is this version of the De Moivre-Laplace theorem that motivates elegant generalizations. The first was given in 1887 by Chebyshev. But since its proof was not rigorous enough, his student Markov worked on the proof and made it both rigorous and simpler. In 1901, Lyapunov, another student of Chebyshev, weakened the conditions of Chebyshev and proved the following generalization of the De Moivre-Laplace theorem, which is now called *the central limit theorem*, arguably the most celebrated theorem in both probability and statistics.

Theorem 7.2 (Central Limit Theorem) *Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed random variables, each with expectation μ and variance σ^2 . Then the distribution of*

$$Z_n = \frac{X_1 + X_2 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

converges to the distribution of a standard normal random variable. That is,

$$\begin{aligned} \lim_{n \rightarrow \infty} P(Z_n \leq x) &= \lim_{n \rightarrow \infty} P\left(\frac{X_1 + X_2 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \leq x\right) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy. \end{aligned}$$

Remark 7.2 Note that Z_n is simply $X_1 + X_2 + \cdots + X_n$ standardized. ♦

Remark 7.3 In Theorem 7.2, let $\bar{X}_n$ be the mean of the random variables $X_1, X_2, \dots, X_n$. The central limit theorem is equivalent to

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy.$$

It is this version of the central limit theorem that holds immense significance and finds extensive applications in statistics. ♦

The central limit theorem shows that many natural phenomena obey approximately a standard normal distribution. In natural processes, the changes that take place are often the result of a sum of many insignificant random factors. Although the effect of each of these factors, separately, may be ignored, the effect of their sum in general may not. Therefore, the study of distributions of sums of a large number of independent random variables is essential. In this connection, the central limit theorem shows that in an enormous number of random phenomena, the distribution functions of such sums are approximately normal. Some examples are the weight of a man; the height of a woman; the error made in a measurement; the position and velocity of a molecule of a gas; the quantity of diffused material; the growth distribution of plants, animals, or their organs; and so on. The weight of a man, for example, is the result of many environmental and genetic factors, more or less unrelated but each contributing a small amount to his weight.

The proof to the central limit theorem will be presented in [Chapter 10](#), once the necessary tools are developed. In this chapter, we will only discuss some of its applications.

Example 7.10 The lifetime of a TV tube (in years) is an exponential random variable with mean 10. What is the probability that the average lifetime of a random sample of 36 TV tubes is at least 10.5?

Solution: The parameter of the exponential density function of the lifetime of a tube is $\lambda = 1/10$. For $1 \leq i \leq 36$, let X_i be the lifetime of the i th TV tube in the sample. Clearly, for $1 \leq i \leq 36$, $E(X_i) = 1/\lambda = 10$ and $\sigma_{X_i} = 1/\lambda = 10$. By the version of the central limit theorem expressed in Remark 7.3, the desired probability is approximately

$$\begin{aligned} P(\bar{X} \geq 10.5) &= P\left(\frac{\bar{X} - 10}{10/\sqrt{36}} \geq \frac{10.5 - 10}{10/\sqrt{36}}\right) = P\left(\frac{\bar{X} - 10}{10/\sqrt{36}} \geq 0.30\right) \\ &\approx 1 - \Phi(0.30) = 1 - 0.6179 = 0.3821. \quad \blacklozenge \end{aligned}$$

Example 7.11 The time it takes for a student to finish an aptitude test (in hours) has the probability density function

$$f(x) = \begin{cases} 6(x-1)(2-x) & \text{if } 1 < x < 2 \\ 0 & \text{otherwise.} \end{cases}$$

Approximate the probability that the average length of time it takes for a random sample of 15 students to complete the test is less than 1 hour and 25 minutes.

Solution: For $1 \leq i \leq 15$, let X_i be the time that it will take for the i th student in the sample to complete the test. Let $\bar{X} = (1/15) \sum_{i=1}^{15} X_i$. Direct calculations show that, for $1 \leq i \leq 15$,

$$E(X_i) = \int_1^2 6x(x-1)(2-x) dx = 3/2 = 1.5,$$

$$E(X_i^2) = \int_1^2 6x^2(x-1)(2-x) dx = 23/10 = 2.3,$$

$$\sigma_{X_i} = \sqrt{(23/10) - (3/2)^2} = 1/\sqrt{20} = 0.2236.$$

Using the central limit theorem, the desired probability is approximately equal to

$$\begin{aligned} P(\bar{X} < 17/12) &= P\left(\frac{\bar{X} - 1.5}{0.2236/\sqrt{15}} < \frac{(17/12) - 1.5}{0.2236/\sqrt{15}}\right) \\ &= P\left(\frac{\bar{X} - 1.5}{0.2236/\sqrt{15}} < -1.44\right) \approx \Phi(-1.44) = 0.0749. \quad \blacklozenge \end{aligned}$$

Example 7.12 If 20 random numbers are selected independently from the interval $(0, 1)$, what is the approximate probability that the sum of these numbers is at least eight?

Solution: For $1 \leq i \leq 20$, let X_i be the i th number selected. To calculate $P(\sum_{i=1}^{20} X_i \geq 8)$, we use the central limit theorem. Since $\forall i, 1 \leq i \leq 20$, $E(X_i) = (0 + 1)/2 = 1/2$ and $\text{Var}(X) = (1 - 0)^2/12 = 1/12$,

$$\begin{aligned} P\left(\sum_{i=1}^{20} X_i \geq 8\right) &= P\left(\frac{\sum_{i=1}^{20} X_i - 20(1/2)}{\sqrt{1/12} \sqrt{20}} \geq \frac{8 - 20(1/2)}{\sqrt{1/12} \sqrt{20}}\right) \\ &= P\left(\frac{\sum_{i=1}^{20} X_i - 10}{\sqrt{5/3}} \geq -1.55\right) \approx 1 - \Phi(-1.55) \approx 0.9394. \quad \blacklozenge \end{aligned}$$

Example 7.13 A biologist wants to estimate ℓ , the life expectancy of a certain type of insect. To do so, he takes a sample of size n and measures the lifetime from birth to death of each insect. Then he finds the average of these numbers. If he believes that the lifetimes of these insects are independent random variables with variance 1.5 days, how large a sample should he choose to be 98% sure that his average is accurate within ± 0.2 (± 4.8 hours)?

Solution: For $i = 1, 2, \dots, n$, let X_i be the lifetime of the i th insect of the sample. We want to determine n , so that

$$P\left(-0.2 < \frac{X_1 + X_2 + \dots + X_n}{n} - \ell < 0.2\right) \approx 0.98.$$

Since $E(X_i) = \ell$ and $\text{Var}(X_i) = 1.5$, by the central limit theorem,

$$\begin{aligned} P\left(-0.2 < \frac{\sum_{i=1}^n X_i}{n} - \ell < 0.2\right) &= P((-0.2)n < \sum_{i=1}^n X_i - n\ell < (0.2)n) \\ &= P\left(\frac{-(0.2)n}{\sqrt{1.5n}} < \frac{\sum_{i=1}^n X_i - n\ell}{\sqrt{1.5n}} < \frac{(0.2)n}{\sqrt{1.5n}}\right) \\ &\approx \Phi\left[\frac{(0.2)n}{\sqrt{1.5n}}\right] - \Phi\left[\frac{-(0.2)n}{\sqrt{1.5n}}\right] = 2\Phi\left(\frac{0.2\sqrt{n}}{\sqrt{1.5}}\right) - 1. \end{aligned}$$

Thus the quantity $2\Phi(0.2\sqrt{n}/\sqrt{1.5}) - 1$ should approximately equal 0.98; that is, $\Phi(0.2\sqrt{n}/\sqrt{1.5}) \approx 0.99$. From [Table 1](#) of the Appendix Tables, we find that $0.2\sqrt{n}/\sqrt{1.5} = 2.33$. This gives $n = 203.58$. Therefore, the biologist should choose a sample of size 204. ♦

Remark 7.4: In the solution of Example 7.13, we have assumed implicitly that, for a sample of size 204, normal distribution is a good approximation. Although usually such assumptions are reasonable, the size of n depends on the distribution function of X_i 's, which is not known in most cases. If the biologist desires complete certainty regarding the analysis results, instead of relying on the central limit theorem, he or she should opt for Chebyshev's inequality, which is discussed in [Chapter 10](#). The results obtained by applying Chebyshev's inequality are safe but, in most cases, much larger than needed. For this example, the answer obtained by applying Chebyshev's inequality is 1875, a number much larger than the 204 obtained by using the central limit theorem. (See Example 10.33.) ♦

Example 7.14 Suppose that, with equal probabilities, the value of a specific stock on any trading day increases 30% or decreases 25%, independent of the fluctuations of the stock value on the past and future trading days. Let

$$r_i = \begin{cases} 0.30 & \text{with probability } 1/2 \\ -0.25 & \text{with probability } 1/2. \end{cases}$$

Then r_i is the rate of return on the i th trading day. Schiller, a very conservative investor, invests A dollars in this stock thinking that

$$E(r_i) = 0.30 \cdot \frac{1}{2} + (-0.25) \cdot \frac{1}{2} = 0.025$$

implies that, on average, every day the value of his investment increases by 2.5% of its value on the previous day. The fact that, by the end of the first trading day, the value of his stock shares is $A + Ar_1 = A(1 + r_1)$, and

$$E[A(1 + r_1)] = A[1 + E(r_1)] = A(1.025) = 1.025A,$$

makes this investment attractive to Schiller. In this example, we show that Schiller is mistaken, and if he holds his shares of this stock long enough, eventually he will lose the entire value of

his investment in this stock. To do so, note that at the end of the second trading day, the value of Schiller's stock shares is

$$A(1 + r_1) + A(1 + r_1)r_2 = A(1 + r_1)(1 + r_2);$$

at the end of the third trading day, its value is

$$A(1 + r_1)(1 + r_2) + A(1 + r_1)(1 + r_2)r_3 = A(1 + r_1)(1 + r_2)(1 + r_3);$$

and, in general, at the end of the n th trading day its value is

$$A(1 + r_1)(1 + r_2) \cdots (1 + r_n).$$

Let $Y_i = 1 + r_i$, then

$$Y_i = \begin{cases} 1.30 & \text{with probability } 1/2 \\ 0.75 & \text{with probability } 1/2. \end{cases}$$

The sequence $\{Y_1, Y_2, \dots\}$ is an independent sequence of identically distributed random variables. Using the central limit theorem, we will now find n , the number of trading days after which, with probability 0.99, the value of the stock decreases to 10% of its original value. That is, we will find n so that

$$\begin{aligned} 0.99 &\leq P(A(1 + r_1)(1 + r_2) \cdots (1 + r_n) \leq (0.10)A) \\ &= P(Y_1 Y_2 \cdots Y_n \leq 0.10) = P(\ln Y_1 + \ln Y_2 + \cdots + \ln Y_n \leq \ln 0.10). \end{aligned}$$

Note that $\ln 0.10 = -2.303$, and $\{\ln Y_1, \ln Y_2, \dots\}$ is a sequence of identically distributed and independent random variables with

$$\begin{aligned} E(\ln Y_i) &= \ln(1.30) \cdot \frac{1}{2} + \ln(0.75) \cdot \frac{1}{2} = -0.127, \\ E[(\ln Y_i)^2] &= [\ln(1.30)]^2 \cdot \frac{1}{2} + [\ln(0.75)]^2 \cdot \frac{1}{2} = 0.0758, \\ \text{Var}(\ln Y_i) &= E[(\ln Y_i)^2] - [E(\ln Y_i)]^2 = (0.0758) - (-0.127)^2 = 0.0597 \\ \sigma_{\ln Y_i} &= \sqrt{0.0597} = 0.244. \end{aligned}$$

Since by the central limit theorem,

$$\begin{aligned} &P(\ln Y_1 + \ln Y_2 + \cdots + \ln Y_n \leq -2.303) \\ &= P\left(\frac{\ln Y_1 + \ln Y_2 + \cdots + \ln Y_n - n(-0.127)}{0.244\sqrt{n}} \leq \frac{-2.303 - n(-0.127)}{0.244\sqrt{n}}\right) \\ &\approx \Phi\left(\frac{-2.303 + 0.127n}{0.244\sqrt{n}}\right), \end{aligned}$$

for this probability to be at least 0.99, by [Table 2](#) of the Appendix Tables, we must have

$$\frac{-2.303 + 0.127n}{0.244\sqrt{n}} = 2.33,$$

or, equivalently,

$$0.127n - 0.569\sqrt{n} - 2.303 = 0.$$

This is a quadratic equation in $\sqrt{n}$. Solving it for $\sqrt{n}$ yields $\sqrt{n} = 7.052$, or $n = 49.73$. This shows that, with probability 0.99, after 50 trading days, the value of the stock reduces to 10% of its original value in spite of the fact that the expected return, on each trading day, is positive. The assertion that, if Schiller holds his shares of the stock long enough, eventually he will lose the entire value of his investment in this stock follows from the preceding observation. ♦

EXERCISES

A

1. What is the probability that the average of 150 random points from the interval $(0, 1)$ is within 0.02 of the midpoint of the interval?
2. For the scores on an achievement test given to a certain population of students, the expected value is 500 and the standard deviation is 100. Let $\bar{X}$ be the mean of the scores of a random sample of 35 students from the population. Estimate $P(460 < \bar{X} < 540)$.
3. A random sample of size 24 is taken from a distribution with probability density function

$$f(x) = \begin{cases} \frac{1}{9}\left(x + \frac{5}{2}\right) & 1 < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

Let $\bar{X}$ be the sample mean. Approximate $P(2 < \bar{X} < 2.15)$.

4. A random sample of size n ($n \geq 1$) is taken from a distribution with the following probability density function:

$$f(x) = \frac{1}{2}e^{-|x|}, \quad -\infty < x < \infty.$$

What is the probability that the sample mean is positive?

5. Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables, and let $S_n = X_1 + X_2 + \dots + X_n$. For large n , what is the approximate probability that S_n is between $E(S_n) - \sigma_{S_n}$ and $E(S_n) + \sigma_{S_n}$?
6. Each time that Jim charges an item to his credit card, he rounds the amount to the nearest dollar in his records. If he has used his credit card 300 times in the last 12 months, what is the probability that his record differs from the total expenditure by, at most, 10 dollars?
7. Carla has invested \$5,000 in a stock. Suppose that, each month, the investment increases by \$377 with probability 0.49 and decreases by \$315 with probability 0.51. If the change in stock prices in different months are independent, what is the probability that at the end of year 10, the value of Carla's investment in this stock is at least \$8,000?

8. A physical quantity is measured 50 times, and the average of these measurements is taken as the result. If each measurement has a random error uniformly distributed over $(-1, 1)$, what is the probability that our result differs from the actual value by less than 0.25?
9. Suppose that, whenever invited to a party, the probability that a person attends with his or her guest is $1/3$, attends alone is $1/3$, and does not attend is $1/3$. A company has invited all 300 of its employees and their guests to a Christmas party. What is the probability that at least 320 will attend?
10. Consider a distribution with mean μ and probability density function

$$f(x) = \begin{cases} \frac{1}{x \ln(3/2)} & 4 \leq x \leq 6 \\ 0 & \text{elsewhere.} \end{cases}$$

Determine the values of n for which the probability is at least 0.98 that the mean of a random sample of size n from the population is within ± 0.07 of μ ?

B

11. An investor buys 1000 shares of the XYZ Corporation at \$50.00 per share. Subsequently, the stock price varies by \$0.125 ($1/8$) every day, but unfortunately it is just as likely to move down as up. What is the most likely value of his holdings after 60 days?
Hint: First calculate the distribution of the *change* in the stock price after 60 days.
12. A fair coin is tossed successively. Using the central limit theorem, find an approximation for the probability of obtaining at least 25 heads before 50 tails.
13. Let $\{X_1, X_2, \dots\}$ be a sequence of independent Poisson random variables, each with parameter 1. By applying the central limit theorem to this sequence, prove that

$$\lim_{n \rightarrow \infty} \frac{1}{e^n} \sum_{k=0}^n \frac{n^k}{k!} = \frac{1}{2}.$$



Self-Quiz on Section 7.3

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. An actuary wants to determine a new estimate for the average cost, to her insurance company, of the vehicles that are involved in accidents and are considered total losses. Based on the previous sets of data, the actuary has estimated that the standard deviation of future samples of such vehicles, per totaled vehicle, is \$4050. How many totaled vehicles should the actuary sample to make sure, with a probability of at least 0.95, that her estimate is only within \$300 of the actual mean of the population of all totaled vehicles insured by her company?

2. In August of 2017, in the wake of Hurricane Harvey, a group of social workers established an online relief fund, and in only 48 hours, 7000 contributions were made to the fund. Suppose that contributions were identically distributed, independent random variables with mean \$220 and standard deviation \$90. Find an approximate 95th percentile for F , the distribution function of the total amount contributed to the relief fund. That is, find the value of x for which $F(x) = 0.95$.



7.4 EXPONENTIAL RANDOM VARIABLES

Let $\{N(t): t \geq 0\}$ be a Poisson process. Then, as discussed in [Section 6.2](#), $N(t)$ is the number of “events” that have occurred at or prior to time t . Let X_1 be the time of the first event, X_2 be the elapsed time between the first and the second events, X_3 be the elapsed time between the second and third events, and so on. The sequence of random variables $\{X_1, X_2, \dots\}$ is called the **sequence of interarrival times** of the Poisson process $\{N(t): t \geq 0\}$. Let $\lambda = E[N(1)]$, then

$$P(N(t) = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}.$$

This enables us to calculate the distribution functions of the random variables X_i , $i \geq 1$. For $t \geq 0$,

$$P(X_1 > t) = P(N(t) = 0) = e^{-\lambda t}.$$

Therefore,

$$P(X_1 \leq t) = 1 - P(X_1 > t) = 1 - e^{-\lambda t}.$$

Since a Poisson process is stationary and possesses independent increments, at any time t , the process probabilistically starts all over again. Hence the interarrival time of any two consecutive events has the same distribution as X_1 ; that is, the sequence $\{X_1, X_2, \dots\}$ is independent and identically distributed. Therefore, for all $n \geq 1$,

$$P(X_n \leq t) = P(X_1 \leq t) = \begin{cases} 1 - e^{-\lambda t} & t \geq 0 \\ 0 & t < 0. \end{cases}$$

Let

$$F(t) = \begin{cases} 1 - e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

for some $\lambda > 0$. Then F is the distribution function of X_n for all $n \geq 1$. It is called **exponential distribution** and is one of the most important distributions of pure and applied probability. Since

$$f(t) = F'(t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (7.2)$$

and

$$\int_0^{\infty} \lambda e^{-\lambda t} dt = \lim_{b \rightarrow \infty} \int_0^b \lambda e^{-\lambda t} dt = \lim_{b \rightarrow \infty} \left[-e^{-\lambda t} \right]_0^b = \lim_{b \rightarrow \infty} (1 - e^{-\lambda b}) = 1,$$

f is a density function.

Definition 7.4 A continuous random variable X is called **exponential** with parameter $\lambda > 0$ if its density function is given by (7.2).

Because the interarrival times of a Poisson process are exponential, the following are examples of random variables that might be exponential.

1. The interarrival time between two customers at a post office
2. The duration of Jim's next telephone call
3. The time between two consecutive earthquakes in California
4. The time between two accidents at an intersection
5. The time until the next baby is born in a hospital
6. The time until the next crime in a certain town
7. The time to failure of the next fiber segment in a large group of such segments when all of them are initially fault free
8. The time interval between the observation of two consecutive shooting stars on a summer evening
9. The time between two consecutive fish caught by a fisherman from a large lake with lots of fish

From [Section 6.2](#) we know that λ is the average number of the events in one time unit; that is, $E[N(1)] = \lambda$. Therefore, we should expect an average time $1/\lambda$ between two consecutive events. To prove this, let X be an exponential random variable with parameter λ , then

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x (\lambda e^{-\lambda x}) dx.$$

Using integration by parts with $u = x$ and $dv = \lambda e^{-\lambda x} dx$, we obtain

$$E(X) = \left[-x e^{-\lambda x} \right]_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx = 0 - \left[\frac{1}{\lambda} e^{-\lambda x} \right]_0^{\infty} = \frac{1}{\lambda}.$$

A similar calculation shows that

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 (\lambda e^{-\lambda x}) dx \\ &= \left[-\left(x^2 + \frac{2}{\lambda} x + \frac{2}{\lambda^2} \right) e^{-\lambda x} \right]_0^{\infty} = \frac{2}{\lambda^2}. \end{aligned}$$

Hence

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2},$$

and therefore $\sigma_X = 1/\lambda$. We have shown that

For an exponential random variable with parameter λ ,

$$E(X) = \sigma_X = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Figures 7.11 and 7.12 represent the graphs of the exponential density and exponential distribution functions, respectively.

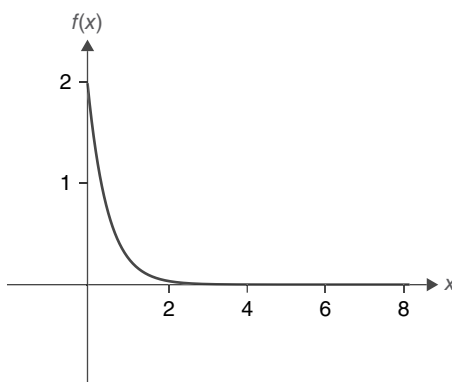


Figure 7.11 Exponential density function with parameter $\lambda = 2$.

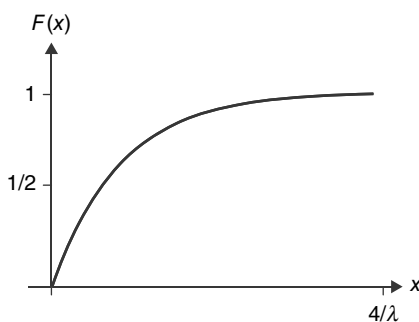


Figure 7.12 Exponential distribution function with parameter λ .

Example 7.15 Suppose that every 3 months, on average, an earthquake occurs in California. What is the probability that the next earthquake occurs after three but before 7 months?

Solution: Let X be the time (in months) until the next earthquake; it can be assumed that X is an exponential random variable with $1/\lambda = 3$ or $\lambda = 1/3$. To calculate $P(3 < X < 7)$, note that since F , the distribution function of X , is given by

$$F(t) = P(X \leq t) = 1 - e^{-t/3} \quad \text{for } t > 0,$$

we can write

$$P(3 < X < 7) = F(7) - F(3) = (1 - e^{-7/3}) - (1 - e^{-1}) \approx 0.27. \quad \blacklozenge$$

Example 7.16 At an intersection there are two accidents per day, on average. What is the probability that after the next accident there will be no accidents at all for the next two days?

Solution: Let X be the time (in days) between the next two accidents. It can be assumed that X is exponential with parameter λ , satisfying $1/\lambda = 1/2$, so that $\lambda = 2$. To find $P(X > 2)$, note that F , the distribution function of X , is given by $F(t) = 1 - e^{-2t}$, $t > 0$. Hence

$$P(X > 2) = 1 - P(X \leq 2) = 1 - F(2) = e^{-4} \approx 0.02. \quad \blacklozenge$$

An important feature of exponential distribution is its *memoryless property*. A nonnegative random variable X is called **memoryless** if, for all $s, t \geq 0$,

$$P(X > s + t \mid X > t) = P(X > s). \quad (7.3)$$

If, for example, X is the lifetime of some type of instrument, then (7.3) means that there is no deterioration with age of the instrument. The probability that a new instrument will last more than s years is the same as the probability that a used instrument that has lasted more than t years will last at least another s years. In other words, the probability that such an instrument will deteriorate in the next s years does not depend on the age of the instrument.

To show that an exponential distribution is memoryless, note that (7.3) is equivalent to

$$\frac{P(X > s + t, X > t)}{P(X > t)} = P(X > s)$$

and

$$P(X > s + t) = P(X > s)P(X > t). \quad (7.4)$$

Now since

$$\begin{aligned} P(X > s + t) &= 1 - [1 - e^{-\lambda(s+t)}] = e^{-\lambda(s+t)}, \\ P(X > s) &= 1 - (1 - e^{-\lambda s}) = e^{-\lambda s}, \end{aligned}$$

and

$$P(X > t) = 1 - (1 - e^{-\lambda t}) = e^{-\lambda t},$$

we have that (7.4) follows. Hence X is memoryless. It can be shown that exponential is the only continuous distribution which possesses a memoryless property (see Exercise 27).

Example 7.17 The lifetime of a TV tube (in years) is an exponential random variable with mean 10. If Jim bought his TV set 10 years ago, what is the probability that its tube will last another 10 years?

Solution: Let X be the lifetime of the tube. Since X is an exponential random variable, there is no deterioration with age of the tube. Hence

$$P(X > 20 \mid X > 10) = P(X > 10) = 1 - [1 - e^{(-1/10)10}] \approx 0.37. \quad \blacklozenge$$

Example 7.18 Suppose that, on average, two earthquakes occur in San Francisco and two in Los Angeles every year. If the last earthquake in San Francisco occurred 10 months ago and the last earthquake in Los Angeles occurred 2 months ago, what is the probability that the next earthquake in San Francisco occurs after the next earthquake in Los Angeles?

Solution: It can be assumed that the number of earthquakes in San Francisco and Los Angeles are both Poisson processes with common rate $\lambda = 2$. Hence the times between two consecutive earthquakes in Los Angeles and two consecutive earthquakes in San Francisco are both exponentially distributed with the common mean $1/\lambda = 1/2$. Because of the memoryless property of the exponential distribution, it does not matter when the last earthquakes in San Francisco and Los Angeles have occurred. The times between now and the next earthquake in San Francisco and the next earthquake in Los Angeles both have the same distribution. Since these time periods are exponentially distributed with the same parameter, by symmetry, the probability that the next earthquake in San Francisco occurs after that in Los Angeles is $1/2$. $\blacklozenge$

Example 7.19 A system has n components, whose lifetimes are exponential random variables with parameters $\lambda_1, \lambda_2, \dots, \lambda_n$, respectively. Suppose that the lifetimes of the components are independent random variables, and the system fails as soon as one of its components fails. Find the probability density function and the expected value of the time until the system fails.

Solution: Let $X_1, X_2, \dots, X_n$ be the lifetimes of the n components, respectively. Then $X_1, X_2, \dots, X_n$ are independent random variables and for $i = 1, 2, \dots, n$,

$$P(X_i \leq t) = 1 - e^{-\lambda_i t}, \quad t \geq 0.$$

Letting X be the time until the system fails, we have $X = \min(X_1, X_2, \dots, X_n)$. Therefore,

$$\begin{aligned} P(X > t) &= P(\min(X_1, X_2, \dots, X_n) > t) = P(X_1 > t, X_2 > t, \dots, X_n > t) \\ &= P(X_1 > t)P(X_2 > t) \cdots P(X_n > t) \\ &= (e^{-\lambda_1 t})(e^{-\lambda_2 t}) \cdots (e^{-\lambda_n t}) = e^{-(\lambda_1 + \lambda_2 + \cdots + \lambda_n)t}, \quad t \geq 0. \end{aligned}$$

Let f be the probability density function of X , then

$$\begin{aligned} f(t) &= \frac{d}{dt}P(X \leq t) = \frac{d}{dt}(1 - e^{-(\lambda_1 + \lambda_2 + \cdots + \lambda_n)t}) \\ &= (\lambda_1 + \lambda_2 + \cdots + \lambda_n)e^{-(\lambda_1 + \lambda_2 + \cdots + \lambda_n)t}, \quad t \geq 0. \end{aligned}$$

This shows that X is an exponential random variable with parameter $\lambda_1 + \lambda_2 + \cdots + \lambda_n$. Hence

$$E(X) = \frac{1}{\lambda_1 + \lambda_2 + \cdots + \lambda_n}. \quad \blacklozenge$$

Remark 7.5 In Example 7.19, we have shown the following important property of exponential random variables.

If $X_1, X_2, \dots, X_n$ are n independent exponential random variables with parameters $\lambda_1, \lambda_2, \dots$, and λ_n , respectively, then $\min(X_1, X_2, \dots, X_n)$ is an exponential random variable with parameter $\lambda_1 + \lambda_2 + \cdots + \lambda_n$. Hence

$$E[\min(X_1, X_2, \dots, X_n)] = \frac{1}{\lambda_1 + \lambda_2 + \cdots + \lambda_n}.$$

Relationship between Exponential and Geometric: Recall that if a Bernoulli trial is performed successively and independently, then the number of trials until the *first success* occurs is geometric. Furthermore, the number of trials *between two consecutive successes* is also geometric. Sometimes exponential is considered to be the continuous analog of geometric because, for a Poisson process, the time it will take until the *first event* occurs is exponential, and the time between *two consecutive events* is also exponential. Moreover, exponential is the only memoryless continuous distribution (see Exercise 27), and geometric is the only memoryless discrete distribution (refer to Exercise 26, [Section 6.3](#)). It is also interesting to know that if X is an exponential random variable, then $[X]$, the integer part of X ; i.e., the greatest integer less than or equal to X , is geometric (see Exercise 26). $\blacklozenge$

Remark 7.6 In this section, we showed that if $\{N(t) : t \geq 0\}$ is a Poisson process with rate λ , then the interarrival times of the process form an independent sequence of identically distributed exponential random variables with mean $1/\lambda$. Using the tools of an area of probability called *renewal theory*, we can prove that the converse of this fact is also true:

If, for some process, $N(t)$ is the number of “events” occurring in $[0, t]$, and if the times between consecutive events form a sequence of independent and identically distributed exponential random variables with mean $1/\lambda$, then $\{N(t) : t \geq 0\}$ is a Poisson process with rate λ . $\blacklozenge$

EXERCISES

A

1. Customers arrive at a postoffice at a Poisson rate of three per minute. What is the probability that the next customer does not arrive during the next 3 minutes?
2. Find the median of an exponential random variable with rate λ . Recall that for a continuous distribution F , the median $Q_{0.5}$ is the point at which $F(Q_{0.5}) = 1/2$.
3. Let X be an exponential random variable with mean 1. Find the probability density function of $Y = -\ln X$.
4. The time between the first and second heart attacks for a certain group of people is an exponential random variable. If 50% of those who have had a heart attack will have another one within the next 5 years, what is the probability that a person who had one heart attack 5 years ago will not have another one in the 5 five years?
5. Guests arrive at a hotel, in accordance with a Poisson process, at a rate of five per hour. Suppose that for the last 10 minutes no guest has arrived. What is the probability that (a) the next one will arrive in less than 2 minutes; (b) from the arrival of the tenth to the arrival of the eleventh guest takes no more than 2 minutes?
6. Let X be an exponential random variable with parameter λ . Find

$$P(|X - E(X)| \geq 2\sigma_X).$$

7. Suppose that losses under an insurance policy are distributed exponentially with mean \$6000. If the amount of deductible for each loss is \$500, what is the median of the insurance company's claim payment under this policy? Note that the median of a continuous random variable Y is a number m , which satisfies $P(Y \leq m) = P(Y \geq m) = 0.5$.
8. Suppose that, at an Italian restaurant, the time, in minutes, between two customers ordering pizza is exponential with parameter λ . What is the probability that (a) no customer orders pizza during the next t minutes; (b) the next pizza order is placed in at least t minutes but no later than s minutes ($t < s$)?
9. Suppose that the time it takes for a novice secretary to type a document is exponential with mean 1 hour. If at the beginning of a certain eight-hour working day the secretary receives 12 documents to type, what is the probability that she will finish them all by the end of the day?
10. The time between consecutive arrival times of two buses at a station is exponential with mean 20 minutes. Every day, during the operating hours of the buses, Kayla arrives at the station at a random time and waits for a bus to arrive. However, while she expects that her average waiting time be only 10 minutes, she complains that, on the average, she must wait much longer for a bus to arrive. What is the fallacy in Kayla's argument?

11. The profit is \$350 for each computer assembled by a certain person. Suppose that the assembler guarantees his computers for 1 year and the time between two failures of a computer is exponential with mean 18 months. If it costs the assembler \$40 to repair a failed computer, what is the expected profit per computer?
Hint: Let $N(t)$ be the number of times that the computer fails in $[0, t]$. Then $\{N(t): t \geq 0\}$ is a Poisson process with parameter $\lambda = 1/18$.
12. Mr. Jones is waiting to make a phone call at a train station. There are two public telephone booths next to each other, occupied by two persons, say A and B . If the duration of each telephone call is an exponential random variable with $\lambda = 1/8$, what is the probability that among Mr. Jones, A , and B , Mr. Jones will not be the last to finish his call?
13. The service times at a two-teller bank are independent exponential random variables each with parameter λ . At a random time Jim is being served by teller 2, Jeff is being served by teller 1, Judy and Julie are in the queue waiting to be served, and Judy is ahead of Julie. What is the probability that when Julie's service is completed Jim is still being served?
14. In a factory, a certain machine operates for a period which is exponentially distributed with parameter λ . Then it breaks down and will be in repair shop for a period, which is also exponentially distributed with mean $1/\lambda$. The operating and the repair times are independent. For this machine, we say that a change of "state" occurs each time that it breaks down, or each time that it is fixed. In a time interval of length t , find the probability mass function of the number of times a change of state occurs.
15. In a huge office building, the alarm system has n sensors, the lifetime of each being exponential with mean $1/\lambda$, independently of the other ones. If for the next t units of time the alarm system is not checked, what is the probability that at time t only k sensors are working? Assume that without inspection there is no way for the technical crew to know that a sensor is down.
16. Suppose that the lifetimes of radio transistors are independent exponential random variables with mean 5 years. Arnold buys a radio and decides to replace its transistor upon failure two times: once when the original transistor dies and once when the replacement dies. He stops using the radio when the second replacement of the transistor goes out of order. Assuming that Arnold repairs the radio if it fails for any other reason, find the probability that he uses the radio at least 15 years.
17. Let $X_1, X_2, \dots, X_n$ be identically distributed, independent, exponential random variables with parameters $\lambda_1, \lambda_2, \dots, \lambda_n$. Prove that

$$E[\min(X_1, \dots, X_n)] < \min(E(X_1), \dots, E(X_n)).$$

18. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} 2e^{-(x+2y)} & \text{if } x \geq 0, \quad y \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X^2Y)$.

19. Andy buys an MP3 player and six new AAA batteries. The player is operated with two AAA batteries, and each time a battery dies, Andy replaces that battery with a new one. If the lifetimes of the batteries are independent exponential random variables each with parameter λ , find the expected length of time Andy can listen to his MP3 player with the six batteries.
Hint: Let X_1 and X_2 be the lifetimes of the first two batteries Andy will use to operate his MP3. The first time a battery dies is after $\min(X_1, X_2)$ units of time. Begin by calculating $E[\min(X_1, X_2)]$.
20. The lifetime of a particular type of tablet PC is exponential with mean 8 years. The manufacturer sells each tablet PC for \$960.00 and offers a warranty to fully refund the customer's money if the PC fails within a year of purchase and to refund half of the customer's money if it fails during the second year following the purchase. There will be no refund after the second year. If the manufacturer sells 11,000 tablet PC's every year, what is the expected value of the total refund amount it pays every year? Assume that the set of the lifetimes of all such PC's is an independent set of random variables.
21. Let X and Y be independent exponential random variables with parameters λ and μ , respectively. What is the probability that $X < Y$?
22. Let X and Y be independent exponential random variables both with mean 1. Find $E[\max(X, Y)]$.

B

23. A parallel system with two components is a system that functions if and only if at least one of its components functions. Let X and Y , the lifetimes of the components, be independent exponential random variables with parameters λ and μ , respectively. Find the probability density function of $|X - Y|$, the time between the failures of the two components.
24. In data communication, messages are usually combinations of characters, and each character consists of a number of bits. A bit is the smallest unit of information and is either 1 or 0. Suppose that L , the length of a character (in bits) is a geometric random variable with parameter p . If a sender emits messages at the rate of 1000 bits per second, what is the distribution of T , the time it takes the sender to emit a character?
25. The random variable X is said to be a **Laplace random variable** or **double exponentially distributed** if its probability density function is given by

$$f(x) = ce^{-|x|}, \quad -\infty < x < +\infty.$$

- (a) Find the value of c .
- (b) Prove that $E(X^{2n}) = (2n)!$ and $E(X^{2n+1}) = 0$.
26. Let X , the lifetime (in years) of a radio tube, be exponentially distributed with mean $1/\lambda$. Prove that $[X]$, the integer part of X , which is the complete number of years that the tube works, is a geometric random variable.

27. Prove that if X is a positive, continuous, memoryless random variable with distribution function F , then $F(t) = 1 - e^{-\lambda t}$ for some $\lambda > 0$. This shows that the exponential is the only distribution on $(0, \infty)$ with the memoryless property.
28. An item has n parts, each with an exponentially distributed lifetime with mean $1/\lambda$. If the failure of one part makes the item fail, what is the average lifetime of the item?
Hint: Use the result of Example 7.19.
29. Suppose that X and Y are independent, identically distributed exponential random variables with mean $1/\lambda$. Prove that $X/(X + Y)$ is uniform over $(0, 1)$. ■

Self-Quiz on Section 7.4

Time allotted: 20 Minutes

Each problem is worth 5 points.

- When solid state hard drives fail to function, it is not possible to repair them. They should be replaced. Such drives, manufactured by Seaportal Technology, have lifetimes that are exponentially distributed with mean 8 years. An online company, which sells this brand of hard drives, offers its own warranty that fully refunds the price of the hard drive if it fails within the first year, refunds half of its price if it fails during the second year, and offers no refund if it fails thereafter. If the online company sells each hard drive for \$250, what is the expected value of its loss due to warranty claims per hard drive?
- Patients arrive at a pharmacy for flu shots at a Poisson rate of 3 per hour. If the last patient, who had a flu shot at the pharmacy, arrived 45 minutes ago, what is the probability that the next patient does not arrive for a flu shot within the next 15 minutes? ■

7.5 GAMMA DISTRIBUTIONS

Let $\{N(t) : t \geq 0\}$ be a Poisson process, X_1 be the time of the first event, and for $n \geq 2$, let X_n be the time between the $(n - 1)$ st and n th events. As we explained in [Section 7.4](#), $\{X_1, X_2, \dots\}$ is a sequence of identically distributed exponential random variables with mean $1/\lambda$, where λ is the rate of $\{N(t) : t \geq 0\}$. For this Poisson process let X be the time of the n th event. Then X is said to have a **gamma distribution** with parameters (n, λ) . Therefore, exponential is the time we will wait for the first event to occur, and gamma is the time we will wait for the n th event to occur. Clearly, a gamma distribution with parameters $(1, \lambda)$ is identical with an exponential distribution with parameter λ .

Let X be a gamma random variable with parameters (n, λ) . To find f , the density function of X , note that $\{X \leq t\}$ occurs if the time of the n th event is in $[0, t]$, that is, if the number of events occurring in $[0, t]$ is at least n . Hence F , the distribution function of X , is given by

$$F(t) = P(X \leq t) = P(N(t) \geq n) = \sum_{i=n}^{\infty} \frac{e^{-\lambda t} (\lambda t)^i}{i!}.$$

Differentiating F , the density function f is obtained:

$$\begin{aligned} f(t) &= \sum_{i=n}^{\infty} \left[-\lambda e^{-\lambda t} \frac{(\lambda t)^i}{i!} + \lambda e^{-\lambda t} \frac{(\lambda t)^{i-1}}{(i-1)!} \right] \\ &= \sum_{i=n}^{\infty} -\lambda e^{-\lambda t} \frac{(\lambda t)^i}{i!} + \left[\lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!} + \sum_{i=n+1}^{\infty} \lambda e^{-\lambda t} \frac{(\lambda t)^{i-1}}{(i-1)!} \right] \\ &= \left[-\sum_{i=n}^{\infty} \lambda e^{-\lambda t} \frac{(\lambda t)^i}{i!} \right] + \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!} + \left[\sum_{i=n}^{\infty} \lambda e^{-\lambda t} \frac{(\lambda t)^i}{i!} \right] \\ &= \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}. \end{aligned}$$

The density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} \frac{(\lambda x)^{n-1}}{(n-1)!} & \text{if } x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

is called the **gamma** (or **n -Erlang**) **density** with parameters (n, λ) .

Now we extend the definition of the gamma density from parameters (n, λ) to (r, λ) , where $r > 0$ is not necessarily a positive integer. As we shall see later, this extension has useful applications in probability and statistics. In the formula of the gamma density function, the term $(n-1)!$ is defined only for positive integers. So the only obstacle in such an extension is to find a function of r that has the basic property of the factorial function, namely, $n! = n \cdot (n-1)!$, and coincides with $(n-1)!$ when n is a positive integer. The function with these properties is $\Gamma: (0, \infty) \rightarrow \mathbb{R}$ defined by

$$\Gamma(r) = \int_0^{\infty} t^{r-1} e^{-t} dt.$$

The property analogous to $n! = n \cdot (n-1)!$ is

$$\Gamma(r+1) = r\Gamma(r), \quad r > 1,$$

which is obtained by an integration by parts applied to $\Gamma(r+1)$ with $u = t^r$ and $dv = e^{-t} dt$:

$$\begin{aligned} \Gamma(r+1) &= \int_0^{\infty} t^r e^{-t} dt = \left[-t^r e^{-t} \right]_0^{\infty} + r \int_0^{\infty} t^{r-1} e^{-t} dt \\ &= r \int_0^{\infty} t^{r-1} e^{-t} dt = r\Gamma(r). \end{aligned}$$

To show that $\Gamma(n)$ coincides with $(n-1)!$ when n is a positive integer, note that

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = 1.$$

Therefore,

$$\begin{aligned}\Gamma(2) &= (2-1)\Gamma(2-1) = 1 = 1!, \\ \Gamma(3) &= (3-1)\Gamma(3-1) = 2 \cdot 1 = 2!, \\ \Gamma(4) &= (4-1)\Gamma(4-1) = 3 \cdot 2 \cdot 1 = 3!.\end{aligned}$$

Repetition of this process or a simple induction implies that $\Gamma(n+1) = n!$. Hence $\Gamma(r+1)$ is the natural generalization of $n!$ for a noninteger $r > 0$. This motivates the following definition.

Definition 7.5 A random variable X with probability density function

$$f(x) = \begin{cases} \frac{\lambda e^{-\lambda x} (\lambda x)^{r-1}}{\Gamma(r)} & \text{if } x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

is said to have a **gamma** distribution with parameters (r, λ) , $\lambda > 0$, $r > 0$.

Let X be a gamma random variable with parameters (r, λ) . To find $E(X)$ and $\text{Var}(X)$, note that for all $n \geq 0$,

$$E(X^n) = \int_0^{\infty} x^n \frac{\lambda e^{-\lambda x} (\lambda x)^{r-1}}{\Gamma(r)} dx = \frac{\lambda^r}{\Gamma(r)} \int_0^{\infty} x^{n+r-1} e^{-\lambda x} dx.$$

Let $t = \lambda x$, then $dt = \lambda dx$, so

$$E(X^n) = \frac{\lambda^r}{\Gamma(r)} \int_0^{\infty} \frac{t^{n+r-1}}{\lambda^{n+r-1}} e^{-t} \frac{1}{\lambda} dt = \frac{\lambda^r}{\Gamma(r)\lambda^{n+r}} \int_0^{\infty} t^{n+r-1} e^{-t} dt = \frac{\Gamma(n+r)}{\Gamma(r)\lambda^n}.$$

For $n = 1$, this gives

$$E(X) = \frac{\Gamma(r+1)}{\Gamma(r)\lambda} = \frac{r\Gamma(r)}{\lambda\Gamma(r)} = \frac{r}{\lambda}.$$

For $n = 2$,

$$E(X^2) = \frac{\Gamma(r+2)}{\Gamma(r)\lambda^2} = \frac{(r+1)\Gamma(r+1)}{\lambda^2\Gamma(r)} = \frac{(r+1)r\Gamma(r)}{\lambda^2\Gamma(r)} = \frac{r^2 + r}{\lambda^2}.$$

Thus

$$\text{Var}(X) = \frac{r^2 + r}{\lambda^2} - \left(\frac{r}{\lambda}\right)^2 = \frac{r}{\lambda^2}.$$

We have shown that

For a gamma random variable with parameters r and λ ,

$$E(X) = \frac{r}{\lambda}, \quad \text{Var}(X) = \frac{r}{\lambda^2}, \quad \sigma_X = \frac{\sqrt{r}}{\lambda}.$$

Figures 7.13 and 7.14 demonstrate the shape of the gamma density function for several values of r and λ .

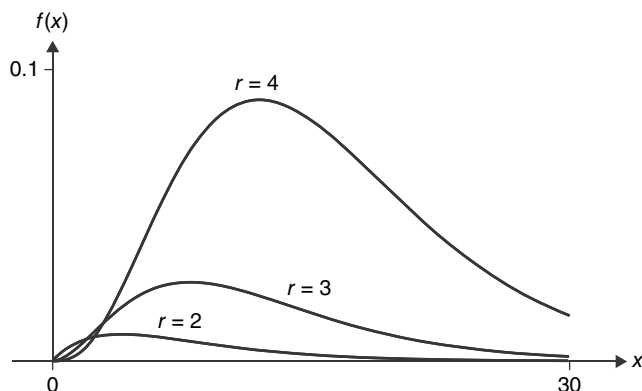


Figure 7.13 Gamma densities for $\lambda = 1/4$.

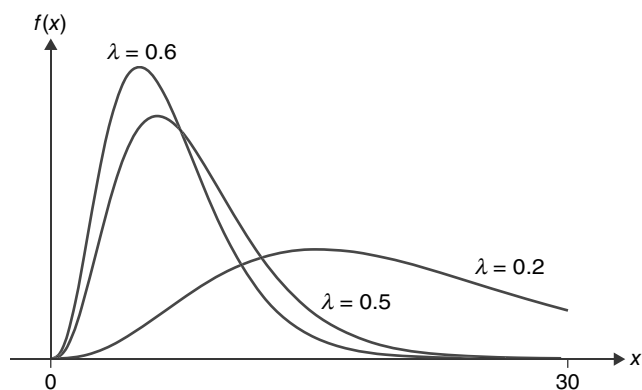


Figure 7.14 Gamma densities for $r = 4$.

Example 7.20 Suppose that, on average, the number of β -particles emitted from a radioactive substance is four every second. What is the probability that it takes at least 2 seconds before the next two β -particles are emitted?

Solution: Let $N(t)$ denote the number of β -particles emitted from a radioactive substance in $[0, t]$. It is reasonable to assume that $\{N(t) : t \geq 0\}$ is a Poisson process. Let 1 second be the

time unit, then $\lambda = E[N(1)] = 4$. X , the time between now and when the second β -particle is emitted, has a gamma distribution with parameters $(2, 4)$. Therefore,

$$\begin{aligned} P(X \geq 2) &= \int_2^\infty \frac{4e^{-4x}(4x)^{2-1}}{\Gamma(2)} dx = \int_2^\infty 16xe^{-4x} dx \\ &= \left[-4xe^{-4x} \right]_2^\infty - \int_2^\infty -4e^{-4x} dx = 8e^{-8} + e^{-8} \approx 0.003. \end{aligned}$$

Note that an alternative solution to this problem is

$$\begin{aligned} P(X \geq 2) &= P(N(2) \leq 1) = P(N(2) = 0) + P(N(2) = 1) \\ &= \frac{e^{-8}(8)^0}{0!} + \frac{e^{-8}(8)^1}{1!} = 9e^{-8} \approx 0.003. \quad \blacklozenge \end{aligned}$$

Example 7.21 There are 100 questions in a test. Suppose that, for all $s > 0$ and $t > 0$, the event that it takes t minutes to answer one question is independent of the event that it takes s minutes to answer another one. If the time that it takes to answer a question is exponential with mean $1/2$, find the distribution, the average time, and the standard deviation of the time it takes to do the entire test.

Solution: Let X be the time to answer a question and $N(t)$ the number of questions answered by time t . Then $\{N(t) : t \geq 0\}$ is a Poisson process at the rate of $\lambda = 1/E(X) = 2$ per minute. Therefore, the time that it takes to complete all the questions is gamma with parameters $(100, 2)$. The average time to finish the test is $r/\lambda = 100/2 = 50$ minutes with standard deviation $\sqrt{r/\lambda^2} = \sqrt{100/4} = 5$. $\blacklozenge$

Relationship between Gamma and Negative Binomial: Recall that if a Bernoulli trial is performed successively and independently, then the number of trials until the r th success occurs is negative binomial. Sometimes gamma is viewed as the continuous analog of negative binomial, one reason being that, for a Poisson process, the time it will take until the r th event occurs is gamma. There are more serious relationships between the two distributions that we will not discuss. For example, in some sense, in limit, negative binomial's distribution approaches gamma distribution. $\blacklozenge$

EXERCISES

A

1. Show that the gamma density function with parameters (r, λ) has a unique maximum at $(r - 1)/\lambda$.
2. Let X be a gamma random variable with parameters (r, λ) . Find the distribution function of cX , where c is a positive constant.

3. In a hospital, babies are born at a Poisson rate of 12 per day. What is the probability that it takes at least seven hours before the next three babies are born?
4. Let X be the time to failure of a mobile robot that is used as an automated guide vehicle. Suppose that $E(X) = 7$ years, and X has a gamma distribution with $r = 4$. Find $\text{Var}(X)$ and $P(5 \leq X \leq 7)$.
5. Let f be the density function of a gamma random variable X , with parameters (r, λ) . Prove that $\int_{-\infty}^{\infty} f(x) dx = 1$.
6. Customers arrive at a restaurant at a Poisson rate of 12 per hour. If the restaurant makes a profit only after 30 customers have arrived, what is the expected length of time until the restaurant starts to make profit?
7. Experience has shown that the bugs of a certain smartphone app cause battery drainage. The time that it will take for the app developers to locate and remove the bugs is a gamma random variable with mean 300 and standard deviation 100 days. What is the probability that it will take less than 250 days for the app developers to get rid of its bugs?
8. Suppose that the lifetime of a certain wheel bearing manufactured by a company is a gamma random variable with mean 75,000 and standard deviation 15,000 miles. If the manufacturer offers a free repair warranty of 60,000 miles on its wheel bearings, what percentage of its wheel bearings must be repaired free of charge?
9. A manufacturer produces light bulbs at a Poisson rate of 200 per hour. The probability that a light bulb is defective is 0.015. During production, the light bulbs are tested one by one, and the defective ones are put in a special can that holds up to a maximum of 25 light bulbs. On average, how long does it take until the can is filled?
10. The lifetime of the alarm system installed in Herman Hall at a private University is exponential with mean $1/\lambda$. Beginning at time T , the facility management crew inspects the system every T units of time. Assuming that without inspection there is no way for the facility management to know whether the system is up or down, find the expected value of the time until the system is *found* to be down.

B

11. For $n = 0, 1, 2, 3, \dots$, calculate $\Gamma(n + 1/2)$.
12. (a) Let Z be a standard normal random variable. Show that the random variable $Y = Z^2$ is gamma and find its parameters.
 (b) Let X be a normal random variable with mean μ and standard deviation σ . Find the distribution function of $W = \left(\frac{X - \mu}{\sigma}\right)^2$.
13. Howard enters a bank that has n tellers. All the tellers are busy serving customers, and there is exactly one queue being served by all tellers, with one customer ahead of Howard waiting to be served. If the service time of a customer is exponential with parameter λ , find the distribution of the waiting time for Howard in the queue.

14. In data communication, messages are usually combinations of characters, and each character consists of a number of bits. A bit is the smallest unit of information and is either 1 or 0. Suppose that the length of a character (in bits) is a geometric random variable with parameter p . Suppose that a sender emits messages at the rate of 1000 bits per second. What is the distribution of T , the time it takes the sender to emit a message combined of k characters of independent lengths?
Hint: Let $N(t)$ be the number of characters emitted at or prior to t . First argue that $\{N(t): t \geq 0\}$ is a Poisson process and find its parameter.
15. Let $\{X_1, X_2, \dots\}$ be a sequence of independent standard normal random variables. Let $S_n = X_1^2 + X_2^2 + \dots + X_n^2$. Find

$$\lim_{n \rightarrow \infty} P(S_n \leq n + \sqrt{2n}).$$



Self-Quiz on Section 7.5

Time allotted: 15 Minutes

Each problem is worth 5 points.

1. A vacuum tube is a device that facilitates the free passage of electric current between electrodes in an evacuated container. The lifetime, in thousands of hours, of a vacuum tube manufactured by a certain company is a random variable with probability density function

$$f(x) = \frac{1}{96} x^3 e^{-x/2}, \quad x \geq 0.$$

Find the mean and the standard deviation of a randomly selected vacuum tube manufactured by this company.

2. A company is behind in manufacturing upper fuser rolling bearings for its photocopiers and needs to produce 500 of them right away. If the company produces these items at a Poisson rate of 5 per hour, what is the expected period and the standard deviation until it manufactures the needed items?



7.6 BETA DISTRIBUTIONS

A random variable X is called **beta** with parameters (α, β) , $\alpha > 0$, $\beta > 0$ if f , its density function, is given by

$$f(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} & \text{if } 0 < x < 1 \\ 0 & \text{otherwise,} \end{cases}$$

where

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx.$$

$B(\alpha, \beta)$ is related to the gamma function by the relation

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

Beta distributions are often appropriate models for random variables that vary between two finite limits—an upper and a lower. For this reason the following are examples of random variables that might be beta.

1. The fraction of people in a community who use a certain product in a given period of time
2. The percentage of total farm acreage that produces healthy watermelons
3. The distance from one end of a tree to that point where it breaks in a severe storm

In these three instances the random variables are restricted between 0 and 1, 0 and 100, and 0 and the length of the tree, respectively.

Beta density also occurs in a natural way in the study of the median of a sample of random points from $(0, 1)$. Let $X_{(1)}$ be the smallest of these numbers, $X_{(2)}$ be the second smallest, $\dots$, $X_{(i)}$ be the i th smallest, $\dots$, and $X_{(n)}$ be the largest of these numbers. If $n = 2k + 1$ is odd, $X_{(k+1)}$ is called the **median** of these n random numbers, whereas if $n = 2k$ is even, $[X_{(k)} + X_{(k+1)}]/2$ is called the **median**. It can be shown that the median of $(2n + 1)$ random numbers from the interval $(0, 1)$ is a *beta* random variable with parameters $(n + 1, n + 1)$. As [Figures 7.15](#) and [7.16](#) show, by changing the values of parameters α and β , beta densities cover a wide range of different shapes. If $\alpha = \beta$, the median is $x = 1/2$, and the density of the beta random variable is symmetric about the median. In particular, if $\alpha = \beta = 1$, the uniform density over the interval $(0, 1)$ is obtained.

To find the expected value and the variance of a beta random variable X , with parameters (α, β) , note that for $n \geq 1$,

$$E(X^n) = \frac{1}{B(\alpha, \beta)} \int_0^1 x^{\alpha+n-1} (1-x)^{\beta-1} dx = \frac{B(\alpha + n, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha + n)\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\alpha + \beta + n)}.$$

Letting $n = 1$ and $n = 2$ in this relation, we find tha

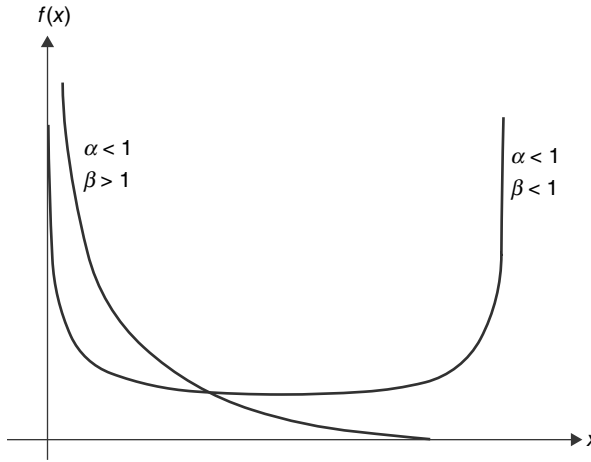


Figure 7.15 Beta densities for $\alpha < 1, \beta < 1$ and $\alpha < 1, \beta > 1$.

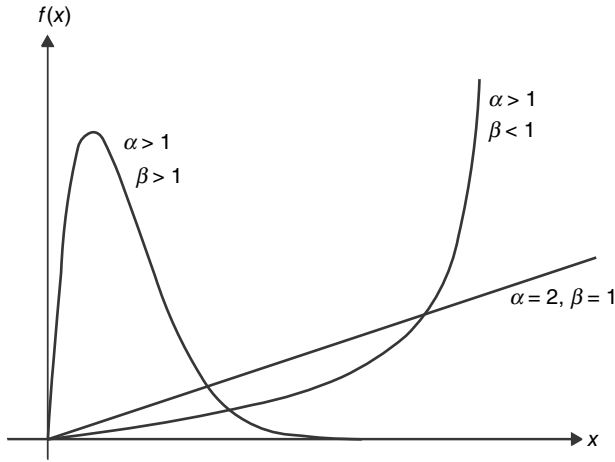


Figure 7.16 Beta densities for the indicated values of α and β .

$$E(X) = \frac{\Gamma(\alpha + 1)\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\alpha + \beta + 1)} = \frac{\alpha\Gamma(\alpha)\Gamma(\alpha + \beta)}{\Gamma(\alpha)(\alpha + \beta)\Gamma(\alpha + \beta)} = \frac{\alpha}{\alpha + \beta},$$

$$E(X^2) = \frac{\Gamma(\alpha + 2)\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\alpha + \beta + 2)} = \frac{(\alpha + 1)\alpha}{(\alpha + \beta + 1)(\alpha + \beta)}.$$

Thus

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2}.$$

We have established the following formulas:

For a beta random variable with parameters (α, β) ,

$$E(X) = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2}.$$

There is also an interesting relation between beta and binomial distributions that the following theorem explains.

Theorem 7.3 *Let α and β be positive integers, X be a beta random variable with parameters α and β , and Y be a binomial random variable with parameters $\alpha + \beta - 1$ and p , $0 < p < 1$. Then*

$$P(X \leq p) = P(Y \geq \alpha).$$

Proof: From the definition,

$$\begin{aligned} P(X \leq p) &= \frac{(\alpha + \beta - 1)!}{(\alpha - 1)! (\beta - 1)!} \int_0^p x^{\alpha-1} (1-x)^{\beta-1} dx \\ &= (\alpha + \beta - 1) \binom{\alpha + \beta - 2}{\alpha - 1} \int_0^p x^{\alpha-1} (1-x)^{\beta-1} dx \\ &= \sum_{i=\alpha}^{\alpha+\beta-1} \binom{\alpha + \beta - 1}{i} p^i (1-p)^{\alpha+\beta-1-i} \\ &= P(Y \geq \alpha), \end{aligned}$$

where the third equality follows from using integration by parts $(\alpha - 1)$ times. $\blacklozenge$

Example 7.22 The proportion of the stocks that will increase in value tomorrow is a beta random variable with parameters α and β . Today, these parameters were determined to be $\alpha = 5$ and $\beta = 4$, by a financial analyst, after she assessed the current political, social, economical, and financial factors. What is the probability that tomorrow the values of at least 70% of the stocks will move up?

Solution: Let X be the proportion of the stocks that will move up in value tomorrow; X is beta with parameters $\alpha = 5$ and $\beta = 4$. Hence the desired probability is

$$P(X \geq 0.70) = \int_{0.70}^1 \frac{1}{B(5, 4)} x^4 (1-x)^3 dx = 280 \int_{0.70}^1 x^4 (1-x)^3 dx = 0.194. \quad \blacklozenge$$

Example 7.23 A beam of length ℓ , rigidly supported at both ends, is hit suddenly at a random point. This leads to a break in the beam at a position X units from the right end. If X/ℓ is beta with parameters $\alpha = \beta = 3$, find $E(X)$, $\text{Var}(X)$, and $P(\ell/5 < X < \ell/4)$.

Solution: Since

$$E(X/\ell) = \frac{\alpha}{\alpha + \beta} = \frac{3}{6} = \frac{1}{2},$$

$$\text{Var}(X/\ell) = \frac{\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2} = \frac{9}{7(6)^2} = \frac{1}{28},$$

we have that $E(X) = \ell/2$, $\text{Var}(X) = \ell^2/28$, and

$$\begin{aligned} P(\ell/5 < X < \ell/4) &= P(1/5 < X/\ell < 1/4) = \int_{1/5}^{1/4} \frac{1}{B(3, 3)} x^2(1-x)^2 dx \\ &= 30 \int_{1/5}^{1/4} x^2(1-x)^2 dx = \frac{72953}{1600000} = 0.046. \quad \blacklozenge \end{aligned}$$

Example 7.24 Suppose that all 25 passengers of a commuter plane, departing at 2:00 P.M., arrive at random times between 12:45 and 1:45 P.M. Find the probability that the median of the arrival times of the passengers is at or prior to 1:12 P.M.

Solution: Let the arrival time of a randomly selected passenger be W hours past 12:45. Then W is a uniform random variable over the interval $(0, 1)$. Let X be the median of the arrival times of the 25 passengers. Since the median of $(2n + 1)$ random numbers from $(0, 1)$ is beta with parameters $(n + 1, n + 1)$, we have that X is beta with parameters $(13, 13)$. Since the length of the time interval from 12:45 to 1:12 is 27 minutes, the desired quantity is $P(X \leq 27/60)$. To calculate this, let Y be a binomial random variable with parameters 25 and $p = 27/60 = 0.45$. Then, by Theorem 7.3,

$$P(X \leq 0.45) = P(Y \geq 13) = \sum_{i=13}^{25} \binom{25}{i} (0.45)^i (0.55)^{25-i} \approx 0.306. \quad \blacklozenge$$

EXERCISES

A

1. Is the following the probability density function of some beta random variable X ? If so, find $E(X)$ and $\text{Var}(X)$.

$$f(x) = \begin{cases} 12x(1-x)^2 & 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

2. Is the following a probability density function? Why or why not?

$$f(x) = \begin{cases} 120x^2(1-x)^4 & 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

3. For what value of c is the following a probability density function of some random variable X ? Find $E(X)$ and $\text{Var}(X)$.

$$f(x) = \begin{cases} cx^4(1-x)^5 & 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

4. Suppose that new blood pressure medicines introduced are effective on $100p\%$ of the patients, where p is a beta random variable with parameters $\alpha = 20$ and $\beta = 13$. What is the probability that a new blood pressure medicine is effective on at least 60% of the hypertensive population?
5. The proportion of resistors a procurement office of an engineering firm orders every month, from a specific vendor, is a beta random variable with mean $1/3$ and variance $1/18$. What is the probability that next month, the procurement office orders at least $7/12$ th of its purchase from this vendor?
6. At a certain university, the fraction of students who get a C in any section of a certain course is uniform over $(0, 1)$. Find the probability that the median of these fractions for the 13 sections of the course that are offered next semester is at least 0.40.
7. Suppose that while daydreaming, the fraction X of the time that one commits brave deeds is beta with parameters $(5, 21)$. What is the probability that next time Jeff is daydreaming, he commits brave deeds at least $1/4$ of the time?
8. A beam of length ℓ is rigidly supported at both ends. Experience shows that whenever the beam is hit at a random point, it breaks at a position X units from the right end, where X/ℓ is a beta random variable. If $E(X) = 3\ell/7$ and $\text{Var}(X) = 3\ell^2/98$, find $P(\ell/7 < X < \ell/3)$.
9. For complicated projects such as construction of spacecrafts, project managers estimate two quantities, a and b , the minimum and maximum lengths of time it will take for a project to be completed, respectively. In estimating b , they consider all of the possible complications that might delay the completion date. Experience shows that the actual length of time it takes for a project to be completed is a random variable defined by

$$Y = a + (b - a)X,$$

where X is beta with parameters α and β ($\alpha > 0$, $\beta > 0$) that can be determined for the project.

(a) Find $E(Y)$ and $\text{Var}(Y)$.

(b) Find the probability density function of Y .

- (c) Suppose that it takes Y years to complete a specific project, where $Y = 2 + 4X$, and X is beta with parameters $\alpha = 2$ and $\beta = 3$. What is the probability that it takes less than 3 years to complete the project?

B

10. Under what conditions and about which point(s) is the probability density function of a beta random variable symmetric?
11. For $\alpha, \beta > 0$, show that

$$B(\alpha, \beta) = 2 \int_0^\infty t^{2\alpha-1} (1+t^2)^{-(\alpha+\beta)} dt.$$

Hint: Make the substitution $x = t^2/(1+t^2)$ in

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx.$$

12. Prove that

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

13. For an integer $n \geq 3$, let X be a random variable with the probability density function

$$f(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}, \quad -\infty < x < \infty.$$

Such random variables have significant applications in statistics. They are called ***t*-distributed with n degrees of freedom**. Using the previous two exercises, find $E(X)$ and $\text{Var}(X)$.

14. For $\alpha > 0$, $\beta > 0$, and $\gamma > 0$, the following function is called the **bivariate Dirichlet probability density function**

$$f(x, y) = \frac{\Gamma(\alpha + \beta + \gamma)}{\Gamma(\alpha)\Gamma(\beta)\Gamma(\gamma)} x^{\alpha-1} y^{\beta-1} (1-x-y)^{\gamma-1}$$

if $x \geq 0$, $y \geq 0$, and $x + y \leq 1$; $f(x, y) = 0$, otherwise. Prove that f_X , the marginal probability density function of X , is beta with parameters $(\alpha, \beta + \gamma)$; and f_Y is beta with the parameters $(\beta, \alpha + \gamma)$.

Hint: Note that

$$\frac{\Gamma(\alpha + \beta + \gamma)}{\Gamma(\alpha)\Gamma(\beta)\Gamma(\gamma)} = \left[\frac{\Gamma(\alpha + \beta + \gamma)}{\Gamma(\alpha)\Gamma(\beta + \gamma)} \right] \left[\frac{\Gamma(\beta + \gamma)}{\Gamma(\beta)\Gamma(\gamma)} \right] = \frac{1}{B(\alpha, \beta + \gamma)} \frac{1}{B(\beta, \gamma)}.$$



Self-Quiz on Section 7.6

Time allotted: 15 Minutes

Each problem is worth 5 points.

1. Suppose that each time the price of a commodity decreases, X , the proportion of change in the price, is a beta random variable with parameters $\alpha = 1$ and $\beta = 10$. That is, if, the price of commodity changes, say, from q_1 to q_2 , $q_2 < q_1$, then $X = (q_1 - q_2)/q_1$ is beta with parameters 1 and 10. Find the probability that next time the price falls, the proportion of change is at least 0.12.
2. Let X be a random number between 0 and 1. Show that $Y = X^2$ is beta with parameters $1/2$ and 1.



★ 7.7 SURVIVAL ANALYSIS AND HAZARD FUNCTIONS

In this section, we will study the risk or rate of failure, per unit of time, of lifetimes that have already survived a certain length of time. The term *lifetime* is broad and applies to appropriate quantities in various models in science, engineering, and business. Depending on the context of a study, by a *lifetime*, we might mean the lifetime of a machine, an electrical component, a living organism, an organism that is under an experimental medical treatment, a financial contract such as a mortgage, the waiting time until a customer arrives at a bank, or the time it takes until a customer is served at a post office. The failure for a living organism is its death. For a mortgage, it is when the mortgage is paid off. For the time until a customer arrives at a bank, failure occurs when a customer enters the bank.

In the following discussion, for convenience, we will talk about the lifetime of a system. However, our definitions and results are general and apply to other examples such as the ones just stated. Let the lifetime of a system be X , where X is a nonnegative, continuous random variable with distribution function F and probability density function f . The function defined by

$$\bar{F}(t) = 1 - F(t) = P(X > t)$$

is called the **survival function** of X . For $t > 0$, $\bar{F}(t)$ is the probability that the system has already survived at least t units of time. Furthermore, the following relation, shown in Remark 5.4, enables us to calculate $E(X)$, the expected value of the lifetime of the system, using $\bar{F}(t)$:

$$E(X) = \int_0^{\infty} \bar{F}(t) dt.$$

The probability that a system, which has survived at least t units of time, fails on the time interval $(t, t + \Delta_t]$ is given by

$$P(X \leq t + \Delta_t \mid X > t) = \frac{P(t < X \leq t + \Delta_t)}{P(X > t)} = \frac{F(t + \Delta_t) - F(t)}{\bar{F}(t)}.$$

To find the instantaneous failure rate of a system of age t at time t , note that during the interval $(t, t + \Delta_t]$, the system fails at a rate of

$$\frac{1}{\Delta_t} P(X \leq t + \Delta_t \mid X > t) = \frac{1}{\bar{F}(t)} \cdot \frac{F(t + \Delta_t) - F(t)}{\Delta_t}$$

per unit of time. As $\Delta_t \rightarrow 0$, this quantity approaches the instantaneous failure rate of the system at time t , given that it has already survived t units of time. Let

$$\begin{aligned} \lambda(t) &= \lim_{\Delta_t \rightarrow 0} \frac{1}{\bar{F}(t)} \cdot \frac{F(t + \Delta_t) - F(t)}{\Delta_t} \\ &= \frac{1}{\bar{F}(t)} \cdot \lim_{\Delta_t \rightarrow 0} \frac{F(t + \Delta_t) - F(t)}{\Delta_t} \\ &= \frac{F'(t)}{\bar{F}(t)} = \frac{f(t)}{\bar{F}(t)}. \end{aligned}$$

Then $\lambda(t)$ is called the **hazard function** of the random variable X . It is the instantaneous failure rate at t , per unit of time, given that the system has already survived until time t . Note that $\lambda(t) \geq 0$, but it is *not* a probability density function.

Remark 7.7 An alternate term used for $\bar{F}(t)$, the survival function of X , is the *reliability function*. Other terms used for $\lambda(t)$, the hazard function, are *hazard rate*, *failure rate function*, *failure rate*, *intensity rate*, and *conditional failure rate*; sometimes actuarial scientists call it *force of mortality*. ♦

We know that if $\Delta_t > 0$ is very small,

$$P(t < X \leq t + \Delta_t) = \int_t^{t+\Delta_t} f(x) dx$$

is the area under f from t to $t + \Delta_t$. This area is almost equal to the area of a rectangle with sides of length Δ_t and $f(t)$. Thus

$$P(t < X \leq t + \Delta_t) \approx f(t)\Delta_t.$$

The smaller Δ_t , the closer $f(t)\Delta_t$ is to the probability that the system fails in $(t, t + \Delta_t]$. This approximation implies that, for infinitesimal $\Delta_t > 0$,

$$\lambda(t)\Delta_t = \frac{f(t)\Delta_t}{\bar{F}(t)} \approx \frac{P(t < X \leq t + \Delta_t)}{P(X > t)} = P(X \leq t + \Delta_t \mid X > t).$$

Therefore,

For very small values of $\Delta_t > 0$, the quantity $\lambda(t)\Delta_t$ is approximately the conditional probability that the system fails in $(t, t + \Delta_t]$, given that it has lasted at least until t . That is,

$$P(X \leq t + \Delta_t \mid X > t) \approx \lambda(t)\Delta_t. \quad (7.5)$$

Example 7.25 Dr. Hirsch has informed one of his employees, Dr. Kizanis, that she will receive her next year's employment contract at a random time between 10:00 A.M. and 3:00 P.M. Suppose that Dr. Kizanis will receive her contract X minutes past 10:00 A.M. Then X is a uniform random variable over the interval $(0, 300)$. Hence the probability density function of X is given by

$$f(t) = \begin{cases} 1/300 & \text{if } 0 < t < 300 \\ 0 & \text{otherwise.} \end{cases}$$

Straightforward calculations show that the survival function of X is given by

$$\bar{F}(t) = P(X > t) = \begin{cases} 1 & \text{if } t < 0 \\ \frac{300 - t}{300} & \text{if } 0 \leq t < 300 \\ 0 & \text{if } t \geq 300. \end{cases}$$

The hazard function, $\lambda(t) = f(t)/\bar{F}(t)$, is defined only for $t < 300$. It is unbounded for $t \geq 300$. We have

$$\lambda(t) = \begin{cases} 0 & \text{if } t < 0 \\ \frac{1}{300 - t} & \text{if } 0 \leq t < 300. \end{cases}$$

Since $\lambda(0) = 0.003333$, at 10:00 A.M. the instantaneous arrival rate of the contract (the failure rate in this context) is 0.003333 per minute. At noon this rate will increase to $\lambda(120) = 0.0056$, at 2:00 P.M. to $\lambda(240) = 0.017$, at 2:59 P.M. it reaches $\lambda(299) = 1$. One second before 3:00 P.M., the instantaneous arrival of the contract is, approximately, $\lambda(299.983) = 58.82$. This shows that if Dr. Kizanis has not received her contract by one second before 3:00 P.M., the instantaneous arrival rate at that time is very high. That rate approaches ∞ , as the time approaches 3:00 P.M. To translate all these into probabilities, let $\Delta_t = 1/60$. Then $f(t)\Delta_t$ is approximately the probability that the contract will arrive within one second after t , whereas $\lambda(t)\Delta_t$ is approximately the probability that the contract will arrive within one second after t , given that it has not yet arrived by time t . The following table shows these probabilities at the indicated times:

t	0	120	240	299	299.983
$f(t)\Delta_t$	0.000056	0.000056	0.000056	0.000056	0.000056
$\lambda(t)\Delta_t$	0.000056	0.000093	0.000283	0.0167	0.98

The fact that $f(t)\Delta_t$ is constant is expected because X is uniformly distributed over $(0, 300)$, and all of the intervals under consideration are subintervals of $(0, 300)$ of equal lengths. ♦

Let X be a nonnegative continuous random variable with distribution function F , probability density function f , survival function $\bar{F}$, and hazard function $\lambda(t)$. We will now calculate $\bar{F}$ and f in terms of $\lambda(t)$. The formulas obtained are very useful in various applications. Let $G(t) = -\ln [\bar{F}(t)]$. Then

$$G'(t) = \frac{f(t)}{\bar{F}(t)} = \lambda(t).$$

Consequently,

$$\int_0^t G'(u) du = \int_0^t \lambda(u) du.$$

Hence

$$G(t) - G(0) = \int_0^t \lambda(u) du.$$

Since X is nonnegative, $G(0) = -\ln[1 - F(0)] = -\ln 1 = 0$. So

$$-\ln[\bar{F}(t)] - 0 = G(t) - G(0) = \int_0^t \lambda(u) du$$

implies that

$$\bar{F}(t) = \exp\left(-\int_0^t \lambda(u) du\right). \quad (7.6)$$

By this equation,

$$F(t) = 1 - \exp\left[-\int_0^t \lambda(u) du\right].$$

Differentiating both sides of this relation, with respect to t , yields

$$f(t) = \lambda(t) \exp\left[-\int_0^t \lambda(u) du\right]. \quad (7.7)$$

This demonstrates that the hazard function *uniquely* determines the probability density function.

In reliability theory, a branch of engineering, it is often observed that $\lambda(t)$, the hazard function of the lifetime of a manufactured machine, is initially large due to undetected defective components during testing. Later, $\lambda(t)$ will decrease and remains more or less the same until a time when it increases again due to aging, which makes worn-out components more likely to fail. A random variable X is said to have **an increasing failure rate** if $\lambda(t)$ is increasing. It is said to have **a decreasing failure rate** if $\lambda(t)$ is decreasing. In the next example, we will show that if $\lambda(t)$ is neither increasing nor decreasing; that is, if $\lambda(t)$ is a constant, then X is an exponential random variable. In such a case, the fact that aging does not change the failure rate is consistent with the memoryless property of exponential random variables. To summarize, the lifetime of a manufactured machine is likely to have a decreasing failure rate in the beginning, a constant failure rate later on, and an increasing failure rate due to wearing out after an aging process. Similarly, lifetimes of living organisms, after a certain age, have increasing failure rates. However, for a newborn baby, the longer he or she survives, the chances of surviving is higher. This means that as t increases, $\lambda(t)$ decreases. That is, $\lambda(t)$ is decreasing.

Example 7.26 In this example, we will prove the following important theorem:

Let $\lambda(t)$, the hazard function of a continuous, nonnegative random variable X , be a constant λ . Then X is an exponential random variable with parameter λ .

To show this theorem, let f be the probability density function of X . By (7.7),

$$f(t) = \lambda \exp\left(-\int_0^t \lambda \, du\right) = \lambda e^{-\lambda t}.$$

This is the density function of an exponential random variable with parameter λ . Thus a random variable with constant hazard function is exponential. As mentioned previously, this result is not surprising. The fact that an exponential random variable is memoryless implies that age has no effect on the distribution of the remaining lifetimes of exponentially distributed random variables. For systems that have exponential lifetime distribution, failures are not due to aging and wearing out. In fact, such systems do not wear out at all. Failures occur abruptly. There is no transition from the previous “state” of the system and no preparation for, or gradual approach to, failure. ♦

EXERCISES

1. Suppose that the lifetime of a device, in hundreds of hours, is a random variable with probability density function

$$f(t) = \begin{cases} 3t^2 & 0 \leq t \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find $\lambda(t)$, the hazard function of the lifetime of the device.
 - (b) Find the instantaneous failure rates of such a device if it has survived 50 hours, 90 hours, and 99 hours, respectively. Using these, calculate the approximate probabilities that the device fails within one hour if it has survived 50 hours, 90 hours, and 99 hours, respectively.
2. The hazard function of a random variable X with the set of possible values $[1, 3)$, is given by

$$\lambda(t) = \frac{3t^2}{27 - t^3}, \quad 1 \leq t < 3.$$

Find the probability density function of X .

3. Suppose that the lifetime of a cell phone chip manufactured by a wireless company, in years, is gamma with parameters $r = 4$ and $\lambda = 1$. Find $\lambda(t)$, the hazard function of such a chip, and its instantaneous failure rate if it has survived for 4 years. Using this, find an approximate value for the probability that an exactly 4-year-old chip fails within one week. Assume that a year is 52 weeks.
4. Experience shows that the failure rate of a certain electrical component is a linear function. Suppose that after two full days of operation, the failure rate is 10% per hour and after three full days of operation, it is 15% per hour.

- (a) Find the probability that the component operates for at least 30 hours.
 - (b) Suppose that the component has been operating for 30 hours. What is the probability that it fails within the next hour?
5. One of the most popular distributions used to model the lifetimes of electric components is the **Weibull distribution**, whose probability density function is given by

$$f(t) = \alpha t^{\alpha-1} e^{-t^\alpha}, \quad t > 0, \alpha > 0.$$

Determine for which values of α the hazard function of a Weibull random variable is increasing, for which values it is decreasing, and for which values it is constant. ■

Self-Quiz on Section 7.7

Time allotted: 15 Minutes

Each problem is worth 5 points.

1. A batch of cell phones manufactured by a wireless company's plant in the U.S.A. has been ordered by the company's main facility in Portugal. Suppose that the hazard function of the delivery lead time, in weeks, is $\lambda(t) = 1/8, t \geq 0$. If the order for the batch of cell phones was placed four weeks ago, what is the probability that it will arrive in less than two weeks?
2. The hazard function of a random variable X with the set of possible values $(0, 1)$, is given by

$$\lambda(t) = \begin{cases} \frac{2}{1-t} & 0 < t < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability density function of X . ■

★ 7.8 SIMULATING SPECIAL CONTINUOUS DISTRIBUTIONS USING R

In this section, we will explore the simulation of the continuous random variables that we have studied in this chapter.

Uniform Random Variables

Consider a probability density function f representing a uniform random variable, and let F denote its corresponding probability distribution function. For a point x within the interval (a, b) , we can calculate $f(x)$ using the command **dunif(x, a, b)** and $F(x)$ using

punif(x, a, b). To generate a sample of n realizations from this distribution, we can employ the command **runif(n, a, b)**. Consequently, **runif(n, a, b)** generates a collection of n uniformly distributed points chosen from the interval (a, b) .

In the case of the uniform distribution, the default behavior assumes uniformity over the interval $(0,1)$. Therefore, for instance, **dunif(0.7)** is equivalent to **dunif(0.7, 0, 1)**, and **punif(0.4)** is equivalent to **punif(0.4, 0, 1)**. Furthermore, the code **runif(100)** generates a sample of 100 randomly selected numbers from the interval $(0,1)$.

Normal Random Variables

Consider a probability density function denoted as f , representing a normal random variable with a mean of μ and a standard deviation of σ . This particular normal random variable is typically denoted as $N(\mu, \sigma^2)$. For a given point x within the range $(-\infty, \infty)$, we can calculate $f(x)$ using the command **dnorm(x, mu, sigma)**, and $F(x)$ can be computed using **pnorm(x, mu, sigma)**. To generate a sample of n realizations from this distribution, we can utilize the command **rnorm(n, mu, sigma)**. Consequently, **rnorm(n, mu, sigma)** generates a sample of size n from a normal random variable with a mean of μ and a standard deviation of σ . It is important to note that although we use σ^2 , the variance, in the notation $N(\mu, \sigma^2)$, in the R commands, we employ σ (**sigma**), the standard deviation.

For the normal distribution, the default assumption is a standard normal distribution, denoted as $N(0, 1)$. Therefore, for example, **dnorm(1.3)** is the same as **dnorm(1.3, 0, 1)**, and **pnorm(-1.5)** is equivalent to **pnorm(-1.5, 0, 1)**. Moreover, the code **rnorm(55)** generates a sample of 55 randomly chosen numbers from the standard normal distribution $N(0, 1)$. To generate, say, 55 randomly chosen numbers from a normal random variable with mean μ and standard deviation σ , in addition to **rnorm(55, mu, sigma)**, we have the option to use the code **mu+sigma*rnorm(55)**.

R generates the curve of the probability density function for a standard normal random variable using the following code (refer to [Figure 7.17](#)):

```
> curve(dnorm, -3.8, 3.8)
> abline(h=0)
> abline(v=0)
> par(pin=c(4,2))
```

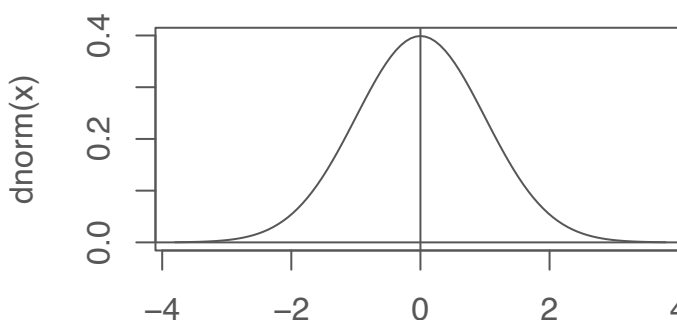


Figure 7.17 An R plot of the standard normal density function.

Similarly, by utilizing the following code, R plots the curve of the probability distribution function for a standard normal random variable (refer to [Figure 7.18](#)):

```
> curve(pnorm, -3, 5)
> abline(h=0)
> abline(v=0)
> abline(h=1)
```

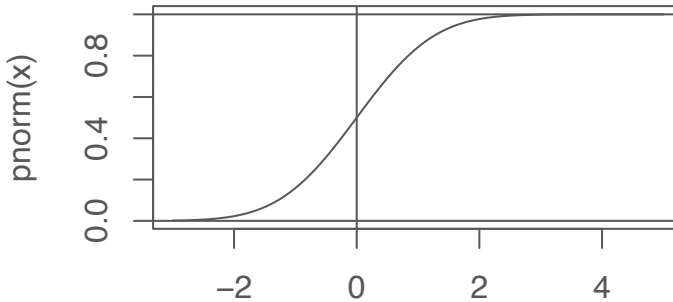


Figure 7.18 An R plot of the standard normal distribution function.

Exponential Random Variables

For an exponential random variable with parameter λ , probability density function f , and probability distribution function F , we can calculate $f(x)$ using the command `dexp(x, lambda)`, and $F(x)$ using `pexp(x, lambda)`. To generate a sample of n realizations from this distribution, we can utilize the command `rexp(n, lambda)`.

The provided code below utilizes R simulation to compute the expected value (mean), variance, and standard deviation of an exponential random variable. This is achieved using the commands `mean()`, `var()`, `sd()`, and the function `rexp()` for generating random samples. Here is how it is done:

```
> lambda <- 0.5 > v <- rexp(2000, lambda)
> ev <- mean(v)
> va <- var(v)
> st <- sd(v)
> print(ev)
[1] 1.995176
> print(va)
[1] 4.14277
> print(st)
[1] 2.03538
```


Simulation of Poisson Process

Consider $N(t)$: $t \geq 0$ as a Poisson process, where $N(t)$ represents the count of “events” or “arrivals” that have occurred up to and including time t . An event’s occurrence time is referred to as an “arrival time,” and the duration between two consecutive events is known as an “interarrival time.” It is important to note that in a Poisson process, the sequence of interarrival times follows independent, identical exponential distributions with a parameter of λ . Consequently, the expected value of the interarrival time between consecutive arrivals is $1/\lambda$.

For all $t > 0$, $N(t)$ itself follows a Poisson distribution with a parameter of λt . To accurately simulate a Poisson process, it is essential to simulate the arrival times. This process can be illustrated through a simple example. Consider $\lambda = 8$, with the assumption that the time unit is a minute. This means, on average, arrivals occur every (1/8)th of a minute. Suppose we aim to simulate the arrival times of the first 30 events in this Poisson process. We begin by simulating the interarrival times for all these events. The code `v <- rexp(30, 8)` generates a vector `v` containing 30 independent, identically distributed realizations of an exponential random variables with a parameter of 8. Specifically, `v[1]`, `v[2]`, ..., `v[30]` represent these interarrival times. Next, we calculate the arrival times by taking the cumulative sum of `v`. Using `t <- cumsum(v)`, we obtain a vector `t` such that, for $1 \leq i \leq 30$, `t[i] = v[1] + v[2] + ... + v[i]` represents the simulated arrival time of the i th arrival.

Simulation of Gamma and Beta Random Variables

For a gamma random variable characterized by parameters r and λ , we can calculate its probability density function $f(x)$ using the command `dgamma(x, r, lambda)`. Additionally, we can compute its cumulative distribution function $F(x)$ using `pgamma(x, r, lambda)`. To generate a sample of n realizations from this distribution, we employ the command `rgamma(n, r, lambda)`.

For a beta random variable with parameter α and β , probability density function f , and probability distribution function F , we can calculate $f(x)$ using the command `dbeta(x, alpha, beta)`, and $F(x)$ using `pbeta(x, alpha, beta)`. To generate a sample of n realizations from this distribution, we utilize the command `rbeta(n, alpha, beta)`.

Example 7.27 The grade distributions for students in five different courses, namely calculus, statistics, music, computer programming, and physical education, are modeled by normal distributions with the following parameters: $N(70, 400)$, $N(75, 225)$, $N(80, 400)$, $N(75, 400)$, and $N(85, 100)$, respectively. Craft an R simulation to approximate the probability that the median of the grades for a randomly selected student taking these five courses is at least 75.

Solution: Let `num` represent the number of simulations, `m` denote the count of simulations in which the median grade is at least 75, and `v` be a vector consisting of realizations from the five specified normal random variables. Each `v` represents a simulation of a set of possible grades for the students. By setting `w` equal to `sort(v)`, `w[3]` will correspond to the median grade. Below is an R code for conducting this simulation:

```
> lambda <- 0.5 > num <- 70000
> m <- 0
```

```

> i <- 1
> while (i <= num) {
+ v <- c(rnorm(1, 70, 20), rnorm(1, 75, 15), rnorm(1, 80, 20),
+       rnorm(1, 75, 20), rnorm(1, 85, 10))
+ w <- sort(v)
+ if (w[3] >= 75) {
+ m <- m + 1
+ }
+ i <- i + 1
+ }
> p <- m / num
> print(p)
[1] 0.6280286  ♦

```

Example 7.28 Passengers arrive at a train station following a Poisson process with a parameter of λ . If the arrival time of the next train is uniformly distributed within the interval $(0, T)$, create an R simulation to estimate the probability that there will be at least n passengers at the station by the time the next train arrives.

Solution: We will conduct the following procedure **num** times to estimate the desired probability:

- Simulate the arrival time $\mathbf{t}$ of the train, which follows a uniform random distribution between 0 and T .
- Generate a Poisson random variable $\mathbf{x}$ with parameter λt . This represents the number of passengers who arrive during the time interval $(0, \mathbf{t})$.
- Examine whether $\mathbf{x}$ is greater than or equal to $\mathbf{n}$. If it is, we increment $\mathbf{m}$, which keeps count of the times $\mathbf{x}$ is greater than or equal to $\mathbf{n}$, by 1. Otherwise, we repeat the simulation until all **num** simulations are completed.
- The counter $\mathbf{i}$ counts the number of simulations performed.

This process is repeated **num** times to estimate the desired probability, $\mathbf{p}=\mathbf{m}/\mathbf{num}$.

```

> num <- 80000
> n <- 70
> T <- 30
> lambda <- 5
> i <- 1
> m <- 0
> while(i<=num) {

```

```

+ t <- runif(1,0,T)
+ x <- rpois(1,lambda*t)
+ if(x>=n){
+ m <- m+1
+ }
+ i <- i+1
+ }
> p <- m/num
> print(m)
[1] 42528
> print(p)
[1] 0.5316      ♦

```

Example 7.29 Consider two towns, town A and town B. The time between any two earthquakes in town A follows an exponential distribution with a mean of $1/\lambda_1$, and in town B, it follows an exponential distribution with a mean of $1/\lambda_2$. Write an R simulation code to approximate the probability that the next earthquake in town B occurs before the next earthquake in town A.

Solution: To approximate the probability that the next earthquake in town B occurs prior to the next earthquake in town A, we will simulate x and y , the arrival times of earthquakes in towns A and B, respectively, using exponential distributions and compare the times. Here is an R simulation code for this scenario:

```

> num <- 65000
> lambda1 <- 1/5
> lambda2 <- 1/7
> m <- 0
> i <- 1
> while(i<=num){
+ x <- rexp(1,lambda1)
+ y <- rexp(1,lambda2)
+ if(x>y){
+ m <- m+1
+ }
+ i <- i+1
+ }
> p <- m/num
> print(p)
[1] 0.4216308

```

In this code, **lambda1** and **lambda2** represent the mean time between earthquakes in town A and town B, respectively. **num** determines the number of simulations to run. The code simulates the arrival times of earthquakes for both towns, checks whether or not town B experiences an earthquake first, and counts the successful simulations where town B experiences an earthquake before town A. Finally, it calculates and prints the estimated probability. The exact answer is $5/12 \approx 0.4166667$, demonstrating that the result of this simulation is highly accurate. ♦

CLT for Exponential Distribution

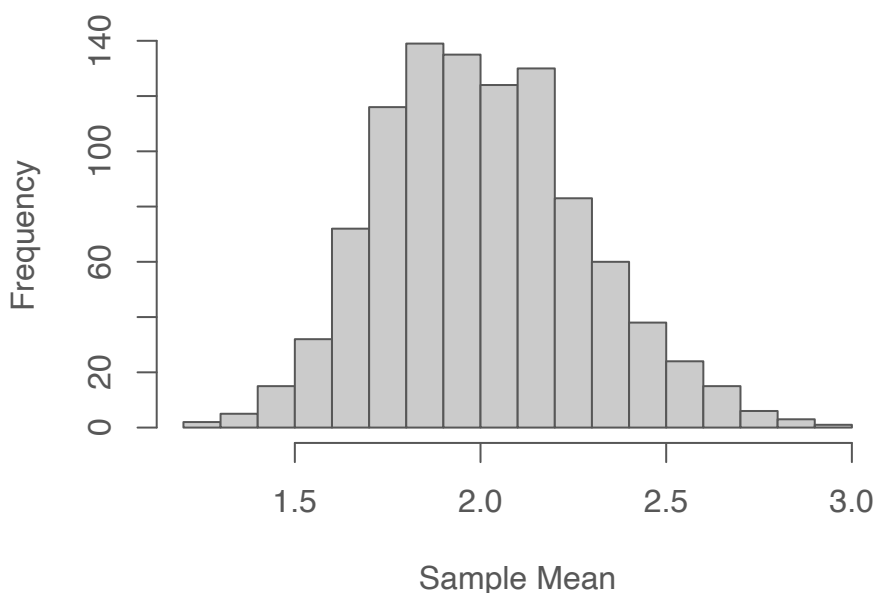


Figure 7.19 Histogram of Example 7.30.

Example 7.30 (R Simulation of the Central Limit Theorem) In this example, we will simulate the central limit theorem for the exponential distribution with parameter $\lambda = 0.5$. Each sample represents a set of random numbers drawn from this distribution. As the sample size increases and we compute sample means, we will observe that the distribution of sample means becomes approximately normal, in line with the central limit theorem.

Let **n** be the size of each sample, and **num** be the number of simulations to run (the number of samples to generate). The command **sm <- numeric(num)** allows us to initialize an empty vector **sm** of length **num**, which can later be filled with numeric values. Initially, all of its **num** entries are set to 0. **sm** is a vector for which **sm[i]** is the mean of the sample generated in the *i*th simulation. The histogram of the data in vector **sm** visually suggests a distribution that resembles a normal distribution, as expected from the central limit theorem.

The code below takes 1000 samples, each of size 50, from an exponential random variable with a parameter of 0.5. Subsequently, it computes the sample means for each of these 1000 simulations and generates a histogram based on these sample means. The resulting histogram is presented in [Figure 7.19](#).

```

> lambda <- 0.5
> n <- 50
> num <- 1000
> sm <- numeric(num)
> for(i in 1:num){
+ v <- rexp(n,lambda)
+ sm[i] <- mean(v)
+ }
> hist(sm,breaks=20,main ="CLT for Exponential Distribution",
+ xlab="Sample Mean", ylab="Frequency")

```

EXERCISES

In this chapter, there are a diverse set of exercises in nearly every section, offering valuable opportunities for simulation practice using R. Professors have the flexibility to assign specific exercises to students, allowing them to delve into R simulations. Similarly, students can autonomously choose exercises to enhance their skills in simulation techniques using R.

CHAPTER 7 SUMMARY

► **Uniform Random Variables** Suppose that X is the value of the random point selected from an interval (a, b) . Then X is called a *uniform random variable* over (a, b) . f , the probability density functions of X , $E(X)$, $\text{Var}(X)$, and σ_X are given by $f(t) = 1/(b - a)$ if $a < t < b$ and $f(t) = 0$, otherwise. $E(X) = (a + b)/2$, $\text{Var}(X) = (b - a)^2/12$, and $\sigma_X = (b - a)/\sqrt{12}$.

► **De Moivre-Laplace Theorem** Let X be a binomial random variable with parameters n and p . Then for any numbers a and b , $a < b$,

$$\lim_{n \rightarrow \infty} P\left(a < \frac{X - np}{\sqrt{np(1-p)}} < b\right) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2} dt.$$

By this theorem, if X is a binomial random variable with parameters (n, p) , the sequence of probabilities

$$P\left(\frac{X - np}{\sqrt{np(1-p)}} \leq t\right), \quad n = 1, 2, 3, \dots,$$

converges to $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx$, where the function $\Phi(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx$ is a distribution function itself.

► **Standard Normal Random Variables** A random variable X is called *standard normal* if its distribution function is Φ . The density function of a standard normal random variable, is given by

$$f(x) = \Phi'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

The standard normal density function is a bell-shaped curve that is symmetric about the y -axis, and $E(X) = 0$, $\text{Var}(X) = \sigma_X = 1$.

► $\Phi(t)$, the distribution function of the standard normal random variable, is the area under the curve of f from $-\infty$ to t . Because $\Phi(\infty) = 1$ and the curve is symmetric about the y -axis, $\Phi(0) = 1/2$, and $\Phi(-t) = 1 - \Phi(t)$.

► **Normal Random Variables** A random variable X is called *normal*, with parameters μ and σ , if its density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right], \quad -\infty < x < \infty.$$

The parameters μ and σ that appear in the formula of the density function of X are its expected value and standard deviation, respectively.

► If $X \sim N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma$ is $N(0, 1)$. That is, if $X \sim N(\mu, \sigma^2)$, the standardized X is $N(0, 1)$.

► **Central Limit Theorem** Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed random variables, each with expectation μ and variance σ^2 . Then

$$\lim_{n \rightarrow \infty} P\left(\frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy.$$

► Let $\bar{X}_n$ be the mean of the random variables $X_1, X_2, \dots, X_n$. The central limit theorem is equivalent to

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy.$$

► **Exponential Random Variables** A continuous random variable X is called *exponential* with parameter $\lambda > 0$ if its density function is given by $f(t) = \lambda e^{-\lambda t}$, $t \geq 0$, and $f(t) = 0$ if $t < 0$. For an exponential random variable with parameter λ , $E(X) = \sigma_X = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$.

► In a Poisson process, the time between two consecutive events is an exponential random variable.

► An important feature of exponential distribution is its *memoryless property*. A nonnegative random variable X is called *memoryless* if, for all $s, t \geq 0$,

$$P(X > s + t \mid X > t) = P(X > s).$$

This shows that if, for example, X is the lifetime of some type of instrument, then there is no deterioration with age of the instrument.

► **Gamma Distributions** We call a random variable X with probability density function

$$f(x) = \begin{cases} \frac{\lambda e^{-\lambda x} (\lambda x)^{r-1}}{\Gamma(r)} & \text{if } x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

a *gamma* distribution with parameters (r, λ) , $\lambda > 0$, $r > 0$, where, $\Gamma(r) = \int_0^\infty t^{r-1} e^{-t} dt$. For a gamma random variable with parameters r and λ , $E(X) = r/\lambda$, $\text{Var}(X) = r/\lambda^2$, and $\sigma_X = \sqrt{r}/\lambda$.

► For a Poisson process with rate λ , the time of the n th event is gamma with parameters (n, λ) .

► **Beta Distributions** A random variable X is called *beta* with parameters (α, β) , $\alpha > 0$, $\beta > 0$ if f , its probability density function, is given by

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1,$$

and $f(x) = 0$, otherwise, where $B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$. $B(\alpha, \beta)$ is related to the gamma function by the relation $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha + \beta)$. For a beta random variable with parameters (α, β) ,

$$E(X) = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2}.$$

► Let α and β be positive integers, X be a beta random variable with parameters α and β , and Y be a binomial random variable with parameters $\alpha + \beta - 1$ and p , $0 < p < 1$. Then

$$P(X \leq p) = P(Y \geq \alpha).$$

► **Survival Analysis and Hazard Functions** Let X be a continuous random variable with probability density function f and distribution function F . The function defined by $\bar{F}(t) = 1 - F(t) = P(X > t)$ is called the *survival function* of X . Let $\lambda(t) = f(t)/\bar{F}(t)$. Then $\lambda(t)$ is called the *hazard function* of the random variable X . If X is the lifetime of a “system,” then $\lambda(t)$ is the instantaneous failure rate at t , per unit of time, given that the system has already survived until time t . For very small values of $\Delta_t > 0$, the quantity $\lambda(t)\Delta_t$ is approximately the conditional probability that the system fails in $(t, t + \Delta_t)$, given that it has lasted at least until t . That is, $P(X \leq t + \Delta_t \mid X > t) \approx \lambda(t)\Delta_t$.

► Let X be a nonnegative continuous random variable with hazard function $\lambda(t)$. Then the hazard function *uniquely* determines the probability density function:

$$f(t) = \lambda(t) \exp \left[- \int_0^t \lambda(u) du \right].$$

In particular, by this formula, if $\lambda(t)$, the hazard function of a continuous, nonnegative random variable X , be a constant λ . Then X is an exponential random variable with parameter λ .

REVIEW PROBLEMS

1. For a restaurant, the time it takes to deliver pizza (in minutes) is uniform over the interval (25, 37). Determine the proportion of deliveries that are made in less than half an hour.
2. It is known that the weight of a random woman from a community is normal with mean 130 pounds and standard deviation 20. Of the women in that community who weigh above 140 pounds, what percent weigh over 170 pounds?
3. One thousand random digits are generated. What is the probability that digit 5 is generated at most 93 times?
4. Let X , the lifetime of a light bulb, be an exponential random variable with parameter λ . Is it possible that X satisfies the following relation?

$$P(X \leq 2) = 2P(2 < X \leq 3).$$

If so, for what value of λ ?

5. Yearly salaries paid to the salespeople employed by a certain company are normally distributed with mean \$27,000 and standard deviation \$4900. What is the probability that the average wage of a random sample of 10 employees of this company is at least \$30,000?
6. The time that it takes for a computer system to fail is exponential with mean 1700 hours. If a lab has 20 such computer systems, what is the probability that at least two fail before 1700 hours of use?
7. Let X be a uniform random variable over the interval $(0, 1)$. Calculate $E(-\ln X)$.
8. Suppose that the diameter of a randomly selected disk produced by a certain manufacturer is normal with mean 4 inches and standard deviation 1 inch. Find the distribution function of the diameter of a randomly chosen disk, in centimeters.
9. Let X be an exponential random variable with parameter λ . Prove that

$$P(\alpha \leq X \leq \alpha + \beta) \leq P(0 \leq X \leq \beta).$$

10. The time that it takes for a calculus student to answer all the questions on a certain exam is an exponential random variable with mean 1 hour and 15 minutes. If all 10 students of a calculus class are taking that exam, what is the probability that at least one of them completes it in less than one hour?
11. Determine the value(s) of k for which the following is a density function.

$$f(x) = ke^{-x^2+3x+2}, \quad -\infty < x < \infty.$$

12. In a certain town the length of residence of a family in a home is normal with mean 80 months and variance 900. What is the probability that of 12 independent families, living on a certain street of that town, at least three will have lived there more than 8 years?
13. A randomly selected book from Vernon's library is X centimeters thick, where $X \sim N(3, 1)$. Vernon has an empty shelf 87 centimeters long. What is the probability that he can fit 31 randomly selected books on it?
14. A fair die is rolled 20 times. What is the approximate probability that the sum of the outcomes is between 65 and 75?
15. The number of minutes that a train from Milan to Rome is late is an exponential random variable X with parameter λ . Find $P(X > E(X))$.
16. In a mayoral election in a small island in Greece, of 5000 voters only 1000 voted for Tasoula. What is the probability that, in a random sample of 100 voters, more than 15 voted for this mayoral candidate?
17. Suppose that the weights of passengers taking an elevator in a certain building are normal with mean 175 pounds and standard deviation 22. What is the minimum weight for a passenger who outweighs at least 90% of the other passengers?
18. The breaking strength of a certain type of yarn produced by a certain vendor is normal with mean 95 and standard deviation 11. What is the probability that, in a random sample of size 10 from the stock of this vendor, the breaking strengths of at least two are over 100?
19. The number of phone calls to a specific exchange is a Poisson process with rate 23 per hour. Calculate the probability that the time until the 91st call is at least 4 hours.
20. Let X be a uniform random variable over the interval $(1 - \theta, 1 + \theta)$, where $0 < \theta < 1$ is a given parameter. Find a function of X , say $g(X)$, so that $E[g(X)] = \theta^2$.
21. A system consists of n components whose lifetimes form an independent sequence of random variables. Suppose that the system functions as long as at least one of its components functions. Let $F_1, F_2, \dots, F_n$ be the distribution functions of the lifetimes of the components of the system. In terms of $F_1, F_2, \dots, F_n$, find the survival function of the lifetime of the system.
22. A black and white (monochrome) CRT monitor uses a single electron gun to create images. A color monitor uses three electron guns (red, green, blue) to create the same quality images, but in color. Dale-Marie buys a monochrome CRT monitor, and Dan

buys a color monitor both manufactured by the same company with the same lifetime distribution for the electron guns. Suppose that each electron gun lasts, independently of other electron guns, for an exponential period with parameter λ . Naturally, Dale-Marie will discard her monitor when it does not produce images; that is, when the electron gun of the monitor dies. Dan discards his monitor when it malfunctions; that is, when for the first time one of the electron guns of his monitor dies. Suppose that Dale-Marie's monitor works for X units of time, and Dan's monitor will be discarded after working for Y units of time. Find $P(X < Y)$, $E[\max(X, Y)]$, and $E[\min(X, Y)]$.

Hint: Let Y_1, Y_2 , and Y_3 be the lifetimes of the electron guns of Dan's monitor. Note that $Y = \min(Y_1, Y_2, Y_3)$.



Self-Test on Chapter 7

Time allotted: 120 Minutes

Each problem is worth 10 points.

1. In a city, a passenger transit bus operates two routes that both stop at the Downtown Center and at the Embarcadero stations. Route B1 operates everyday between 6:00 A.M. and 9:30 P.M. and arrives at Embarcadero every 15 minutes starting at 6:15 A.M. Route B26 operates during the same period but arrives at Embarcadero every 20 minutes starting at 6:20 A.M. To go to the Downtown Center station, Nolan arrives at the Embarcadero station at a random time between 1:05 P.M. and 1:42 P.M., and he takes either B1 or B26, whichever arrives first. What is the probability that he ends up taking bus route B26?
2. While in flight, the fraction X of the time that Nolan sleeps is beta with parameters $(6, 10)$. What is the probability that next time Nolan flies from Chicago to Shanghai, he sleeps longer than 6 hours during the 14-hour flight?
3. In a huge office building, the alarm system has n sensors, the lifetime of each being exponential with mean $1/\lambda$, independently of the other ones. If for the next t units of time the alarm system is not checked, what is the probability that at time t only k sensors are working? Assume that without inspection there is no way for the technical crew to know that a sensor is down.
4. Let $\bar{X}$ denote the mean of a random sample of size 28 from a distribution with $\mu = 1$ and $\sigma^2 = 4$. Approximate $P(0.95 < \bar{X} < 1.05)$.
5. Suppose that buses arrive at a station every 15 minutes. During the operating hours of the buses, Kayla arrives at the station at a random time. (a) If she has been waiting for the bus for 9 minutes, what is the probability that she has to wait at least another 4 minutes? (b) Calculate the same probability if the time between consecutive arrival times of two buses is exponential with mean 15 minutes.

6. Suppose that, everyday, all over the world, 10 million people search the Web for film review websites. Suppose that, among many other sites, the search always turns up a specific website that aggregates reviews of movies and TV shows, independently of all the other sites. If 0.2% of such people visit this site, find the probability that, on a given day, at least 20,200 people visit this website.
7. Let X be the weight of a randomly selected adult from a certain community, and assume that X is a normal random variable. If it is known that, in that community, the mean of the weights of the adults is 160 pounds, and 85% of the adult population has a weight between 120 and 200 pounds, find $\text{Var}(X)$.
8. To improve the reliability of a system, sometimes manufacturers add one or more operating components in parallel so that if one of the components fails, the system still continues to function. Therefore, a parallel system functions if and only if at least one of its components functions. Suppose that the probability density function of X , the operating period, in tens of thousands of hours, of a parallel system is given by

$$f(x) = 2e^{-2x} + 4e^{-4x} - 6e^{-6x}, \quad x \geq 0.$$

- (a) Find $\lambda(t)$, the hazard function of an operating period of this system.
- (b) Find the instantaneous failure rate of such a parallel system if it has survived 6,000 hours. Using this, calculate an approximate probability for the system to fail within 100 hours if it has survived 6,000 hours.
9. A psychologist wants to estimate μ , the mean IQ of the students of a university. To do so, she takes a sample of size n of the students and measures their IQ's. Then she finds the average of these numbers. If she believes that the IQ's of these students are independent random variables with variance 170, using the central limit theorem, how large a sample should she choose to be 98% sure that her average is accurate within ± 0.2 ?
10. A wireless phone company manufactures cell phone chips for 40% of the cell phones it produces and purchases chips for the remaining 60% of its cell phone production from a supplier. Suppose that the lifetime of a chip manufactured by the company's own plant is gamma with mean 4 and variance 4 years whereas the lifetime of a chip purchased from the supplier is gamma with mean 6 and standard deviation 6 years. The company offers its customers with a 1 year replacement warranty. Find the percentage of the cell phones that must be replaced by the company due to chip failure.
Hint: Let X be the lifetime of the chip of a randomly selected cell phone manufactured by the wireless phone company. Let A be the event that the chip was manufactured by the company's own plant. Use the law of total probability to find $P(X \leq 1)$.

■

Chapter 8

Understanding Relationships: Covariance, Correlations, and Conditional Distributions

8.1 INTRODUCTION

In this chapter, we will explore the relationships and dependencies among variables in the context of probability. The central theme is to delve into the interconnections and associations between different variables within the realm of probability. By understanding how various random variables can be dependent on each other, we can gain valuable insights into probability distributions and statistical relationships.

Two key concepts, covariance, and correlations, play a fundamental role in assessing the relationship between two random variables. Covariance quantifies the degree of joint variability between the variables, while correlations standardize the covariance, providing a normalized measure of the strength and direction of the linear relationship between the variables. These measures enable us to comprehend how two variables change together and aid in identifying patterns in the data.

Conditional distributions serve as a powerful tool in probability, allowing us to focus on the behavior of one random variable given specific conditions or values of another variable. By studying conditional distributions, we can develop a deeper understanding of how the presence of one variable influences the distribution and behavior of another.

Transformations of Two Random Variables involve applying mathematical transformations to two random variables to establish new relationships or explore their joint behavior from a different perspective. Such transformations can simplify the analysis or reveal concealed patterns and dependencies between variables that may not be immediately apparent. Through these methods, we can better comprehend the intricate nature of probability relationships and make more informed decisions based on the data at hand.

8.2 COVARIANCE

In [Sections 4.5](#) and [5.3](#), we studied the notion of the variance of a random variable X . We showed that $E[(X - E(X))^2]$, the variance of X , measures the average magnitude of the fluctuations of the random variable X from its expectation, $E(X)$. We mentioned that this quantity measures the dispersion, or spread, of the distribution of X about its expectation. Now suppose that X and Y are two jointly distributed random variables. Then $\text{Var}(X)$ and $\text{Var}(Y)$ determine the dispersions of X and Y independently rather than jointly. In fact, $\text{Var}(X)$ measures the spread, or dispersion, along the x -direction, and $\text{Var}(Y)$ measures the spread, or dispersion, along the y -direction in the plane. We now calculate $\text{Var}(aX + bY)$, the joint spread, or dispersion, of X and Y along the $(ax + by)$ -direction for arbitrary real numbers a and b :

$$\begin{aligned}
& \text{Var}(aX + bY) \\
&= E[(aX + bY) - E(aX + bY)]^2 \\
&= E[(aX + bY) - aE(X) - bE(Y)]^2 \\
&= E[a(X - E(X)) + b(Y - E(Y))]^2 \\
&= E[a^2(X - E(X))^2 + b^2(Y - E(Y))^2 + 2ab(X - E(X))(Y - E(Y))] \\
&= a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2abE[(X - E(X))(Y - E(Y))]. \tag{8.1}
\end{aligned}$$

This formula shows that the joint spread, or dispersion, of X and Y can be measured in *any* direction ($ax + by$) if the quantities $\text{Var}(X)$, $\text{Var}(Y)$, and $E[(X - E(X))(Y - E(Y))]$ are known. On the other hand, the joint spread, or dispersion, of X and Y depends on these three quantities. However, $\text{Var}(X)$ and $\text{Var}(Y)$ determine the dispersions of X and Y independently; therefore, $E[(X - E(X))(Y - E(Y))]$ is the quantity that gives information about the joint spread, or dispersion, of X and Y . It is called the **covariance** of X and Y , is denoted by $\text{Cov}(X, Y)$, and determines how X and Y covary jointly. For example, by relation (8.1), if for random variables X , Y , and Z , $\text{Var}(Y) = \text{Var}(Z)$ and $ab > 0$, then the joint dispersion of X and Y along the $(ax + by)$ -direction is greater than the joint dispersion of X and Z along the $(ax + bz)$ -direction if and only if $\text{Cov}(X, Y) > \text{Cov}(X, Z)$.

Definition 8.1 *Let X and Y be jointly distributed random variables, then the **covariance** of X and Y is defined by*

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))].$$

Note that

$$\text{Cov}(X, X) = \text{Var}(X).$$

Also, by the Cauchy-Schwarz inequality (Theorem 5.7),

$$\begin{aligned}
\text{Cov}(X, Y) &= E[(X - E(X))(Y - E(Y))] \\
&\leq \sqrt{E[X - E(X)]^2 E[Y - E(Y)]^2} \\
&= \sqrt{\sigma_X^2 \sigma_Y^2} = \sigma_X \sigma_Y,
\end{aligned}$$

which shows that if $\sigma_X < \infty$ and $\sigma_Y < \infty$, then $\text{Cov}(X, Y) < \infty$.

Rewriting relation (8.1) in terms of $\text{Cov}(X, Y)$, we obtain the following important theorem:

Theorem 8.1 *Let a and b be real numbers; for random variables X and Y ,*

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

In particular, if $a = 1$ and $b = 1$, this gives

$$\mathbf{Var}(X + Y) = \mathbf{Var}(X) + \mathbf{Var}(Y) + 2 \mathbf{Cov}(X, Y). \quad (8.2)$$

Similarly, if $a = 1$ and $b = -1$, it gives

$$\mathbf{Var}(X - Y) = \mathbf{Var}(X) + \mathbf{Var}(Y) - 2 \mathbf{Cov}(X, Y). \quad (8.3)$$

Letting $\mu_X = E(X)$, and $\mu_Y = E(Y)$, an alternative formula for

$$\mathbf{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

is calculated by the expansion of $E[(X - E(X))(Y - E(Y))]$:

$$\begin{aligned} \mathbf{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E(XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y) \\ &= E(XY) - \mu_X E(Y) - \mu_Y E(X) + \mu_X \mu_Y \\ &= E(XY) - \mu_X \mu_Y - \mu_Y \mu_X + \mu_X \mu_Y \\ &= E(XY) - \mu_X \mu_Y = E(XY) - E(X)E(Y). \end{aligned}$$

Therefore,

$$\mathbf{Cov}(X, Y) = E(XY) - E(X)E(Y). \quad (8.4)$$

Using this relation, we get

$$\begin{aligned} \mathbf{Cov}(aX + b, cY + d) &= E[(aX + b)(cY + d)] - E(aX + b)E(cY + d) \\ &= E(acXY + bcY + adX + bd) - [aE(X) + b][cE(Y) + d] \\ &= ac[E(XY) - E(X)E(Y)] = ac \mathbf{Cov}(X, Y). \end{aligned}$$

Hence, for arbitrary real numbers a, b, c, d and random variables X and Y ,

$$\mathbf{Cov}(aX + b, cY + d) = ac \mathbf{Cov}(X, Y), \quad (8.5)$$

which can be generalized as follows: Let a_i 's and b_j 's be constants. For random variables $X_1, X_2, \dots, X_n$ and $Y_1, Y_2, \dots, Y_m$,

$$\mathbf{Cov}\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \mathbf{Cov}(X_i, Y_j). \quad (8.6)$$

(See Exercise 23.)

For random variables X and Y , $\mathbf{Cov}(X, Y)$ might be positive, negative, or zero. It is positive if the expected value of $[X - E(X)][Y - E(Y)]$ is positive, that is, if X and Y decrease together or increase together. It is negative if X increases while Y decreases, or vice versa. If $\mathbf{Cov}(X, Y) > 0$, we say that X and Y are **positively correlated**. If $\mathbf{Cov}(X, Y) < 0$, we say that they are **negatively correlated**. If $\mathbf{Cov}(X, Y) = 0$, we say that X and Y are **uncorrelated**. For example, the blood cholesterol level of a person is positively correlated

with the amount of saturated fat consumed by that person, whereas the amount of alcohol in the blood is negatively correlated with motor coordination. Generally, the more saturated fat a person ingests, the higher his or her blood cholesterol level will be. The more alcohol a person drinks, the poorer his or her level of motor coordination becomes. As another example, let X be the weight of a person before starting a health fitness program and Y be his or her weight afterward. Then X and Y are negatively correlated because the effect of fitness programs is that, usually, heavier persons lose weight, whereas lighter persons gain weight. The best examples for uncorrelated random variables are independent ones. If X and Y are independent, then

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0.$$

However, as the following example shows, the converse of this is not true; that is,

Two dependent random variables might be uncorrelated.

Example 8.1 Let X be uniformly distributed over $(-1, 1)$ and $Y = X^2$. Then

$$\text{Cov}(X, Y) = E(X^3) - E(X)E(X^2) = 0,$$

since $E(X) = 0$ and $E(X^3) = 0$. Thus the perfectly related random variables X and Y are uncorrelated. ♦

Example 8.2 Let X and Y be two random variables with $\text{Var}(X) = 2$ and $\text{Var}(Y) = 5$.

- (a) Find $\text{Var}(3X + 2Y)$ if X and Y are independent.
- (b) Find $\text{Var}(3X + 2Y)$ if $\text{Cov}(X, Y) = 1$.
- (c) Find $\text{Var}(3X + 2Y)$ if $\text{Cov}(X, Y) = -1$.

Solution: By Theorem 8.1,

$$\text{Var}(3X + 2Y) = 9 \text{Var}(X) + 4 \text{Var}(Y) + 12 \text{Cov}(X, Y).$$

- (a) By this formula, if X and Y are independent, $\text{Var}(3X + 2Y) = 9 \cdot 2 + 4 \cdot 5 + 12 \cdot 0 = 38$.
- (b) If $\text{Cov}(X, Y) = 1$, $\text{Var}(3X + 2Y) = 9 \cdot 2 + 4 \cdot 5 + 12 \cdot 1 = 50$.
- (c) If $\text{Cov}(X, Y) = -1$, $\text{Var}(3X + 2Y) = 9 \cdot 2 + 4 \cdot 5 + 12 \cdot -1 = 26$. ♦

Example 8.3 There are 300 cards in a box numbered 1 through 300. Therefore, the number on each card has one, two, or three digits. A card is drawn at random from the box. Suppose that the number on the card has X digits of which Y are 0. Determine whether X and Y are positively correlated, negatively correlated, or uncorrelated.

Solution: Note that, between 1 and 300, there are 9 one-digit numbers none of which is 0; there are 90 two-digit numbers of which 81 have no 0s, 9 have one 0, and none has two 0s; and there are 201 three-digit numbers of which 162 have no 0s, 36 have one 0, and 3 have two 0s. These facts show that as X increases so does Y . Therefore, X and Y are positively correlated. To show this mathematically, let $p(x, y)$ be the joint probability mass function of X and Y . Simple calculations will yield the following table for $p(x, y)$.

x	y			$p_X(x)$
	0	1	2	
1	9/300	0	0	9/300
2	81/300	9/300	0	90/300
3	162/300	36/300	3/300	201/300
$p_Y(y)$	252/300	45/300	3/300	

To see how we calculated the entries of the table, as an example, consider $p(3, 0)$. This quantity is 162/300 because there are 162 three-digit numbers with no 0s. Now from this table we have that

$$E(X) = 1 \cdot \frac{9}{300} + 2 \cdot \frac{90}{300} + 3 \cdot \frac{201}{300} = 2.64.$$

$$E(Y) = 0 \cdot \frac{252}{300} + 1 \cdot \frac{45}{300} + 2 \cdot \frac{3}{300} = 0.17.$$

$$E(XY) = \sum_{x=1}^3 \sum_{y=0}^2 xyp(x, y) = 2 \cdot \frac{9}{300} + 3 \cdot \frac{36}{300} + 6 \cdot \frac{3}{300} = 0.48.$$

Therefore,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0.48 - (0.17)(2.64) = 0.0312 > 0,$$

which shows that X and Y are positively correlated. ♦

Example 8.4 Ann cuts an ordinary deck of 52 cards and displays the exposed card. Andy cuts the remaining stack of cards and displays his exposed card. Counting jack, queen, and king as 11, 12, and 13, let X and Y be the numbers on the cards that Ann and Andy expose, respectively. Find $\text{Cov}(X, Y)$ and interpret the result.

Solution: Observe that the number of cards in Ann's stack after she cuts the deck, and the number of cards in Andy's stack after he cuts the remaining cards will not change the probabilities we are interested in. The problem is equivalent to choosing two cards at random and without replacement from an ordinary deck of 52 cards, and letting X be the number on one card and Y be the number on the other card. Let $p(x, y)$ be the joint probability mass function of X and Y . For $1 \leq x, y \leq 13$,

$$\begin{aligned} p(x, y) &= P(X = x, Y = y) = P(Y = y \mid X = x)P(X = x) \\ &= \begin{cases} \frac{1}{12} \cdot \frac{1}{13} = \frac{1}{156} & x \neq y \\ 0 & x = y. \end{cases} \end{aligned}$$

Therefore,

$$p_X(x) = \sum_{\substack{y=1 \\ y \neq x}}^{13} p(x, y) = \frac{12}{156} = \frac{1}{13}, \quad x = 1, 2, \dots, 13;$$

$$p_Y(y) = \sum_{\substack{x=1 \\ x \neq y}}^{13} p(x, y) = \frac{12}{156} = \frac{1}{13}, \quad x = 1, 2, \dots, 13.$$

By these relations,

$$E(X) = \sum_{x=1}^{13} \frac{x}{13} = \frac{1}{13} \cdot \frac{13 \times 14}{2} = 7;$$

$$E(Y) = \sum_{y=1}^{13} \frac{y}{13} = \frac{1}{13} \cdot \frac{13 \times 14}{2} = 7.$$

By Theorem 4.7,

$$\begin{aligned} E(XY) &= \sum_{x=1}^{13} \sum_{\substack{y=1 \\ y \neq x}}^{13} \frac{xy}{156} = \frac{1}{156} \sum_{x=1}^{13} \sum_{y=1}^{13} xy - \frac{1}{156} \sum_{x=1}^{13} x^2 \\ &= \frac{1}{156} \left(\sum_{x=1}^{13} x \right) \left(\sum_{y=1}^{13} y \right) - \frac{1}{156} \cdot 819 \\ &= \frac{1}{156} \cdot \frac{13 \times 14}{2} \cdot \frac{13 \times 14}{2} - \frac{819}{156} = \frac{287}{6}. \end{aligned}$$

Therefore,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{287}{6} - 49 = -\frac{7}{6}.$$

This shows that X and Y are negatively correlated. That is, if X increases, then Y decreases; if X decreases, then Y increases. These facts should make sense intuitively. ♦

Example 8.5 Let X be the lifetime of an electronic system and Y be the lifetime of one of its components. Suppose that the electronic system fails if the component does (but not necessarily vice versa). Furthermore, suppose that the joint probability density function of X and Y (in years) is given by

$$f(x, y) = \begin{cases} \frac{1}{49} e^{-y/7} & \text{if } 0 \leq x \leq y < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Determine the expected value of the remaining lifetime of the component when the system dies.

- (b) Find the covariance of X and Y .

Solution:

- (a) The remaining lifetime of the component when the system dies is $Y - X$. So the desired quantity is

$$\begin{aligned} E(Y - X) &= \int_0^\infty \int_0^y (y - x) \frac{1}{49} e^{-y/7} dx dy \\ &= \frac{1}{49} \int_0^\infty e^{-y/7} \left(y^2 - \frac{y^2}{2} \right) dy = \frac{1}{98} \int_0^\infty y^2 e^{-y/7} dy = 7, \end{aligned}$$

where the last integral is calculated using integration by parts twice.

- (b) To find $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$, note that

$$\begin{aligned} E(XY) &= \int_0^\infty \int_0^y (xy) \frac{1}{49} e^{-y/7} dx dy \\ &= \frac{1}{49} \int_0^\infty y e^{-y/7} \left(\int_0^y x dx \right) dy \\ &= \frac{1}{98} \int_0^\infty y^3 e^{-y/7} dy = \frac{14,406}{98} = 147, \end{aligned}$$

where the last integral is calculated, using integration by parts three times. We also have

$$\begin{aligned} E(X) &= \int_0^\infty \int_0^y x \frac{1}{49} e^{-y/7} dx dy = 7, \\ E(Y) &= \int_0^\infty \int_0^y y \frac{1}{49} e^{-y/7} dx dy = 14. \end{aligned}$$

Therefore, $\text{Cov}(X, Y) = 147 - 7(14) = 49$. Note that $\text{Cov}(X, Y) > 0$ is expected because X and Y are positively correlated. ♦

As Theorem 8.1 shows, one important application of the covariance of two random variables X and Y is that it enables us to find $\text{Var}(aX + bY)$ for constants a and b . By direct calculations similar to (8.1), that theorem is generalized as follows: Let $a_1, a_2, \dots, a_n$ be real numbers; for random variables $X_1, X_2, \dots, X_n$,

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j). \quad (8.7)$$

In particular, for $a_i = 1$, $1 \leq i \leq n$,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j). \quad (8.8)$$

By (8.7) and (8.8),

If $X_1, X_2, \dots, X_n$ are *pairwise independent* or, more generally, *pairwise uncorrelated*, then

$$\mathbf{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \mathbf{Var}(X_i) \quad (8.9)$$

$$\mathbf{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \mathbf{Var}(X_i), \quad (8.10)$$

the reason being that $\text{Cov}(X_i, X_j) = 0$, for all $i, j, i \neq j$.

Example 8.6 Let X be the number of 6s in n rolls of a fair die. Find $\text{Var}(X)$.

Solution: Let $X_i = 1$ if on the i th roll the die lands 6, and $X_i = 0$, otherwise. Then $X = X_1 + X_2 + \dots + X_n$. Since $X_1, X_2, \dots, X_n$ are independent,

$$\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n).$$

But for $i = 1, 2, \dots, n$,

$$E(X_i) = 1 \cdot \frac{1}{6} + 0 \cdot \frac{5}{6} = \frac{1}{6},$$

$$E(X_i^2) = (1)^2 \cdot \frac{1}{6} + 0^2 \cdot \frac{5}{6} = \frac{1}{6},$$

and hence

$$\text{Var}(X_i) = \frac{1}{6} - \frac{1}{36} = \frac{5}{36}.$$

Therefore, $\text{Var}(X) = n(5/36) = (5n)/36$. ♦

Example 8.7 Using relation (8.10), calculate the variance of a binomial random variable X with parameters (n, p) .

Solution: Recall that X is the number of successes in n independent Bernoulli trials. Thus for $i = 1, 2, \dots, n$, letting

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{otherwise,} \end{cases}$$

we obtain

$$X = X_1 + X_2 + \dots + X_n,$$

where X_i is a Bernoulli random variable for $i = 1, 2, \dots, n$. Note that $E(X_i) = p$ and $\text{Var}(X_i) = p(1 - p)$. Since $\{X_1, X_2, \dots, X_n\}$ is an independent set of random variables, by (8.10),

$$\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n) = np(1 - p). \quad \blacklozenge$$

Example 8.8 Using relation (8.10), calculate the variance of a negative binomial random variable X , with parameter (r, p) .

Solution: Recall that in a sequence of independent Bernoulli trials, X is the number of trials until the r th success. Let X_1 be the number of trials until the first success, X_2 the number of additional trials to get the second success, X_3 the number of additional ones to obtain the third success, and so on. Then $X = X_1 + X_2 + \dots + X_r$, where for $i = 1, 2, \dots, r$, the random variable X_i is geometric with parameter p . Since $X_1, X_2, \dots, X_r$ are independent,

$$\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_r).$$

But $\text{Var}(X_i) = (1 - p)/p^2$, for $i = 1, 2, \dots, r$ (refer to [Section 6.3](#)). Thus $\text{Var}(X) = r(1 - p)/p^2$. $\blacklozenge$

Let F be the distribution function of a certain characteristic of the elements of some population. For example, let F be the distribution function of lifetimes of light bulbs manufactured by a company, a Scottish soldier's chest size, or the score on an achievement test of a random student. Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution F . For example, let $X_1, X_2, \dots, X_n$ be the chest sizes of n Scottish soldiers selected at random and independently. Then the following lemma enables us to find the expected value and variance of the sample mean in terms of the expected value and variance of the distribution function F (i.e., the population mean and variance).

Lemma 8.1 Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution F with mean μ and variance σ^2 . Let $\bar{X}$ be the mean of the random sample. Then

$$E(\bar{X}) = \mu \quad \text{and} \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Proof: By Theorem 4.9,

$$E(\bar{X}) = E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] = \frac{1}{n} \cdot n\mu = \mu.$$

Since $X_1, X_2, \dots, X_n$ are independent random variables, by (8.9),

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{1}{n^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] \\ &= \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}. \quad \blacklozenge \end{aligned}$$

EXERCISES

A

1. Let X and Y be two random variables with $\text{Var}(X) = 2$ and $\text{Var}(Y) = 5$.
 - (a) Find $\text{Var}(3X - 2Y)$ if X and Y are independent.
 - (b) Find $\text{Var}(3X - 2Y)$ if $\text{Cov}(X, Y) = 1$.
 - (c) Find $\text{Var}(3X - 2Y)$ if $\text{Cov}(X, Y) = -1$.
2. Ann cuts an ordinary deck of 52 cards and displays the exposed card. After Ann places her stack back on the deck, Andy cuts the same deck and displays the exposed card. Counting jack, queen, and king as 11, 12, and 13, let X and Y be the numbers on the cards that Ann and Andy expose, respectively. Find $\text{Cov}(X, Y)$.
3. Let the joint probability mass function of random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{70} x(x + y) & \text{if } x = 1, 2, 3, \quad y = 3, 4 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $\text{Cov}(X, Y)$.

4. Roll a balanced die and let the outcome be X . Then toss a fair coin X times and let Y denote the number of tails. Find $\text{Cov}(X, Y)$ and interpret the result.
Hint: Let $p(x, y)$ be the joint probability mass function of X and Y . To save time, use the table for $p(x, y)$ constructed in Example 4.28.
5. Thieves stole four animals at random from a farm that had seven sheep, eight goats, and five burros. Calculate the covariance of the number of sheep and goats stolen.
6. In n independent Bernoulli trials, each with probability of success p , let X be the number of successes and Y the number of failures. Calculate $E(XY)$ and $\text{Cov}(X, Y)$.
7. For random variables X , Y , and Z , prove that
 - (a) $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$.
 - (b) $\text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z)$.
8. Show that if X and Y are independent random variables, then for all random variables Z ,

$$\text{Cov}(X, Y + Z) = \text{Cov}(X, Z).$$
9. For random variables X and Y , show that

$$\text{Cov}(X + Y, X - Y) = \text{Var}(X) - \text{Var}(Y).$$

10. Prove that

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2 \text{Cov}(X, Y).$$

11. Let X and Y be two independent random variables.

(a) Show that $X - Y$ and $X + Y$ are uncorrelated if and only if $\text{Var}(X) = \text{Var}(Y)$.

(b) Show that $\text{Cov}(X, XY) = E(Y)\text{Var}(X)$.

12. Prove that if Θ is a random number from the interval $[0, 2\pi]$, then the dependent random variables $X = \sin \Theta$ and $Y = \cos \Theta$ are uncorrelated.

13. Let X and Y be the coordinates of a random point selected uniformly from the unit disk $\{(x, y) : x^2 + y^2 \leq 1\}$. Are X and Y independent? Are they uncorrelated? Why or why not?

14. Mr. Jones has two jobs. Next year, he will get a salary raise of X thousand dollars from one employer and a salary raise of Y thousand dollars from his second. Suppose that X and Y are independent random variables with probability density functions f and g , respectively, where

$$f(x) = \begin{cases} 8x/15 & \text{if } 1/2 < x < 2 \\ 0 & \text{elsewhere,} \end{cases}$$

$$g(y) = \begin{cases} 6\sqrt{y}/13 & \text{if } 1/4 < y < 9/4 \\ 0 & \text{elsewhere.} \end{cases}$$

What are the expected value and variance of the total raise that Mr. Jones will get next year?

15. Let X and Y be independent random variables with expected values μ_1 and μ_2 , and variances σ_1^2 and σ_2^2 , respectively. Show that

$$\text{Var}(XY) = \sigma_1^2 \sigma_2^2 + \mu_1^2 \sigma_2^2 + \mu_2^2 \sigma_1^2.$$

16. A voltmeter is used to measure the voltage of voltage sources, such as batteries. Every time this device is used, a random error is made, independent of other measurements, with mean 0 and standard deviation σ . Suppose that we want to measure the voltages, V_1 and V_2 , of two batteries. If a measurement with smaller error variance is preferable, determine which of the following methods should be used:

(a) To measure V_1 and V_2 separately.

(b) To measure $V = V_1 + V_2$ and $W = V_1 - V_2$, and then find V_1 and V_2 from $V_1 = (V + W)/2$ and $V_2 = (V - W)/2$. There are methods available for engineers to do these. For example, to measure $V_1 + V_2$, they attach batteries in series so that each battery pushes current in the same direction. To measure $V_1 - V_2$, they attach the batteries in series so that they push current in opposite directions. (Assume that the internal resistances of the batteries are negligible.)

17. Let X and Y have the following joint probability density function

$$f(x, y) = \begin{cases} 8xy & \text{if } 0 < x \leq y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Calculate $\text{Var}(X + Y)$.
- (b) Show that X and Y are not independent. Explain why this does not contradict Exercise 14 of Section 5.5.
18. Find the variance of a sum of n randomly and independently selected points from the interval $(0, 1)$.
19. Let X and Y be jointly distributed with joint probability density function

$$f(x, y) = \begin{cases} \frac{1}{2}x^3e^{-xy-x} & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Determine if X and Y are positively correlated, negatively correlated, or uncorrelated.

Hint: Note that for all $a > 0$, $\int_0^\infty x^n e^{-ax} dx = n!/a^{n+1}$.

20. Let X be a random variable. Prove that $\text{Var}(X) = \min_t E[(X - t)^2]$.
- Hint:* Let $\mu = E(X)$ and look at the expansion of

$$E[(X - t)^2] = E[(X - \mu + \mu - t)^2].$$

B

21. Let S be the sample space of an experiment. Let A and B be two events of S . Let I_A and I_B be the indicator variables for A and B . That is,

$$I_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A, \end{cases}$$

$$I_B(\omega) = \begin{cases} 1 & \text{if } \omega \in B \\ 0 & \text{if } \omega \notin B. \end{cases}$$

Show that I_A and I_B are positively correlated if and only if $P(A | B) > P(A)$, and if and only if $P(B | A) > P(B)$.

22. Show that for random variables X, Y, Z , and W and constants a, b, c , and d ,

$$\begin{aligned} \text{Cov}(aX + bY, cZ + dW) \\ = ac \text{Cov}(X, Z) + bc \text{Cov}(Y, Z) + ad \text{Cov}(X, W) + bd \text{Cov}(Y, W). \end{aligned}$$

Hint: For a simpler proof, use the results of Exercise 7.

23. Prove the following generalization of Exercise 22:

$$\text{Cov}\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{Cov}(X_i, Y_j).$$

24. A fair die is thrown n times. What is the covariance of the number of 1s and the number of 6s obtained?

Hint: Use the result of Exercise 23.

25. Show that if $X_1, X_2, \dots, X_n$ are random variables and $a_1, a_2, \dots, a_n$ are constants, then

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

26. Let X be a hypergeometric random variable with probability mass function

$$p(x) = P(X = x) = \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}},$$

$$n \leq \min(D, N-D), \quad x = 0, 1, 2, \dots, n.$$

Recall that X is the number of defective items among n items drawn randomly and without replacement from a box containing D defective and $N - D$ nondefective items. Show that

$$\text{Var}(X) = \frac{nD(N-D)}{N^2} \left(1 - \frac{n-1}{N-1}\right).$$

Hint: Let A_i be the event that the i th item drawn is defective. Also for $i = 1, 2, \dots, n$, let

$$X_i = \begin{cases} 1 & \text{if } A_i \text{ occurs} \\ 0 & \text{otherwise.} \end{cases}$$

Then $X = X_1 + X_2 + \dots + X_n$.

27. Exactly n married couples are living in a small town. What is the variance of the surviving couples after m deaths occur among them? Assume that the deaths occur at random, there are no divorces, and there are no new marriages.

Note: This situation involves the Daniel Bernoulli problem discussed in Example 4.36.



Self-Quiz on Section 8.2

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. We draw 8 cards, one at a time, randomly, and without replacement from an ordinary deck of 52 cards. Let

$$X_i = \begin{cases} 1 & \text{if the } i\text{th card drawn is a heart} \\ 0 & \text{otherwise.} \end{cases}$$

For $1 \leq i < j \leq 8$, find $\text{Cov}(X_i, X_j)$.

2. Let X and Y be independent and identically distributed exponential random variables with parameter λ . Let $U = \max(X, Y)$ and $V = \min(X, Y)$. Using the relation

$$\text{Cov}(U, V) = \text{Cov}(U - V + V, V) = \text{Cov}(U - V, V) + \text{Cov}(V, V),$$

Calculate $\text{Cov}(U, V)$ in terms of $\text{Var}(V)$.



8.3 CORRELATION

Although for random variables X and Y , $\text{Cov}(X, Y)$ provides information about how X and Y vary jointly, it has a major shortcoming: It is not independent of the units in which X and Y are measured. For example, suppose that for random variables X and Y , when measured in (say) centimeters, $\text{Cov}(X, Y) = 0.15$. For the same random variables, if we change the measurements to millimeters, then $X_1 = 10X$ and $Y_1 = 10Y$ will be the new observed values, and by relation (8.5) we get

$$\text{Cov}(X_1, Y_1) = \text{Cov}(10X, 10Y) = 100 \text{Cov}(X, Y) = 15,$$

showing that $\text{Cov}(X, Y)$ is sensitive to the units of measurement. From [Section 4.6](#) we know that for a random variable X , the standardized X , $X^* = [X - E(X)]/\sigma_X$, is independent of the units in which X is measured. Thus, to define a measure of association between X and Y , independent of the scales of measurements, it is appropriate to consider $\text{Cov}(X^*, Y^*)$ rather than $\text{Cov}(X, Y)$. Using relation (8.5), we obtain

$$\begin{aligned} \text{Cov}(X^*, Y^*) &= \text{Cov}\left(\frac{X - E(X)}{\sigma_X}, \frac{Y - E(Y)}{\sigma_Y}\right) \\ &= \text{Cov}\left(\frac{1}{\sigma_X}X - \frac{1}{\sigma_X}E(X), \frac{1}{\sigma_Y}Y - \frac{1}{\sigma_Y}E(Y)\right) \end{aligned}$$

$$= \frac{1}{\sigma_X} \cdot \frac{1}{\sigma_Y} \text{Cov}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Definition 8.2 Let X and Y be two random variables with $0 < \sigma_X^2 < \infty$ and $0 < \sigma_Y^2 < \infty$. The covariance between the standardized X and the standardized Y is called the **correlation coefficient** between X and Y and is denoted by $\rho = \rho(X, Y)$. Therefore,

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

The quantity $\text{Cov}(X, Y)/(\sigma_X \sigma_Y)$ gives all the important information that $\text{Cov}(X, Y)$ provides about how X and Y covary and, at the same time, it is not sensitive to the scales of measurement. Clearly, $\rho(X, Y) > 0$ if and only if X and Y are positively correlated; $\rho(X, Y) < 0$ if and only if X and Y are negatively correlated; and $\rho(X, Y) = 0$ if and only if X and Y are uncorrelated. Moreover, $\rho(X, Y)$ roughly measures the amount and the sign of linear relationship between X and Y . It is -1 if $Y = aX + b$, $a < 0$, and $+1$ if $Y = aX + b$, $a > 0$. Thus $\rho(X, Y) = \pm 1$ in the case of perfect linear relationship and 0 in the case of independence of X and Y . The key to the proof of “ $\rho(X, Y) = \pm 1$ if and only if $Y = aX + b$ ” is the following lemma.

Lemma 8.2 For random variables X and Y with correlation coefficient $\rho(X, Y)$,

$$\begin{aligned} \text{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) &= 2 + 2\rho(X, Y); \\ \text{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) &= 2 - 2\rho(X, Y). \end{aligned}$$

Proof: We prove the first relation; the second can be shown similarly. By Theorem 8.1,

$$\text{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) = \frac{1}{\sigma_X^2} \text{Var}(X) + \frac{1}{\sigma_Y^2} \text{Var}(Y) + 2 \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = 2 + 2\rho(X, Y). \quad \blacklozenge$$

Theorem 8.2 For random variables X and Y with correlation coefficient $\rho(X, Y)$,

- (a) $-1 \leq \rho(X, Y) \leq 1$.
- (b) With probability 1, $\rho(X, Y) = 1$ if and only if $Y = aX + b$ for some constants a, b , $a > 0$.
- (c) With probability 1, $\rho(X, Y) = -1$ if and only if $Y = aX + b$ for some constants a, b , $a < 0$.

Proof:

- (a) Since the variance of a random variable is nonnegative,

$$\text{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) \geq 0 \quad \text{and} \quad \text{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) \geq 0.$$

Therefore, by Lemma 8.2, $2 + 2\rho(X, Y) \geq 0$ and $2 - 2\rho(X, Y) \geq 0$. That is, $\rho(X, Y) \geq -1$ and $\rho(X, Y) \leq 1$.

(b) First, suppose that $\rho(X, Y) = 1$. In this case, by Lemma 8.2,

$$\text{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) = 2[1 - \rho(X, Y)] = 0.$$

Therefore, with probability 1,

$$\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} = c,$$

for some constant c . Hence, with probability 1,

$$Y = \frac{\sigma_Y}{\sigma_X}X - c\sigma_Y \equiv aX + b,$$

with $a = \sigma_Y/\sigma_X > 0$ and $b = -c\sigma_Y$. Next, assume that $Y = aX + b$, $a > 0$. We have that

$$\begin{aligned}\rho(X, Y) &= \rho(X, aX + b) = \frac{\text{Cov}(X, aX + b)}{\sigma_X \sigma_{aX+b}} \\ &= \frac{a \text{Cov}(X, X)}{\sigma_X (a\sigma_X)} = \frac{a \text{Var}(X)}{a \text{Var}(X)} = 1.\end{aligned}$$

(c) The proof of this statement is similar to that of part (b). ♦

Example 8.9 Show that if X and Y are continuous random variables with the joint probability density function

$$f(x, y) = \begin{cases} x + y & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise,} \end{cases}$$

then X and Y are not linearly related.

Solution: Since X and Y are linearly related if and only if $\rho(X, Y) = \pm 1$ with probability 1, it suffices to prove that $\rho(X, Y) \neq \pm 1$. To do so, note that

$$E(X) = \int_0^1 \int_0^1 x(x + y) dx dy = \frac{7}{12},$$

$$E(XY) = \int_0^1 \int_0^1 xy(x + y) dx dy = \frac{1}{3}.$$

Also, by symmetry, $E(Y) = 7/12$; therefore,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{3} - \frac{7}{12} \frac{7}{12} = -\frac{1}{144}.$$

Similarly,

$$E(X^2) = \int_0^1 \int_0^1 x^2(x+y) dx dy = \frac{5}{12},$$

$$\sigma_X = \sqrt{E(X^2) - [E(X)]^2} = \sqrt{\frac{5}{12} - \left(\frac{7}{12}\right)^2} = \frac{\sqrt{11}}{12}.$$

Again, by symmetry, $\sigma_Y = \sqrt{11}/12$. Thus

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-1/144}{\sqrt{11}/12 \cdot \sqrt{11}/12} = -\frac{1}{11} \neq \pm 1. \quad \blacklozenge$$

The following example shows that even if X and Y are dependent through a nonlinear relationship such as $Y = X^2$, still, statistically, there might be a strong linear association between X and Y . That is, $\rho(X, Y)$ might be very close to 1 or -1 , indicating that the points (x, y) are tightly clustered around a line.

Example 8.10 Let X be a random number from the interval $(0, 1)$ and $Y = X^2$. The probability density function of X is

$$f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{elsewhere,} \end{cases}$$

and for $n \geq 1$,

$$E(X^n) = \int_0^1 x^n dx = \frac{1}{n+1}.$$

Thus $E(X) = 1/2$, $E(Y) = E(X^2) = 1/3$,

$$\sigma_X^2 = \text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12},$$

$$\sigma_Y^2 = \text{Var}(Y) = E(X^4) - [E(X^2)]^2 = \frac{1}{5} - \left(\frac{1}{3}\right)^2 = \frac{4}{45},$$

and, finally,

$$\text{Cov}(X, Y) = E(X^3) - E(X)E(X^2) = \frac{1}{4} - \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{12}.$$

Therefore,

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{1/12}{1/2\sqrt{3} \cdot 2/3\sqrt{5}} = \frac{\sqrt{15}}{4} = 0.968. \quad \blacklozenge$$

In situations where it is challenging to compute the joint probability density function or the joint probability mass function of two random variables X and Y , statisticians often turn to sampling methods to collect data. Take, for instance, if X represents the height of a randomly selected individual from a population, and Y denotes her or his weight. Statisticians would measure or survey a sufficiently large random sample from the population and then systematically arrange the collected data. A prevalent first step in understanding the potential relationship or correlation between X and Y is visualizing the data using a scatter plot. When we plot the dataset for X and Y , it provides an intuitive understanding of the magnitude and nature of their relationship. Different patterns in a scatter plot can suggest the following interpretations:

- **Positive Correlation:** The data points tend to align from the bottom left to the top right in a cohesive manner, indicating a clear linear trend. As X values increase, Y values typically rise as well.
- **Negative Correlation:** The data points tend to align from the top left to the bottom right in a cohesive manner, showcasing a clear linear trend. As X values increase, Y values generally decline.
- **Independence:** The data points appear scattered without any discernible linear trend.
- **Non-linear Relationships:** At times, the data might trace a curved pattern, such as parabolic or exponential shapes. Even if there isn't a linear correlation, X and Y can still be related in such instances.

While scatter plots give a visual sense of correlation, it's essential to complement this visual inspection with quantitative measures like the correlation coefficient. This is especially true when the relationship isn't clear-cut, or when you want to quantify the strength of the relationship. From elementary statistics courses, we learn that if the data comprises n pairs of observed values (x_i, y_i) for X and Y , the sample correlation coefficient, denoted by r , is given by the following formula. This r often serves as a reliable approximation for $\rho(X, Y)$.

$$r = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{\sqrt{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} \sqrt{n \sum_{i=1}^n y_i^2 - \left(\sum_{i=1}^n y_i \right)^2}}.$$

The sample correlation coefficient r , measures the strength and direction of the linear relationship between two sets of data. The value of r always lies between -1 and 1 , inclusive. Here is a general interpretation of the ranges:

1. $r = 1$: Perfect positive linear correlation.
2. $0.7 \leq r < 1$: Strong positive linear correlation.
3. $0.3 \leq r < 0.7$: Moderate positive linear correlation.
4. $0.2 \leq r < 0.3$: Weak positive linear correlation.

5. $-0.2 < r < 0.2$: No linear correlation. It is worth noting that this does not mean there is no relationship; it simply means there is no *linear* relationship.
6. $-0.3 < r \leq -0.2$: Weak negative linear correlation.
7. $-0.7 < r \leq -0.3$: Moderate negative linear correlation.
8. $-1 < r \leq -0.7$: Strong negative linear correlation.
9. $r = -1$: Perfect negative linear correlation.

Note that while these ranges offer a general guideline, the exact boundaries between “weak,” “moderate,” and “strong” can vary depending on the field of study or the context in which the analysis is being performed. Also, we should remember that correlation (visual or quantitative) does not imply causation. Even if two variables appear to move together, it doesn’t necessarily mean that one causes the other. There could be other lurking variables or confounding factors influencing the relationship.

Figure 8.1 illustrates sample data drawn from the joint distribution of (X, Y) for different dependence structures.

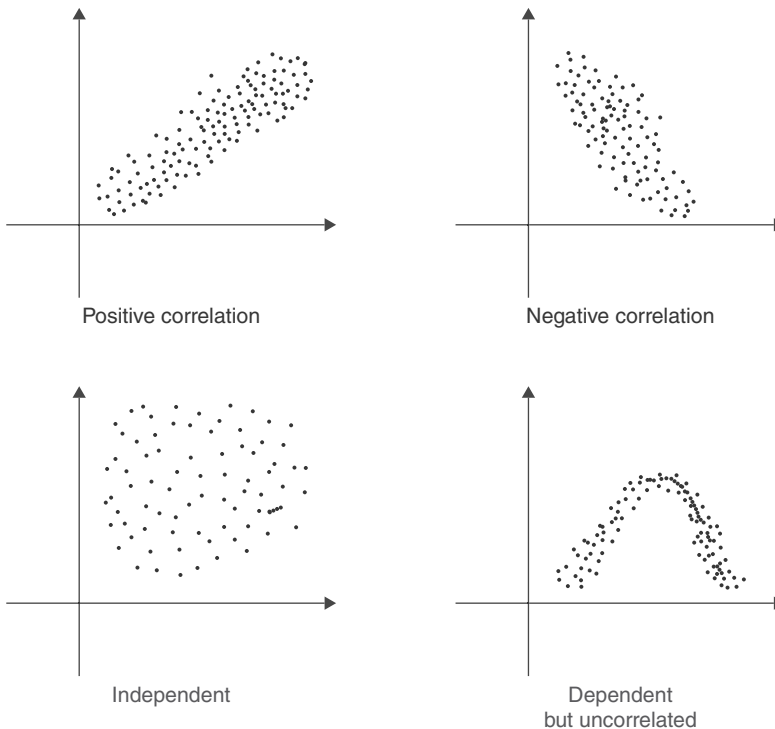


Figure 8.1 Correlation between X and Y visualized by a scatter plot.

EXERCISES

A

1. Let X and Y be jointly distributed, with $\rho(X, Y) = 1/2$, $\sigma_X = 2$, $\sigma_Y = 3$. Find $\text{Var}(2X - 4Y + 3)$.
2. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} \sin x \sin y & \text{if } 0 \leq x \leq \pi/2, \quad 0 \leq y \leq \pi/2 \\ 0 & \text{otherwise.} \end{cases}$$

Calculate the correlation coefficient of X and Y .

3. A stick of length 1 is broken into two pieces at a random point. Find the correlation coefficient and the covariance of these pieces.
4. For real numbers α and β , let

$$\text{sgn}(\alpha\beta) = \begin{cases} 1 & \text{if } \alpha\beta > 0 \\ 0 & \text{if } \alpha\beta = 0 \\ -1 & \text{if } \alpha\beta < 0. \end{cases}$$

Prove that for random variables X and Y ,

$$\rho(\alpha_1 X + \alpha_2, \beta_1 Y + \beta_2) = \rho(X, Y) \text{sgn}(\alpha_1 \beta_1).$$

5. Is it possible that for some random variables X and Y , $\rho(X, Y) = 3$, $\sigma_X = 2$, and $\sigma_Y = 3$?
6. Prove that if $\text{Cov}(X, Y) = 0$, then

$$\rho(X + Y, X - Y) = \frac{\text{Var}(X) - \text{Var}(Y)}{\text{Var}(X) + \text{Var}(Y)}.$$

B

7. Show that if the joint probability density function of X and Y is

$$f(x, y) = \begin{cases} \frac{1}{2} \sin(x + y) & \text{if } 0 \leq x \leq \frac{\pi}{2}, \quad 0 \leq y \leq \frac{\pi}{2} \\ 0 & \text{elsewhere,} \end{cases}$$

then there exists no linear relation between X and Y .



Self-Quiz on Section 8.3

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. In a probability exam, for two random variables X and Y with a given joint probability density function, Fiona's calculations resulted in $E(X) = E(Y) = 2$, $E(X^2) = 13$, $E(Y^2) = 40$, and $E(XY) = 23$. However, Fiona got no points for these answers. Under her solution, the professor wrote, impossible results. Show that, in fact, for no joint probability density function such results are possible.
2. For random variables $X_1, X_2, \dots, X_n$, we have that $\text{Var}(X_i) = 4$, $1 \leq i \leq n$, and for $1 \leq i \neq j \leq n$, $\rho(X_i, X_j) = -1/16$. Find $\text{Var}(X_1 + X_2 + \dots + X_n)$.
Hint: Note that $\sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{Cov}(X_i, X_j)$ has $(n-1) + (n-2) + \dots + 1 = [(n-1)n]/2$ terms.



8.4 CONDITIONAL DISTRIBUTIONS

Let X and Y be two discrete or two continuous random variables. In this section we study the distribution function and the expected value of the random variable X given that $Y = y$.

Conditional Distributions: Discrete Case

Let X be a discrete random variable with set of possible values A , and let Y be a discrete random variable with set of possible values B . Let $p(x, y)$ be the joint probability mass function of X and Y , and let p_X and p_Y be the marginal probability mass functions of X and Y . When no information is given about the value of Y ,

$$p_X(x) = P(X = x) = \sum_{y \in B} P(X = x, Y = y) = \sum_{y \in B} p(x, y) \quad (8.11)$$

is used to calculate the probabilities of events concerning X . However, if the value of Y is known, then instead of $p_X(x)$, the **conditional probability mass function of X given that $Y = y$** is used. This function, denoted by $p_{X|Y}(x|y)$, is defined as follows:

$$p_{X|Y}(x|y) = P(X = x | Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p(x, y)}{p_Y(y)},$$

where $x \in A$, $y \in B$, and $p_Y(y) > 0$. Hence, **for $x \in A$, $y \in B$, and $p_Y(y) > 0$,**

$$p_{X|Y}(x|y) = \frac{p(x, y)}{p_Y(y)}. \quad (8.12)$$

Note that

$$\sum_{x \in A} p_{X|Y}(x|y) = \sum_{x \in A} \frac{p(x, y)}{p_Y(y)} = \frac{1}{p_Y(y)} \sum_{x \in A} p(x, y) = \frac{1}{p_Y(y)} p_Y(y) = 1.$$

Hence, for any fixed $y \in B$, $p_{X|Y}(x|y)$ is itself a probability mass function with the set of possible values A . If X and Y are independent, $p_{X|Y}$ coincides with p_X because

$$\begin{aligned} p_{X|Y}(x|y) &= \frac{p(x, y)}{p_Y(y)} = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{P(X = x)P(Y = y)}{P(Y = y)} \\ &= P(X = x) = p_X(x). \end{aligned}$$

Note that, **for $x \in A$, $y \in B$, and $p_X(x) > 0$,**

$$p_{Y|X}(y|x) = \frac{p(x, y)}{p_X(x)}. \quad (8.13)$$

Similar to $p_{X|Y}(x|y)$, the **conditional distribution function of X , given that $Y = y$** is defined as follows:

$$F_{X|Y}(x|y) = P(X \leq x | Y = y) = \sum_{t \leq x} P(X = t | Y = y) = \sum_{t \leq x} p_{X|Y}(t|y).$$

Example 8.11 Let the joint probability mass function of X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{15}(x + y) & \text{if } x = 0, 1, 2, y = 1, 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find $p_{X|Y}(x|y)$ and $P(X = 0 | Y = 2)$.

Solution: To use $p_{X|Y}(x|y) = p(x, y)/p_Y(y)$, we must first calculate $p_Y(y)$.

$$p_Y(y) = \sum_{x=0}^2 p(x, y) = \frac{1+y}{5}.$$

Therefore,

$$p_{X|Y}(x|y) = \frac{(x+y)/15}{(1+y)/5} = \frac{x+y}{3(1+y)}, \quad x = 0, 1, 2 \text{ when } y = 1 \text{ or } 2.$$

$$\text{In particular, } P(X = 0 | Y = 2) = \frac{0+2}{3(1+2)} = \frac{2}{9}. \quad \blacklozenge$$

Example 8.12 Let $N(t)$ be the number of males who enter a certain post office at or prior to time t . Let $M(t)$ be the number of females who enter a certain post office at or prior to t .

Suppose that $\{N(t): t \geq 0\}$ and $\{M(t): t \geq 0\}$ are independent Poisson processes with rates λ and μ , respectively. So for all $t > 0$ and $s > 0$, $N(t)$ is independent of $M(s)$. If at some instant t , $N(t) + M(t) = n$, what is the conditional probability mass function of $N(t)$?

Solution: For simplicity, let $K(t) = N(t) + M(t)$ and $p(x, n)$ be the joint probability mass function of $N(t)$ and $K(t)$. Then $p_{N(t)|K(t)}(x|n)$, the desired probability mass function, is found as follows:

$$\begin{aligned} p_{N(t)|K(t)}(x|n) &= \frac{p(x, n)}{p_{K(t)}(n)} = \frac{P(N(t) = x, K(t) = n)}{P(K(t) = n)} \\ &= \frac{P(N(t) = x, M(t) = n - x)}{P(K(t) = n)}. \end{aligned}$$

Since $N(t)$ and $M(t)$ are independent random variables and $K(t) = N(t) + M(t)$ is a Poisson random variable with rate $\lambda t + \mu t$ (see Example 6.18),

$$\begin{aligned} p_{N(t)|K(t)}(x|n) &= \frac{P(N(t) = x)P(M(t) = n - x)}{P(K(t) = n)} = \frac{\frac{e^{-\lambda t}(\lambda t)^x}{x!} \frac{e^{-\mu t}(\mu t)^{n-x}}{(n-x)!}}{\frac{e^{-(\lambda t + \mu t)}(\lambda t + \mu t)^n}{n!}} \\ &= \frac{n!}{x!(n-x)!} \frac{\lambda^x \mu^{n-x}}{(\lambda + \mu)^n} = \binom{n}{x} \left(\frac{\lambda}{\lambda + \mu}\right)^x \left(\frac{\mu}{\lambda + \mu}\right)^{n-x} \\ &= \binom{n}{x} \left(\frac{\lambda}{\lambda + \mu}\right)^x \left(1 - \frac{\lambda}{\lambda + \mu}\right)^{n-x}, \quad x = 0, 1, \dots, n. \end{aligned}$$

This shows that the conditional probability mass function of the number of males who have entered the post office at or prior to t , given that altogether n persons have entered the post office during this period, is binomial with parameters n and $\lambda/(\lambda + \mu)$. ♦

We now generalize the concept of mathematical expectation to the conditional case. Let X and Y be discrete random variables, and let the set of possible values of X be A . The **conditional expectation of the random variable X given that $Y = y$** is as follows:

$$E(X | Y = y) = \sum_{x \in A} xP(X = x | Y = y) = \sum_{x \in A} xp_{X|Y}(x|y),$$

where $p_Y(y) > 0$. Hence, by definition, for $p_Y(y) > 0$,

$$E(X | Y = y) = \sum_{x \in A} xp_{X|Y}(x|y). \quad (8.14)$$

Similarly, for $p_X(x) > 0$,

$$E(Y | X = x) = \sum_{y \in B} yp_{Y|X}(y|x). \quad (8.15)$$

Note that conditional expectations are simply ordinary expectations computed relative to conditional distributions. For this reason, they satisfy the same properties that ordinary expectations do. For example, if h is an ordinary function from $\mathbb{R}$ to $\mathbb{R}$, then for the discrete random variables X and Y with set of possible values A for X , the expected value of $h(X)$ is obtained from

$$E[h(X) | Y = y] = \sum_{x \in A} h(x)p_{X|Y}(x|y). \quad (8.16)$$

Example 8.13 Calculate the expected number of aces in a randomly selected poker hand that is found to have exactly two jacks.

Solution: Let X and Y be the number of aces and jacks in a random poker hand, respectively. Then

$$\begin{aligned} E(X | Y = 2) &= \sum_{x=0}^3 xp_{X|Y}(x|2) = \sum_{x=0}^3 x \frac{p(x, 2)}{p_Y(2)} \\ &= \sum_{x=0}^3 x \frac{\frac{\binom{4}{2} \binom{4}{x} \binom{44}{3-x}}{\binom{52}{5}}}{\frac{\binom{4}{2} \binom{48}{3}}{\binom{52}{5}}} = \sum_{x=0}^3 x \frac{\binom{4}{x} \binom{44}{3-x}}{\binom{48}{3}} \approx 0.25. \quad \blacklozenge \end{aligned}$$

Example 8.14 While rolling a balanced die successively, the first 6 occurred on the third roll. What is the expected number of rolls until the first 1?

Solution: Let X and Y be the number of rolls until the first 1 and the first 6, respectively. The required quantity, $E(X | Y = 3)$, is calculated from

$$E(X | Y = 3) = \sum_{x=1}^{\infty} xp_{X|Y}(x|3) = \sum_{x=1}^{\infty} x \frac{p(x, 3)}{p_Y(3)}.$$

Clearly,

$$\begin{aligned} p_Y(3) &= \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right) = \frac{25}{216}, & p(1, 3) &= \left(\frac{1}{6}\right) \left(\frac{5}{6}\right) \left(\frac{1}{6}\right) = \frac{5}{216}, \\ p(2, 3) &= \left(\frac{4}{6}\right) \left(\frac{1}{6}\right) \left(\frac{1}{6}\right) = \frac{4}{216}, & p(3, 3) &= 0, \end{aligned}$$

and for $x > 3$,

$$p(x, 3) = \left(\frac{4}{6}\right)^2 \left(\frac{1}{6}\right) \left(\frac{5}{6}\right)^{x-4} \left(\frac{1}{6}\right) = \frac{1}{81} \left(\frac{5}{6}\right)^{x-4}.$$

Therefore,

$$\begin{aligned} E(X \mid Y = 3) &= \sum_{x=1}^{\infty} x \frac{p(x, 3)}{p_Y(3)} = \frac{216}{25} \sum_{x=1}^{\infty} x p(x, 3) \\ &= \frac{216}{25} \left[1 \cdot \frac{5}{216} + 2 \cdot \frac{4}{216} + \sum_{x=4}^{\infty} x \cdot \frac{1}{81} \left(\frac{5}{6}\right)^{x-4} \right] \\ &= \frac{13}{25} + \frac{8}{75} \sum_{x=4}^{\infty} x \left(\frac{5}{6}\right)^{x-4} \\ &= \frac{13}{25} + \frac{8}{75} \sum_{x=4}^{\infty} (x - 4 + 4) \left(\frac{5}{6}\right)^{x-4} \\ &= \frac{13}{25} + \frac{8}{75} \left[\sum_{x=4}^{\infty} (x - 4) \left(\frac{5}{6}\right)^{x-4} + 4 \sum_{x=4}^{\infty} \left(\frac{5}{6}\right)^{x-4} \right] \\ &= \frac{13}{25} + \frac{8}{75} \left[\sum_{x=0}^{\infty} x \left(\frac{5}{6}\right)^x + 4 \sum_{x=0}^{\infty} \left(\frac{5}{6}\right)^x \right] \\ &= \frac{13}{25} + \frac{8}{75} \left[\frac{5/6}{(1 - 5/6)^2} + 4 \left(\frac{1}{1 - 5/6} \right) \right] = 6.28. \quad \blacklozenge \end{aligned}$$

Example 8.15 (The Box Problem: To Switch or Not to Switch) Suppose that there are two identical closed boxes, one containing twice as much money as the other. David is asked to choose one of these boxes for himself. He picks a box at random and opens it. If, after observing the content of the box, he has the option to exchange it for the other box, should he exchange or not?

Solution: Suppose that David finds $\$x$ in the box. Then the other box has $\$(2x)$ with probability $1/2$ and $\$(x/2)$ with probability $1/2$. Therefore, the expected value of the dollar amount in the other box is

$$\frac{x}{2} \cdot \frac{1}{2} + 2x \cdot \frac{1}{2} = 1.25x.$$

This implies that David should switch no matter how much he finds in the box he picks at random. But as Steven J. Brams and D. Marc Kilgour observe in their paper “The Box Problem: To Switch or Not to Switch” (*Mathematics Magazine*, February 1995, Volume 68, Number 1),

It seems paradoxical that it would always be better to switch to the second box, no matter how much money you found in the first one.

They go on to explain,

What is needed to determine whether or not switching is worthwhile is some prior notion of the amount of money to be found in each box. For particular, knowing the prior probability distribution would enable you to calculate whether the amount found in the first box is less than the expected value of what is in the other box—and, therefore, whether a switch is profitable.

Let L be the event that David picked the box with the larger amount. Let X be the dollar amount David finds in the box he picks. According to a general exchange condition, discovered by Brams and Kilgour, on the average, an exchange is profitable if and only if for the observed value x ,

$$P(L \mid X = x) < 2/3.$$

That is, David should switch if and only if the conditional probability that he picked the box with the larger amount given that he found $\$x$ is less than $2/3$. To show this, let Y be the amount in the box David did not pick, and let S be the event that David picked the box with the smaller amount. Then

$$E(Y \mid X = x) = \frac{x}{2} \cdot P(L \mid X = x) + 2x \cdot P(S \mid X = x).$$

On average, exchange is profitable if and only if $E(Y \mid X = x) > x$, or, equivalently, if and only if

$$\frac{x}{2} \cdot P(L \mid X = x) + 2x \cdot P(S \mid X = x) > x.$$

Given that $P(S \mid X = x) = 1 - P(L \mid X = x)$, this relation yields

$$\frac{1}{2} P(L \mid X = x) + 2[1 - P(L \mid X = x)] > 1,$$

which implies that $P(L \mid X = x) < 2/3$.

As an example, suppose that the box with the larger amount contains \$1, \$2, \$4, or \$8 with equal probabilities. Then, for $x = 1/2, 1, 2$, and 4 , David should switch, for $x = 8$ he should not. This is because, for example, by Bayes' formula (Theorem 3.5),

$$\begin{aligned} P(L \mid X = 2) &= \frac{P(X = 2 \mid L)P(L)}{P(X = 2 \mid L)P(L) + P(X = 2 \mid S)P(S)} \\ &= \frac{\frac{1}{4} \times \frac{1}{2}}{\frac{1}{4} \times \frac{1}{2} + \frac{1}{4} \times \frac{1}{2}} = \frac{1}{2} < \frac{2}{3}, \end{aligned}$$

whereas $P(L \mid X = 8) = 1 > 2/3$. ♦

Conditional Distributions: Continuous Case

Now let X and Y be two continuous random variables with the joint probability density function $f(x, y)$. Again, when no information is given about the value of Y , $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$ is used to calculate the probabilities of events concerning X . However, when the value of Y is known, to find such probabilities, $f_{X|Y}(x|y)$, the **conditional probability density function of X given that $Y = y$** is used. Similar to the discrete case, $f_{X|Y}(x|y)$ is defined as follows:

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}, \quad (8.17)$$

provided that $f_Y(y) > 0$. Note that

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y) dx = \int_{-\infty}^{\infty} \frac{f(x, y)}{f_Y(y)} dx = \frac{1}{f_Y(y)} \int_{-\infty}^{\infty} f(x, y) dx = \frac{1}{f_Y(y)} f_Y(y) = 1,$$

showing that for a fixed y , $f_{X|Y}(x|y)$ is itself a probability density function. If X and Y are independent, then $f_{X|Y}$ coincides with f_X because

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x).$$

Similarly, the **conditional probability density function of Y given that $X = x$** is defined by

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}, \quad (8.18)$$

provided that $f_X(x) > 0$. Also, as we expect, $F_{X|Y}(x|y)$, the **conditional distribution function of X given that $Y = y$** is defined as follows:

$$F_{X|Y}(x|y) = P(X \leq x | Y = y) = \int_{-\infty}^x f_{X|Y}(t|y) dt.$$

Therefore,

$$\frac{d}{dx} F_{X|Y}(x|y) = f_{X|Y}(x|y). \quad (8.19)$$

Example 8.16 Let X and Y be continuous random variables with joint probability density function

$$f(x, y) = \begin{cases} \frac{3}{2}(x^2 + y^2) & \text{if } 0 < x < 1, \quad 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $f_{X|Y}(x|y)$.

Solution: By definition, $f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$, where

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{3}{2}(x^2 + y^2) dx = \frac{3}{2}y^2 + \frac{1}{2}.$$

Thus

$$f_{X|Y}(x|y) = \frac{3/2(x^2 + y^2)}{(3/2)y^2 + 1/2} = \frac{3(x^2 + y^2)}{3y^2 + 1}$$

for $0 < x < 1$ and $0 < y < 1$. Everywhere else, $f_{X|Y}(x|y) = 0$. ♦

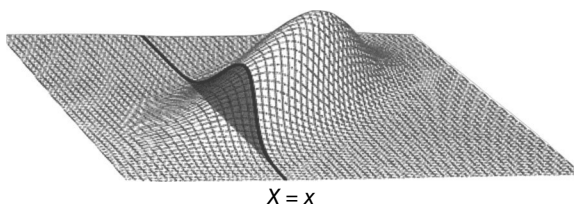


Figure 8.2 Conditional probability density function of Y given $X = x$.

Figure 8.2 shows the conditional probability density function of Y given that $X = x$. Note that $f_{Y|X}(y|x)$ is a vertical slice taken from the surface of $f(x, y)$ at the given point x . Thus the graph of $f_{Y|X}(y|x)$ is obtained from the intersection of the plane $X = x$ and the surface $f(x, y)$.

Example 8.17 First, a point Y is selected at random from the interval $(0, 1)$. Then another point X is chosen at random from the interval $(0, Y)$. Find the probability density function of X .

Solution: Let $f(x, y)$ be the joint probability density function of X and Y . Then

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy,$$

where from $f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$, we obtain

$$f(x, y) = f_{X|Y}(x|y)f_Y(y).$$

Therefore,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y)f_Y(y) dy.$$

Since Y is uniformly distributed over $(0, 1)$,

$$f_Y(y) = \begin{cases} 1 & \text{if } 0 < y < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Since given $Y = y$, X is uniformly distributed over $(0, y)$,

$$f_{X|Y}(x|y) = \begin{cases} 1/y & \text{if } 0 < y < 1, \quad 0 < x < y \\ 0 & \text{elsewhere.} \end{cases}$$

Thus

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y)f_Y(y) dy = \int_x^1 \frac{dy}{y} = \ln 1 - \ln x = -\ln x.$$

Therefore,

$$f_X(x) = \begin{cases} -\ln x & \text{if } 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases} \quad \blacklozenge$$

Example 8.18 Let the conditional probability density function of X , given that $Y = y$, be

$$f_{X|Y}(x|y) = \frac{x+y}{1+y} e^{-x}, \quad 0 < x < \infty, \quad 0 < y < \infty.$$

Find $P(X < 1 \mid Y = 2)$.

Solution: The probability density function of X given $Y = 2$ is

$$f_{X|Y}(x|2) = \frac{x+2}{3} e^{-x}, \quad 0 < x < \infty.$$

Therefore,

$$\begin{aligned} P(X < 1 \mid Y = 2) &= \int_0^1 \frac{x+2}{3} e^{-x} dx = \frac{1}{3} \left[\int_0^1 x e^{-x} dx + \int_0^1 2 e^{-x} dx \right] \\ &= \frac{1}{3} \left[-x e^{-x} - e^{-x} \right]_0^1 - \frac{2}{3} \left[e^{-x} \right]_0^1 = 1 - \frac{4}{3} e^{-1} \approx 0.509. \end{aligned}$$

Note that while we calculated $P(X < 1 \mid Y = 2)$, it is not possible to calculate probabilities such as $P(X < 1 \mid 2 < Y < 3)$ using $f_{X|Y}(x|y)$. This is because the probability density function of X , given that $2 < Y < 3$, cannot be found from the conditional probability density function of X , given $Y = y$. $\blacklozenge$

Similar to the case where X and Y are discrete, for continuous random variables X and Y with joint probability density function $f(x, y)$, the **conditional expectation of X given that $Y = y$** is as follows:

$$E(X \mid Y = y) = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx, \quad (8.20)$$

where $f_Y(y) > 0$. Similarly, the **conditional expectation of Y given that $X = x$** is given by

$$E(Y | X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy, \quad (8.21)$$

where $f_X(x) > 0$.

As explained in the discrete case, conditional expectations are simply ordinary expectations computed relative to conditional distributions. For this reason, they satisfy the same properties that ordinary expectations do. For example, if h is an ordinary function from $\mathbb{R}$ to $\mathbb{R}$, then, for continuous random variables X and Y , with joint probability density function $f(x, y)$,

$$E[h(X) | Y = y] = \int_{-\infty}^{\infty} h(x) f_{X|Y}(x|y) dx. \quad (8.22)$$

In particular, this implies that the conditional variance of X given that $Y = y$ is given by

$$\sigma_{X|Y=y}^2 = \int_{-\infty}^{\infty} [x - E(X | Y = y)]^2 f_{X|Y}(x|y) dx. \quad (8.23)$$

Example 8.19 Let X and Y be continuous random variables with joint probability density function

$$f(x, y) = \begin{cases} e^{-y} & \text{if } y > 0, \quad 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X | Y = 2)$.

Solution: From the definition,

$$E(X | Y = 2) = \int_{-\infty}^{\infty} x f_{X|Y}(x|2) dx = \int_0^1 x \frac{f(x, 2)}{f_Y(2)} dx = \int_0^1 x \frac{e^{-2}}{f_Y(2)} dx.$$

But $f_Y(2) = \int_0^1 f(x, 2) dx = \int_0^1 e^{-2} dx = e^{-2}$; therefore,

$$E(X | Y = 2) = \int_0^1 x \frac{e^{-2}}{e^{-2}} dx = \frac{1}{2}. \quad \blacklozenge$$

Example 8.20 The lifetimes of batteries manufactured by a certain company are identically distributed with probability distribution and probability density functions F and f , respectively. In terms of F , f , and s , find the expected value of the lifetime of an s -hour-old battery.

Solution: Let X be the lifetime of the s -hour-old battery. We want to calculate $E(X | X > s)$. Let

$$F_{X|X>s}(t) = P(X \leq t | X > s),$$

and $f_{X|X>s}(t) = F'_{X|X>s}(t)$. Then

$$E(X | X > s) = \int_0^{\infty} t f_{X|X>s}(t) dt.$$

Now

$$\begin{aligned} F_{X|X>s}(t) &= P(X \leq t | X > s) = \frac{P(X \leq t, X > s)}{P(X > s)} \\ &= \begin{cases} 0 & \text{if } t \leq s \\ \frac{P(s < X \leq t)}{P(X > s)} & \text{if } t > s. \end{cases} \end{aligned}$$

Therefore,

$$F_{X|X>s}(t) = \begin{cases} 0 & \text{if } t \leq s \\ \frac{F(t) - F(s)}{1 - F(s)} & \text{if } t > s. \end{cases}$$

Differentiating $F_{X|X>s}(t)$ with respect to t , we obtain

$$f_{X|X>s}(t) = \begin{cases} 0 & \text{if } t \leq s \\ \frac{f(t)}{1 - F(s)} & \text{if } t > s. \end{cases}$$

This yields

$$E(X | X > s) = \int_0^{\infty} t f_{X|X>s}(t) dt = \frac{1}{1 - F(s)} \int_s^{\infty} t f(t) dt. \quad \blacklozenge$$

★ Remark 8.1 Suppose that, in Example 8.20, a battery manufactured by the company is installed at time 0 and begins to operate. If at time s an inspector finds the battery dead, then the expected lifetime of the dead battery is $E(X | X < s)$. Similar to the calculations in that example, we can show that

$$E(X | X < s) = \frac{1}{F(s)} \int_0^s t f(t) dt.$$

(See Exercise 28.) $\blacklozenge$

Conditional Distributions in Higher Dimensions

Let the discrete random variables $X_1, X_2, \dots, X_n$ have a joint probability mass function denoted by $p(x_1, x_2, \dots, x_n)$. In [Section 4.7](#), we explored the concept of $p_{X_i}(x_i)$, the marginal probability mass function for the random variable X_i . Now, we will extend this concept and

introduce the **joint probability mass function marginalized** over a given set of k of these random variables. This marginalization is obtained by summing all terms of $p(x_1, x_2, \dots, x_n)$ over all possible values of the remaining $n - k$ random variables. For example, if $p(x, y, z)$ denotes the joint probability mass function of random variables X , Y , and Z , then

$$p_{X,Y}(x, y) = \sum_z p(x, y, z)$$

is the joint probability mass function marginalized over X and Y , whereas

$$p_{Y,Z}(y, z) = \sum_x p(x, y, z)$$

is the joint probability mass function marginalized over Y and Z . Similarly, let the continuous random variables $X_1, X_2, \dots, X_n$ have a joint probability density function denoted by $f(x_1, x_2, \dots, x_n)$. To find the **joint probability density function marginalized** over a given set of k of these random variables, we integrate $f(x_1, x_2, \dots, x_n)$ over all possible values of the remaining $n - k$ random variables. For example, if $f(x, y, z, t)$ denotes the joint probability density function of random variables X , Y , Z , and T , then

$$f_{Y,T}(y, t) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y, z, t) dx dz$$

is the joint probability density function marginalized over Y and T , whereas

$$f_{X,Z,T}(x, z, t) = \int_{-\infty}^{+\infty} f(x, y, z, t) dy$$

is the joint probability density function marginalized over X , Z , and T .

Example 8.21 Suppose that the joint probability density function of the continuous random variables X , Y , Z , and T is given by

$$f(x, y, z, t) = \begin{cases} \frac{1}{xyz} & \text{if } 0 < t \leq z \leq y \leq x \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $f_{Y,Z,T}(y, z, t)$ and $f_{X,T}(x, t)$.

Solution: For $0 < t \leq z \leq y \leq 1$,

$$f_{Y,Z,T}(y, z, t) = \int_y^1 \frac{1}{xyz} dx = \frac{1}{yz} \ln x \Big|_y^1 = -\frac{\ln y}{yz}.$$

Therefore,

$$f_{Y,Z,T}(y, z, t) = \begin{cases} -\frac{\ln y}{yz} & \text{if } 0 < t \leq z \leq y \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

To find $f_{X,T}(x, t)$, we have that for $0 < t \leq x \leq 1$,

$$\begin{aligned} f_{X,T}(x, t) &= \int_t^x \int_t^y \frac{1}{xyz} dz dy = \int_t^x \left[\frac{\ln z}{xy} \right]_t^y dy \\ &= \int_t^x \left(\frac{\ln y}{xy} - \frac{\ln t}{xy} \right) dy = \left[\frac{1}{2x} (\ln y)^2 - \frac{\ln t}{x} \ln y \right]_t^x \\ &= \frac{1}{2x} (\ln x)^2 - \frac{1}{x} (\ln t)(\ln x) + \frac{1}{2x} (\ln t)^2 = \frac{1}{2x} (\ln x - \ln t)^2 \\ &= \frac{1}{2x} \ln^2 \frac{x}{t}. \end{aligned}$$

Therefore,

$$f_{X,T}(x, t) = \begin{cases} \frac{1}{2x} \ln^2 \frac{x}{t} & \text{if } 0 < t \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases} \quad \blacklozenge$$

Example 8.22 (Marginals of Multinomials) Let $X_1, X_2, \dots, X_r$ ($r \geq 4$) have the joint multinomial probability mass function $p(x_1, x_2, \dots, x_r)$ with parameters n and $p_1, p_2, \dots, p_r$. Find the joint probability mass function marginalized over X_1, X_2 , and X_3 .

Solution: To find the joint probability mass function marginalized over X_1, X_2 , and X_3 , let $k = n - x_1 - x_2 - x_3$, then

$$\begin{aligned} p_{X_1, X_2, X_3}(x_1, x_2, x_3) &= \sum_{x_4 + \dots + x_r = k} \frac{n!}{x_1! x_2! x_3! x_4! \dots x_r!} p_1^{x_1} p_2^{x_2} p_3^{x_3} p_4^{x_4} \dots p_r^{x_r} \\ &= \frac{n!}{x_1! x_2! x_3! k!} p_1^{x_1} p_2^{x_2} p_3^{x_3} \sum_{x_4 + \dots + x_r = k} \frac{k!}{x_4! \dots x_r!} p_4^{x_4} \dots p_r^{x_r} \\ &= \frac{n!}{x_1! x_2! x_3! k!} p_1^{x_1} p_2^{x_2} p_3^{x_3} (p_4 + \dots + p_r)^k \\ &= \frac{n!}{x_1! x_2! x_3! (n - x_1 - x_2 - x_3)!} p_1^{x_1} p_2^{x_2} p_3^{x_3} (1 - p_1 - p_2 - p_3)^{n - x_1 - x_2 - x_3}. \end{aligned}$$

This shows that *the joint probability mass function marginalized over X_1, X_2 , and X_3 is multinomial with parameters n and $p_1, p_2, p_3, 1 - p_1 - p_2 - p_3$.* $\blacklozenge$

Remark 8.2 The method of Example 8.22 can be extended to prove the following theorem:

Let the joint distribution of the random variables $X_1, X_2, \dots, X_r$ be multinomial with parameters n and $p_1, p_2, \dots, p_r$. The joint probability mass function marginalized over a subset $X_{i_1}, X_{i_2}, \dots, X_{i_k}$ of k ($k > 1$) of these r random variables is multinomial with parameters n and $p_{i_1}, p_{i_2}, \dots, p_{i_k}, 1 - p_{i_1} - p_{i_2} - \dots - p_{i_k}$. $\blacklozenge$

Conditional probability mass functions and **conditional probability density functions** in higher dimensions are defined similarly to dimension two. For example, if X , Y , and Z are three continuous random variables with the joint probability density function f , then

$$f_{X,Y|Z}(x, y|z) = \frac{f(x, y, z)}{f_Z(z)}$$

at all points z for which $f_Z(z) > 0$. As another example, let X , Y , Z , V , and W be five continuous random variables with the joint probability density function f . Then

$$f_{X,Z,W|Y,V}(x, z, w|y, v) = \frac{f(x, y, z, v, w)}{f_{Y,V}(y, v)}$$

at all points (y, v) for which $f_{Y,V}(y, v) > 0$.

Example 8.23 Let X be a random point from the interval $(0, 1)$, Y be a random point from $(0, X)$, and Z be a random point from $(X, 1)$. Find f , the joint probability density function of X , Y , and Z .

Solution: Note that

$$\begin{aligned} f(x, y, z) &= f_X(x) \cdot \frac{f_{X,Y}(x, y)}{f_X(x)} \cdot \frac{f(x, y, z)}{f_{X,Y}(x, y)} \\ &= f_X(x) \cdot f_{Y|X}(y|x) \cdot f_{Z|X,Y}(z|x, y). \end{aligned}$$

Clearly,

$$\begin{aligned} f_X(x) &= \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases} \\ f_{Y|X}(y|x) &= \begin{cases} 1/x & 0 < y < x \\ 0 & \text{otherwise.} \end{cases} \\ f_{Z|X,Y}(z|x, y) &= \begin{cases} 1/(1-x) & x < z < 1 \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Thus

$$f(x, y, z) = \begin{cases} \frac{1}{x(1-x)} & 0 < y < x < z < 1 \\ 0 & \text{otherwise.} \end{cases} \quad \blacklozenge$$

The relationship

$$f(x, y, z) = f_X(x)f_{Y|X}(y|x)f_{Z|X,Y}(z|x, y), \quad (8.24)$$

which was established in Example 8.23, can be generalized to n continuous random variables. Let f be the joint probability density function of $X_1, X_2, \dots, X_n$. We have

$$\begin{aligned} f(x_1, x_2, \dots, x_n) &= f_{X_1}(x_1)f_{X_2|X_1}(x_2|x_1)f_{X_3|X_1, X_2}(x_3|x_1, x_2) \\ &\quad \cdots f_{X_n|X_1, X_2, \dots, X_{n-1}}(x_n|x_1, x_2, \dots, x_{n-1}). \end{aligned} \quad (8.25)$$

Example 8.24 Let X_1 be a random point from the interval $(0, 1)$, X_2 be a random point from the interval $(0, X_1)$, X_3 be a random point from the interval $(0, X_2)$, $\dots$, and X_n be a random point from the interval $(0, X_{n-1})$. Find the joint probability density function of $X_1, X_2, \dots, X_n$.

Solution: Let f be the joint probability density function of $X_1, X_2, \dots, X_n$. By (8.25),

$$\begin{aligned} f(x_1, x_2, \dots, x_n) &= f_{X_1}(x_1)f_{X_2|X_1}(x_2|x_1)f_{X_3|X_1, X_2}(x_3|x_1, x_2) \\ &\quad \cdots f_{X_n|X_1, X_2, \dots, X_{n-1}}(x_n|x_1, x_2, \dots, x_{n-1}). \end{aligned}$$

Therefore, for $0 < x_n < x_{n-1} < \dots < x_2 < x_1 < 1$,

$$f(x_1, x_2, \dots, x_n) = 1 \cdot \frac{1}{x_1} \cdot \frac{1}{x_2} \cdot \dots \cdot \frac{1}{x_{n-1}} = \frac{1}{x_1 x_2 \cdots x_{n-1}},$$

and $f(x_1, x_2, \dots, x_n) = 0$, otherwise. $\blacklozenge$

EXERCISES

A

1. Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{25}(x^2 + y^2) & \text{if } x = 1, 2, \quad y = 0, 1, 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find $p_{X|Y}(x|y)$, $P(X = 2 | Y = 1)$, and $E(X | Y = 1)$.

2. Regions A and B are prone to dust storms. An actuary has calculated that the number of dust storms that strike region A, in a year, is binomial with parameters 4 and 0.7. Furthermore, she has observed that if, in a year, n dust storms strike region A, then the number of dust storms striking region B is n with probability $2/3$ and $n + 1$ with probability $1/3$. What is the expected number of dust storms in region A in a year in which 3 dust storms strike region B.
3. Let the joint probability density function of continuous random variables X and Y be given by

$$f(x, y) = \begin{cases} 2 & \text{if } 0 < x < y < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $f_{X|Y}(x|y)$.

4. An unbiased coin is flipped until the sixth head is obtained. If the third head occurs on the fifth flip, what is the probability mass function of the number of flips?
5. Let the conditional probability density function of X given that $Y = y$ be given by

$$f_{X|Y}(x|y) = \frac{3(x^2 + y^2)}{3y^2 + 1}, \quad 0 < x < 1, \quad 0 < y < 1.$$

Find $P(1/4 < X < 1/2 \mid Y = 3/4)$.

6. Let X and Y be independent discrete random variables. Prove that for all y , $E(X \mid Y = y) = E(X)$. Do the same for continuous random variables X and Y .
7. Let X and Y be continuous random variables with joint probability density function

$$f(x, y) = \begin{cases} x + y & \text{if } 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Calculate $f_{X|Y}(x|y)$.

8. Let X and Y be continuous random variables with joint probability density function given by

$$f(x, y) = \begin{cases} e^{-x(y+1)} & \text{if } x \geq 0, \quad 0 \leq y \leq e - 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Calculate $E(X \mid Y = y)$.

9. A box contains 5 blue, 10 green, and 5 red chips. We draw 4 chips at random and without replacement. If exactly one of them is blue, what is the probability mass function of the number of green balls drawn?
10. First a point Y is selected at random from the interval $(0, 1)$. Then another point X is selected at random from the interval $(Y, 1)$. Find the probability density function of X .

11. Let (X, Y) be a random point from a unit disk centered at the origin. Find $P(0 \leq X \leq 4/11 \mid Y = 4/5)$.
12. An actuary working for an insurance company has calculated that the time that it will take for an insured driver to report an accident he or she has been involved in, in days, is a continuous random variable X , with the probability density function

$$f(x) = \begin{cases} \frac{196 - x^2}{579} & 0 < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

Furthermore, the actuary has discovered that if the accident is reported x days after its occurrence, and if the insured is entitled to claim payment, the time that it will take until he or she receives payment is a uniform random variable between $x + 7$ and 21 days. What is the probability that a driver involved in an accident and entitled to claim payment will receive his or her payment within 14 days from the time of the accident?

13. Let $p(x, y, z) = (xyz)/162$, $x = 4, 5$, $y = 1, 2, 3$, and $z = 1, 2$, be the joint probability mass function of the random variables X, Y, Z . Calculate the joint marginal probability mass function of X and Y , the joint marginal probability mass function of Y and Z , and the joint marginal probability mass function of X and Z .
14. Let the joint probability density function of X, Y , and Z be given by

$$f(x, y, z) = \begin{cases} 6e^{-x-y-z} & \text{if } 0 < x < y < z < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

Calculate the joint marginal probability density functions of X and Y , the joint marginal probability density functions of Y and Z , and the joint marginal probability density functions of X and Z .

15. The joint probability density function of X and Y is given by

$$f(x, y) = \begin{cases} ce^{-x} & \text{if } x \geq 0, \quad |y| < x \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine the constant c .
- (b) Find $f_{X|Y}(x|y)$ and $f_{Y|X}(y|x)$.
- (c) Calculate $E(Y \mid X = x)$ and $\text{Var}(Y \mid X = x)$.
16. Leon leaves his office every day at a random time between 4:30 P.M. and 5:00 P.M. If he leaves t minutes past 4:30, the time it will take him to reach home is a random number between 20 and $20 + (2t)/3$ minutes. Let Y be the number of minutes past 4:30 that Leon leaves his office tomorrow and X be the number of minutes it takes him to reach home. Find the joint probability density function of X and Y .

17. Let the joint probability density function of X , Y , and Z be given by

$$f(x, y, z) = \begin{cases} -\frac{\ln x}{xy} & \text{if } 0 < z \leq y \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $f_{X,Y}(x, y)$.

18. Show that if $\{N(t) : t \geq 0\}$ is a Poisson process, the conditional distribution of the first arrival time given $N(t) = 1$ is uniform on $(0, t)$.
19. In a sequence of independent Bernoulli trials, let X be the number of successes in the first m trials and Y be the number of successes in the first n trials, $m < n$. Show that the conditional distribution of X , given $Y = y$, is hypergeometric. Also, find the conditional distribution of Y given $X = x$.

B

20. A point is selected at random and uniformly from the region

$$R = \{(x, y) : |x| + |y| \leq 1\}.$$

Find the conditional probability density function of X given $Y = y$.

21. Let $\{N(t) : t \geq 0\}$ be a Poisson process. For $s < t$ show that the conditional distribution of $N(s)$ given $N(t) = n$ is binomial with parameters n and $p = s/t$. Also find the conditional distribution of $N(t)$ given $N(s) = k$.
22. Cards are drawn from an ordinary deck of 52, one at a time, randomly and with replacement. Let X and Y denote the number of draws until the first ace and the first king are drawn, respectively. Find $E(X | Y = 5)$.
23. A box contains 10 red and 12 blue chips. Suppose that 18 chips are drawn, one by one, at random and with replacement. If it is known that 10 of them are blue, show that the expected number of blue chips in the first nine draws is five.
24. Let X and Y be continuous random variables with joint probability density function

$$f(x, y) = \begin{cases} n(n-1)(y-x)^{n-2} & \text{if } 0 \leq x \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the conditional expectation of Y given that $X = x$.

25. A point (X, Y) is selected randomly from the triangle with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$.
- Find the joint probability density function of X and Y .
 - Calculate $f_{X|Y}(x|y)$.
 - Evaluate $E(X | Y = y)$.

26. Let X and Y be discrete random variables with joint probability mass function

$$p(x, y) = \frac{1}{e^2 y! (x - y)!}, \quad x = 0, 1, 2, \dots, \quad y = 0, 1, 2, \dots, x,$$

$p(x, y) = 0$, elsewhere. Find $E(Y \mid X = x)$.

27. Suppose that the lifetimes of a certain brand of transistor are identically distributed and independent random variables with probability distribution function F . These transistors are randomly selected, one at a time, and their lifetimes are measured. Let the N th be the first transistor that will last longer than s hours. Let X_N be the lifetime of this transistor. Are N and X_N independent random variables?
28. The lifetimes of batteries manufactured by a certain company are identically distributed with probability distribution and density functions F and f , respectively. Suppose that a battery manufactured by this company is installed at time 0 and begins to operate. If at time s an inspector finds the battery dead, in terms of F , f , and s , find the expected lifetime of the dead battery.



Self-Quiz on Section 8.4

Time allotted: 20 Minutes

Each problem is worth 5 points.

- There are 16 pea seeds in a packet available to a gardener. Seven of the peas are Wando, 4 are Maestro, and 5 are Lincoln. To test the soil of a specific garden for growing peas, the gardener picks 8 of these seeds at random to plant in that garden. Let X be the number of Lincoln seeds and Y be the number of Wando seeds chosen. Find the joint probability mass function of X and Y , the conditional probability mass function of X given that $Y = y$, and $P(X = 3 \mid Y = 3)$.
- A bivariate probability density function is called **copula** if its marginal probability density functions are uniform random variables over the interval $(0, 1)$. For a constant α between -1 and 1 , let X and Y be continuous random variables with the joint probability density function given by

$$f(x, y) = 1 + \alpha(1 - 2x)(1 - 2y), \quad 0 < x < 1, \quad 0 < y < 1.$$

The function f is an example of a copula bivariate probability density function. Copula distributions have significant applications in portfolio optimization and financial risk assessment. Find $f_{X|Y}(x|y)$ and $E(X \mid Y = y)$.



8.5 TRANSFORMATIONS OF TWO RANDOM VARIABLES

In our preceding discussions of random variables, cases have arisen where we have calculated the distribution and the density functions of a function of a random variable X . Functions such as X^2 , e^X , $\cos X$, $X^3 + 1$, and so on. In particular, in [Section 5.2](#) we explained how, in general, density functions and distribution functions of such functions can be obtained. In this section, we demonstrate a method for finding the joint density function of functions of two random variables. The key is the following theorem, which is the analog of the change of variable theorem for functions of several variables. For extending this method to more than two random variables, please consult Section 8A on the Instructor and Student website associated with this book.

Theorem 8.3 *Let X and Y be continuous random variables with joint probability density function $f(x, y)$. Let h_1 and h_2 be real-valued functions of two variables, $U = h_1(X, Y)$ and $V = h_2(X, Y)$. Suppose that*

- (a) *$u = h_1(x, y)$ and $v = h_2(x, y)$ defines a one-to-one transformation of a set R in the xy -plane onto a set Q in the uv -plane. That is, for $(u, v) \in Q$, the system of two equations in two unknowns,*

$$\begin{cases} h_1(x, y) = u \\ h_2(x, y) = v, \end{cases} \quad (8.26)$$

has a unique solution $x = w_1(u, v)$ and $y = w_2(u, v)$ for x and y , in terms of u and v , and

- (b) *the functions w_1 and w_2 have continuous partial derivatives, and the Jacobian of the transformation $x = w_1(u, v)$ and $y = w_2(u, v)$ is nonzero at all points $(u, v) \in Q$; that is, the following 2×2 determinant is nonzero on Q :*

$$\mathbf{J} = \begin{vmatrix} \frac{\partial w_1}{\partial u} & \frac{\partial w_1}{\partial v} \\ \frac{\partial w_2}{\partial u} & \frac{\partial w_2}{\partial v} \end{vmatrix} = \frac{\partial w_1}{\partial u} \frac{\partial w_2}{\partial v} - \frac{\partial w_1}{\partial v} \frac{\partial w_2}{\partial u} \neq 0.$$

Then the random variables U and V are jointly continuous with the joint probability density function $g(u, v)$ given by

$$g(u, v) = \begin{cases} f(w_1(u, v), w_2(u, v)) |\mathbf{J}| & (u, v) \in Q \\ 0 & \text{elsewhere.} \end{cases} \quad (8.27)$$

Theorem 8.3 is a result of the change of a variable theorem in double integrals. To see this, let B be a subset of Q in the uv -plane. Suppose that in the xy -plane, $A \subseteq R$ is the set that is transformed to B by the one-to-one transformation (8.26). Clearly, the events $(U, V) \in B$ and $(X, Y) \in A$ are equiprobable. Therefore,

$$P((U, V) \in B) = P((X, Y) \in A) = \iint_A f(x, y) dx dy.$$

Using the change of variable formula for double integrals, we have

$$\iint_A f(x, y) dx dy = \iint_B f(w_1(u, v), w_2(u, v)) |\mathbf{J}| du dv.$$

Hence

$$P((U, V) \in B) = \iint_B f(w_1(u, v), w_2(u, v)) |\mathbf{J}| du dv.$$

Since this is true for all subsets B of Q , it shows that $g(u, v)$, the joint density of U and V , is given by (8.27).

Example 8.25 Let X and Y be positive independent random variables with the identical probability density function e^{-x} for $x > 0$. Find the joint probability density function of $U = X + Y$ and $V = X/Y$.

Solution: Let $f(x, y)$ be the joint probability density function of X and Y . Then

$$f_X(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0, \end{cases}$$

$$f_Y(y) = \begin{cases} e^{-y} & \text{if } y > 0 \\ 0 & \text{if } y \leq 0. \end{cases}$$

Therefore,

$$f(x, y) = f_X(x)f_Y(y) = \begin{cases} e^{-(x+y)} & \text{if } x > 0 \text{ and } y > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Let $h_1(x, y) = x + y$ and $h_2(x, y) = x/y$. Then the system of equations

$$\begin{cases} x + y = u \\ \frac{x}{y} = v \end{cases}$$

has the unique solution $x = (uv)/(v + 1)$, $y = u/(v + 1)$, and

$$\mathbf{J} = \begin{vmatrix} \frac{v}{v+1} & \frac{u}{(v+1)^2} \\ \frac{1}{v+1} & -\frac{u}{(v+1)^2} \end{vmatrix} = -\frac{u}{(v+1)^2} \neq 0,$$

since $x > 0$ and $y > 0$ imply that $u > 0$ and $v > 0$; that is,

$$Q = \{(u, v) : u > 0 \text{ and } v > 0\}.$$

Hence, by Theorem 8.3, $g(u, v)$, the joint probability density function of U and V , is given by

$$g(u, v) = e^{-u} \frac{u}{(v+1)^2}, \quad u > 0 \text{ and } v > 0. \quad \blacklozenge$$

The following example demonstrates the widely recognized theorem known as **Box-Muller's theorem**. We will provide a detailed explanation of this theorem in Section 12.4 of Chapter 12, which can be accessed on the Instructor and Student website for this text. It is worth noting that this theorem is utilized for simulating normal random variables, as outlined in Theorem 12.3 within the same section.

Example 8.26 Let X and Y be two independent uniform random variables over $(0, 1)$. Show that the random variables $U = \cos(2\pi X)\sqrt{-2 \ln Y}$ and $V = \sin(2\pi X)\sqrt{-2 \ln Y}$ are independent standard normal random variables.

Solution: Let $h_1(x, y) = \cos(2\pi x)\sqrt{-2 \ln y}$ and $h_2(x, y) = \sin(2\pi x)\sqrt{-2 \ln y}$. Then the system of equations

$$\begin{cases} \cos(2\pi x)\sqrt{-2 \ln y} = u \\ \sin(2\pi x)\sqrt{-2 \ln y} = v \end{cases}$$

defines a one-to-one transformation of the set

$$R = \{(x, y) : 0 < x < 1, 0 < y < 1\}$$

onto

$$Q = \{(u, v) : -\infty < u < \infty, -\infty < v < \infty\};$$

hence it can be solved uniquely in terms of x and y . To see this, square both sides of these equations and sum them up. We obtain $-2 \ln y = u^2 + v^2$, which gives

$$y = \exp \left[-(u^2 + v^2)/2 \right].$$

Putting $-2 \ln y = u^2 + v^2$ back into these equations, we get

$$\cos 2\pi x = \frac{u}{\sqrt{u^2 + v^2}} \quad \text{and} \quad \sin 2\pi x = \frac{v}{\sqrt{u^2 + v^2}},$$

which enable us to determine the unique value of x . For example, if $u > 0$ and $v > 0$, then $2\pi x$ is uniquely determined in the first quadrant from $2\pi x = \arccos(u/\sqrt{u^2 + v^2})$. Hence the first condition of Theorem 8.3 is satisfied. To check the second condition, note that for $u > 0$ and $v > 0$,

$$w_1(u, v) = \frac{1}{2\pi} \arccos \left(\frac{u}{\sqrt{u^2 + v^2}} \right),$$

$$w_2(u, v) = \exp \left[-(u^2 + v^2)/2 \right].$$

Hence

$$\begin{aligned} \mathbf{J} &= \begin{vmatrix} \frac{\partial w_1}{\partial u} & \frac{\partial w_1}{\partial v} \\ \frac{\partial w_2}{\partial u} & \frac{\partial w_2}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{-v}{2\pi(u^2 + v^2)} & \frac{u}{2\pi(u^2 + v^2)} \\ -u \exp[-(u^2 + v^2)/2] & -v \exp[-(u^2 + v^2)/2] \end{vmatrix} \\ &= \frac{1}{2\pi} \exp[-(u^2 + v^2)/2] \neq 0. \end{aligned}$$

Now X and Y being two independent uniform random variables over $(0, 1)$ imply that f , their joint probability density function is

$$f(x, y) = f_X(x)f_Y(y) = \begin{cases} 1 & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Hence, by Theorem 8.3, $g(u, v)$, the joint probability density function of U and V is given by

$$g(u, v) = \frac{1}{2\pi} \exp[-(u^2 + v^2)/2], \quad -\infty < u < \infty, -\infty < v < \infty.$$

The probability density function of U is calculated as follows:

$$g_U(u) = \int_{-\infty}^{\infty} \frac{1}{2\pi} \exp\left(-\frac{u^2 + v^2}{2}\right) dv = \frac{1}{2\pi} \exp\left(-\frac{u^2}{2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{v^2}{2}\right) dv,$$

where $\int_{-\infty}^{\infty} (1/\sqrt{2\pi}) \exp(-v^2/2) dv = 1$ implies that $\int_{-\infty}^{\infty} \exp(-v^2/2) dv = \sqrt{2\pi}$.

Therefore,

$$g_U(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right),$$

which shows that U is standard normal. Similarly,

$$g_V(v) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{v^2}{2}\right).$$

Since $g(u, v) = g_U(u)g_V(v)$, U and V are independent standard normal random variables. $\blacklozenge$

As an application of Theorem 8.3, we now prove the following theorem, an excellent resource for calculation of density and distribution functions of sums of continuous *independent* random variables.

Theorem 8.4 (Convolution Theorem) *Let X and Y be continuous independent random variables with probability density functions f_1 and f_2 and probability distribution functions F_1 and F_2 , respectively. Then g and G , the probability density and distribution functions of $X + Y$, respectively, are given by*

$$g(t) = \int_{-\infty}^{\infty} f_1(x)f_2(t - x) dx,$$

$$G(t) = \int_{-\infty}^{\infty} f_1(x)F_2(t-x) dx.$$

Proof: Let $f(x, y)$ be the joint probability density function of X and Y . Then $f(x, y) = f_1(x)f_2(y)$. Let $U = X + Y$, $V = X$, $h_1(x, y) = x + y$, and $h_2(x, y) = x$. Then the system of equations

$$\begin{cases} x + y = u \\ x = v \end{cases}$$

has the unique solution $x = v$, $y = u - v$, and

$$\mathbf{J} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1 \neq 0.$$

Hence, by Theorem 8.3, the joint probability density function of U and V , $\psi(u, v)$, is given by

$$\psi(u, v) = f_1(v)f_2(u-v)|\mathbf{J}| = f_1(v)f_2(u-v).$$

Therefore, the marginal probability density function of $U = X + Y$ is

$$g(u) = \int_{-\infty}^{\infty} \psi(u, v) dv = \int_{-\infty}^{\infty} f_1(v)f_2(u-v) dv,$$

which is the same as

$$g(t) = \int_{-\infty}^{\infty} f_1(x)f_2(t-x) dx.$$

To find $G(t)$, the distribution function of $X + Y$, note that

$$\begin{aligned} G(t) &= \int_{-\infty}^t g(u) du = \int_{-\infty}^t \left(\int_{-\infty}^{\infty} f_1(x)f_2(u-x) dx \right) du \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^t f_2(u-x) du \right) f_1(x) dx \\ &= \int_{-\infty}^{\infty} F_2(t-x)f_1(x) dx, \end{aligned}$$

where, letting $s = u - x$, the last equality follows from

$$\int_{-\infty}^t f_2(u-x) du = \int_{-\infty}^{t-x} f_2(s) ds = F_2(t-x). \quad \blacklozenge$$

Note that by symmetry we can also write

$$g(t) = \int_{-\infty}^{\infty} f_2(y)f_1(t-y) dy,$$

$$G(t) = \int_{-\infty}^{\infty} f_2(y)F_1(t-y) dy.$$

Definition 8.3 Let f_1 and f_2 be two probability density functions. Then the function $g(t)$, defined by

$$g(t) = \int_{-\infty}^{\infty} f_1(x)f_2(t-x) dx,$$

is called the **convolution** of f_1 and f_2 .

Theorem 8.4 shows that

If X and Y are independent continuous random variables, the probability density function of $X + Y$ is the convolution of the probability density functions of X and Y .

Theorem 8.4 is also valid for discrete random variables. Let p_X and p_Y be probability mass functions of two discrete random variables X and Y . Then the function

$$p(z) = \sum_x p_X(x)p_Y(z-x)$$

is called the **convolution** of p_X and p_Y . It is readily seen that if X and Y are independent, the probability mass function of $X + Y$ is the convolution of the probability mass functions of X and Y :

$$\begin{aligned} P(X+Y=z) &= \sum_x P(X=x, Y=z-x) = \sum_x P(X=x)P(Y=z-x) \\ &= \sum_x p_X(x)p_Y(z-x). \end{aligned}$$

Exercise 6, a famous example given by W. J. Hall, shows that the converse of Theorem 8.4 is not valid. That is, it may happen that the probability mass function of two dependent random variables X and Y is the convolution of the probability mass functions of X and Y .

Example 8.27 Let X and Y be independent exponential random variables, each with parameter λ . Find the distribution function of $X + Y$.

Solution: Let f_1 and f_2 be the probability density functions of X and Y , respectively. Then

$$f_1(x) = f_2(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Hence

$$f_2(t-x) = \begin{cases} \lambda e^{-\lambda(t-x)} & \text{if } 0 \leq x \leq t \\ 0 & \text{otherwise.} \end{cases}$$

By convolution theorem, h , the probability density function of $X + Y$, is given by

$$h(t) = \int_{-\infty}^{\infty} f_2(t-x)f_1(x) dx = \int_0^t \lambda e^{-\lambda(t-x)} \cdot \lambda e^{-\lambda x} dx = \lambda^2 t e^{-\lambda t}, \quad t \geq 0.$$

This is the density function of a gamma random variable with parameters 2 and λ . Hence $X + Y$ is gamma with parameters 2 and λ . We will study a generalization of this important theorem in [Section 10.2](#). ♦

EXERCISES

A

1. Let X and Y be independent random numbers from the interval $(0, 1)$. Find the joint probability density function of $U = -2 \ln X$ and $V = -2 \ln Y$.
2. Let X and Y be two positive independent continuous random variables with the probability density functions $f_1(x)$ and $f_2(y)$, respectively. Find the probability density function of $U = X/Y$.
Hint: Let $V = X$; find the joint probability density function of U and V . Then calculate the marginal probability density function of U .
3. Let $X \sim N(0, 1)$ and $Y \sim N(0, 1)$ be independent random variables. Find the joint probability density function of $R = \sqrt{X^2 + Y^2}$ and $\Theta = \arctan(Y/X)$. Show that R and Θ are independent. Note that (R, Θ) is the polar coordinate representation of (X, Y) .
4. Let X and Y be continuous random variables with the joint probability density function given by

$$f(x, y) = \begin{cases} 8xy & 0 < y \leq x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability density function of $U = XY$.

5. From the interval $(0, 1)$, two random numbers X and Y are selected independently. Show that the probability density function of their sum is given by

$$g(t) = \begin{cases} t & \text{if } 0 \leq t < 1 \\ 2 - t & \text{if } 1 \leq t < 2 \\ 0 & \text{otherwise.} \end{cases}$$

Because of the shape of the probability density function of $X + Y$, the random variable $X + Y$ is said to have a **triangular distribution**. (See [Figure 8.3](#).)

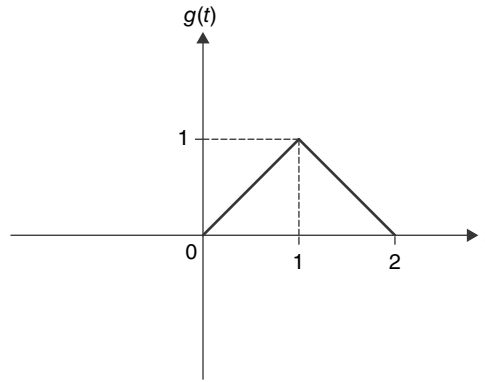


Figure 8.3 The shape of a triangular probability density function.

6. Let $-1/9 < c < 1/9$ be a constant. Let $p(x, y)$, the joint probability mass function of the random variables X and Y , be given by the following table:

x	y		
	-1	0	1
-1	$1/9$	$1/9 - c$	$1/9 + c$
0	$1/9 + c$	$1/9$	$1/9 - c$
1	$1/9 - c$	$1/9 + c$	$1/9$

- (a) Show that the probability mass function of $X + Y$ is the convolution function of the probability mass functions of X and Y for all c .
- (b) Show that X and Y are independent if and only if $c = 0$.
7. All international passengers arriving at a U.S. airport must go through the first stage of an immigration process, which takes an exponential length of time with parameter λ . However, experience shows that for some p , $0 < p < 1$, $100p\%$ of the passengers have discrepancies in their travel related documents and must go through a second stage, which is exponential with parameter μ independently of stage 1. Determine the probability density function of the time a randomly selected international passenger spends in the Airport Immigration Office upon arrival to the United States. Such a probability density function is called **Coxian**.

B

8. Let X and Y be independent random variables with common probability density function

$$f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \geq 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Calculate the joint probability density function of $U = X/Y$ and $V = XY$.

9. Let X and Y be independent random variables with common probability density function

$$f(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Find the joint probability density function of $U = X + Y$ and $V = e^X$.

10. Prove that if X and Y are independent standard normal random variables, then $X + Y$ and $X - Y$ are independent random variables. This is a special case of the following important theorem.

Let X and Y be independent random variables with a common distribution F . The random variables $X + Y$ and $X - Y$ are independent if and only if F is a normal distribution function.

11. Let X and Y be independent (strictly positive) gamma random variables with parameters (r_1, λ) and (r_2, λ) , respectively. Define $U = X + Y$ and $V = X/(X + Y)$.
- (a) Find the joint probability density function of U and V .
 - (b) Prove that U and V are independent.
 - (c) Show that U is gamma and V is beta.
12. Let X and Y be independent gamma random variables with parameters $(n/2, 1/2)$ and $(m/2, 1/2)$, respectively, where n and m are positive integers. The random variable $U = (X/n)/(Y/m)$ is said to have **F-distribution** with n and m degrees of freedom, and has significant applications in statistics. Find the probability density function of U .
13. Let X and Y be independent (strictly positive) exponential random variables each with parameter λ . Are the random variables $X + Y$ and X/Y independent? ■

Self-Quiz on Section 8.5

Time allotted: 25 Minutes

Each problem is worth 5 points.

- Let X and Y be two independently selected random numbers from the interval $(0, 1)$. Find the joint probability density function of $U = X + Y$ and $V = X - Y$.
- The joint probability density function of the continuous random variables X and Y is given by

$$f(x, y) = \begin{cases} 2e^{-(x+2y)} & x \geq 0, y \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find h , the probability density function of $X + Y$.



CHAPTER 8 SUMMARY

► **Covariance** Let X and Y be jointly distributed random variables; then the *covariance* of X and Y is defined by $\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$.

Note that $\text{Cov}(X, X) = \text{Var}(X)$.

► For real numbers a and b and random variables X, Y ,

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y),$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y),$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2 \text{Cov}(X, Y),$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

► In general, for random variables $X_1, X_2, \dots, X_n$ and $Y_1, Y_2, \dots, Y_m$, and real numbers $a, b, c, d, a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_m$,

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$$

$$\text{Cov}\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{Cov}(X_i, Y_j)$$

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

► **Sample Mean** Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution F with mean μ and variance σ^2 . Let $\bar{X}$ be the mean of the random sample. Then

$$E(\bar{X}) = \mu \quad \text{and} \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

► **Correlation** If $\text{Cov}(X, Y) > 0$, we say that X and Y are *positively correlated*. If $\text{Cov}(X, Y) < 0$, we say that they are *negatively correlated*. If $\text{Cov}(X, Y) = 0$, we say that X and Y are *uncorrelated*. If X and Y are independent, then they are uncorrelated. However, the converse of this is not true; that is, *two dependent random variables might be uncorrelated*. If $X_1, X_2, \dots, X_n$ are *pairwise independent* or, more generally, *pairwise uncorrelated*, then

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i).$$

► **Correlation Coefficient** Let X and Y be two random variables with $0 < \sigma_X^2 < \infty$ and $0 < \sigma_Y^2 < \infty$. The *correlation coefficient* between X and Y , denoted by $\rho = \rho(X, Y)$, is defined by

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

and satisfies the following important properties

- (a) $-1 \leq \rho(X, Y) \leq 1$.
- (b) With probability 1, $\rho(X, Y) = 1$ if and only if $Y = aX + b$ for some constants a, b , $a > 0$.
- (c) With probability 1, $\rho(X, Y) = -1$ if and only if $Y = aX + b$ for some constants a, b , $a < 0$.

► **Conditional Distributions: Discrete Case** Let X be a discrete random variable with set of possible values A , and let Y be a discrete random variable with set of possible values B . Let $p(x, y)$ be the joint probability mass function of X and Y , and let p_X and p_Y be the marginal probability mass functions of X and Y . The *conditional probability mass function of X given that $Y = y$* , denoted by $p_{X|Y}(x|y)$, is defined as follows:

$$p_{X|Y}(x|y) = P(X = x | Y = y) = \frac{p(x, y)}{p_Y(y)}, \quad x \in A, y \in B,$$

where $p_Y(y) > 0$. The *conditional expectation of the random variable X given that $Y = y$* is as follows:

$$E(X | Y = y) = \sum_{x \in A} xP(X = x | Y = y) = \sum_{x \in A} xp_{X|Y}(x|y),$$

where $p_Y(y) > 0$. Similarly, for $p_X(x) > 0$,

$$E(Y | X = x) = \sum_{y \in B} yp_{Y|X}(y|x),$$

and if h is an ordinary function from $\mathbb{R}$ to $\mathbb{R}$, then

$$E[h(X) | Y = y] = \sum_{x \in A} h(x)p_{X|Y}(x|y).$$

► **Conditional Distributions: Continuous Case** Let X and Y be two continuous random variables with the joint probability density function $f(x, y)$. The *conditional probability density function of X given that $Y = y$* , $f_{X|Y}(x|y)$, is defined as follows:

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)},$$

provided that $f_Y(y) > 0$. Similarly, The *conditional probability density function of Y given that $X = x$* , $f_{Y|X}(y|x)$, is defined as follows:

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)},$$

provided that $f_X(x) > 0$. The *conditional expectation of X given that $Y = y$* is as follows:

$$E(X | Y = y) = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx,$$

where $f_Y(y) > 0$. Similarly, the *conditional expectation of Y given that $X = x$* is given by

$$E(Y | X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy,$$

where $f_X(x) > 0$. If h is an ordinary function from $\mathbb{R}$ to $\mathbb{R}$, then,

$$E[h(X) | Y = y] = \int_{-\infty}^{\infty} h(x) f_{X|Y}(x|y) dx.$$

► Let the discrete random variables $X_1, X_2, \dots, X_n$ have a joint probability mass function denoted by $p(x_1, x_2, \dots, x_n)$. The *joint probability mass function marginalized* over a given set of k of these random variables is obtained by summing all terms of $p(x_1, x_2, \dots, x_n)$ over all possible values of the remaining $n - k$ random variables. Similarly, let the continuous random variables $X_1, X_2, \dots, X_n$ have a joint probability density function denoted by $f(x_1, x_2, \dots, x_n)$. To find the *joint probability density function marginalized* over a given set of k of these random variables, integrate $f(x_1, x_2, \dots, x_n)$ over all possible values of the remaining $n - k$ random variables.

► Let f be the joint probability density function of $X_1, X_2, \dots, X_n$. We have

$$\begin{aligned} f(x_1, x_2, \dots, x_n) &= f_{X_1}(x_1) f_{X_2|X_1}(x_2|x_1) f_{X_3|X_1, X_2}(x_3|x_1, x_2) \\ &\quad \cdots f_{X_n|X_1, X_2, \dots, X_{n-1}}(x_n|x_1, x_2, \dots, x_{n-1}). \end{aligned}$$

In particular, for continuous random variables X, Y , and Z with joint probability density function f ,

$$f(x, y, z) = f_X(x) f_{Y|X}(y|x) f_{Z|X, Y}(z|x, y),$$

► **Transformations of Two Random Variables** Let X and Y be continuous random variables with joint probability density function $f(x, y)$. Let h_1 and h_2 be real-valued functions of two variables, $U = h_1(X, Y)$ and $V = h_2(X, Y)$. Suppose that

- (a) $u = h_1(x, y)$ and $v = h_2(x, y)$ defines a *one-to-one* transformation of a set R in the xy -plane *onto* a set Q in the uv -plane. That is, for $(u, v) \in Q$, the system of two equations in two unknowns,

$$\begin{cases} h_1(x, y) = u \\ h_2(x, y) = v, \end{cases}$$

has a unique solution $x = w_1(u, v)$ and $y = w_2(u, v)$ for x and y , in terms of u and v ; and

- (b) the functions w_1 and w_2 have continuous partial derivatives, and the Jacobian of the transformation $x = w_1(u, v)$ and $y = w_2(u, v)$ is nonzero at all points $(u, v) \in Q$; that is, the following 2×2 determinant is nonzero on Q :

$$\mathbf{J} = \begin{vmatrix} \frac{\partial w_1}{\partial u} & \frac{\partial w_1}{\partial v} \\ \frac{\partial w_2}{\partial u} & \frac{\partial w_2}{\partial v} \end{vmatrix} = \frac{\partial w_1}{\partial u} \frac{\partial w_2}{\partial v} - \frac{\partial w_1}{\partial v} \frac{\partial w_2}{\partial u} \neq 0.$$

Then the random variables U and V are jointly continuous with the joint probability density function $g(u, v)$ given by

$$g(u, v) = \begin{cases} f(w_1(u, v), w_2(u, v)) |\mathbf{J}| & (u, v) \in Q \\ 0 & \text{elsewhere.} \end{cases}$$

► **Convolution Theorem** Let X and Y be continuous independent random variables with probability density functions f_1 and f_2 and probability distribution functions F_1 and F_2 , respectively. Then g and G , the probability density and distribution functions of $X + Y$, respectively, are given by

$$g(t) = \int_{-\infty}^{\infty} f_1(x) f_2(t - x) dx,$$

$$G(t) = \int_{-\infty}^{\infty} f_1(x) F_2(t - x) dx.$$

REVIEW PROBLEMS

1. Let the joint probability density function of X , Y , and Z be given by

$$f(x, y, z) = \begin{cases} 8xyz & \text{if } 0 < x < 1, \quad 0 < y < 1, \quad 0 < z < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $\rho(X, Y)$, $\rho(X, Z)$, and $\rho(Y, Z)$.

2. Let X and Y be jointly distributed with $\rho(X, Y) = 2/3$, $\sigma_X = 1$, $\text{Var}(Y) = 9$. Find $\text{Var}(3X - 5Y + 7)$.
3. Let the joint probability mass function of discrete random variables X and Y be given by

$$p(x, y) = \begin{cases} \frac{1}{25}(x^2 + y^2) & \text{if } (x, y) = (1, 1), (1, 3), (2, 3) \\ 0 & \text{otherwise.} \end{cases}$$

Find $\text{Cov}(X, Y)$.

4. Two dice are rolled. The sum of the outcomes is denoted by X and the absolute value of their difference by Y . Calculate the covariance of X and Y . Are X and Y uncorrelated? Are they independent?
5. A fair coin is flipped 20 times. If the total number of heads is 12, what is the expected number of heads in the first 10 flips?
6. Let X and Y be continuous random variables with the joint probability density function

$$f(x, y) = \begin{cases} e^{-y} & \text{if } y > 0, \quad 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X^n \mid Y = y)$, $n \geq 1$.

7. A bar of length ℓ is broken into three pieces at two random spots. What is the probability that the length of at least one piece is less than $\ell/20$?
8. Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} 1 & \text{if } |y| < x, \quad 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Show that $E(Y \mid X = x)$ is a linear function of x while $E(X \mid Y = y)$ is not a linear function of y .

9. Two green and two blue dice are rolled. If X and Y are the numbers of 6s on the green and on the blue dice, respectively, calculate the correlation coefficient of $|X - Y|$ and $X + Y$.
10. In terms of the means, variances, and the covariance of the random variables X and Y , find α and β for which $E(Y - \alpha - \beta X)^2$ is minimum. This is the **method of least squares**; it fits the “best” line $y = \alpha + \beta x$ to the distribution of Y .
11. **(The Wallet Paradox)** Consider the following “paradox” given by Martin Gardner in his book *Aha! Gotcha* (W. H. Freeman and Company, New York, 1981).

Each of two persons places his wallet on the table. Whoever has the smallest amount of money in his wallet, wins all the money in the other wallet. Each of the players reason as follows: “I may lose what I have but I may also win more than I have. So the game is to my advantage.”

As Kent G. Merryfield, Ngo Viet, and Saleem Watson have observed in their paper “The Wallet Paradox” in the August–September 1997 issue of the *American Mathematical Monthly*,

Paradoxically, it seems that the game is to the advantage of both players. . . . However, the inference that “the game is to my advantage” is the source of the apparent paradox, because it does not take into account the *probabilities* of winning or losing. In other words, if the game is played many times, how often does a player win? How often does he lose? And by how much?

Following the analysis of Kent G. Merryfield, Ngo Viet, and Saleem Watson, let X and Y be the amount of money in the wallets of players A and B, respectively. Let W_A and W_B be the amount of money that player A and B will win, respectively. $W_A(X, Y) = -W_B(X, Y)$ and

$$W_A(X, Y) = \begin{cases} -X & \text{if } X > Y \\ Y & \text{if } X < Y \\ 0 & \text{if } X = Y. \end{cases}$$

Suppose that the distribution function of the money in each player’s wallet is the same; that is, X and Y are independent, identically distributed random variables on some interval $[a, b]$ or $[a, \infty)$, $0 \leq a < b < \infty$. Show that

$$E(W_A) = E(W_B) = 0. \quad \blacksquare$$

Self-Test on Chapter 8

Time allotted: 150 Minutes

Each problem is worth 10 points.

- For an integer $n > 1$, let $X_1, X_2, \dots, X_n$ be identically distributed random variables with $\text{Var}(X_i) = \sigma^2$, $1 \leq i \leq n$. For i, j , $1 \leq i \neq j \leq n$, let $\rho(X_i, X_j) = a$, for a constant a . Show that $a \geq -\frac{1}{n-1}$.

Hint: Start with $\text{Var}(X_1 + X_2 + \dots + X_n) \geq 0$.

- A point is selected randomly from the unit disk $D = \{(x, y) : x^2 + y^2 \leq 1\}$.
 - Find the conditional probability density function of X , the x -coordinate of the point selected, given that Y , its y -coordinate, is y .
 - Find the conditional expected value and the conditional variance of X given that $Y = y$.

3. A point (X, Y) is selected randomly from the annulus

$$R = \{(x, y): 4 < x^2 + y^2 < 9\}.$$

That is, the random point is inside the region between the concentric circles with radii 2 and 3. Show that X and Y are dependent but uncorrelated.

4. Let the joint probability density function of the continuous random variables X and Y be given by

$$f(x, y) = \begin{cases} xy/16 & 1 < x < 3, 1 < y < 3 \\ 0 & \text{otherwise.} \end{cases}$$

Using the convolution theorem, find the probability density function of $X + Y$.

5. Let X be the amount of time a patient with a non-life threatening condition has to spend in an emergency room before being seen by a doctor. Suppose that the emergency room serves such patients with rate Y , where Y is an exponential random variable with mean $1/2$. Also suppose that the conditional distribution for X given $Y = y$ is exponential with mean $1/y$. Find the probability density function, the mean, and the standard deviation of the rate at which the emergency room serves the non-life threatening patients when the waiting time is the constant x .

6. Amber has decided to ask randomly selected people what their birthdays are until she has met people with birthdays on each and every day of the year. Ignoring February 29, find the expected value and the standard deviation of the number of people that Amber will need to meet in order to achieve her goal.

Hint: Let $X_1 = 1$ for counting the first person that Amber will meet. Let X_2 be the number of people Amber will meet until she finds a person whose birthday is different from the birthday of the first person she met. Let X_3 be the number of people Amber will meet until she finds a person whose birthday is different from the first two birthdays that she has already recorded, and so on. Clearly, X , the number of people that Amber will need to meet in order to achieve her goal, satisfies the following equation.

$$X = X_1 + X_2 + X_3 + \cdots + X_{365}.$$

7. A fair die is tossed four times. Let X be the number of sixes obtained, and let Y be the length of the longest run of consecutive sixes. So if, for example, an outcome is $N6N6$; that is, a non-6, 6, a non-6, and 6, respectively, then $X = 2$, and $Y = 1$. Similarly, if an outcome is $66N6$, then $X = 3$ and $Y = 2$. Find the correlation coefficient of X and Y .

Hint: First find the set of possible outcomes of this experiment. Then construct a table, which represents $p(x, y)$, the joint probability mass function of X and Y , $p_X(x)$ and $p_Y(y)$. Then calculate the probability mass function of XY .

8. Let the joint probability density function of the continuous random variable X and Y be given by

$$f(x, y) = \begin{cases} 4e^{-2(x+y)} & x > 0, y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability density function of the random variable $U = \frac{X}{X + Y}$.

9. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} e^{-x} & \text{if } 0 < y < x < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Find the marginal probability density functions of X and Y .
- (b) Determine the correlation coefficient of X and Y .
10. Let X and Y be two independently selected random numbers from the interval $(0, 1)$.
- (a) Find the joint probability density function of $U = X + Y$ and $V = X/(X + Y)$.
- (b) Find the marginal probability density function of V .



Chapter 9

More Bivariate and Multivariate Topics

9.1 CONDITIONING ON RANDOM VARIABLES

An important application of conditional expectations is that ordinary expectations and probabilities can be calculated by conditioning on appropriate random variables. First, to explain this procedure, we state a definition.

Definition 9.1 *Let X and Y be two random variables. By $E(X|Y)$ we mean a function of Y that is defined to be $E(X | Y = y)$ when $Y = y$.*

Recall that a function of a random variable Y , say $h(Y)$, is defined to be $h(a)$ at all sample points at which $Y = a$. For example, if $Z = \log Y$, then at a sample point ω , where $Y(\omega) = a$, we have that $Z(\omega) = \log a$. In this definition $E(X | Y)$ is a function of Y , which at a sample point ω is defined to be $E(X | Y = y)$, where $y = Y(\omega)$. Since $E(X|Y)$ is defined only when $p_Y(y) > 0$, it is defined at all sample points ω , where $p_Y(y) > 0$, $y = Y(\omega)$. $E(X|Y)$, being a function of Y , is a random variable. Its expectation, whenever finite, is equal to the expectation of X . This extremely important property sometimes enables us to calculate expectations which are otherwise, if possible, very difficult to find. To see why the expected value of $E(X|Y)$ is $E(X)$, let X and Y be discrete random variables with sets of possible values A and B , respectively. Let $p(x, y)$ be the joint probability mass function of X and Y . On the one hand,

$$\begin{aligned} E(X) &= \sum_{x \in A} xp_X(x) = \sum_{x \in A} x \sum_{y \in B} p(x, y) \\ &= \sum_{y \in B} \sum_{x \in A} xp(x, y) = \sum_{y \in B} \sum_{x \in A} xp_{X|Y}(x|y)p_Y(y) \\ &= \sum_{y \in B} \left(\sum_{x \in A} xp_{X|Y}(x|y) \right) p_Y(y) \\ &= \sum_{y \in B} E(X | Y = y)P(Y = y), \end{aligned} \tag{9.1}$$

showing that $E(X)$ is the weighted average of the conditional expectations of X (given $Y = y$) over all possible values of Y . On the other hand, we know that, for a real-valued function h ,

$$E[h(Y)] = \sum_{y \in B} h(y)P(Y = y).$$

Applying this formula to $h(Y) = E(X|Y)$ yields

$$E[E(X|Y)] = \sum_{y \in B} E(X | Y = y)P(Y = y). \quad (9.2)$$

Comparing (9.1) and (9.2), we obtain

$$E(X) = E[E(X|Y)] = \sum_{y \in B} E(X | Y = y)P(Y = y). \quad (9.3)$$

Therefore, $E[E(X|Y)] = E(X)$ is a condensed way to say that $E(X)$ is the weighted average of the conditional expectations of X (given $Y = y$) over all possible values of Y . We have proved the following theorem for the discrete case.

Theorem 9.1 *Let X and Y be two random variables. Then*

$$E[E(X | Y)] = E(X).$$

Proof: We have already proven this for the discrete case. Now we will prove it for the case where X and Y are continuous random variables with joint probability density function, $f(x, y)$.

$$\begin{aligned} E[E(X|Y)] &= \int_{-\infty}^{\infty} E(X|Y = y)f_Y(y) dy \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx \right) f_Y(y) dy \\ &= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y) dy \right) dx \\ &= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} \frac{f(x, y)}{f_Y(y)} f_Y(y) dy \right) dx \\ &= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f(x, y) dy \right) dx \\ &= \int_{-\infty}^{\infty} x f_X(x) dx = E(X). \quad \blacklozenge \end{aligned}$$

Example 9.1 Suppose that $N(t)$, the number of people who pass by a museum at or prior to t , is a Poisson process having rate λ . If a person passing by enters the museum with probability p , what is the expected number of people who enter the museum at or prior to t ?

Solution: Let $M(t)$ denote the number of people who enter the museum at or prior to t , then

$$E[M(t)] = E[E(M(t) | N(t))] = \sum_{n=0}^{\infty} E[M(t) | N(t) = n]P(N(t) = n).$$

Given that $N(t) = n$, the number of people who enter the museum at or prior to t , is a binomial random variable with parameters n and p . Thus $E[M(t) \mid N(t) = n] = np$. Therefore,

$$E[M(t)] = \sum_{n=1}^{\infty} np \frac{e^{-\lambda t} (\lambda t)^n}{n!} = pe^{-\lambda t} \lambda t \sum_{n=1}^{\infty} \frac{(\lambda t)^{n-1}}{(n-1)!} = pe^{-\lambda t} \lambda t e^{\lambda t} = p\lambda t. \quad \blacklozenge$$

Example 9.2 Let X and Y be continuous random variables with joint probability density function

$$f(x, y) = \begin{cases} \frac{3}{2}(x^2 + y^2) & \text{if } 0 < x < 1, \quad 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X|Y)$.

Solution: $E(X|Y)$ is a random variable that is defined to be $E(X \mid Y = y)$ when $Y = y$. First, we calculate $E(X \mid Y = y)$:

$$E(X \mid Y = y) = \int_0^1 x f_{X|Y}(x|y) dx = \int_0^1 x \frac{f(x, y)}{f_Y(y)} dx,$$

where

$$f_Y(y) = \int_0^1 \frac{3}{2}(x^2 + y^2) dx = \frac{3}{2}y^2 + \frac{1}{2}.$$

Hence

$$\begin{aligned} E(X \mid Y = y) &= \int_0^1 x \frac{(3/2)(x^2 + y^2)}{(3/2)y^2 + 1/2} dx = \int_0^1 x \frac{3(x^2 + y^2)}{3y^2 + 1} dx \\ &= \frac{3}{3y^2 + 1} \int_0^1 (x^3 + xy^2) dx = \frac{3(2y^2 + 1)}{4(3y^2 + 1)}. \end{aligned}$$

Thus, if $Y = y$, then

$$E(X \mid Y = y) = \frac{3(2y^2 + 1)}{4(3y^2 + 1)}.$$

Now since the random variable $E(X|Y)$ coincides with $E(X \mid Y = y)$ if $Y = y$, we have

$$E(X|Y) = \frac{3(2Y^2 + 1)}{4(3Y^2 + 1)}. \quad \blacklozenge$$

Example 9.3 What is the expected number of random digits that should be generated to obtain three consecutive zeros?

Solution: Let X be the number of random digits to be generated until three consecutive zeros

are obtained. Let Y be the number of random digits to be generated until the first nonzero digit is obtained. Then

$$\begin{aligned} E(X) &= E[E(X|Y)] = \sum_{i=1}^{\infty} E(X | Y = i)P(Y = i) \\ &= \sum_{i=1}^3 E(X | Y = i)P(Y = i) + \sum_{i=4}^{\infty} E(X | Y = i)P(Y = i) \\ &= \sum_{i=1}^3 [i + E(X)] \left(\frac{1}{10}\right)^{i-1} \left(\frac{9}{10}\right) + \sum_{i=4}^{\infty} 3 \left(\frac{1}{10}\right)^{i-1} \left(\frac{9}{10}\right), \end{aligned}$$

which gives

$$E(X) = 1.107 + 0.999 E(X) + 0.003.$$

Solving this for $E(X)$, we find that $E(X) = 1110$. ♦

Example 9.4 Let X and Y be two random variables and f be a real-valued function from $\mathbb{R}$ to $\mathbb{R}$. Prove that

$$E[f(Y)X | Y] = f(Y)E(X|Y).$$

Proof: If $Y = y$, then $f(Y)E(X|Y)$ is $f(y)E(X | Y = y)$. We show that $E[f(Y)X|Y]$ is also equal to this quantity. Let $f_{X|Y}(x|y)$ be the conditional probability density function of X given that $Y = y$, then

$$\begin{aligned} E[f(Y)X | Y = y] &= E[f(y)X | Y = y] = \int_{-\infty}^{\infty} f(y)x f_{X|Y}(x|y) dx \\ &= f(y) \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx = f(y)E(X | Y = y). \quad \blacklozenge \end{aligned}$$

Suppose that a certain airplane breaks down N times a year, where N is a random variable. If the repair time for the i th breakdown is X_i , then the total repair time for this airplane is the random variable $X_1 + X_2 + \cdots + X_N$. To find the expected length of time that, due to breakdowns, the plane cannot fly, we need to calculate $E(\sum_{i=1}^N X_i)$. What is different about this sum is that, not only is each of its terms a random variable, but the number of its terms is also a random variable. The expected values of such sums are calculated using the following theorem, discovered by Abraham Wald, a statistician who is best known for developing the theory of sequential statistical procedures during World War II.

Theorem 9.2 (Wald's Equation) Let $X_1, X_2, \dots$ be independent and identically distributed random variables with the finite mean $E(X)$. Let $N > 0$ be an integer-valued random variable, independent of $\{X_1, X_2, \dots\}$, with $E(N) < \infty$. Then

$$E\left(\sum_{i=1}^N X_i\right) = E(N)E(X).$$

Proof: By (9.3),

$$E\left(\sum_{i=1}^N X_i\right) = E\left[E\left(\sum_{i=1}^N X_i \mid N\right)\right] = \sum_{n=1}^{\infty} E\left(\sum_{i=1}^N X_i \mid N = n\right) P(N = n), \quad (9.4)$$

where

$$E\left(\sum_{i=1}^N X_i \mid N = n\right) = E\left(\sum_{i=1}^n X_i \mid N = n\right) = E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i) = nE(X),$$

since N is independent of $\{X_1, X_2, \dots\}$. Hence, by (9.4),

$$E\left(\sum_{i=1}^N X_i\right) = \sum_{n=1}^{\infty} nE(X)P(N = n) = E(X) \sum_{n=1}^{\infty} nP(N = n) = E(X)E(N). \quad \blacklozenge$$

Example 9.5 Suppose that the average number of breakdowns for a certain airplane is 12.5 times a year. If the expected value of repair time is 7 days for each breakdown, and if the repair times are identically distributed, independent random variables, find the expected total repair time. Assume that repair times are independent of the number of breakdowns.

Solution: Let N be the number of breakdowns in a year and X_i be the repair time for the i th breakdown. Then, by Wald's equation, the expected total repair time is

$$E\left(\sum_{i=1}^N X_i\right) = E(N)E(X_i) = (12.5)(7) = 87.5. \quad \blacklozenge$$

The following theorem gives a formula, analogous to Wald's equation, for variance. We leave its proof as an exercise.

Theorem 9.3 Let $\{X_1, X_2, \dots\}$ be an independent and identically distributed sequence of random variables with finite mean $E(X)$ and finite variance $\text{Var}(X)$. Let $N > 0$ be an integer-valued random variable independent of $\{X_1, X_2, \dots\}$ with $E(N) < \infty$ and $\text{Var}(N) < \infty$. Then

$$\text{Var}\left(\sum_{i=1}^N X_i\right) = E(N)\text{Var}(X) + [E(X)]^2 \text{Var}(N).$$

We now explain a procedure for calculation of probabilities by conditioning on random variables. Let B be an event associated with an experiment and X be a discrete random variable with possible set of values A . Let

$$Y = \begin{cases} 1 & \text{if } B \text{ occurs} \\ 0 & \text{if } B \text{ does not occur.} \end{cases}$$

Then

$$E(Y) = E[E(Y|X)]. \quad (9.5)$$

But

$$E(Y) = 1 \cdot P(B) + 0 \cdot P(B^c) = P(B) \quad (9.6)$$

and

$$\begin{aligned} E[E(Y|X)] &= \sum_{x \in A} E(Y | X = x)P(X = x) \\ &= \sum_{x \in A} P(B | X = x)P(X = x), \end{aligned} \quad (9.7)$$

where the last equality follows since

$$\begin{aligned} E(Y | X = x) &= 1 \cdot P(Y = 1 | X = x) + 0 \cdot P(Y = 0 | X = x) \\ &= \frac{P(Y = 1, X = x)}{P(X = x)} = \frac{P(B \text{ and } X = x)}{P(X = x)} = P(B | X = x). \end{aligned}$$

Relations (9.5), (9.6), and (9.7) imply the following theorem.

Theorem 9.4 *Let B be an arbitrary event and X be a discrete random variable with possible set of values A , then*

$$P(B) = \sum_{x \in A} P(B | X = x)P(X = x). \quad (9.8)$$

If X is a continuous random variable, the relation analogous to (9.8) is

$$P(B) = \int_{-\infty}^{\infty} P(B | X = x)f(x) dx, \quad (9.9)$$

where f is the probability density function of X .

Theorem 9.4 shows that the probability of an event B is the weighted average of the conditional probabilities of B (given $X = x$) over all possible values of X . For the discrete case, this is a generalization of Theorem 3.4 and is the most general version of the law of total probability.

Example 9.6 The time between consecutive earthquakes in Los Angeles and the time between consecutive earthquakes in San Francisco are independent and exponentially distributed with means $1/\lambda$ and $1/\mu$, respectively. What is the probability that the next earthquake occurs in Los Angeles?

Solution: Let X and Y denote the times between now and the next earthquake in Los Angeles and San Francisco, respectively. Because of the memoryless property of exponential

distribution, X and Y are exponentially distributed with means $1/\lambda$ and $1/\mu$, respectively. To calculate $P(X < Y)$, the desired probability, we will condition on X :

$$\begin{aligned} P(X < Y) &= \int_0^\infty P(X < Y \mid X = x) \lambda e^{-\lambda x} dx \\ &= \int_0^\infty P(Y > x) \lambda e^{-\lambda x} dx = \int_0^\infty e^{-\mu x} \lambda e^{-\lambda x} dx \\ &= \lambda \int_0^\infty e^{-(\lambda+\mu)x} dx = \frac{\lambda}{\lambda + \mu} \int_0^\infty (\lambda + \mu) e^{-(\lambda+\mu)x} dx = \frac{\lambda}{\lambda + \mu}, \end{aligned}$$

where $\int_0^\infty (\lambda + \mu) e^{-(\lambda+\mu)x} dx = 1$ since its integrand is the probability density function of an exponential random variable with parameter $\lambda + \mu$, and $P(Y > x)$ is calculated from

$$P(Y > x) = 1 - P(Y \leq x) = 1 - (1 - e^{-\mu x}) = e^{-\mu x}. \quad \blacklozenge$$

Remark 9.1 In Example 9.6, we have shown the following important property of exponential random variables.

Let X and Y be independent exponential random variables with parameters λ and μ , respectively. Then

$$P(X < Y) = \frac{\lambda}{\lambda + \mu}. \quad \blacklozenge$$

Example 9.7 Suppose that Z_1 and Z_2 are independent standard normal random variables. Show that the ratio $Z_1/|Z_2|$ is a Cauchy random variable. That is, $Z_1/|Z_2|$ is a random variable with the probability density function

$$f(t) = \frac{1}{\pi(1+t^2)}, \quad -\infty < t < \infty.$$

Solution: Let $g(x)$ be the probability density function of $|Z_2|$. To find $g(x)$, note that, for $x \geq 0$,

$$P(|Z_2| \leq x) = P(-x \leq Z_2 \leq x) = \int_{-x}^x \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = 2 \int_0^x \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du.$$

Hence

$$g(x) = \frac{d}{dx} P(|Z_2| \leq x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}, \quad x \geq 0.$$

To find the probability density function of $Z_1/|Z_2|$, note that, by Theorem 9.4,

$$\begin{aligned} P\left(\frac{Z_1}{|Z_2|} \leq t\right) &= 1 - P\left(\frac{Z_1}{|Z_2|} > t\right) = 1 - P(Z_1 > t|Z_2|) \\ &= 1 - \int_0^\infty P(Z_1 > t|Z_2| \mid |Z_2| = x) \frac{2}{\sqrt{2\pi}} e^{-x^2/2} dx \end{aligned}$$

$$\begin{aligned}
&= 1 - \int_0^\infty P(Z_1 > tx) \frac{2}{\sqrt{2\pi}} e^{-x^2/2} dx \\
&= 1 - \int_0^\infty \left(\int_{tx}^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du \right) \frac{2}{\sqrt{2\pi}} e^{-x^2/2} dx.
\end{aligned}$$

Now, by the fundamental theorem of calculus,

$$\frac{d}{dt} \int_{tx}^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = \frac{d}{dt} \left(1 - \int_{-\infty}^{tx} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du \right) = -x \frac{1}{\sqrt{2\pi}} e^{-t^2 x^2/2}.$$

Therefore,

$$\frac{d}{dt} P\left(\frac{Z_1}{|Z_2|} \leq t\right) = \int_0^\infty \frac{x}{\sqrt{2\pi}} e^{-t^2 x^2/2} \cdot \frac{2}{\sqrt{2\pi}} e^{-x^2/2} dx = \frac{2}{2\pi} \int_0^\infty x e^{-(t^2+1)x^2/2} dx.$$

Making the change of variable $y = (1 + t^2)x^2/2$ yields

$$\frac{d}{dt} P\left(\frac{Z_1}{|Z_2|} \leq t\right) = \frac{1}{\pi(1+t^2)} \int_0^\infty e^{-y} dy = \frac{1}{\pi(1+t^2)}, \quad -\infty < t < \infty. \quad \blacklozenge$$

Example 9.8 Let X be the natural lifetime of a device that also fails if a catastrophe such as a shock occurs. Let Y be the time until the next catastrophe. Suppose that X and Y are independent exponential random variables with parameters λ and μ , respectively. Given that $X < Y$, find the expected lifetime of the device.

Warning: Since $X < Y$, we might fallaciously think that the expected lifetime of the device is $1/\lambda$.

Solution: Let F be the distribution function of the period in which the device remains operative. We have

$$F(t) = P(X \leq t \mid X < Y) = \frac{P(X \leq t, X < Y)}{P(X < Y)},$$

where,

$$\begin{aligned}
P(X \leq t, X < Y) &= P(X < \min\{t, Y\}) \\
&= \int_0^\infty P(X < \min\{t, Y\} \mid X = x) f_X(x) dx \\
&= \int_0^\infty P(\min\{t, Y\} > x \mid X = x) f_X(x) dx \\
&= \int_0^t P(Y > x \mid X = x) f_X(x) dx \\
&= \int_0^t P(Y > x) f_X(x) dx = \int_0^t e^{-\mu x} \lambda e^{-\lambda x} dx
\end{aligned}$$

$$= \frac{\lambda}{\lambda + \mu} [1 - e^{-(\lambda + \mu)t}],$$

and, by Remark 9.1,

$$P(X < Y) = \frac{\lambda}{\lambda + \mu}.$$

Thus

$$P(X \leq t \mid X < Y) = \frac{\frac{\lambda}{\lambda + \mu} [1 - e^{-(\lambda + \mu)t}]}{\frac{\lambda}{\lambda + \mu}} = 1 - e^{-(\lambda + \mu)t}.$$

This shows that, given $X < Y$, the lifetime of the device is exponential with parameter $\lambda + \mu$. Therefore,

$$E(X \mid X < Y) = \frac{1}{\lambda + \mu}. \quad \blacklozenge$$

Remark 9.2 By Example 9.8, we have established the following important property of exponential random variables.

Let X and Y be independent exponential random variables with parameters λ and μ , respectively. Given that $X < Y$, the distribution function of X is exponential with parameter $\lambda + \mu$. Hence

$$E(X \mid X < Y) = \frac{1}{\lambda + \mu}. \quad \blacklozenge$$

Let B be an event of a sample space with $P(B) > 0$, and X be a random variable defined on the same sample space. Let $F(t) = P(X \leq t \mid B)$, and $f(t) = \frac{d}{dt}F(t)$. Then, by definition, $E(X \mid B) = \int_{-\infty}^{\infty} xf(t) dt$. The following is an important tool for calculating $E(X)$ in terms of $E(X|B)$ and $E(X|B^c)$.

Theorem 9.5 *Let B be an event of a sample space with $P(B) > 0$ and $P(B^c) > 0$. Let X be a random variable defined on the same sample space. Then*

$$E(X) = E(X|B)P(B) + E(X|B^c)P(B^c).$$

Proof: Let N be a discrete random variable with the set of possible values $\{0, 1\}$ defined as follows: $N = 0$ if and only if B occurs, and $N = 1$ if and only if B^c occurs. Then, by Theorem 9.1,

$$\begin{aligned} E(X) &= E[E(X|N)] \\ &= E(X \mid N = 0)P(N = 0) + E(X \mid N = 1)P(N = 1) \\ &= E(X|B)P(B) + E(X|B^c)P(B^c). \quad \blacklozenge \end{aligned}$$

The following is a natural generalization of Theorem 9.5.

Theorem 9.6 *Let $\{B_1, B_2, \dots, B_n\}$ be a partition of the sample space of an experiment with $P(B_i) > 0$ for $i = 0, 1, \dots, n$. Let X be a random variable defined on the same sample space. Then*

$$E(X) = \sum_{i=1}^n E(X|B_i)P(B_i).$$

Let X and Y be two given random variables. Define the new random variable $\mathbf{Var}(X|Y)$ by

$$\mathbf{Var}(X|Y) = E[(X - E(X|Y))^2 | Y].$$

Then the formula analogous to $\mathbf{Var}(X) = E(X^2) - [E(X)]^2$ is given by

$$\mathbf{Var}(X|Y) = E(X^2|Y) - E(X|Y)^2. \quad (9.10)$$

(See Exercise 22.) The following theorem shows that $\mathbf{Var}(X)$ is the sum of the expected value of $\mathbf{Var}(X|Y)$ and the variance of $E(X|Y)$.

Theorem 9.7 $\mathbf{Var}(X) = E[\mathbf{Var}(X|Y)] + \mathbf{Var}(E[X|Y]).$

Proof: By (9.10),

$$\begin{aligned} E[\mathbf{Var}(X|Y)] &= E[E(X^2|Y)] - E[E(X|Y)^2] \\ &= E(X^2) - E[E(X|Y)^2]. \end{aligned}$$

By the definition of variance,

$$\begin{aligned} \mathbf{Var}(E[X|Y]) &= E[E(X|Y)^2] - (E[E(X|Y)])^2 \\ &= E[E(X|Y)^2] - [E(X)]^2. \end{aligned}$$

Adding these two equations, we have the theorem. $\blacklozenge$

Example 9.9 A fisherman catches fish in a large lake with lots of fish, at a Poisson rate of two per hour. If, on a given day, the fisherman spends randomly anywhere between 3 and 8 hours fishing, find the expected value and the variance of the number of fish he catches.

Solution: Let X be the number of hours the fisherman spends fishing. Then X is a uniform random variable over the interval $(3, 8)$. Label the time the fisherman begins fishing on the given day at $t = 0$. Let $N(t)$ denote the total number of fish caught at or prior to t . Then $\{N(t) : t \geq 0\}$ is a Poisson process with parameter $\lambda = 2$. Assuming that X is independent of $\{N(t) : t \geq 0\}$, we have

$$E[N(X) | X = t] = E[N(t)] = 2t.$$

This implies that

$$E[N(X) | X] = 2X.$$

Therefore,

$$E[N(X)] = E[E(N(X) | X)] = E(2X) = 2E(X) = 2 \cdot \frac{8+3}{2} = 11.$$

Similarly,

$$\text{Var}(N(X) | X = t) = \text{Var}(N(t)) = 2t.$$

Thus

$$\text{Var}(N(X) | X) = 2X.$$

By Theorem 9.7,

$$\begin{aligned} \text{Var}(N(X)) &= E[\text{Var}(N(X) | X)] + \text{Var}(E[N(X) | X]) \\ &= E(2X) + \text{Var}(2X) = 2E(X) + 4 \text{Var}(X) \\ &= 2 \cdot \frac{8+3}{2} + 4 \cdot \frac{(8-3)^2}{12} = 19.33. \quad \blacklozenge \end{aligned}$$

EXERCISES

A

1. A fair coin is tossed until two tails occur successively. Find the expected number of the tosses required.
Hint: Let

$$X = \begin{cases} 1 & \text{if the first toss results in tails} \\ 0 & \text{if the first toss results in heads,} \end{cases}$$

and condition on X .

2. The orders received for grain by a farmer add up to X tons, where X is a continuous random variable uniformly distributed over the interval $(4, 7)$. Every ton of grain sold brings a profit of a , and every ton that is not sold is destroyed at a loss of $a/3$. How many tons of grain should the farmer produce to maximize his expected profit?
Hint: Let $Y(t)$ be the profit if the farmer produces t tons of grain. Then

$$E[Y(t)] = E\left[aX - \frac{a}{3}(t - X)\right]P(X < t) + E(at)P(X \geq t).$$

3. In a box, Lynn has b batteries of which d are dead. She tests them randomly and one by one. Every time that a good battery is drawn, she will return it to the box; every time that a dead battery is drawn, she will replace it by a good one.

- (a) Determine the expected value of the number of good batteries in the box after n of them are checked.
- (b) Determine the probability that on the n th draw Lynn draws a good battery.

Hint: Let X_n be the number of good batteries in the box after n of them are checked. Show that

$$E(X_n | X_{n-1}) = 1 + \left(1 - \frac{1}{b}\right)X_{n-1}.$$

Then, by computing the expected value of this random variable, find a recursive relation between $E(X_n)$ and $E(X_{n-1})$. Use this relation and induction to prove that

$$E(X_n) = b - d\left(1 - \frac{1}{b}\right)^n.$$

Note that n should approach ∞ to get $E(X_n) = b$. For part (b), let E_n be the event that on the n th draw she gets a good battery. By conditioning on X_{n-1} prove that $P(E_n) = E(X_{n-1})/b$.

4. For given random variables Y and Z , let

$$X = \begin{cases} Y & \text{with probability } p \\ Z & \text{with probability } 1 - p. \end{cases}$$

Find $E(X)$ in terms of $E(Y)$ and $E(Z)$.

5. If a car is under 1 year old and is totaled, an insurance company replaces that car with a new one. An actuary has calculated that when a \$60,000-car gets involved in an accident, the probability of total loss is 0.015 and the probability of partial loss is 0.038. She has also approximated the probability density function of X , the cost to repair partial damage, in thousands of dollars, to be

$$f(x) = \begin{cases} \frac{1}{8}e^{-x/8} & 0 < x < 60 \\ 0 & \text{otherwise,} \end{cases}$$

where f is only approximately a probability density function. Find the expected value of claim payment per accident for cars that are not yet 1 year old, were purchased for \$60,000 and are insured by this insurance company under policies that have no deductibles.

6. A typist, on average, makes three typing errors in every two pages. If pages with more than two errors must be retyped, on average how many pages must she type to prepare a report of 200 pages? Assume that the number of errors in a page is a Poisson random variable. Note that some of the retyped pages should be retyped, and so on.

Hint: Find p , the probability that a page should be retyped. Let X_n be the number of pages that should be typed at least n times. Show that $E(X_1) = 200p$, $E(X_2) = 200p^2$, $\dots$, $E(X_n) = 200p^n$. The desired quantity is $E(\sum_{i=1}^{\infty} X_i)$, which can be calculated using relation (4.5).

7. From an ordinary deck of 52 cards, cards are drawn at random, one by one and without replacement until a heart is drawn. What is the expected value of the number of cards drawn?

Hint: Consider a deck of cards with 13 hearts and $39 - n$ nonheart cards. Let X_n be the number of cards to be drawn *before* the first heart is drawn. Let

$$Y = \begin{cases} 1 & \text{if the first card drawn is a heart} \\ 0 & \text{otherwise.} \end{cases}$$

By conditioning on Y , find a recursive relation between $E(X_n)$ and $E(X_{n+1})$. Use $E(X_{39}) = 0$ to show that $E(X_i) = (39 - i)/14$. The answer is $1 + E(X_0)$. (For a totally different solution see Exercise 19, [Section 4.7](#).)

8. Suppose that X and Y are independent random variables with probability density functions f and g , respectively. Use conditioning technique to calculate $P(X < Y)$.
9. Prove that, for a Poisson random variable N , if the parameter λ is not fixed and is itself an exponential random variable with parameter 1, then

$$P(N = i) = \left(\frac{1}{2}\right)^{i+1}.$$

10. At Berkeley, California, Lily commutes by bus every work day. To go from home to work, she arrives at the bus stop at the intersection of University and Shattuck Avenues at a random time between 8:30 A.M. and 8:45 A.M. She then takes bus number 67 or bus number 78, whichever arrives first. Suppose that bus 67 arrives at this bus stop according to a Poisson process with rate 15 minutes. If bus number 78 arrives at the bus stop every 15 minutes, 24 hours a day, beginning at midnight, what is the probability that, on the next work day, Lily will take bus 78 to work?
11. At the intersection of two remote roads, the vehicles arriving are either cars or trucks. Suppose that cars arrive at the intersection at a Poisson rate of λ per minute, and trucks arrive at a Poisson rate of μ per minute. Suppose that the arrivals are independent of each other. If we are given that the next vehicle arriving at this intersection is a car, find the expected value of the time until the next arrival.
Warning: Since cars and trucks arrive at this intersection independently, we might fallaciously think that, given the next arrival is a car, the expected value until the next arrival is $1/\lambda$.
12. The lifetimes of batteries manufactured by a certain company are independent exponential random variables each with mean $1/\lambda$. Andy loads his two-battery and his single-battery flashlights with such batteries. What is the probability that he can shine his two-battery flashlight longer than his single-battery flashlight?
13. Two devices, the lifetime of which are independent exponential random variables with parameters λ and μ , are put on a life test. Suppose that the time until a catastrophe, such as a shock, that will fail both devices independently of their lifetimes is exponential with mean $1/\gamma$. Find the probability that both devices will fail at the same time.

B

14. In Rome, tourists arrive at a historical monument according to a Poisson process, on average, one every 5 minutes. There are guided tours that depart (a) whenever there is a group of 10 tourists waiting to take the tour, or (b) one hour has elapsed from the time the previous tour began. It is the policy of the Tourism Department that a tour will only run for less than 10 people if the last guided tour left one hour ago. If in any one hour period, there will always be tourists arriving to take the tour, find the expected value of the time between two consecutive tours.
15. During an academic year, the admissions office of a small college receives student applications at a Poisson rate of 5 per day. It is a policy of this college to double its student recruitment efforts if no applications arrive for two consecutive business days. Find the expected number of business days until a time when the college needs to double its recruitment efforts. Do the admission officers need to worry about this policy at all?
Hint: Let X_1 be the time until the first application arrives. Let X_2 be the time between the first and second applications, and so forth. Let N be the first integer for which

$$X_1 \leq 2, X_2 \leq 2, \dots, X_N \leq 2, X_{N+1} > 2.$$

The time that the admissions office has to wait before doubling its student recruitment efforts is $S_{N+1} = X_1 + X_2 + \dots + X_{N+1}$. Find $E(S_{N+1})$ by conditioning on N .

16. Each time that Steven calls his friend Adam, the probability that Adam is available to talk with him is p independently of other calls. On average, after how many calls has Steven not missed Adam k consecutive times?
17. Recently, Larry taught his daughter Emily how to play backgammon. To encourage Emily to practice this game, Larry decides to play with her until she wins two of the recent three games. If the probability that Emily wins a game is 0.35 independently of all preceding and future games, find the expected number of games to be played.
18. Suppose that a device is powered by a battery. Since an uninterrupted supply of power is needed, the device has a spare battery. When the battery fails, the circuit is altered electronically to connect the spare battery and remove the failed battery from the circuit. The spare battery then becomes the working battery and emits a signal to alert the user to replace the failed battery. When that battery is replaced, it becomes the new spare battery. Suppose that the lifetimes of the batteries used are independent uniform random variables over the interval $(0, 1)$, where the unit of measurement is 1000 hours. For $0 < t \leq 1$, on average, how many batteries are changed by time t ? How many are changed, on average, after 950 hours of operation?
19. For constants α and random variables X_1 , X_2 , and Y , show that
- (a) $E(\alpha X \mid Y) = \alpha E(X \mid Y)$;
 - (b) $E(X_1 + X_2 \mid Y) = E(X_1 \mid Y) + E(X_2 \mid Y)$.
20. Let X and Y be independent and identically distributed random variables. In terms of X and Y , find $E(X \mid X + Y)$.

21. Let X and Y be continuous random variables. Prove that

$$E[(X - E(X|Y))^2] = E(X^2) - E[E(X|Y)^2].$$

Hint: Let $Z = E(X|Y)$. By conditioning on Y and using Example 9.4, first show that $E(XZ) = E(Z^2)$.

22. Let X and Y be two given random variables. Prove that

$$\text{Var}(X|Y) = E[X^2|Y] - E(X|Y)^2.$$

23. Prove Theorem 9.3. ■

Self-Quiz on Section 9.1

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. Niki is babysitting Hannah and Joshua all day today. Hannah cries at a Poisson rate of 10 times per day. Joshua cries at a Poisson rate of 7 times per day independently of Hannah. If it was Hannah who cried the last three times, what is the probability that the next baby who will cry is Joshua?

Hint: Due to the memoryless property of exponential random variables, it does not matter who cried last or how many times before.

2. At a certain location in downtown Chicago, there are ample parking spaces on the street. At that location, cars are parked illegally and independently of the police inspection times according to a Poisson process with rate λ . The duration of the time a car is illegally parked is exponentially distributed with mean 10 minutes. Starting at midnight, Chicago police inspect that location 24 hours a day, every 30 minutes, and fine the cars that are parked illegally since the last inspection. Determine the percentage of cars that are parked illegally and are caught.

Hint: Between 0 and 30, let Y be the time a random car is illegally parked on the street. Suppose that the car parked illegally at Y is left parked for X units of time. Let A be the event that it is caught. Show that $100 \cdot \int_0^{30} P(A | Y = y) \frac{1}{30} dy$ is the desired percentage and calculate it. ■

9.2 ORDER STATISTICS

Definition 9.2 Let $\{X_1, X_2, \dots, X_n\}$ be an independent set of identically distributed continuous random variables with the common density and distribution functions f and F , respectively. Let $X_{(1)}$ be the smallest value in $\{X_1, X_2, \dots, X_n\}$, $X_{(2)}$ be the second smallest value, $X_{(3)}$ be the third smallest, and, in general, $X_{(k)}$ ($1 \leq k \leq n$) be the k th smallest value in $\{X_1, X_2, \dots, X_n\}$. Then $X_{(k)}$ is called the **k th order statistic**, and the set $\{X_{(1)}, X_{(2)}, \dots, X_{(n)}\}$ is said to consist of the **order statistics** of $\{X_1, X_2, \dots, X_n\}$.

By this definition, for example, if at a sample point ω of the sample space, $X_1(\omega) = 8$, $X_2(\omega) = 2$, $X_3(\omega) = 5$, and $X_4(\omega) = 6$, then the order statistics of $\{X_1, X_2, X_3, X_4\}$ is $\{X_{(1)}, X_{(2)}, X_{(3)}, X_{(4)}\}$, where $X_{(1)}(\omega) = 2$, $X_{(2)}(\omega) = 5$, $X_{(3)}(\omega) = 6$, and $X_{(4)}(\omega) = 8$. Continuity of X_i 's implies that $P(X_{(i)} = X_{(j)}) = 0$. Hence

$$P(X_{(1)} < X_{(2)} < X_{(3)} < \dots < X_{(n)}) = 1.$$

Unlike X_i 's, the random variables $X_{(i)}$'s are neither independent nor identically distributed.

There are many useful and practical applications of order statistics in different branches of pure and applied probability, as well as in estimation theory. To show how it arises, we present three examples.

Example 9.10 Suppose that customers arrive at a warehouse from n different locations. Let X_i , $1 \leq i \leq n$, be the time until the arrival of the next customer from location i , then $X_{(1)}$ is the arrival time of the next customer to the warehouse. ♦

Example 9.11 Suppose that a machine consists of n components with the lifetimes $X_1, X_2, \dots, X_n$, respectively, where X_i 's are independent and identically distributed. Suppose that the machine remains operative unless k or more of its components fail. Then $X_{(k)}$, the k th order statistic of $\{X_1, X_2, \dots, X_n\}$, is the time when the machine fails. Also, $X_{(1)}$ is the failure time of the first component. ♦

Example 9.12 Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a population with continuous distribution F . Then the following important statistical concepts are expressed in terms of order statistics:

- (i) The sample range is $X_{(n)} - X_{(1)}$.
- (ii) The sample midrange is $[X_{(n)} + X_{(1)}]/2$.
- (iii) The sample median is

$$m = \begin{cases} X_{(i+1)} & \text{if } n = 2i + 1 \\ \frac{X_{(i)} + X_{(i+1)}}{2} & \text{if } n = 2i. \end{cases} \quad \blacklozenge$$

We will now determine the probability distribution and the probability density functions of $X_{(k)}$, the k th order statistic.

Theorem 9.8 Let $\{X_{(1)}, X_{(2)}, \dots, X_{(n)}\}$ be the order statistics of the independent and identically distributed continuous random variables $X_1, X_2, \dots, X_n$ with the common probability distribution and probability density functions F and f , respectively. Then F_k and f_k , the probability distribution and probability density functions of $X_{(k)}$, respectively, are given by

$$F_k(x) = \sum_{i=k}^n \binom{n}{i} [F(x)]^i [1 - F(x)]^{n-i}, \quad -\infty < x < \infty, \quad (9.11)$$

and

$$f_k(x) = \frac{n!}{(k-1)!(n-k)!} f(x) [F(x)]^{k-1} [1 - F(x)]^{n-k}, \quad -\infty < x < \infty. \quad (9.12)$$

Proof: Let $-\infty < x < \infty$. To calculate $P(X_{(k)} \leq x)$, note that $X_{(k)} \leq x$ if and only if at least k of the random variables $X_1, X_2, \dots, X_n$ are in $(-\infty, x]$. Thus

$$\begin{aligned} F_k(x) &= P(X_{(k)} \leq x) \\ &= \sum_{i=k}^n P(i \text{ of the random variables } X_1, X_2, \dots, X_n \text{ are in } (-\infty, x]) \\ &= \sum_{i=k}^n \binom{n}{i} [F(x)]^i [1 - F(x)]^{n-i}, \end{aligned}$$

where the last equality follows because from the random variables $X_1, X_2, \dots, X_n$ the number of those that lie in $(-\infty, x]$ has binomial distribution with parameters (n, p) , $p = F(x)$.

We will now obtain f_k by differentiating F_k :

$$\begin{aligned} f_k(x) &= \sum_{i=k}^n \binom{n}{i} i f(x) [F(x)]^{i-1} [1 - F(x)]^{n-i} \\ &\quad - \sum_{i=k}^n \binom{n}{i} [F(x)]^i (n-i) f(x) [1 - F(x)]^{n-i-1} \\ &= \sum_{i=k}^n \frac{n!}{(i-1)!(n-i)!} f(x) [F(x)]^{i-1} [1 - F(x)]^{n-i} \\ &\quad - \sum_{i=k}^n \frac{n!}{i!(n-i-1)!} f(x) [F(x)]^i [1 - F(x)]^{n-i-1} \\ &= \sum_{i=k}^n \frac{n!}{(i-1)!(n-i)!} f(x) [F(x)]^{i-1} [1 - F(x)]^{n-i} \\ &\quad - \sum_{i=k+1}^n \frac{n!}{(i-1)!(n-i)!} f(x) [F(x)]^{i-1} [1 - F(x)]^{n-i}. \end{aligned}$$

After cancellations, this gives (9.12). ♦

Remark 9.3 Note that by (9.11) and (9.12), respectively, F_1 and f_1 , the probability distribution and the probability density functions of $X_{(1)} = \min(X_1, X_2, \dots, X_n)$, are found to be

$$\begin{aligned} F_1(x) &= \sum_{i=1}^n \binom{n}{i} [F(x)]^i [1 - F(x)]^{n-i} \\ &= \sum_{i=0}^n \binom{n}{i} [F(x)]^i [1 - F(x)]^{n-i} - [1 - F(x)]^n \\ &= [F(x) + [1 - F(x)]]^n - [1 - F(x)]^n \\ &= 1 - [1 - F(x)]^n, \quad -\infty < x < \infty. \end{aligned}$$

and

$$f_1(x) = nf(x)[1 - F(x)]^{n-1}, \quad -\infty < x < \infty.$$

Also, F_n and f_n , the probability distribution and probability density functions of $X_{(n)} = \max(X_1, X_2, \dots, X_n)$, respectively, are found to be

$$F_n(x) = [F(x)]^n, \quad -\infty < x < \infty,$$

and

$$f_n(x) = nf(x)[F(x)]^{n-1}, \quad -\infty < x < \infty.$$

These quantities can also be calculated directly (see, for example, Example 7.19). $\blacklozenge$

Example 9.13 Let $X_1, X_2, \dots, X_{2n+1}$ be $2n + 1$ random numbers from $(0, 1)$. Then f and F , the respective probability density and distribution functions of X_i 's, are given by

$$\begin{aligned} f(x) &= \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{elsewhere,} \end{cases} \\ F(x) &= \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1. \end{cases} \end{aligned}$$

Using (9.12), the probability density function of $X_{(n+1)}$, the median of these numbers, is found to be

$$f_{n+1}(x) = \frac{(2n+1)!}{n!n!} x^n (1-x)^n, \quad 0 < x < 1;$$

0, elsewhere. Since for the beta function, $B(n+1, n+1) = (n!n!)/(2n+1)!$, we have

$$f_{n+1}(x) = \frac{1}{B(n+1, n+1)} x^n (1-x)^n, \quad 0 < x < 1;$$

0, elsewhere. Hence $X_{(n+1)}$ is beta with parameters $(n+1, n+1)$. In fact, it is straightforward to see that in this example, for $1 \leq k \leq 2n+1$, $X_{(k)}$ is beta with parameters k and $2n-k+2$. $\blacklozenge$

The following theorem gives the joint probability density function of $X_{(i)}$ and $X_{(j)}$. Its proof is similar to that of Theorem 9.8.

Theorem 9.9 *Let $\{X_{(1)}, X_{(2)}, \dots, X_{(n)}\}$ be the order statistics of the independent and identically distributed continuous random variables $X_1, X_2, \dots, X_n$ with the common probability density and probability distribution functions f and F , respectively. Then, for $i < j$ and $x < y$, $f_{ij}(x, y)$, the joint probability density function of $X_{(i)}$ and $X_{(j)}$, is given by*

$$f_{ij}(x, y) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} f(x)f(y)[F(x)]^{i-1}[F(y)-F(x)]^{j-i-1}[1-F(y)]^{n-j}.$$

It is clear that for $x \geq y$, $f_{ij}(x, y) = 0$.

Another important quantity that is used often is the joint probability density function of $X_{(1)}, X_{(2)}, \dots, X_{(n)}$. The following theorem, whose proof we skip, expresses the general form of this function.

Theorem 9.10 *Let $\{X_{(1)}, X_{(2)}, \dots, X_{(n)}\}$ be the order statistics of the independent and identically distributed continuous random variables $X_1, X_2, \dots, X_n$ with the common probability density and distribution functions f and F , respectively. Then $f_{12\dots n}$, the joint probability density function of $X_{(1)}, X_{(2)}, \dots, X_{(n)}$, is given by*

$$f_{12\dots n}(x_1, x_2, \dots, x_n) = \begin{cases} n!f(x_1)f(x_2)\cdots f(x_n) & -\infty < x_1 < x_2 < \cdots < x_n < \infty \\ 0 & \text{otherwise.} \end{cases}$$

Justification: For small $\varepsilon > 0$,

$$\begin{aligned} & P(x_1 - \varepsilon < X_{(1)} < x_1 + \varepsilon, \dots, x_n - \varepsilon < X_{(n)} < x_n + \varepsilon) \\ &= \int_{x_1 - \varepsilon}^{x_1 + \varepsilon} \int_{x_{n-1} - \varepsilon}^{x_{n-1} + \varepsilon} \cdots \int_{x_1 - \varepsilon}^{x_1 + \varepsilon} f_{12\dots n}(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n \\ &\approx 2^n \varepsilon^n f_{12\dots n}(x_1, x_2, \dots, x_n). \end{aligned} \tag{9.13}$$

Now for $x_1 < x_2 < \cdots < x_n$, let $\mathcal{P}$ be the set of all permutations of $\{x_1, x_2, \dots, x_n\}$, then $\mathcal{P}$ has $n!$ elements and we can write

$$\begin{aligned} & P(x_1 - \varepsilon < X_{(1)} < x_1 + \varepsilon, \dots, x_n - \varepsilon < X_{(n)} < x_n + \varepsilon) \\ &= \sum_{\{x_{i_1}, x_{i_2}, \dots, x_{i_n}\} \in \mathcal{P}} P(x_{i_1} - \varepsilon < X_1 < x_{i_1} + \varepsilon, \dots, x_{i_n} - \varepsilon < X_n < x_{i_n} + \varepsilon) \\ &\approx \sum_{\{x_{i_1}, x_{i_2}, \dots, x_{i_n}\} \in \mathcal{P}} 2^n \varepsilon^n f(x_{i_1})f(x_{i_2}) \cdots f(x_{i_n}). \end{aligned} \tag{9.14}$$

This is because $X_1, X_2, \dots, X_n$ are independent, and hence their joint probability density function is the product of their marginal probability density functions. Putting (9.13) and (9.14) together, we obtain

$$\sum_{\{x_{i_1}, x_{i_2}, \dots, x_{i_n}\} \in \mathcal{P}} f(x_{i_1})f(x_{i_2}) \cdots f(x_{i_n}) = f_{12 \cdots n}(x_1, x_2, \dots, x_n). \quad (9.15)$$

But

$$f(x_{i_1})f(x_{i_2}) \cdots f(x_{i_n}) = f(x_1)f(x_2) \cdots f(x_n).$$

Therefore,

$$\begin{aligned} \sum_{\{x_{i_1}, x_{i_2}, \dots, x_{i_n}\} \in \mathcal{P}} f(x_{i_1})f(x_{i_2}) \cdots f(x_{i_n}) &= \\ \sum_{\{x_{i_1}, x_{i_2}, \dots, x_{i_n}\} \in \mathcal{P}} f(x_1)f(x_2) \cdots f(x_n) &= n!f(x_1)f(x_2) \cdots f(x_n). \end{aligned} \quad (9.16)$$

Relations (9.15) and (9.16) imply that

$$f_{12 \cdots n}(x_1, x_2, \dots, x_n) = n!f(x_1)f(x_2) \cdots f(x_n). \quad \blacklozenge$$

Example 9.14 The distance between two towns, A and B , is 30 miles. If three gas stations are constructed independently at randomly selected locations between A and B , what is the probability that the distance between any two gas stations is at least 10 miles?

Solution: Let X_1, X_2 , and X_3 be the locations at which the gas stations are constructed. The probability density function of X_1, X_2 , and X_3 is given by

$$f(x) = \begin{cases} \frac{1}{30} & \text{if } 0 < x < 30 \\ 0 & \text{elsewhere.} \end{cases}$$

Therefore, by Theorem 9.10, f_{123} , the joint probability density function of the order statistics of X_1, X_2 , and X_3 , is as follows.

$$f_{123}(x_1, x_2, x_3) = 3! \left(\frac{1}{30} \right)^3, \quad 0 < x_1 < x_2 < x_3 < 30.$$

Using this, we have that the desired probability is given by the following triple integral.

$$\begin{aligned} &P(X_{(1)} + 10 < X_{(2)} \text{ and } X_{(2)} + 10 < X_{(3)}) \\ &= \int_0^{10} \int_{x_1+10}^{20} \int_{x_2+10}^{30} f_{123}(x_1, x_2, x_3) dx_3 dx_2 dx_1 = \frac{1}{27} \quad \blacklozenge \end{aligned}$$

Order Statistics Simulation with R

To simulate order statistics in R for a set of identically distributed and independent random variables $X_1, X_2, \dots, X_n$, we use the **sort** command. To illustrate this concept, let us consider an example where customers arrive at a warehouse from four different locations. For each $1 \leq i \leq 4$, we denote X_i as the time until the arrival of the next customer from location i . Assuming that X_i follows an exponential random variable with parameter $e[i]$, we can use the following R code to generate one realization of $X_{(1)}, X_{(2)}, X_{(3)}$, and $X_{(4)}$, which represent the order statistics of X_1, X_2, X_3 , and X_4 :

```
sort (c (rexp (1, e [1]) , rexp (1, e [2]) , rexp (1, e [3]) , rexp (1, e [4]) ) )
```

In this context, the resulting values represent the arrival times of customers at the warehouse, ordered according to the sequence of arrivals. Specifically, $X_{(1)}$ is the arrival time of the first customer, $X_{(2)}$ and $X_{(3)}$ are the arrival times of the next two customers, and $X_{(4)}$ is the arrival time of the last customer. To estimate the arrival times of customers in the order of their arrivals, we utilize the following straightforward code to calculate the expected values of $X_{(1)}, X_{(2)}, X_{(3)}$, and $X_{(4)}$. In this code, the vector **e** represents the components **e[1]**, **e[2]**, **e[3]** and **e[4]**. Initially, all components of the vector **w** are set to zero. However, after each simulation run, the ordered arrival times are accumulated in their corresponding components of **w**. Ultimately, the components of **w** will represent the sums of 75,000 simulated arrival times. Let **x** be the vector **w/num**. Then, **x[i]**, the i th entry of **x**, represents the average of the i th arrival times across all simulations. Essentially, it provides a simulation of $X_{(i)}$.

```
> e <- c (1/30, 1/20, 1/45, 1/15)
> w <- c (0, 0, 0, 0)
> num <- 75000
> i <- 1
> while (i<=num) {
+ v <- sort (c (rexp (1, e [1]) , rexp (1, e [2]) , rexp (1, e [3]) , rexp (1, e [4]) ) )
+ w <- w+v
+ i <- i+1
+ }
> x <- w/num
>> options (digits=4)
> print (x)
[1] 5.791 14.080 28.055 62.202
```

EXERCISES

A

1. Let X_1, X_2, X_3 , and X_4 be four independently selected random numbers from $(0, 1)$. Find $P(1/4 < X_{(3)} < 1/2)$.
2. Two random points are selected from $(0, 1)$ independently. Find the probability that one of them is at least three times the other.
3. A box contains 20 identical balls numbered 1 to 20. Seven balls are drawn randomly and without replacement. Find the probability mass function of the median of the numbers on the balls drawn.
4. Let X_1, X_2, X_3 , and X_4 be independent exponential random variables, each with parameter λ . Find $P(X_{(4)} \geq 3\lambda)$.
5. Let $X_1, X_2, X_3, \dots, X_n$ be a sequence of nonnegative, identically distributed, and independent random variables. Let F be the probability distribution function of X_i , $1 \leq i \leq n$. Prove that

$$E[X_{(n)}] = \int_0^\infty (1 - [F(x)]^n) dx.$$

Hint: Use Remark 5.4.

6. Let $X_1, X_2, X_3, \dots, X_m$ be a sequence of nonnegative, independent binomial random variables, each with parameters (n, p) . Find the probability mass function of $X_{(i)}$, $1 \leq i \leq m$.
7. Prove that G , the probability distribution function of $[X_{(1)} + X_{(n)}]/2$, the midrange of a random sample of size n from a population with continuous probability distribution function F and probability density function f , is given by

$$G(t) = n \int_{-\infty}^t [F(2t - x) - F(x)]^{n-1} f(x) dx.$$

Hint: Use Theorem 9.9 to find f_{1n} , then integrate over the region $x + y \leq 2t$ and $x \leq y$.

B

8. Let X_1 and X_2 be two independent exponential random variables each with parameter λ . Show that $X_{(1)}$ and $X_{(2)} - X_{(1)}$ are independent.
9. Let X_1 and X_2 be two independent $N(0, \sigma^2)$ random variables. Find $E[X_{(1)}]$.
Hint: Let $f_{12}(x, y)$ be the joint probability density function of $X_{(1)}$ and $X_{(2)}$. The desired quantity is $\iint x f_{12}(x, y) dx dy$, where the integration is taken over an appropriate region.

10. Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a population with continuous probability distribution function F and probability density function f .
- Calculate the probability density function of the sample range, $R = X_{(n)} - X_{(1)}$.
 - Use (a) to find the probability density function of the sample range of n random numbers from $(0, 1)$.
11. Let $X_1, X_2, \dots, X_n$ be n independently randomly selected points from the interval $(0, \theta)$, $\theta > 0$. Prove that

$$E(R) = \frac{n-1}{n+1}\theta,$$

where $R = X_{(n)} - X_{(1)}$ is the range of these points.

Hint: Use part (a) of Exercise 10. Also compare this with Exercise 23, [Section 7.1](#). ■

Self-Quiz on Section 9.2

Time allotted: 20 Minutes

Each problem is worth 5 points.

- Find the expected value of the distance between two random points selected independently from the interval $(0, 1)$.
- All that we know about a horse race that was held last week in Louisville, Kentucky, is that all of the five horses that were competing reached the finish line, independently, at random times after 2:10 P.M. and before 2:22 P.M.
 - Find the probability that the winning horse passed the finish line between 2:10 P.M. and 2:11 P.M.
 - Find the probability that the time that the last horse crossed the finish line was after 2:21 P.M.



9.3 BIVARIATE NORMAL DISTRIBUTION

Let $f(x, y)$ be the joint probability density function of continuous random variables X and Y ; f is called a **bivariate normal probability density function** if

- The distribution function of X is normal. That is, the marginal probability density function of X is

$$f_X(x) = \frac{1}{\sigma_X \sqrt{2\pi}} \exp \left[-\frac{(x - \mu_X)^2}{2\sigma_X^2} \right], \quad -\infty < x < \infty. \quad (9.17)$$

- (b) The conditional distribution of Y , given that $X = x$, is normal for each $x \in (-\infty, \infty)$. That is, $f_{Y|X}(y|x)$ has a normal density for each $x \in \mathbb{R}$.
- (c) The conditional expectation of Y , given that $X = x$, $E(Y | X = x)$, is a linear function of x . That is, $E(Y | X = x) = a + bx$ for some $a, b \in \mathbb{R}$.
- (d) The conditional variance of Y , given that $X = x$, is constant. That is, $\sigma_{Y|X=x}^2$ is independent of the value of x .

As an example, let X be the height of a man and let Y be the height of his daughter. It is reasonable to assume and is statistically verified, that the joint probability density function of X and Y satisfies (a) to (d) and hence is bivariate normal. As another example, let X and Y be grade point averages of a student in his or her first and senior years, respectively. Then the joint probability density function of X and Y is bivariate normal.

We now prove that if $f(x, y)$ satisfies (a) to (d), it must be of the following form:

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} Q(x, y) \right], \quad (9.18)$$

where ρ is the correlation coefficient of X and Y and

$$Q(x, y) = \left(\frac{x - \mu_X}{\sigma_X} \right)^2 - 2\rho \frac{x - \mu_X}{\sigma_X} \frac{y - \mu_Y}{\sigma_Y} + \left(\frac{y - \mu_Y}{\sigma_Y} \right)^2.$$

To prove (9.18), note that from Lemmas 9.1 and 9.2 (which follow) we have that for each $x \in \mathbb{R}$ the expected value and the variance of the normal density function $f_{Y|X}(y|x)$ are $\mu_Y + \rho(\sigma_Y/\sigma_X)(x - \mu_X)$ and $(1 - \rho^2)\sigma_Y^2$, respectively. Therefore, for every real x ,

$$f_{Y|X}(y|x) = \frac{1}{\sigma_Y\sqrt{2\pi}\sqrt{1-\rho^2}} \exp \left[-\frac{[y - \mu_Y - \rho(\sigma_Y/\sigma_X)(x - \mu_X)]^2}{2\sigma_Y^2(1-\rho^2)} \right], \quad (9.19)$$

$$-\infty < y < \infty.$$

Now $f(x, y) = f_{Y|X}(y|x)f_X(x)$, and $f_X(x)$ has the normal density given by (9.17). Multiplying (9.17) by (9.19), we obtain (9.18). So what remains to be shown is the proofs of the following general lemmas, which are valid even if the joint density function of X and Y is not bivariate normal.

Lemma 9.1 *Let X and Y be two random variables with probability density function $f(x, y)$. If $E(Y | X = x)$ is a linear function of x , that is, if $E(Y | X = x) = a + bx$ for some $a, b \in \mathbb{R}$, then*

$$E(Y | X = x) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X).$$

Proof: By definition,

$$E(Y | X = x) = \int_{-\infty}^{\infty} y \frac{f(x, y)}{f_X(x)} dy = a + bx.$$

Therefore,

$$\int_{-\infty}^{\infty} y f(x, y) dy = a f_X(x) + b x f_X(x). \quad (9.20)$$

This gives

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dy dx = a \int_{-\infty}^{\infty} f_X(x) dx + b \int_{-\infty}^{\infty} x f_X(x) dx, \quad (9.21)$$

which is equivalent to

$$\mu_Y = a + b \mu_X. \quad (9.22)$$

Now, multiplying (9.20) by x and integrating both sides, we obtain

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x y f(x, y) dy dx = a \int_{-\infty}^{\infty} x f_X(x) dx + b \int_{-\infty}^{\infty} x^2 f_X(x) dx,$$

which is equivalent to

$$E(XY) = a \mu_X + b E(X^2). \quad (9.23)$$

Solving (9.22) and (9.23) for a and b , we get

$$b = \frac{E(XY) - \mu_X \mu_Y}{E(X^2) - \mu_X^2} = \frac{\text{Cov}(X, Y)}{\sigma_X^2} = \frac{\rho \sigma_X \sigma_Y}{\sigma_X^2} = \frac{\rho \sigma_Y}{\sigma_X}$$

and

$$a = \mu_Y - \frac{\rho \sigma_Y}{\sigma_X} \mu_X.$$

Therefore,

$$E(Y | X = x) = a + bX = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X). \quad \blacklozenge$$

Lemma 9.2 *Let $f(x, y)$ be the joint probability density function of continuous random variables X and Y . If $E(Y | X = x)$ is a linear function of x and $\sigma_{Y|X=x}^2$ is constant, then*

$$\sigma_{Y|X=x}^2 = (1 - \rho^2) \sigma_Y^2.$$

Proof: We have that

$$\sigma_{Y|X=x}^2 = \int_{-\infty}^{\infty} [y - E(Y | X = x)]^2 f_{Y|X}(y|x) dy.$$

Multiplying both sides of this equation by $f_X(x)$ and integrating on x gives

$$\sigma_{Y|X=x}^2 \int_{-\infty}^{\infty} f_X(x) dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [y - E(Y | X = x)]^2 f_{Y|X}(y|x) f_X(x) dy dx.$$

Now, since $f(x, y) = f_{Y|X}(y|x)f_X(x)$, and

$$E(Y | X = x) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X),$$

we get

$$\sigma_{Y|X=x}^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left[y - \mu_Y - \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X) \right]^2 f(x, y) dx dy,$$

or, equivalently,

$$\sigma_{Y|X=x}^2 = E \left[(Y - \mu_Y) - \rho \frac{\sigma_Y}{\sigma_X}(X - \mu_X) \right]^2.$$

Therefore,

$$\begin{aligned} \sigma_{Y|X=x}^2 &= E[(Y - \mu_Y)^2] - 2\rho \frac{\sigma_Y}{\sigma_X} E[(Y - \mu_Y)(X - \mu_X)] + \rho^2 \frac{\sigma_Y^2}{\sigma_X^2} E[(X - \mu_X)^2] \\ &= \sigma_Y^2 - 2\rho \frac{\sigma_Y}{\sigma_X} \rho \sigma_X \sigma_Y + \rho^2 \frac{\sigma_Y^2}{\sigma_X^2} \sigma_X^2 \\ &= (1 - \rho^2) \sigma_Y^2. \quad \blacklozenge \end{aligned}$$

Figure 9.1 demonstrates the graph of

$$f(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2},$$

a bivariate normal probability density function, in the case where $\rho = 0$, $\mu_X = \mu_Y = 0$, and $\sigma_X = \sigma_Y = 1$.

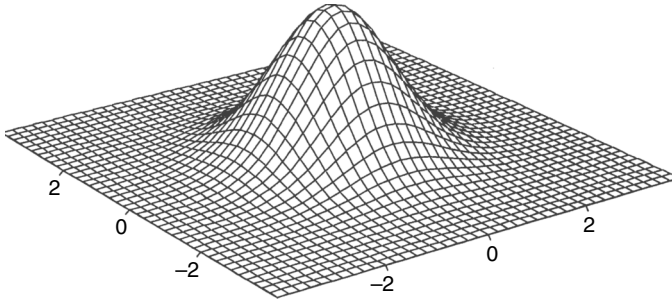


Figure 9.1 Bivariate normal probability density function.

We know that if X and Y are independent random variables, their correlation coefficient is 0. We also know that, in general, the converse of this fact is not true. However, if X and Y

have a bivariate normal distribution, the converse is also true. That is, $\rho = 0$ implies that X and Y are independent, which can be deduced from (9.18). For $\rho = 0$,

$$\begin{aligned} f(x, y) &= \frac{1}{2\pi\sigma_X\sigma_Y} \exp \left[-\frac{(x - \mu_X)^2}{2\sigma_X^2} - \frac{(y - \mu_Y)^2}{2\sigma_Y^2} \right] \\ &= \frac{1}{\sigma_X\sqrt{2\pi}} \exp \left[-\frac{(x - \mu_X)^2}{2\sigma_X^2} \right] \frac{1}{\sigma_Y\sqrt{2\pi}} \exp \left[-\frac{(y - \mu_Y)^2}{2\sigma_Y^2} \right] \\ &= f_X(x)f_Y(y), \end{aligned}$$

showing that X and Y are independent.

Example 9.15 At a certain university, the joint probability density function of X and Y , the grade point averages of a student in his or her first and senior years, respectively, is bivariate normal. From the grades of past years it is known that $\mu_X = 3$, $\mu_Y = 2.5$, $\sigma_X = 0.5$, $\sigma_Y = 0.4$, and $\rho = 0.4$. Find the probability that a student with grade point average 3.5 in his or her first year will earn a grade point average of at least 3.2 in his or her senior year.

Solution: The conditional probability density function of Y , given that $X = 3.5$, is normal with mean

$$\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X) = 2.5 + (0.4)(0.4/0.5)(3.5 - 3) = 2.66$$

and standard deviation

$$\sigma_Y \sqrt{1 - \rho^2} = (0.4)\sqrt{1 - (0.4)^2} = 0.37.$$

Therefore, the desired probability is calculated as follows:

$$\begin{aligned} P(Y \geq 3.2 \mid X = 3.5) &= P\left(\frac{Y - 2.66}{0.37} \geq \frac{3.2 - 2.66}{0.37} \mid X = 3.5\right) \\ &= P(Z \geq 1.46 \mid X = 3.5) = 1 - \Phi(1.46) \approx 1 - 0.9279 = 0.0721, \end{aligned}$$

where Z is a standard normal random variable. ♦

EXERCISES

1. Let X be the height of a man and Y the height of his daughter (both in inches). Suppose that the joint probability density function of X and Y is bivariate normal with the following parameters: $\mu_X = 71$, $\mu_Y = 60$, $\sigma_X = 3$, $\sigma_Y = 2.7$, and $\rho = 0.45$. Find the probability that the height of the daughter, of a man who is 70 inches tall, is at least 59 inches.

2. Based on calculations of an actuary of an insurance company, the company's auto insurance division provides an annual net underwriting income with mean and standard deviation, in millions of dollars, of 200 and 60, respectively. The corresponding dollar amounts for the company's homeowner's insurance divisions are 80 and 25 million dollars, respectively. The actuary has observed that these annual net underwriting incomes follow a bivariate normal distribution, and the correlation coefficient between them is 0.75. In a year that the company's auto insurance division provided an annual net underwriting income of 220 million dollars, what is the probability that the company's homeowner's insurance division provided at least 87 million dollars?
3. The joint probability density function of X and Y is bivariate normal with $\sigma_X = \sigma_Y = 9$, $\mu_X = \mu_Y = 0$, and $\rho = 0$. Find (a) $P(X \leq 6, Y \leq 12)$; (b) $P(X^2 + Y^2 \leq 36)$.
4. Let the joint probability density function of X and Y be bivariate normal. For what values of α is the variance of $\alpha X + Y$ minimum?
5. Let $f(x, y)$ be a joint bivariate normal probability density function. Determine the point at which the maximum value of f is obtained.
6. Let the joint probability density function of two random variables X and Y be given by

$$f(x, y) = \begin{cases} 2 & \text{if } 0 < y < x, 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X | Y = y)$, $E(Y | X = x)$, and $\rho(X, Y)$.

Hint: To find ρ , use Lemma 9.1.

7. Let Z and W be independent standard normal random variables. Let X and Y be defined by

$$\begin{aligned} X &= \sigma_1 Z + \mu_1, \\ Y &= \sigma_2 [\rho Z + \sqrt{1 - \rho^2} W] + \mu_2, \end{aligned}$$

where $\sigma_1, \sigma_2 > 0$, $-\infty < \mu_1, \mu_2 < \infty$, and $-1 < \rho < 1$. Show that the joint probability density function of X and Y is bivariate normal and $\sigma_X = \sigma_1$, $\sigma_Y = \sigma_2$, $\mu_X = \mu_1$, $\mu_Y = \mu_2$, and $\rho(X, Y) = \rho$.

Note: By this exercise, if the joint probability density function of X and Y is bivariate normal, X and Y can be written as sums of independent standard normal random variables.

8. Let the joint probability density function of random variables X and Y be bivariate normal. Show that if $\sigma_X = \sigma_Y$, then $X + Y$ and $X - Y$ are independent random variables.
Hint: Show that the joint probability density function of $X + Y$ and $X - Y$ is bivariate normal with correlation coefficient 0.



Self-Quiz on Section 9.3

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. When a golf ball is dropped on concrete, the ratio of its speed after hitting the ground to its speed before hitting the ground is called its *coefficient of restitution*. Let X and Y be the coefficients of restitution of a certain brand of new golf balls and five-year old golf balls, respectively. Suppose that the joint probability density function of X and Y is bivariate normal. From a very large number of experiments, we know that $\mu_X = 0.95$, $\mu_Y = 0.53$, $\sigma_X = 0.28$, $\sigma_Y = 0.19$, and $\rho = 0.2$. Find the probability that a new golf ball of this brand with a coefficient of restitution of 0.97 will have a coefficient of restitution of at least 0.60 in 5 years.
2. Let $f(x, y)$ be a joint bivariate normal probability density function. Determine the point at which the maximum value of f is obtained.
Hint: Note that $f(x, y)$ is maximum if and only if $Q(x, y)$ is minimum. Show that $Q(x, y)$ is minimum if it is zero.

■

CHAPTER 9 SUMMARY

► **Conditioning on Random Variables** Let X and Y be two random variables. By $E(X|Y)$ we mean a function of Y that is defined to be $E(X | Y = y)$ when $Y = y$. The most important property of this random variable is that $E[E(X | Y)] = E(X)$. Furthermore, the random variable $\text{Var}(X|Y)$, defined by

$$\text{Var}(X|Y) = E[(X - E(X|Y))^2 | Y],$$

satisfies

$$\begin{aligned}\text{Var}(X|Y) &= E(X^2|Y) - E(X|Y)^2 \\ \text{Var}(X) &= E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]).\end{aligned}$$

► **Wald's Equation** Let $X_1, X_2, \dots$ be independent and identically distributed random variables with the finite mean $E(X)$ and finite variance $\text{Var}(X)$. Let $N > 0$ be an integer-valued random variable independent of $\{X_1, X_2, \dots\}$, with $E(N) < \infty$. Then

$$E\left(\sum_{i=1}^N X_i\right) = E(N)E(X).$$

► **Calculating Probabilities by Conditioning on Random Variables** Let B be an arbitrary event and X be a discrete random variable with possible set of values A , then

$$P(B) = \sum_{x \in A} P(B | X = x)P(X = x).$$

If X is a continuous random variable with probability density function f , then

$$P(B) = \int_{-\infty}^{\infty} P(B | X = x)f(x) dx.$$

► **Calculating Expected Values by Conditioning on Events** Let B be an event of a sample space with $P(B) > 0$ and $P(B^c) > 0$, and X be a random variable defined on the same sample space. Let $F(t) = P(X \leq t | B)$, and $f(t) = \frac{d}{dt}F(t)$. Then, by definition,

$$E(X | B) = \int_{-\infty}^{\infty} xf(t) dt, \text{ and}$$

$$E(X) = E(X|B)P(B) + E(X|B^c)P(B^c).$$

In general, let $\{B_1, B_2, \dots, B_n\}$ be a partition of the sample space of an experiment with $P(B_i) > 0$ for $i = 0, 1, \dots, n$. Then

$$E(X) = \sum_{i=1}^n E(X|B_i)P(B_i).$$

► **Order Statistics** Let $\{X_1, X_2, \dots, X_n\}$ be an independent set of identically distributed continuous random variables with the common density and distribution functions f and F , respectively. Let $X_{(1)}$ be the smallest value in $\{X_1, X_2, \dots, X_n\}$, $X_{(2)}$ be the second smallest value, $X_{(3)}$ be the third smallest, and, in general, $X_{(k)}$ ($1 \leq k \leq n$) be the k th smallest value in $\{X_1, X_2, \dots, X_n\}$. Then $X_{(k)}$ is called the k th *order statistic*, and the set $\{X_{(1)}, X_{(2)}, \dots, X_{(n)}\}$ is said to consist of the *order statistics* of $\{X_1, X_2, \dots, X_n\}$. Let F_k and f_k be the probability distribution and the probability density functions of $X_{(k)}$, respectively, and, for $i < j$ and $x < y$, $f_{ij}(x, y)$ be the joint probability density function of $X_{(i)}$ and $X_{(j)}$. Then

$$F_k(x) = \sum_{i=k}^n \binom{n}{i} [F(x)]^i [1 - F(x)]^{n-i}, \quad -\infty < x < \infty,$$

$$f_k(x) = \frac{n!}{(k-1)!(n-k)!} f(x)[F(x)]^{k-1}[1 - F(x)]^{n-k}, \quad -\infty < x < \infty,$$

and

$$f_{ij}(x, y) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} f(x)f(y)[F(x)]^{i-1}[F(y) - F(x)]^{j-i-1}[1 - F(y)]^{n-j}.$$

It is clear that for $x \geq y$, $f_{ij}(x, y) = 0$. Moreover, $f_{12\dots n}$, the joint probability density function of $X_{(1)}, X_{(2)}, \dots, X_{(n)}$, is given by

$$f_{12\dots n}(x_1, x_2, \dots, x_n) = \begin{cases} n!f(x_1)f(x_2)\cdots f(x_n) & -\infty < x_1 < x_2 < \cdots < x_n < \infty \\ 0 & \text{otherwise.} \end{cases}$$

► **Bivariate Normal Distribution** Let $f(x, y)$ be the joint probability density function of continuous random variables X and Y ; f is called a *bivariate normal probability density function* if

- (a) The distribution function of X is normal. That is, the marginal probability density function of X is

$$f_X(x) = \frac{1}{\sigma_X \sqrt{2\pi}} \exp \left[-\frac{(x - \mu_X)^2}{2\sigma_X^2} \right], \quad -\infty < x < \infty.$$

- (b) The conditional distribution of Y , given that $X = x$, is normal for each $x \in (-\infty, \infty)$. That is, $f_{Y|X}(y|x)$ has a normal density for each $x \in \mathbb{R}$.
- (c) The conditional expectation of Y , given that $X = x$, $E(Y | X = x)$, is a linear function of x . That is, $E(Y | X = x) = a + bx$ for some $a, b \in \mathbb{R}$.
- (d) The conditional variance of Y , given that $X = x$, is constant. That is, $\sigma_{Y|X=x}^2$ is independent of the value of x .

If $f(x, y)$ satisfies (a) to (d), it must be of the following form:

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} Q(x, y) \right],$$

where ρ is the correlation coefficient of X and Y and

$$Q(x, y) = \left(\frac{x - \mu_X}{\sigma_X} \right)^2 - 2\rho \frac{x - \mu_X}{\sigma_X} \frac{y - \mu_Y}{\sigma_Y} + \left(\frac{y - \mu_Y}{\sigma_Y} \right)^2.$$

REVIEW PROBLEMS

1. The lifetime of a machine, in years, is a uniform random variable over the interval $(0, 7)$. The machine will be replaced at age 5 or when it fails, if that occurs before age 5. Find the expected value of the age of the machine at the time of replacement.

2. In data communication, usually messages sent are combinations of characters, and each character consists of a number of bits. A bit is the smallest unit of information and is either 1 or 0. Suppose that the length of a character (in bits) is a geometric random variable with parameter p . Suppose that a message is combined of K characters, where K is a random variable with mean μ and variance σ^2 . If the lengths of characters of a message are independent of each other and of K , and if it takes a sender 1000 bits per second to emit a message, find the expected value and variance of T , the time it will take the sender to emit a message.
3. A random point (X, Y) is selected from the rectangle $[0, \pi/2] \times [0, 1]$. What is the probability that it lies below the curve $y = \sin x$?
4. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} ye^{-y(1+x)} & \text{if } x > 0, \quad y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Show that $E(X)$ does not exist.
 - (b) Find $E(X|Y)$.
5. Bus A arrives at a station at a random time between 10:00 A.M. and 10:30 A.M. tomorrow. Bus B arrives at the same station at a random time between 10:00 A.M. and the arrival time of bus A . Find the expected value of the arrival time of bus B .
 6. Let $\{X_1, X_2, X_3, \dots\}$ be a sequence of independent and identically distributed exponential random variables with parameter λ . Let N be a geometric random variable with parameter p independent of $\{X_1, X_2, X_3, \dots\}$. Find the distribution function of $\sum_{i=1}^N X_i$.
 7. Suppose that X and Y represent the amount of money in the wallets of players A and B, respectively. Let X and Y be jointly uniformly distributed on the unit square $[0, 1] \times [0, 1]$. A and B each places his wallet on the table. Whoever has the smallest amount of money in his wallet, wins all the money in the other wallet. Let W_A be the amount of money that player A will win. Show that $E(W_A) = 0$. (For a history of this problem, see the *Wallet Paradox* in Exercise 11, Review Problems, [Chapter 8](#).)
 8. A fair coin is tossed successively. Let K_n be the number of tosses until n consecutive heads occur.
 - (a) Argue that $E(K_n | K_{n-1} = i) = (i + 1)\frac{1}{2} + [i + 1 + E(K_n)]\frac{1}{2}$.
 - (b) Show that $E(K_n | K_{n-1}) = K_{n-1} + 1 + \frac{1}{2}E(K_n)$.
 - (c) By finding the expected values of both sides of (b) find a recursive relation between $E(K_n)$ and $E(K_{n-1})$.
 - (d) Note that $E(K_1) = 2$. Use this and (c) to find $E(K_n)$.
 9. Let X_1, X_2 , and X_3 be independent random variables from $(0, 1)$. Find the probability density function and the expected value of the midrange of these random variables $[X_{(1)} + X_{(3)}]/2$.



Self-Test on Chapter 9

Time allotted: 90 Minutes

Each problem is worth 20 points.

- At the office of a college Provost, calls arrive at extension 1247 according to a Poisson process $\{N_1(t): t \geq 0\}$ with rate λ , and calls arrive at extension 1223 independently according to a Poisson process $\{N_2(t): t \geq 0\}$ with rate μ . Let N be the number of calls arriving at extension 1223 between two consecutive calls made to extension 1247. Find the probability mass function of N .

Hint: Let X be the period between two consecutive calls made to extension 1247. We are interested in $P(N_2(X) = i), i \geq 0$.

- A business has a benefit-limit policy to protect its finances from calamity losses; that is, payments for its indirect costs each time it closes down by a devastating calamity such as hurricane, tornado, flood, and snowstorm. According to this policy, each year, the insurance company pays the business its total calamity losses up to a maximum of \$400,000. Suppose that the probability density function of the total annual calamity loss of the business, in millions of dollars, is

$$f(x) = \begin{cases} \frac{1}{64x^5} & \text{if } x > 0.25 \\ 0 & \text{otherwise.} \end{cases}$$

Find the expected value of the annual calamity loss of the business paid by the insurance company.

- A seismograph is an instrument that detects, measures, and accurately records the magnitude, depth, and location of an earthquake. Suppose that the lifetime of a certain seismograph, placed in the New Mexico desert, is exponential with mean 7 years. Furthermore, suppose that the seismograph will be replaced after working for 5 years or when it fails, whichever occurs first. Find the expected value of the time until the seismograph is replaced.

Hint: Let X be the time until the seismograph placed in the New Mexico desert is replaced. Let L be its lifetime. Let

$$Y = \begin{cases} 1 & \text{if } L \geq 5 \\ 0 & \text{if } L < 5. \end{cases}$$

Find $E(X)$ using $E(X) = E[E(X | Y)]$.

- Each month, the average number of claims received by an insurance company is a gamma random variable with parameters n and λ , n being a positive integer. If claims arrive according to a Poisson process, find the distribution of the total number of claims received by the insurance company during one whole year.

5. A system has 7 components, and it functions if and only if at least one of its components functions. Suppose that the lifetimes of the components are independent, identically distributed exponential random variables with mean 1 year. Furthermore, suppose that currently all the components are functional.
- (a) What is the probability that 2 years from now the system is still functional?
 - (b) What is the probability that 2 years from now the system has at least three functional components?



Chapter 10

Inequalities and Limit Theorems

Some of the most momentous theorems that have a very central role and widespread applications in probability, statistics, and other branches of knowledge are concerning limit theorems. Among those theorems, probably various versions of the laws of large numbers and the central limit theorem are the most prominent ones. In this chapter, we will study those theorems. However, to do so, first we need to study sums of independent random variables and Markov's and Chebyshev's inequalities as well as a prerequisite for the study of sums of independent random variables; namely, the celebrated technique of the moment-generating functions.

10.1 MOMENT-GENERATING FUNCTIONS

For a random variable X , we have demonstrated how important its first moment $E(X)$ and its second moment $E(X^2)$ are. For other values of n also, $E(X^n)$ is a valuable measure both in theory and practice. For example, letting $\mu = E(X)$ and $\sigma^2 = \text{Var}(X)$, we have that the quantity $E[(X - \mu)/\sigma]^3$ is a measure of symmetry of the distribution of X . It is called the measure of **skewness** and is zero if the distribution of X is symmetric, negative if it is skewed to the left, and positive if it is skewed to the right (see [Figure 10.1](#) on the following page).

As another example, $E[(X - \mu)/\sigma]^4$, the measure of **kurtosis**, provides insight into the shape, particularly in the tails and the peak. A distribution with high kurtosis has heavier tails than a standard normal distribution, meaning it has a higher probability of producing extreme values (both large and small). Conversely, a distribution with low kurtosis has lighter tails, indicating a lower chance of extreme values. In addition to the tails, kurtosis also measures the sharpness of the peak. A distribution with high kurtosis will have a more peaked center than a standard normal distribution, whereas one with low kurtosis will be more flat-topped. A distribution with heavier tails assigns more probability to extreme events than a standard normal distribution does. This means that while rare events are still unlikely, they are less so than one would predict if assuming normality.

Since, for a standard normal random variable Z ,

$$E\left(\frac{Z - \mu}{\sigma}\right)^4 = \int_{-\infty}^{\infty} z^4 \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 3,$$

the standard normal distribution has a kurtosis of 3. When discussing kurtosis, often the **excess kurtosis** is referenced, which is the kurtosis of the distribution minus 3. Therefore, for a standard normal distribution function, the excess kurtosis is 0.

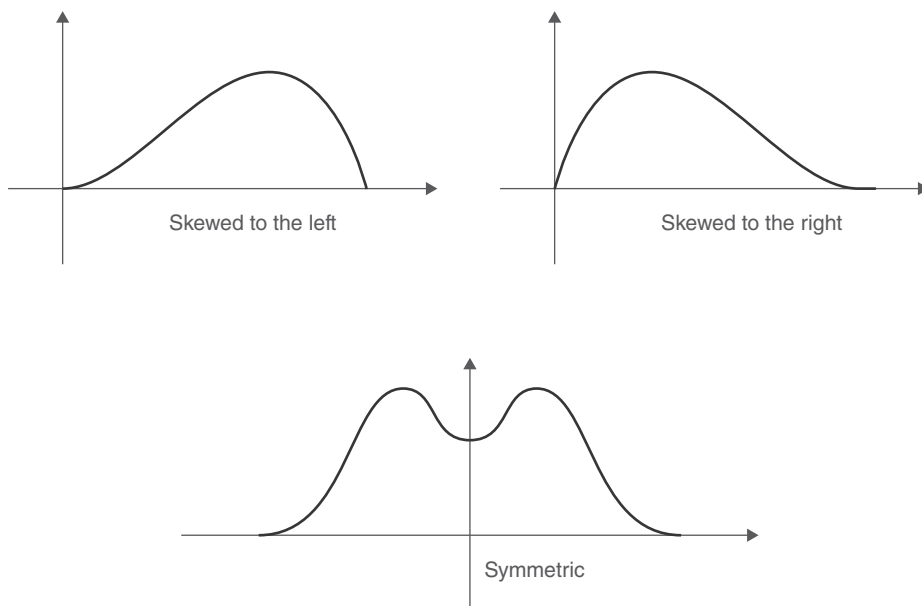


Figure 10.1 Symmetry and skewness: comparing curve shapes.

- If the excess kurtosis is zero, the distribution is called **mesokurtic**. This means its shape, in terms of peakedness and tails, is similar to the standard normal distribution.
- If the excess kurtosis is positive, the distribution is called **leptokurtic**. Such a distribution exhibits heavier tails and a sharper peak compared to the standard normal distribution.
- If the excess kurtosis is negative, the distribution is termed **platykurtic**. Such a distribution has lighter tails and is more flat-topped compared to the standard normal distribution.

For examples of probability density functions for mesokurtic, leptokurtic, and platykurtic distributions, refer to [Figure 10.2](#).

The moments of a random variable X give information of other sorts, too. For example, it can be proven that $E(|X|^k) < \infty$ implies that $\lim_{n \rightarrow \infty} n^k P(|X| > n) \rightarrow 0$, which shows that if $E(|X|^k) < \infty$, then $P(|X| > n)$ approaches 0 faster than $1/n^k$ as $n \rightarrow \infty$.

In this section we study moment-generating functions of random variables. These are real-valued functions with two major properties: They enable us to calculate the moments of random variables readily, and upon existence, they are unique. That is, no two different random variables have the same moment-generating function. Because of this, for proving that a random variable has a certain distribution function F , it is often shown that its moment-generating function coincides with that of F . This method enables us to prove many interesting facts, including the celebrated central limit theorem (see [Section 10.6](#)). Furthermore, Theorem 8.4, the convolution theorem, which is used to find the distribution of the sum of *two* independent random variables, cannot be extended so easily to many random variables. This fact is a major motivation for the introduction of the moment-generating functions, which, by Theorem 10.4,

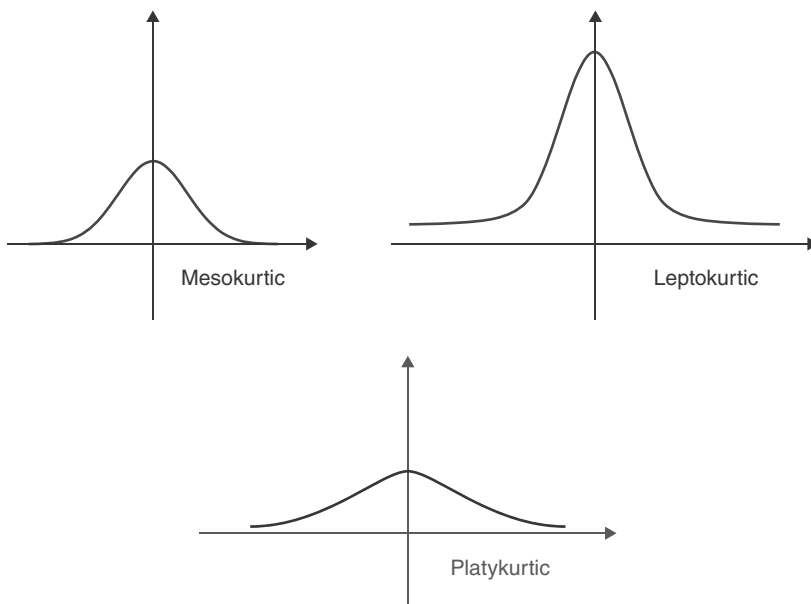


Figure 10.2 Curves highlighting excess kurtosis: zero, positive, and negative.

can be used to find, almost readily, the distribution functions of sums of many independent random variables.

Definition 10.1 For a random variable X , let

$$M_X(t) = E(e^{tX}).$$

If $M_X(t)$ is finite for all values of t in some interval $(-\delta, \delta)$, $\delta > 0$, then $M_X(t)$ is called the **moment-generating function** of X . If not, we say that the moment generating function of the random variable X does not exist.

Hence if X is a discrete random variable with set of possible values A and probability mass function $p(x)$, then

$$M_X(t) = \sum_{x \in A} e^{tx} p(x),$$

and if X is a continuous random variable with probability density function $f(x)$, then

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx.$$

Note that the condition that $M_X(t)$ be finite in some neighborhood of 0, $(-\delta, \delta)$, $\delta > 0$, is an important requirement. Without this condition some moments of X may not exist.

As we mentioned in the beginning and as can be guessed from its name, the moment-generating function of a random variable X can be used to calculate the moments of X . The

following theorem shows that the moments of X can be found by differentiating $M_X(t)$ and evaluating the derivatives at $t = 0$.

Theorem 10.1 *Let X be a random variable with moment-generating function $M_X(t)$. Then*

$$E(X^n) = M_X^{(n)}(0),$$

where $M_X^{(n)}(t)$ is the n th derivative of $M_X(t)$.

Proof: We prove the theorem for continuous random variables. For discrete random variables, the proof is similar. If X is continuous with probability density function f , then

$$\begin{aligned} M_X'(t) &= \frac{d}{dt} \left(\int_{-\infty}^{\infty} e^{tx} f(x) dx \right) = \int_{-\infty}^{\infty} x e^{tx} f(x) dx, \\ M_X''(t) &= \frac{d}{dt} \left(\int_{-\infty}^{\infty} x e^{tx} f(x) dx \right) = \int_{-\infty}^{\infty} x^2 e^{tx} f(x) dx, \\ &\vdots \\ M_X^{(n)}(t) &= \int_{-\infty}^{\infty} x^n e^{tx} f(x) dx, \end{aligned} \quad (10.1)$$

where we have assumed that the derivatives of these integrals are equal to the integrals of the derivatives of their integrands. This property is valid for sufficiently smooth densities. Letting $t = 0$ in (10.1), we get

$$M_X^{(n)}(0) = \int_{-\infty}^{\infty} x^n f(x) dx = E(X^n).$$

Note that since in some interval $(-\delta, \delta)$, $\delta > 0$, $M_X(t)$ is finite, it follows that $M_X^{(n)}(0)$ exists for all $n \geq 1$. ♦

One immediate application of Theorem 10.1 is the following corollary.

Corollary *The MacLaurin's series for $M_X(t)$ is given by*

$$M_X(t) = \sum_{n=0}^{\infty} \frac{M_X^{(n)}(0)}{n!} t^n = \sum_{n=0}^{\infty} \frac{E(X^n)}{n!} t^n. \quad (10.2)$$

Therefore, $E(X^n)$ is the coefficient of $t^n/n!$ in the MacLaurin's series representation of $M_X(t)$.

It is important to know that if M_X is to be finite, then the moments of all orders of X must be finite. But the converse need not be true. That is, all the moments may be finite and yet there is no neighborhood of 0, $(-\delta, \delta)$, $\delta > 0$, on which M_X is finite (see self-test problem 6).

Lemma 10.1 *Let X be a random variable with moment generating function $M_X(t)$. For constants a and b , let $Y = aX + b$. Then the moment generating function of Y is given by*

$$M_Y(t) = e^{bt} M_X(at).$$

Proof: By the definition of moment generating function,

$$M_Y(t) = E(e^{tY}) = E[e^{t(aX+b)}] = e^{bt}E(e^{atX}) = e^{bt}M_X(at) \quad \blacklozenge$$

Example 10.1 Let X be a Bernoulli random variable with parameter p , that is,

$$P(X = x) = \begin{cases} 1 - p & \text{if } x = 0 \\ p & \text{if } x = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Determine $M_X(t)$ and $E(X^n)$.

Solution: From the definition of a moment-generating function,

$$M_X(t) = E(e^{tX}) = (1 - p)e^{t \cdot 0} + pe^{t \cdot 1} = (1 - p) + pe^t.$$

Since $M_X^{(n)}(t) = pe^t$ for all $n > 0$, it follows that $E(X^n) = M_X^{(n)}(0) = p$. $\blacklozenge$

Example 10.2 Let X be a binomial random variable with parameters (n, p) . Find the moment-generating function of X , and use it to calculate $E(X)$ and $\text{Var}(X)$.

Solution: The probability mass function of X , $p(x)$, is given by

$$p(x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n, \quad q = 1 - p.$$

Hence

$$M_X(t) = E(e^{tX}) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x q^{n-x} = \sum_{x=0}^n \binom{n}{x} (pe^t)^x q^{n-x} = (pe^t + q)^n,$$

where the last equality follows from the binomial expansion (see Theorem 2.5). To find the mean and the variance of X , note that

$$\begin{aligned} M'_X(t) &= npe^t(pe^t + q)^{n-1} \\ M''_X(t) &= npe^t(pe^t + q)^{n-1} + n(n-1)(pe^t)^2(pe^t + q)^{n-2}. \end{aligned}$$

Thus

$$\begin{aligned} E(X) &= M'_X(0) = np \\ E(X^2) &= M''_X(0) = np + n(n-1)p^2. \end{aligned}$$

Therefore,

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = np + n(n-1)p^2 - n^2p^2 = npq. \quad \blacklozenge$$

Example 10.3 Let X be an exponential random variable with parameter λ . Using moment-generating functions, calculate the mean and the variance of X .

Solution: The probability density function of X is given by

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

Thus

$$M_X(t) = E(e^{tX}) = \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx = \lambda \int_0^\infty e^{(t-\lambda)x} dx.$$

Since the integral $\int_0^\infty e^{(t-\lambda)x} dx$ converges if $t < \lambda$, restricting the domain of $M_X(t)$ to $(-\infty, \lambda)$, we get

$$M_X(t) = \frac{\lambda}{\lambda - t}.$$

Thus $M'_X(t) = \lambda/(\lambda - t)^2$ and $M''_X(t) = (2\lambda)/(\lambda - t)^3$. We obtain $E(X) = M'_X(0) = 1/\lambda$ and $E(X^2) = M''_X(0) = 2/\lambda^2$. Therefore,

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \quad \blacklozenge$$

Example 10.4 Let X be an exponential random variable with parameter λ . Using moment-generating functions, find $E(X^n)$, where n is a positive integer.

Solution: By Example 10.3,

$$M_X(t) = \frac{\lambda}{\lambda - t} = \frac{1}{1 - (t/\lambda)}, \quad t < \lambda.$$

Now, on the one hand, by the geometric series theorem,

$$M_X(t) = \frac{1}{1 - (t/\lambda)} = \sum_{n=0}^{\infty} \left(\frac{t}{\lambda}\right)^n = \sum_{n=0}^{\infty} \left(\frac{1}{\lambda}\right)^n t^n,$$

and, on the other hand, by (10.2),

$$M_X(t) = \sum_{n=0}^{\infty} \frac{E(X^n)}{n!} t^n.$$

Comparing these two relations, we obtain

$$\frac{E(X^n)}{n!} = \left(\frac{1}{\lambda}\right)^n,$$

which gives $E(X^n) = n!/\lambda^n$. $\blacklozenge$

Example 10.5 Let Z be a standard normal random variable.

- (a) Calculate the moment-generating function of Z .
- (b) Use part (a) to find the moment-generating function of X , where X is a normal random variable with mean μ and variance σ^2 .
- (c) Use part (b) to calculate the mean and the variance of X .

Solution:

- (a) From the definition,

$$M_Z(t) = E(e^{tZ}) = \int_{-\infty}^{\infty} e^{tz} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz - z^2/2} dz.$$

Since

$$tz - \frac{z^2}{2} = \frac{t^2}{2} - \frac{(z - t)^2}{2},$$

we obtain

$$\begin{aligned} M_Z(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \left[\frac{t^2}{2} - \frac{(z - t)^2}{2} \right] dz \\ &= \frac{1}{\sqrt{2\pi}} e^{t^2/2} \int_{-\infty}^{\infty} \exp \left[-\frac{(z - t)^2}{2} \right] dz. \end{aligned}$$

Let $u = z - t$. Then $du = dz$ and

$$\begin{aligned} M_Z(t) &= \frac{1}{\sqrt{2\pi}} e^{t^2/2} \int_{-\infty}^{\infty} e^{-u^2/2} du \\ &= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2} du = e^{t^2/2}, \end{aligned}$$

where the last equality follows since the function $(1/\sqrt{2\pi})e^{-u^2/2}$ is the probability density function of a standard normal random variable, and hence its integral on $(-\infty, +\infty)$ is 1.

- (b) Letting $Z = (X - \mu)/\sigma$, we have that Z is $N(0, 1)$ and $X = \sigma Z + \mu$. Thus, by Lemma 10.1,

$$M_X(t) = e^{\mu t} M_Z(\sigma t) = e^{\mu t} \exp \left(\frac{1}{2} \sigma^2 t^2 \right) = \exp \left(\mu t + \frac{1}{2} \sigma^2 t^2 \right).$$

- (c) Differentiating $M_X(t)$, we obtain

$$M'_X(t) = (\mu + \sigma^2 t) \exp \left(t\mu + \frac{1}{2} \sigma^2 t^2 \right),$$

which upon differentiation gives

$$M_X''(t) = (\mu + \sigma^2 t)^2 \exp\left(t\mu + \frac{1}{2}\sigma^2 t^2\right) + \sigma^2 \exp\left(t\mu + \frac{1}{2}\sigma^2 t^2\right).$$

Therefore,

$$E(X) = M_X'(0) = \mu \quad \text{and} \quad E(X^2) = M_X''(0) = \mu^2 + \sigma^2.$$

Thus

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \mu^2 + \sigma^2 - \mu^2 = \sigma^2. \quad \blacklozenge$$

Example 10.6 A positive random variable X is called **lognormal** with parameters μ and σ^2 if $\ln X \sim N(\mu, \sigma^2)$. Let X be a lognormal random variable with parameters μ and σ^2 .

- (a) For a positive integer r , calculate the r th moment of X .
- (b) Use the r th moment of X to find $\text{Var}(X)$.
- (c) In 1977, a British researcher demonstrated that if X is the loss from a large fire, then $\ln X$ is a normal random variable. That is, X is lognormal. Suppose that the expected loss due to fire in the buildings of a certain industry, in thousands of dollars, is 120 with standard deviation 36. What is the probability that the loss from a fire in such an industry is less than \$100,000?

Solution:

- (a) Let $Y = \ln X$, then $X = e^Y$. Now $Y \sim N(\mu, \sigma^2)$ gives

$$E(X^r) = E(e^{rY}) = M_Y(r) = \exp\left(r\mu + \frac{1}{2}\sigma^2 r^2\right).$$

- (b) By part (a),

$$E(X) = e^{\mu + (1/2)\sigma^2},$$

$$E(X^2) = e^{2\mu + 2\sigma^2},$$

$$\text{Var}(X) = e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1).$$

- (c) Let X be the loss from the fire in the industry. Let the mean and standard deviation of $\ln X$ be μ and σ , respectively. By part (b),

$$\frac{\text{Var}(X)}{[E(X)]^2} = e^{\sigma^2} - 1.$$

Hence $e^{\sigma^2} - 1 = 36^2/120^2 = 0.09$, or, equivalently, $e^{\sigma^2} = 1.09$, which gives $\sigma^2 = \ln 1.09$, or $\sigma = \sqrt{\ln 1.09} = 0.294$. Therefore,

$$120 = e^\mu \cdot e^{(1/2)\sigma^2} = e^\mu \cdot \sqrt{1.09},$$

which gives $e^\mu = 114.939$, or $\mu = \ln(114.939) = 4.744$. Thus the probability we are interested in is

$$\begin{aligned} P(X < 100) &= P(\ln X < \ln 100) = P(\ln X < 4.605) \\ &= P\left(Z < \frac{4.605 - 4.744}{0.294}\right) = P(Z < -0.47) = 0.3192. \quad \blacklozenge \end{aligned}$$

Another important property of a moment-generating function is its uniqueness, which we now state without proof.

Theorem 10.2 *Let X and Y be two random variables with moment-generating functions $M_X(t)$ and $M_Y(t)$. If for some $\delta > 0$, $M_X(t) = M_Y(t)$ for all values of t in $(-\delta, \delta)$, then X and Y have the same distribution.*

Note that the converse of Theorem 10.2 is trivially true.

Theorem 10.2 serves as both a significant tool and a startling discovery. It shows that a moment-generating function determines the distribution function uniquely. That is, if two random variables have the same moment-generating function, then they are identically distributed. As we mentioned before, because of the uniqueness property, in order that a random variable has a certain distribution function F , it suffices to show that its moment-generating function coincides with that of F .

Example 10.7 Let the moment-generating function of a random variable X be

$$M_X(t) = \frac{1}{7}e^t + \frac{3}{7}e^{3t} + \frac{2}{7}e^{5t} + \frac{1}{7}e^{7t}.$$

Since the moment-generating function of a discrete random variable with the probability mass function

i	1	3	5	7	Other values
$p(i)$	1/7	3/7	2/7	1/7	0

is $M_X(t)$ given previously, by Theorem 10.2, the probability mass function of X is $p(i)$. $\blacklozenge$

Example 10.8 Let X be a random variable with moment-generating function $M_X(t) = e^{2t^2}$. Find $P(0 < X < 1)$.

Solution: Comparing $M_X(t) = e^{2t^2}$ with $\exp[\mu t + (1/2)\sigma^2 t^2]$, the moment-generating function of $N(\mu, \sigma^2)$ (see Example 10.5), we have that $X \sim N(0, 4)$ by the uniqueness of the moment-generating function. Let $Z = (X - 0)/2$. Then $Z \sim N(0, 1)$, so

$$\begin{aligned} P(0 < X < 1) &= P(0 < X/2 < 1/2) = P(0 < Z < 0.5) \\ &= \Phi(0.5) - \Phi(0) \approx 0.6915 - 0.5 = 0.1915, \end{aligned}$$

by Table 2 of the Appendix Tables. $\blacklozenge$

One of the key characteristics of the moment-generating function, used to prove the central limit theorem, is that when the moment-generating functions of a sequence of random variables converge to a specific function as the number of variables in the sequence increases, this limiting function uniquely characterizes the probability distribution of the random variable to which the sequence converges. The following theorem, stated without proof, explains this critical phenomenon.

Theorem 10.3 (Lévy Continuity Theorem for Moment-Generating Functions)

Let $X_1, X_2, \dots$ be a sequence of random variables with distribution functions $F_1, F_2, \dots$ and moment-generating functions $M_{X_1}(t), M_{X_2}(t), \dots$, respectively. Let X be a random variable with distribution function F and moment-generating function $M_X(t)$. If for all values of t , $M_{X_n}(t)$ converges to $M_X(t)$, then at the points of continuity of F , F_n converges to F .

Example 10.9 Suppose that for $n = 1, 2, \dots$, the random variable X_n is a uniform random variable over the interval $[-1/n, 1/n]$. Find X , the random variable whose distribution function is the limit of the distributions of the sequence $X_1, X_2, \dots$.

Solution: From Table 4 in the Appendix Tables, we have that the moment-generating function of a uniform random variable over an interval (a, b) is $\frac{e^{tb} - e^{ta}}{t(b-a)}$. Therefore, for $n = 1, 2, \dots$,

$$M_{X_n}(t) = \frac{e^{t/n} - e^{-t/n}}{t\left[\frac{1}{n} - \left(-\frac{1}{n}\right)\right]} = \frac{n}{2t}(e^{t/n} - e^{-t/n}).$$

Clearly, for real numbers x ,

$$\lim_{n \rightarrow \infty} M_{X_n}(t) = \lim_{n \rightarrow \infty} \frac{n}{2t}(e^{t/n} - e^{-t/n}) = \lim_{x \rightarrow \infty} \frac{x}{2t}(e^{t/x} - e^{-t/x}).$$

Using l'Hôpital's rule,

$$\begin{aligned} \lim_{n \rightarrow \infty} M_{X_n}(t) &= \lim_{x \rightarrow \infty} \frac{e^{t/x} - e^{-t/x}}{2t/x} \\ &= \lim_{x \rightarrow \infty} \frac{(-t/x^2)e^{t/x} - (t/x^2)e^{-t/x}}{-2t/x^2} = \lim_{x \rightarrow \infty} \frac{1}{2}(e^{t/x} + e^{-t/x}) = 1. \end{aligned}$$

So, the moment-generating function of the random variable X , whose distribution is the limit of the distributions of X_n 's is given by $M_X(t) = 1$. This corresponds to a random variable that takes the value 0 with probability 1. Therefore, the distribution functions of uniform random variables over $[-1/n, 1/n]$ converges to the distribution function of the random variable X , which is 0 with probability 1. ♦

One of the significant features of moment-generating functions is that they enable us to find distribution functions of sums of independent random variables. We discuss this important property in Section 10.2 and use it in Section 10.6 to prove the central limit theorem.

Often it happens that the moment-generating function of a random variable is known but its distribution function is not. In such cases [Table 4](#) of the Appendix Tables might help us identify the distribution.

EXERCISES

A

1. Let X be a discrete random variable with probability mass function $p(i) = 1/5$, $i = 1, 2, \dots, 5$, zero elsewhere. Find $M_X(t)$.
2. Let X be a random variable with probability density function

$$f(x) = \begin{cases} 1/4 & \text{if } x \in (-1, 3) \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find $M_X(t)$, $E(X)$, and $\text{Var}(X)$.
- (b) Using $M_X(t)$, calculate $E(X)$.

Hint: Note that by the definition of derivative,

$$M'_X(0) = \lim_{h \rightarrow 0} \frac{M_X(h) - M_X(0)}{h}.$$

3. Let X be a discrete random variable with the probability mass function

$$p(i) = 2\left(\frac{1}{3}\right)^i, \quad i = 1, 2, 3, \dots; \quad \text{zero elsewhere.}$$

Find $M_X(t)$ and $E(X)$.

4. Let X be a continuous random variable with probability density function $f(x) = 2x$, if $0 \leq x \leq 1$, zero elsewhere. Find the moment-generating function of X .
5. Let X be a continuous random variable with the probability density function $f(x) = 6x(1-x)$, if $0 \leq x \leq 1$, zero elsewhere.
 - (a) Find $M_X(t)$.
 - (b) Using $M_X(t)$, find $E(X)$.
6. Let $X \sim N(\mu, \sigma^2)$ and $Y = 2X + 3$. Find the distribution of Y .
7. Prove that the function $t/(1-t)$, $t < 1$, cannot be the moment-generating function of a random variable.
8. Let X be a discrete random variable. Prove that $E(X^n) = M_X^{(n)}(0)$.

9. (a) Find $M_X(t)$, the moment-generating function of a Poisson random variable X with parameter λ .
 (b) Use $M_X(t)$ to find $E(X)$ and $\text{Var}(X)$.
10. Let X be a uniform random variable over the interval (a, b) . Find the moment-generating function of X .
11. Let X be a geometric random variable with parameter p . Show that the moment-generating function of X is given by

$$M_X(t) = \frac{pe^t}{1 - qe^t}, \quad q = 1 - p, \quad t < -\ln q.$$

Use $M_X(t)$ to find $E(X)$ and $\text{Var}(X)$.

12. Let $M_X(t) = (1/21) \sum_{n=1}^6 ne^{nt}$. Find the probability mass function of X .
13. Suppose that the moment-generating function of a random variable X is given by

$$M_X(t) = \frac{1}{3}e^t + \frac{4}{15}e^{3t} + \frac{2}{15}e^{4t} + \frac{4}{15}e^{5t}.$$

Find the probability mass function of X .

14. Let $M_X(t) = 1/(1-t)$, $t < 1$ be the moment-generating function of a random variable X . Find the moment-generating function of the random variable $Y = 2X + 1$.
15. For a random variable X , $M_X(t) = [2/(2-t)]^3$. Find $E(X)$ and $\text{Var}(X)$.
16. Suppose that the moment-generating function of X is given by

$$M_X(t) = \frac{e^t + e^{-t}}{6} + \frac{2}{3}, \quad -\infty < t < \infty.$$

Find $E(X^r)$, $r \geq 1$.

17. According to an actuary, the claim size for a certain brand of luxury car when it gets involved in an accident is a continuous random variable with moment generating function

$$M_X(t) = \frac{1}{(1 - 3000t)^5}.$$

Find the expected value and the standard deviation of the claim size for such a car involved in an accident and selected randomly.

18. In each of the following cases $M_X(t)$, the moment-generating function of X , is given. Determine the distribution of X .
- (a) $M_X(t) = \left(\frac{1}{4}e^t + \frac{3}{4}\right)^7$.
- (b) $M_X(t) = e^t/(2 - e^t)$.
- (c) $M_X(t) = [2/(2 - t)]^r$.

(d) $M_X(t) = \exp [3(e^t - 1)]$.

Hint: Use Table 4 of the Appendix Tables.

19. For a random variable X , $M_X(t) = (1/81)(e^t + 2)^4$. Find $P(X \leq 2)$.
20. Let X be a uniform random variable over $(0, 1)$. Let a and b be two positive numbers. Using moment-generating functions, show that $Y = aX + b$ is uniformly distributed over $(b, a + b)$.
21. Let X be the outcome of a biased die when thrown. Suppose that the probability mass function of X is given by

i	1	2	3	4	5	6
$p(i)$	1/12	3/12	2/12	1/12	4/12	1/12

If the die is thrown 8 times, what is the probability that the sum of the outcomes is 11?

Hint: Use the multinomial expansion, Theorem 2.6, to find the coefficient of e^{11t} in the expansion of $[M_X(t)]^8$.

B

22. Let $Z \sim N(0, 1)$. Use $M_Z(t) = e^{t^2/2}$ to calculate $E(Z^n)$, where n is a positive integer.
- Hint:* Use (10.2).
23. Let X be a gamma random variable with parameters r and λ . Derive a formula for $M_X(t)$, and use it to calculate $E(X)$ and $\text{Var}(X)$.
24. Let X be a continuous random variable whose probability density function f is even; that is, $f(-x) = f(x)$, $\forall x$. Prove that (a) the random variables X and $-X$ have the same distribution function; (b) the function $M_X(t)$ is an even function.
25. Let X be a discrete random variable with probability mass function

$$p(i) = \frac{6}{\pi^2 i^2}, \quad i = 1, 2, 3, \dots; \quad \text{zero elsewhere.}$$

Show that the moment-generating function of X does not exist.

Hint: Show that $M_X(t)$ is a divergent series on $(0, \infty)$. This implies that on no interval of the form $(-\delta, \delta)$, $\delta > 0$, $M_X(t)$ exists.

26. Suppose that $\forall n \geq 1$, the n th moment of a random variable X , is given by $E(X^n) = (n + 1)! 2^n$. Find the distribution of X .
27. Suppose that A dollars are invested in a bank that pays interest at a rate of X per year, where X is a random variable.
- (a) Show that if a year is divided into k equal periods, and the bank pays interest at the end of each of these k periods, then after n such periods, with probability 1, the investment will grow to $A \left(1 + \frac{X}{k}\right)^n$.

- (b) For an infinitesimal $\varepsilon > 0$, suppose that the interest is compounded at the end of each period of length ε . If $\varepsilon \rightarrow 0$, then the interest is said to be **compounded continuously**. Suppose that at time t , the investment has grown to $A(t)$. By demonstrating that

$$A(t + \varepsilon) = A(t) + A(t) \cdot \varepsilon X,$$

show that, with probability 1,

$$A'(t) = XA(t).$$

- (c) Using part (b), prove that,

If the bank compounds interest continuously, then, on average, the money will grow by a factor of $M_X(t)$, the moment-generating function of the interest rate.

28. Let the joint probability mass function of $X_1, X_2, \dots, X_r$ be multinomial with parameters n and $p_1, p_2, \dots, p_r$ ($p_1 + p_2 + \dots + p_r = 1$). Find $\rho(X_i, X_j)$, $1 \leq i \neq j \leq r$. *Hint:* Note that X_i and X_j are binomial random variables and the joint marginal probability mass function of X_i and X_j is multinomial. To find $E(X_i X_j)$, calculate

$$M(t_1, t_2) = E(e^{t_1 X_i + t_2 X_j})$$

for all values of t_1 and t_2 and note that

$$E(X_i X_j) = \frac{\partial^2 M(0, 0)}{\partial t_1 \partial t_2}.$$

The function $M(t_1, t_2)$ is called the **moment-generating function of the joint distribution** of X_i and X_j . ■

Self-Quiz on Section 10.1

Time allotted: 25 Minutes

Each problem is worth 2.5 points.

- Let X be a binomial random variable with parameters n and p . Using moment generating functions, find the probability mass function of $Y = n - X$.
- Can the function $f(t) = \left(\frac{1}{3}e^t + \frac{1}{4}\right)^5$ be the moment generating function of a random variable X ?
- For some $\alpha > 0$, the moment generating function of X is given by

$$M_X(t) = \frac{1}{8}e^{\alpha t} + \frac{3}{8}e^{2\alpha t} + \frac{1}{2}e^{3\alpha t}.$$

For what value of α does $\text{Var}(X) = 279/64$?

4. Suppose that for a random variable X , $E(X^n) = 2^n$, $n = 1, 2, 3, \dots$. Calculate the moment-generating function and the probability mass function of X .

Hint: Note that $\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$, $-\infty < x < \infty$.



10.2 FURTHER APPLICATIONS OF MOMENT-GENERATING FUNCTIONS

As mentioned before, the convolution theorem cannot be extended so easily to find the distribution function of sums of more than two independent random variables. For this reason, in this section, we study the distribution functions of such sums using moment-generating functions. To begin, we prove the following theorem, which together with the uniqueness property of moment-generating functions (see Theorem 10.2) are two of the main tools of this section.

Theorem 10.4 *Let $X_1, X_2, \dots, X_n$ be independent random variables with moment-generating functions $M_{X_1}(t), M_{X_2}(t), \dots, M_{X_n}(t)$. The moment-generating function of $X_1 + X_2 + \dots + X_n$ is given by*

$$M_{X_1+X_2+\dots+X_n}(t) = M_{X_1}(t)M_{X_2}(t) \cdots M_{X_n}(t).$$

Proof: Let $W = X_1 + X_2 + \dots + X_n$, by definition,

$$\begin{aligned} M_W(t) &= E(e^{tW}) = E(e^{tX_1+tX_2+\dots+tX_n}) \\ &= E(e^{tX_1}e^{tX_2} \cdots e^{tX_n}) \\ &= E(e^{tX_1})E(e^{tX_2}) \cdots E(e^{tX_n}) \\ &= M_{X_1}(t)M_{X_2}(t) \cdots M_{X_n}(t), \end{aligned}$$

where the next-to-last equality follows from the independence of $X_1, X_2, X_3, \dots, X_n$. 

We now prove that sums of independent binomial random variables are binomial. Let X and Y be two independent binomial random variables with parameters (n, p) and (m, p) , respectively. Then X and Y are the numbers of successes in n and m independent Bernoulli trials with parameter p , respectively. Thus $X + Y$ is the number of successes in $n + m$ independent Bernoulli trials with parameter p . Therefore, $X + Y$ is a binomial random variable with parameters $(n + m, p)$. An alternative for this probabilistic argument is the following analytic proof.

Theorem 10.5 *Let $X_1, X_2, \dots, X_r$ be independent binomial random variables with parameters $(n_1, p), (n_2, p), \dots, (n_r, p)$, respectively. Then $X_1 + X_2 + \dots + X_r$ is a binomial random variable with parameters $n_1 + n_2 + \dots + n_r$ and p .*

Proof: Let, as usual, $q = 1 - p$. We know that

$$M_{X_i}(t) = (pe^t + q)^{n_i}, \quad i = 1, 2, 3, \dots, n$$

(see Example 10.2). Let $W = X_1 + X_2 + \dots + X_r$, then, by Theorem 10.4,

$$\begin{aligned} M_W(t) &= M_{X_1}(t)M_{X_2}(t) \cdots M_{X_r}(t) \\ &= (pe^t + q)^{n_1}(pe^t + q)^{n_2} \cdots (pe^t + q)^{n_r} \\ &= (pe^t + q)^{n_1+n_2+\dots+n_r}. \end{aligned}$$

Since $(pe^t + q)^{n_1+n_2+\dots+n_r}$ is the moment-generating function of a binomial random variable with parameters $(n_1 + n_2 + \dots + n_r, p)$, the uniqueness property of moment-generating functions implies that $W = X_1 + X_2 + \dots + X_r$ is binomial with parameters $(n_1 + n_2 + \dots + n_r, p)$. $\blacklozenge$

We just showed that if X and Y are independent binomial random variables with parameters (n, p) and (m, p) , respectively, then $X + Y$ is a binomial random variable with parameters $(n + m, p)$. For large values of n and m , small p , and moderate $\lambda_1 = np$ and $\lambda_2 = mp$, Poisson probability mass functions with parameters λ_1 , λ_2 , and $\lambda_1 + \lambda_2$ approximate probability mass functions of X , Y , and $X + Y$, respectively. Therefore, it is reasonable to expect that for independent Poisson random variables X and Y with parameters λ_1 and λ_2 , $X + Y$ is a Poisson random variable with parameter $\lambda_1 + \lambda_2$. The proof of this interesting fact follows for n random variables.

Theorem 10.6 *Let $X_1, X_2, \dots, X_n$ be independent Poisson random variables with means $\lambda_1, \lambda_2, \dots, \lambda_n$, respectively. Then $X_1 + X_2 + \dots + X_n$ is a Poisson random variable with mean $\lambda_1 + \lambda_2 + \dots + \lambda_n$.*

Proof: Let Y be a Poisson random variable with mean λ . Then

$$\begin{aligned} M_Y(t) &= E(e^{tY}) = \sum_{y=0}^{\infty} e^{ty} \frac{e^{-\lambda} \lambda^y}{y!} = e^{-\lambda} \sum_{y=0}^{\infty} \frac{(e^t \lambda)^y}{y!} \\ &= e^{-\lambda} \exp(\lambda e^t) = \exp[\lambda(e^t - 1)]. \end{aligned}$$

Let $W = X_1 + X_2 + \dots + X_n$, then, by Theorem 10.4,

$$\begin{aligned} M_W(t) &= M_{X_1}(t)M_{X_2}(t) \cdots M_{X_n}(t) \\ &= \exp[\lambda_1(e^t - 1)] \exp[\lambda_2(e^t - 1)] \cdots \exp[\lambda_n(e^t - 1)] \\ &= \exp[(\lambda_1 + \lambda_2 + \dots + \lambda_n)(e^t - 1)]. \end{aligned}$$

Now, since $\exp[(\lambda_1 + \lambda_2 + \dots + \lambda_n)(e^t - 1)]$ is the moment-generating function of a Poisson random variable with mean $\lambda_1 + \lambda_2 + \dots + \lambda_n$, the uniqueness property of moment-generating functions implies that $X_1 + X_2 + \dots + X_n$ is Poisson with mean $\lambda_1 + \lambda_2 + \dots + \lambda_n$. $\blacklozenge$

Example 10.10 **A (nonrigorous) proof of Stirling's formula (Theorem 2.7):**

$$n! \sim \sqrt{2\pi n} n^n e^{-n}.$$

That is,

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi n} n^n e^{-n}} = 1.$$

To justify this theorem, let $\{X_1, X_2, \dots\}$ be a sequence of independent and identically distributed Poisson random variables with $\lambda = 1$. Then $\mu = E(X_i) = 1$ and $\sigma^2 = \text{Var}(X_i) = 1$, $i \geq 1$. Let $S_n = \sum_{i=1}^n X_i$, then, by Theorem 10.6, S_n is Poisson with parameter n . Therefore,

$$P(S_n = n) = \frac{e^{-n} \cdot n^n}{n!}. \quad (10.3)$$

Since $\mu = \sigma = 1$, by the central limit theorem,

$$\begin{aligned} P(S_n = n) &= P(n-1 < S_n \leq n) = P\left(\frac{-1}{\sqrt{n}} < \frac{S_n - n}{\sqrt{n}} \leq 0\right) \\ &= P\left(\frac{-1}{\sqrt{n}} < \frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq 0\right) \approx \Phi(0) - \Phi\left(\frac{-1}{\sqrt{n}}\right). \end{aligned}$$

By the mean value theorem of calculus, there exists $t_n \in (-1/\sqrt{n}, 0)$ such that

$$\Phi(0) - \Phi\left(\frac{-1}{\sqrt{n}}\right) = \Phi'(t_n) \left[0 - \left(-\frac{1}{\sqrt{n}}\right)\right] = \frac{1}{\sqrt{2\pi}} e^{-t_n^2/2} \cdot \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{2\pi n}} e^{-t_n^2/2}.$$

Clearly, the sequence $\{t_n\}$ converges to 0 as $n \rightarrow \infty$. Therefore, for large n ,

$$P(S_n = n) \approx \Phi(0) - \Phi\left(\frac{-1}{\sqrt{n}}\right) \approx \frac{1}{\sqrt{2\pi n}}. \quad (10.4)$$

Comparing (10.3) and (10.4), we have

$$\lim_{n \rightarrow \infty} \frac{1/\sqrt{2\pi n}}{e^{-n} \cdot n^n / n!} = 1,$$

or, equivalently,

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi n} n^n e^{-n}} = 1;$$

that is, $n! \sim \sqrt{2\pi n} n^n e^{-n}$. ♦

Theorem 10.7 Let $X_1 \sim N(\mu_1, \sigma_1^2)$, $X_2 \sim N(\mu_2, \sigma_2^2)$, $\dots$, $X_n \sim N(\mu_n, \sigma_n^2)$ be independent random variables. Then

$$X_1 + X_2 + \dots + X_n \sim N(\mu_1 + \mu_2 + \dots + \mu_n, \sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2).$$

Proof: In Example 10.5, we showed that if X is normal with parameters μ and σ^2 , then $M_X(t) = \exp[\mu t + (1/2)\sigma^2 t^2]$. Let $W = X_1 + X_2 + \cdots + X_n$, then

$$\begin{aligned} M_W(t) &= M_{X_1}(t)M_{X_2}(t) \cdots M_{X_n}(t) \\ &= \exp\left(\mu_1 t + \frac{1}{2}\sigma_1^2 t^2\right) \exp\left(\mu_2 t + \frac{1}{2}\sigma_2^2 t^2\right) \cdots \exp\left(\mu_n t + \frac{1}{2}\sigma_n^2 t^2\right) \\ &= \exp\left[(\mu_1 + \mu_2 + \cdots + \mu_n)t + \frac{1}{2}(\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_n^2)t^2\right]. \end{aligned}$$

This implies that

$$X_1 + X_2 + \cdots + X_n \sim N(\mu_1 + \mu_2 + \cdots + \mu_n, \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_n^2). \quad \blacklozenge$$

The technique used to show Theorems 10.5, 10.6, and 10.7 can also be used to prove the following and similar theorems.

- The sum of n independent geometric random variables each with parameter p is negative binomial with parameters (n, p) .
- The sum of n independent negative binomial random variables with parameters $(r_1, p), (r_2, p), \dots, (r_n, p)$ is negative binomial with parameters (r, p) , $r = r_1 + r_2 + \cdots + r_n$.
- The sum of n independent exponential random variables with the identical mean $1/\lambda$ is gamma with parameters (n, λ) .
- The sum of n independent gamma random variables with parameters $(r_1, \lambda), (r_2, \lambda), \dots, (r_n, \lambda)$ is gamma with parameters $(r_1 + r_2 + \cdots + r_n, \lambda)$.

These important theorems are discussed in the exercises.

If X is a normal random variable with parameters (μ, σ^2) , then for $\alpha \in \mathbb{R}$, by Lemma 10.1 and Example 10.5, $M_{\alpha X}(t) = \exp[\alpha\mu t + (1/2)\alpha^2\sigma^2 t^2]$, implying that $\alpha X \sim N(\alpha\mu, \alpha^2\sigma^2)$. This and Theorem 10.7 imply the following important theorem.

Linear combinations of independent normal random variables are normal:

Theorem 10.8 Let $\{X_1, X_2, \dots, X_n\}$ be a set of independent random variables and $X_i \sim N(\mu_i, \sigma_i^2)$ for $i = 1, 2, \dots, n$, then for constants $\alpha_1, \alpha_2, \dots, \alpha_n$,

$$\sum_{i=1}^n \alpha_i X_i \sim N\left(\sum_{i=1}^n \alpha_i \mu_i, \sum_{i=1}^n \alpha_i^2 \sigma_i^2\right).$$

In particular, this theorem implies that if $X_1, X_2, \dots, X_n$ are independent normal random variables all with the same mean μ , and the same variance σ^2 , then $S_n = X_1 + X_2 + \cdots + X_n$ is $N(n\mu, n\sigma^2)$ and the sample mean, $\bar{X} = S_n/n$, is $N(\mu, \sigma^2/n)$. That is,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \text{ the sample mean of } n \text{ independent } N(\mu, \sigma^2), \text{ is } N\left(\mu, \frac{\sigma^2}{n}\right).$$

Example 10.11 Suppose that the distribution of students' grades in a probability test is normal, with mean 72 and variance 25.

- (a) What is the probability that the average of grade of such a probability class with 25 students is 75 or more?
- (b) If a professor teaches two different sections of this course, each containing 25 students, what is the probability that the average of one class is at least three more than the average of the other class?

Solution:

- (a) Let $X_1, X_2, \dots, X_{25}$ denote the grades of the 25 students. Then $X_1, X_2, \dots, X_{25}$ are independent random variables all being normal, with $\mu = 72$ and $\sigma^2 = 25$. The average of the grades of the class, $\bar{X} = (1/25) \sum_{i=1}^{25} X_i$, is normal, with mean $\mu = 72$ and variance $\sigma^2/n = 25/25 = 1$. Hence

$$P(\bar{X} \geq 75) = P\left(\frac{\bar{X} - 72}{1} \geq \frac{75 - 72}{1}\right) = P(\bar{X} - 72 \geq 3) = 1 - \Phi(3) \approx 0.0013.$$

- (b) Let $\bar{X}$ and $\bar{Y}$ denote the means of the grades of the two classes. Then, as seen in part (a), $\bar{X}$ and $\bar{Y}$ are both $N(72, 1)$. Since a student does not take two sections of the same course, $\bar{X}$ and $\bar{Y}$ are independent random variables. By Theorem 10.8, $\bar{X} - \bar{Y}$ is $N(0, 2)$. Hence, by symmetry,

$$\begin{aligned} P(|\bar{X} - \bar{Y}| > 3) &= P(\bar{X} - \bar{Y} > 3 \text{ or } \bar{Y} - \bar{X} > 3) \\ &= 2P(\bar{X} - \bar{Y} > 3) \\ &= 2P\left(\frac{\bar{X} - \bar{Y} - 0}{\sqrt{2}} > \frac{3 - 0}{\sqrt{2}}\right) \\ &= 2P\left(\frac{\bar{X} - \bar{Y}}{\sqrt{2}} > 2.12\right) = 2[1 - \Phi(2.12)] \approx 0.034. \quad \blacklozenge \end{aligned}$$

As we mentioned previously, by using moment-generating functions, it can be shown that a sum of n independent exponential random variables, each with parameter λ , is gamma with parameters n and λ (see Exercise 2). We now present an alternative proof for this important fact: Let X be a gamma random variable with parameters (n, λ) , where n is a positive integer. Consider a Poisson process $\{N(t): t \geq 0\}$ with rate λ . $\{X_1, X_2, \dots\}$, the set of interarrival times of this process [X_i is the time between the $(i-1)$ st event and the i th event, $i = 1, 2, \dots$], is an independent, identically distributed sequence of exponential random variables with $E(X_i) = 1/\lambda$ for $i = 1, 2, \dots$. Since the distribution of X is the same as the time of the occurrence of the n th event,

$$X = X_1 + X_2 + \dots + X_n.$$

Hence

A gamma random variable with parameters (n, λ) is the sum of n independent exponential random variables, each with mean $1/\lambda$, and vice versa.

Using moment-generating functions, we can also prove that if $X_1, X_2, \dots, X_n$ are n independent gamma random variables with parameters $(r_1, \lambda), (r_2, \lambda), \dots, (r_n, \lambda)$, respectively, then $X_1 + X_2 + \dots + X_n$ is gamma with parameters $(r_1 + r_2 + \dots + r_n, \lambda)$ (see Exercise 4). The following example is an application of this fact. It gives an interesting class of gamma random variables with parameters $(r, 1/2)$, where r is not necessarily an integer.

Example 10.12 Office fire insurance policies by a certain company have a \$1000 deductible. The company has received three claims, independent of each other, for damages caused by office fire. If reconstruction expenses for such claims are exponentially distributed, each with mean \$45,000, what is the probability that the total payment for these claims is less than \$120,000?

Solution: Let X be the total reconstruction expenses for the three claims in thousands of dollars; X is the sum of three independent exponential random variables, each with parameter \$45. Therefore, it is a gamma random variable with parameters 3 and $\lambda = 1/45$. Hence its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{45} e^{-x/45} \frac{(x/45)^2}{2} = \frac{1}{182,250} x^2 e^{-x/45} & \text{if } x \geq 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Considering the deductibles, the probability we are interested in is

$$\begin{aligned} P(X < 123) &= \frac{1}{182,250} \int_0^{123} x^2 e^{-x/45} dx \\ &= \frac{1}{182,250} (-45x^2 - 4050x - 182,250)e^{-x/45} \Big|_0^{123} = 0.5145. \quad \blacklozenge \end{aligned}$$

Example 10.13 Let $X_1, X_2, \dots, X_n$ be independent standard normal random variables. Then $X = X_1^2 + X_2^2 + \dots + X_n^2$, referred to as **chi-squared random variable with n degrees of freedom**, is gamma with parameters $(n/2, 1/2)$. An example of such a gamma random variable is the error of hitting a target in n -dimensional Euclidean space when the error of each coordinate is individually normally distributed.

Proof: Since the sum of n independent gamma random variables, each with parameters $(1/2, 1/2)$, is gamma with parameters $(n/2, 1/2)$, it suffices to prove that for all i , $1 \leq i \leq n$, X_i^2 is gamma with parameters $(1/2, 1/2)$. To prove this assertion, note that

$$P(X_i^2 \leq t) = P(-\sqrt{t} \leq X_i \leq \sqrt{t}) = \Phi(\sqrt{t}) - \Phi(-\sqrt{t}).$$

Let the probability density function of X_i^2 be f . Differentiating this equation yields

$$f(t) = \left(\frac{1}{2\sqrt{t}} \frac{1}{\sqrt{2\pi}} e^{-t/2} \right) - \left(-\frac{1}{2\sqrt{t}} \frac{1}{\sqrt{2\pi}} e^{-t/2} \right).$$

Therefore,

$$f(t) = \frac{1}{\sqrt{2\pi t}} e^{-t/2} = \frac{(1/2)e^{-t/2}(t/2)^{1/2-1}}{\sqrt{\pi}}.$$

Since $\sqrt{\pi} = \Gamma(1/2)$ (see Exercise 11 of Section 7.5), this relation shows that X_i^2 is a gamma random variable with parameters $(1/2, 1/2)$. ♦

Remark 10.1 If $X_1, X_2, \dots, X_n$ are independent random variables and, for $1 \leq i \leq n$, $X_i \sim N(\mu_i, \sigma_i^2)$, then, by Example 10.13, $\sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2$ is gamma with parameters $(n/2, 1/2)$. This holds since $(X_i - \mu_i)/\sigma_i$ is standard normal for $1 \leq i \leq n$.

EXERCISES

A

1. Let $X_1, X_2, \dots, X_n$ be independent geometric random variables each with parameter p . Using moment-generating functions, prove that $X_1 + X_2 + \dots + X_n$ is negative binomial with parameters (n, p) .
2. Let $X_1, X_2, \dots, X_n$ be n independent exponential random variables with the identical mean $1/\lambda$. Use moment-generating functions to show that the distribution of $X_1 + X_2 + \dots + X_n$ is gamma with parameters n and λ .
3. Using moment-generating functions, show that the sum of n independent negative binomial random variables with parameters $(r_1, p), (r_2, p), \dots, (r_n, p)$ is negative binomial with parameters (r, p) , $r = r_1 + r_2 + \dots + r_n$.
4. Let $X_1, X_2, \dots, X_n$ be n independent gamma random variables with parameters $(r_1, \lambda), (r_2, \lambda), \dots, (r_n, \lambda)$, respectively. Use moment-generating functions to show that the distribution of $X_1 + X_2 + \dots + X_n$ is gamma with parameters $(r_1 + r_2 + \dots + r_n, \lambda)$.
5. The probability is 0.15 that a bottle of a certain soda is underfilled, independent of the amount of soda in other bottles. If machine one fills 100 bottles and machine two fills 80 bottles of this soda per hour, what is the probability that tomorrow, between 10:00 A.M. and 11:00 A.M., both of these machines will underfill exactly 27 bottles altogether?
6. Let X and Y be independent binomial random variables with parameters (n, p) and (m, p) , respectively. Calculate $P(X = i \mid X + Y = j)$ and interpret the result.
7. Let X, Y , and Z be three independent Poisson random variables with parameters λ_1, λ_2 , and λ_3 , respectively. For $y = 0, 1, 2, \dots, t$, calculate $P(Y = y \mid X + Y + Z = t)$.
8. Mr. Watkins is at a train station, waiting to make a phone call. There is only one public telephone booth, and it is being used by someone. Another person ahead of Mr. Watkins

is also waiting to call. If the duration of each telephone call is an exponential random variable with $\lambda = 1/8$, find the probability that Mr. Watkins should wait at least 12 minutes before being able to call.

9. Let $X \sim N(1, 2)$ and $Y \sim N(4, 7)$ be independent random variables. Find the probability of the following events: (a) $X + Y > 0$, (b) $X - Y < 2$, (c) $3X + 4Y > 20$.
10. The distribution of the IQ of a randomly selected student from a certain college is $N(110, 16)$. What is the probability that the average of the IQ's of 10 randomly selected students from this college is at least 112?
11. Vicki owns two department stores. Delinquent charge accounts at store 1 show a normal distribution, with mean \$90 and standard deviation \$30, whereas at store 2 they show a normal distribution with mean \$100 and standard deviation \$50. If 10 delinquent accounts are selected randomly at store 1 and 15 at store 2, what is the probability that the average of the accounts selected at store 1 exceeds the average of those selected at store 2?
12. Let the joint probability density function of X and Y be bivariate normal. Prove that any linear combination of X and Y , $\alpha X + \beta Y$, is a normal random variable.
Hint: Use Theorem 10.8 and the result of Exercise 7, [Section 9.3](#).
13. Let X be the height of a man, and let Y be the height of his daughter (both in inches). Suppose that the joint probability density function of X and Y is bivariate normal with the following parameters: $\mu_X = 71$, $\mu_Y = 60$, $\sigma_X = 3$, $\sigma_Y = 2.7$, and $\rho = 0.45$. Find the probability that the man is at least 8 inches taller than his daughter.
Hint: Use the result of Exercise 12.
14. The capacity of an elevator is 2700 pounds. If the weight of a random athlete is normal with mean 225 pounds and standard deviation 25, what is the probability that the elevator can safely carry 12 random athletes?
15. The distributions of the grades of the students of probability and calculus at a certain university are $N(65, 418)$ and $N(72, 448)$, respectively. Dr. Olwell teaches a calculus section with 28 and a probability section with 22 students. What is the probability that the difference between the averages of the final grades of these two classes is at least 2?
16. Suppose that car mufflers last random times that are normally distributed with mean 3 years and standard deviation 1 year. If a certain family buys two new cars at the same time, what is the probability that (a) they should change the muffler of one car at least $1\frac{1}{2}$ years before the muffler of the other car; (b) one car does not need a new muffler for a period during which the other car needs two new mufflers?
17. On a certain day, the first customer of a single-teller bank arrives at 9:00 A.M. when the bank opens. The second customer arrives 10 minutes later at 9:10 A.M. Suppose that the service times of customers are independent exponential random variables each with mean 10 minutes, customers wait in a single queue if the teller is busy serving others, and they leave the bank as soon as they are served. Find the probability density function and the expected value of the amount of time the second customer will spend in the bank.

B

18. An elevator can carry up to 3500 pounds. The manufacturer has included a safety margin of 500 pounds and lists the capacity as 3000 pounds. The building's management seeks to avoid accidents by limiting the number of passengers on the elevator. If the weight of the passengers using the elevator is $N(155, 625)$, what is the maximum number of passengers who can use the elevator if the odds against exceeding the rated capacity (3000 pounds) are to be greater than 10,000 to 3?
19. Let the joint probability mass function of $X_1, X_2, \dots, X_r$ be multinomial, that is,

$$p(x_1, x_2, \dots, x_r) = \frac{n!}{x_1! x_2! \cdots x_r!} p_1^{x_1} p_2^{x_2} \cdots p_r^{x_r},$$

where $x_1 + x_2 + \cdots + x_r = n$, and $p_1 + p_2 + \cdots + p_r = 1$. Show that for $k < r$, $X_1 + X_2 + \cdots + X_k$ has a binomial distribution.

20. Kim is at a train station, waiting to make a phone call. Two public telephone booths, next to each other, are occupied by two callers, and 11 persons are waiting in a single line ahead of Kim to call. If the duration of each telephone call is an exponential random variable with $\lambda = 1/3$, what are the distribution and the expectation of the time that Kim must wait until being able to call?



Self-Quiz on Section 10.2

Time allotted: 15 Minutes

Each problem is worth 5 points.

- Suppose that a Massachusetts-based insurance company insures residential properties in Boston, Worcester, and Springfield. Suppose that, during a given year, X_1 , X_2 , and X_3 are the losses of this company due to house fires in the aforementioned areas, respectively. If, in millions of dollars, $X_1 \sim N(3, 2)$, $X_2 \sim N(1, 1)$, and $X_3 \sim N(2, 2)$, find the probability that next year, the total loss of the company from house fires in these regions is at least 4 million but not more than 6 million dollars. Note that since Boston, Worcester, and Springfield are sufficiently distant from one another, it is reasonable to assume that X_1 , X_2 , and X_3 are independent random variables.
- Let $Z_1, Z_2, \dots, Z_n$ be independent standard normal random variables. For what value of α , $\alpha > 0$, is $\alpha(Z_1 + Z_2 + \cdots + Z_n)$ also a standard normal random variable?



10.3 CHARACTERISTIC FUNCTIONS

In [Section 10.1](#), we learned that $M_X(t)$, the moment-generating function of a random variable X may not exist for all values of t . To ensure the existence of $M_X(t)$, the function must be finite for each t in some interval $(-\delta, \delta)$, where $\delta > 0$. If not, the moment-generating function of X does not exist. Additionally, if $M_X(t)$ is finite, then the moments of all orders of X are finite. However, the converse is not necessarily true; that is, even if all the moments of all orders of X are finite, there may not exist a neighborhood of 0, $(-\delta, \delta)$ with $\delta > 0$, on which M_X is finite.

In this section, we will delve into characteristic functions, which serve purposes similar to moment-generating functions, but they always exist for any random variable, regardless of whether its higher moments exist or not. Therefore, in cases where moment-generating functions fail to provide useful information, as they may be undefined or infinite, characteristic functions are still well-defined and can be used to study the properties of the random variable. The characteristic function of a given random variable is unique, whereas moment-generating functions may not be unique if certain moments are infinite. In other words, the characteristic function uniquely determines the distribution of a random variable. This property is known as the *Inversion Theorem*, which states that if two random variables have the same characteristic functions, they have the same distribution. Consequently, we can uniquely identify the distribution of a random variable from its characteristic function.

Proving the strong law of large numbers using moment generating functions is complex, and using it to prove the central limit theorem relies on the Lévy Continuity Theorem, whose proof is somewhat complicated. Therefore, we will state and use it without providing a proof. However, there are much simpler proofs for these celebrated theorems using characteristic functions.

Characteristic functions provide a broad and versatile framework for studying random variables, offering advantages in terms of existence, uniqueness, convergence, transformation properties, and inversion. Nevertheless, as we have seen in [Sections 10.1](#) and [10.2](#), moment-generating functions also have their own merits and strengths. They are often easier to compute than characteristic functions, especially for discrete random variables, involving simple algebraic manipulations instead of complex trigonometric functions and integrals.

The moment-generating function has a direct connection to the moments of a random variable through its derivatives. Taking the n th derivative of the moment-generating function at zero yields the n th moment of the random variable, which can be useful in certain situations. Furthermore, moment-generating function uniquely determine the moments of a random variable. If two random variables have the same moment-generating function, they share the same moments up to a certain order. This property can be valuable in determining the distribution of a random variable, as demonstrated in many applications in [Section 10.2](#).

Before we begin with a discussion of characteristic functions, let us recall a few basic facts about complex numbers. The conjugate of a complex number $z = a + bi$ changes the sign of its imaginary part. Thus, the conjugate of $a + bi$ is $a - bi$. The conjugate of a complex number z is denoted by $\bar{z}$. So

$$\overline{a + bi} = a - bi.$$

Clearly, for a complex number $z = a + bi$,

$$z \cdot \bar{z} = a + bi \cdot \overline{a + bi} = (a + bi) \cdot (a - bi) = a^2 + b^2 = |z|^2,$$

where $|z|$ is the magnitude of z , the distance from the origin to the point (a, b) representing the complex number z in the complex plane. Thus, $|z| = \sqrt{a^2 + b^2}$. Using Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, we have

$$|e^{i\theta}| = |\cos \theta + i \sin \theta| = \sqrt{\cos^2 \theta + \sin^2 \theta} = \sqrt{1} = 1.$$

In particular, for real numbers t and x , the relation $|e^{itx}| = 1$ will be frequently used.

Definition 10.2 Let X be a random variable, and $\mathbb{C}$ be the set of all complex numbers; that is, $\mathbb{C} = \{a + bi, \text{ where } a, b \in \mathbb{R} \text{ and } i = \sqrt{-1}\}$. The **characteristic function** of X , denoted by $\phi_X(t)$ is a function $\phi_X: \mathbb{R} \rightarrow \mathbb{C}$ defined by the expected value of e^{itX} :

$$\phi_X(t) = E(e^{itX}).$$

By Euler's formula,

$$\phi_X(t) = E(e^{itX}) = E(\cos tX + i \sin tX) = E(\cos tX) + iE(\sin tX).$$

Let X be a continuous random variable with probability density function f . Then

$$\phi_X(t) = E(e^{itX}) = \int_{-\infty}^{\infty} e^{itx} f(x) dx.$$

For a discrete random variable X with set of possible values A and probability mass function p ,

$$\phi_X(t) = E(e^{itX}) = \sum_{x \in A} e^{itx} p(x).$$

Example 10.14 Let X be a Bernoulli random variable with parameter p . That is, $X = 1$ with probability p , and $X = 0$ with probability $1 - p$. Then

$$\phi_X(t) = E(e^{itX}) = e^{it(1)} \cdot p + e^{it(0)} \cdot (1 - p) = pe^{it} + 1 - p. \quad \blacklozenge$$

Example 10.15 Let X be a Poisson random variable with parameter λ . Find the Characteristic function of X .

Solution: By the definition of characteristic function,

$$\begin{aligned} \phi_X(t) &= E(e^{itX}) = \sum_{n=0}^{\infty} e^{itn} \cdot P(X = n) = \sum_{n=0}^{\infty} e^{itn} \cdot \frac{e^{-\lambda} \lambda^n}{n!} \\ &= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda e^{it})^n}{n!} = e^{-\lambda} \cdot e^{\lambda e^{it}} = e^{-\lambda(1 - e^{it})}. \quad \blacklozenge \end{aligned}$$

Example 10.16 Find the Characteristic function of X , an exponential random variable with parameter λ .

Solution: Recall that $\lambda > 0$. The probability density function of X is $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Therefore,

$$\begin{aligned}\phi_X(t) &= E(e^{itX}) = \int_0^\infty e^{itx} \lambda e^{-\lambda x} dx = \lambda \int_0^\infty e^{(it-\lambda)x} dx \\ &= \left[\frac{\lambda}{it-\lambda} e^{(it-\lambda)x} \right]_0^\infty = 0 - \frac{\lambda}{it-\lambda} = \frac{\lambda}{\lambda - it},\end{aligned}$$

where $\frac{\lambda}{it-\lambda} e^{(it-\lambda)x} \rightarrow 0$ as $x \rightarrow \infty$ because

$$\left| \frac{\lambda}{it-\lambda} e^{(it-\lambda)x} \right| = \left| \frac{\lambda}{it-\lambda} \right| \cdot |e^{itx}| \cdot |e^{-\lambda x}| = \left| \frac{\lambda}{it-\lambda} \right| \cdot |e^{-\lambda x}| \rightarrow 0 \text{ as } x \rightarrow \infty. \quad \blacklozenge$$

★ Remark 10.2 For many integrals, especially when the integral doesn't involve complex functions or variables, $i = \sqrt{-1}$ (the imaginary unit) can be treated much like a constant. As in Example 10.16, any multiple of i can be treated as a scalar multiple in integration, similar to how a real number constant would be treated. However, when working with functions of complex variables, or when the domain of integration is in the complex plane, then the situation is more intricate. Functions of complex variables have their own special properties and techniques for integration, such as contour integration and residues. In these cases, the simple treatment of i as a constant will not suffice, and a more comprehensive understanding of complex analysis is required. To emphasize, for integrals that only have i appearing as a multiplicative factor (and are not otherwise dealing with complex functions or domains), treating i as a constant will give the correct result.

The principle explained for integrals holds for derivatives as well. When you differentiate with respect to a real variable, the imaginary unit i can be treated as a constant. But, as with integration, when you're dealing with complex functions of a complex variable, the situation is different. The derivative in the context of complex functions, often referred to as the complex derivative or holomorphic derivative, requires the function to be differentiable in a more stringent sense than in the real case. When a function is complex differentiable (holomorphic), it not only has partial derivatives but also meets the Cauchy-Riemann equations, which are a fundamental set of equations in complex analysis. In the realm of functions of a complex variable, differentiation can have very different properties than its real counterpart. For instance, once a function is complex differentiable at a point, it's infinitely differentiable at that point, which is not the case for functions of a real variable. Again to emphasize, for derivatives involving real functions with i treated as a scalar, you handle i just as you would a constant. But for complex functions of a complex variable, the differentiation process and its properties are more nuanced and require an understanding of complex analysis. $\blacklozenge$

Some of the basic properties of the characteristic function are as follows:

1. $\phi_X(0) = 1$.

$$\text{Clearly, } \phi_X(0) = E[e^{i(0)X}] = E(1) = 1.$$

2. $|\phi_X(t)| \leq 1$.

If X is a continuous random variable with probability density function f , then

$$\begin{aligned} |\phi_X(t)| &= \left| \int_{-\infty}^{\infty} e^{itx} f(x) dx \right| \leq \int_{-\infty}^{\infty} |e^{itx} f(x)| dx \\ &= \int_{-\infty}^{\infty} |e^{itx}| |f(x)| dx = \int_{-\infty}^{\infty} f(x) dx = 1. \end{aligned}$$

Similarly, if X is a discrete random variable with set of possible values A and probability mass function p , then

$$\begin{aligned} |\phi_X(t)| &= \left| \sum_{x \in A} e^{itx} p(x) \right| \leq \sum_{x \in A} |e^{itx} p(x)| \\ &= \sum_{x \in A} |e^{itx}| |p(x)| = \sum_{x \in A} p(x) = 1. \end{aligned}$$

3. $\phi_X(-t) = \overline{\phi_X(t)}$, where $\overline{\phi_X(t)}$ is the complex conjugate of $\phi_X(t)$.

The proof is straightforward:

$$\begin{aligned} \phi_X(-t) &= E[e^{i(-t)X}] = E[\cos(-tX)] + iE[\sin(-tX)] \\ &= E[\cos(tX)] - iE[\sin(tX)] = \overline{\phi_X(t)}. \end{aligned}$$

4. Let X be a random variable, then for constants $a, b \in \mathbb{R}$,

$$\phi_{aX+b}(t) = e^{itb} \phi_X(at) \quad (10.5)$$

This follows from the definition of characteristic function:

$$\phi_{aX+b}(t) = E[e^{it(aX+b)}] = E[e^{i(at)X} e^{itb}] = e^{itb} E[e^{i(at)X}] = e^{itb} \phi_X(at).$$

5. Consider a random variable X with characteristic function denoted by $\phi_X(t)$. From (10.5), we have

$$\phi_{-X}(t) = \phi_X(-t).$$

By Property 3 mentioned above, this is equivalent to $\overline{\phi_X(t)}$. Therefore,

$$\phi_{-X}(t) = \overline{\phi_X(t)}.$$

Example 10.17 Let Z be a standard normal random variable, and let X be a normal random variable with mean μ and variance σ^2 . First, determine $\phi_Z(t)$, the characteristic function of Z . Then, use this to find $\phi_X(t)$.

Solution: By definition of characteristic function,

$$\begin{aligned}\phi_Z(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{itx} \cdot e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{(-x^2/2)+itx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x^2-2itx)/2} dx.\end{aligned}$$

By completing the square for $x^2 - 2itx$, we get

$$x^2 - 2itx = x^2 - 2itx + (it)^2 - (it)^2 = (x - it)^2 + t^2.$$

Therefore,

$$\begin{aligned}\phi_Z(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-it)^2/2} \cdot e^{-t^2/2} dx = e^{-t^2/2} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-it)^2/2} dx \\ &= e^{-t^2/2}.\end{aligned}$$

To elucidate the last equality, consider that if i were a real number then $(1/\sqrt{2\pi})e^{-(x-it)^2/2}$ would represent the probability density function of a normal random variable with mean it and variance 1. Under such circumstances, we could express this as

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-it)^2/2} dx = 1.$$

Nevertheless, from a purely mathematical standpoint, the complex nature of it has no impact on the integral's computation.

To find $\phi_X(t)$, note that, by Lemma 7.1, $X = \sigma Z + \mu$. Thus, by (10.5),

$$\phi_X(t) = e^{it\mu} \phi_Z(\sigma t) = e^{it\mu - (1/2)\sigma^2 t^2}. \quad \blacklozenge$$

Remark 10.3 Let X be a random variable. By comparing $M_X(t) = E(e^{tX})$ to $\phi_X(t) = E(e^{itx})$, it appears intuitive to deduce that $\phi_X(t) = M_X(it)$. Under specific conditions, this relationship holds true. One such condition is as follows: If there exists a real number $\alpha > 0$ such that $|M_X(t)| < \infty$ for $|t| < \alpha$, then we have

$$\phi_X(t) = M_X(it). \quad \blacklozenge$$

Example 10.18 Consider Z , the standard normal random variable. We aim to demonstrate that $\phi_Z(t) = M_Z(it)$. This equality is established under the conditions discussed in Remark 10.3. By Example 10.5, $M_Z(t) = e^{t^2/2}$. For $|M_X(t)|$ to be finite, the exponent must be finite. The term $t^2/2$ can become infinite if t becomes large or negative large. However, near $t = 0$, the exponent is small. Specifically, for any small t , t^2 is much smaller than t , meaning the exponential will be close to 1. Thus, there exists some $\alpha > 0$ such that for all t in the interval $(-\alpha, \alpha)$ the value $t^2/2$ is sufficiently small to make $e^{t^2/2}$ finite. ($\alpha = 1/2$ is one of

many choices for α .) We have shown that the condition discussed in Remark 10.3 is valid for a standard normal random variable. Thus,

$$\phi_Z(t) = M_Z(it) = e^{(it)^2} = e^{-t^2/2}.$$

This is the same result obtain in Example 10.17. $\blacklozenge$

Theorem 10.9 *If $X_1, X_2, \dots, X_n$ are independent random variables, then*

$$\phi_{X_1+X_2+\dots+X_n}(t) = \phi_{X_1}(t)\phi_{X_2}(t)\cdots\phi_{X_n}(t).$$

Proof: Since $X_1, X_2, \dots, X_n$ are independent, so are $e^{itX_1}, e^{itX_2}, \dots, e^{itX_n}$. Therefore,

$$\begin{aligned}\phi_{X_1+X_2+\dots+X_n}(t) &= E[e^{it(X_1+X_2+\dots+X_n)}] = E(e^{itX_1}e^{itX_2}\cdots e^{itX_n}) \\ &= E(e^{itX_1})E(e^{itX_2})\cdots E(e^{itX_n}) = \phi_{X_1}(t)\phi_{X_2}(t)\cdots\phi_{X_n}(t). \quad \blacklozenge\end{aligned}$$

Example 10.19 Let X be a binomial random variable with parameters n and p . Find the characteristic function of X .

Solution: The random variable X is the sum of n independent Bernoulli random variables each with parameter p . So, by Example 10.14 and Theorem 10.9,

$$\phi_X(t) = (pe^{it} + 1 - p)^n. \quad \blacklozenge$$

Example 10.20 Let X be a gamma random variable with parameters $\lambda > 0$ and n , where n is a positive integer. Calculate the characteristic function of X .

Solution: It follows that, for n independent exponential random variables, $X_1, X_2, \dots, X_n$,

$$X = X_1 + X_2 + \cdots + X_n,$$

where each X_k (for $1 \leq k \leq n$) has a parameter of λ . Now, By Theorem 10.9,

$$\phi_X(t) = \phi_{X_1+X_2+\dots+X_n}(t) = \phi_{X_1}(t)\phi_{X_2}(t)\cdots\phi_{X_n}(t).$$

By Example 10.16,

$$\phi_{X_k}(t) = \frac{\lambda}{\lambda - it}, \quad k = 1, 2, \dots, n.$$

Therefore

$$\phi_X(t) = \phi_{X_1}(t)\phi_{X_2}(t)\cdots\phi_{X_n}(t) = \left(\frac{\lambda}{\lambda - it}\right)^n. \quad \blacklozenge$$

Table 5 in the Appendix lists common distributions along with their characteristic functions.

Theorem 10.10 Let X be a random variable with characteristic function ϕ_X . If $\phi_X^{(n)}(0)$, the n th derivative of ϕ_X at 0, exists, then

$$\phi_X^{(n)}(0) = i^n E(X^n).$$

Proof: We will prove this for a continuous random variable. For the discrete case, the proof follows a nearly identical approach. Let f be the probability density function of X . Then

$$\begin{aligned}\phi_X(t) &= \int_{-\infty}^{\infty} e^{itx} f(x) dx. \\ \phi_X'(t) &= \frac{d}{dt} \int_{-\infty}^{\infty} e^{itx} f(x) dx = \int_{-\infty}^{\infty} \frac{d}{dt} e^{itx} f(x) dx.\end{aligned}$$

To avoid technical details, in this relation, we have interchanged derivative and integral without justification. Therefore,

$$\begin{aligned}\phi_X'(t) &= \int_{-\infty}^{\infty} ix e^{itx} f(x) dx, \\ \phi_X''(t) &= \int_{-\infty}^{\infty} (ix)^2 e^{itx} f(x) dx, \\ &\vdots \\ \phi_X^{(n)}(t) &= \int_{-\infty}^{\infty} (ix)^n e^{itx} f(x) dx.\end{aligned}$$

Therefore,

$$\phi_X^{(n)}(0) = \int_{-\infty}^{\infty} (ix)^n e^{i(0)x} f(x) dx = i^n \int_{-\infty}^{\infty} x^n f(x) dx = i^n E(X^n). \quad \blacklozenge$$

Example 10.21 Using characteristic function, find the expected value and the variance of a binomial random variable X with parameters n and p .

Solution: By Example 10.19,

$$\phi_X(t) = (pe^{it} + 1 - p)^n.$$

This gives

$$\phi_X'(t) = npe^{it}(pe^{it} + 1 - p)^{n-1}.$$

By Theorem 10.10,

$$iE(X) = \phi_X'(0) = npi(p + 1 - p)^{n-1} = npi.$$

So,

$$E(X) = np.$$

Now,

$$\phi_X''(t) = npi^2 e^{it}(pe^{it} + 1 - p)^{n-1} + npie^{it} \cdot (n-1)pie^{it}(pe^{it} + 1 - p)^{n-2}$$

gives

$$i^2 E(X^2) = \phi_X''(0) = -np[1 + (n-1)p],$$

which implies that

$$E(X^2) = np[1 + (n-1)p].$$

Hence,

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = np[1 + (n-1)p] - (np)^2 = np(1-p). \quad \blacklozenge$$

Example 10.22 Using characteristic function, find the expected value and the variance of an exponential random variable X with parameter λ .

Solution: By Example 10.16,

$$\phi_X(t) = \frac{\lambda}{\lambda - it}.$$

Differentiating ϕ_X and then ϕ_X' , we have

$$\phi_X'(t) = \frac{\lambda i}{(\lambda - it)^2}, \quad \phi_X''(t) = \frac{2\lambda i^2}{(\lambda - it)^3}.$$

Now, $iE(X) = \phi_X'(0) = \lambda i/\lambda^2$ implies that $E(X) = 1/\lambda$, and $i^2 E(X^2) = \phi_X''(0) = 2i^2/\lambda^2$ implies that $E(X^2) = 2/\lambda^2$. Therefore,

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \quad \blacklozenge$$

Inversion Theorem

As we mentioned in the beginning of this section, **the characteristic function uniquely determines the distribution of a random variable**. This property is known as the **Inversion Theorem**, which states that if two random variables have the same characteristic functions, they have the same distribution.

Those who are familiar with Fourier transform know that, for a random variable with probability density function f , the characteristic function $\phi_X(t) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$ represents a variation of the conventional Fourier transform applied to the function f . Hence the familiar Fourier inversion yields

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_X(t) dt,$$

at each x where the function f is differentiable.

While the inversion formula can be readily deduced from the techniques of Fourier transform, we take an alternative approach. In an effort to avoid dependence on said theory, we present a derivation of the formula solely in the realm of probability. This self-contained perspective holds significant intrinsic value and fascination.

The derivation can be found in the sixth edition of Boris v. Gnedenko's book *Theory of Probability* (Gordon and Breach Science Publishers, 1997). However, for a more accessible proof, T. T. Soong offers a friendly version in his book *Fundamentals of Probability and Statistics for Engineering* (John Wiley & Sons Ltd, 2004). Here, we present this simplified derivation for convenience.

Theorem 10.11 (The Inversion Theorem) (a) *Let X be a continuous random variable with probability density function f and characteristic function ϕ_X . Then*

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_X(t) dt,$$

at each x where the function f is differentiable.

(b) *Let X be a discrete random variable with probability mass function p and characteristic function ϕ_X . Then*

$$p(x) = \lim_{a \rightarrow \infty} \frac{1}{2a} \int_{-a}^a e^{-itx} \phi_X(t) dt.$$

Proof: (a) First, note that, for $\alpha \in \mathbb{R}$,

$$\int_{-\infty}^{\infty} \frac{1 - \cos \alpha t}{t} dt = 0$$

since the integrand, being odd with respect to t , results in an integral of 0 over a symmetric range. Next, remember the following well-known integral:

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin \alpha t}{t} dt = \begin{cases} -1 & \text{if } \alpha < 0 \\ 0 & \text{if } \alpha = 0 \\ 1 & \text{if } \alpha > 0. \end{cases}$$

Multiplying the first integral by i and adding it to the second integral, we obtain

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin \alpha t + i(1 - \cos \alpha t)}{t} dt = \begin{cases} -1 & \text{if } \alpha < 0 \\ 0 & \text{if } \alpha = 0 \\ 1 & \text{if } \alpha > 0. \end{cases} \quad (10.6)$$

Let x be a fixed real number. Replacing α by $X - x$, the numerator of this fraction becomes

$$\begin{aligned}
 \sin(X - x)t + i[1 - \cos(X - x)t] &= \sin(X - x)t - i \cos(X - x)t + i \\
 &= i[1 - \cos(X - x)t + \frac{1}{i} \sin(X - x)t] \\
 &= i[1 - \cos(X - x)t - i \sin(X - x)t] \\
 &= i[1 - e^{i(X-x)t}].
 \end{aligned}$$

This and (10.6) imply that

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{i[1 - e^{i(X-x)t}]}{t} dt = \begin{cases} -1 & \text{if } X < x \\ 0 & \text{if } X = x \\ 1 & \text{if } X > x. \end{cases}$$

Therefore,

$$\frac{i}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{i(X-x)t}}{t} dt = \begin{cases} -1/2 & \text{if } X < x \\ 0 & \text{if } X = x \\ 1/2 & \text{if } X > x, \end{cases}$$

which yields

$$\frac{1}{2} - \frac{i}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{i(X-x)t}}{t} dt = \begin{cases} 1 & \text{if } X < x \\ 1/2 & \text{if } X = x \\ 0 & \text{if } X > x. \end{cases}$$

Now, let Y be a discrete random variable with set of possible values $\{0, 1/2, 1\}$, where $P(Y = 0) = P(X > x)$, $P(Y = 1/2) = P(X = x)$, and $P(Y = 1) = P(X < x)$. Then

$$Y = \frac{1}{2} - \frac{i}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{i(X-x)t}}{t} dt. \quad (10.7)$$

Clearly,

$$E(Y) = 0 \cdot P(X > x) + \frac{1}{2} \cdot P(X = x) + 1 \cdot P(X < x).$$

Since X is a continuous random variable, $P(X = x) = 0$ and

$$E(Y) = P(X < x) = P(X \leq x) = F(x),$$

where F is the probability distribution function of X . By (10.7),

$$\begin{aligned} F(x) = E(Y) &= \frac{1}{2} - \frac{i}{2\pi} \int_{-\infty}^{\infty} \frac{1 - E[e^{i(X-x)t}]}{t} dt \\ &= \frac{1}{2} - \frac{i}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{-ixt} E[e^{iXt}]}{t} dt \\ &= \frac{1}{2} - \frac{i}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{-ixt} \phi_X(t)}{t} dt. \end{aligned}$$

By differentiating both sides of this relation, we have established the inversion formula:

$$f(x) = F'(x) = -\frac{i}{2\pi} \int_{-\infty}^{\infty} \frac{ite^{-ixt} \phi_X(t)}{t} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_X(t) dt.$$

(b) Note that, for $a, \alpha \in \mathbb{R}$,

$$\int_{-a}^a e^{i\alpha t} dt = \int_{-a}^a \cos \alpha t dt + i \int_{-a}^a \sin \alpha t dt = \int_{-a}^a \cos \alpha t dt.$$

Since $\sin \alpha t$ is an odd function, the integral $\int_{-a}^a \sin \alpha t dt$ is 0. Now, for $\alpha \neq 0$,

$$\int_{-a}^a e^{i\alpha t} dt = \int_{-a}^a \cos \alpha t dt = \frac{2}{\alpha} \sin \alpha t.$$

Putting these together, we have

$$\frac{1}{2a} \int_{-a}^a e^{i\alpha t} dt = \begin{cases} \frac{\sin \alpha a}{\alpha a} & \text{if } \alpha \neq 0 \\ 1 & \text{if } \alpha = 0. \end{cases}$$

For a fixed $x \in \mathbb{R}$, let $\alpha = X - x$. Then

$$\frac{1}{2a} \int_{-a}^a e^{i(X-x)t} dt = \begin{cases} \frac{\sin(X-x)a}{(X-x)a} & \text{if } X \neq x \\ 1 & \text{if } X = x. \end{cases}$$

Clearly,

$$\lim_{a \rightarrow \infty} \left| \frac{\sin(X-x)a}{(X-x)a} \right| \leq \left| \frac{1}{(X-x)a} \right| = 0.$$

So,

$$\lim_{a \rightarrow \infty} \frac{1}{2a} \int_{-a}^a e^{i(X-x)t} dt = \begin{cases} 0 & \text{if } X \neq x \\ 1 & \text{if } X = x. \end{cases} \quad (10.8)$$

Let Y be a discrete random variable with set of possible values $\{0, 1\}$ defined by $P(Y = 0) = P(X \neq x)$ and $P(Y = 1) = P(X = x)$. By (10.8)

$$Y = \lim_{a \rightarrow \infty} \frac{1}{2a} \int_{-a}^a e^{i(X-x)t} dt.$$

Since

$$E(Y) = 0 \cdot P(Y = 0) + 1 \cdot P(Y = 1) = P(X = x) = p(x),$$

$$\begin{aligned} p(x) = E(Y) &= \lim_{a \rightarrow \infty} \frac{1}{2a} \int_{-a}^a E[e^{i(X-x)t}] dt \\ &= \lim_{a \rightarrow \infty} \frac{1}{2a} \int_{-a}^a e^{-itx} E[e^{itX}] = \lim_{a \rightarrow \infty} \frac{1}{2a} \int_{-a}^a e^{-itx} \phi_X(t) dt. \end{aligned}$$

This establishes the inversion theorem for discrete random variables. $\blacklozenge$

Example 10.23 Let X be a continuous random variable whose characteristic function, $\phi_X(t)$, is given by $\phi_X(t) = e^{-|t|}$ for all t in the range $-\infty < t < \infty$. Determine the probability density function, f , of X .

Solution: By the inversion theorem,

$$\begin{aligned} f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} e^{-|t|} dt = \frac{1}{2\pi} \int_{-\infty}^0 e^{-itx} e^t dt + \frac{1}{2\pi} \int_0^{\infty} e^{-itx} e^{-t} dt \\ &= \frac{1}{2\pi} \left[\frac{1}{1-ix} e^{(1-ix)t} \right]_{-\infty}^0 + \frac{1}{2\pi} \left[\frac{1}{-1-ix} e^{(-1-ix)t} \right]_0^{\infty} \\ &= \frac{1}{2\pi} \left(\frac{1}{1-ix} + \frac{1}{1+ix} \right) = \frac{1}{\pi(1+x^2)}, -\infty < x < \infty. \end{aligned}$$

Note that,

$$\left| \frac{1}{1-ix} e^{(1-ix)t} \right| = \left| \frac{1}{1-ix} \right| \cdot |e^t| \cdot |e^{-ixt}| = \left| \frac{1}{1-ix} \right| \cdot |e^t|$$

This inequality shows that

$$\left| \frac{1}{1-ix} e^{(1-ix)t} \right| \rightarrow 0 \text{ as } t \rightarrow -\infty, \text{ and hence, } \lim_{t \rightarrow -\infty} \frac{1}{1-ix} e^{(1-ix)t} = 0.$$

Similarly,

$$\lim_{t \rightarrow \infty} \frac{1}{-1-ix} e^{(-1-ix)t} = 0.$$

Observe that $f(x) = 1/[\pi(1+x^2)]$ is the probability density function of the well-known Cauchy distribution (see Exercise 8, [Section 5.2](#) and Example 5.9). Thus, by the uniqueness property of characteristic function, X is a Cauchy random variable. $\blacklozenge$

Example 10.24 Let X be distributed as $N(\mu_1, \sigma_1^2)$ and Y be distributed as $N(\mu_2, \sigma_2^2)$, where X and Y are independent random variables. Calculate the characteristic function of $X + Y$.

Solution: We know that

$$\phi_X(t) = e^{i\mu_1 t - (1/2)\sigma_1^2 t^2}, \quad \phi_Y(t) = e^{i\mu_2 t - (1/2)\sigma_2^2 t^2}.$$

Therefore,

$$\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t) = e^{i(\mu_1 + \mu_2)t - (1/2)(\sigma_1^2 + \sigma_2^2)t^2}.$$

This is the characteristic function of a normal random variable with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$. Given that the characteristic function uniquely determines the distribution of a random variable, it follows that $X + Y$ is a normal random variable with the same mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$.

Remark: Let $\{X_1, X_2, \dots, X_n\}$ be independent random variables such that $X_k \sim N(\mu_k, \sigma_k^2)$ for $k = 1, 2, \dots, n$. Using characteristic functions, it can be shown (refer to Exercise 12) that for constants $\alpha_1, \alpha_2, \dots, \alpha_n$, the distribution of the linear combination is given by:

$$\sum_{k=1}^n \alpha_k X_k \sim N\left(\sum_{k=1}^n \alpha_k \mu_k, \sum_{k=1}^n \alpha_k^2 \sigma_k^2\right). \quad \blacklozenge$$

Example 10.25 Let $X_1, X_2, \dots, X_n$ be independent Poisson random variables with parameters $\lambda_1, \lambda_2, \dots, \lambda_n$, respectively. Using characteristic functions, show that $X_1 + X_2 + \dots + X_n$ is Poisson with parameter $\lambda_1 + \lambda_2 + \dots + \lambda_n$.

Solution: By Example 10.15, for $k = 1, 2, \dots, n$, $\phi_{X_k}(t) = e^{-\lambda_k(1-e^{it})}$. Therefore,

$$\begin{aligned} \phi_{X_1+X_2+\dots+X_n}(t) &= \phi_{X_1}(t)\phi_{X_2}(t)\cdots\phi_{X_n}(t) \\ &= e^{-\lambda_1(1-e^{it})}e^{-\lambda_2(1-e^{it})}\cdots e^{-\lambda_n(1-e^{it})} = e^{-(\lambda_1+\lambda_2+\dots+\lambda_n)(1-e^{it})}, \end{aligned}$$

which is the characteristic function of a Poisson random variable with parameter $\lambda_1 + \lambda_2 + \dots + \lambda_n$. Given that the characteristic function uniquely determines the distribution of a random variable, it follows that $X_1 + X_2 + \dots + X_n$ is Poisson with parameter $\lambda_1 + \lambda_2 + \dots + \lambda_n$. $\blacklozenge$

Lévy Continuity Theorem for Characteristic Functions

As in the case of the moment-generating function, a pivotal feature of characteristic functions—which aids in proving multiple limit theorems—is the convergence of these functions. Specifically, as the characteristic functions of a sequence of random variables approach a defined function with an increasing number of variables, this function distinctively represents the probability distribution of the resulting random variable. The subsequent theorem, presented without its proof, elucidates this crucial behavior.

Theorem 10.12 (Lévy Continuity Theorem for Characteristic Functions) *Let $X_1, X_2, \dots$ be a sequence of random variables with distribution functions $F_1, F_2, \dots$ and characteristic functions $\phi_{X_1}(t), \phi_{X_2}(t), \dots$, respectively. Let X be a random variable with distribution function F and characteristic function $\phi_X(t)$. If for all values of t , $\phi_{X_n}(t)$ converges to $\phi_X(t)$, then at the points of continuity of F , F_n converges to F .*

Example 10.26 Consider a sequence of Bernoulli random variables $X_1, X_2, \dots$, where, for $n \geq 1$,

$$X_n = \begin{cases} 1 & \text{with probability } \frac{1}{n} \\ 0 & \text{with probability } 1 - \frac{1}{n}. \end{cases}$$

We are interested in the convergence of the distribution functions of X_n . By Example 10.14,

$$\phi_{X_n}(t) = \frac{1}{n}e^{it} + \left(1 - \frac{1}{n}\right).$$

It is evident that $\lim_{n \rightarrow \infty} \phi_{X_n}(t) = 1$. Thus, the characteristic function of the random variable X , whose distribution is the limit of the distributions of X_n 's, is given by $\phi_X(t) = 1$. This corresponds to a random variable that is 0 with probability 1. Consequently, the distribution functions of X_n converge to the distribution function of the random variable X , which also takes the value 0 with probability 1. ♦

EXERCISES

A

1. For a certain random variable X and some constant c , the characteristic function of X is given by $\phi_X(t) = 3/(c - it)$. Determine the value of c .
2. Let X be a random variable with the characteristic function $\phi_X(t) = 1/(1 + t^2)$. Find $E(X)$ and $\text{Var}(X)$.
3. Let $X \sim N(\mu, \sigma^2)$. Using $\phi_X(t)$, show that $\text{Var}(X) = \sigma^2$.
4. Let X be a gamma random variable with $r = 3$ and $\lambda = 1$. Using characteristic functions, find $E(X^3)$.
5. Let X be a discrete random variable with set of possible values $\{-2, -1, 0, 1\}$ and probability mass function given by

x	-2	-1	0	1
$p(x)$	1/4	1/8	1/2	1/8

Find $\phi_X(t)$ and $\phi_X(2\pi)$

6. Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{2} - \frac{1}{4}|x - 3| & \text{if } 1 \leq x \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

Find $\phi_X(t)$.

7. In many engineering systems, we encounter scenarios where, for independent continuous random variables X and Y , while the characteristic functions of X and $Z = XY$ are known, we need the characteristic functions of Y and its probability density function to perform certain calculations. Given that $\phi_X(t) = 1/(1 + t^2)$ and $\phi_Z(t) = e^{-t^2}/(1 + t^2)$, determine $\phi_Y(t)$ and g , the probability density function of Y .
8. Is $\cos t$ the characteristic function of some random variable X ? If so, which random variable? What about $(\cos t)^{13}$, is it a characteristic function?
9. Let $X_1, X_2, \dots, X_n$ be independent random variables that are identically distributed. For each k , $1 \leq k \leq n$, the characteristic function of X_k is given by $\phi(t)$. In terms of ϕ , determine the characteristic function of

$$Y = \frac{X_1 + X_2 + \dots + X_n}{n},$$

which represents the mean of the random variables $X_1, X_2, \dots, X_n$.

10. Find the Characteristic function of X , an exponential random variable with parameter $\lambda > 0$. Achieve this by expanding the exponential term e^{itx} into its cosine and sine components, and then integrating the real and imaginary parts of $\phi_X(t)$.
11. Consider Theorem 10.9, but restrict it to just two random variables. The Theorem states that if X and Y are independent random variables, then $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$. To Prove this theorem, expand each exponential term into its cosine and sine components. Then, perform the multiplication, using the independence between X and Y . Finally recombine the terms to arrive at the desired result.
12. Let $\{X_1, X_2, \dots, X_n\}$ be a set of independent random variables and $X_k \sim N(\mu_k, \sigma_k^2)$ for $k = 1, 2, \dots, n$. Using characteristic functions, show that, for constants $\alpha_1, \alpha_2, \dots, \alpha_n$,

$$\sum_{k=1}^n \alpha_k X_k \sim N\left(\sum_{k=1}^n \alpha_k \mu_k, \sum_{k=1}^n \alpha_k^2 \sigma_k^2\right).$$

13. Let X and Y be independent, identically distributed Cauchy random variables. The probability density function of both X and Y is given by:

$$f(x) = \frac{1}{\pi(x^2 + 1)}, \quad -\infty < x < \infty.$$

As discussed in Example 10.23, we have that $\phi_X(t) = \phi_Y(t) = e^{-|t|}$ for all t in the range of $-\infty < x < \infty$. Show that the random variables $X + Y$ and $X + X$ share

the same characteristic function. This presents an intriguing case where the sum of two independent, identically distributed random variables has the same distribution as the sum of two fully dependent random variables.

14. Let X be a continuous random variable with characteristic function $\phi_X(t) = \sin t/t$. Find f , the probability density function of X .
15. Show that the function $g(t) = e^{-t^4}$ cannot be the characteristic function of a random variable X .
Hint: Suppose it is and calculate $E(X^2)$.
16. Let Z be a standard normal random variable. Find the characteristic function of $X = Z^2$.
Hint: X is a gamma random variable.
17. Consider a sequence of random variables $X_1, X_2, \dots$ defined as

$$X_n = \begin{cases} \frac{1}{n} & \text{with probability } 1 - \frac{1}{n} \\ n & \text{with probability } \frac{1}{n}. \end{cases}$$

Determine the random variable X such that its distribution function is the limit of the distribution functions of the sequence $X_1, X_2, \dots$

18. Let ϕ_X be the characteristic function of a random variable X . Show that

$$\lim_{h \rightarrow 0} |\phi_X(t+h) - \phi_X(t)| = 0.$$

This proves that ϕ_X is absolutely continuous.

19. Let $X_1, X_2, \dots$ be independent and identically distributed random variables, each with the same characteristic function given by $\phi(t)$. Let N be a positive integer-valued random variable that is independent of the sequence $\{X_1, X_2, \dots\}$. Given that $P(N = k) = \alpha_k$ and $\sum_{k=1}^{\infty} \alpha_k = 1$, determine the characteristic function of $Y = \sum_{k=1}^N X_k$ in terms of $\phi(t)$ and the α_k values.
20. Let X be a continuous random variable with probability density function f . Show that $\phi_X(t)$ is real if and only if f is symmetric. That is, $\phi_X(t) = \overline{\phi_X(t)}$ if and only if $f(-x) = f(x)$. ■

Self-Quiz on Section 10.3

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. Let X be a continuous random variable with the probability density function

$$f(x) = e^{-2|x|}, \quad -\infty < x < \infty.$$

Calculate the characteristic function of X .

2. Show that if $\phi(t)$ is the characteristic function of a random variable, then both $[\phi(t)]^2$ and $|\phi(t)|^2$ also represent the characteristic functions of certain random variables. ■

10.4 MARKOV AND CHEBYSHEV INEQUALITIES

Thus far, we have seen that, to calculate probabilities, we need to know distribution functions, probability mass functions, or probability density functions. It frequently happens that, for some random variables, we cannot determine any of these three functions, but we can calculate their expected values and/or variances. In such cases, although we cannot calculate exact probabilities, using Markov and Chebyshev inequalities, we are able to derive bounds on probabilities. These inequalities have useful applications and significant theoretical values. Moreover, Chebyshev's inequality is a further indication of the importance of the concept of variance. Chebyshev's inequality was first discovered by the French mathematician Irénée Bienaymé (1796–1878). For this reason some authors call it the Chebyshev-Bienaymé inequality. In the middle of the nineteenth century, Chebyshev discovered the inequality independently, in connection with the laws of large numbers (see [Section 10.5](#)). He used it to give an elegant and short proof for the law of large numbers, discovered by James Bernoulli early in the eighteenth century. Since the usefulness and applicability of the inequality were demonstrated by Chebyshev, most authors call it Chebyshev's inequality.

Theorem 10.13 (Markov's Inequality) *Let X be a nonnegative random variable, then for any $t > 0$,*

$$P(X \geq t) \leq \frac{E(X)}{t}.$$

Proof: We prove the theorem for a discrete random variable X with probability mass function $p(x)$. For continuous random variables the proof is similar. Let A be the set of possible values of X and $B = \{x \in A: x \geq t\}$. Then

$$E(X) = \sum_{x \in A} xp(x) \geq \sum_{x \in B} xp(x) \geq t \sum_{x \in B} p(x) = tP(X \geq t).$$

Thus

$$P(X \geq t) \leq \frac{E(X)}{t}. \quad \blacklozenge$$

Example 10.27 A post office, on average, handles 10,000 letters per day. What can be said about the probability that it will handle (a) at least 15,000 letters tomorrow; (b) less than 15,000 letters tomorrow?

Solution: Let X be the number of letters that this post office will handle tomorrow. Then $E(X) = 10000$.

(a) By Markov's inequality,

$$P(X \geq 15000) \leq \frac{E(X)}{15000} = \frac{10000}{15000} = \frac{2}{3}.$$

(b) Using the inequality obtained in (a), we have

$$P(X < 15000) = 1 - P(X \geq 15000) \geq 1 - \frac{2}{3} = \frac{1}{3}. \quad \blacklozenge$$

Theorem 10.14 (Chebyshev's Inequality) *If X is a random variable with expected value μ and variance σ^2 , then for any $t > 0$,*

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

Proof: Since $(X - \mu)^2 \geq 0$, by Markov's inequality

$$P((X - \mu)^2 \geq t^2) \leq \frac{E[(X - \mu)^2]}{t^2} = \frac{\sigma^2}{t^2}.$$

Chebyshev's inequality follows since $(X - \mu)^2 \geq t^2$ is equivalent to $|X - \mu| \geq t$. $\blacklozenge$

Letting $t = k\sigma$ in Chebyshev's inequality, we get that

$$P(|X - \mu| \geq k\sigma) \leq 1/k^2.$$

That is,

The probability that X deviates from its expected value at least k standard deviations is less than $1/k^2$.

Thus, for example,

$$\begin{aligned} P(|X - \mu| \geq 2\sigma) &\leq 1/4, \\ P(|X - \mu| \geq 4\sigma) &\leq 1/16, \\ P(|X - \mu| \geq 10\sigma) &\leq 1/100. \end{aligned}$$

On the other hand,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

implies that

$$P(|X - \mu| < k\sigma) \geq 1 - 1/k^2.$$

Therefore,

The probability that X deviates from its mean less than k standard deviations is at least $1 - 1/k^2$.

In particular, this implies that, for any set of data, at least a fraction $1 - 1/k^2$ of the data are within k standard deviations on either side of the mean. Thus, for any data, at least $1 - 1/2^2 = 3/4$, or 75% of the data, lie within two standard deviations on either side of the mean. This implication of Chebyshev's inequality is true for any set of real numbers. That is, let $\{x_1, x_2, \dots, x_n\}$ be a set of real numbers, and define

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2;$$

then at least a fraction $1 - 1/k^2$ of the x_i 's are between $\bar{x} - ks$ and $\bar{x} + ks$ (see Exercise 22).

Example 10.28 Suppose that, on average, a post office handles 10,000 letters a day with a variance of 2000. What can be said about the probability that this post office will handle between 8000 and 12,000 letters tomorrow?

Solution: Let X denote the number of letters that this post office will handle tomorrow. We want to calculate $P(8000 < X < 12000)$. Note that $\mu = E(X) = 10000$, $\sigma^2 = \text{Var}(X) = 2000$. Hence

$$\begin{aligned} P(8000 < X < 12000) &= P(-2000 < X - 10000 < 2000) \\ &= P(|X - 10000| < 2000) \\ &= 1 - P(|X - 10000| \geq 2000). \end{aligned}$$

By Chebyshev's inequality,

$$P(|X - 10000| \geq 2000) \leq \frac{2000}{(2000)^2} = 0.0005.$$

Therefore,

$$P(8000 < X < 12000) = P(|X - 10000| < 2000) \geq 1 - 0.0005 = 0.9995.$$

Note that this answer is consistent with our intuitive understanding of the concepts of expectation and variance. ♦

Example 10.29 A blind will fit Myra's bedroom's window if its width is between 41.5 and 42.5 inches. Myra buys a blind from a store that has 30 such blinds. What can be said about the probability that it fits her window if the average of the widths of the blinds is 42 inches with standard deviation 0.25?

Solution: Let X be the width of the blind that Myra purchased. We know that

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}.$$

Therefore,

$$P(41.5 < X < 42.5) = P(|X - 42| < 2(0.25)) \geq 1 - \frac{1}{4} = 0.75. \quad \blacklozenge$$

It should be mentioned that the bounds that are obtained on probabilities by Markov's and Chebyshev's inequalities are not usually very close to the actual probabilities. The following example demonstrates this.

Example 10.30 Roll a die and let X be the outcome. Clearly,

$$\begin{aligned} E(X) &= \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6}, \\ E(X^2) &= \frac{1}{6}(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = \frac{91}{6}. \end{aligned}$$

Thus $\text{Var}(X) = 91/6 - 441/36 = 35/12$. By Markov's inequality,

$$P(X \geq 6) \leq \frac{21/6}{6} \approx 0.583.$$

By Chebyshev's inequality,

$$P\left(\left|X - \frac{21}{6}\right| \geq \frac{3}{2}\right) \leq \frac{35/12}{9/4} \approx 1.296,$$

a trivial bound because we already know that $P(|X - 21/6| \geq 3/2) \leq 1$. However, the exact values of these probabilities are much smaller than these bounds: $P(X \geq 6) = 1/6 \approx 0.167$ and

$$P\left(\left|X - \frac{21}{6}\right| \geq \frac{3}{2}\right) = P(X \leq 2 \text{ or } X \geq 5) = \frac{4}{6} \approx 0.667. \quad \blacklozenge$$

The following example is an elegant proof of the fact, shown previously, that if $\text{Var}(X) = 0$, then X is constant with probability 1. It is a good application of Chebyshev's inequality.

Example 10.31 Let X be a random variable with mean μ and variance 0. In this example, we will show that X is a constant. That is,

$$P(X = \mu) = 1.$$

To prove this, let $E_i = \{|X - \mu| < 1/i\}$, then

$$E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots \supseteq E_n \supseteq E_{n+1} \supseteq \cdots$$

That is, $\{E_i, i = 1, 2, 3, \dots\}$ is a decreasing sequence of events. Therefore,

$$\lim_{n \rightarrow \infty} E_n = \bigcap_{n=1}^{\infty} E_n = \bigcap_{n=1}^{\infty} \left\{|X - \mu| < \frac{1}{n}\right\} = \{X = \mu\}.$$

Hence, by the continuity property of the probability function (Theorem 1.8),

$$\lim_{n \rightarrow \infty} P(E_n) = P(\lim_{n \rightarrow \infty} E_n) = P(X = \mu).$$

Now, by Chebyshev's inequality,

$$P\left(|X - \mu| \geq \frac{1}{n}\right) \leq \frac{\text{Var}(X)}{1/n^2} = 0,$$

implying that $\forall n \geq 1$,

$$P(E_n) = P\left(|X - \mu| < \frac{1}{n}\right) = 1.$$

Thus

$$P(X = \mu) = \lim_{n \rightarrow \infty} P(E_n) = 1. \quad \blacklozenge$$

Chebyshev's Inequality and Sample Mean

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a (continuous or discrete) distribution function F with mean μ and variance σ^2 . Let $\bar{X}$ be the mean of the sample. Then, by Lemma 8.1,

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \mu, \\ \text{Var}(\bar{X}) &= \text{Var}\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) = \frac{\sigma^2}{n}. \end{aligned}$$

Applying Chebyshev's inequality to $\bar{X}$, we have that, for any $\varepsilon > 0$,

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2 n}. \quad (10.9)$$

Therefore, in other words, if $\{X_1, X_2, \dots\}$ is a sequence of independent and identically distributed random variables with mean μ and variance σ^2 , then, for all $n \geq 1$ and $\varepsilon > 0$, (10.9) holds. Inequality (10.9) has many important applications, some of which are discussed next.

Example 10.32 For the scores on an achievement test given to a certain population of students, the expected value is 500 and the standard deviation is 100. Let $\bar{X}$ be the mean of the scores of a random sample of 10 students. Find a lower bound for $P(460 < \bar{X} < 540)$.

Solution: Since

$$\begin{aligned} P(460 < \bar{X} < 540) &= P(-40 < \bar{X} - 500 < 40) \\ &= P(|\bar{X} - 500| < 40) \\ &= 1 - P(|\bar{X} - 500| \geq 40), \end{aligned}$$

by (10.9),

$$P(|\bar{X} - 500| \geq 40) \leq \frac{100^2}{(40)^2 10} = \frac{5}{8}.$$

Hence

$$P(460 < \bar{X} < 540) \geq 1 - \frac{5}{8} = \frac{3}{8}. \quad \blacklozenge$$

Example 10.33 A biologist wants to estimate ℓ , the life expectancy of a certain type of insect. To do so, he takes a sample of size n and measures the lifetime from birth to death of each insect. Then he finds the average of these numbers. If he believes that the lifetimes of these insects are independent random variables with variance 1.5 days, how large a sample should he choose to be at least 98% sure that his average is accurate within ± 0.2 (± 4.8 hours)?

Solution: For $i = 1, 2, \dots, n$, let X_i be the lifetime of the i th insect of the sample. We want to determine n , so that

$$P\left(-0.2 < \frac{X_1 + X_2 + \dots + X_n}{n} - \ell < 0.2\right) \geq 0.98;$$

that is,

$$P(|\bar{X} - \ell| < 0.2) \geq 0.98.$$

This is equivalent to

$$P(|\bar{X} - \ell| \geq 0.2) \leq 0.02.$$

Since $E(X_i) = \ell$ and $\text{Var}(X_i) = 1.5$, by (10.9),

$$P(|\bar{X} - \ell| \geq 0.2) \leq \frac{1.5}{(0.2)^2 n} = \frac{37.5}{n}.$$

Therefore, all we need to do is to find n for which $37.5/n \leq 0.02$. This gives $n \geq 1875$. Hence the biologist should choose a sample of size 1875. $\blacklozenge$

In statistics, an important application of Chebyshev's inequality is approximation of the parameter of a Bernoulli random variable. Suppose that, for a Bernoulli trial, p , the probability of success is unknown, and we are interested in estimating it. Suppose that, for small $\varepsilon > 0$ and small $\alpha > 0$, we want the probability to be at least $1 - \alpha$ that the error of estimation is less than ε . A simple way to do this is to repeat the Bernoulli trial independently and, for $i \geq 1$, let

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{otherwise.} \end{cases}$$

Then $(X_1 + X_2 + \dots + X_n)/n$ is the average number of successes in the first n trials. The goal is to find the smallest n for which $\hat{p} = \frac{X_1 + X_2 + \dots + X_n}{n}$ satisfies

$$P(|\hat{p} - p| < \varepsilon) \geq 1 - \alpha.$$

Note that, for $i \geq 1$, $E(X_i) = p$ and $\text{Var}(X_i) = p(1 - p)$. Applying (10.9) to the sequence $\{X_1, X_2, \dots\}$ yields

$$P(|\hat{p} - p| \geq \varepsilon) \leq \frac{p(1 - p)}{\varepsilon^2 n},$$

or, equivalently,

$$P(|\hat{p} - p| < \varepsilon) \geq 1 - \frac{p(1 - p)}{\varepsilon^2 n}. \quad (10.10)$$

Thus we need to find the smallest n for which

$$1 - \frac{p(1 - p)}{\varepsilon^2 n} \geq 1 - \alpha,$$

or

$$n \geq \frac{p(1 - p)}{\varepsilon^2 \alpha}.$$

The difficulty with this inequality is that p is unknown. To overcome this, note that the function $f(p) = p(1 - p)$ is maximum for $p = 1/2$. Hence, for all $0 < p < 1$,

$$p(1 - p) \leq \frac{1}{4}.$$

Thus if we choose n to satisfy

$$n \geq \frac{1}{4\varepsilon^2 \alpha}, \quad (10.11)$$

then

$$n \geq \frac{1}{4\varepsilon^2 \alpha} \geq \frac{p(1 - p)}{\varepsilon^2 \alpha},$$

and we have the desired inequality; namely, $P(|\hat{p} - p| < \varepsilon) \geq 1 - \alpha$.

Example 10.34 To estimate the percentage of defective nails manufactured by a specific company, a quality control officer decides to take a random sample of nails produced by the company and calculate the fraction of those that are defective. If the officer wants to be sure, with a probability of at least 98%, that the error of estimation is less than 0.03, how large should the sample size be?

Solution: In (10.11), let $\varepsilon = 0.03$ and $\alpha = 0.02$. We have

$$n \geq \frac{1}{4(0.03)^2(0.02)} = 13,888.89.$$

Therefore, at least 13,889 nails should be tested. Note that if the quality control officer reduces the confidence level from 98% to 95%, then $\alpha = 0.05$, and we obtain $n \geq 5555.56$. That is, the number of nails to be tested, in that case, is $n = 5556$, a number considerably lower than the number required when the confidence level is to be 98%. ♦

Example 10.35 For a coin, p , the probability of heads is unknown. To estimate p , we flip the coin 3000 times and let $\hat{p}$ be the fraction of times it lands heads up. Show that the probability is at least 0.90 that $\hat{p}$ estimates p within ± 0.03 .

Solution: By (10.10),

$$P(|\hat{p} - p| < 0.03) \geq 1 - \frac{p(1-p)}{(0.03)^2 3000} \geq 1 - \frac{1}{4(0.03)^2 3000} = 0.907,$$

since $p(1-p) \leq 1/4$ implies that $-p(1-p) \geq -1/4$. ♦

EXERCISES

A

- According to the Board of Governors of the Federal Reserve System, October 10, 2017, <https://www.federalreserve.gov/faqs/how-long-is-the-life-span-of-us-paper-money.htm>, the average life of a one-dollar Federal Reserve note is 5.8 years. Show that at most 39% of one-dollar bills last for at least 15 years.
- Show that if for a nonnegative random variable X , $P(X < 2) = 3/5$, then $E(X) \geq 4/5$.
- Let X be a nonnegative random variable with $E(X) = 5$ and $E(X^2) = 42$. Find an upper bound for $P(X \geq 11)$ using (a) Markov's inequality, (b) Chebyshev's inequality.
- The average and standard deviation of lifetimes of light bulbs manufactured by a certain factory are, respectively, 800 hours and 50 hours. What can be said about the probability that a random light bulb lasts, at most, 700 hours?
- Suppose that the average number of accidents at an intersection is two per day.
 - Use Markov's inequality to find a bound for the probability that at least five accidents will occur tomorrow.
 - Using Poisson random variables, calculate the probability that at least five accidents will occur tomorrow. Compare this value with the bound obtained in part (a).
 - Let the variance of the number of accidents be two per day. Use Chebyshev's inequality to find a bound on the probability that tomorrow at least five accidents will occur.
- The average IQ score on a certain campus is 110. If the variance of these scores is 15, what can be said about the percentage of students with an IQ above 140?
- The number of days from the time a book is ordered until it is received (including the day the book is ordered and the day that it is received) is a random variable with mean seven days and standard deviation two days. If Helen wants to be 95% sure that she receives a book by certain date, how early should she order the book?

8. Show that for a nonnegative random variable X with mean μ , $P(X \geq 2\mu) \leq 1/2$.
9. Suppose that X is a random variable with $E(X) = \text{Var}(X) = \mu$. What does Chebyshev's inequality say about $P(X > 2\mu)$?
10. From a distribution with mean 42 and variance 60, a random sample of size 25 is taken. Let $\bar{X}$ be the mean of the sample. Show that the probability is at least 0.85 that $\bar{X} \in (38, 46)$.
11. The mean IQ of a randomly selected student from a specific university is μ and its variance is 150. A psychologist wants to estimate μ . To do so, for some n , she takes a sample of size n of the students at random and independently and measures their IQ's. Then she finds the average of these numbers. How large a sample should she choose to make at least 92% sure that the average is accurate within ± 3 points?
12. For a distribution, the mean of a random sample is taken as estimation of the expected value of the distribution. How large should the sample size be so that, with a probability of at least 0.98, the error of estimation is less than 2 standard deviations of the distribution?
13. To determine p , the proportion of time that an airline operator is busy answering customers, a supervisor observes the operator at times selected randomly and independently from other observed times. Let $X_i = 1$ if the i th time the operator is observed, he is busy; let $X_i = 0$ otherwise. For large values of n , if $(1/n) \sum_{i=1}^n X_i$'s are taken as estimates of p , for what values of n is the error of estimation at most 0.05 with probability 0.96?
14. For a coin, p , the probability of heads is unknown. To estimate p , for some n , we flip the coin n times independently. Let $\hat{p}$ be the proportion of heads obtained. Determine the value of n for which $\hat{p}$ estimates p within ± 0.05 with a probability of at least 0.94.
15. Let X be a random variable with mean μ . Show that if $E[(X - \mu)^{2n}] < \infty$, then for $\alpha > 0$,

$$P(|X - \mu| \geq \alpha) \leq \frac{1}{\alpha^{2n}} E[(X - \mu)^{2n}].$$

16. Let X be a random variable and k be a constant. Prove that

$$P(X > t) \leq \frac{E(e^{kX})}{e^{kt}}.$$

17. Let X be a random variable with $E(X^2) = E(X) = 1$. Find $E(X^3)$.
18. Prove that if the random variables X and Y satisfy $E[(X - Y)^2] = 0$, then with probability 1, $X = Y$.
19. Let X and Y be two randomly selected numbers from the set of positive integers $\{1, 2, \dots, n\}$. Prove that $\rho(X, Y) = 1$ if and only if $X = Y$ with probability 1.
Hint: First prove that $E[(X - Y)^2] = 0$, then use Exercise 18.
20. Let X be a random variable; show that for $\alpha > 1$ and $t > 0$,

$$P\left(X \geq \frac{1}{t} \ln \alpha\right) \leq \frac{1}{\alpha} M_X(t).$$

B

21. Let the probability density function of a random variable X be

$$f(x) = \frac{x^n}{n!} e^{-x}, \quad x \geq 0.$$

Show that

$$P(0 < X < 2n + 2) > \frac{n}{n + 1}.$$

Hint: Note that $\int_0^\infty x^n e^{-x} dx = \Gamma(n + 1) = n!$. Use this to calculate $E(X)$ and $\text{Var}(X)$. Then apply Chebyshev's inequality.

22. Let $\{x_1, x_2, \dots, x_n\}$ be a set of real numbers and define

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Prove that at least a fraction $1 - 1/k^2$ of the x_i 's are between $\bar{x} - ks$ and $\bar{x} + ks$.

Sketch of a Proof: Let N be the number of $x_1, x_2, \dots, x_n$ that fall in $A = [\bar{x} - ks, \bar{x} + ks]$. Then

$$\begin{aligned} s^2 &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \geq \frac{1}{n-1} \sum_{x_i \notin A} (x_i - \bar{x})^2 \\ &\geq \frac{1}{n-1} \sum_{x_i \notin A} k^2 s^2 = \frac{n-N}{n-1} k^2 s^2. \end{aligned}$$

This gives $(N-1)/(n-1) \geq 1 - (1/k^2)$. The result follows since $N/n \geq (N-1)/(n-1)$. ■

Self-Quiz on Section 10.4

Time allotted: 25 Minutes

Each problem is worth 5 points.

1. Dr. O'Neill is a proficient, caring, and popular physician. When a patient calls for a non-urgent appointment, the number of days before he can see the patient is a random variable with mean 10 days and standard deviation 3 days. (That includes the day the patient calls, but not the day of the appointment.) If Margie, a patient of Dr. O'Neill, wants to be at least 95% sure that she can get an appointment to see him by a certain date, how early should she call Dr. O'Neill's clinic?

2. A fair die is tossed 120 times. Let Y be the sum of the outcomes. Show that

$$P(330 < Y < 510) \geq \frac{155}{162} \approx 0.96.$$



10.5 LAWS OF LARGE NUMBERS

Let A be an event of some experiment that can be repeated. In [Chapter 1](#) we mentioned that the mathematicians of the eighteenth and nineteenth centuries observed that, in a series of sequential or simultaneous repetitions of the experiment, the proportion of times that A occurs approaches a constant. As a result, they were motivated to define the probability of A to be the number $p = \lim_{n \rightarrow \infty} n(A)/n$, where $n(A)$ is the number of times that A occurs in the first n repetitions. We also mentioned that this relative frequency interpretation of probability, which to some extent satisfies one's intuition, is mathematically problematic and cannot be the basis of a rigorous probability theory. We now show that despite these facts, for repeatable experiments, the relative frequency interpretation is valid. It is the special case of one of the most celebrated theorems of probability and statistics: the **strong law of large numbers**. As a consequence of another important theorem called the **weak law of large numbers**, we have that, for sufficiently large n , it is very likely that $\left| \frac{n(A)}{n} - P(A) \right|$ is very small. Therefore, for repeatable experiments, the laws of large numbers confirm our intuitive interpretation of the concept of probability as relative frequency of the occurrence of an event. Moreover, they play a pivotal role in the study of probability and statistics. It should be noted that, even though the weak law of large numbers is mathematically weaker than the strong law of large numbers, strangely enough, it has more applications in certain areas of statistics.

Recall that a sequence $X_1, X_2, \dots$ of random variables is called **identically distributed** if all of them have the same distribution function. If $X_1, X_2, \dots$ are identically distributed, then since they have the same distribution function, their set of possible values is the same. Also, in the discrete case their probability mass functions, and in the continuous case their probability density functions, are identical. Moreover, in both cases X_i 's have the same expectation, the same standard deviation, and the same variance.

Theorem 10.15 (Weak Law of Large Numbers) *Let $X_1, X_2, X_3, \dots$ be a sequence of independent and identically distributed random variables with $\mu = E(X_i)$ and $\sigma^2 = \text{Var}(X_i) < \infty, i = 1, 2, \dots$. Then $\forall \varepsilon > 0$,*

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) = 0.$$

Note that, in the standard statistical models, $(X_1 + X_2 + \dots + X_n)/n$ is $\bar{X}$, the sample mean.

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample of a population. The weak law of large numbers implies that if the sample size is large enough ($n \rightarrow \infty$), then μ , the mean of the population, is very close to $\bar{X} = (X_1 + X_2 + \dots + X_n)/n$. That is, for sufficiently large n , it is very likely that $|\bar{X} - \mu|$ is very small.

Note that the type of convergence in the weak law of large numbers is not the usual point-wise convergence discussed in calculus. This type of convergence is called **convergence in probability** and is defined as follows.

Definition 10.3 Let $X_1, X_2, \dots$ be a sequence of random variables defined on a sample space S . We say that X_n **converges to a random variable X in probability** if, for each $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0.$$

So if X_n converges to X in probability, then for sufficiently large n it is very likely that $|X_n - X|$ is very small. By this definition, the weak law of large numbers states that if $X_1, X_2, \dots$ is a sequence of independent and identically distributed random variables with $\mu = E(X_i)$ and $\sigma^2 = \text{Var}(X_i) < \infty$, $i = 1, 2, \dots$, then $(X_1 + X_2 + \dots + X_n)/n$ converges to μ in probability. To prove Theorem 10.15, we use the fact that the variance of a sum of independent random variables is the sum of the variances of the random variables [see relation (8.9)].

Proof of Theorem 10.15: Since

$$E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{1}{n}[E(X_1) + E(X_2) + \dots + E(X_n)] = \frac{1}{n}n\mu = \mu$$

and

$$\text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{1}{n^2}[\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] = \frac{1}{n^2}n\sigma^2 = \frac{\sigma^2}{n},$$

by Chebyshev's inequality we get

$$P\left(\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) \leq \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2 n}.$$

This shows that

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) = 0. \quad \blacklozenge$$

To show how the weak law of large numbers relates to the relative frequency interpretation of probability, in a repeatable experiment with sample space S , let A be an event of S . Suppose that the experiment is repeated independently, and let $X_i = 1$ if A occurs on the i th repetition, and $X_i = 0$ if A does not occur on the i th repetition. Then, for $i = 1, 2, \dots$,

$$E(X_i) = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A)$$

and

$$X_1 + X_2 + \dots + X_n = n(A), \tag{10.12}$$

where $n(A)$, as before, is the number of times that A occurs in the first n repetitions of the experiment. Thus, according to the weak law of large numbers,

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{n(A)}{n} - P(A)\right| > \varepsilon\right) = \lim_{n \rightarrow \infty} P\left(\left|\frac{X_1 + X_2 + \dots + X_n}{n} - P(A)\right| > \varepsilon\right) = 0.$$

This shows that, for large n , the relative frequency of occurrence of A [that is, $n(A)/n$], is very likely to be close to $P(A)$.

It is important to know that, historically, the weak law of large numbers was shown in an ingenious way by Jacob Bernoulli (1654–1705) for Bernoulli random variables. Later, this law was generalized by Chebyshev for a certain class of sequences of identically distributed and independent random variables. Chebyshev discovered his inequality in this connection. Theorem 10.15 is Chebyshev's version of the weak law of large numbers. This is weaker than the theorem that the Russian mathematician Khintchine proved; that is, its conclusions are the same as Khintchine's result, but its assumptions are more restrictive. In Khintchine's proof, it is not necessary to assume that $\text{Var}(X_i) < \infty$.

Theorem 10.16 (Strong Law of Large Numbers) *Let $X_1, X_2, \dots$ be an independent and identically distributed sequence of random variables with $\mu = E(X_i)$, $i = 1, 2, \dots$. Then*

$$P\left(\lim_{n \rightarrow \infty} \frac{X_1 + X_2 + \cdots + X_n}{n} = \mu\right) = 1.$$

As before, in a repeatable experiment, let A be an event of the sample space S . Then, by (10.12),

$$\frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{n(A)}{n},$$

which, according to the strong law of large numbers, converges to $P(A)$ with probability 1 as $n \rightarrow \infty$.

Note that, in Theorem 10.16, convergence of $(X_1 + X_2 + \cdots + X_n)/n$ to μ is neither the usual pointwise convergence studied in calculus, nor is it convergence in probability as discussed earlier. This type of convergence is called **almost sure convergence** and is defined as follows: Let X and the sequence $X_1, X_2, X_3, \dots$ be random variables defined on a sample space S . Let V be the set of all points ω , in S , at which $X_n(\omega)$ converges to $X(\omega)$. That is,

$$V = \left\{ \omega \in S : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \right\}.$$

If $P(V) = 1$, then we say that X_n converges **almost surely** to X .

Therefore, almost sure convergence implies that, on a set whose probability is 1, at all sample points ω , $X_n(\omega) - X(\omega)$ gets smaller and smaller as n gets larger and larger. Whereas convergence in probability implies that the probability that $X_n - X$ remains nonnegligible gets smaller and smaller as n gets larger and larger. It is not hard to show that, if X_n converges to X almost surely, it converges to X in probability. However, the converse of this is not true (see Exercise 7).

To clarify the definition of almost sure convergence, we now give an example of a sequence that converges almost surely but is not pointwise convergent. Consider an experiment in which we choose a random point from the interval $[0, 1]$. The sample space of this experiment is $[0, 1]$. For all $\omega \in [0, 1]$, let $X_n(\omega) = \omega^n$ and $X(\omega) = 0$. Then, clearly,

$$V = \left\{ \omega \in [0, 1] : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \right\} = [0, 1].$$

Since $P(V) = P([0, 1)) = 1$, X_n converges almost surely to X . However, this convergence is not pointwise because, at 1, X_n converges to 1 and not to $X(1) = 0$.

Example 10.36 At a large international airport, a currency exchange bank with only one teller is open 24 hours a day, 7 days a week. Suppose that at some time $t = 0$, the bank is free of customers and new customers arrive at random times $T_1, T_1 + T_2, T_1 + T_2 + T_3, \dots$, where $T_1, T_2, T_3, \dots$ are identically distributed and independent random variables with $E(T_i) = 1/\lambda$. When the teller is free, the service time of a customer entering the bank begins upon arrival. Otherwise, the customer joins the queue and waits to be served on a first-come, first-served basis. The customer leaves the bank after being served. The service time of the i th new customer is S_i , where $S_1, S_2, S_3, \dots$ are identically distributed and independent random variables with $E(S_i) = 1/\mu$. Therefore, new customers arrive at the rate λ , and while the teller is busy, they are served at the rate μ . Show that if $\lambda < \mu$, that is, if new customers are served at a higher rate than they arrive, then with probability 1, eventually, for some period, the bank will be empty of customers again.

Solution: Suppose that the bank will never again be empty of customers. We will show a contradiction. Let $U_n = T_1 + T_2 + \dots + T_n$. Then U_n is the time the n th new customer arrives. Let $Z_n = T_1 + S_1 + S_2 + \dots + S_n$. Since the bank will never be empty of customers again, and customers are served on a first-come, first-served basis, Z_n is the departure time of the n th new customer. By the strong law of large numbers,

$$\lim_{n \rightarrow \infty} \frac{U_n}{n} = \frac{1}{\lambda}$$

and

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{Z_n}{n} &= \lim_{n \rightarrow \infty} \left(\frac{T_1}{n} + \frac{S_1 + S_2 + \dots + S_n}{n} \right) \\ &= \lim_{n \rightarrow \infty} \frac{T_1}{n} + \lim_{n \rightarrow \infty} \frac{S_1 + S_2 + \dots + S_n}{n} = 0 + \frac{1}{\mu} = \frac{1}{\mu}. \end{aligned}$$

Clearly, the bank will never remain empty of customers again if and only if $\forall n$,

$$U_{n+1} < Z_n.$$

This implies that

$$\frac{U_{n+1}}{n} < \frac{Z_n}{n},$$

or, equivalently,

$$\frac{n+1}{n} \cdot \frac{U_{n+1}}{n+1} < \frac{Z_n}{n}.$$

Thus

$$\lim_{n \rightarrow \infty} \frac{n+1}{n} \cdot \frac{U_{n+1}}{n+1} \leq \lim_{n \rightarrow \infty} \frac{Z_n}{n}. \quad (10.13)$$

Since $\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$, and with probability 1, $\lim_{n \rightarrow \infty} \frac{U_{n+1}}{n+1} = \frac{1}{\lambda}$ and $\lim_{n \rightarrow \infty} \frac{Z_n}{n} = \frac{1}{\mu}$, (10.13) implies that $\frac{1}{\lambda} \leq \frac{1}{\mu}$ or $\lambda \geq \mu$. This contradicts the fact that $\lambda < \mu$. Hence, with probability 1, eventually, for some period, the bank will be empty of customers again. ♦

Example 10.37 Suppose that an “immortal monkey” is constantly typing on a word processor that is not breakable, lasts forever, and has infinite memory. Suppose that the keyboard of the wordprocessor has $m - 1$ keys, a space bar for blank spaces, and separate keys for different symbols. If each time the monkey presses one of the m symbols (including the space bar) at random, and if at the end of each line and at the end of each page the word processor advances to a new line and a new page by itself, what is the probability that the monkey eventually will produce the complete works of Shakespeare in chronological order and with no errors?

Solution: The complete works of Shakespeare in chronological order and with no errors are the result of typing N specific symbols (including the space bar) for some large N . For $i = 1, 2, \dots$, let A_i be the event that the symbols numbered $(i-1)N + 1$ to number iN , typed by the monkey, form the complete works of Shakespeare, in chronological order with no errors. Also for $i = 1, 2, \dots$, let

$$X_i = \begin{cases} 1 & \text{if } A_i \text{ occurs} \\ 0 & \text{otherwise;} \end{cases}$$

then $E(X_i) = P(A_i) = (1/m)^N$. Now, since $\{X_1, X_2, \dots\}$ is a sequence of independent and identically distributed random variables, by the strong law of large numbers we have that

$$P\left(\lim_{n \rightarrow \infty} \frac{X_1 + X_2 + \dots + X_n}{n} = \left(\frac{1}{m}\right)^N\right) = 1.$$

This relation shows that $\sum_{i=1}^{\infty} X_i = \infty$, the reason being that otherwise $\sum_{i=1}^{\infty} X_i < \infty$ implies that $\lim_{n \rightarrow \infty} (X_1 + X_2 + \dots + X_n)/n$ is 0, which contradicts the fact that, with probability 1, the limit is $(1/m)^N > 0$. Now $\sum_{i=1}^{\infty} X_i = \infty$ implies that an infinite number of X_i 's are equal to 1, meaning that infinitely many of A_i 's will occur. Therefore, not only once, but an infinite number of times, the monkey will produce the complete works of Shakespeare in chronological order with no errors. However, if the monkey types 200 symbols a minute, then for $m = 88$ and $N = 10^7$, it is easily shown that, on average, it will take him T years to produce the complete works of Shakespeare for the first time, where T is a number with tens of millions of digits. ♦

Example 10.38 Suppose that $f: [0, 1] \rightarrow [0, 1]$ is a continuous function. We will now present a probabilistic method for evaluation of the integral $\int_0^1 f(x) dx$, using the strong law of large numbers. Let $\{X_1, Y_1, X_2, Y_2, \dots\}$ be a sequence of independent uniform random numbers from the interval $[0, 1]$. Then $\{(X_1, Y_1), (X_2, Y_2), \dots\}$ is a sequence of independent random points from $[0, 1] \times [0, 1]$. For $i = 1, 2, 3, \dots$, let

$$Z_i = \begin{cases} 1 & \text{if } Y_i < f(X_i) \\ 0 & \text{otherwise.} \end{cases}$$

Note that Z_i is 1 if (X_i, Y_i) falls under the curve $y = f(x)$ and is 0 if it falls above $y = f(x)$. Therefore, from the first n points selected, $Z_1 + Z_2 + \cdots + Z_n$ of them fall below the curve $y = f(x)$, and thus $\lim_{n \rightarrow \infty} (Z_1 + Z_2 + \cdots + Z_n)/n$ is the limit of the relative frequency of the points that fall below the curve $y = f(x)$. We will show that this limit converges to $\int_0^1 f(x) dx$. To do so, note that Z_i 's are identically distributed and independent. Moreover, since g , the joint probability density function of (X_i, Y_i) , $i \geq 0$, is given by

$$g(x, y) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0 & \text{otherwise,} \end{cases}$$

we have that

$$\begin{aligned} E(Z_i) &= P(Y_i < f(X_i)) = \iint_{\{(x, y) : y < f(x)\}} g(x, y) dx dy \\ &= \int_0^1 \left(\int_0^{f(x)} dy \right) dx = \int_0^1 f(x) dx < \infty. \end{aligned}$$

Hence, by the strong law of large numbers, with probability 1,

$$\lim_{n \rightarrow \infty} \frac{Z_1 + Z_2 + \cdots + Z_n}{n} = \int_0^1 f(x) dx.$$

This gives a new method for calculation of $\int_0^1 f(x) dx$. Namely, to find an approximate value for $\int_0^1 f(x) dx$ it suffices that, for a large n , we choose n random points from $[0, 1] \times [0, 1]$ and calculate m , the number of those falling below the curve $y = f(x)$. The quotient m/n is then approximately $\int_0^1 f(x) dx$. This concept is further explained in our discussion of the Monte Carlo procedure and simulation in Section 12.5 of this book, which can be accessed on the Instructor and Student Resources website for this text. ♦

Example 10.39 Suppose that F , the distribution function of the elements of a class of random variables (say, a population), is unknown and we want to estimate it at a point x . For this purpose we take a random sample from the population; that is, we find independent random variables $X_1, X_2, \dots, X_n$, each with distribution function F . Then we let $n(x)$ = the number of X_i 's $\leq x$ and $\hat{F}_n(x) = n(x)/n$. Clearly, $\hat{F}_n(x)$ is the relative frequency of the number of data $\leq x$ [in statistics, $\hat{F}_n(x)$ is called the **empirical distribution function of the sample**]. We will show that $\lim_{n \rightarrow \infty} \hat{F}_n(x) = F(x)$. To do so, let

$$Y_i = \begin{cases} 1 & \text{if } X_i \leq x \\ 0 & \text{otherwise.} \end{cases}$$

Then Y_i 's are identically distributed, independent, and

$$E(Y_i) = P(X_i \leq x) = F(x) \leq 1 < \infty.$$

Thus, by the strong law of large numbers,

$$\lim_{n \rightarrow \infty} \hat{F}_n(x) = \lim_{n \rightarrow \infty} \frac{Y_1 + Y_2 + \cdots + Y_n}{n} = E(Y_i) = F(x).$$

Therefore, for large n , $\hat{F}_n(x)$ is an approximation of $F(x)$. ♦

Example 10.40 A number between 0 and 1 is called **normal** if for every d , $0 \leq d \leq 9$, the “frequency” of the digit d in the decimal expansion of the number is $1/10$. In the first version of the strong law of large numbers, in 1909, Émile Borel proved that, if X is a random number from the interval $(0, 1)$, then, with probability 1, X is a normal number. In other words, the probability that a number selected at random from $(0, 1)$ is not normal is 0—a fact known as **Borel’s normal number theorem**. It is interesting to know that in spite of this fact, *constructing specific normal numbers is an extremely challenging problem*. To prove Borel’s normal number theorem, let X be a random number from the interval $(0, 1)$. The probability that X has two distinct decimal expansions is 0, since the set of such numbers is countable (see Exercise 12, [Section 1.7](#)). Therefore, with probability 1, X has a unique decimal expansion $0.X_1X_2 \cdots X_n \cdots$. For a fixed d , $0 \leq d \leq 9$, let

$$Y_n = \begin{cases} 1 & \text{if } X_n = d \\ 0 & \text{if } X_n \neq d. \end{cases}$$

Then $\sum_{i=1}^n Y_i$ is the number of times that d appears among the first n digits of the decimal expansion of X . Therefore, $\frac{1}{n} \sum_{i=1}^n Y_i$ is the relative frequency of d . The fact that $\{Y_1, Y_2, \dots\}$ is a sequence of identically distributed random variables with $\mu = E(Y_n) = P(X_n = d) = 1/10$ is trivial. If we show that this sequence is independent, then by the strong law of large numbers, with probability 1,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n Y_n = \mu = \frac{1}{10},$$

showing that X is normal. Therefore, all that is left is to show that $\{Y_1, Y_2, \dots\}$ is a sequence of independent random variables. To do so, we need to prove that for any subsequence $\{Y_{i_1}, Y_{i_2}, \dots, Y_{i_m}\}$ of $\{Y_1, Y_2, \dots\}$, and any set of numbers $\alpha_i \in \{0, 1\}$, $1 \leq i \leq m$, we have

$$P(Y_{i_1} = \alpha_1, Y_{i_2} = \alpha_2, \dots, Y_{i_m} = \alpha_m) = P(Y_{i_1} = \alpha_1)P(Y_{i_2} = \alpha_2) \cdots P(Y_{i_m} = \alpha_m).$$

For $1 \leq i \leq m$, let

$$\beta_i = \begin{cases} 9 & \text{if } \alpha_i = 0 \\ 1 & \text{if } \alpha_i = 1. \end{cases}$$

Then, by the generalized counting principle (Theorem 2.2),

$$\begin{aligned} P(Y_{i_1} = \alpha_1, Y_{i_2} = \alpha_2, \dots, Y_{i_m} = \alpha_m) &= \frac{\beta_1 \beta_2 \cdots \beta_m}{10^m} \\ &= \frac{\beta_1}{10} \cdot \frac{\beta_2}{10} \cdots \frac{\beta_m}{10} = P(Y_{i_1} = \alpha_1)P(Y_{i_2} = \alpha_2) \cdots P(Y_{i_m} = \alpha_m). \end{aligned}$$

This establishes Borel's normal number theorem. ♦

Example 10.41 Suppose that there exist N families on the earth and that the maximum number of children a family has is c . Let α_j , $j = 0, 1, \dots, c$, $\sum_{j=0}^c \alpha_j = 1$, be the fraction of families with j children. Determine the fraction of the children in the world who are from a family with k ($k = 1, 2, \dots, c$) children.

Solution: Let X_i denote the number of children of family i ($1 \leq i \leq N$) and

$$Y_i = \begin{cases} 1 & \text{if family } i \text{ has } k \text{ children} \\ 0 & \text{otherwise.} \end{cases}$$

Then the fraction of the children in the world who are from a family with k children is

$$\frac{k \sum_{i=1}^N Y_i}{\sum_{i=1}^N X_i} = \frac{k \frac{1}{N} \sum_{i=1}^N Y_i}{\frac{1}{N} \sum_{i=1}^N X_i}.$$

Now $(1/N) \sum_{i=1}^N Y_i$ is the fraction of the families with k children; therefore, it is equal to α_k . The quantity $(1/N) \sum_{i=1}^N X_i$ is the average number of children in a family; hence it is the same as $\sum_{j=0}^c j \alpha_j$. So the desired fraction is $k \alpha_k / \sum_{j=0}^c j \alpha_j$.

For large N , the strong law of large numbers can be used to show the same thing. Note that it is reasonable to assume that $\{X_1, X_2, \dots\}$ and hence $\{Y_1, Y_2, \dots\}$ are sequences of independent random variables. So we can write that

$$\lim_{N \rightarrow \infty} \frac{k \frac{1}{N} \sum_{i=1}^N Y_i}{\frac{1}{N} \sum_{i=1}^N X_i} = \frac{k E(Y_i)}{E(X_i)} = \frac{k \alpha_k}{\sum_{j=0}^c j \alpha_j}. \quad \blacklozenge$$

EXERCISES

A

- Let $\{X_1, X_2, \dots\}$ be a sequence of nonnegative independent random variables and, for all i , suppose that the probability density function of X_i is

$$f(x) = \begin{cases} 6x(1-x) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find

$$\lim_{n \rightarrow \infty} \frac{X_1 + X_2 + \dots + X_n}{n}.$$

2. Let $\{X_1, X_2, \dots\}$ be a sequence of independent, identically distributed random variables with positive expected value. Show that for all $M > 0$,

$$\lim_{n \rightarrow \infty} P(X_1 + X_2 + \dots + X_n > M) = 1.$$

3. Let X be a nonnegative continuous random variable with probability density function $f(x)$. Define

$$Y_n = \begin{cases} 1 & \text{if } X > n \\ 0 & \text{otherwise.} \end{cases}$$

Prove that Y_n converges to 0 in probability.

4. Let $\{X_1, X_2, \dots\}$ be a sequence of independent, identically distributed random variables. In other words, for all n , let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean $\mu < \infty$. Let $S_n = X_1 + X_2 + \dots + X_n$, $\bar{X}_n = S_n/n$. Show that S_n grows at rate n . That is,

$$\lim_{n \rightarrow \infty} P(n(\mu - \varepsilon) \leq S_n \leq n(\mu + \varepsilon)) = 1.$$

B

5. Suppose that in Example 10.36 rather than customers being served in their arrival order, they are served on a last-come, first-served basis, or they are simply served in a random order. Show that if $\lambda < \mu$, then still with probability 1, eventually, for some period, the bank will be empty of customers again.
6. In Example 10.36, suppose that at $t = 0$ the bank is not free, there are $m > 0$ customers waiting in a queue to be served, and a customer is being served. Show that, with probability 1, eventually, for some period, the bank will be empty of customers.
7. For a positive integer n , let $\tau(n) = (2^k, i)$, where i is the remainder when we divide n by 2^k , the largest possible power of 2. For example, $\tau(10) = (2^3, 2)$, $\tau(12) = (2^3, 4)$, $\tau(19) = (2^4, 3)$, and $\tau(69) = (2^6, 5)$. In an experiment a point is selected at random from $[0, 1]$. For $n \geq 1$, $\tau(n) = (2^k, i)$, let

$$X_n = \begin{cases} 1 & \text{if the outcome is in } \left[\frac{i}{2^k}, \frac{i+1}{2^k} \right] \\ 0 & \text{otherwise.} \end{cases}$$

Show that X_n converges to 0 in probability while it does not converge at any point, let alone almost sure convergence. ■

Self-Quiz on Section 10.5

Time allotted: 20 Minutes

Each problem is worth 5 points.

1. Let $X_1, X_2, \dots, X_n$ be independent random numbers from the interval (a, b) , $a < b$. Find the value of α for which

$$\lim_{n \rightarrow \infty} \frac{X_1 + X_2 + \dots + X_n}{X_1^2 + X_2^2 + \dots + X_n^2} = \alpha$$

with probability 1.

2. For $n \geq 1$, let X_n be a binomial random variable with parameters n and p . Since X_n is the number of successes in n independently performed Bernoulli trials, X_n/n is the fraction of successes in the first n trials. Show that X_n/n converges to p in probability. That is, for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{X_n}{n} - p\right| > \varepsilon\right) = 0.$$

■

10.6 TWO PROOFS FOR THE CENTRAL LIMIT THEOREM

In this section, we will provide two proofs for Theorem 7.2, the Central Limit Theorem: one utilizing moment-generating functions and the other employing characteristic functions. For coherence with the numbering in [Chapter 10](#), Theorem 7.2 is renumbered as Theorem 10.17.

Theorem 10.17 (Central Limit Theorem) *Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed random variables, each with expectation μ and variance σ^2 . Then the distribution of*

$$Z_n = \frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}}$$

converges to the distribution of a standard normal random variable. That is,

$$\begin{aligned} \lim_{n \rightarrow \infty} P(Z_n \leq x) &= \lim_{n \rightarrow \infty} P\left(\frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq x\right) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy. \end{aligned}$$

Remark Note that, as mentioned in [Chapter 7](#), Z_n is simply $X_1 + X_2 + \dots + X_n$ standardized. Also, note that, in Theorem 10.17, letting $\bar{X}_n$ be the mean of the random variables $X_1, X_2, \dots, X_n$, the central limit theorem is equivalent to

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy.$$

This version of the central limit theorem is notably significant for its applications in statistics. $\blacklozenge$

Using Moment-Generating Functions to Prove Theorem 10.17, the Central Limit Theorem:

Let $Y_n = X_n - \mu$, then $E(Y_n) = 0$ and $\text{Var}(Y_n) = \sigma^2$. We want to prove that the sequence of distribution functions of the random variables $Z_n = (Y_1 + Y_2 + \cdots + Y_n)/(\sigma\sqrt{n})$, $n \geq 1$, converges to the distribution of Z . Since $Y_1, Y_2, \dots$ are identically distributed, they have the same moment-generating function M . By independence of $Y_1, Y_2, \dots, Y_n$,

$$\begin{aligned} M_{Z_n}(t) &= E\left[\exp\left(\frac{Y_1 + Y_2 + \cdots + Y_n}{\sigma\sqrt{n}}t\right)\right] = M_{Y_1+Y_2+\cdots+Y_n}\left(\frac{t}{\sigma\sqrt{n}}\right) \\ &= M_{Y_1}\left(\frac{t}{\sigma\sqrt{n}}\right)M_{Y_2}\left(\frac{t}{\sigma\sqrt{n}}\right)\cdots M_{Y_n}\left(\frac{t}{\sigma\sqrt{n}}\right) \\ &= \left[M\left(\frac{t}{\sigma\sqrt{n}}\right)\right]^n. \end{aligned} \quad (10.14)$$

By the Lévy continuity theorem for moment-generating functions, it suffices to prove that $M_{Z_n}(t)$ converges to $\exp(t^2/2)$, the moment-generating function of Z . Equivalently, it is enough to show that

$$\lim_{n \rightarrow \infty} \ln M_{Z_n}(t) = \frac{t^2}{2}. \quad (10.15)$$

Let $h = t/(\sigma\sqrt{n})$, then $n = t^2/(\sigma^2 h^2)$. Thus (10.14) gives

$$\ln M_{Z_n}(t) = n \ln M(h) = \frac{t^2}{\sigma^2 h^2} \ln M(h) = \frac{t^2}{\sigma^2} \left[\frac{\ln M(h)}{h^2} \right].$$

Therefore,

$$\lim_{n \rightarrow \infty} \ln M_{Z_n}(t) = \frac{t^2}{\sigma^2} \lim_{h \rightarrow 0} \left[\frac{\ln M(h)}{h^2} \right]. \quad (10.16)$$

Since $M(0) = 1$, $\lim_{h \rightarrow 0} [\ln M(h)/h^2]$ is indeterminate. To determine its value, we apply l'Hôpital's rule twice. We get

$$\begin{aligned} \lim_{h \rightarrow 0} \left[\frac{\ln M(h)}{h^2} \right] &= \lim_{h \rightarrow 0} \frac{M'(h)/M(h)}{2h} = \lim_{h \rightarrow 0} \frac{M'(h)}{2hM(h)} \\ &= \lim_{h \rightarrow 0} \frac{M''(h)}{2M(h) + 2hM'(h)} = \frac{M''(0)}{2M(0)} = \frac{\sigma^2}{2}, \end{aligned}$$

where $M''(0) = \sigma^2$ since $E(Y_i) = 0$. Thus, by (10.16),

$$\lim_{n \rightarrow \infty} \ln M_{Z_n}(t) = \frac{t^2}{\sigma^2} \frac{\sigma^2}{2} = \frac{t^2}{2}.$$

This establishes (10.15) and proves the theorem. $\blacklozenge$

Using Characteristic Functions to Prove Theorem 10.17, the Central Limit Theorem:

The proof is almost identical to the proof using the moment-generating functions. Let $Y_n = X_n - \mu$, then $E(Y_n) = 0$ and $\text{Var}(Y_n) = \sigma^2$. We want to prove that the sequence of distribution functions of the random variables $Z_n = (Y_1 + Y_2 + \cdots + Y_n)/(\sigma\sqrt{n})$, $n \geq 1$, converges to the distribution of Z . Since $Y_1, Y_2, \dots$ are identically distributed, they have the same characteristic function ϕ . By independence of $Y_1, Y_2, \dots, Y_n$,

$$\begin{aligned}\phi_{Z_n}(t) &= E\left[\exp\left(\frac{Y_1 + Y_2 + \cdots + Y_n}{\sigma\sqrt{n}} it\right)\right] = \phi_{Y_1+Y_2+\cdots+Y_n}\left(\frac{t}{\sigma\sqrt{n}}\right) \\ &= \phi_{Y_1}\left(\frac{t}{\sigma\sqrt{n}}\right)\phi_{Y_2}\left(\frac{t}{\sigma\sqrt{n}}\right)\cdots\phi_{Y_n}\left(\frac{t}{\sigma\sqrt{n}}\right) \\ &= \left[\phi\left(\frac{t}{\sigma\sqrt{n}}\right)\right]^n.\end{aligned}\tag{10.17}$$

By the Lévy continuity theorem for characteristic functions, it suffices to prove that $\phi_{Z_n}(t)$ converges to $\exp(-t^2/2)$, the characteristic function of a standard normal random variable. Equivalently, it is enough to show that

$$\lim_{n \rightarrow \infty} \ln \phi_{Z_n}(t) = -\frac{t^2}{2}.\tag{10.18}$$

Let $h = t/(\sigma\sqrt{n})$, then $n = t^2/(\sigma^2 h^2)$. Thus (10.17) gives

$$\ln \phi_{Z_n}(t) = n \ln \phi(h) = \frac{t^2}{\sigma^2 h^2} \ln \phi(h) = \frac{t^2}{\sigma^2} \left[\frac{\ln \phi(h)}{h^2} \right].$$

Therefore,

$$\lim_{n \rightarrow \infty} \ln \phi_{Z_n}(t) = \frac{t^2}{\sigma^2} \lim_{h \rightarrow 0} \left[\frac{\ln \phi(h)}{h^2} \right].\tag{10.19}$$

Since $\phi(0) = 1$, $\lim_{h \rightarrow 0} [\ln \phi(h)/h^2]$ is indeterminate. To determine its value, we apply l'Hôpital's rule twice. We get

$$\begin{aligned}\lim_{h \rightarrow 0} \left[\frac{\ln \phi(h)}{h^2} \right] &= \lim_{h \rightarrow 0} \frac{\phi'(h)/\phi(h)}{2h} = \lim_{h \rightarrow 0} \frac{\phi'(h)}{2h\phi(h)} \\ &= \lim_{h \rightarrow 0} \frac{\phi''(h)}{2\phi(h) + 2h\phi'(h)} = \frac{\phi''(0)}{2\phi(0)} = -\frac{\sigma^2}{2},\end{aligned}$$

where $\phi''(0) = i^2 E(y^2) = i^2 \sigma^2 = -\sigma^2$ since $E(Y_i) = 0$. Thus, by (10.19),

$$\lim_{n \rightarrow \infty} \ln \phi_{Z_n}(t) = \frac{t^2}{\sigma^2} \frac{-\sigma^2}{2} = -\frac{t^2}{2}.$$

This establishes (10.18) and proves the theorem. ♦

10.7 MORE ON R SIMULATIONS

Moments In the R programming language, it is straightforward to calculate moments of continuous random variables. For this illustration, consider a random variable X that follows an exponential distribution with a parameter of $\lambda = 5$. To calculate the fourth moment, denoted as $E(X^4)$, we use the R code:

```
> f=function(x) x^4*dexp(x,rate=5)
> integrate(f,0,Inf)
0.0384 with absolute error <1e-09
```

Similarly, the code below computes the third moment of a uniform random variable over the interval $(0, 1)$ and the second moment of a Poisson random variable with a parameter of $\lambda = 4.5$.

```
> g=function(x) x^3*dunif(x,0,1)
> integrate(g,lower=0,upper=1)
0.25 with absolute error<2.8e-15
> h=function(n) n^2*dpois(n,4.5)
> sum(h(0:50))
[1] 24.75
> sum(h(0:100))
[1] 24.75
```

Regarding the Poisson random variable, we included only the initial 50 terms in the calculation because beyond that point, the contributions of additional terms become negligible. Notably, it's worth noting that even when we extended the summation to the first 100 terms, the result remained unchanged.

The strong law of large numbers: To visually illustrate the convergence of sample means toward the population mean, it is essential to observe how the sample means evolve as we progressively enlarge the sample size. Here is a clear example that employs an R simulation to vividly demonstrate the Strong Law of Large Numbers:

Example 10.42 Let us define **num** as the number of simulations. In this context, **num** also represents the number of samples we will generate from our chosen distribution, which, in this example, is the standard normal distribution. Our goal is to investigate the convergence of sample means toward the population mean as we increase the sample size **n**. This convergence is a fundamental concept of the strong law of large numbers.

To ensure reproducibility, we will start by setting a random seed using the **set.seed()** function. This step is not mandatory, but it allows us to specify the initial state of the random number generator. This means that if we or someone else runs the same code with the same seed, we will obtain identical random results. Reproducibility is crucial for replicating experiments and sharing code with others.

In our code below, we use **sm** to represent the sample means, **ss** for the sample size, and **sda** for the sample data. For each simulation (indexed by **i**), the sample data is generated as **rnorm(i)**. The population mean, denoted as **mu** in this case, is set to 0, representing the expected value of a standard normal random variable.

To summarize, our approach involves varying the sample size from 1 to **num** and computing the sample mean for each sample size. We then create a plot that displays the relationship between the sample means and the sample size. This visual representation enables us to observe the convergence of sample means toward the population mean as the sample size progressively increases (see [Figure 10.3](#)). This graphical illustration effectively demonstrates the strong law of large numbers, illustrating how sample means tend to approach the population mean as the sample size becomes larger. Below is the R code for conducting this simulation:

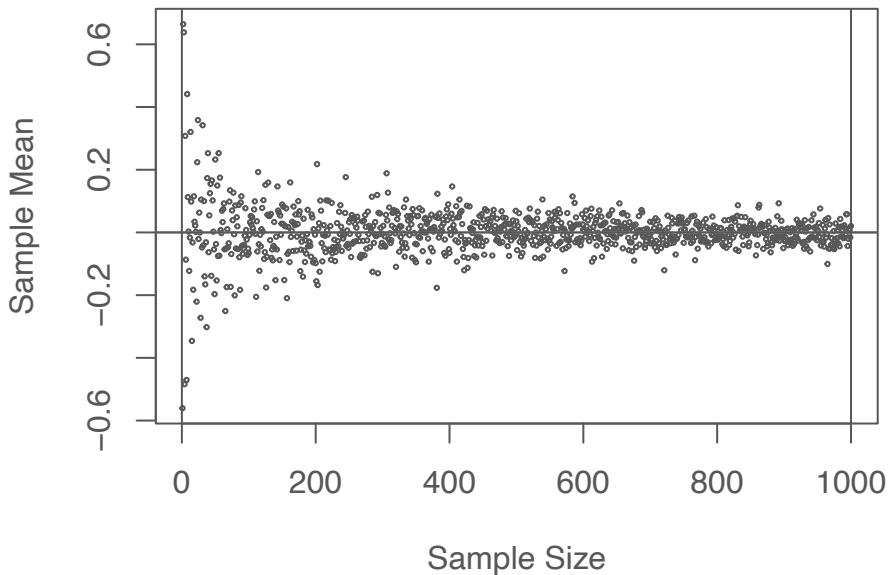


Figure 10.3 An R simulation of strong law of large numbers.

```
> set.seed(123)
> num <- 1000
> sm <- numeric(num)
> ss <- 1:num
> for(i in 1:num){
+   sda <- rnorm(i)
+   sm[i] <- mean(sda)
+ }
> mu <- 0
```

```

> plot(ss,sm, cex=0.3, xlab = "Sample Size", ylab = "Sample Mean",
+ main = "Strong Law of Large Numbers Simulation")
> abline(h=mu, col="red")
>abline(v=0)
abline(v=1000)

```

Monte Carlo Method

As explained in [Section 5.4](#), if S is a subset of the plane with area $A(S)$ and R is a subset of S with area $A(R)$, the probability that a random point from S falls in R is equal to $A(R)/A(S)$. This important fact gives an excellent algorithm, called the **Monte Carlo method**, for finding the area under a bounded curve $y = f(x)$ by simulation. Suppose that we want to estimate I , the area under $y = f(x)$, from $x = a$ to $x = b$ of [Figure 10.4](#). To do so, we first construct a rectangle $[a, b] \times [0, d]$ that includes the region under $y = f(x)$ from a to b as a subset. Then for a large integer n , we choose n random points from $[a, b] \times [0, d]$ and count m , the number of those that lie below the curve $y = f(x)$. Now the probability that a random point from $[a, b] \times [0, d]$ is under the curve $y = f(x)$ is approximately m/n . Thus

$$\frac{m}{n} \approx \frac{\text{area under } f(x) \text{ from } a \text{ to } b}{\text{area of the rectangle } [a, b] \times [0, d]} = \frac{I}{(b-a)(d-0)} = \frac{I}{(b-a)d},$$

and hence

$$I \approx \frac{md(b-a)}{n}.$$

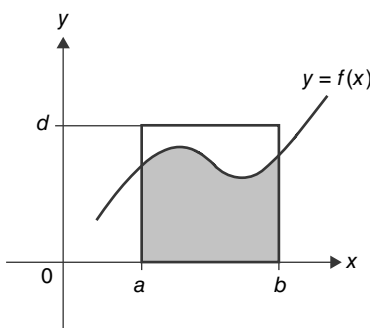


Figure 10.4 Estimating area under f with Monte Carlo simulation.

This method of simulation was introduced by the Hungarian-born American mathematician John von Neumann (1903–1957), and Stanislaw Ulam (1909–1984). They used it during World War II to study the extent that neutrons can travel through various materials. Since their studies were classified, von Neumann gave it the code name **Monte Carlo method**.

We will now apply this method to compute the area under the curve $y = \sqrt{1 - x^2}$, bounded by the lines $y = 0$, $x = -1$, and $x = 1$. This area corresponds to a semicircle inscribed inside

the rectangle defined by $y = 0$, $y = 1$, $x = -1$, and $x = 1$ (see Figure 10.5). For this semicircle, the radius is $r = 1$, and therefore, its area is half of πr^2 , which simplifies to $\pi/2$.

To estimate this area using the Monte Carlo method, we will randomly select 1000 points from the region $[-1, 1] \times [0, 1]$ and count m , the number of points that lie below the curve. Let I represent the area under the curve, we have that

$$\frac{I}{\text{area}([-1, 1] \times [0, 1])} = \frac{m}{1000}.$$

This equation yields $\frac{I}{2} = \frac{m}{1000}$, which simplifies to $I = 2m/1000$. Consequently, $\pi/2 = 2m/1000$, and we can approximate π as $\pi \approx 4m/1000$.

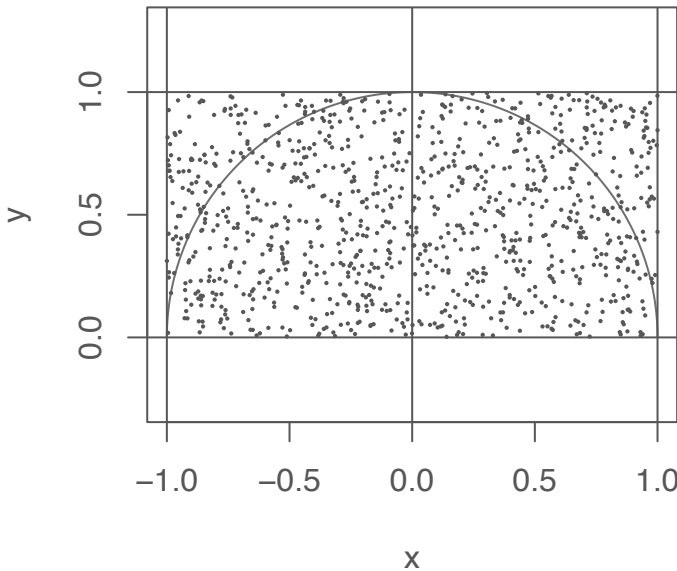


Figure 10.5 Utilizing Monte Carlo simulation to compute π .

On the following page, you'll find an R code that calculates the estimation of $4m/1000$, which serves as an approximation for π . The initial lines of code plot the rectangle $[-1, 1] \times [0, 1]$ and the semicircle $y = \sqrt{1 - x^2}$ using its parametric equation: $x = \cos \theta$, $y = \sin \theta$. To randomly select points within the rectangle and determine whether they fall under the curve, vectorized operations are employed. Random **x** and **y** values are generated simultaneously using the **runif** function. The resulting **x** and **y** are vectors of length **num=1000**, containing random numbers from the intervals $(-1, 1)$ and $(0, 1)$, respectively.

A logical vector named **condition** is created, where each element corresponds to whether the condition $y \leq \sqrt{1 - x^2}$ is satisfied for each pair of **(x[i], y[i])**, $1 \leq i \leq \text{num}$. The variable **m** is calculated by summing the TRUE values in the **condition** vector, effectively counting the number of points that meet the condition. Subsequently, the code computes **I** and

$\mathbf{p}$ based on the count $\mathbf{m}$. This vectorized approach avoids the necessity of using a loop and proves to be more efficient for a large number of iterations.

```
> plot(NA,xlim=c(-1,1),ylim=c(0,1),asp=1,type="n",
      xlab="x", ylab="y",main="Semicircle")
> theta <- seq(0,pi,length.out=1000)
> x <- cos(theta)
> y <- sin(theta)
> lines(x,y,col="blue")
> abline(h=1)
> abline(h=0)
> abline(v=-1)
> abline(v=0)
> abline(v=1)
> num <- 1000
> x <- runif(num,-1,1)
> y <- runif(num,0,1)
> points(x,y,pch=16,col="black",cex=0.3)
> condition <- y<=sqrt(1-x^2)
> m <- sum(condition)
> p <- 4*m/num
> print(p)
[1] 3.124
```

The outcome of this simulation, even with only 1000 points selected, is quite satisfactory. When we conducted the simulation with a larger number of points, specifically for `num = 50,000`, the result improved considerably, yielding an approximation of approximately 3.1384 for π .

Example 10.43 (Buffon's Needle Problem Revisited) In Example 5.21, we explored one of the fascinating problems in geometric probability, known as *Buffon's needle problem*. In this scenario, a plane is divided by parallel lines at a distance of d apart. If a needle with a length of ℓ , where $\ell < d$, is randomly tossed onto the plane, the probability that the needle intersects one of the parallel lines can be determined as $2\ell/\pi d$. This formula has played a significant role in the history of probability, as it provides a means to approximate the value of π .

To apply this formula, with fixed and predefined values of d and ℓ , where $\ell < d$, one can rule a plane with parallel lines spaced at a distance of d apart and toss a needle of length ℓ multiple times, denoted as n . By counting the instances, represented as m , where the needle intersects a line, we can estimate the probability of such an occurrence. When n is sufficiently large, m/n provides an approximate value for the probability that the needle intersects a line. Thus, we have the approximation $2\ell/\pi d \approx m/n$, which leads to the estimation $\pi \approx 2n\ell/md$.

Historically, in 1850, Swiss scientist Rudolf Wolf (1816–1893) conducted an experiment involving 5000 needle tosses, which yielded an approximation of π as 3.1596. In 1846, Belgian astronomer and mathematician Adolphe Quetelet (1796–1874) performed a similar experiment with 3204 tosses and obtained an approximation of 3.1553. In 1860, De Morgan repeated the experiment and, with 600 tosses, arrived at an estimate of 3.137 for π . In subsequent years, mathematician Fox conducted similar experiments in 1894 using 1120 tosses, resulting in a value of 3.1419, and Italian mathematician Mario Lazzarini conducted 3408 experiments in 1901, acquiring the result of 3.1415929 for π . It is important to note that the accuracy of these later results is met with skepticism because the probability of obtaining such close approximations to π with relatively low numbers of needle tosses is exceedingly small.

For more detailed insights into this topic, please refer to N. T. Gridgeman's paper titled "Geometric Probability and the Number π ," published in *Scripta Mathematica* in 1960, Volume 25, Number 3.

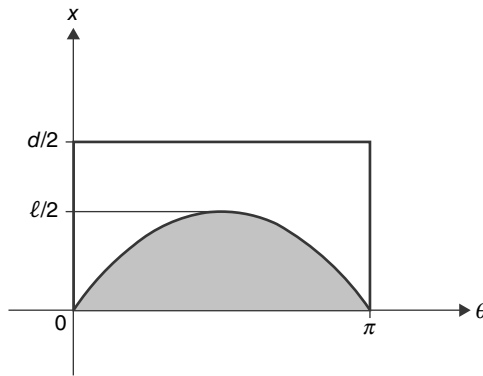


Figure 10.6 The curve $x = (\ell/2) \sin \theta$.

To find more accurate estimates for π , using the result of Buffon's needle problem, we may use the Monte Carlo Method. As we explained in Example 5.21, every time that the needle is tossed, a point (Θ, X) is selected from the square $[0, \pi] \times [0, d/2]$, and the needle intersects the line if and only if $X/\sin \Theta < \ell/2$. Therefore, if for a large number n , we select n random points from the square $[0, \pi] \times [0, d/2]$ and find m , the number of those that fall below the curve $x = (\ell/2) \sin \theta$ (see Figure 10.6), then m/n is approximately the probability that the needle intersects a line; that is, $m/n \approx 2\ell/\pi d$, which gives $\pi \approx 2n\ell/md$. ♦

CHAPTER 10 SUMMARY

► **Moment-Generating Functions** For a random variable X , let $M_X(t) = E(e^{tX})$. If $M_X(t)$ is finite for all values of t in some interval $(-\delta, \delta)$, $\delta > 0$, then $M_X(t)$ is called the *moment-generating function* of X . If not, we say that the moment generating function of the

random variable X does not exist. Hence if X is a discrete random variable with set of possible values A and probability mass function $p(x)$, then

$$M_X(t) = \sum_{x \in A} e^{tx} p(x),$$

and if X is a continuous random variable with probability density function $f(x)$, then

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx.$$

► Let X be a random variable with moment-generating function $M_X(t)$. Then

$$E(X^n) = M_X^{(n)}(0),$$

where $M_X^{(n)}(t)$ is the n th derivative of $M_X(t)$. It follows that $E(X^n)$ is the coefficient of $t^n/n!$ in the MacLaurin's series representation of $M_X(t)$.

► Let X be a random variable with moment generating function $M_X(t)$. For constants a and b , let $Y = aX + b$. Then the moment generating function of Y is given by

$$M_Y(t) = e^{bt} M_X(at).$$

► **Lévy Continuity Theorem for Moment-Generating Functions** Let $X_1, X_2, \dots$ be a sequence of random variables with distribution functions $F_1, F_2, \dots$ and moment-generating functions $M_{X_1}(t), M_{X_2}(t), \dots$, respectively. Let X be a random variable with distribution function F and moment-generating function $M_X(t)$. If for all values of t , $M_{X_n}(t)$ converges to $M_X(t)$, then at the points of continuity of F , F_n converges to F .

► Let X and Y be two random variables with moment-generating functions $M_X(t)$ and $M_Y(t)$. If for some $\delta > 0$, $M_X(t) = M_Y(t)$ for all values of t in $(-\delta, \delta)$, then X and Y have the same distribution.

► Let $X_1, X_2, \dots, X_n$ be independent random variables with moment-generating functions $M_{X_1}(t), M_{X_2}(t), \dots, M_{X_n}(t)$. The moment-generating function of $X_1 + X_2 + \dots + X_n$ is given by

$$M_{X_1+X_2+\dots+X_n}(t) = M_{X_1}(t) M_{X_2}(t) \cdots M_{X_n}(t).$$

This theorem implies the following important facts:

- The sum of n independent binomial random variables with parameters $(n_1, p), (n_2, p), \dots, (n_r, p)$ is binomial with parameters $n_1 + n_2 + \dots + n_r$ and p .
- The sum of n independent Poisson random variables with means $\lambda_1, \lambda_2, \dots, \lambda_n$ is Poisson with mean $\lambda_1 + \lambda_2 + \dots + \lambda_n$.
- The sum of n independent normal random variables with parameters $(\mu_1, \sigma_1^2), (\mu_2, \sigma_2^2), \dots, (\mu_n, \sigma_n^2)$ is normal with parameters $\mu_1 + \mu_2 + \dots + \mu_n$ and $\sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2$.

- The sum of n independent geometric random variables each with parameter p is negative binomial with parameters (n, p) .
- The sum of n independent negative binomial random variables with parameters $(r_1, p), (r_2, p), \dots, (r_n, p)$ is negative binomial with parameters $(r, p), r = r_1 + r_2 + \dots + r_n$.
- The sum of n independent exponential random variables with the identical mean $1/\lambda$ is gamma with parameters (n, λ) .
- The sum of n independent gamma random variables with parameters $(r_1, \lambda), (r_2, \lambda), \dots, (r_n, \lambda)$ is gamma with parameters $(r_1 + r_2 + \dots + r_n, \lambda)$.

► Let $\{X_1, X_2, \dots, X_n\}$ be a set of independent random variables and $X_i \sim N(\mu_i, \sigma_i^2)$ for $i = 1, 2, \dots, n$, then for constants $\alpha_1, \alpha_2, \dots, \alpha_n$,

$$\sum_{i=1}^n \alpha_i X_i \sim N\left(\sum_{i=1}^n \alpha_i \mu_i, \sum_{i=1}^n \alpha_i^2 \sigma_i^2\right).$$

► If $X_1, X_2, \dots, X_n$ are independent normal random variables all with the same mean μ , and the same variance σ^2 , then $S_n = X_1 + X_2 + \dots + X_n$ is $N(n\mu, n\sigma^2)$ and the sample mean, $\bar{X} = S_n/n$, is $N(\mu, \sigma^2/n)$.

► **Characteristic Functions** Let X be a random variable, and $\mathbb{C}$ be the set of all complex numbers; that is, $\mathbb{C} = \{a + bi, \text{ where } a, b \in \mathbb{R} \text{ and } i = \sqrt{-1}\}$. The *characteristic function* of X , denoted by $\phi_X(t)$ is a function $\phi_X: \mathbb{R} \rightarrow \mathbb{C}$ defined by the expected value of e^{itX} :

$$\phi_X(t) = E(e^{itX}).$$

► For a random variable X , we have that $\phi_X(0) = 1$, $|\phi_X(t)| \leq 1$, $\phi_X(-t) = \overline{\phi_X(t)}$, $\phi_{aX+b}(t) = e^{itb} \phi_X(at)$, and $\phi_{-X}(t) = \overline{\phi_X(t)}$

► If there exists a real number $\alpha > 0$ such that $|M_X(t)| < \infty$ for $|t| < \alpha$, then we have $\phi_X(t) = M_X(it)$.

► If $X_1, X_2, \dots, X_n$ are independent random variables, then

$$\phi_{X_1+X_2+\dots+X_n}(t) = \phi_{X_1}(t) \phi_{X_2}(t) \cdots \phi_{X_n}(t).$$

► If $\phi_X^{(n)}(0)$, the n th derivative of ϕ_X at 0, exists, then $\phi_X^{(n)}(0) = i^n E(X^n)$.

► **The Inversion Theorem (a)** Let X be a continuous random variable with probability density function f and characteristic function ϕ_X . Then

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_X(t) dt,$$

at each x where the function f is differentiable.

(b) Let X be a discrete random variable with probability mass function p and characteristic function ϕ_X . Then

$$p(x) = \lim_{a \rightarrow \infty} \frac{1}{2a} \int_{-a}^a e^{-itx} \phi_X(t) dt.$$

► **Lévy Continuity Theorem for Characteristic Functions** Let $X_1, X_2, \dots$ be a sequence of random variables with distribution functions $F_1, F_2, \dots$ and characteristic functions $\phi_{X_1}(t), \phi_{X_2}(t), \dots$, respectively. Let X be a random variable with distribution function F and characteristic function $\phi_X(t)$. If for all values of t , $\phi_{X_n}(t)$ converges to $\phi_X(t)$, then at the points of continuity of F , F_n converges to F .

► **Markov's Inequality** Let X be a nonnegative random variable, then for any $t > 0$,

$$P(X \geq t) \leq \frac{E(X)}{t}.$$

► **Chebyshev's Inequality** If X is a random variable with expected value μ and variance σ^2 , then for any $t > 0$,

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

Letting $t = k\sigma$ in Chebyshev's inequality, we get that

$$P(|X - \mu| \geq k\sigma) \leq 1/k^2.$$

$$P(|X - \mu| < k\sigma) \geq 1 - 1/k^2.$$

► Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a (continuous or discrete) distribution function F with mean μ and variance σ^2 . Let $\bar{X}$ be the mean of the sample. Then, for any $\varepsilon > 0$,

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2 n}.$$

► **Weak Law of Large Numbers** Let $X_1, X_2, X_3, \dots$ be a sequence of independent and identically distributed random variables with $\mu = E(X_i)$ and $\sigma^2 = \text{Var}(X_i) < \infty$, $i = 1, 2, \dots$. Then $\forall \varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) = 0.$$

► **Strong Law of Large Numbers** Let $X_1, X_2, \dots$ be an independent and identically distributed sequence of random variables with $\mu = E(X_i)$, $i = 1, 2, \dots$. Then

$$P\left(\lim_{n \rightarrow \infty} \frac{X_1 + X_2 + \dots + X_n}{n} = \mu\right) = 1.$$

► **Central Limit Theorem** Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed random variables, each with expectation μ and variance σ^2 . Then

$$\lim_{n \rightarrow \infty} P\left(\frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy.$$

► Let $\bar{X}_n$ be the mean of the random variables $X_1, X_2, \dots, X_n$. The central limit theorem is equivalent to

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy.$$

REVIEW PROBLEMS

1. The moment-generating function of a random variable X is given by

$$M_X(t) = \left(\frac{1}{3} + \frac{2}{3}e^t\right)^{10}.$$

Find $\text{Var}(X)$ and $P(X \geq 8)$.

2. The moment-generating function of a random variable X is given by

$$M_X(t) = \frac{1}{6}e^t + \frac{1}{3}e^{2t} + \frac{1}{2}e^{3t}.$$

Find the distribution function of X .

3. For a random variable X , suppose that $M_X(t) = \exp(2t^2 + t)$. Find $E(X)$ and $\text{Var}(X)$.
4. Let the moment-generating function of a random variable X be given by

$$M_X(t) = \begin{cases} \frac{1}{t}(e^{t/2} - e^{-t/2}) & \text{if } t \neq 0 \\ 1 & \text{if } t = 0. \end{cases}$$

Find the distribution function of X .

5. The moment-generating function of X is given by

$$M_X(t) = \exp\left(\frac{e^t - 1}{2}\right).$$

Find $P(X > 0)$.

6. The moment-generating function of a random variable X is given by

$$M_X(t) = \frac{1}{(1-t)^2}, \quad t < 1.$$

Find the moments of X .

7. Suppose that in a community the distributions of heights of men and women (in centimeters) are $N(173, 40)$ and $N(160, 20)$, respectively. Calculate the probability that the average height of 10 randomly selected men is at least 5 centimeters larger than the average height of six randomly selected women.
8. In a clinical trial, the probability of success for a treatment is to be estimated. If the error of estimation is not allowed to exceed 0.01 with probability 0.94, how many patients should be chosen independently and at random for the treatment group?
9. For a coin, p , the probability of heads is unknown. To estimate p , we flip the coin 5000 times and let $\hat{p}$ be the fraction of times it lands heads up. Show that the probability is at least 0.98 that $\hat{p}$ estimates p within ± 0.05 .
10. Each time that Ed charges an expense to his credit card, he omits the cents and records only the dollar value. If this month he has charged his credit card 20 times, using Chebyshev's inequality, find an upper bound on the probability that his record shows at least \$15 less than the actual amount charged.
11. In a multiple-choice test with false answers receiving negative scores, the mean of the grades of the students is 0 and its standard deviation is 15. Find an upper bound for the probability that a student's grade is at least 45.
12. Show that for a nonnegative random variable X with mean μ , $\forall n$, $nP(X \geq n\mu) \leq 1$.
13. Consider a sequence of random variables $X_1, X_2, \dots$. Let the probability density function of X_n be given by

$$f_{X_n}(x) = ne^{-nx}, \quad x > 0.$$

Determine the random variable X such that its distribution function is the limit of the distributions of the sequence $X_1, X_2, \dots$

14. An ordinary deck of 52 cards is divided randomly into 26 pairs. Using Chebyshev's inequality, find an upper bound for the probability that, at most, 10 pairs consist of a black and a red card.
Hint: For $i = 1, 2, \dots, 26$, let $X_i = 1$ if the i th red card is paired with a black card and $X_i = 0$ otherwise. Find an upper bound for

$$P\left(\sum_{i=1}^{26} X_i \leq 10\right).$$



Self-Test on Chapter 10

Time allotted: 180 Minutes

Each problem is worth 10 points.

1. The probability that next month an ombudsman of a university receives no complaints from a first-year student is 0.73. Otherwise, the number of complaints from the freshmen is a normal random variable with mean 2 and standard deviation 3. The probability that next month the ombudsman receives no complaints from a sophomore is 0.53. Otherwise, the number of complaints from the sophomores is a normal random variable with mean 3 and standard deviation 4. Assuming that complaints from freshmen are independent of complaints from sophomores, find the probability that next month the number of complaints from freshmen is greater than the number of complaints from sophomores.

Hint: Let A be the event that next month the ombudsman will receive at least one complaint from a first-year student. Let B be the event that next month he or she will receive at least one complaint from a sophomore. Let C be the event that next month the number of complaints by the freshmen is greater than the number of complaints by the sophomores. The probability that we are interested in is $P(AB^c) + P(ABC)$. Use the multiplication rule to calculate $P(ABC)$.

2. As defined in Exercise 25, [Section 7.4](#), a random variable X is said to be **Laplace**, if its probability density function is given by

$$f(x) = \frac{1}{2} e^{-|x|}, \quad -\infty < x < \infty.$$

Let U and V be independent and identically distributed exponential random variables with $\lambda = 1$. Show that $W = U - V$ is a Laplace random variable.

Hint: Show that for $|t| < 1$, $M_X(t) = M_W(t)$.

3. Let $X_1, X_2, \dots$ be a sequence of independently selected random numbers from the interval (a, b) , $a < b$. For $n \geq 1$, let $Y_n = \max(X_1, X_2, \dots, X_n)$. Show that Y converges to the constant random variable $Y = b$ in probability. That is, for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|Y_n - Y| > \varepsilon) = 0.$$

4. For a large n , let $X_1, X_2, \dots, X_n$ be identically distributed and independent exponential random variables, each with mean $1/5$. Find the approximate distribution of the

$$\text{random variable } Y = \frac{1}{n} \sum_{i=1}^n e^{X_i}.$$

5. In a country, which is the target of more natural disasters than most other countries, the time between two consecutive natural disasters is a gamma random variable with parameters $r = 3$ and $\lambda = 2$, independently of the times of the occurrences of other natural disasters in that country. Label the time point at which a natural disaster occurs

as $t = 0$. Let $X_1, X_1 + X_2, X_1 + X_2 + X_3, \dots$, be the times of the occurrences of the subsequent disasters, respectively. Define and find the long-term rate at which natural disasters occur in this country.

Hint: Argue that the rate at which natural disasters occur should be defined by

$$\lim_{n \rightarrow \infty} \frac{n}{X_1 + X_2 + \dots + X_n}.$$

6. Let X be a lognormal random variable with parameters μ and σ^2 . By definition, $\ln X \sim N(\mu, \sigma^2)$. In Example 10.6, we showed that for any positive r ,

$$E(X^r) = \exp\left(\mu r + \frac{1}{2}\sigma^2 r^2\right).$$

Show that even though moments of all orders of X exist, in no neighborhood $(-\delta, \delta)$, $\delta > 0$, around 0, $M_X(t)$ exists. That is, the moment generating function of X does not exist.

Hint: Let $Y = \ln X$, then $X = e^Y$. Show that, for any fixed $t > 0$, $M_X(t) \geq (t^n/n!) M_Y(n)$ for all positive integers n , and $(t^n/n!) M_Y(n) \rightarrow \infty$ as $n \rightarrow \infty$.

7. (a) For $m \geq 2$, let $X_1, X_2, \dots, X_m$ be a random sample from a Bernoulli distribution with parameter p . Let $\bar{X} = (X_1 + X_2 + \dots + X_m)/m$. Show that $\text{Var}(\bar{X}) \leq 1/(4m) \leq 1/8$.
- (b) For a study, a sociologist needs to estimate p , the proportion of foreign-born residents of California. To do so, she takes n ($n \geq 2$) random samples from the residents of California independently and measures $p_1, p_2, \dots, p_n$, the proportions of foreign-born residents in those samples. Determine how large n should be so that, with a probability of at least 0.96, we have

$$\left| \frac{p_1 + p_2 + \dots + p_n}{n} - p \right| \leq 0.05.$$

Note that the sizes of the n samples taken need not be equal.

- (c) Do part (b) if the sizes of the n samples taken are all equal to 50.
8. For a large positive integer n , let X be a Poisson random variable with parameter n ; let Y be a gamma random variable with parameters n and λ , and let W be a binomial random variable with parameters n and p . Show that X is approximately $N(n, n)$, Y is approximately $N(n/\lambda, n/\lambda^2)$, and W is approximately $N(np, np(1-p))$.
- Hint:* Note that X is the sum of n independent Poisson random variables, each with parameter 1; Y is the sum of n independent exponential random variables, each with parameter λ ; and W is the sum of n independent Bernoulli random variables, each with parameter p .
9. Let X be a geometric random variable with parameter p . Then the probability mass function of X is given by $P(X = n) = (1-p)^{n-1}p$, $n = 1, 2, \dots$. Find the characteristic function of X and use it to calculate its expected value.
10. A recently built bridge was carefully structured to survive the strengths of the most probable earthquakes. Suppose that (1) the bridge will be maintained properly, and no

catastrophe other than earthquake would ruin it; (2) for a small $p > 0$, the probability is p that an earthquake occurring in the seismic region of the bridge is so intensive that the bridge cannot withstand it; and (3) the time until the next earthquake and the times between consecutive earthquakes are independent exponential random variables each with parameter λ . Determine the distribution of the lifetime of the bridge.

Hint: Let X_1 be the time until the first earthquake. For $i \geq 2$, let X_i be the time between the $(i - 1)$ st and the i th earthquakes. Let N be the number of earthquakes until the one that the bridge cannot withstand. The lifetime of the bridge is $L = \sum_{k=1}^N X_k$. To identify the distribution function of L , calculate its moment-generating function. ■



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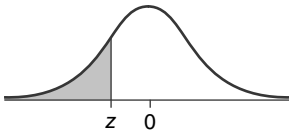
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Appendix Tables

1. Area under the Standard Normal Distribution to the Left of Z : Negative z
2. Area under the Standard Normal Distribution to the Left of Z : Positive z
3. Expectations and Variances of Common Distributions
4. Moment-Generating Functions of Common Distributions
5. Characteristic Functions of Common Distributions

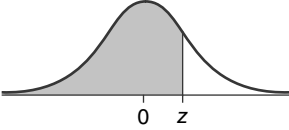
Table 1 Area under the Standard Normal Distribution to the Left of z : Negative z

$$\Phi(z) = P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-x^2/2} dx$$


Note that for $z \leq -3.90$, $\Phi(z) = P(Z \leq z) \approx 0$

z	0	1	2	3	4	5	6	7	8	9
−3.8	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
−3.7	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
−3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
−3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Table 2 Area under the Standard Normal Distribution to the Left of z : Positive z

$$\Phi(z) = P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-x^2/2} dx$$


z	0	1	2	3	4	5	6	7	8	9
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7703	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.5	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998
3.6	.9998	.9998	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999
3.7	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999
3.8	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999

Note that for $z > 3.89$, $\Phi(z) = P(Z \leq z) \approx 1$.

Table 3 Expectations and Variances of Some Common Distributions

Distribution of X	Probability Function of X	$E(X)$	$\text{Var}(X)$
Bernoulli with parameter p	$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$	p	$p(1 - p)$
Binomial with parameters n and p	$\binom{n}{x} p^x (1 - p)^{n-x}, x = 0, 1, \dots, n$	np	$np(1 - p)$
Poisson with parameter $\lambda > 0$	$\frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$	λ	λ
Geometric with parameter p	$p(1 - p)^{x-1}, x = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
Negative binomial with parameters r and p	$\binom{x-1}{r-1} p^r (1 - p)^{x-r}, x = r, r + 1, \dots$	$\frac{r}{p}$	$\frac{r(1 - p)}{p^2}$
Uniform over (a, b)	$\frac{1}{b - a}, a < x < b$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$
Standard Normal	$\frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2}, -\infty < x < \infty$	0	1
Normal with parameters μ and σ^2	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right], -\infty < x < \infty$	μ	σ^2
Exponential with parameter λ	$\lambda e^{-\lambda x}, 0 \leq x < \infty$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma with parameters r and λ	$\frac{\lambda e^{-\lambda x} (\lambda x)^{r-1}}{\Gamma(r)}, x \geq 0$	$\frac{r}{\lambda}$	$\frac{r}{\lambda^2}$

Table 4 The Moment-Generating Functions of Some Common Distributions

Distribution of X	Probability Function of X	$M_X(t)$
Bernoulli with parameter p	$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$	$pe^t + 1 - p$
Binomial with parameters n and p	$\binom{n}{x} p^x (1 - p)^{n-x}, x = 0, 1, \dots, n$	$(pe^t + 1 - p)^n$
Poisson with parameter $\lambda > 0$	$\frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$	$\exp[\lambda(e^t - 1)]$
Geometric with parameter p	$p(1 - p)^{x-1}, x = 1, 2, \dots$	$\frac{pe^t}{1 - (1 - p)e^t}$
Negative binomial with parameters r and p	$\binom{x-1}{r-1} p^r (1 - p)^{x-r}, x = r, r + 1, \dots$	$\left[\frac{pe^t}{1 - (1 - p)e^t} \right]^r$
Uniform over (a, b)	$\frac{1}{b - a}, a < x < b$	$\frac{e^{tb} - e^{ta}}{t(b - a)}$
Standard Normal	$\frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2}, -\infty < x < \infty$	$e^{t^2/2}$
Normal with parameters μ and σ^2	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right], -\infty < x < \infty$	$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$
Exponential with parameter λ	$\lambda e^{-\lambda x}, 0 \leq x < \infty$	$\frac{\lambda}{\lambda - t}$
Gamma with parameters r and λ	$\frac{\lambda e^{-\lambda x} (\lambda x)^{r-1}}{\Gamma(r)}, x \geq 0$	$\left(\frac{\lambda}{\lambda - t}\right)^r$

Table 5 The Characteristic Functions of Some Common Distributions

Distribution of X	Probability Function of X	$\phi_X(t)$
Bernoulli with parameter p	$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$	$pe^{it} + 1 - p$
Binomial with parameters n and p	$\binom{n}{x} p^x (1 - p)^{n-x}, x = 0, 1, \dots, n$	$(pe^{it} + 1 - p)^n$
Poisson with parameter $\lambda > 0$	$\frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$	$\exp[-\lambda(1 - e^{it})]$
Geometric with parameter p	$p(1 - p)^{x-1}, x = 1, 2, \dots$	$\frac{pe^{it}}{1 - (1 - p)e^{it}}$
Negative binomial with parameters r and p	$\binom{x-1}{r-1} p^r (1 - p)^{x-r}, x = r, r + 1, \dots$	$\left[\frac{pe^{it}}{1 - (1 - p)e^{it}} \right]^r$
Uniform over (a, b)	$\frac{1}{b - a}, a < x < b$	$\frac{e^{itb} - e^{ita}}{it(b - a)}$
Standard Normal	$\frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2}, -\infty < x < \infty$	$e^{-t^2/2}$
Normal with parameters μ and σ^2	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right], -\infty < x < \infty$	$\exp\left(i\mu t - \frac{\sigma^2 t^2}{2}\right)$
Exponential with parameter λ	$\lambda e^{-\lambda x}, 0 \leq x < \infty$	$\frac{\lambda}{\lambda - it}$
Gamma with parameters r and λ	$\frac{\lambda e^{-\lambda x} (\lambda x)^{r-1}}{\Gamma(r)}, x \geq 0$	$\left(\frac{\lambda}{\lambda - it}\right)^r$

Answers to Odd-Numbered Exercises

Section 1.2

1. $\{M, I, S, P\}$ and $\{I\}$. 3. The event of at least two heads. 5. (a) Yes; (b) no.
7. $\{x: 0 < x < 20\}; \{1, 2, 3, \dots, 19\}$. 9. One 1 and one even; one 1 and one odd; both even or both belong to $\{3, 5\}$. 11. $\{x: 7 \leq x \leq 9\frac{1}{6}\}; \{x: 7 \leq x \leq 7\frac{1}{4}\} \cup \{x: 7\frac{3}{4} \leq x \leq 8\frac{1}{4}\} \cup \{x: 8\frac{3}{4} \leq x \leq 9\frac{1}{6}\}$. 13. The experiment of flipping a coin until two tails appear consecutively. 15. For $1 \leq i \leq 3, 1 \leq j \leq 3$, by $a_i b_j$ we mean passenger a gets off at hotel i and passenger b gets off at hotel j . The answers are $\{a_i b_j: 1 \leq i \leq 3, 1 \leq j \leq 3\}$ and $\{a_1 b_1, a_2 b_2, a_3 b_3\}$, respectively.
17. $S = \{(x, y, z): 30 \leq x \leq 36, 28 \leq y \leq 33, 300 \leq z \leq 600\}; \{(x, y, z) \in S: x \geq 33, z < 435\}$. 19. (a) $F \cup EG$; (b) FE . 21. $E = \cap_{i=1}^n A_i$.
23. (a) True; (b) false; (c) true; (d) true. 25. (a) $S = \{H, TH, TTH, TTTH, \dots\}$; (b) The desired event is $E = \{H, TTH, TTTH, TTTTH, TTTTTH, \dots\}$. 27. $(-1/2, 1/2); \{0\}$. 39. For $n \geq 1$, let $B_n = A_n - \bigcup_{i=1}^{n-1} A_i$.

Section 1.4

1. No! 3. 0.40. 5. 0.9673. 7. 0.5. 9. 6/13, 4/13, and 3/13, respectively.
11. 0.769 13. 35%. 15. No, it is not! 17. 0.06811 and 0.1169, respectively.
19. (a) 0.41, (b) 0.21, (c) 0.38. 21. 15/29. 23. (a) 1/13; (b) 8/13; (c) 9/13.
25. 65% and 10%, respectively. 29. 7/11. 31. 3/11. 33. 19/36.
35. (a) 267/1000; (b) 533/1000. 37. 1/4. 39. 0.36. 41. 99/200.
45. Neither Q nor R is a probability on S .

Section 1.7

1. 2/3. 3. (a) False; (b) false. 5. 0. 7. 3/5. 15. Let $E_t = (0, 1) - \{t\}$, $0 < t < 1$.

Review Problems for Chapter 1

1. 0.54. 3. (a) $\{1, 3, 5, 7, 9\}$; (b) $\{3, 6, 9\}$. 5. $E_1 E_2 E_3 \cup E_1^c E_2 E_3 \cup E_1 E_2^c E_3 \cup E_1 E_2 E_3^c$. 7. (a) False; (b) true. 9. Let ao be the outcome that the Audi is operative at the random time, and let at be the outcome that it is totaled at that time, etc.

$S = \{(ao, jo), (ao, jt), (at, jo), (at, jt)\}$; $E = \{(ao, jo), (ao, jt), (at, jo)\}$. **11.** In both cases, in the experiment of tossing a die, let $A = \{1, 3, 5\}$; $B = \{5\}$. **13.** 50%. **15.** 18. **19.** 0.3. **21.** 6/13, 3/13, and 1/13, respectively. **23.** No, it is 1/4. **27.** Denote a box of books from publisher i by a_i , $i = 1, 2, 3$. Then the sample space is $S = \{x_1 x_2 x_3 x_4 x_5 x_6: \text{two } x_i\text{'s are } a_1, \text{two are } a_2, \text{and the remaining two are } a_3\}$. The desired event is $\{x_1 x_2 x_3 x_4 x_5 x_6 \in S: x_5 = x_6\}$. **29.** 1/43. **31.** 4/15. **33.** (a) $U_i^c D_i^c$, (b) $U_1 U_2 \cdots U_n$, (c) $(U_1^c D_1^c) \cup \cdots \cup (U_n^c D_n^c)$, (d) $(U_1 D_2 U_3^c D_3^c) \cup (U_1 U_2^c D_2^c D_3) \cup (D_1 U_2 U_3^c D_3^c) \cup (D_1 U_2^c D_2^c U_3) \cup (D_1^c U_1^c D_2 U_3) \cup (D_1^c U_1^c U_2 D_2) \cup (D_1^c U_1^c D_2^c U_2^c D_3^c U_3^c)$, (e) $D_1^c D_2^c \cdots D_n^c$. **35.** 173/216.

Section 2.2

1. 900,000; 427,608. **3.** Yes, it is. **5.** 0.00000024. **7.** $1/6^8$. **9.** 1/13. **11.** 0.00000000093. **13.** Yes, there are. **15.** 36. **17.** 0.89998. **19.** 4096. **21.** 5040; (a) 4536; (b) 4984. **23.** 0.0026. **25.** 30.43%. **27.** 4.77×10^{-13} . **29.** 0.469. **31.** 0.868. **33.** 0.031. **35.** 0.067.

Section 2.3

1. 0.0417. **3.** 6. **5.** 0.625. **7.** 0.875. **9.** 30. **11.** 3.04×10^{64} ; 2.41×10^{50} . **13.** 20,160. **15.** 0.985. **17.** 0.000333. **19.** 6.043×10^{-13} . **21.** 50,450,400. **23.** 0.05. **25.** 0.6. **27.** 2080. **29.** 0.00068. **31.** 0.0079. **33.** 0.985. **35.** 326,998,056 **37.** 0.003. **39.** 3.23×10^{-16} and 2.25×10^{-6} .

Section 2.4

1. 38,760. **3.** 6,864,396,000. **5.** 0.066. **7.** n/N . **9.** 10. **11.** 21,624. **13.** 2/33. **15.** 72,072. **17.** 71,680. **19.** $10,500x^3$. **21.** 0.548. **23.** 0.194. **25.** (a) 46,200; (b) 1/11. **27.** (a) 120,173/120,175; (b) 10,914/10,925. **29.** $2^{\binom{n}{2}}$. **31.** (a) 3^n ; (b) $(x+1)^n$. **33.** 0.00151. **35.** (a) 1820; (b) 6188. **37.** Royal Flush: 0.0000015; straight flush: 0.000014; four of a kind: 0.00024; full house: 0.0014; flush: 0.002; straight: 0.0039; three of a kind: 0.021; two pairs: 0.048; one pair: 0.42; none of the above: 0.5034445. **39.** 0.023333. **41.** -7,560. **43.** 4.47×10^{-28} . **45.** 36. **47.** (a) 0.665; (b) 0.619; (c) 0.597. **49.** 0.000346. **51.** 0.218. **53.** (a) $(1/n^m) \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)^m$; (b) $(1/n^m) \binom{n}{r} \sum_{i=0}^{n-r} (-1)^i \binom{n-r}{i} (n-r-i)^m$. **55.** 0.0625. **59.** $[1/(n+1)](2^{n+1} - 1)$. **61.** $\frac{\binom{t}{m} \binom{N-t}{n-m}}{\binom{N}{n}}$. **67.** $[1/(10 \times 2^{2n})] [(5 + 3\sqrt{5})(1 + \sqrt{5})^n + (5 - 3\sqrt{5})(1 - \sqrt{5})^n]$.

Section 2.5

1. (a) $1/\sqrt{\pi n}$; (b) $\sqrt{2}/4^n$.

Review Problems for Chapter 2

1. 127. 3. 31. 5. 0.278. 7. 7560. 9. 5/9. 11. 91573.
 13. 618,413,435,640,000. 15. 82,944. 17. 0.83. 19. In one way.
 21. 720. 25. 0.00036, 0.0095, 0.075, and 0.915, respectively. 27. Yes, it is because the probability of what happened is 0.02. 29. (a) 18/35; (b) 1/35. 31. 2/3243.
 33. 0.000396. 35. 0.0002112. 37. 617,512,896. 39. 369,600. 41. 0.00495.

Section 3.1

1. 60%. 3. 0.625. 5. 2/3. 7. 2/5. 9. 0.239 11. (a) 1/14; (b) 13/42; (c) 3/7.
 17. 0.0263. 19. 5%. 21. 0.615. 23. 0.883. 25. The crucial difference is reflected in the implicit assumption that both girls cannot be Mary.

Section 3.2

1. 0.1625. 3. 0.72. 5. 33.5%. 7. (a) 1/210; (b) 1/28. 9. 0.94. 11. 92.8%
 13. 0.059.

Section 3.3

1. 0.02625. 3. (a) and (c) are not, but (b) and (d) are. 5. 1/4. 7. 0.290. 9. 0.7.
 11. 0.467. 13. 0.73. 15. 21/31st 17. 58.76%. 23. $1/(n-1)$. 25. 7/10.
 27. 5/12.

Section 3.4

1. 3/5. 3. 0.98. 5. 0.77. 7. 0.607. 9. 0.2, 0.45, 0.35, respectively.
 11. 0.98. 13. 23.88%. 15. 0.59. 17. 0.056. 19. 0.21. 21. 0.36.
 23. 0.571. 25. 0.61.

Section 3.5

1. You should not agree! 3. Neither Mia is right nor Jim. 5. No, it is not!
 7. 0.00503. 9. (a) 0.526; (b) 0.076. 13. 0.9936. 19. 0.021. 21. 0.079.

- 23.** 0.226. **25.** 13/14. **29.** (a) $1 - [(n-1)/n]^n$; (b) this approaches $1 - (1/e)$.
31. 1 if $p = 1$; 0 if $p < 1$. **33.** For families with three children, they are independent. For families with two children and families with four children, they are not independent.
35. (a) $1 - (1-p_1)(1-p_2) \cdots (1-p_n)$; (b) $(1-p_1)(1-p_2) \cdots (1-p_n)$. **37.** $2p^4 - p^6$.
39. 0.329. **41.** (a) $1 - (1/2)^n$; (b) $\binom{n}{k}(1/2)^n$; (c) 0. **43.** (a) 0.168; (b) 0.286.
45. 0.4929. **47.** 0.00020997237. **49.** $2p^2 + 2p^3 - 5p^4 + p^5$. **51.** The only quantity not calculated in the hint is $P(U_i | R_m)$, which equals to $(i/n)^m / \sum_{k=0}^n (k/n)^m$.

Review Problems for Chapter 3

- 1.** 0.347. **3.** 0.167. **5.** 65%. **7.** 0.44. **9.** (a) 0.783; (b) 0.999775; (c) 0.217; (d) 0.000225. **11.** 11/16. **13.** 0.929. **15.** 0.615. **17.** 0.54. **19.** 3/4.
21. 0.42. **23.** No, regardless of the value of α , Lorna's chances are 50% to guess correctly. **25.** The answer is that it is the same either way.

Section 4.2

- 1.** Possible values are 0, 1, 2, 3, 4, and 5. Probabilities associated with these values are 6/36, 10/36, 8/36, 6/36, 4/36, and 2/36, respectively. **3.** $(1-p)^{i-1}p$ if $1 \leq i \leq N$; $(1-p)^N$ if $i = 0$. **5.** The answers are 1/2, 1/6, 1/4, 1/2, 0, and 1/3, respectively.
7. (a) 1/33; (b) 25/33; (c) 9/33; (d) 5/6. **13.** F is a distribution function.
15. $P(X = 0) = p_1 p_2 p_3 \cdots p_9$, and for $1 \leq i \leq 9$, $P(X = i) = p_1 p_2 p_3 \cdots p_{i-1} (1 - p_i)$.
19. 2/3. **21.** 26, 63, and 125, respectively.

Section 4.3

- 1.** $F(x) = 0$ if $x < 1$, 1/15 if $1 \leq x < 2$, 3/15 if $2 \leq x < 3$, 6/15 if $3 \leq x < 4$, 10/15 if $4 \leq x < 5$, and 1 if $x \geq 5$. **3.** Possible values are 2, 3, ..., 12. Probabilities associated with these values are 1/36, 2/36, 3/36, 4/36, 5/36, 6/36, 5/36, 4/36, 3/36, 2/36, 1/36, respectively. **5.** (a) $(9/10)^{i-1}(1/10)$, $i = 1, 2, \dots$; (b) $(9/10)^{(j-3)/2}(1/10)$, $j = 3, 5, 7, \dots$.
7. (a) 1/15; (b) 1/15; (c) 8; (d) $2/[n(n+1)]$; (e) $6/[n(n+1)(2n+1)]$.
9. $F(x) = 0$ if $x < 0$; $1 - (1/4)^{n+1}$ if $n \leq x < n+1$, $n = 0, 1, 2, \dots$. **11.** 0.1024.
13. $(1/2)^{n-1}$, $n \geq 2$. **15.** Possible values are 0, 1, 2, 3, 4, 5. Probabilities associated with these values are 42/1001, 252/1001, 420/1001, 240/1001, 45/1001, 2/1001, respectively. **17.** $(5/6)^{n-1} - 5(4/6)^{n-1} + 10(3/6)^{n-1} - 10(2/6)^{n-1} + 5(1/6)^{n-1}$, $n \geq 6$.

Section 4.4

- 1.** Yes, of course. **3.** -13. **5.** On the average, the player loses 38 cents per game.

7. 0.606. 9. Either 4 or 5 magazines should be ordered. 11. (b) 0, 44/27, 349/27, respectively. 13. $(79/28)a$. 15. 22. 17. 3. 19. 10.6. 21. $E(X)$ does not exist.

Section 4.5

1. The first business. 3. 3. 5. 12.234 and 3.498, respectively. 7. 55%. 9. 36.
13. (a) n and $n(n-1)$ (b) $(n+1)/2$ and $(n^2-1)/12$.

Section 4.6

1. Mr. Norton should hire the salesperson who worked in store 2.

Section 4.7

1. (a) $2/9$; (b) $p_X(x) = x/3$, $x = 1, 2$; $p_Y(y) = 2/(3y)$, $y = 1, 2$; (c) $2/3$; (d) $E(X) = 5/3$, $E(Y) = 4/3$. 3. (a) $k = 1/25$; (b) $p_X(x)$ is $12/25$ if $x = 1$, $13/25$ if $x = 2$; $p_Y(y)$ is $2/25$ if $y = 1$, $23/25$ if $y = 3$; (c) $E(X) = 38/25$, $E(Y) = 71/25$.
5. $\left[\binom{7}{x} \binom{8}{y} \binom{5}{4-x-y} \right] / \binom{20}{4}$, $0 \leq x \leq 4$, $0 \leq y \leq 4$, $0 \leq x+y \leq 4$. 7. $p(1, 1) = 0$, $p(1, 0) = 0.30$, $p(0, 1) = 0.50$, $p(0, 0) = 0.20$. 9. $\left[\binom{13}{h} \binom{13}{d} \binom{13}{c} \binom{13}{s} \right] / \binom{52}{13}$. 11. The desired quantity is 1. 13. 6. 15. $n/32$. 17. 23.41. 19. 3.786

Section 4.8

1. No, they are not. 3. $4/81$ and $4/27$, respectively. 5. 0.0179. 7. Show that for all x, y , $p(x, y) = p_X(x)p_Y(y)$, 9. 26. 11. $P(X = Y) = 1/(n+1)$, $P(X \leq Y) = 1 - [n/2(n+1)]$.

Review Problems for Chapter 4

1. It is $1/45$ for $i = 1, 2, 16, 17$; $2/45$ for $i = 3, 4, 14, 15$; $3/45$ for $i = 5, 6, 12, 13$; $4/45$ for $i = 7, 8, 10, 11$; and $5/45$ for $i = 9$. 3. 35. 5. 1.067. 7. They are 16, 16, 0.013, and 0.008, respectively. 9. (a) e^{-2t} ; (b) $e^{-2t}[1 + 2t + 2t^2 + (4t^3/3)]$ and $1 - e^{-2t} - 2te^{-2t}$, respectively. 11. 415.58. 13. $\left[\binom{13}{x} \binom{26}{9-x} \right] / \binom{39}{9}$, $0 \leq x \leq 9$.
15. $\left[\binom{13}{x} \binom{13}{6-x} \right] / \binom{26}{6}$, $0 \leq x \leq 6$. 17. $p(x) = [(7-x)/6]^n - [(6-x)/6]^n$, $x = 1, 2, 3, 4, 5, 6$. 19. 0 if $t < 1$, $[t]/n$ if $1 \leq t < n$, and 1 if $t \geq n$.
21. $(0.90)^{x+y-2}(0.10)^2$.

Section 5.1

1. (a) 3; (b) 0.78. 3. (a) 6; (b) 0 if $x < 1$, $-2x^3 + 9x^2 - 12x + 5$ if $1 \leq x < 2$, and 1 if $x \geq 2$; (c) $5/32$ and $1/2$, respectively. 5. (a) $1/\pi$; (b) 0 if $x < -1$, $\frac{1}{\pi} \arcsin x + \frac{1}{2}$ if $-1 \leq x < 1$, 1 if $x \geq 1$. 7. (a) $k = 1/16$, (c) $\alpha = 4$, $\beta = 6.735$. 9. 0.6836
11. Not a distribution. 13. 0.3327

Section 5.2

1. The probability density function of Y is $(1/12)y^{-2/3}$ if $-8 \leq y < 8$; 0, otherwise. The probability density function of Z is $(1/8)z^{-3/4}$ if $0 < z \leq 16$; 0, otherwise.
3. It is $\frac{1}{t}f(\ln t)$ if $t > 0$, 0 if $t \leq 0$. 5. $(3 \ln 2)2^y e^{-3(2^y)}$, $y \in (-\infty, \infty)$. 7. It is $\frac{1}{2\sqrt{t}}[f(\sqrt{t}) + f(-\sqrt{t})]$ if $t \geq 0$; 0, otherwise. 9. $e^{-t} + \frac{1}{t^2}e^{-1/t}$ if $0 < t < 1$; 0, otherwise.

Section 5.3

1. (a) 8. 3. The muffler of company B . 5. \$100,000. 7. 0.
9. $(5/9)h((t-32)/1.8)$; $1.8 \int_{-\infty}^{\infty} xh(x)dx + 32$. 11. 8.4. 17. (a) c_1 is arbitrary, $c_n = n^{-1/(n-1)}$; (b) ∞ if $n = 1$, and $n^{(n-2)/(n-1)}/(n-1)$ if $n > 1$; (c) $c_n e^{-nt}$, $t \geq \ln c_n$; (d) it exists if and only if $m-n < -1$. 23. $[(2\alpha - \alpha^2)/\lambda^2] + [(2\beta - \beta^2)/\mu^2] - [2\alpha\beta/\lambda\mu]$.

Section 5.4

1. (a) $2(1-y)$, $0 \leq y \leq 1$, (b) $2/3$ and $1/3$, (c) $1/4$, $1/2$, 0. 3. e^{-x} , $x > 0$; $(1/2)y$, $0 < y < 2$. 5. 0.417. 7. $1 - F_X(t) - F_Y(u) + F(t, u)$. 9. 0.36. 11. $3/4$.
13. F is not a joint distribution function. 15. $f_X(x) = (3/2)x^2 + 1/2$, $f_Y(y) = (3/2)y^2 + 1/2$. 17. $f_Y(y) = \frac{1}{2}(\ln y)^2$, $0 \leq y \leq 1$. 19. Yes, it is. 21. 12.5.
23. For $1 \leq i \leq n$, $E(Z_i) = 1/n$. 29. For $i = 0$, it is $1/8$; for $i = 1$, it is $3/4$; for $i = 2$, it is $1/8$. 31. $P(E) = (\frac{3\alpha}{\ell} - 1)^2$, if $\frac{\ell}{3} \leq \alpha \leq \frac{\ell}{2}$; $1 - 3(1 - \frac{\alpha}{\ell})^2$ if $\frac{\ell}{2} \leq \alpha \leq \ell$.

Section 5.5

1. $F(t)G(t)$; $F(t) + G(t) - F(t)G(t)$. 3. 0. 5. Not true. 7. Not independent.
9. Not independent. 11. For $r < R$, the probability is $(1 - \frac{r^2}{R^2})^n$; for $r \geq R$, it is 0.
13. 0.12. 17. They are not. 19. $1/2, 880$.

Review Problems for Chapter 5

1. $c = 14$; 0.27; 0 if $t < 8$, $2 - \frac{14}{t-1}$ if $8 \leq t < 15$, 1 if $t \geq 15$. 3. 0.467.

- 5.** $E(X) = 2$, $\text{Var}(X)$ does not exist. **7.** 0.748 **11.** The relation is valid.
13. $(r_2^2 - r_3^2)/r_1^2$. **15.** $f_X(x) = -\frac{3}{2}x^2 + \frac{3}{2}$, $0 < x < 1$, $f_Y(y) = -\frac{3}{2}y^2 + \frac{3}{2}$, $0 < y < 1$;
 29/64 **17.** They are not independent. **19.** 511/270. **21.** 0.2775
23. $[1 - \frac{\pi}{6}(\frac{r}{a})^3]^n$. **25.** $f_Y(y) = \frac{4(\ln y)^3}{15y}$, $y \in [e, e^2]$. $f_Z(z) = \frac{2z}{15}$, $z \in [1, 4]$.
 $f_W(w) = \frac{2(1+\sqrt{w})^3}{15\sqrt{w}}$, $w \in [0, 1]$.

Section 6.1

- 1.** 0.087. **3.** 0.054. **5.** 0.098. **7.** It is not fair to reject company's claim since there is 16.2% chance that two or more defective nails are found.
9. $p(x) = \binom{4}{x}(0.60)^x(0.4)^{4-x}$, $x = 0, 1, 2, 3, 4$;
 $q(y) = \binom{4}{(y-1)/2}(0.60)^{(y-1)/2}(0.4)^{(9-y)/2}$, $y = 1, 3, 5, 7, 9$. **11.** 0.108.
13. They are $np + p$ and $np + p - 1$. **15.** 0.219. **17.** 0.095. **19.** $k = 4$ and the maximum probability is 0.238. **21.** For $i = 0, 1, 4, 9$, $P(Y = i) = 250/7776$, $1275/7776$, $3126/7776$, and $3125/7776$, respectively. **23.** It is preferable to send in a single parcel. **25.** 6. **27.** No, it is not! **29.** $(\frac{1}{2})^n \sum_{i=0}^{\lfloor \frac{n+(b-a)}{2} \rfloor} \binom{n}{i}$.
33. 0.000028. **35.** 90,072. **39.** 0.317.

Section 6.2

- 1.** 0.9502. **3.** 0.594. **5.** 0.823. **7.** 0.063. **9.** 83%. **11.** For 0, 1, 2, and 3, using the binomial, the results are 0.93115, 0.06651, 0.00228, and 0.00005, respectively. Using the Poisson, they are 0.93125, 0.06634, 0.00236, and 0.00006, respectively.
13. 2/5 and 0.63, respectively. **15.** 0.00248. **17.** 0.307. **19.** 0.0028.
21. 0.013. **23.** 0.325. **25.** 0.273. **27.** 3.06% **29.** (a) $\binom{n}{k}(n-1)^{n-k}/n^n$.
33. (a) $e^{-\lambda\pi d^2}$, (b) $\sum_{i=0}^{n-1} [e^{-\lambda\pi d^2}(\lambda\pi d^2)^i]/i!$.

Section 6.3

- 1.** $\{NNN, DNN, NDN, NND, NDD, DND, DDN\}$. **3.** (a) 12; (b) 0.07.
5. 0.055. **7.** 0.42. **9.** x/n . **11.** 49. **13.** $(1/2)^{N+M-m+1} [\binom{N+M-m}{N} + \binom{N+M-m}{M}]$. **15.** 0.969. **17.** 0.74. **19.** 0.987. **21.** 0.6346. **23.** 11.42 **25.** 254.80 **27.** Negative binomial with parameters k and p . **29.** $\binom{2N-M-1}{N-1} (\frac{1}{2})^{2N-M-1}$.
31. 1/7. **33.** (a) D/N , (b) $(D-1)/(N-1)$.

Section 6.4

- 1.** 0.028. **3.** 0.171. **5.** 0.19. **7.** $p_{X|Y}(x|2) = \binom{8}{x}(5/7)^x(2/7)^{8-x}$, $0 \leq x \leq 8$.

Review Problems for Chapter 6

1. 0.0009. 3. 1.067. 5. 0.179. 7. 0.244. 9. 0.285. 11. 2 and 2, respectively.
 13. 0.967. 15. X is binomial with parameters 6 and 0.255; $E(X) = 1.53$; $\text{Var}(X) = 1.13985$.
 17. 0.772. 19. 0.115 21. 0.10111. 23. (a) 0.000368. (b) 0.000224.
 25. (a) $\left(\frac{w}{w+b}\right)^{n-1} \left(\frac{b}{w+b}\right)$. (b) $\left(\frac{w}{w+b}\right)^{n-1}$. 27. k/n . 31. 71.9548. 33. 0.1029.
 35. 0.22.

Section 7.1

1. $3/7$. 3. $a = -6, b = 6$. 5. $1/2$. 7. \$1000. 9. a/b . 11. $[nX]$ is a random number from the set $\{0, 1, 2, \dots, n-1\}$.
 13. $-\ln t, 0 < t < 1$; 0, otherwise.
 15. 0 if $t < 0$; 1 if $0 \leq t < 1/2$; $1/(4t^2)$, if $t \geq 1/2$. 17. $g(X) = 3X^2 - 4X + 1$.
 23. $n/(n+1)$; $1/(n+1)$.

Section 7.2

1. 0.4963. 3. 0.0735. 5. They are all equal to 0. 7. 0.2%, 4.96%, 30.04%, 45.86%, and 18.94%, respectively.
 9. 0.4796. 11. 0.388. 13. 0.000016.
 17. 0.7047. 19. 1 and $1/4$. 21. π . 23. $e^{\alpha^2/2}$ 25. $\frac{1}{\sigma t \sqrt{2\pi}} \exp \left[-\frac{(\ln t - \mu)^2}{2\sigma^2} \right]$, $t \geq 0$; 0, otherwise. 27. 10 copies. 33. No, it is not reasonable.

Section 7.3

1. 0.6046. 3. 0.4806. 5. 0.6826. 7. 0.488. 9. 0.0793. 11. 50,000.

Section 7.4

1. 0.0001234. 3. $\exp(-y - e^{-y}), -\infty < y < \infty$. 5. (a) 0.1535; (b) 0.1535.
 7. \$3658.88. 9. 0.112. 11. \$323.33. 13. $1/8$.
 15. $\binom{n}{k} (e^{-\lambda t})^k (1 - e^{-\lambda t})^{n-k}$. 19. $5/(2\lambda)$. 21. $\lambda/(\lambda + \mu)$
 23. $\frac{\lambda\mu}{\lambda + \mu} (e^{-\lambda t} + e^{-\mu t}), t \geq 0$.

Section 7.5

3. 0.3208. 7. 0.338. 9. 8 hours and 20 minutes. 11. $(2n)! \sqrt{\pi} / (4^n \cdot n!)$.
 13. Howard's waiting time in the queue is gamma with parameters 2 and $n\lambda$.
 15. 0.8413.

Section 7.6

1. Yes, it is. 3. $c = 1260$; $E(X) = 5/11$; $\text{Var}X = 5/242$. 5. 0.17. 7. 0.214.
 9. (a) $E(Y) = a + (b - a)\frac{\alpha}{\alpha + \beta}$, $\text{Var}(Y) = \frac{(b-a)^2\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2}$. (b) The probability density function of Y is $f(y) = \frac{1}{b-a} \cdot \frac{1}{B(\alpha, \beta)} \left(\frac{y-a}{b-a}\right)^{\alpha-1} \left(\frac{b-y}{b-a}\right)^{\beta-1}$, $a < y < b$. (c) 0.26.
 13. $E(X) = 0$ and $\text{Var}(X) = n/(n-2)$.

Section 7.7

1. (a) $\lambda(t) = 3t^2/(1 - t^3)$ if $0 \leq t < 1$; 0, if $t < 0$. (b) If the device has survived 50, 90, and 99 hours, the instantaneous failure rates are 0.857, 8.07, and 98.997, respectively. The corresponding *approximate* probabilities are 0.00857, 0.0807, and 0.98997, respectively.
 3. $\lambda(t) = (1/6)(t^3 e^{-t}) / [-1 + e^{-t}(1 + t + (1/2)t^2 + (1/6)t^3)]$ if $t \geq 0$; 0 if $t < 0$; The desired instantaneous failure rate is 0.433, and the corresponding probability is 0.0083.
 5. It is decreasing for $\alpha < 1$; it is increasing for $\alpha > 1$; and it is constant for $\alpha = 1$, in which case $\lambda(t) = 1$ and so the Weibull is exponential with parameter 1.

Review Problems for Chapter 7

1. 5/12. 3. 0.248. 5. 0.0262. 7. 1. 9. 0.99966. 11. $(1/\pi)e^{-17/4}$.
 13. 0.1401. 15. 0.36788. 17. 203.38 19. 0.44458. 21. $1 - F_1(t)F_2(t) \cdots F_n(t)$.

Section 8.2

1. (a) 38, (b) 26, (c) 50. 3. $-1/245$. 5. $-224/475$. 13. They are dependent but uncorrelated. 17. $1/9$. 19. Negatively Correlated.

Section 8.3

1. 112. 3. They are -1 and $-1/12$, respectively. 5. No! it is not. 7. Show that $\rho(X, Y) = -0.245 \neq \pm 1$.

Section 8.4

1. $p_{X|Y}(x|y) = (x^2 + y^2)/(2y^2 + 5)$, $x = 1, 2$, $y = 0, 1, 2$. The desired conditional probability and conditional expectation are $5/7$ and $12/7$, respectively. 3. $1/y$, $0 < x < y$, $0 < y < 1$. 5. $17/86$. 7. $\frac{x+y}{(1/2)+y}$, $0 \leq x \leq 1$, $0 \leq y \leq 1$; 0, otherwise.
 9. $[(\binom{10}{g})(\binom{5}{3-g})]/(\binom{15}{3})$, $g = 0, 1, 2, 3$. 11. $10/33$. 13. $\frac{xy}{54}$, $x = 4, 5$, $y = 1, 2, 3$.

$p_{Y,Z}(y,z) = \frac{yz}{18}$, $y = 1, 2, 3$, $z = 1, 2$. $p_{X,Z}(x,z) = \frac{xz}{27}$, $x = 4, 5$, $z = 1, 2$.

15. (a) $1/2$; (b) $f_{X|Y}(x|y) = e^{-x+|y|}$, $x > |y|$; $f_{Y|X}(y|x) = \frac{1}{2x}$, $-x < y < x$; (c) $E(Y|X = x) = 0$ and $\text{Var}(Y|X = x) = x^2/3$. **17.** $\frac{1}{2}(\ln y)^2$, $0 \leq y \leq 1$.

19. $P(X = x | Y = y) = \frac{\binom{m}{x}\binom{n-m}{y-x}}{\binom{n}{y}}$. The conditional distribution of Y given that $X = x$ is binomial with parameters $n - m$ and p . **21.** The conditional distribution of $N(t)$ given $N(s) = k$ is Poisson with parameter $\lambda(t - s)$.

25. (a) $f(x, y) = 2$ if $x \geq 0$, $y \geq 0$, $x + y \leq 1$; 0, elsewhere. (b) $f_{X|Y}(x|y) = \frac{1}{1-y}$, $0 \leq x \leq 1 - y$, $0 \leq y < 1$. (c) $E(X | Y = y) = (1 - y)/2$, $0 < y < 1$.

27. They are independent.

Section 8.5

1. $g(u, v) = (1/4)e^{-(u+v)/2}$, $v > 0$, $u > 0$. **3.** $(1/2\pi)re^{-r^2/2}$, $0 < \theta < 2\pi$, $r > 0$.

7. For $\lambda \neq \mu$, it is $f(t) = \frac{\mu - \lambda + \lambda p}{\mu - \lambda} \lambda e^{-\lambda t} + \frac{\lambda p}{\lambda - \mu} \mu e^{-\mu t}$. For $\lambda = \mu$, it is $f(t) = p\lambda t e^{-\lambda t} + (1 - p)\lambda e^{-\lambda t}$.

9. $g(u, v) = (1/v)e^{-u}$, $u > 0$, $1 < v < e^u$. **11.** (a) $g(u, v) = [\lambda^{r_1+r_2} u^{r_1+r_2-1} e^{-\lambda u} v^{r_1-1} (1-v)^{r_2-1}] / [\Gamma(r_1)\Gamma(r_2)]$, $u > 0$, $0 < v < 1$; (c) U is gamma with parameters $r_1 + r_2$ and λ ; V is beta with parameters r_1 and r_2 .

13. The joint probability density function of $X + Y$ and X/Y is the product of their marginal probability density functions. So they are independent.

Review Problems for Chapter 8

1. $\rho(X, Y) = \rho(Y, Z) = \rho(X, Z) = 0$. **3.** 52/625. **5.** 6. **7.** 0.7225. **9.** 0.563.

Section 9.1

1. 6. **5.** \$1,202.48. **7.** 3.786. **11.** $1/(\lambda + \mu)$. **13.** $\gamma/(\lambda + \mu + \gamma)$. **15.** It will take 4,407.286 business days, on average, until there is a lapse of two days between two consecutive applications. Since an academic year is 9 months long the admission officers should not be concerned about this rule at all.

17. 7.805.

Section 9.2

1. 0.26172. **3.** $p(x) = \frac{\binom{x-1}{3}\binom{1}{1}\binom{20-x}{3}}{\binom{20}{7}}$, $x = 4, 5, \dots, 17$. **9.** $-\sigma/\sqrt{\pi}$.

Section 9.3

1. 0.5987. **3.** (a) 0.68; (b) 0.8. **5.** (μ_X, μ_Y) is the point at which the maximum of f is obtained.

Review Problems for Chapter 9

1. 3.21. 3. $2/\pi$. 5. 7.5 minutes past 10:00 A.M. 9. Let $U = [X_{(1)} + X_{(3)}]/2$. $g_U(u)$ is $12u^2$ for $0 < u < 1/2$ and $12(u-1)^2$ for $1/2 \leq u < 1$. $E(U) = 1/2$.

Section 10.1

1. $(1/5)(e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t})$. 3. $2e^t/(3 - e^t)$, $t < \ln 3$; $E(X) = 3/2$.
 5. (a) $M_X(t) = (12/t^3)(1 - e^t) + (6/t^2)(1 + e^t)$, $t \neq 0$; $M_X(t) = 1$ if $t = 0$.
 (b) $E(X) = 1/2$. 9. (a) $M_X(t) = \exp[\lambda(e^t - 1)]$; (b) $E(X) = \text{Var}(X) = \lambda$.
 11. $E(X) = 1/p$, $\text{Var}(X) = (1-p)/p^2$. 13. $p(1) = 5/15$, $p(3) = 4/15$, $p(4) = 2/15$,
 $p(5) = 4/15$; $p(i) = 0$, $i \notin \{1, 3, 4, 5\}$. 15. $E(X) = 3/2$, $\text{Var}(X) = 3/4$.
 17. 15,000 and $3000\sqrt{5}$, respectively. 19. 8/9. 21. 0.000004316. 23. $(\frac{\lambda}{\lambda-t})^r$,
 r/λ , r/λ^2 .

Section 10.2

5. 0.083. 7. $(\frac{t}{y})(\frac{\lambda_2}{\lambda_1 + \lambda_2 + \lambda_3})^y(\frac{\lambda_1 + \lambda_3}{\lambda_1 + \lambda_2 + \lambda_3})^{t-y}$. 9. (a) 0.9525; (b) 0.9525; (c) 0.4641.
 11. 0.2676. 13. 0.8413. 15. 0.8671.
 17. It is $f(t) = \frac{1}{100}te^{-t/10} \cdot e^{-1} + \frac{1}{10}e^{-t/10} \cdot (1 - e^{-1})$. The desired expected value is 13.68
 minutes. 19. $X_1 + X_2 + \cdots + X_k$ is binomial with parameters n and $p = p_1 + p_2 + \cdots + p_k$.

Section 10.3

1. 3. 5. $\phi_X(t) = \frac{1}{4}e^{-2it} + \frac{1}{8}e^{-it} + \frac{1}{8}e^{it} + \frac{1}{2}$, $\phi_X(2\pi) = 1$. 7. $\phi_Y(t) = e^{-t^2}$.
 $g(x) = \frac{1}{4\pi}e^{-x^2/4}$. 9. $\phi_Y(t) = [\phi(t/n)]^n$. 13. The characteristic function of both
 sums is $e^{-2|t|}$. 17. X is 0 with probability 1. 19. $\phi_Y(t) = \sum_{n=1}^{\infty} [\phi(t)]^n \alpha_n$.

Section 10.4

3. (a) 0.4545; (b) 0.472. 5. (a) 0.4; (b) 0.053; (c) 0.222. 7. 15 days earlier. 9. It
 is $\leq 1/\mu$. 11. A sample of size 209. 13. 2500 or higher. 17. 1.

Section 10.5

1. The limit is $1/2$ with probability 1.

Review Problems for Chapter 10

1. 2.22 and 0.299. 3. 1 and 4, respectively. 5. 0.393. 7. 0.9984. 11. $1/9$.
 13. X is 0 with probability 1.



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Index

The following *abbreviations* are used in this index: r.v. for random variable, p.d.f. for probability density function, p.m.f. for probability mass function, and m.g.f. moment-generating function, and CF for characteristic function.

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